Feedzai Platform

Feedzai is a platform for managing financial risk. It relies on a cutting-edge enterprise Data Science platform that integrates big data analytics and Machine Learning techniques with a native runtime production environment, and a Risk Operations platform focused on boosting the efficiency of financial crime prevention and increase detection accuracy.

Follow these guides to learn how to install and operate the Feedzai platform.

Learn more about Feedzai platform

Skip to main content

Loading ...

Default Page

Markdown page example

You don't need React to write simple standalone pages.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/api-reference/authentication

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Authentication

Feedzai allows you to authenticate and protect your data through JSON Web Tokens (JWTs). A JSON Web Token (JWT) is a piece of information that is transmitted as a JSON object from clients to Feedzai, and its purpose is to ensure that client requests (e.g. transactions for processing) have not been compromised.

These tokens are generated using the clientâs private key, which can be verified with the corresponding public key. The public key must be shared with Feedzai beforehand to verify that the token sent with the request belongs to the client.

Overviewâ

JWTs protect against different attacks using the following mechanisms:

- **Encryption/signing** so that only authenticated users can access sensitive data.
- **Expiration dates** to protect against replay attacks.
- **JWT revocation** to block exposed JWTs as soon as possible.

You can see more details on integrating JWTs with your software in the sub-sections below, but in general you will:

1. Generate a private-public key pair to sign the JWT.
2. Provide the public key to Feedzai in [JSON Web Key Set] format.
3. Frequently generate new JWTs with a reduced validity period (e.g. 5 minutes). Each JWT is signed with your private key generated in step 1.
4. For each request to Feedzai, include a JWT in the `Authorization` header.

Given the recommended algorithm and parameters, you can choose the tool that best fits your system. A comprehensive list of tools is available at the [libraries section of jwt.io].

danger

Third-party security vulnerabilities: When choosing a library, make sure it has no known security vulnerabilities.

JSON Web Keyâ

A key pair is necessary to cryptographically sign and verify the JWTs sent with your requests.

- You will use the private key to sign your JWTs.
- Feedzai will use the public key to verify the signature, and therefore the authenticity of the requests.

To create a JWK set, you need to choose the signing algorithm and the JWK parameters. Feedzai recommends the `RS512` algorithm (âRSASSA-PKCS1-v1_5 using SHA-512â) with a 4096-bit key. See the sections below to learn how to generate a JWK set with standard tools.

danger

Beware of private key parameters: Construct the JWK set so that you **do not** send any of the private key parameters.

Your JWK set should look similar to this:

```
{
    "keys": [
        {
            "e": "AQAB",
            "n":
```

```
"187EWmXQwXi08P8bqSOSeOiyOyJGsojjmeL6qwLLjTBfxxeNgTtivzG8YdCgrLnBQWy4j9FqQ5wbqyOwf6CBv6xj9ZnqyyS
cle7ubAKRt4RKtv4yrxVb1g5ZWcfD6yWDIH1jRy5oGQEwWRg9918Z73OS4SlYe8rMDiWkFPybt1vfGBdGqbvm374XNfLRb0s
F7cPCRcBa3Vdzmg5BgoN9kFUXIvFKjXdgZQGEEIonLL8ZSmqq993VBVLOPh02OiJ91DVvb9JMIFAqROHQNyzchbOWgEPUKoR
BRDXSCus9ZSpHuUl7iL7qKLaQdPyO1ffIG0o7kF6allFG6OBcnzqX8r",
              "kty": "RSA",
              "kid": "a28db6df-f2f4-4267-be46-371502768aa1sig",
              "use": "sig"
         }
     ]
}

{
    "kid":"mykey2",
    "kty":"RSA",
    "alg":"RS512",
    "use":"sig",
    "e":"AQAB",
    "n":"k51ZPyu9gocn508aXzob_hSKaIAnJApY4pY3h1oym1Z15-6DN3WxB3YHUyXPRd-iBzF7177XjQAywTzAaz-
PKLZ4bC52noanLx7Ihyf3nQAOwhEv16zaJLI-
HqMEMp6JKPjKJ8LKanVnvxWNI0zUBwcBOwt9cxph0evyVO94cXA0RZxUZxa7C3FuSYkO2zWorMH0ZkC3harkJx0C9Q7nBiGW
PQmB8ZkLx04iWlm8ldyxDJPKl-PC4-
gS3y2mgPfZVAaK_VGod88PUmF4vFax7a03v2VLtn9wm8UWs9SXATxJ5cExgIIvFkdsZAcS3hWnhUe3Of4zAEbKh2Ah5DtfY9
xvtsU8FGlak7wLCwVnubWWglI2hmFvFU7Jsyzb_L-
jD7XCYrg7QuFV8mX2jVcTbsSxnd8GLRNbvmztvsjfaQHXEHraTRJFoGO1Y3JXDiY8Kj40OQXxBNXaFxHda9lBfzWfDkgWvcy
eiTPeWpKc6cS1zcDaQNRzZr3aZR-
gI4U8YTAXcIKDdom2GCQSx9WfmFq3uAj95duL5CgyGfw7L-02c6ckRr6kBAKebURkSsZXOyTO8bdbS8GzPMD59CrRL2SnJJe
qHyrEav8JCu0x3UCXbyqlfqQgzVsXNpFUMIS-sUhe3nyVQOeJ_WoYq7oAbjLpx_Bw5elQJf4zJbJd2Pk"
}
```

Copy

In the aforementioned example, the following parameters are used:

Name	Extended name	Description
`e`	Exponent	RSA public exponent e.
`n`	Modulus	RSA public modulus n.
`kty`	Key Type	Should be `RSA` to symbolize you are using the RS512 algorithm.
`kid`	Key Id	Should specify a key identifier of your choice.
`use`	Public Key Use	Should be `sig` to note this key will be used to verify JWTs.

How to generate a key-pair and JWKâ

1. Download Feedzai's JWT Generator.
2. Unpack the tar.gz artifact.
3. Check generator options:

   ```
   ./bin/generate-asymmetric --help
   ```

   Copy

   info

   Make sure you repeat steps 1 through 3 for a total of two private-public key pairs: one for `prod` and another one for `qa`.

4. Run the JWT Generator using the following command. Adjust the parameter values according to your needs:

   ```
   ./bin/generate-asymmetric \
     --mode JWK \
     --generateKeys \
     --privateKey "/path/to/privateKey" \
   ```

```
  --publicKey "/path/to/publicKey" \
  --kid "key_id"
```

Copy

The JWK Set JSON will be displayed on your standard output. Alternatively, you can use the `--jwk` parameter to specify a file where it should be stored instead.

5. Store the generated public and private key files securely as they will be required to generate JWTs. **Never share your private key.**
6. Manage JWKs in Pulse.

JSON Web Tokensâ

Once you have a private key, you can generate your JWTs. All JWTs are composed of a set of headers, a set of claims and a signature.

JWT Headersâ

Besides the algorithm (`alg`), the only mandatory header for JWTs is the `kid` (the key ID), which indicates which key was used to sign the token and **must** match a valid key in the JWK Set used for validation. The set of headers should be similar to:

```
{
  "kid": "keyID",
  "alg": "RS512"
}
```

Copy

JWT Claimsâ

Typically, the following claims should be present on the JWTs:

Claim name	Extended name	Description
`iss`	Issuer name	Identifies the issuer of the JWT. This value will be provided by Feedzai, and it should match the tenant id, if applicable.
`exp`	Expiration Time	Date after which the JWT must be rejected.
`nbf`	Not before Time	Date before which the JWT must be rejected.
`jti`	JWT id	Identifier used to revoke JWTs.
`sub`	Subject	Identifies the user of the JWT.

Your final set of claims should be similar to:

```
{
    "iss": "feedzai",
    "exp": 1523022355,
    "nbf": 1523022055,
    "jti": "a00cd6dd-4e55-4694-aeb5-4050e7d063b6",
    "sub": "700e55f9-15b0-4f95-8094-960138a0d886"
}
```

Copy

How to generate a JWTâ

1. Download Feedzai's JWT Generator.
2. Unpack the tar.gz artifact.
3. Check generator options:

```
./bin/generate-asymmetric --help
```

Copy

4. Run the JWT Generator using the following command. Adjust the parameter values according to your needs:

```
./bin/generate-asymmetric \
  --mode JWT \
  --privateKey "/path/to/privateKey" \
  --publicKey "/path/to/publicKey" \
  --iss "issuer" \
  --sub "subject" \
  --exp "$((EPOCHSECONDS + 300))" \
  --kid "key_id"
```

Copy

The JWT will be displayed on your standard output. Alternatively, you can use the `--jwt` parameter to specify a file where it should be stored instead.

How to send the signed JWTâ

danger

Note that you can only send the signed JWT after the corresponding JWK is uploaded to Pulse (see Manage JWKs in Pulse).

Send the signed JWT in the `Authorization` header of HTTP requests made to Feedzai's API, as follows:

```
Authorization: Bearer <your_jwt>
```

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Manage JWKs in Pulseâ

Once you create the JWK set, the following JWK steps must be taken:

1. Log into Pulse and confirm the current list of Authorization keys. Before revoking any key, make sure you are no longer using it.
2. Add the new key to Pulse.
3. Start signing messages with JWTs based on the new key.
4. Revoke the old key.
5. Delete revoked Authorization keys.

List Authorization keysâ

To list the existing Authorization keys, access the **Administration** page and select the **Authorization Keys** tab on the sidebar.

Select a label name or expand to see the Authorization keys associated with that label:

Add Authorization keysâ

To add a new Authorization key, select a label name and upload the JSON file that contains the new key:

info

The new key must be associated with an existing label.

tip

Make sure you start signing the new messages with the new key.

Revoke Authorization keysâ

The following animation illustrates how to add a new key and revoke the current one.

danger

An Authorization key should only be revoked once a new Authorization key has been added (see previous step) and JWTs are already being signed with the new key.

Delete revoked Authorization keysâ

By deleting an Authorization key, you are permanently removing it from the list of keys. Only revoked keys can be deleted.

API Resources

This section provides you with the information you need to work with the API.

Endpoints

Learn all the operations you can do with Feedzai's API and how to communicate with its multiple endpoints.

Additional Resources

Find information about authentication, response and errors, generated fields, among other resources necessary to successfully work with the API.

Postman Suite

Postman collection and Environment for testing purposes.

Postman

- Download Postman Collection
- Download Postman Local Environment

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Responses and Errors

When a client makes a request to an HTTP server and the server successfully receives their request, the server notifies the client if the request was successfully processed or not. Read this page to understand what each response and error means.

Successful responseâ

Successful calls to Feedzai's API result in a response with HTTP 200 code. Please check each endpoint's documentation to see the structure of a successful response.

Warning responseâ

A warning response is a subtype of a successful response which is sent with the HTTP 200 code. The transaction is also processed. The response returns:

- `status` field with the response 'warning'.
- `warnings` field with the reasons for this outcome..

Error responseâ

When a call to Feedzai's API results in an error, the response object returned always has the same fields. In particular, the `status` field is shared with the successful response objects of all endpoints. The **Error Response** object is defined as follows:

Name	Field Type	Description
`code`	String	An indicative code for the error. See possible values below.
`errors`	Array of strings	A list of free text explanations for the error.
`status`	String	Always with value `error`.

Generic Error codes - Possible Valuesâ

Error code	Description	HTTP Code
`authenticationError`	The request could not be authenticated.	`401`
`authorizationError`	The authenticated user does not have permission to access the endpoint.	`403`
`parseError`	The request could not be parsed.	`400`
`invalidParameterError`	The URL parameter provided is invalid.	`400`
`invalidTimestampError`	The timestamp provided in the request is invalid.	`400`
`nonexistentEndpoint`	The endpoint being called does not exist.	`404`
`unsupportedMethod`	The HTTP method to call this endpoint is not supported.	`405`
`tooManyRequests`	The allowed request rate was exceeded.	`429`
`internalError`	Feedzai's system produced an internal error.	`500`
`timeoutError`	Feedzai's system failed to produce a timely response.	`504`

Please check each endpoint's documentation for specific error codes.

Responses for timestamps out of rangeâ

This subsection describes responses for requests with timestamps that are out of the valid range.

Endpoint	Timestamp	Response HTTP code	Error or Warning	Is transaction processed?
Authorization	`event_occurred_at` in the past	400	Error	No
Authorization	`event_occurred_at` in the future	400	Error	No
Info, Clearing, Feedback	`event_occurred_at` in the past	200	Warning	Yes
Info, Clearing, Feedback	`event_occurred_at` in the future	400	Error	No

Error Response Example:

```
{
    "status": "error",
    "code": "invalidTimestampError",
    "errors": [
        "Event has timestamp in the future: <EVENT_OCCURRED_AT> | Now is:
'2018-09-04T16:41:30.774Z'"
    ]
}
```

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Warning response example:

```
{
    "lifecycle_id": <LIFECYCLE_ID>,
    "warnings": [
        "Event has timestamp in the past: <EVENT_OCCURRED_AT> | Now is:
'2018-09-04T16:40:40.522Z'"
    ],
    "status": "warning"
}
```

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Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/concepts/concepts-overview

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Overview

Welcome to the Feedzai Concepts Page. In this document you will find useful information regarding conceptual elements.

Conceptsâ

- FIV markers
- Historic Events
- Multi-Tenancy
- Feedback Update Strategy
- PB Delegate
- Transaction Block

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Feedback Update Strategy

As of version 2.8.0 (and 2.5.5, 2.7.2) we have introduced a configurable strategy to control how events are updated when a feedback is received.

This page focuses on the concepts of feedback and labeling strategies. For configuration details, refer to the Pulse Configuration Reference.

How feedback labeling worksâ

When a feedback is received by Pulse there is a process in place to label events related with the feedback. Each of the labeled events will generate a new feedback event that is injected into the workflow and later to be later consumed by Case Manager.

In previous versions this strategy was not configurable and it meant that a feedback would generate `n` feedback events, where `n` is the number of related events (e.g. initiation, info).

In this new approach this strategy is configurable and supports two strategies:

- `ALL` : all events sharing the same `lifecycle_id`
- `SINGLE` : a feedback update per Case Manager event with the same `lifecycle_id` , according to the event type precedence

info

For Cards all clearings will receive feedback labels as they are independent Case Manager events

Strategy examplesâ

To better understanding how the feedback strategy works take the following example:

- A transfer initiation reaches Pulse
- Two info requests are made to update the initiation
- A feedback event is made

Using the `ALL` feedback strategy would mean three feedback events would be reinjected into the workflow and later processed by Case Manager. This ensures consistency across all related events by propagating the label to all events with the same `lifecycle_id` .

In the database we would be able to see all related events with the feedback label:

Event date	lifecycle_id	event_type	FDZ_FEEDBACKS
Tue, 28 Jan 2025 00:56:27	b8298fd6-7b96-4cb8-b5e6-3f0c13de489d	feedback	{}
Tue, 28 Jan 2025 00:56:18	b8298fd6-7b96-4cb8-b5e6-3f0c13de489d	info	{"fraud": {"@type": "bincla", "metadata": {"inject": false, "lifecycle_id": "b8298fd6-7b96-4cb8-b5e6-3f0c13de489d"}, "realOutcome": false, "reportInstant": 1738025787583}}
Tue, 28 Jan 2025 00:56:17	b8298fd6-7b96-4cb8-b5e6-3f0c13de489d	info	{"fraud": {"@type": "bincla", "metadata": {"inject": false, "lifecycle_id": "b8298fd6-7b96-4cb8-b5e6-3f0c13de489d"}, "realOutcome": false, "reportInstant": 1738025787583}}
		transfer_initiation	

Event date	lifecycle_id	event_type	FDZ_FEEDBACKS
Tue, 28 Jan 2025 00:56:08	b8298fd6-7b96-4cb8-b5e6-3f0c13de489d		{"fraud": {"@type": "bincla", "metadata": {"inject": false, "lifecycle_id": "b8298fd6-7b96-4cb8-b5e6-3f0c13de489d"}, "realOutcome": false, "reportInstant": 1738025787583}}

On the other hand, using the `SINGLE` feedback strategy would mean only one event would be reinjected into the workflow, minimizing the number of events that would be processed by Case Manager.

To choose the event to update a event type precedence is taken into account. In outbound the event to updated is calculated as follows:

- Choose the event type according to a precedence of (from highest to lowest): info, transfer_initiation, settlement
- If two events have matching precedence the most recent one is selected

In the database we would be able to see the most recent event with the feedback label:

Event date	lifecycle_id	event_type	FDZ_FEEDBACKS
Tue, 28 Jan 2025 00:56:27	b8298fd6-7b96-4cb8-b5e6-3f0c13de489d	feedback	{}
Tue, 28 Jan 2025 00:56:18	b8298fd6-7b96-4cb8-b5e6-3f0c13de489d	info	{"fraud": {"@type": "bincla", "metadata": {"inject": false, "lifecycle_id": "b8298fd6-7b96-4cb8-b5e6-3f0c13de489d"}, "realOutcome": false, "reportInstant": 1738025787583}}
Tue, 28 Jan 2025 00:56:17	b8298fd6-7b96-4cb8-b5e6-3f0c13de489d	info	{}
Tue, 28 Jan 2025 00:56:08	b8298fd6-7b96-4cb8-b5e6-3f0c13de489d	transfer_initiation	{}

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FIV (Financial Inclusion Vulnerability) markers

Business Contextâ

To ensure fair treatment and fairness for all, inclusion measures for all customers, especially those in vulnerable circumstances are required to be incorporated within a companyâs strategy, services and products.

A vulnerable customer is recognized as someone who, due to their personal circumstances is particularly susceptible to detriment, especially when a firm is not acting with appropriate levels of care.

Feedzai Implementationâ

`customer_fiv_markers` is a new HSBC custom field in the Customer Reference Data.

This way, the Payments and the Digital activity data streams get enriched with this data, and it is thus available to be used in Feedzai Pulse Rules and Metrics.

Since a user might have multiple FIV markers a custom UDF is available to fetch an FIV marker if it exists and has not expired.

Due to the dynamic nature of this field - new FIV markers might be created at any time and number of markers per customer is variable - it has been implemented as a single String variable that Feedzai expects to contain an encoded JSON array, where each entry has the following schema:

```
{
  "fiv_marker_code": "FM1",
  "fiv_marker_exp_date": "2023-08-28"
}
```

Copy

note

Since the field is simply a String that by agreement is in the JSON format, the quotes inside have to escaped as `\"`

Additionally, since FIV markers are included in the same field they can't be `PATCH`ed individually as other attributes, meaning if we want to add a single FIV marker to the existing list, the `PATCH`ing logic as to be done on the caller side.

In practice, this might be equivalent to fetching the Customer Reference Data (via the `GET` method), appending to the array and submitting an update to the Customer Reference Data (via the `PUT` method).

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/concepts/historic-events

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Historic Events

As of version 1.2.0, the system supports receiving historic events for Digital Activity and Fraud Tagging. These events have a workflow field `is_historic` set to true and are dropped during the workflow, not contributing to realtime metrics. Furthermore, historic events are inserted in the database with the `event_occurred_at` as the `FDZ_timestamp` value, instead of the Pulse clock.

note

In this page, please have in consideration that Transfers applies to both Inbound, Outbound Payments. Cards also follows the same logic as Transfers

How are events classified as historic?â

Historic events are determined by a set of *slack* properties and the following picture describes their interaction with the different event types.

What are the current slack values?â

The values chosen for the past, historic and future slacks influence how events are processed and which events are accepted by the system. Currently, they following values are in use:

	Past Slack	Historic Slack	Future Slack
Transfers	30 min	30 min	30 min
Cards	30 min	30 min	30 min
Digital Activity	â	30 min	30 min

Taking those values into consideration, we can affirm that the system will:

- **reject** any request (regardless of the channel) with timestamp more than **30 minutes in the future**
- **not reject** any **Digital Activity** request in the past
- treat any Digital Activity or Fraud Feedback request as **historic** if it is more than **30 minutes in the past**

info

What happens to Rejected events?

When an event is Rejected, the API will return an a Bad Request status code and the event will **not reach the workflow**. These events won't be processed in any way, nor stored in any database.

How are historic events handled by the ETL extractions?â

The ETL extraction is a process that extracts Event Storage events and runs every hour, 10 minutes after the hour (i.e. at 00:10am, 01:10am, etc.). When the ETL runs, all events stored in the Event Storage with `FDZ_timestamp` for the previous hour are extracted, regardless of whether they are historic or not.

As mentioned above, historic events differ from regular events in the fact that they are stored in the Event Storage with the `event_occurred_at` as the `FDZ_timestamp` value. As the current configuration for historic slack is just 30 minutes, some historic events may appear in the extractions while other may fall outside the extraction window. The following examples illustrate both scenarios.

Example of an historic event not caught by the ETL extractionâ

Consider the current time is 15:45. In this scenario, historic events must have an `event_occurred_at` value before 15:15. The next ETL execution will be at 16:10 and extract events with `FDZ_timestamp` between 15:00 and 16:00.

An event received now (15:45) with `event_occurred_at` set to 14:30 will be considered historic and stored in the database with FDZ_timestamp equal to 14:30. It falls outside the next ETL extraction interval and therefore will not be extracted.

Example of an historic event caught by the ETL extractionâ

Keeping the same invariants as the previous example, consider the current time is 15:45, meaning historic events must have an `event_occurred_at` value before 15:15, and the next ETL execution will be at 16:10 and extract events with `FDZ_timestamp` between 15:00 and 16:00.

An event received now (15:45) with `event_occurred_at` set to 15:10 will be considered historic as it falls before the historic slack. Being historic, that event will be stored with `FDZ_timestamp` equal to its `event_occurred_at` (15:10). Unlike the previous example, despite being an historic event, the timestamp still falls inside the next ETL extraction window, and the event will be extracted.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/concepts/pb-delegate

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PB Delegate

As of version 2.8.0 we have introduced a set of changes to make the PB Delegate use case ready for use, to allow a customer to delegate another customer to make payments on their behalf.

This page focuses on explaining the changes made to both Pulse and Case Manager to support this feature.

Pulse API Changesâ

In this section you can find a description of the API changes to support this feature.

All of the fields described in this section were also added to the `Payments` and `Payments + Outcomes` schemas to allow events to be enriched when received by Pulse and also to enable them to be used in the fraud strategy.

Channel fieldsâ

As described in the 2.8.0 Release Notes we have added a set of new fields to Pulse Transfers and Digital Activity APIs.

Please find below a list of the new added fields:

Field Name	Datatype	Endpoints
delegate_id	String	Transfers & Digital Activity
delegate_initiated	Boolean	Transfers & Digital Activity
approver_id	String	Transfers & Digital Activity
transaction_id	String	Transfers & Digital Activity
account_holder_customer_id	String	Transfers & Digital Activity

Reference Data fieldsâ

The Reference Data API was also changed with the addition of the following fields:

Field Name	Datatype	Description
delegate_of	List of Objects	List of delegate ids (their customer id in HSBC) and the date when the delegate was assigned
delegates	List of Objects	List of customers whom the customer serves as a delegate

Each entry of `delegate_of` and `delegates` has the following fields:

Field Name	Datatype
customer_id	String
date_added	Long

In the below snippet you can find an example of customer data that acts as a delegate for two different customers:

```
{
  "delegate_of": [
    {
      "customer_id": "customer_id_1",
      "date_added": 1603207359060
    },
```

```
    {
      "customer_id": "customer_id_2",
      "date_added": 1603207359061
    }
  ]
}
```

Copy

A customer can have multiple delegates and a single delegate can also represent multiple customers. The following example illustrates a scenario where a customer has more than one delegate acting on their behalf:

```
{
  "delegates": [
    {
      "customer_id": "customer_id_3",
      "date_added": 1603207359060
    },
    {
      "customer_id": "customer_id_4",
      "date_added": 1603207359061
    }
  ]
}
```

Copy

Case Managerâ

Similarly to Pulse, we also made changes to Case Manager in order to support this feature. In this section you can find a description of the changes.

Event Details Pageâ

In the Event Details Page we have created a new field mapping (`By Delegate`) linked to the delegate id that acted on behalf of a customer, added to the following widgets:

- Transfers `Transaction` widget
- Digital Activity `Event Summary` widget

In the same page, the `Transaction History` widget now features a new Delegate CIN column and filter.

note

In order for an event to appear when filtering by `Delegate CIN` , the `customer_id` must match the `delegate_id` field of an event

Customer Details Pageâ

In the Customer Details Page we have added the `Transaction History Delegate` widget, similar to `Transaction History` , where events in which a customer acted as a delegate will be shown.

This new widget will only show the `Transfers` and `Digital Activity` channels and have specific filters for `Delegate CIN` .

note

In order for an event to appear when filtering by `Delegate CIN` , the `customer_id` must match the `delegate_id` field of an event

In addition we have also added two new widgets to show the current delegations the customer has in place and also to which customers it is delegated to.

Each of the entries links to the given customer page.

Enable new features in the Case Manager UIâ

The new widgets can be enabled/disabled via a new Ansible variable `show_delegate_transaction_history_widget`, set to `true` by default.

This can be used, for instance, to hide the widgets if a specific region or market does not require the PB Delegate use case.

Please refer to Case Manager Configuration Reference for more information.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/concepts/transaction-block

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Transaction Block

In version 3.0.0 we have introduced a set of changes to accommodate the Cards CCM use case requirements.

This page focuses on explaining the changes made to both Pulse and Case Manager to support this feature.

Pulseâ

In this section you can find the changes introduced in Pulse to support the use case.

Transaction Block Outcomeâ

We've added the transaction block outcome to the Cards Authorization and Non-Mon endpoints.

The new block includes two new fields, that were also added to the Cards workflow:

- `transaction_block_code_outcome` : the outcome value calculated in the workflow
- `transaction_block_computed_date` : calculated from the time of the response plus the amount of days configured based on the card's type

info

The `transaction_block_computed_date` field is only added to the response if the `card_type` is `debit` or `credit` .

Find below some examples of API responses:

Response that has card_type as credit


```
{
    "lifecycle_id": "2498c54f-4731-4d2b-a955-87fecfca42da",
    "status": "ok",
    "action_codes": [],
    "secondary_action_codes": [],
    "alert": false,
    "score": 0,
    "decision": "approve",
    "sca_required": true,
    "wallet_decision": "none",
    "transaction_block_code_outcome": "none",
    "transaction_block_computed_date": 1774978027914
}
```


Copy

Response that has a card_type that is not credit or debit


```
{
    "lifecycle_id": "9b160255-5fa8-43b9-8682-d3dae28046a0",
    "status": "ok",
    "action_codes": [],
    "secondary_action_codes": [],
    "alert": false,
    "score": 0,
    "decision": "approve",
    "sca_required": true,
    "wallet_decision": "none",
```

```
    "transaction_block_code_outcome": "none"
}
```

Copy

The offset can be configured via Ansible using the `cards_offset_days` variable, as shown below:

```
cards_offset_days:
  debit: 4
  credit: 5
```

Copy

Schema Changesâ

We've added the following fields to the Cards endpoints and workflow schemas:

Field Name	Datatype
system_block_code1	String
system_block_code1_datetime	Long
system_block_code2	String
system_block_code2_datetime	Long
transaction_block_code	String
transaction_block_code_datetime	Long

The Cards Reference Data API was extended to include these new fields. Please refer to the [Cards Reference Data API](#) for more information.

ⓘ**note**

We've also added support for Cards enrichment in the Cards workflow. By default Cards enrichment will be done using `card_pan` but the field can be changed during deployment using the Ansible variable `cards_card_enrichment_key`.

Case Managerâ

In this section you can find the changes introduced in Case Manager to support the use case.

Block Code actionsâ

We've added a new set of Case Manager actions in the Cards channel to support the new Block Code state machine. The `Fraud`, `Not-Fraud`, `Move to Unresolved` and `Move to On-Hold` actions were adjusted to support the use case. Please find below the conditions in which the actions appear:

Action	Previous Condition	New Condition
Set Block Code	N/A	`Assigned`
Skip Block Code	N/A	`Assigned` and Block Code is `Undefined`
Clear Block Code	N/A	`Assigned` and Block Code is not `Undefined/Skipped`
Fraud	`Assigned` and Operational Status is `In Progress/Unresolved/On Hold`	`Assigned` and Operational Status is `In Progress/Unresolved/On Hold` and Block Code is not `Undefined`
Not Fraud	`Assigned` and Operational Status is `In Progress/Unresolved/On Hold`	`Assigned` and Operational Status is `In Progress/Unresolved/On Hold` and Block Code is not `Undefined`
	N/A	Block Code is `Undefined`

Action	Previous Condition	New Condition
Move to Unresolved		
Move to On-Hold	N/A	Block Code is `Undefined`

Please find below an example:

- While the event is `Unassigned`, the option to select a Block Code is not available
- After the event is assigned to a user, the option to select a Block Code iss available (it meets the condition stated in the first row of the table above)

info

States added during deployment (using the Ansible variable `cm_block_code_machine_states`) will show if the event is `Assigned` and the user has the required permission.

warning

Bulk actions do not support this feature. Analysts will be able to mark events as Fraud/Not Fraud without setting the block code as long as they have the required permissions.

Extension Changesâ

There is a set of changes to the Case Manager extensions that needs to be taken into consideration.

Please find below a list of the fields added to the different extensions:

Extension	Fields Added

Cards Fulfilment	`cm_transaction_block`
	`cm_transaction_block_action`
	`cm_transaction_block_date`
	`cm_transaction_block_timezone`
	`cm_transaction_block_expiration`

Cards 2-WAY SMS	`cm_transaction_block_date` `cm_transaction_block_expiration`
Cards 1-WAY SMS	`cm_transaction_block_date` `cm_transaction_block_expiration`
Send Reminder SMS	`cm_transaction_block_date` `cm_transaction_block_expiration`

Regarding `cm_transaction_block_date` and `cm_transaction_block_expiration`, both are calculated from the time the transaction block was set plus the configured offset (using `cm_transaction_block_date_offset_days`) with the following format:

- `cm_transaction_block_date`: date time string format (e.g. `2025-10-21T08:14:00`)
- `cm_transaction_block_expiration`: UNIX timestamp (e.g. `1772638079711`)

Note that the base values should reflect the time when the analyst sets the block code. Consequently, if the block code status is undefined or clear, the date and expiration values won't be calculated.

The offset can be set during deployment using the following Ansible variable:

- `cm_transaction_block_date_offset_days` (default: `5`)

For more details on the fields description and examples of payloads please refer to the [AFIS extension documentation](#).

ⓘnote

The logic of `cm_transaction_block_action` is not yet implemented as of release 3.0.0, so the expected value for this field is always `null`.

Block Code State Machineâ

We've added the `Block Code` state machine to Case Manager.

By default this state machine is only available to Cards with the states `Undefined` (default value), `Clear` and `Skip`. In contrast with other state machines, the `Block Code` state machine is configurable during deployment using the Ansible variable `cm_block_code_machine_states`.

For example, to add two extra states the variable would have to be configured as follows:

```
cm_block_code_machine_states:
 - sample_1
 - sample_2
```

Copy

After deploying Case Manager the new states will show in the Reasons configuration page:

ⓘinfo

UI translations for new states have to be managed in the self serviceable translations page.

Similar to other state machines we've also created the corresponding permissions:

- `cm:state-machine:event-status:write:block_code`
- `cm:state-machine:event:read:block_code`
- `cm:state-machine:status-reason:delete:block_code`
- `cm:state-machine:status-reason:read:block_code`
- `cm:state-machine:status-reason:write:block_code`

Please refer to the Case Manager Configuration Reference and to the Permissions Guide for more details.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/features/hsbc_status_reason

On this page

Feedback API `external_status_reason`

Source:

- Requirement FTMT-1085
- HSBC-823

A new field `external_status_reason` was added in the Feedback API to support SAS EFM as the primary decision maker. This new field is optional and will not be enforced. Although it can receive any value - since we expect this list to be updated fairly regularly and don't want to drop any events - we'll only update Case Manager Status Reason in case we receive the following values:

Possible values
Fraud - Telephone Banking Automated
Fraud - Telephone Call Centre Agent
Fraud - Branch
Fraud - Email/Fax/Mail
Fraud - Direct Payment
Fraud - Other
Fraud - Token Reactivation
Fraud - Remote Access
Fraud - Credential Stuffing
Fraud - Compromise Point Unknown
Fraud - Smishing
Fraud - Malware
Fraud - Token Reactivation
Fraud - Remote Access
Fraud - Credential Stuffing
Fraud - Compromise Point Unknown
Fraud - Smishing
Fraud - Malware
Scam - CEO/Business Email Compromise
Scam - Impersonation (Police/Bank)
Scam - Impersonation (Other)
Scam - Other
Scam - Invoice and Mandate
Scam - Investment
Scam - Advance Fee
Scam - Purchase
Scam - Romance

Possible values
Scam - Other
Fraud - Account Takeover - Secure Key Activation
Fraud - Account Take Over
Fraud - Payment Beneficiary/Mule
Fraud - 1st Party Fraud

Referencesâ

Requirement definition and specification

On this page

Base Requirements

The purpose of this guide is to walk you through the requirements needed for setting up each Feedzai component.

Furthermore, this guide also includes the requirements for installing or configuring the following components:

- Pulse
- Case Manager
- Reference Data Service
- Cassandra
- RabbitMQ
- ZooKeeper

info

The requirements below are applicable to all servers where Feedzai services will be installed, as well as other third-party services that are part of the product stack.

Operating Systemâ

Feedzai deployments officially support the following operating systems:

- Red Hat Enterprise Linux 7.9 (64 bit)
- CentOS 7.9 (64 bit)

info

Installing on AWS

If you are installing Feedzai products on AWS, the Amazon Linux 2 distribution is also supported.

To ensure Feedzaiâs services work properly, and the Feedzai team is able to investigate or remediate any issues, the following must be ensured:

- All servers need to have `ntp` daemon configured.
- Turn-off the `swap` file (disable all swap lines on the `/etc/fstab file`);
- All servers need to have an Operating System group and user named `feedzai` .
- All the servers should be accessible via SSH for the `feedzai` user.
- All Feedzai Pulse servers must have `ulimit` set to `65535` . Example:

```
sudo tee /etc/security/limits.d/99-feedzaiproc.conf << 'EOF'
feedzai soft nofile 65535
feedzai hard nofile 65535
feedzai soft nproc 65535
feedzai hard nproc 65535
EOF
```

Copy

- The following packages must be installed:
    - `which`
    - `tar`
    - `bzip2`
    - `less`

    - `nc`
    - `unzip`
    - `vim`
    - `telnet`
    - `wget`
    - `Python3.6`
    - `Python2.7`
    - `libselinux-python3`

info

It is recommended that both `Python2.7` and `Python3.6` are installed because there may be dependencies that require different Python versions.

tip

It is recommended that the servers are kept up to date with the latest Operating System and security patches. Our recommendaton is that servers have access to a CentOS mirror with at least daily updates. In case virtual machines are provisioned based on immutable images, this should be ensured for the base image.

Web Browserâ

Feedzai products that have a Web UI (Pulse, Case Manager, and Genome) officially support the following browsers:

- Chrome (version 92 or later)
- Firefox (version 91 or later)
- Edge (version 92 or later)

Data Extraction Pipelines

In this section you will find a set of pages with information about the Data Extraction Pipelines:

- Data Extraction Pipelines Overview
- Data Extraction Pipelines Requirements
- Installing Data Extraction Pipelines with Ansible
- Data Extraction Pipelines Configurations
- Data Schema

Logstash Recovery tool

Welcome to the Logstash Recovery tool Guide.

In this section you will find information about the Logstash recovery tool process:

- Logstash Pulse Workflow Audit Pipeline Recovery Tool

On this page

Installation Guide

Welcome to the Feedzai Installation Guide. In this guide you will find useful information on how to install Feedzaiâs products.

Artifacts

Name	Version	Note
Installer	3.0.2	
Project Pulse Configurations	3.0.2	
Project Pulse Data Science Configurations	3.0.2	
Project Case Manager Configurations	3.0.2	
Project ETL Configurations	3.0.2	
Empty Application for DSE Environments	3.0.2	
Pulse	25.1.9	
Spark Libs	25.1.9	
Spark Dist	3.5.7-hadoop-3.3.5	
Case Manager	13.31.0	
Reference Data Server	3.7.12	
Keycloak Feedzai Theme	2.3.0	
Event Storage ETL	18.3.0	
JupyterLab Docker Image	25.0.49-gcs-2.2.29	**Load Docker Image as** `docker.feedzai.com/feedzai/jupyterlab:25.0.49-gcs-2.2.29-gcp`

Installing Feedzai Servicesâ

- Pulse Requirements
- Case Manager Requirements
- Reference Data Service Requirements

Configuring Third-Party Servicesâ

- Cassandra Recommendations
- RabbitMQ Recommendations
- ZooKeeper Recommendations

Feedzai Ansible Toolkitâ

- Installing Feedzai with Ansible

Database Setupâ

- Retention Scripts

Schema Migrator

Welcome to the Schema Migrator Guide.

In this section you will find a set of pages with information about the Schema Migrator process:

- Schema Migrator overview
- Fix Pulse Clock
- Drop database index

How-to Guides

Welcome to the Operations How-to Guides.

In this section you will find the following set of tutorials with information on how to authenticate via the Pulse API, changing log levels at runtime and how to import/export Pulse applications:

- API Authentication
- Change Log Levels at Runtime
- Export and Import Pulse applications
- Export HAR files
- Use Enrichers in Case Manager
- Migrate Cassandra Profile Store

On this page

Operations Guide

Welcome to the Feedzai Operations Guide. In this document you will find useful information on how to operate Feedzaiâs products.

Monitoringâ

- Building a Monitoring Solution
- Monitoring Feedzai Services
- Monitoring Third-Party Services
- Monitoring the Infrastructure
- Pulse Application cleanup

Permissions Guide

Â

Functionality Tool	Action	Permission	Description
Case Manager and Pulse Common Permissions	Login	cm:login pulseviews:login	Lets the user log into pulse and case r
Case Manager Events	View	cm:resource:event:read:*	Lets the user read an event.
	Update	cm:resource:event:update:*	Lets the user update events.
	Assign/Unassign Event	cm:resource:event-user:create:* - **Backend** cm:action:event-user:create:* - **Frontend**	Lets the user assign users to event.
		cm:resource:assign:*	Gives the user rights to assign an eve already assigned to another user to its Permission only relevant for Assign ar
		cm:action:event-case:delete:*	Lets the user unlink events from cases
	Notes	cm:resource:event-note:create:* - **Backend** cm:action:event-note:create:* - **Frontend**	Lets the user create notes in events.
		(1) cm:action:event-note:read:* (2) cm:resource:event-note:read:*	(1) Lets the user see Event Notes wid events details page. (2) Lets the user see notes in events.
	Move to Queue	cm:resource:event-queue:create:* - **Backend** cm:action:event-queue:create:* - **Frontend**	Lets the user move events to queues.
	Move to Close	cm:action:manual-close-alert	Lets the user manually move events to status
	Resource Manager	cm:resource:manager:*	Gives the user rights to have all action permissions even if is not assigned the Permission only relevant for Assign ar
	Export/download search results	cm:resource:event-export:read:*	Lets the user export the results.
		cm:action:event-export:create:*	Lets the user see the export button.

	Status	cm:action:event-status:create:*	Lets the user set event statuses
		cm:resource:status:read:*	Lets the user read event status option
	Files	(1) cm:resource:event-file:create:* (2) cm:resource:file:create:*	(1) Lets the user create N-to-N relatio between events and files. (2) Lets the user create files.
		cm:resource:file:read:*	Lets the user read files.
		cm:resource:file:update:*	Lets the user update an existing file.
		cm:resource:file:delete:*	Lets the user delete an existing file
		(1) cm:resource:file:event-file:create:* (2) cm:resource:file:event-file:update:*	Lets the user upload an existing file in Details page (Attached Files widget). Lets the user update an existing file in Details page(widget: Attached Files).
		cm:resource:file:event-file:read:*	Lets the user list all the files associate event.
	Locale	cm:resource:locale:create:*	Lets the user create locales.
		cm:resource:locale:read:*	Lets the user read locales.
		cm:resource:locale:update:*	Lets the user update locales.
		cm:resource:locale:delete:*	Lets the user delete locales.
	Users	cm:resource:user:read:*	Lets the user read available users.
Case Manager Lists	View	cm:resource:list:search:*	Lets the user search for entities in list
		cm:section:settings-lists:read:*	Lets the user see List Management op the menu Settings section.
		cm:action:view-entity-audit:read:*	Lets the user view the "view-entity-au action.
	Update	cm:resource:list:update:*	Lets the user update lists content and
	Create	cm:action:add-entity:create:*	Lets the user add a EntityValue to a L
		cm:action:event-list:create:*	Lets the user add events to lists.
	Delete	cm:action:remove-entity:delete:*	Lets the user remove an EntityValue f List.
	Posneg	cm:action:event-posneg:create:*	Lets the user add events to posneg lis
		cm:resource:event-posneg:create:*	Lets the user create posneg lists for e
		cm:resource:event-posneg:read:*	Lets the user read posneg lists for eve

Case Manager SLAs	View SLAs	cm:resource:sla:read:* - **Backend** cm:section:settings-slas:read:* - **Frontend**	Lets the user read SLAs.
	Create SLAs	cm:resource:sla:create:* - **Backend** cm:section:settings-slas:create:* - **Frontend**	Lets the user create SLAs.
	Edit SLAs	cm:resource:sla:update:* - **Backend** cm:section:settings-slas:update:* - **Frontend**	Lets the user update SLAs.
	Delete SLAs	cm:resource:sla:delete:* - **Backend** cm:section:settings-slas:delete:* - **Frontend**	Lets the user delete SLAs.
Case Manager Cases	View Cases	cm:resource:case:read:* - **Backend** cm:section:case:read:* - **Frontend**	Lets the user read cases.

		cm:section:case-details:read:*	Lets the user see the Case Details pag
		cm:section:case-details:*	Lets the user access all case-details p and functionalities.
	Search Cases	cm:section:case-search:*	Lets the user search for cases
	Create Cases	cm:resource:case:create:* - **Backend** cm:section:case:create:* - **Frontend**	Lets the user create cases.
	Update case	cm:resource:case:update:* - **Backend** cm:section:case:update:* - **Frontend**	Lets the user update cases.
	Delete case	cm:resource:case:delete:*	Lets the user delete cases.
	Assign Case	cm:resource:case-user:create:*	Lets the user assign users to cases.
	Link events/alerts to cases	cm:action:event-case:create:*	Lets the user link events to case.
		cm:resource:event-case:create:*	Lets the user add events to cases.
		cm:resource:event-case:delete:*	Lets the user delete events in cases.
		cm:resource:event-case:read:*	Lets the user read events in cases.
		cm:resource:event-case:update:*	Lets the user update events in cases.
	Move case to queue	cm:resource:case-queue:create:*	Lets the user create queues for cases
	Set case status	cm:action:case-state:create:*	Lets the user open or close case state cases.
	Upload/Download Files	(1) cm:resource:case-file:create:* (2) cm:resource:file:case-file:read:*	(1) Lets the user upload file in Case D (2) Lets the user download file in Case
	Files	cm:resource:case-details:create:*	Lets user list all the files associated wi current case.
		cm:resource:file:case-file:update:*	Lets the user update file in Case Deta
	Channels	cm:action:case*-global:read:*	Lets the user see channels in cases.
		cm:action:case-global:create:*	Lets the user select channels in cases Note that, for Omnichannel Multichann channel field should be hidden and for Omnichannel Single channel (Default)
	Export	(1) cm:action:case-export:create:* (2) cm:resource:case-export:read:*	(1) Lets the user see the export button (2) Lets the user export the results fro cases search page, or any other table only uses those cases fields.
		(1) cm:action:case-note:create:* - **Frontend**	

	Notes	cm:resource:case-note:create:* - **Backend** (2) cm:action:case-note:read:* - **Frontend** cm:resource:case-note:read - **Backend** (3) cm:resource:case-note:update (4) cm:resource:case-note:delete	(1) Lets the user add a note to a case Case Details page. (2) Lets the user see case notes on N widget. (3) Lets the user update notes in case (4) Lets the user delete notes in cases
	States	cm:resource:case-state:create:*	Lets the user create states for cases.
	Casetype	cm:resource:casetype:<CASE_TYPE_NAME>:read (e.g cm:resource:casetype:fraud:read)	Gives read-level permission over the c given by <CASE_TYPE_NAME>, whe case type name is given by the config case type name without the type. prefi (example: for a configured case type o type.aml, the permission is cm:resource:casetype:aml:read)
		cm:resource:casetype:<CASE_TYPE_NAME>:write (e.g cm:resource:casetype:fraud:write)	Gives both read and write-level permis over the case type given by <CASE_TYPE_NAME>, where the ca name is given by the configured case name without the type. Prefix (exampl configured case type of type.fraud, the permission is cm:resource:casetype:fraud:write).
Case Manager Reasons	View Reasons	cm:resource:status-reason:read:* - **Backend** cm:section:settings-reason:read:* - **Frontend**	Enables the user to list the available r
	Create Reasons	cm:section:settings-reason:create:*	Enables the user to create new reaso
	Delete Reasons	cm:action:reason:delete:*	Enables the user to remove reasons.
Case Manager Automation Rules	View Automation Rule	(1) cm:resource:automation:read:* (2) cm:resource:configuration-automation:read: (3) cm:section:settings-automation:read:*	(1) Lets the user read automation rule (2) Lets the user read the automation configuration. (3) Lets the user see the Automation F page
	Create Automation Rule	cm:section:settings-automation:create:* - **Frontend** cm:resource:automation:create:* - **Backend**	Lets the user create automation rules Automation Rules page
	Edit Automation Rule	cm:resource:automation:update:* - **Backend** cm:section:settings-automation:update:* - **Frontend**	Lets the user update automation rules Automation Rules page
	Delete Automation Rule	cm:resource:automation:delete:* - **Backend** cm:section:settings-automation:delete:* - **Frontend**	Lets the user delete automation rules Automation Rules page
		(1) cm:resource:queue:read:* (2) cm:resource:queue-user:read:*	(1) Lets the user read queues. (2) Lets the user read its own queues.

	View Queue	(3) cm:resource:queue-user:* (4) cm:section:settings-queue:read:*	(3) Lets the user perform CRUD opera over the current queues assigned to a (4) Lets the user see the queues page Settings section.
	Create Queue	cm:resource:queue:create:* - **Backend** cm:section:settings-queue:create:* - **Frontend**	Lets the users create queues in the Se section.
	Edit Queue	cm:resource:queue:update:* - **Backend** cm:section:settings-queue:update:* - **Frontend**	Lets the user update queues in the Se section.
Case Manager **Queue**	Delete Queue	cm:resource:queue:delete:* - **Backend** cm:section:settings-queue:delete:* - **Frontend**	Lets the user delete queues in the Set section.
	Assign Queues	cm:resource:queue-user:assign:*	Lets the user assign queues to users.
	Import and Export Queues	cm:resource:settings:import-export:* - **Backend** cm:action:settings:import-export:* - **Frontend**	Lets the user import and export queue
	Get Next	cm:resource:user-queue:*	Lets the user see the working queues, context of the review next feature.
	View SSR	cm:section:settings-rules:read:*	Lets the user see rules in the Self-Ser Rules page.
	Create SSR	cm:section:settings-rules:create:*	Lets the user create rules in the Self-S Rules page.
	Edit SSR	cm:section:settings-rules:update:*	Lets the user update rules in the Self-S Rules page.
	Delete SSR	cm:section:settings-rules:delete:*	Lets the user delete rules in the Self-S Rules page.
	Publish SSR	cm:section:settings-rules:publish:*	Lets the user publish rules in the Self- Rules page.
Case Manager **Self Service Rules**	Test SSR	(1) cm:resource:settings-rules-custom-test:create:* (2) cm:resource:settings-rules-test:create:* cm:section:settings-rules-test:create:*	(1) Lets the user test rules with custon properties, either the user is a landlord contains the permission. (2) Lets the user test rules, either the u landlord or contains the permission.
		cm:section:settings-rules-test:create:*	Lets the user test rules, either the use landlord or contains the permission.
	Rules Audit	cm:resource:self-service-rules:audit:*	Lets the user access rules audit entrie
Case Manager **Reports**	View reports	cm:resource:report:read:* - **Backend** cm:section:report:read:* - **Frontend**	Lets the user read reports.

	Create reports	cm:resource:report:create:* - **Backend** cm:section:report:create:* - **Frontend**	Lets the user create reports.
	Update reports	cm:resource:report:update:*	Lets the user update reports.
	Delete reports	cm:resource:report:delete:* - **Backend** cm:action:delete-generated-report:delete:* - **Frontend**	Lets the user delete reports.
	Download reports	cm:resource:file:report:read:*	Lets the user download reports.
Case Manager **Report Templates**	View	cm:resource:report-template:read:* - **Backend** cm:section:settings-report-template:read:*- **Frontend**	Lets the user read report templates.
	Create	cm:resource:report-template:create:* - **Backend** cm:section:settings-report-template:create:*- **Frontend**	Lets the user create report templates.
	Update	cm:resource:report-template:update:* - **Backend** cm:section:settings-report-template:update:*- **Frontend**	Lets the user update report templates.
	Delete	cm:resource:report-template:delete:* - **Backend** cm:section:settings-report-template:delete:*- **Frontend**	Lets the user delete report templates.
	Download	cm:resource:file:report-template:read:*	Lets the user download report templat
Case Manager **Entity Notes**	View	cm:action:entity-note:read:* - **Frontend** cm:resource:entity-note:read:* - **Backend**	Allows the user to see Notes widget (e customer notes widget) on an entity de page (e.g., Customer details page).
	Create	cm:action:entity-note:create:* - **Frontend** cm:resource:entity-note:create:* - **Backend**	Allows the user to assign notes to data entities.
Case Manager **Entity Data**	View	(1) cm:section:customer-details:read:* - **Frontend** (2) cm:entity-data:<entity_type>:read:* - **Backend** (entity_type available right now are accounts, cards and customers)	(1) Lets the user see the customer pag (2) Lets the user read data about an e a given type (e.g. customers, cards, et places such as the customer page
Case Manager **Entity Files**	View	cm:resource:file:entity-file:read:*	Lets the user read files associated wit (e.g. customers, cards, etc.) in places the customer page.
Case Manager **Cases Sources**	View	cm:resource:case-source:read:* - **Backend** cm:section:settings-case-source:read:* - **Frontend**	Lets the user have access to the case page and list case sources
	Create	cm:resource:case-source:create:* - **Backend** cm:section:settings-case-source:create:* - **Frontend**	Lets the user create new case sources
	Update		

		cm:resource:case-source:update:* - **Backend** cm:section:settings-case-source:update:* - **Frontend**	Lets the user edit (enable/disable) exi case sources.
Case Manager **Custom Extensions**	onEventStatusUpdate	cm:action:extension:onEventStatusUpdate	Lets the user execute the custom exte given by onEventStatusUpdate
Case Manager **Channel Based Permissions**	Read and Write	cm:channel:<channel_id>:read-write	Lets the users see and edit <channel_ events and artifacts (cards, digital acti transfers, inbound-payments, non-mo
	Read Only	cm:channel:<channel_id:resource:group:readel_id>:read-only	Users can see <channel_id>'s events artifacts, but cannot edit them (except notes andfile-uploading)
	None	cm:channel:<channel_id:resource:group:readel_id>:none	Users do not have access to <channe
Case Manager **Access Groups**	Search	cm:resource:group:read:*	Lets the user search groups.
Case Manager **Multi-tenancy**	Landlord	cm:resource:landlord:*	Lets the user have landlord permissio
	Tenant	cm:resource:tenant:read:*	Lets the user read the tenant list.
Case Manager **Multiple State Machines**	Change Status	cm:state-machine:event-status:write:<STATEMACHINE>	Lets the user change the status for a state machine (whose id is <STATEMACHINE>)
	Read	cm:state-machine:event:read:<state-machine-id>	Enables the user to read the state ma information.
	List Reasons	cm:state-machine:status-reason:read:<state-machine-id>	Enables the user to list the status rea a state machine with a given id.
	Create Reasons	cm:state-machine:status-reason:write:<state-machine-id>	Enables the user to create a status re a state machine with a given id.
Case Manager **Tokenizer**	Reveal	(1) cm:action:reveal-<field_path> (2) cm:resource:reveal-all (3) cm:resource:reveal-<FIELD> (4) cm:resource:reveal:* (5) cm:resource:search-by:<FIELD>	(1) Lets the user see the reveal icon t a field. (2) Lets the user automatically reveal encrypted fields when entering an eve details page. (3)Lets the user reveal a field value us tokenizer. (The <field_path> has to be replaced with the field path in the obje (4) Give user all permissions regardin (5) Lets the user perform a search ope by a specified field in plain text using tokenizer. This permission combined w user permission cm:resource:search-b allows users to search for all tokenize elements.
Case Manager **Details Page**	Widgets	(1) cm:action:self-service-config:advanced-mode (2) cm:resource:self-service-config	(1) Lets the user have access to the A edit mode in the UI. (2) Lets the user list active preference

Case Manager **Login and Logout**	Login and Logout	(1) security:ownership:read (2) security:permissions:read	(1) Lets the user see the ownership de Pulse user groups (toggle "advanced" (2) Lets the user read the permissions
Case Manager **Manage Message**	Manage Message	(1) cm:section:settings-message:read:* (2) cm:resource:message:read:* (3) cm:action:settings-message:create:* (4) cm:resource:message:create:* (5) cm:action:settings-message:update:* (6) cm:resource:message:update:* (7) cm:action:settings-message:delete:* (8) cm:resource:message:delete:*	(1) Lets the user see the Messages pa the Settings section. (2) Lets the user read messages. (3) Lets the user create new message Settings section. (4) Lets the user create new message (5) Lets the user update messages in Settings section. (6) Lets the user update messages. (7) Lets the user delete messages in t Settings section. (8) Lets the user delete messages.
Case Manager General Permissions	Dashboards	cm:section:dashboard-analyst-performance:* cm:section:dashboard-business-overview:* cm:section:dashboard-psd2:* cm:section:rules-monitoring-rules-filter:* cm:resource:dashboards-user-sync:update:* cm:section:rules-monitoring-transactions-filter:*	Allows the users to see the dashboard (Requires enabling Insights)
	Get Next	cm:section:review-next:read:*	Lets the user see the review next func
	UI Configuration	cm:resource:configuration:read:* cm:resource:configuration:create:* cm:resource:configuration:update:* cm:resource:configuration:delete:* cm:action:gui-config:edit cm:resource:gui-config:edit cm:resource:gui-config:reset-to-default cm:resource:gui-config:list-active:*	Lets the user read and modify the UI configuration.
	Language Configuration	cm:resource:language:read:* cm:resource:language:create:* cm:resource:language:delete:* cm:resource:language:update:*	Lets the user read and modify the ava languages.
	Time Configuration	cm:resource:timezone:read:* cm:resource:timezone:create:*	Lets the user read and modify timezor

	cm:resource:timezone:delete:* cm:resource:timezone:update:*	
Translated Messages	cm:resource:message:read:*	Lets the user read the translated mess
Event Status	cm:resource:event-status:create:*	Lets the user create status for events.
Edit Resource	cm:action:edit-resource:update:*	Lets the user edit the resource.
Custom Link	cm:action:link-to	Lets the user navigate to a custom link
Pulse App	(1) cm:resource:pulse-app:read:* (2) cm:resource:pulse-rules:read:*	(1) Gets the Pulse application status a (2) Gets the Pulse application rules
Session	cm:resource:session:read:* cm:resource:session:create:* cm:resource:session:update:* cm:resource:session:delete:*	Lets the user read and modify session information in Case Manager.
Event Visibility	cm:resource:event-visibility cm:resource:event-visibility:create:*	Gives the user full event visibility.
Customer Page	cm:section:customer:read:*	Lets the user have access to Custome details
Event Details	cm:section:event-details:*	Lets the user have access to all Event operations
	cm:section:event-details:read:* cm:section:event:*	Lets the user have access to read Eve Details
	cm:resource:event-users:read:*	Retrieves a list of users that can be us assignment according to the tenant pr
	cm:resource:event-queues:read:*	Retrieves a list of queues that are ass with an event's tenant.
	cm:section:event:read:*	Lets the user have access to read Eve Details
	cm:section:event:create:*	Lets the user have access to create E Details
	cm:section:event:delete:*	Lets the user have access to delete E Details
	cm:section:event:update:*	Lets the user have access to update E Details
Fraud Alerts	cm:section:fraud-alerts:*	Lets the user have access to all Fraud operations
	cm:section:fraud-alerts:read:*	Lets the user have access to read Fra
Rules Block	cm:resource:configuration-rules-block:read:*	Lets the user read the configuration of blocks.
Search	cm:action:search-by-schema.card_pan	Lets the user search by Card Pan
Assign	cm:resource:assign-no-steal:*	

			Allows users to assign unassigned eve themselves and to others.
	Export Entity	cm:resource:entity-export:read:* cm:action:entity-export:create:*	Lets the user export the results from a entity's search page.
	Edit Mode	cm:action:self-service-config:normal-mode	Lets the user have access to the Norm mode in the UI.
Pulse App Management	Create	application:apps:create	Permission to create new applications
	Delete	application:apps:delete	Permission to delete applications.
	Manage	application:apps:manage	Permission to start/stop/publish/expor applications.
	Manage	application:apps:all-nodes-publish	Permission to publish applications in a non-rolling fashion (all nodes at the sa time).
	View	application:apps:read	Permission to read applications.
	Update	application:apps:update	Permission to update applications.
	Import	application:apps:partialImport	Permission to partially import applicati
	Export	application:apps:partialExport	Permission to partially export applicati
	All CRUD Permissions	application:*	Permission to perform all CRUD actio application models.
Read Applications	application:*:read	Permission to read applications.	**No**
Pulse Sandbox Evaluations	View	application:sandbox_evaluations:read	Permission to read sandbox evaluatio
	Create	application:sandbox_evaluations:create	Permission to create sandbox evaluat
	Update	application:sandbox_evaluations:update	Permission to update sandbox evalua
	Delete	application:sandbox_evaluations:delete	Permission to delete sandbox evaluati
Pulse Custom Services	View	application:dependencies:read	Permission to read custom service.
	Create	application:dependencies:create	Permission to create new custom serv
	Update	application:dependencies:update	Permission to update custom service.
	Delete	application:dependencies:delete	Permission to delete custom service.
Pulse Application History	View	application:history:global:read application:history:resource:read	Permission to read history log applica wide. Permission to read history log of a sin resource.
Pulse Job Executions	Create	application:jobexecutions:create	Permission to create new job executio
	View	application:jobexecutions:read	Permission to read job execution.
	Update	application:jobexecutions:update	Permission to update job execution.
	Delete	application:jobexecutions:delete	Permission to delete job execution.
	Start	application:jobexecutions:start	Permission to start job executions
	Abort	application:jobexecutions:abort	Permission to abort job executions
	View	application:job_profiles:read	Permission to read job profiles.

Pulse Job Profiles	Create	application:job_profiles:create	Permission to create job profiles.
	Update	application:job_profiles:update	Permission to update job profiles.
	Delete	application:job_profiles:delete	Permission to delete job profiles.
Pulse Managed Lists	View	application:managedlists:read	Permission to read managed lists.
	Create	application:managedlists:create	Permission to create managed lists.
	Update	application:managedlists:update	Permission to update managed lists.
	Delete	application:managedlists:delete	Permission to delete managed lists.
Pulse Managed List Items	View	application:managedlistitems:read	Permission to read managed list items
	Create	application:managedlistitems:create	Permission to create managed list iten
	Update	application:managedlistitems:update	Permission to update managed list iter
	Delete	application:managedlistitems:delete	Permission to delete managed list iten
	Decrypt	application:managedlistitems:decrypt	Permission to download encrypted val managed list items in plain text format plain text download needs to be enabl the pulse.global.lists.allowListDownloadIn property.
	View Revision	application:managedlistitems:revision:read	Permission to read managed list revisi
	Update Revision	application:managedlistitems:revision:update	Permission to update managed list rev
	Create Revision	application:managedlistitems:revision:create	Permission to create managed list rev
Pulse Outcome Combinators	View	application:outcome_combinators:read	Permission to read outcome combinat
	Create	application:outcome_combinators:create	Permission to create a new outcome combinator.
	Update	application:outcome_combinators:update	Permission to update outcome combir
	Delete	application:outcome_combinators:delete	Permission to delete a outcome comb
Pulse Workflows	View	application:rte_workflows:read	Permission to read workflow.
	Create	application:rte_workflows:create	Permission to create new workflow.
	Update	application:rte_workflows:update	Permission to update workflow.
	Delete	application:rte_workflows:delete	Permission to delete workflow.
Pulse User Management	View	security:users:read	Permission to read user.
	Create	security:users:create	Permission to create new user.
	Update	security:users:update	Permission to update user.
	Delete	security:users:delete	Permission to delete user.
Pulse Admin Rights	Admin	admin:view	Permission to access the administratio screens.
	Read Logs	admin:logs:read	Permission that grants access to the lo
	Manage Cluster	admin:cluster:manage	Permission that grants management a the cluster.
	Read Cluster	admin:cluster:read	Permission that grants read access to cluster.
	Delete Configuration	admin:configuration:delete	Permission to delete configurations or administration screens.

	Read Configuration	admin:configuration:read	Permission to read configurations on t administration screens.
	Update Configuration	admin:configuration:update	Permission to update configurations o administration screens.
	Deploy Services	admin:deploy	Permission to deploy services through REST API.
	Reload Security	admin:security:reload	Permission to reload security through API.
	Control Services	admin:service:control	Permission to start and stop services Administration API.
Pulse Tabs	Management Tabs	pulseviews:management:tabs:*	Permission to allow the user to access pulseviews tabs.
Pulse SSR Audit	View	(1) application:ssr_audit:privilegedView (2) application:ssr_audit:read	(1) Permission to read Self-Service Ru Audit entries in privileged views, for in retrieving the audit respective to all Ru present in a given application. (2) Permission to read self-service rule entries.
Pulse SSR Test	View	application:ssr_test:read	Permission to read self-service rule te results.
	Run	application:ssr_test:run	Permission to run a self-service rule te
	Update	application:ssr_test:update	Permission to update self-service rule results
Pulse Security	Security	application:streams:read:*	Permission to read streams (stream corresponds to a schema)
		security:tenants:read	Permission to read tenants.
		security:audit:*	Permission to read audit logs.
		useride:management	Permission to manage users
		security:roles:read	Permission to read role
		security:logging:update	Permission to update the logging level
		cm:resource:secure-metadata	Lets the user see encrypted metadata
		cm:element:tenant:read:*	Lets the user access UI elements acc to landlords.
Pulse Datascience Datasources	View	application:datascience_datasources:read	Permission to read data science data
	Create	application:datascience_datasources:create	Permission to create data science dat sources.
	Update	application:datascience_datasources:update	Permission to update data science dat sources.
	Delete	application:datascience_datasources:delete	Permission to delete data science dat sources.
Pulse Datascience Evaluation Metrics	View	application:datascience_evaluationmetrics:read	Permission to read data science evalu metrics.
	Create	application:datascience_evaluationmetrics:create	Permission to create data science eva metrics.
	Update	application:datascience_evaluationmetrics:update	

			Permission to update data science eva metrics.
	Delete	application:datascience_evaluationmetrics:delete	Permission to delete data science eva metrics.
Pulse Datascience Model Projects	View	application:datascience_modelprojects:read	Permission to read data science mode projects.
	Create	application:datascience_modelprojects:create	Permission to create data science mod projects.
	Update	application:datascience_modelprojects:update	Permission to update data science mo projects.
	Delete	application:datascience_modelprojects:delete	Permission to delete data science mod projects.
Pulse Datascience Model Projects Evaluations	View	application:datascience_modelprojects_evaluations:read	Permission to read data science mode evaluations.
	Create	application:datascience_modelprojects_evaluations:create	Permission to create data science mod evaluations.
	Update	application:datascience_modelprojects_evaluations:update	Permission to update data science mo evaluations.
	Delete	application:datascience_modelprojects_evaluations:delete	Permission to delete data science mod evaluations.
Pulse Datascience Model Projects Feature Importance	View	application:datascience_modelprojects_featureimportance:read	Permission to read feature importance
	Create	application:datascience_modelprojects_featureimportance:create	Permission to create feature importan
	Update	application:datascience_modelprojects_featureimportance:update	Permission to update feature importan
	Delete	application:datascience_modelprojects_featureimportance:delete	Permission to delete feature importan
Pulse Datascience Model Projects Hyperparameter Tuning	View	application:datascience_modelprojects_hyperparam_tuning:read	Permission to read hyperparameter tu
	Create	application:datascience_modelprojects_hyperparam_tuning:create	Permission to create hyperparameter
	Update	application:datascience_modelprojects_hyperparam_tuning:update	Permission to update hyperparameter
	Delete	application:datascience_modelprojects_hyperparam_tuning:delete	Permission to delete hyperparameter
Pulse Datascience Model Projects Hyperparameter Tuning Eval	View	application:datascience_modelprojects_hyperparam_tuning_eval:read	Permission to read hyperparameter tu eval.
	Create	application:datascience_modelprojects_hyperparam_tuning_eval:create	Permission to create hyperparameter eval.
	Update	application:datascience_modelprojects_hyperparam_tuning_eval:update	Permission to update hyperparameter eval.
	Delete	application:datascience_modelprojects_hyperparam_tuning_eval:delete	Permission to delete hyperparameter eval.
Pulse Datascience Model Projects Models	View	application:datascience_modelprojects_models:read	Permission to read Data Science mod
	Create	application:datascience_modelprojects_models:create	Permission to create Data Science mo
	Update	application:datascience_modelprojects_models:update	Permission to update Data Science m
	Delete	application:datascience_modelprojects_models:delete	Permission to delete Data Science mo

Pulse Datascience Model Projects Targets	View	application:datascience_modelprojects_targets:read	Permission to read Data Science targe
	Create	application:datascience_modelprojects_targets:create	Permission to create Data Science tar
	Update	application:datascience_modelprojects_targets:update	Permission to update Data Science ta
	Delete	application:datascience_modelprojects_targets:delete	Permission to delete Data Science tar
Pulse Datascience Plans	View	application:datascience_plans:read	Permission to read Data Science Plan
	Create	application:datascience_plans:create	Permission to create Data Science Pla
	Update	application:datascience_plans:update	Permission to update Data Science Pl
	Delete	application:datascience_plans:delete	Permission to delete Data Science Pla
	Run	application:datascience_plans:run	Permission to run Data Science Plans
Pulse Datascience Plans Exectutions	View	application:datascience_plans_executions:read	Permission to read Data Science Plan Execution.
	Create	application:datascience_plans_executions:create	Permission to create Data Science Pla Execution.
	Update	application:datascience_plans_executions:update	Permission to update Data Science Pl Execution.
	Delete	application:datascience_plans_executions:delete	Permission to delete Data Science Pla Execution.
Pulse Datascience Rules Projects	View	application:datascience_rulesprojects:read	Permission to read data science rules
	Create	application:datascience_rulesprojects:create	Permission to create data science rule projects.
	Update	application:datascience_rulesprojects:update	Permission to update data science rul project.
	Delete	application:datascience_rulesprojects:delete	Permission to delete data science rule projects.
	Self Service Publish	application:datascience_rulesprojects:self-service-publish	Permission to publish self-service rule
Pulse Datascience Rules Projects Evaluations	View	application:datascience_rulesprojects_evaluations:read	Permission to read data science rules evaluations.
	Create	application:datascience_rulesprojects_evaluations:create	Permission to create data science rule evaluations.
	Update	application:datascience_rulesprojects_evaluations:update	Permission to update data science rul project evaluations.
	Delete	application:datascience_rulesprojects_evaluations:delete	Permission to delete data science rule evaluations.
Pulse Datascience Rules Projects Rules	View	application:datascience_rulesprojects_rules:read	Permission to read data science rules.
	Create	application:datascience_rulesprojects_rules:create application:datascience_rulesprojects_rules:create:*	Permission to create Data Science Ru
	Update	application:datascience_rulesprojects_rules:update application:datascience_rulesprojects_rules:update:*	Permission to update data science rul
	Delete	application:datascience_rulesprojects_rules:delete	Permission to delete data science rule
	View	application:datascience_rulesprojects_snapshots:read	Permission to read data science rules snapshots.

Pulse Datascience Rules Projects Snapshots	Create	application:datascience_rulesprojects_snapshots:create	Permission to create data science rule snapshots.
	Update	application:datascience_rulesprojects_snapshots:update	Permission to update data science rul project snapshots.
	Delete	application:datascience_rulesprojects_snapshots:delete	Permission to delete data science rule snapshots.
Pulse Datascience Rules Projects Templates	View	application:datascience_rulesprojects_templates:read	Permission to read data science rules templates.
	Create	application:datascience_rulesprojects_templates:create	Permission to create data science rule templates.
	Update	application:datascience_rulesprojects_templates:update	Permission to update data science rul project templates.
	Delete	application:datascience_rulesprojects_templates:delete	Permission to delete data science rule templates.
Pulse Datascience Schemas	View	application:datascience_schemas:read	Permission to read Data Science Sche
	Create	application:datascience_schemas:create	Permission to create Data Science Sc
	Update	application:datascience_schemas:update	Permission to update Data Science Sc
	Delete	application:datascience_schemas:delete	Permission to delete Data Science Sc
Pulse Datascience Tag Groups	View	application:datascience_tag_groups:read	Permission to read Data Science Tag
	Create	application:datascience_tag_groups:create	Permission to create Data Science Ta Groups.
	Update	application:datascience_tag_groups:update	Permission to update Data Science Ta Groups.
	Delete	application:datascience_tag_groups:delete	Permission to delete Data Science Ta Groups.
Pulse Datascience Tag Types	View	application:datascience_tag_types:read	Permission to read Data Science Tag
	Create	application:datascience_tag_types:create	Permission to create Data Science Ta
	Update	application:datascience_tag_types:update	Permission to update Data Science Ta
	Delete	application:datascience_tag_types:delete	Permission to delete Data Science Ta
Pulse Datascience White Box Explanations	View	application:datascience_white_box_explanations:read	Permissions to read White Box Explar configuration.
	Create	application:datascience_white_box_explanations:create	Permissions to create White Box Expl configuration.
	Update	application:datascience_white_box_explanations:update	Permissions to update White Box Exp configuration.
	Delete	application:datascience_white_box_explanations:delete	Permissions to delete White Box Expl configuration.

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Custom UDFs

`Tmx` â

`Tmx.risk(String reasonCodes)`

Receives a String `reasonCodes`, which is a concatenation of 3 character reason codes, fetches the risk score of each and sums them all, returning a transaction risk score.

`FivMarkers` â

```
/**
 * Checks whether the FIV marker is present and not expired.
 *
 * @param fivMarkerCode  The FIV marker code being looked for.
 * @param fivMarkersJson Customer FIV markers list, encoded as JSON string.
 *                       In practice this should always be the value of the same
 *                       Pulse field - 'customer_fiv_markers'
 * @return true if the FIV marker is part of the Customer FIV markers list and has not expired,
 *         false otherwise.
 */
FivMarkers.getIfNotExpired(String fivMarkerCode, String fivMarkersJson)
```

Copy

Checks whether the specified `FivMarker` exists in the customer's FIV markers JSON data and, if present, validates its expiration date if present.

Exampleâ

In Pulse Rule variables:

```
pip_fiv_marker_present = FivMarkers.getIfNotExpired("PIP", customer_fiv_markers)
```

Copy

Where "PIP" is the FIV marker code we're looking for in `customer_fiv_markers`.

note

In practice we always expect this UDF to be called with `customer_fiv_markers` as the second argument.

Implementation Notesâ

- A FIV marker is considered expired as soon as the current date reaches expiration date, meaning if its expiration date is today, it is considered expired.
- We don't expect multiple entries with the same FIV marker code in `customer_fiv_markers`, but, if we do and in order to have consistent behavior, we only consider it not expired if none of the entries have expired.
- In entries in the `customer_fiv_markers` list are malformed according to the expected schema they will be disregarded. In that case, we'll still make a best effort to consider other valid entries in the array.

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Release Notes

Please find below the complete list of releases.â

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Overview

Multi-Tenancyâ

Multi-Tenancy allows Feedzai clients to host and support multiple tenants in a single environment. This allows them to provide at-scale solutions to their own clients, who can use the product on their own.

For **HSBC**, tenants will correspond to **markets**. For example, within the APAC region, there might be a `singapore` and `australia` tenants, which can be used to segregate access to these specific regions, even if they're serviced by the same application/channel in Pulse and Case Manager (outbound payments, for instance).

You can refer this link for more on [Rules and Lists](#).

Landlordâ

A **landlord** is a client who has a Fraud Prevention installation that they can offer to other clients, also known **tenants**. Landlords can carry out the following actions:

- Create, deploy and customize tenant instances, along with all the configuration/data integration and mapping required.
- Write simple and advanced rules.
- Create Lists.
- Edit lists: Add/Remove items to/from lists.
- Create applications.
- Create models, workflows, dashboards and engineer features that can be deployed to one or more tenants.
- Create case management queues, administer tenant users.

For example, **HSBC Hong Kong** will have the main installation so it can be considered as landlord.

Rules and Listsâ

Landlords can create rules and lists that can be applied to all tenants. In addition, landlords can create customized rules and lists for each tenant, and allow tenants to access and modify these resources.

From a multi-tenancy perspective, rules and lists can be of different natures:

- Rules can be **Global** (applying to all tenants and state shared by all tenants), **Common to all tenants** (applying to all tenants, and state specific for each tenant), or **Custom per tenant** (rules customized individually).
- Lists can also be **Global** (Used by all tenants, for instance, a blacklist shared by all tenants), **Common to all tenants** (a list with the same motivation but with different items for different tenants) and **Custom per tenant** (a list that is specifically created for a tenant).

There are also other resources (namely **Plans** and **Models**) that can be defined and/or customized for tenants, with a similar procedure to that of rules and lists.

info

Refer the below links for more on Multi-Tenancy.

- [Multi-Tenancy Concepts](#)
- [Tenant Management](#)

Rules and Lists

Multi-tenancy is the ability to support multiple tenants in the same environment offering a practical solution to apply rules to several tenants at the same time, as well as to customize rules for each tenant.

When creating a rules project, the user can select from the following options:

- **Global** is used for rules that are not multi-tenanted.
- With **Duplicated for Tenant**, the same rule applies to all tenants, but rule conditions can be computed using tenant-specific data.
- With **By Tenant**, custom rules apply to individual tenants.

info

A **landlord** is a client who has a fraud prevention installation that he can offer to other clients, also known as tenants, acting as a reseller. A **tenant** is a client that uses a fraud prevention application provided by a landlord. Tenants have the possibility to create, manage and deploy their own rules and lists depending on the permissions the landlord has given to them.

Duplicated for Tenantâ

When a rules project is set as **Duplicated for Tenant** it means that the set of rules applies to all the tenants in the same way. In terms of deployment, a rule snapshot is created and applied in runtime, the events are then grouped by each tenant and all rules defined in the rule snapshot are applied for each tenant individually, as illustrated in the following image.

By Tenantâ

When a rules project is set as **By Tenant**, it means that a set of rules applies to specific tenants. In terms of deployment, a rule snapshot is created and applied in runtime, the events are then grouped by tenant and the corresponding set of rules are applied for each tenant, as illustrated in the following image.

Advanced vs. simple rulesâ

Users can set the type to advanced or simple, the former being the default type.

This setting has the following meaning:

- **Advanced rules projects** contain rules that can be written using a template and which may have an environment and state.
- **Simple rules projects** contain rules that are not editable inside Pulse and may be used for self-service rules created in Case Manager.

When a rules project is set as Simple, the State input is hidden, the label for the Actions becomes Secondary Actions, and the multi-tenancy options are limited to Global and By Tenant.

Self-service rulesâ

With **self-service rules** (SSR), fraud analysts can write and deploy rules themselves very quickly.

The characteristics of self-service rules in a nutshell are as follows:

- They are deployed directly to the Production environment, in real time.
- They are created in Case Manager by analysts and/or their managers.
- Governance can audit and support self-service rules.

This lets you implement completely new processes and achieve higher operational efficiencies while dynamically following changes in fraud patterns.

Testing self-service rulesâ

Self-service rules can be tested in Case Manager, allowing production environments to execute Spark jobs processing the most recent events that have gone through a self-service rule block of a given workflow. These events will be tested against an updated version of that rule block and measure Hit Rate, Money Captured, etc.

In order for Pulse to process the data in the same way as when going through the live workflow, those events are stored in a custom data store used only for self-service rule testing.

This data store thus receives every event that goes through a self-service rule block in the workflow. Depending on the testing configuration in Case Manager, this data store keeps the data of the time interval that is necessary for rule testing.

Auditing self-service rulesâ

Pulse can be configured to keep audit logs of the creation, edition, deletion, ordering, testing, and publishing operations made to rules. The raw logs are accessible via a REST API through a query that specifies a rule, a workflow element or an application. Get in touch with your contact person at Feedzai to get more information on the audit logs and the Self-Service Rules REST API.

info

Refer below for more.

- Multi tenancy in rules
- Configuring Multi-Tenancy

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Tenant Management

Create new Tenantsâ

You can create Tenants that you can later associate to your users. Do so in the **Multi-Tenancy** section in Pulse:

1. Click on the **Administration** button [] in Pulse
2. Click on **Multi-Tenancy** [] on the left hand pane
3. Click on **Add** [] a new Tenant
4. Populate its **Name** and **ID** as well as any comment you may want to add
5. **Save** and you will be able to see your tenants in the **Multi-Tenancy** page:

Create users associated with tenantsâ

1. Click on the **Security** Icon [ ] in Pulse
2. The **Users** Section should be visible as shown below
3. Click on **Add** [] which will show an user form
4. Fill in the required fields, select the **Tenant** to which the user should be associated and **Save**
5. The user should now be available in the previous page

Multi-Tenancy in Listsâ

Lists in Pulse also support the multi-tenancy concept:

- **Global** lists are aimed to be used on non multi-tenanted environments or if shared across all tenants
- In **By Tenant** and **By Tenant Group** lists each Tenant (or group of tenants) will be able to see and manage a subset of all the items without impacting other Tenants

The list configuration will should under the `Tenancy` column in the List Management page:

info

Refer below for more information

- [Creating new Tenants]
- [Multi tenancy in lists]

On this page

Automation Rules

Automation rules are used to automatically handle the distribution of workload in Case Manager. This page provides you with the first version of HSBC operations flows with the corresponding Automation Rules.

note

Download Full Diagram

Event arrival workflowâ

Wait to Contact Customerâ

This rule is triggered when an alert arrives to Case Manager with a customer contact strategy defined, for example to send a push notification. The rule starts the Delay to Send Customer Notification to send a push notification to the customer when the SLA is breached.

Reviewâ

This rule is triggered when an alert arrives to Case Manager without customer contact strategy defined. The event is moved to the Manual queue to be reviewed by a Fraud Analyst.

Event contact update workflowâ

Unable to Send Push Notificationâ

When an event receives a feedback code `E82`, meaning that the external service was unable to send a push notification to the customer, this rule sets the event breach status to *Not Breached* and adds a 24 hour SLA.

Customer Notification - Fraudâ

When the workflow receives an alert feedback with code `E85`, meaning that the customer marked the transaction as Fraud, this rule sets the event to hold and moves it to the Manual Queue to be reviewed by a Fraud Analyst.

Customer Notification - Genuineâ

When the workflow receives an alert feedback with code `E84`, meaning that the customer marked the transaction as Genuine, this rule marks the event as Not Fraud and closes it.

Event update workflowâ

Handle new assigneeâ

This rule sets an event status to *In Progress* when an Analyst starts working on the transaction.

Mark Fraudâ

This rule removes an alert from its queue, closes and marks it as Fraud when an Analyst marks the event as Fraud. An external notification with the status update is also sent to the customer.

Mark Not Fraudâ

This rule removes an alert from its queue, closes and marks it as Not Fraud when an Analyst marks the event as Not Fraud. An external notification with the status update is also sent to the customer.

Set event on Holdâ

This rule adds a 2 hours SLA when an event is set to On Hold by a Fraud Analyst.

Manual Queue Set SLAâ

This rule adds a 24 hours SLA when an event is moved to the Manual Queue.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/guides/casemanager/event-enrichment

On this page

Event Enrichment

Find in this page a detailed guide on how Case Manager enrichment via Pulse Lists work and how it can be configured.

Overviewâ

When a new event reaches Pulse it can be enriched in the following way:

- It goes through an enrichment plan
- The plan searches for a specific field match in a Pulse list and generates a new field with the matching value
- Pulse picks up the generated fields and adds them to the event's `additional_fields`
- The event is sent to Case Manager with the additional fields. These fields can be mapped in the UI to show the original field enriched value.

Configurationâ

Extension Configurationâ

The inclusion of the enriched fields can be configured by two variables:

- `alerttransformations.cmenrichment.enabled` : indicates whether events should be enriched before being sent to Case Manager
- `alerttransformations.cmenrichment.enrichedFieldPrefix` : indicates which prefix the fields generated by the enrichment plan should have

These can be changed during deployment. Please check [Pulse Configuration Reference] for more information.

Workflow Configurationâ

warning

Please note that although this guide applies to all applications, default resources are only present in the Outbound application.

Before creating the plan, the lists which will be used to enrich events must be created in Pulse Lists tab. By default the following enrichment lists are available:

- Account Type
- Authentication Method
- Beneficiary Bank Details
- CAM Level
- Channel Type
- Event Description
- Event Type
- Sender Transaction Cop Result
- Transaction Status
- Transaction Type

The enrichment plan can then be created and added to a workflow so events are enriched by mapping a set of events fields to their corresponding value in a list.

This can be achieved in Pulse Data Science plans view. By default a `Case Manager Enrichment Plan` exists and holds the following configuration:

Original Field	List	Generated Field	List Field	Fallback Value
auth_method	Authentication Method	cmd_auth_method	value	auth_method
channel_type	Channel Type	cmd_channel_type	value	channel_type
payment_type	Transaction Type	cmd_payment_type	value	payment_type
payment_status	Transaction Status	cmd_payment_status	value	payment_status
sender_account_type	Account Type	cmd_sender_account_type	value	sender_account_type
sender_transaction_cop_result	Sender Transaction Cop Result	cmd_sender_transaction_cop_result	value	sender_transaction_cop_result
auth_cam_level	CAM Level	cmd_auth_cam_level	value	auth_cam_level
payment_type_sub_type	Event Type	cmd_payment_type_sub_type	value	payment_type_sub_type
event_description	Event Description	cmd_event_description	value	sender_transaction_code
sender_bank_sort_code	Beneficiary Bank Details	cmd_sender_bank_name	bank_name	sender_bank_name
sender_bank_sort_code	Beneficiary Bank Details	cmd_sender_bank_address	bank_address	sender_bank_address
sender_bank_sort_code	Beneficiary Bank Details	cmd_sender_bank_post_code	bank_post_code	sender_bank_postal_code
receiver_bank_sort_code	Beneficiary Bank Details	cmd_receiver_bank_name	bank_name	receiver_bank_name
receiver_bank_sort_code	Beneficiary Bank Details	cmd_receiver_bank_address	bank_address	receiver_bank_address
receiver_bank_sort_code	Beneficiary Bank Details	cmd_receiver_bank_post_code	bank_post_code	receiver_bank_postal_code

Please find below a visual example of the configuration.

And a visual example of how the fallback values are computed, in a map operator that comes after the list enrichment operator. When a mapping is not found in the list enrichment for that entry, the map operator *coalesces* the new field using the `??` syntax, to fallback to the specified field from the event payload.

Note: by default all generated fields have the `cmd_` prefix which matches the default prefix defined in `alerttransformations.cmenrichment.enrichedFieldPrefix`.

After the lists and plan are created the plan can be added to the workflow.

Case Manager Configurationâ

The previous mapping was changed to reflect in the enrichment done with the default configuration.

Page	Channel	Widget	Field Description	Previous Mapping	New Mapping
Event Details	Transfers	Beneficiary Account Details	Bank Name	receiver_bank_name	cmd_receiver_bank_name
Event Details	Transfers	Beneficiary Account Details	Bank Address	receiver_bank_address	cmd_receiver_bank_address
Event Details	Transfers	Beneficiary Account Details	Bank City	receiver_bank_city	cmd_receiver_bank_post_code

Page	Channel	Widget	Field Description	Previous Mapping	New Mapping
Event Details	Transfers	Originator Account Details	Bank Name	sender_bank_name	cmd_sender_bank_name
Event Details	Transfers	Originator Account Details	Bank Address	sender_bank_address	cmd_sender_bank_address
Event Details	Transfers	Transaction History	Transaction Type	payment_type	cmd_payment_type
Event Details	Transfers	Transaction	Transaction Type	payment_type	cmd_payment_type
Event Details	Transfers	Transaction	Transaction Status	payment_status	cmd_payment_status
Event Details	Transfers	Transaction	Channel	channel_type	cmd_channel_type
Event Details	Transfers	Confirmation of Payee Details	Payment CoP Description	sender_transaction_cop_result	cmd_sender_transaction_cop_result
Event Details	Digital Activity	Event Summary	Event Description	sender_transaction_code	cmd_event_description
Event Details	Digital Activity	Session	Login Last Authentication	auth_method	cmd_auth_method
Event Details	Digital Activity	Transaction History	Event Description	sender_transaction_code	cmd_event_description
Event Details	Digital Activity	Transaction History	Login Last Authentication	auth_method	cmd_auth_method
Event Details	Digital Activity	Confirmation of Payee Details	Result of the Payment CoP	sender_transaction_cop_result	cmd_sender_transaction_cop_result
Search Page	Transfers	Search Results	Transaction Type	payment_type	cmd_payment_type
Search Page	Digital Activity	Search Results	Event Description	sender_transaction_code	cmd_event_description
Alerts Page	Transfers	Search Results	Transaction Type	payment_type	cmd_payment_type
Alerts Page	Digital Activity	Search Results	Event Description	sender_transaction_code	cmd_event_description

This configuration can be changed by editing Case Manager UI and replacing the mapping to the generated field (all generated fields will be under `event.additional_fields`).

Overview

This page provides you with a set of Case Manager Guides.

Automation Rules

This page provides you with the first version of HSBC operations flows with the corresponding Automation Rules.

SLAs

This page provides you with the first version of HSBC operations flows with the corresponding SLAs.

Case Manager Event Enrichment via Pulse Lists

This page provides you with a guide on how to use Event Enrichment via Pulse Lists.

On this page

SLAs

Service Level Agreements (SLAs) allow you to automate the organization of events into queues based on a time period. This page provides you with the first version of HSBC operations flows with the corresponding SLA's.

Delay to Send Customer Notificationâ

This SLA delays a notification dispatch for 1 minute. Upon breach, the SLA sets the event status to `Contact Customer`, starts the Wait for Customer Response SLA and sends an external notification to the customer. During this SLA if Case Manager receives a notification from Condor to not notify the customer this SLA will not be breached.

Wait for Customer Responseâ

This SLA adds events to the Manual Queue when the customer fails to reply after 2 hours.

24h to Reviewâ

This SLA closes and marks as Not Fraud events that are currently assigned to an analyst but there was no decision made within a 24 hour period.

2h on Holdâ

This SLA sets events to an Unreviewed state after being on hold for a 2 hour period.

On this page

Installing Feedzai with Ansible

The Feedzai Ansible Toolkit refers to a series of Ansible Playbooks created by Feedzai to automate the services installation, and **manage configurations** across multiple environments.

In order to use it, a server (or container) must be provisioned for the purpose of having an [Ansible control node](). This node must contain all the necessary binaries and dependencies for installing each component. Ansible needs to have the ability to connect to each server where the Feedzai platform will be installed, which means that all servers must be able to resolve the IP address (or hostnames) of each other, and the control node must be able to connect to them via SSH. For more information about the connectivity requirements check the [Ansible documentation]().

warning

The Ansible Playbooks provided by Feedzai can be used solely for the purpose of **configuration management**, and are not a solution for deploying Feedzaiâ□□s products.

Requirementsâ□□

- All servers hostnames must resolve their own IP's.
- The servers should be able to resolve each others hostnames, please update the `/etc/hosts` file if necessary, or use a DNS service.
  - Use `ping` to ensure to confirm that each server is able to ping other servers.
- A Linux-based Control Node (preferably dedicated to this task), with a deployment user configured. The node must have `Python3` for running Ansible.
- A deployment user on each server with superuser (`sudo`) permissions (some deployment steps include installing system services, and changing ownership of folders).
- SSH public key authentication between the deployment user in the control node and the deployment user in all servers.

Services Installationâ□□

To install a `<service>` (i.e.: `pulse` , `case-manager` , `reference-data`) with default configurations:

```
python3.6 ansible-playbook \
    --connection=local \
    -i project-default \
    -i project-hsbc \
    -i gcp \
    <service>.yml \
    --tags=<service>.install
```

Copy

Services Configurationâ□□

warning

The following steps are currently under review and assume we can run ansible locally to configure a `<services>` for a given environment `<env>` .

To configure a `<service>` for a given `<region>` (`uk` , `hk` , `asp`), portfolio (`payments` , `cards` , `amh` , `hase`) and `<env>` (e.g.: `dev` , `qa` , `prod` ...):

```
python3.6 ansible-playbook \
    --connection=local \
    -i project-default \
    -i project-hsbc \
    -i project-<env> \
    -i <region>/project-<env> \
    -i <region>/project-<env>/<portfolio> \
    <service>.yml \
    --tags=<service>.configure
```

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Notesâ

The `ansible_user` property needs to exist in the machine that will execute the ansible roles prior to the execution, by default, the `ansible_user` is `feedzai`. This applies to all playbook executions.

In the above command-line invocation of `ansible-playbook`, each corresponding inventories need to exist and contain a `variables.yml` file. If the same variable is defined in multiple inventories, the later ones appearing in the command-line invocation take precedence over the earlier ones. In this way, by adding the very specific variables at the end (for example, a region-specific, portfolio-specific, environment-specific set of variables), they can overwrite any values inherited from earlier inventories such as the default ones (e.g. `project-default`, ...).

Inventory	Directory that must exist with variables.yml
`project-default`	`project-default/group_vars/<service>/variables.yml`
`project-hsbc`	`project-hsbc/group_vars/<service>/variables.yml`
`<region>/project-<env>/<portfolio>`	`<region>/project-<env>/<portfolio>/group_vars/<service>/variables.yml`
`<region>/project-<env>/<portfolio>`	`<region>/project-<env>/<portfolio>/group_vars/<service>/variables.yml`

More inventories may be added as needed, in the same manner.

Follow-up planâ

Embed the previous command in a script to be called in the systemd service with `ExecStartPre` (add `TimeoutSec=30m in` accordingly) so it configures the service before starting it.

The script is able to gather the environment from the machine metadata and configure our service.

Feedzai Ansible Toolkit

Welcome to the Feedzai Ansible Toolkit Guide.

In this section you will find information about the Feedzai Ansible Toolkit:

- Installing Feedzai with Ansible

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/installation-guide/casemanager/afis-extensions

On this page

AFIS Extensions

There are a few extensions in Case Manager able to send data to AFIS that have different triggers and channels with which they work.

Extension	Trigger	Channels
Send Mobile Payment Push	when configured as an **Automation Rule action**	*Transfers*, *Digital Activity*, *Cards*
Send 2 WAY / 1 WAY SMS	when configured as an **Automation Rule action**	*Transfers*, *Digital Activity*
Send 2-WAY SMS	when configured as an **Automation Rule action**	*Cards*
Send 1-WAY SMS	when configured as an **Automation Rule action**	*Cards*, *Non-Mon*
Send Reminder SMS	when configured as an **Automation Rule action**	*Cards*, *Non-Mon*
Inbound Payment Notification	when configured as an **Automation Rule action**	*Inbound Payments*
Card Status Update	when configured as an **Automation Rule action**	*Cards*, *Non-Mon*
Payment Fulfillment	automatically when there is an **Event Status update**	*Transfers*, *Inbound Payments*, *Cards*, *Non-Mon*

Each extension has its own schema which can be seen below. The field `notification_type` is present in all extensions schemas and allows downstream systems to differentiate between the different extensions.

Example Payloadsâ

- Send Mobile Payment Push (Transfers)
- Send Mobile Payment Push (Digital Activity)
- Send Mobile Payment Push (Cards)
- Send 2 WAY / 1 WAY SMS
- Send 2-WAY SMS (Cards)
- Send 1-WAY SMS (Cards)
- Send 1-WAY SMS (Non-Mon)
- Send Reminder SMS (Cards)
- Send Reminder SMS (Non-Mon)
- Inbound Payment Notification
- Card Status Update
- Payment Fulfillment
- Cards Fulfillment


```
{
    "notification_type": "payment_mobile_push",
    "sender_transaction_amount": "15500.0",
    "sender_transaction_amount_dbl": "400.75",
    "sender_transaction_currency": "GPB",
    "sender_transaction_original_amount": "16600.0",
    "sender_transaction_original_currency": "EUR",
    "sender_transaction_bank_account_number": "4b5731a0-aca7-44a7-a641-cea50d32b633",
    "receiver_name": "Receiver Joe Doe",
    "text_for_receiver": "text_for_receiver_sms",
    "lifecycle_id": "7cf0e27f-db8c-4a96-b3e2-f345437b4366",
    "system_decision": "approve",
    "office_id": "",
    "customer_portfolio_country": "",
```

```
    "receiver_transaction_bank_account_number": "9a08f316-df1f-4096-9a2d-a904a82bb65e",
    "customer_portfolio_region": "",
    "sender_initiated_at": "1554196919000",
    "customer_id": "acc_10022917",
    "customer_portfolio_product": "",
    "customer_portfolio_channel": "",
    "customer_portfolio_type": "",
    "customer_portfolio_class": "",
    "operational_status": "remind_customer",
    "customer_id_number": "4be9074c-8018-4755-8776-5135b96e5d22",
    "alert_id": "437b4366",
    "treatment_strategy": "mobile_push_notification",
    "self_serve_strategy": "ivr_self_no",
    "customer_name": "Fraudulent Joe",
    "customer_surname": "Joe",
    "customer_mobile_phone": "0571265381238",
    "customer_email": "[email protected]",
    "customer_preferred_language": "en",
    "payment_message_source": "HK_CMOP" ,
    "payment_sub_type": "T-GTFF" ,
    "thailand_fps_ind": "MOBN",
    "customer_alias": "",
    "user_id": "jbu3470",
    "customer_id_global": "jbu3470",
    "channel_user_id": "[email protected]",
    "device_ip_address": "216.215.104.205",
    "tmx.ip_address": "8.8.8.8",
    "tmx.ip_city": "Wageningen",
    "tmx.ip_country": "Netherlands",
    "tmx.true_isp": "Hutchison Telephone Company Limited",
    "tmx.proxy_isp": "Packethub s.a",
    "tmx.dns_isp": "China Mobile hongkong company limited",
    "device_banking_type": "A",
    "device_os": "Windows",
    "device_user_agent": "Mozilla/5.0 (Windows; U; Windows NT 5.1) AppleWebKit/532.0.2 (KHTML, 
like Gecko) Chrome/19.0.833.0 Safari/532.0.2",
    "receiver_account_bank_id": "ecf6f689-1317-4ed6-ab30-44d03298ab5e",
    "receiver_bank_bic": "4ad3fe88-317b-4b31-ad9b-9136424c17ad",
    "sender_account_type": "CREDIT",
    "receiver_account_type": "CREDIT",
    "customer_account_type": "CREDIT",
    "account_type": "CREDIT",
    "product_code": "ZPP",
    "sender_product_type": "UFAAE"
}
```


Copy

Schemasâ

- Send Mobile Payment Push (Transfers)
- Send Mobile Payment Push (Digital Activity)
- Send Mobile Payment Push (Cards)
- Send 2 WAY / 1 WAY SMS
- Send 2-WAY SMS (Cards)
- Send 1-WAY SMS (Cards)
- Send 1-WAY SMS (Non-Mon)
- Send Reminder SMS (Cards)
- Send Reminder SMS (Non-Mon)
- Inbound Payment Notification
- Card Status Update
- Payment Fulfillment
- Cards Fulfillment

Field	Type	Description
notification_type	String	This field always takes the value `payment_mobile_push` in this extension
text_for_receiver	String	Payment description, destination narrative (payment ref included)
sender_transaction_amount	String	Transaction amount in transaction currency and in scientific notation. This should always be a positive number
sender_transaction_amount_dbl	String	Transaction amount in transaction currency and in scientific notation based on the new payload amount field. This should always be a positive number
sender_transaction_currency	String	Three-digit numeric ISO code of transaction currency (Do not truncate leading zeroes.)
sender_transaction_original_amount	String	Transaction original amount in transaction original currency and in scientific notation. This should always be a positive number
sender_transaction_original_currency	String	Three-digit numeric ISO code of transaction original currency (Do not truncate leading zeroes.)
sender_transaction_bank_account_number	String	Account number of the primary interest e.g. The transacting account for payment card transactions
receiver_name	String	Beneficiary name in non-latin characters
system_decision	String	The original decision provided by the fraud decision system
office_id	String	The RPS/CONDOR queue name sent as part of the payment initiation API
customer_portfolio_country	String	Customer Account Level Country including markets e.g. UK FD, M&S
receiver_transaction_bank_account_number	String	Customer (beneficiary) Account number
customer_portfolio_region	String	Customer Account Level Region - Europe, Latin America, Middle East, Asia Pacific, and North America
customer_portfolio_channel	String	Customer Account Level Channel - Debit Cards, Credit Cards, Digital Banking, Telephone Banking, Physical Banking, Manual Processing, Unknown
customer_portfolio_class	String	Customer Account Level Classification e.g. Mass, Premier, Business, and N/A
sender_initiated_at	String	Effective date of the transaction (in UTC format when known). Payment Card Transactions: This date reflects when the transaction first entered the issuer's system. This should be derived from the issuer's own timestamp. Payments via online banking payment application: This date reflects when the payment is released from payment application for payment processing.
customer_portfolio_product	String	Customer Account Level Product - Plastic and Non-Plastic cards
customer_id	String	Customer number
customer_id_number	String	Customer ID number
customer_portfolio_type	String	Customer Account Level Type e.g. General, Jade, Prem, Basic, Advanced etc.
alert_id	String	Unique reference number generated whenever the system decides to generate an alert on a particular transaction
lifecycle_id	String	Identification number of transaction Unique ID/Reference Generated for Payment by Bank
operational_status	String	Operational Status of the Event (Possible values Contact Customer, Unreviewed Etc Etc)
treatment_strategy	String	Treatment Strategy
self_serve_strategy	String	Self Serve Strategy
customer_name	String	Customer Name
customer_surname	String	Customer Surname

Field	Type	Description
customer_mobile_phone	String	Mobile Phone
customer_email	String	Email Address
customer_preferred_language	String	customerPreferredLanguage
payment_message_source	String	Unique identifier of the source system sending data
payment_sub_type	String	Payment Sub Type
thailand_fps_ind	String	Payment FPS Indicator
customer_alias	String	Customer Chinese Name
user_id	String	Online Banking User ID
customer_id_global	String	Global Customer ID
channel_user_id	String	Online/Internet Banking User ID
device_ip_address	String	IP address
tmx.ip_address	String	True IP Address
tmx.ip_city	String	True IP City
tmx.ip_country	String	True IP Country
tmx.true_isp	String	The legitimate Internet Service Provider that assigned the original IP address used by an end-user to access the internet. This ISP is tied directly to the user's physical location or device - without any intermediate.
tmx.proxy_isp	String	It is the service provider offering proxy IP addresses that mask the user's actual IP address. These includes VPNs, residential proxies, datacenter proxies, and anonymizers. In essence, the ISP visible to the internet is that of the proxy provider, not the user's real ISP.
tmx.dns_isp	String	It refers to the ISP seen during Domain Name System (DNS) resolution - the process of converting human-readable domain names into IP addresses. This is typically the ISP of the DSN resolver the user is.
device_banking_type	String	Online Banking device
device_os	String	Operating System Version
device_user_agent	String	User Agent HTTP Header
sender_account_type	String	Type of sender account
receiver_account_type	String	Type of receiver account
customer_account_type	String	Type of customer account
account_type	String	Account type
product_code	String	The code of the product offered by the bank that was used to initiate the transfer
sender_product_type	String	Type of product held

Previous Transactionsâ

The `previous_transactions` field is a special field that contains information about previous transactions that relate to the event that triggered an extension.

The default criteria to list related transactions is the following:

- Same `card_pan` value
- Same `customer_id` value
- Happened in the last 24 hours (calculate from the time the extension was triggered)
- Is not the event that triggered the extension

If a transaction is found, its information is then added to the extension payload, e.g:


```json
{
  "notification_type": "2_way_1_way_sms",
  ...
  "previous_transactions": [
    {
      "amount_dbl": "11000.0",
      "lifecycle_id": "9a3c955c-4add-4664-8954-818639e59b7e",
      "local_datetime": "20110123",
      "acceptor_name": "GOLFSTORE3DS",
      "currency_code": "634",
      "event_occurred_at": "1771927092140"
    }
  ]
}
```


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Case Manager Installation Guide

Welcome to the Case Manager Installation Guide.

In this section you will find a set of pages with information about the Case Manager installation process:

On this page

Configuration Recommendations

tip

For a complete reference check the official Configuration Guide.

Ansibleâ

note

Available ansible variables for running the ansible playbook.

Variable	Default Value
ansible_user	feedzai
ansible_local_user	{{ ansible_user }}
cm_install_path	"/opt/feedzai"
cm_home_path	"{{ cm_install_path }}/product-cm-{{ cm_version }}"
cm_java_options	"-Xms4g -Xmx4g"
database_casemanager_hostname	"postgres"
database_casemanager_port	"5432"
database_casemanager_database_name	"feedzai"

Variable	Default Value
database_casemanager_ro_hostname	"{{ database_casemanager_hostname }}"
database_casemanager_ro_port	"{{ database_casemanager_port }}"
database_casemanager_ro_database_name	"{{ database_casemanager_database_name }}"
database_casemanager_jdbc_opts	[currentSchema={{ database_casemanager_schema }}, loginTimeout=10, connectTimeout=2,
database_casemanager_jdbc	"jdbc:postgresql://{{ database_casemanager_hostname }}:{{ database_casemanager_port }}/{
database_casemanager_jdbc_ro_opts	[currentSchema={{ database_casemanager_schema }}, loginTimeout=10, connectTimeout=2,
database_casemanager_ro_jdbc	"jdbc:postgresql://{{ database_casemanager_ro_hostname }}:{{ database_casemanager_ro_p
database_casemanager_username	casemanager
database_casemanager_password	secret
database_casemanager_schema	casemanager
database_casemanager_ro_username	{{ database_casemanager_username }}

Variable	Default Value
database_casemanager_ro_password	{{ database_casemanager_password }}
database_casemanager_ro_schema	{{ database_casemanager_schema }}
cm_url	"http://localhost:8090"
reference_data_servers	reference-data
reference_data_url	"http://{{reference_data_servers}}:{{reference_data_port}}""
reference_data_casemanager_jwt_token	"eyJraWQiOiJkZWZhdWx0IiwidHlwIjoiSldUIiwiYWxnIjoiSFMyNTYifQ.eyJzdWIiOiJoc2JjIiwibm y9bm8isg4"
cm_jetty_http_port	8080
pulse_casemanager_username	casemanager
pulse_casemanager_password	Secret1!
rabbitmq_casemanager_brokers	{{ rabbitmq_brokers }}
rabbitmq_casemanager_username	cm
rabbitmq_casemanager_password	secret
rabbitmq_casemanager_vhost	cm
jks_store_password	N/A
cm_truststore_path	"{{ cm_home_path }}/conf/trustedca.jks"

Variable	Default Value
pulse_internal_url	"http://pulse:8080"
pulse_url	{{ pulse_internal_url }}
cm_secret_value	N/A
cm_properties_to_encrypt	[]
afis_scheme	https
afis_scheme_send_sms	{{ afis_scheme }}
afis_payment_fulfillment_cm_extension_enabled	true
afis_mobile_outbound_cm_extension_enabled	true
afis_cm_extension_enabled_send_sms	true

Variable	Default Value
send_2_way_sms_extension_enabled	true
send_1_way_sms_extension_enabled	true
send_reminder_sms_extension_enabled	true
inbound_payment_notification_extension_enabled	true
card_status_update_extension_enabled	true
afis_host	
afis_host_send_sms	{{ afis_host }}
afis_port	443
afis_port_send_sms	{{ afis_port }}

Variable	Default Value
afis_api_endpoint	feedzai-event-notification
afis_api_endpoint_send_sms	{{ afis_api_endpoint }}
afis_timeout_in_seconds_send_sms	{{ afis_timeout_in_seconds }}
afis_waitstrategy_in_seconds_send_sms	{{ afis_waitstrategy_in_seconds }}
afis_stopstrategy_attempts_send_sms	{{ afis_stopstrategy_attempts }}
afis_proxy_enabled_send_sms	{{ afis_proxy_enabled }}
afis_proxy_host_send_sms	{{ afis_proxy_host }}
afis_proxy_port_send_sms	{{ afis_proxy_port }}

Variable	Default Value
dsp_endpoint	
dsp_username	{{ lookup('gcp_secret_manager', secret='dsp_username', version=secret_manager_secret_ve
dsp_password	{{ lookup('gcp_secret_manager', secret='dsp_password', version=secret_manager_secret_ver
dsp_endpoint_send_sms	{{ dsp_endpoint }}
dsp_username_send_sms	{{ dsp_username }}

Variable	Default Value
casemanager_api_url	{{ cm_url }}
casemanager_api_username	{{ pulse_casemanager_username}}
casemanager_api_password	{{ pulse_casemanager_password }}
cm_liquibase_environment	
getnext_time_frame	604800
casemanager_setup_thread_dumps_cron	false
casemanager_thread_dumps_cron_job	[]
cm_liquibase_rollback	false

Variable	Default Value
apps_enabled	apps_enabled: Â Â - uk_outbound_payments
cm_multi_tenancy_enabled	false
payment_fulfillment_exclude_events_without_status_reason	true
show_delegate_transaction_history_widget	true
cm_loggers	{}
cm_probes_log_level	"OFF"

Variable	Default Value
cm_probes_output_file_name	"cm-probes.log"
use_rsync	true
show_hsbcnet_search_fields	false
fulfillment_channels_state_machines	{"transfers": [{"name": "status"}], "inbound-payments": [{"name": "status"}], "cards": [{"name": "s

Variable	Default Value
cm_transaction_block_date_offset_days	5
rabbitmq_casemanager_use_quorum_queues	false

Cron typeâ

Variable	type	Mandatory	Default Value	Description
name	STRING	X		Name of the cron job
hour	STRING		*	The cron expression for `hour`
minute	STRING		*	The cron expression for `minute`
day	STRING		*	The cron expression for `day`
weekday	STRING		*	The cron expression for `weekday`
month	STRING		*	The cron expression for `month`

PDB channel specific databaseâ

Variable	Default Value	Description
pdb_active_channels		Identifies the usage of the dedicated databases. The first channel of the list identifies the main database. Available values: Â Â - cards Â Â - digital-activity Â Â - inbound-payments Â Â - transfers Â Â - non-mon

Each of the following variables needs a suffix _<channel> that identifies the channel and the configuration of the dedicated database.

Variable	Default Value
database_casemanager_engine_<channel>	"{{ database_casemanager_engine }}"
database_casemanager_jdbc_<channel>	"{{ database_casemanager_jdbc }}"
database_casemanager_username_<channel>	"{{ database_casemanager_username }}"
database_casemanager_password_<channel>	"{{ database_casemanager_password }}"
database_casemanager_schema_<channel>	"{{ database_casemanager_schema }}"
database_casemanager_schemapolicy_<channel>	"{{ database_casemanager_schemapolicy }}"
database_casemanager_pool_size_<channel>	"{{ database_casemanager_pool_size }}"
database_casemanager_pool_lazy_<channel>	"{{ database_casemanager_pool_lazy }}"
database_casemanager_reconnect_on_lost_<channel>	"{{ database_casemanager_reconnect_on_lost }}"
database_casemanager_max_number_retries_<channel>	"{{ database_casemanager_max_number_retries }}"
database_casemanager_retry_interval_<channel>	"{{ database_casemanager_retry_interval }}"
database_casemanager_query_select_timeout_<channel>	"{{ database_casemanager_query_select_timeout }}"
database_casemanager_ro_jdbc_<channel>	"{{ database_casemanager_ro_jdbc }}"
database_casemanager_ro_pool_size_<channel>	"{{ database_casemanager_ro_pool_size }}"
database_casemanager_ro_username_<channel>	"{{ database_casemanager_ro_username }}"
database_casemanager_ro_password_<channel>	"{{ database_casemanager_ro_password }}"
database_casemanager_ro_schema_<channel>	"{{ database_casemanager_ro_schema }}"
database_casemanager_ro_reconnect_on_lost_<channel>	"{{ database_casemanager_ro_reconnect_on_lost }}"
database_casemanager_ro_max_number_retries_<channel>	"{{ database_casemanager_ro_max_number_retries }}"
database_casemanager_ro_retry_interval_<channel>	"{{ database_casemanager_ro_retry_interval }}"
database_casemanager_ro_query_select_timeout_<channel>	"{{ database_casemanager_ro_query_select_timeout }}"
database_casemanager_ro_pool_lazy_<channel>	"{{ database_casemanager_ro_pool_lazy }}"
database_casemanager_ro_schemapolicy_<channel>	"{{ database_casemanager_ro_schemapolicy }}"

Performance configurationâ

With the inclusion of new use cases it is expected that Case Manager sizing to be updated. Aside from bare metal sizing updates, Case Manager also provides performance configurations that allow fine tuning of the various nodes.

warning

Changing any of these configurations can have a performance impact in Case Manager. They should only be changed in agreement with the project team.

Please find below a description of each of the available configurations.

Variable	Default Value	Description
cm_performance_event_worker_threads	8	The number of workers that should be used to asynchronously store events into event storage.
cm_performance_queue_prefetch_size	200	The number of messages Case Manager fetches from RabbitMQ.
cm_performance_batch_storage_size	300	The size of the batch, i.e. the number of entries the batch can hold after which a flush will occur.
cm_performance_batch_storage_timeout	1000	

Variable	Default Value	Description
		The batch timeout, i.e. the time after which a batch flush will occur, regardless of the amount of entries present in it.
cm_performance_batch_flush_workers	4	The number of workers that should be used to asynchronously execute batch flush callback operations.
cm_performance_batch_flush_queue_size	5000	The size of the buffer that holds events to be added to the batch.
cm_performance_event_storage_queue_size	200	The size of the buffer that holds events to be added to the batch.
cm_performance_event_storage_batch_size	500	The size of the batch, i.e. the number of entries the batch can hold after which a flush will occur.
cm_performance_event_storage_batch_timeout	1000	The batch timeout, i.e. the time after which a batch flush will occur, regardless of the amount of entries present in it.
cm_performance_event_storage_batch_threads	1	The batch threads, i.e. the number of threads getting entries of the queue and writing to the batch.
cm_performance_related_max_last_flush_secs	10	Maximum period (in seconds) that notifier should wait before sending the RTS notification.
cm_performance_ignore_event_updates	false	Whether event updates should be ignored, and thus, the event ingestion process optimized.
cm_performance_disable_auditing_without_users	false	Whether the audit for actions without users should be skipped in the database.
cm_performance_max_queued_explanations	5000	The Number of queued explanations for persistence by the explanations batch.
cm_performance_max_related_transactions	500	The maximum number of related transactions to an event to fetch.
cm_performance_explanations_batch_size	300	The size of the batch, i.e. the number of entries the batch can hold after which a flush will occur.
cm_performance_automation_worker_threads	4	The batch threads, i.e. the number of threads getting entries of the queue and writing to the batch.
cm_performance_automation_batches_queue_size	200	The size of the buffer that holds events to be added to the batch.
cm_performance_automation_batches_batch_size	500	The size of the batch, i.e. the number of entries the batch can hold after which a flush will occur.
cm_performance_automation_batches_batch_timeout	1000	The batch timeout, i.e. the time after which a batch flush will occur, regardless of the amount of entries present in it.
cm_performance_automation_batches_batch_threads	1	The batch threads, i.e. the number of threads getting entries of the queue and writing to the batch.
cm_performance_automation_rule_cache	0	Automation cache value until automation rules are refreshed in memory.
cm_performance_max_queued_automations	5000	Maximum number of queued automations for execution by the automation worker threads
cm_performance_audit_batches_queue_size	200	The size of the buffer that holds events to be added to the batch.
cm_performance_audit_batches_batch_size	300	The size of the batch, i.e. the number of entries the batch can hold after which a flush will occur.

Variable	Default Value	Description
cm_performance_audit_batches_batch_timeout	1000	The batch timeout, i.e. the time after which a batch flush will occur, regardless of the amount of entries present in it.
cm_performance_audit_batches_batch_threads	1	The batch threads, i.e. the number of threads getting entries of the queue and writing to the batch.
cm_performance_enable_db_heuristic_for_event_queue_search	false	Whether an explicit `queueId neq null` should be added to improve queue searches.

How To

Welcome to the Case Manager How To.

In this section you will find the following set of tutorials with information on how to configure Case Manager to support global rollout:

- Deploy new channel
- Use channel specific databases

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/installation-guide/casemanager/multi-tenancy-configuration

On this page

Multi-tenancy Configuration

This page describes the configuration required in order to enable Multi-tenancy in Case Manager.

Enabling Multi-tenancyâ

Multi-tenancy can be enabled during deployment by setting `cm_multi_tenancy_enabled` to `true`. This will enable tenancy for the following features:

- Event segregation
- List management
- Queues management

Tenant fieldâ

The tenant field to which Case Manager looks to define the tenant will always be the same `schema.tenant` in Case Manager since its value is based on a field configured in Pulse.

Check Pulse Configuration Reference for more details.

Permissionsâ

For users to be able to use multi-tenancy features in Case Manager, their role should have the `cm:resource:landlord:*` permission.

This permission reduces tenancy configuration, thus not requiring system administrators to manually add each tenant to a landlord user.

info

Refer below for more

- Multi tenancy configuration

On this page

Network and Connectivity

info

This page is based on the architecture of Feedzai standard solutions. Specific clients or custom solutions may have additional requirements, so this is not a definitive guide.

Firewall Rulesâ

tip

For a complete list of the the Firewall Configurations required for Case Manager, please refer to the [product documentation].

Inboundâ

note

Inbound â other services connect to this service.

Service	Port	Description
HTTP	TCP 8080	Case Manager UI HTTP
HTTPS	TCP 8443	Case Manager UI HTTPS
SSH	TCP 22	SSH access (if available)

Outboundâ

note

Outbound â this service connects to external services.

Service	Port	Description
RBDMS (PostgreSQL)	TCP 5432	
RabbitMQ	TCP 5761-5762	
Elasticsearch	TCP 9200	
Pulse	8080	Communication with Pulse (via HTTP)
Pulse	8443	Communication with Pulse (via HTTPS)
Reference Data Service	8443-8444	Communication with Reference Data Service

Load Balancersâ

For high available environments Load Balancers must be set as follow:

Service	Description	Algorithm	Healthcheck URL	Response Codes	Policy Example
Case Manager	Entry to access the web UI.	Least connections (or round robin)	`HTTPS:8443/api/configuration/latest` `HTTP:8080/api/configuration/latest`	200 - Node is available	Checks should be performed every 30 seconds with a timeout of 5 seconds. Nodes should be removed after the second failure.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/installation-guide/casemanager/notifications-schema

Notifications Schema

In this page you can find the notification schema and examples for Case Manager external notifications feature.

Please find below the notification schema for Outbound events.

Field	Type	Description
self_serve_strategy	String	Self Serve Strategy field.
lifecycle_id	String	Identification number of transaction Unique ID/Reference Generated for Payment by Bank.
customer_portfolio_region	String	Customer Account Level Region - Europe, Latin America, Middle East, Asia Pacific, and North America.
customer_portfolio_channel	String	Customer Account Level Channel - Debit Cards, Credit Cards, Digital Banking, Telephone Banking, Physical Banking, Manual Processing, Unknown.
customer_portfolio_country	String	Customer Account Level Country including markets e.g. UK FD, M&S.
event_type	String	Field that indicates the endpoint from which the event entered.
premier_marker	String	Premier Marker field.
customer_portfolio_type	String	Customer Account Level Type e.g. General, Jade, Prem, Basic, Advanced etc.
treatment_strategy	String	Treatment Strategy field.
alert_id	String	Unique reference number generated whenever the system decides to generate an alert on a particular transaction.
pib_suspended	String	PIB Suspended.
customer_portfolio_class	String	Customer Account Level Classification e.g. Mass, Premier, Business, and N/A.
customer_id	String	Customer number.
customer_id_number	String	Hong Kong Identification Number.
customer_portfolio_product	String	Customer Account Level Product - Plastic and Non-Plastic cards.
tenant	String	Tenant id field.

An example of an event state update notification would be:


```
{
    "ids": [
        {
            "identifier": "lc_id_outbound_payments_3",
            "timestamp": 1736322305157,
            "channelId": "transfers",
            "payload": {
                "schema": {
                    "self_serve_strategy": "ivr_self_no",
                    "lifecycle_id": "lc_id_outbound_payments_3",
                    "customer_portfolio_region": "Europe",
                    "customer_portfolio_channel": "Digital Banking",
                    "customer_portfolio_country": "UK",
                    "event_type": "transfer_initiation",
                    "premier_marker": "premier1",
                    "customer_portfolio_type": "GENERAL",
                    "treatment_strategy": "mobile_push_notification",
                    "alert_id": "yments_3",
                    "pib_suspended": "YESD",
                    "customer_portfolio_class": "PREMIER",
```

```
                    "customer_id": "customer_id_1",
                    "customer_id_number": "0HKHSXXXXXX796256",
                    "customer_portfolio_product": "Non-plastic",
                    "sender_transaction_extra_1": "Sender Transaction Additional Info 1",
                    "sender_transaction_extra_2": "Sender Transaction Additional Info 2",
                    "sender_transaction_extra_3": "Sender Transaction Additional Info 3",
                    "customer_additional_info_1": "Customer Additional Info 1",
                    "customer_additional_info_2": "Customer Additional Info 2",
                    "customer_additional_info_3": "Customer Additional Info 3"
                    "customer_additional_info_4": "Customer Additional Info 4",
                    "tenant": "feedzai"
                }
            }
        }
    ],
    "notificationVersion": "1",
    "stateMachineId": "status",
    "state": {
        "id": "fraud"
    },
    "reasons": [
        {
            "id": "23f7348e-19f4-4f1d-8968-01862b1c5a44",
            "statusId": "fraud",
            "name": "Fraud - 1st Party Fraud"
        }
    ],
    "updatedAt": 1736322343970,
    "userId": "mWVznM0cV8dLzVXMXUVH_w",
    "username": "pulse",
    "note": "Outbound Payments 1st Party Fraud"
}
```


Copy

Please find below the notification schema for Inbound events.

Field	Type	Description
event_type	String	Field that indicates the endpoint from which the event entered.
alert_id	String	Unique reference number generated whenever the system decides to generate an alert on a particular transaction.
lifecycle_id	String	Identification number of transaction Unique ID/Reference Generated for Payment by Bank.
customer_id	String	Customer number.
customer_id_number	String	Hong Kong Identification Number.
treatment_strategy	String	Treatment Strategy.
self_serve_strategy	String	Self Serve Strategy.
pib_suspended	String	PIB Suspended.
tenant	String	Tenant id field.

An example of an event state update notification would be:


```
{
    "ids": [
        {
            "identifier": "lc_id_inbound_payments_1",
            "timestamp": 1736332401699,
            "channelId": "inbound-payments",
            "payload": {
                "schema": {
```

```
                    "self_serve_strategy": "none",
                    "event_type": "inbound_payment_initiation",
                    "treatment_strategy": "none",
                    "alert_id": "yments_1",
                    "pib_suspended": "YESD",
                    "lifecycle_id": "lc_id_inbound_payments_1",
                    "customer_id": "customer_id_1",
                    "customer_id_number": "0HKHSXXXXXX796256",
                    "tenant": "feedzai"
                }
            }
        }
    ],
    "notificationVersion": "1",
    "stateMachineId": "status",
    "state": {
        "id": "fraud"
    },
    "reasons": [
        {
            "id": "b8b1c2dc-7b4b-43d9-bbb4-66c676f1498a",
            "statusId": "fraud",
            "name": "Inbound Fraud - 1st Party Fraud"
        }
    ],
    "updatedAt": 1736332720278,
    "userId": "mWVznM0cV8dLzVXMXUVH_w",
    "username": "pulse",
    "note": "Inbound Payments - 1st Party Fraud"
}
```


Copy

Please find below the notification schema for Card events.

Field	Type	Description
self_serve_strategy	String	Self Serve Strategy field.
event_type	String	Field that indicates the endpoint from which the event entered.
alert_id	String	Unique reference number generated whenever the system decides to generate an alert on a particular transaction.
lifecycle_id	String	Identification number of transaction Unique ID/Reference Generated for Payment by Bank.
tenant	String	Tenant ID field.
lifecycle_id_alt	String	SO Mandate ID + Schedule Number.
card_transaction_ref_number	String	Unique reference number received from source.
card_transaction_component_type	String	Customer Type, Convention.
source	String	Source field.
customer_portfolio_region	String	Customer Portfolio Region.
customer_portfolio_country	String	Customer Portfolio Country.
customer_portfolio_channel	String	Customer Portfolio Channel.
customer_portfolio_product	String	Customer Portfolio Product.
customer_portfolio_type	String	Customer Portfolio Type.
customer_portfolio_class	String	Customer Portfolio Class.
customer_portfolio	String	Customer Portfolio.

Field	Type	Description
account_id	String	Bank account ID.
account_number	String	The International Bank Account Number (IBAN), or any other unique identifier used as bank account.
card_phone_banking_user_id	String	Phone Banking User ID.
customer_id	String	Customer ID.
customer_email	String	Customer email.
customer_name	String	Customer name.
customer_id_personal	String	Personal Customer Identification Number (CIN) for HINV Business Debit Cards.
customer_id_business	String	Business Customer Identification Number (CIN) for HINV Business Debit Cards.
customer_id_number	String	Stores the unique Country ID that is issued to the customer as part of their National Identification.
customer_id_global	String	Can be used to map GUCI or another global customer identifier.
card_txn_sub_type	String	Card transaction sub type
trans_message_type	String	Trans message type field.
acquirer_id	String	Acquirer ID.
amount_dbl	Number	Amount field.
trans_model_amt	Number	Trans model amount field.
cardholder_bill_amount	Number	Cardholder bill amount field.
cardholder_bill_currency_code	String	Cardholder bill currency code.
trans_currency_code	String	Trans currency code field.
currency_code	String	Currency code field.
treatment_strategy	String	Treatment strategy.
oob_comm_type	String	The type of communication made with the client.
oob_notes	String	Notes related to the out-of-band communication.
oob_feedback_notes	String	Notes related to the out-of-band feedback event.
oob_response_type	String	The type of out-of-band response made by the client.
feedback_type	String	Identifies the different types of feedback.
feedback_label	String	The feedback label being provided for the transaction (e.g. 'fraud').
external_status_reason	String	The status of the transaction as manually reviewed by a fraud analyst.
channel_type	String	Channel type.
card_pan	String	Payment card number.
saf_ind	String	SAF Indicator.
sender_initiated_at	Timestamp	Timestamp for when the transaction is initiated by the user.
sender_initiated_at_local	String	Local timestamp for when the transaction is initiated by the user.
local_datetime	String	Time and date for when payment originated.
event_occurred_at	Timestamp	Timestamp corresponding to the moment when the current event occurred (in milliseconds since the Unix epoch).
crypto_transaction_info	String	Indicates if the transaction is being done to purchase crypto assets.
card_type	String	Specifies whether the card is black, platinum, gold, silver, etc.
outcome_decision	String	The original decision provided by the fraud decision system.

Field	Type	Description
wallet_decision	String	Wallet decision field.

An example of an event state update notification would be:


```
{
    "ids": [
        {
            "identifier": "lc_id_cards_1",
            "timestamp": 1736333077961,
            "channelId": "cards",
            "payload": {
                "schema": {
                    "self_serve_strategy": "ivr_self_yes",
                    "event_type": "authorization",
                    "alert_id": "_cards_1",
                    "lifecycle_id": "lc_id_cards_1",
                    "tenant": "feedzai",
                    "lifecycle_id_alt": "",
                    "card_transaction_ref_number": "1234",
                    "card_transaction_component_type": "TRX",
                    "source": "MSC",
                    "customer_portfolio_region": "US-EAST",
                    "customer_portfolio_country": "feedzai",
                    "customer_portfolio_channel": "portfoliochannel",
                    "customer_portfolio_product": "product",
                    "customer_portfolio_type": "portfoliotype",
                    "customer_portfolio_class": "elite",
                    "customer_portfolio": "EU_PC_CR_MS_GNRL",
                    "account_id": "account_id",
                    "account_number": "1005103001044120",
                    "card_phone_banking_user_id": "A5435454435",
                    "customer_id": "customer_1",
                    "customer_email": "[email protected]",
                    "customer_name": "John Doe",
                    "customer_id_personal": "",
                    "customer_id_business": "",
                    "customer_id_number": "",
                    "customer_id_global": "",
                    "card_txn_sub_type": "",
                    "trans_message_type": "14",
                    "acquirer_id": "12345678912",
                    "amount_dbl": 245.0,
                    "trans_model_amt": 10.0,
                    "cardholder_bill_amount": 11000.0,
                    "cardholder_bill_currency_code": "951",
                    "trans_currency_code": "951",
                    "currency_code": "951",
                    "treatment_strategy": "none",
                    "oob_comm_type": "sms",
                    "oob_notes": "ContactedJohnDoetoconfirm.",
                    "oob_feedback_notes": "Theaccountholderdeniedtheauthenticityofthetransaction.
",
                    "oob_response_type": "call",
                    "feedback_type": "out_of_band",
                    "feedback_label": "",
                    "external_status_reason": "",
                    "channel_type": "mobile",
                    "card_pan": "",
                    "saf_ind": "N",
                    "sender_initiated_at": 1752066662,
                    "sender_initiated_at_local": "2025-07-12 00:00:00.000000 UTC",
                    "local_datetime": "20110123",
```

```
                    "event_occurred_at": 1769686995757,
                    "crypto_transaction_info": "",
                    "card_type": "credit",
                    "outcome_decision": "fraud",
                    "wallet_decision": "RED"
                }
            }
        }
    ],
    "notificationVersion": "1",
    "stateMachineId": "status_cards",
    "state": {
        "id": "fraud_cards"
    },
    "reasons": [
        {
            "id": "6e5fb943-febb-47dc-a460-dede19dae61a",
            "statusId": "fraud_cards",
            "name": "Cards Fraud - 1st Party Fraud"
        }
    ],
    "updatedAt": 1736333159273,
    "userId": "mWVznM0cV8dLzVXMXUVH_w",
    "username": "pulse",
    "note": "Cards - 1st Party Fraud"
}
```


Copy

An example of a **transaction block** update would be:


```
{
    "ids": [
        {
            "identifier": "lc_id_cards_2",
            "timestamp": 1774006677854,
            "channelId": "cards",
            "payload": {
                "schema": {
                    "self_serve_strategy": "ivr_self_yes",
                    "event_type": "authorization",
                    "alert_id": "_cards_1",
                    "lifecycle_id": "lc_id_cards_1",
                    "tenant": "feedzai",
                    "lifecycle_id_alt": "",
                    "card_transaction_ref_number": "1234",
                    "card_transaction_component_type": "TRX",
                    "source": "MSC",
                    "customer_portfolio_region": "US-EAST",
                    "customer_portfolio_country": "feedzai",
                    "customer_portfolio_channel": "portfoliochannel",
                    "customer_portfolio_product": "product",
                    "customer_portfolio_type": "portfoliotype",
                    "customer_portfolio_class": "elite",
                    "customer_portfolio": "EU_PC_CR_MS_GNRL",
                    "account_id": "account_id",
                    "account_number": "1005103001044120",
                    "card_phone_banking_user_id": "A5435454435",
                    "customer_id": "customer_1",
                    "customer_email": "[email protected]",
                    "customer_name": "John Doe",
                    "customer_id_personal": "",
                    "customer_id_business": "",
                    "customer_id_number": "",
                    "customer_id_global": "",
```

```
                    "card_txn_sub_type": "",
                    "trans_message_type": "14",
                    "acquirer_id": "12345678912",
                    "amount_dbl": 245.0,
                    "trans_model_amt": 10.0,
                    "cardholder_bill_amount": 11000.0,
                    "cardholder_bill_currency_code": "951",
                    "trans_currency_code": "951",
                    "currency_code": "951",
                    "treatment_strategy": "none",
                    "oob_comm_type": "sms",
                    "oob_notes": "ContactedJohnDoetoconfirm.",
                    "oob_feedback_notes": "Theaccountholderdeniedtheauthenticityofthetransaction.
",
                    "oob_response_type": "call",
                    "feedback_type": "out_of_band",
                    "feedback_label": "",
                    "external_status_reason": "",
                    "channel_type": "mobile",
                    "card_pan": "",
                    "saf_ind": "N",
                    "sender_initiated_at": 1752066662,
                    "sender_initiated_at_local": "2025-07-12 00:00:00.000000 UTC",
                    "local_datetime": "20110123",
                    "event_occurred_at": 1769686995757,
                    "crypto_transaction_info": "",
                    "card_type": "credit",
                    "outcome_decision": "fraud",
                    "wallet_decision": "RED"
                }
            }
        }
    ],
    "notificationVersion": "1",
    "stateMachineId": "block_code",
    "state": {
        "id": "sample_1"
    },
    "reasons": [
        {
            "id": "29989fc9-a3df-4e81-aed5-b587d079846e",
            "statusId": "sample_1",
            "name": "Block code - Sample 1"
        }
    ],
    "updatedAt": 1774006715513,
    "userId": "mWVznM0cV8dLzVXMXUVH_w",
    "username": "pulse",
    "note": null
}
```

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Requirements

tip

For a complete list of the Case Manager requirements, please refer to the official product documentation.

Service Requirementsâ

This service requires the following software to run.

Java Environmentâ

OpenJDK SE Development Kit version 1.8, with the following vendors (with their latest hotfix versions):

- Azul Zulu rpm zip tar.gz
- Azul Platform Prime (formerly Zing)
- Amazon Corretto
- Oracle

info

Product tests are done with the **Azul Zulu JVM**.

The following versions are not supported, and are known to have problems:

- 1.8.0u242
- 1.8.0u121
- 1.8.0u60
- 1.8.0u45 and earlier

Non-listed releases are either not supported or haven't been put to extensive testing.

Setup Requirementsâ

RDBMSâ

Case Manager uses PulseDB for database access, meaning that Case Manager can be deployed on top of:

- Oracle 19.3 (or higher)
- PostgreSQL 10.1 (or higher) - driver version 42.2.18 should be used.
- SQL Server 2017 GA (or higher)
- H2 1.4.200 (or higher) - only for test/evaluation purposes.

info

Feedzai validates the product primarily against H2 and PostgresSQL.

caution

Before selecting a database engine, make sure you understand how different database systems are supported by our PulseDB.

RabbitMQâ

RabbitMQ is the messaging platform Case Manager uses to ingest events from Pulse, and also to propagate events to other downstream services.

The following RabbitMQ and Erlang versions are supported:

- RabbitMQ - 3.8.23 with Erlang - 23.3.4+
- RabbitMQ - 3.9.8 with Erlang - 23.3.4+
- RabbitMQ - 3.13.7 with Erlang - 26.2.5+
- RabbitMQ - 4.2.3 with Erlang - 27.3.4+

Before starting Pulse you must configure the messaging cluster:

1. Provision a RabbitMQ cluster and ensure that Pulse is able to establish a connection to that any node of the cluster.
2. Create a new vhost for Case Manager.

```
rabbitmqctl rabbitmqctl add_vhost casemanager
```

Copy

1. Create a new user for Case Manager.

```
rabbitmqctl add_user casemanager <password>
```

Copy

1. Update the vhost policies to grant all permissions to the Case Manager user.

```
# sets write and read permissions on every resource
rabbitmqctl set_permissions casemanager -p casemanager ".*" ".*" ".*"
```

Copy

Pulseâ

Case Manager retrieves data from Pulse, and therefore requires an accessible Pulse cluster to start. Read more about how to install and configure Pulse in the installation guide.

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Rolling Upgrades

The recommended approach for zero downtime deployments in Case Manager is a technique known as rolling upgrades.

Rolling upgrades leverage Case Manager's maintenance mode in the following way:

- The existing cluster is put into maintenance mode which will stop all RabbitMQ message consumption, e.g. new events.
- The existing Case Manager cluster (or a new one) upgrade is performed
- When the upgrade is finished maintenance mode is stopped allowing the cluster to start processing messages again

Please note that this technique should only be used when an existing Case Manager cluster exists.

Requirementsâ

- Maintenance mode is enabled via Case Manager API. This requires a user with the following permissions:
  - `*` : includes the permission to login in Case Manager
  - `cm:resource:maintenance-start-stop:admin` : permission to enable/disable maintenance mode
- A dedicated machine to run database patches with access to Case Manager API. This can be either direct access to a Case Manager instance or access to a Load Balancer

---

Rolling upgradesâ

warning

This approach is currently under discussion and may not represent the final solution

To prevent database locking issues when running in high available environments the orchestrator (control-node) should be responsible for running the database changelogs.

The pipeline is defined as follows:

1. Maintenance mode is enabled in all Case Manager instances
2. A machine is created to apply patches
3. Case Manager images are created with the latest Feedzai release
4. Case Manager instances are configured but do not apply patches
5. After validating the new cluster, load balancers are updated to redirect traffic to the new cluster
   1. If validation fails a rollback command can be issued to bring the database back to its previous state
6. Maintenance mode is disabled allowing the new cluster to start processing events

This solution follows a similar approach to [Blue-Green deployments] where SSH to Case Manager machines is not available.

To support maintenance mode we have created two new playbooks, `case-manager-enable-maintenance-mode.yml` and `case-manager-disable-maintenance-mode.yml` . The required configuration parameters can be found in [Case Manager Configuration] with the description tag `Maintenance mode only` .

These playbooks use Case Manager Ansible variables and hosts inventory and all playbook tasks run only once even if the hosts file contains more than one Case Manager instance. This is due to the fact that we only require a request to a single Case Manager instance to enable/disable maintenance since the maintenance status will be synchronized across all instances.

Step 1 consists in running the following Ansible task that will enable Case Manager maintenance mode.

```
python3.6 ansible-playbook \
    -i project-default \
    -i project-hsbc \
    -i gcp \
    -i project-<env> \
    case-manager-enable-maintenance-mode.yml
```

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In step 2 only tags that are responsible for extracting and setting up the required components to run Liquibase patches should be included. This step can be run in the same machines as step 1 in which case only tags `case-manager.750` and `case-manager.755` are required.

```
python3.6 ansible-playbook \
    -i project-default \
    -i project-hsbc \
    -i gcp \
    -i project-<env> \
    case-manager.yml \
    --tags case-manager.050,case-manager.060,case-manager.110,case-manager.120,case-manager.215,case-manager.220,case-manager.410,case-manager.415,case-manager.425,case-manager.440,case-manager.710,case-manager.730,case-manager.750,case-manager.755
```

Copy

In step 3 all tags should be included except tags that are responsible applying patches which were executed in the previous step.

```
python3.6 ansible-playbook \
    -i project-default \
    -i project-hsbc \
    -i gcp \
    -i project-<env> \
    case-manager.yml \
    --tags case-manager.configure \
    --skip-tags case-manager.750,case-manager.755
```

Copy

In case any issues are detected during validation, the database can be reverted to its previous state by rollbacking the database changes.

For this cases, there is a need to override the variable `cm_liquibase_rollback` to `true` and then run the proper Ansible task

```
python3.6 ansible-playbook \
    -i project-default \
    -i project-hsbc \
    -i gcp \
    -i project-<env> \
    case-manager.yml \
    --tags case-manager.rollback
```

Copy

After validation of the new cluster/installation is done, maintenance mode can be disabled (step 6) by running the corresponding tag.

```
python3.6 ansible-playbook \
    -i project-default \
    -i project-hsbc \
    -i gcp \
```

```
    -i project-<env> \
    case-manager-disable-maintenance-mode.yml
```

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Installing Data Extraction Pipelines with Ansible

Data Extraction Pipelinesâ

The Feedzai Ansible Toolkit refers to a series of Ansible Playbooks created by Feedzai to automate the data extraction pipelines installation and manage configurations across multiple environments.

Requirementsâ

- All servers hostnames must resolve their own IP's.
- The servers should be able to resolve each others hostnames, please update the /etc/hosts file if necessary, or use a DNS service.
- Use ping to ensure to confirm that each server is able to ping other servers.
- A Linux-based Node (preferably dedicated to this task), with a deployment user configured. The node must have Python3 for running Ansible.
- A deployment user on each server with superuser (sudo) permissions (some deployment steps include installing system services, and changing ownership of folders).

Data Extraction Pipelines Installationâ

To install a the data extraction pipelines component, execute the following command:

```
python3 ansible-playbook \
    --connection=local \
    -i project-default \
    -i project-hsbc \
    -i gcp \
    logstash.yml \
    --tags external-notification-exporter.install
```

Copy

To configure data extraction pipelines component for a given `<env>` (e.g.: `dev` , `qa` , `prod` ...):

```
python3 ansible-playbook \
    --connection=local \
    -i project-default \
    -i project-hsbc \
    -i project-<env> \
    logstash.yml \
    --tags external-notification-exporter.configure
```

Copy

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Data Extraction Pipelines Configurations

Ansibleâ

Variable	type	Mandatory	Default Value	Description
rabbitmq_casemanager_brokers	STRING	X		RabbitMQ host from where the pipelines will read the Case Manager notifications (same as configured in Case Manager)
rabbitmq_casemanager_username	STRING	X		RabbitMQ user from where the pipelines will read the Case Manager notifications (same as configured in Case Manager)
rabbitmq_casemanager_password	STRING	X		RabbitMQ password from where the pipelines will read the Case Manager notifications (same as configured in Case Manager)
rabbitmq_casemanager_vhost_blue	STRING	X		RabbitMQ Blue vhost from where the pipelines will read the Case Manager notifications (same as configured in Case Manager)
rabbitmq_casemanager_vhost_green	STRING	X		RabbitMQ Green vhost from where the pipelines will read the Case Manager notifications (same as configured in Case Manager)
rabbitmq_casemanager_queue_type_blue	STRING		`CLASSIC`	RabbitMQ Blue Queue Type (can be one of `CLASSIC` / `QUORUM`
rabbitmq_casemanager_queue_type_green	STRING		`CLASSIC`	RabbitMQ Green Queue Type (can be one of `CLASSIC` / `QUORUM`
pulse_elb	STRING	X		Pulse host name (to be used in Rules Metadata pipeline) - in the format `http(s)://<host>:<port>`
pulse_app_id	STRING		599912dbf381eb06	Pulse Payments application identifier (to be used in Rules Metadata pipeline)

Variable	type	Mandatory	Default Value	Description
pulse_workflow_id	STRING		daef64964ee3b9b9	Pulse Payments workflow identifier (to be used in Rules Metadata pipeline)
pulse_insights_logstash_username	STRING	X		Pulse user name (to be used in Rules Metadata pipeline)
pulse_insights_logstash_password	STRING	X		Pulse password (to be used in Rules Metadata pipeline)
gcs_tmp_directory	STRING		/tmp/logstash/	Directory where temporary files are stored
case-manager-cases-case.workers	NUMBER		1	Setting determines how many threads to run for filter and output processing for the specified pipeline
case-manager-cases-event.workers	NUMBER		1	Setting determines how many threads to run for filter and output processing for the specified pipeline
case-manager-cases-queue.workers	NUMBER		1	Setting determines how many threads to run for filter and output processing for the specified pipeline
case-manager-cases-status.workers	NUMBER		1	Setting determines how many threads to run for filter and output processing for the specified pipeline
case-manager-cases-user.workers	NUMBER		1	Setting determines how many threads to run for filter and output processing for the specified pipeline
case-manager-events-event.workers	NUMBER		1	Setting determines how many threads to run for filter and output processing for the specified pipeline
case-manager-events-notes.workers	NUMBER		1	Setting determines how many threads to run for filter and output processing for the specified pipeline
case-manager-events-queue.workers	NUMBER		1	Setting determines how many threads to run for filter and output processing for the specified pipeline
case-manager-events-status.workers	NUMBER		1	Setting determines how many threads to run for

Variable	type	Mandatory	Default Value	Description
				filter and output processing for the specified pipeline
case-manager-events-user.workers	NUMBER		1	Setting determines how many threads to run for filter and output processing for the specified pipeline
rules-metadata.workers	NUMBER		1	Setting determines how many threads to run for filter and output processing for the specified pipeline
pulse-workflow-audit.workers	NUMBER		1	Setting determines how many threads to run for filter and output processing for the specified pipeline
pulse-workflow-audit-recovery.workers	NUMBER		1	Setting determines how many threads to run for filter and output processing for the specified pipeline
logstash_heap_size	STRING		{{ (ansible_memtotal_mb * 0.5)	int
gcs_bucket_name	STRING	X		Bucket to where the pipelines will export the data
gcs_bucket_name_cases_case	STRING		{{ gcs_bucket_name }}	Bucket to where the New Case pipeline will export the data
gcs_bucket_name_cases_event	STRING		{{ gcs_bucket_name }}	Bucket to where the Case events linked/ unlinked pipeline will export the data
gcs_bucket_name_cases_queue	STRING		{{ gcs_bucket_name }}	Bucket to where the Case queue changed pipeline will export the data
gcs_bucket_name_cases_status	STRING		{{ gcs_bucket_name }}	Bucket to where the Case state change pipeline will export the data
gcs_bucket_name_cases_user	STRING		{{ gcs_bucket_name }}	Bucket to where the Case assignee changed pipeline will export the data
gcs_bucket_name_events_event	STRING		{{ gcs_bucket_name }}	Bucket to where the Event arrival pipeline will export the data
gcs_bucket_name_events_notes	STRING		{{ gcs_bucket_name }}	Bucket to where the Add note pipeline will export the data
gcs_bucket_name_events_queue	STRING		{{ gcs_bucket_name }}	

Variable	type	Mandatory	Default Value	Description
				Bucket to where the Event queue changed pipeline will export the data
gcs_bucket_name_events_status	STRING		{{ gcs_bucket_name }}	Bucket to where the Event state changed pipeline will export the data
gcs_bucket_name_events_user	STRING		{{ gcs_bucket_name }}	Bucket to where the Event assignee changed pipeline will export the data
gcs_bucket_name_rules_metadata	STRING		{{ gcs_bucket_name }}	Bucket to where the rules metadata pipeline will export the data
gcs_bucket_name_pulse_workflow_audit	STRING		{{ gcs_bucket_name }}	Bucket to where the pulse workflow audit pipeline will export the data
gcs_bucket_name_pulse_workflow_audit_recovery	STRING		{{ gcs_bucket_name }}	Bucket to where the pulse workflow audit pipeline will export the data
max_file_size_kbytes	NUMBER		1024 (from defaults)	Sets max file size in kbytes. 0 disable max file check
uploader_interval_secs	NUMBER		60	Uploader interval when uploading new files to GCS. Adjust time based on your time pattern (for example, for hourly files, this interval can be around one hour)
logstash_pipelines_tmp_path	STRING		/tmp/staging/logstash/pipelines	Source directory of the pipeline template files
dead_letter_queue_enable	BOOLEAN		False	Enable Logstash Dead Letter Queueing System
allow_superuser_execution	BOOLEAN		True	Allows superuser execution

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/installation-guide/data-extraction-pipelines/data-schema

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Data Schema

Pulse Workflow Audit Schemaâ

The Pulse Workflow Audit pipeline extracts workflow information when there is a publish and has the following schema:

Field	type	Description
workflow.name	STRING	The workflow name.
workflow.timestamp	NUMBER	UNIX timestamp of when the workflow publish was triggered.
workflow.username	STRING	The user who triggered the publish.
workflow.elements	ARRAY of OBJECT	A list of workflow elements consisting of rules, plans and models.
workflow.elements[].type	STRING	The type of the workflow element. Possible values are `plan`, `rules` and `model`.
workflow.elements[].name	STRING	The name of the workflow element.
workflow.elements[type = plan].planExecutionId	STRING	The plan execution id that was published.
workflow.elements[type = plan].planDesc	STRING	The name of the plan (in the data science section) that was publishedt.
workflow.elements[type = rules].rulesSnapshotId	STRING	The rules snapshot id that was published.
workflow.elements[type = rules].outcomes	ARRAY of STRING	The outcomes that are affected by the rules block.
workflow.elements[type = rules].rulesProjectDesc	STRING	The name of the rules project that was published.
workflow.elements[type = rules].isMultiTenant	BOOLEAN	`true` if the workflow element belongs to a multi tenant rules project (`By Tenant`), `false` otherwise.
workflow.elements[type = model].modelId	STRING	The model id that was published.
workflow.elements[type = model].weight	NUMBER	The weight of the model.
workflow.elements[type = model].outcomes	ARRAY of STRING	The outcomes that are affected by the model.
workflow.rules	ARRAY of OBJECT	A list of rules present in the workflow.
workflow.rules[].actions	ARRAY of STRING	The actions that could be triggered by the rule.
workflow.rules[].createdAt	NUMBER	NIX timestamp of when the rule was created.
workflow.rules[].createdBy	STRING	The user who created the rule.
workflow.rules[].comment	STRING	The rule's comment.
workflow.rules[].condition	STRING	The rule's condition, as define in its implementation.
workflow.rules[].description	STRING	The rule's name, as shown in Pulse UI.
workflow.rules[].id	STRING	The rule's identifier.

Field	type	Description
workflow.rules[].name	STRING	The rule's internal name, generated by Pulse.
workflow.rules[].originalId	STRING	The rule's original identifier.
workflow.rules[].rulesProjectId	STRING	The rule project's identifier.
workflow.rules[].rulesProjectSnapshotId	STRING	The rules snapshot id that was published.
workflow.rules[].rulesProjectSnapshotDesc	STRING	The rules snapshot name, as shown in the Pulse UI.
workflow.rules[].rulesProjectDesc	STRING	The name of the rules project that was published.
workflow.rules[].scoreEnabled	BOOLEAN	If the rule contributes to the score or not.
workflow.rules[].targetVarValue	STRING	The rule's configured target variable.
workflow.rules[].variables	STRING	The rule's variables, as define in its implementation.
workflow.rules[].workflowRuleBlockName	STRING	The name of the workflow element where the rule is present.
workflow.rules[].isMultiTenant	BOOLEAN	`true` if the rule snapshot belongs to a multi tenant rules project (`By Tenant`), `false` otherwise.
workflow.rules[].tenantId	STRING	The tenant key of the rules snapshot. If the rules project is not multi tenanted it will show as `null` .

As an example, the resulting file in the Google Cloud Storage bucket can contain an element such as:


```
{
  "workflow": {
    "name": "Payments",
    "username": "pulse",
    "rules": [
      {
        "tenantId": "feedzai",
        "originalId": "m2sybxqP4mvBdOvsq7VNsg",
        "rulesProjectId": "n5a7jucCv-W_a5m4i-lHTA",
        "createdAt": 1744801197337,
        "description": "Rule_1_FDZ",
        "id": "q12BrwF1hyE5ad561_JGnw",
        "comment": "Feedzai tenant rule",
        "rulesProjectSnapshotDesc": "MultiTenantRulesProject_20250416_1207_7240",
        "targetVarValue": "true",
        "rulesProjectSnapshotId": "vdzChIy2gW5MTul65h5OIg",
        "isMultiTenant": true,
        "name": "rule_1_fdz",
        "actions": ["alert"],
        "scoreEnabled": false,
        "condition": "event.customer_name == 'some_customer_for_tenant_feedzai'",
        "variables": "",
        "workflowRuleBlockName": "MT Rules",
        "createdBy": "mWVznM0cV8dLzVXMXUVH_w",
        "rulesProjectDesc": "MultiTenantRulesProject"
      },
      {
        "tenantId": "hsbc",
        "originalId": "ig_ZEgDl79sqP0uEq8JAlA",
        "rulesProjectId": "n5a7jucCv-W_a5m4i-lHTA",
        "createdAt": 1744801251499,
        "description": "Rule_1_HSBC",
        "id": "l-yXzv5FeTmXaeuO0AVHbA",
        "comment": "Rule for HSBC tenant",
        "rulesProjectSnapshotDesc": "MultiTenantRulesProject_20250416_1207_7240",
        "targetVarValue": "true",
        "rulesProjectSnapshotId": "vdzChIy2gW5MTul65h5OIg",
        "isMultiTenant": true,
        "name": "rule_1_hsbc",
```

```
        "actions": ["alert"],
        "scoreEnabled": false,
        "condition": "event.customer_name == 'some_customer_for_tenant_hsbc'",
        "variables": "",
        "workflowRuleBlockName": "MT Rules",
        "createdBy": "mWVznM0cV8dLzVXMXUVH_w",
        "rulesProjectDesc": "MultiTenantRulesProject"
      },
      {
        "tenantId": "feedzai",
        "originalId": "mkTCBmUdjU_NA1NFt3NPSA",
        "rulesProjectId": "n5a7jucCv-W_a5m4i-lHTA",
        "createdAt": 1744801633345,
        "description": "Rule_2_FDZ",
        "id": "kTtVz6TkgVBbnLl57T1EPg",
        "comment": "",
        "rulesProjectSnapshotDesc": "MultiTenantRulesProject_20250416_1207_7240",
        "targetVarValue": "true",
        "rulesProjectSnapshotId": "vdzChIy2gW5MTul65h5OIg",
        "isMultiTenant": true,
        "name": "rule_2_fdz",
        "actions": ["alert"],
        "scoreEnabled": false,
        "condition": "event.customer_type == 'non paying cust'",
        "variables": "",
        "workflowRuleBlockName": "MT Rules",
        "createdBy": "mWVznM0cV8dLzVXMXUVH_w",
        "rulesProjectDesc": "MultiTenantRulesProject"
      },
      {
        "tenantId": null,
        "originalId": "mH-d55yjrqgubeSgWvJPGw",
        "rulesProjectId": "ngi1d406V8lKibPNavdHmA",
        "createdAt": 1744791446639,
        "description": "Sample Rule",
        "id": "nr4W190_9rrjRGQLT_1H3g",
        "comment": "Sample rule to ensure a rule exists, so that Pulse commands won't fail when 
changing its schema",
        "rulesProjectSnapshotDesc": "RP-20250416 081953.577000",
        "targetVarValue": null,
        "rulesProjectSnapshotId": "kfbIM4OCNrhVHCUDSU1GAQ",
        "isMultiTenant": false,
        "name": "sample_rule",
        "actions": [],
        "scoreEnabled": false,
        "condition": "false",
        "variables": "",
        "workflowRuleBlockName": "Self Serve Strategy Rule",
        "createdBy": "system",
        "rulesProjectDesc": "Self Serve Strategy"
      },
      {
        "tenantId": null,
        "originalId": "lP7bp45ioG9Og22W9YBJMw",
        "rulesProjectId": "ngi1d406V8lKibPNavdHmA",
        "createdAt": 1744791446639,
        "description": "Self Serve Strategy template",
        "id": "gBha-NbPG86ldaTKJntGng",
        "comment": "Template rule to target Self Serve strategy variable",
        "rulesProjectSnapshotDesc": "RP-20250416 081953.577000",
        "targetVarValue": "ivr_self_no",
        "rulesProjectSnapshotId": "kfbIM4OCNrhVHCUDSU1GAQ",
        "isMultiTenant": false,
        "name": "self_serve_strategy_template",
```

```
        "actions": ["Alert"],
        "scoreEnabled": false,
        "condition": "event.customer_name == 'Fraudulent Joe'\n",
        "variables": "score = 0;\n",
        "workflowRuleBlockName": "Self Serve Strategy Rule",
        "createdBy": "system",
        "rulesProjectDesc": "Self Serve Strategy"
      },
      {
        "tenantId": null,
        "originalId": "p_r3fVtuZiwFcewhzQhK_Q",
        "rulesProjectId": "qizzCVvEskCENH_jqE5PYA",
        "createdAt": 1744791446639,
        "description": "FIV Rule",
        "id": "vLfO0fPfh0uzfVWKfNpMHw",
        "comment": "Trigger alert for specific FIV Rule",
        "rulesProjectSnapshotDesc": "RP-20250416 081949.730000",
        "targetVarValue": "true",
        "rulesProjectSnapshotId": "kkd106V_W21rjivbQopMYQ",
        "isMultiTenant": false,
        "name": "fiv_rule",
        "actions": ["Alert"],
        "scoreEnabled": false,
        "condition": "scammed_last_5days_fiv_marker_present\n",
        "variables": "scammed_last_5days_fiv_marker_present = 
FivMarkers.getIfNotExpired(\"scammed_last_5days_fiv_marker\",event.customer_fiv_markers);\n",
        "workflowRuleBlockName": "FIV Rule",
        "createdBy": "system",
        "rulesProjectDesc": "FIV rule project"
      },
      {
        "tenantId": null,
        "originalId": "oluV-b950dUtgm14RxRM6g",
        "rulesProjectId": "icNVZ58QiirqchKT9lFEaA",
        "createdAt": 1744791446639,
        "description": "Template",
        "id": "hO0noVXJPMNZtg6So8lJZw",
        "comment": "Template rule to target treatment strategy variable",
        "rulesProjectSnapshotDesc": "RP-20250416 081947.630000",
        "targetVarValue": "mobile_push_notification",
        "rulesProjectSnapshotId": "rU1OqxIr60m_c_Bd6FVJBg",
        "isMultiTenant": false,
        "name": "template",
        "actions": [],
        "scoreEnabled": false,
        "condition": "false\n",
        "variables": "score = 0;\n",
        "workflowRuleBlockName": "Treatment Strategy Rule",
        "createdBy": "system",
        "rulesProjectDesc": "Treatment Strategy"
      },
      {
        "tenantId": null,
        "originalId": "lWdQy83xgSh81tiMYg9EDA",
        "rulesProjectId": "icNVZ58QiirqchKT9lFEaA",
        "createdAt": 1744791446639,
        "description": "Treatment Strategy template",
        "id": "olvdMVIPBbeOktX2YfpNOg",
        "comment": "Template rule to target treatment strategy variable",
        "rulesProjectSnapshotDesc": "RP-20250416 081947.630000",
        "targetVarValue": "mobile_push_notification",
        "rulesProjectSnapshotId": "rU1OqxIr60m_c_Bd6FVJBg",
        "isMultiTenant": false,
        "name": "treatment_strategy_template",
```

```
        "actions": ["Alert"],
        "scoreEnabled": false,
        "condition": "event.customer_name == 'Fraudulent Joe'\n",
        "variables": "score = 0;\n",
        "workflowRuleBlockName": "Treatment Strategy Rule",
        "createdBy": "system",
        "rulesProjectDesc": "Treatment Strategy"
      },
      {
        "tenantId": null,
        "originalId": "mI_NayHmjUkeQiG8ByNK1g",
        "rulesProjectId": "kUMP2zWl8Fg3-o-NALtNiw",
        "createdAt": 1744791446639,
        "description": "Dummy",
        "id": "k-Vc9MGE3iyz04oBLBxHqg",
        "comment": "Trigger alert for specific customer name",
        "rulesProjectSnapshotDesc": "RP-20250416 081947.430000",
        "targetVarValue": "true",
        "rulesProjectSnapshotId": "sOeZy89xhDo7EIOQE6NAKA",
        "isMultiTenant": false,
        "name": "dummy",
        "actions": ["Alert"],
        "scoreEnabled": false,
        "condition": "event.customer_name == 'Fraudulent Joe'\n",
        "variables": "score = 0;\n",
        "workflowRuleBlockName": "Dummy rule",
        "createdBy": "system",
        "rulesProjectDesc": "Dummy rule project"
      }
    ],
    "timestamp": 1744807284620,
    "elements": [
      {
        "name": "List Enrichment Plan",
        "planDesc": "Case Manager Enrichment Plan",
        "type": "plan",
        "planExecutionId": "qF3GPTA0ZyaJf-a6JIVKWQ"
      },
      {
        "name": "ArSLA Plan",
        "planDesc": "ArSLA",
        "type": "plan",
        "planExecutionId": "n1H1_Uxbe3H3cpCHXEZPSQ"
      },
      {
        "name": "Dummy Metrics Plan",
        "planDesc": "Dummy Metrics Plan",
        "type": "plan",
        "planExecutionId": "jFwXtRRfqvliUIQpGdpB_A"
      },
      {
        "name": "PIB Plan",
        "planDesc": "PIB",
        "type": "plan",
        "planExecutionId": "nrACgXfcV2bq42WNt2xHkw"
      },
      {
        "type": "rules",
        "isMultiTenant": false,
        "name": "FIV Rule",
        "outcomes": [],
        "rulesSnapshotId": "kkd106V_W21rjivbQopMYQ",
        "rulesProjectDesc": "FIV rule project"
      },
```

```
      {
        "type": "rules",
        "isMultiTenant": false,
        "name": "Self Serve Strategy Rule",
        "outcomes": ["self_serve_strategy"],
        "rulesSnapshotId": "kfbIM4OCNrhVHCUDSU1GAQ",
        "rulesProjectDesc": "Self Serve Strategy"
      },
      {
        "type": "rules",
        "isMultiTenant": false,
        "name": "Treatment Strategy Rule",
        "outcomes": ["treatment_strategy"],
        "rulesSnapshotId": "rU1OqxIr60m_c_Bd6FVJBg",
        "rulesProjectDesc": "Treatment Strategy"
      },
      {
        "type": "rules",
        "isMultiTenant": false,
        "name": "Dummy rule",
        "outcomes": ["fraud"],
        "rulesSnapshotId": "sOeZy89xhDo7EIOQE6NAKA",
        "rulesProjectDesc": "Dummy rule project"
      },
      {
        "type": "rules",
        "isMultiTenant": true,
        "name": "MT Rules",
        "outcomes": ["fraud"],
        "rulesSnapshotId": "vdzChIy2gW5MTul65h5OIg",
        "rulesProjectDesc": "MultiTenantRulesProject"
      }
    ]
  }
}
```

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Data Extraction Pipelines Overview

Data Extraction Pipelinesâ

In order to feed the FDR (Fraud Date Repository - HSBC data lake solution), it is required to export data from Case Manager.

The information will be used for reporting / dashboard and audit purposes.

The designed mechanism is based on Case Manager External Notifications. Case Manager has the ability of notifying external systems of modifications made to its data (i.e. events and cases). These notifications are sent in the form of messages via a message broker (e.g. RabbitMQ) and then read by a set of logstash pipelines that will store the information in Google Cloud Storage bucket.

Logstash Pipelinesâ

Case Manager real-time data exportâ

Case Manager "actions" export pipeline. Reads data from RabbitMQ and stores the data in a GCS bucket. To store the data, this pipeline will use the json_lines logstash plugin.

Pipeline Name	Description	Schedule
case-manager-cases-case	Action: new case creation	near-real-time
case-manager-cases-event	Action: case event(s) update	near-real-time
case-manager-cases-queue	Action: case queue update	near-real-time
case-manager-cases-status	Action: case status updates	near-real-time
case-manager-cases-user	Action: case assignee changes	near-real-time
case-manager-events-event	Action: new event arrival	near-real-time
case-manager-events-queue	Action: event queue change	near-real-time
case-manager-events-status	Action: event status change	near-real-time
case-manager-events-user	Action: event assignee change	near-real-time

Rules Metadata informationâ

Runs periodically and extract the rules metadata currently deployed in payments workflow.

Pipeline Name	Schedule
rules-metadata	0 * * * *

Pulse Workflow Audit informationâ

Runs periodically and extracts the workflow information for each app publish performed in the last hour.

Pipeline Name	Schedule
pulse-workflow-audit	0 * * * *

Pulse Workflow Audit Recovery informationâ

Runs based on events that were not correctly processed and were spilled to disk. This pipeline uses Logstash Dead Letter Queueing System (DLQ).

The pipeline will attempt to process the events again, taking their original timestamps to gather the necessary resource information from Pulse.

Pipeline Name	Schedule
pulse-workflow-audit	On DLQ event

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Data Extraction Pipelines Requirements

info

This guide is just a quick reference of the requirements to install Pulse. For a comprehensive guide, please refer to the official documentation.

Requirementsâ

This service requires the following software to run.

Java Environmentâ

OpenJDK SE Development Kit, with the following vendors (with their latest hotfix versions):

- Azul Zulu
- Amazon Corretto
- Oracle

info

Tests are done with the **Azul Zulu JVM** (version 17 or higher). **Java 8 is no longer supported** as this service now uses Logstash 9.

info

Java is required to run execute the ansible plugin that fetches the secrets from Google Secret Manager.

Serviceâ

The machine should be configured with a service account with access to the configured bucket.

Database Setup

Welcome to the Database Setup Guide.

In this section you will find information about the Retention scripts:

- Retention scripts

On this page

Retention Scripts

To enhance scalability, minimize contention, and optimize performance, it is advisable to partition the Pulse and Case Manager databases. Additionally, in accordance with Feedzai system requirements, data should not be retained for more than 6 months. All scripts enforcing the data retention period are developed and included in the installation bundle.

List of Retention Scriptsâ

- create_drop_old_partition_function.sql
- create_drop_old_partitions_function.sql
- drop_7_month_old_partitions.sql

How to execute the deletion scriptâ

The deletion script `drop_old_partitions` only deletes one partition per execution. This means that in order to delete multiple partitions the script has to be run multiple times with a different value set in the parameter it takes.

The script should be run as follows:

```
SELECT drop_old_partitions(MONTHS);
```

Copy

Where `MONTHS` indicates how far back the script will calculate the partition to delete, based on the formula `current month - MONTHS`.

As an example and assuming the current month is June, to delete the previous month's partition (May) `MONTHS` would be set to `1`:

```
SELECT drop_old_partitions(1);
```

Copy

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/installation-guide/dtp/configuration-dtp

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DTP Configurations

In this page you can find the configuration reference for the Data Transfer Pipeline (DTP).

info

The Ansible variables described in this page are used to configure DTP for all applications. If different configurations for an application are needed use the `<app_slug>_<property_name>` definition.

Service propertiesâ

Property	Ansible Variable	Type	Mandatory	Default Value	Descrip
enabled	dtp_enabled	BOOLEAN		false	Boolean enable/c the DTP service.
deployment.color	deployment_color	STRING		N/A	The curr Pulse cl color. M the Ansi variable `oyment` `or` .
enableFeatures	dtp_enable_features	BOOLEAN		true	If the fea should b stored o
gcp.projectId	dtp_gcp_project_id	STRING	X		The GC project I
pubsub.topic.enableCreation	dtp_pubsub_topic_enable_creation	BOOLEAN		false	Enables DTP to c the Pub/ topic wh starting.
pubsub.topic.id	dtp_pubsub_topic_id	STRING	When `pubsub.topic.enableCreation` is `false`		The GC Sub topi used wh topic cre is disabl
pubsub.topic.namePrefix	dtp_pubsub_topic_name_prefix	STRING	When `pubsub.topic.enableCreation` is `true`		The GC Sub topi prefix us generate topic id v topic cre is enable
pubsub.schema.idPrefix	dtp_pubsub_schema_id_prefix	STRING	X		The GC Sub sch prefix.

Property	Ansible Variable	Type	Mandatory	Default Value	Descrip
pubsub.subscription.id	dtp_pubsub_subscription_id	STRING	When `pubsub.subscription.enableCreation` is `false`		The GC Sub subscrip used wh subscrip creation disabled
pubsub.subscription.enableCreation	dtp_pubsub_subscription_enable_creation	BOOLEAN		false	Enables DTP to c the Pub/ subscrip when sta
publisher.enableCompression	dtp_pub_enable_compression	BOOLEAN		true	Enables Sub pub messag compres
publisher.compressionBytesThreshold	dtp_pub_compression_threshold	NUMBER		10	Pub/Sub publishe messag compres threshol bytes.
publisher.batch.elementCountThreshold	dtp_pub_batch_element_threshold	NUMBER		10	Pub/Sub publishe request messag count.
publisher.batch.requestBytesThreshold	dtp_pub_batch_request_threshold	NUMBER		1000000	Pub/Sub publishe request request
publisher.batch.delayThreshold	dtp_pub_batch_delay_threshold	NUMBER		100	Pub/Sub publishe request time sinc publish i millisecc
publisher.flow.maxOutstandingElements	dtp_pub_flow_max_outstanding_elements	NUMBER		10000	Max me count th Sub pub client ca in memo
publisher.flow.maxOutstandingRequestBytes	dtp_pub_flow_max_outstanding_request_bytes	NUMBER		1500000	Max req size in b the Pub/ publishe can hold memory
batchexecutorservice.batcherImpl	N/A	STRING		com.*.SimpleBatcher	Impleme of the Ba that the should u
batchexecutorservice.batchSize	dtp_bes_batch_size	NUMBER		500	Maximu number operatio will be b

Property	Ansible Variable	Type	Mandatory	Default Value	Descrip
					and flus together
batchexecutorservice.batchTimeout	dtp_bes_batch_timeout	NUMBER		1000	How ofte batches be exec automat
batchexecutorservice.consumerWaitingTime	dtp_bes_consumer_waiting_time	NUMBER		10	Time in milliseco that the consum threads willing to before ta an elem the queu
batchexecutorservice.flushTimeout	dtp_bes_flush_timeout	NUMBER		86400000	Maximu one is w wait for flush operatio take.
batchexecutorservice.monitoring.timeUnit	dtp_bes_monitoring_time_unit	STRING		MINUTES	The time used for metrics reservoi
batchexecutorservice.monitoring.ttl	dtp_bes_monitoring_ttl	NUMBER		1	The win size use the metr reservoi
batchexecutorservice.pendingQueueSize	dtp_bes_pending_queue_size	NUMBER		10000	Capacity service's pending size.
batchexecutorservice.pool.consumerThreads	dtp_bes_consumer_threads	NUMBER		1	Number threads consum tasks fro pending queue.
batchexecutorservice.pool.processingThreads	dtp_bes_pool_processing_threads	NUMBER		1	Number threads process tasks tal from the pending queue.
batchexecutorservice.pool.queueSize	dtp_bes_pool_queue_size	NUMBER		1000	Size of t process pool-bac queue. Expecte the sam the num threads, only one is submi

Property	Ansible Variable	Type	Mandatory	Default Value	Descrip
					each av thread.
batchexecutorservice.pool.threadFactoryName	dtp_bes_pool_thread_factory_name	STRING		batch-executor-service	The nam used on thread factories pools cr by the s
batchexecutorservice.shutdowntimeout	dtp_bes_shutdown_time	NUMBER		5000	Maximu one is w wait for service t shutdow
dumpstorageconfig.enabled	dtp_dump_enabled	BOOLEAN		false	Boolean enable/d the dum storage system.
dumpstorageconfig.dumpDirectory	dtp_dump_directory	STRING		var/data-transfer-pipeline	The loca where to the even
dumpstorageconfig.flushingInterval	dtp_dump_flush_interval	NUMBER		1000	The flus interval millisecо used to the write disk.
dumpstorageconfig.recoveryCronExpression	dtp_dump_recovery_expression	STRING		0 0 0/2 ? * * *	The reco cron express defines t the auto recovery run. By d every 2
security.kmsKeyName	dtp_security_kms_key_name	STRING			Specifie custome manage encrypti (CMEK) to encry handled Pub/Sub and Dat pipeline resource Should f the form describe Configur topic wit CMEK.
security.allowedRegions	dtp_security_allowed_regions	STRING			A comm separate of allowe regions data ma stored in

Property	Ansible Variable	Type	Mandatory	Default Value	Descrip
					Sub. Ex "europe- west1,as east2"
security.enforceInTransit	dtp_security_enforce_in_transit	BOOLEAN		false	A boolea that enfo in-transi restrictio Pub/Sub same re defined storage.
security.regionalEndpoint	dtp_security_regional_endpoint	STRING			The regi Pub/Sub service endpoin use. Mandato `nforce ansit` enabled Complet of endpo [here]().

Dataflow job propertiesâ

info

If a Dataflow job is already running, changes to these properties will not be applied when republishing the Pulse app if there are no changes in the schema.

Property	Ansible Variable	Type	Mandatory	Default Value	Description
dataflow.region	dtp_dataflow_region	STRING	X		GCP Dataflow job region, e.g., 'europe-west1'.
dataflow.jobNamePrefix	dtp_dataflow_job_name_prefix	STRING	X		GCP Dataflow job name prefix.
dataflow.tempLocation	dtp_dataflow_temp_location	STRING	X		GCP Dataflow GCS temp bucket path, e.g., 'gs://dtp/temp'.
dataflow.stagingLocation	dtp_dataflow_staging_location	STRING	X		GCP Dataflow GCS staging bucket path, e.g., 'gs://dtp/staging'.
dataflow.outputLocation	dtp_dataflow_output_location	STRING	X		GCP Dataflow GCS output bucket path,

Property	Ansible Variable	Type	Mandatory	Default Value	Description
					e.g., 'gs://dtp/output'.
dataflow.fileToStage	N/A	STRING		var/data-transfer-pipeline-service/data-transfer-pipeline-service-pipeline-dependencies.jar	GCP Dataflow job dependencies file to stage.
dataflow.batchWindow	dtp_dataflow_batch_window	NUMBER		1	GCP Dataflow job batch writes window in minutes.
dataflow.numShards	dtp_dataflow_number_of_shards	NUMBER		0	GCP Dataflow job maximum number of output file shards per batch window (0=auto).
dataflow.streamingEnabled	N/A	BOOLEAN		true	Boolean flag to enable streaming mode for the GCP Dataflow job.
dataflow.skipPipelineCreation	dtp_dataflow_skip_pipeline_creation	BOOLEAN		false	Boolean flag to skip the GCP Dataflow job creation.
dataflow.enableAutoscaling	dtp_dataflow_enable_autoscaling	BOOLEAN		true	Boolean flag to enable/disable the dataflow job autoscaling.
dataflow.minimumNumberOfWorkers	dtp_dataflow_minimum_number_of_workers	NUMBER		1	Defines the minimum number of workers the Dataflow job needs to keep.
dataflow.maximumNumberOfWorkers	dtp_dataflow_maximum_number_of_workers	NUMBER		10	Defines the maximum number of workers the Dataflow job can use.
dataflow.numberOfWorkers	dtp_dataflow_number_of_workers	NUMBER		0	Set the number of workers the Dataflow job uses when the autoscaling is

Property	Ansible Variable	Type	Mandatory	Default Value	Description
					disabled (0=auto).
dataflow.workerServiceAccount	dtp_dataflow_worker_service_account	STRING			Set the service account that the worker instances use. By default, workers use the Compute Engine default service account of your project.
dataflow.network	dtp_dataflow_network	STRING			GCP Dataflow job network. If omitted along with `dataflow.subNetwork` the jobs will run on the `default` network.
dataflow.subNetwork	dtp_dataflow_subnetwork	STRING			GCP Dataflow job subnetwork. If specified, GCP will determine the network and `dataflow.network` can be omitted. This configuration has to be set in the format `regions/REGION_NAME/subnetworks/SUBNETWORK_NAME`.
dataflow.usePublicIps	dtp_dataflow_use_public_ips	BOOLEAN		false	Boolean flag to indicate if the worker pools should be created with public IP addresses.

Inactivity service propertiesâ

info

If a Dataflow job is already running, changes to these properties will not be applied when republishing the Pulse app if there are no changes in the schema.

Property	Ansible Variable	Type	Default Value	Description
inactivity.topic.enableCreation	dtp_inactivity_topic_enable_creation	BOOLEAN	false	Enables the DTP to create the Pub/Sub inactivity topic when starting.
inactivity.topic.id	dtp_inactivity_topic_id	STRING	dtp-inactivity	GCP Pub/Sub inactivity topic ID.
inactivity.subscription.id	dtp_inactivity_sub_id	STRING	dtp-inactivity-sub	GCP Pub/Sub inactivity topic subscription ID.
inactivity.subscription.ackDeadlineSeconds	dtp_inactivity_sub_ack_deadline	NUMBER	60	GCP Pub/Sub inactivity topic subscription message ack deadline in seconds.
inactivity.delayMinutes	dtp_inactivity_delay	NUMBER	30	Delay in minutes for the service to start monitoring inactivity.
inactivity.thresholdMinutes	dtp_inactivity_threshold	NUMBER	60	Threshold in minutes for a GCP Dataflow job to be considered inactive.
inactivity.enabled	dtp_inactivity_enabled	BOOLEAN	true	Boolean flag to enable/disable the inactivity service.
inactivity.subscription.enableCreation	dtp_inactivity_subscription_enable_creation	BOOLEAN	false	Enables the DTP to create the Pub/Sub inactivity subscription when starting.

Data Transfer Pipeline

Welcome to the Data Transfer Pipeline (DTP) Guide.

In this section you will find a set of pages with information about the DTP service:

- DTP service
- DTP configuration
- GCP setup
- DTP extractions schema
- Migrating from ETL to DTP

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/installation-guide/dtp/extractions-dtp

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DTP Extractions Schema

The DTP extracts two types of messages in `JSON` format:

- Workflow messages
- Feedback updates

This page provides information about the schema of both types of messages when they are published to the GCP Pub/Sub topic.

Workflow Messagesâ

The workflow messages are published in JSON format and contain the schema fields plus fields present in extra-fields, according to the following snippet:

```
{
  "timestamp": 1721125213611,
  "uuid": "aa628b9a-fcb8-4114-875b-6c2436eadd65",
  "lifecycle_id": "cc891616-745d-42e2-8e7e-",
  "bulk_id": "b398a9e1-3900-",
  "payment_type": "sepa",
  "payment_sub_type": "credit_transfer",
  "payment_status": "entered",
  "created_at": 1554198919000,
  ...
  "metricA": 0.5,
  "metricB": 100,
  "decision": "approve",
  "decision_scores": [0.0, 0.0, 0.0],
  "risk_level": "low",
  "risk_level_scores": [0.0, 0.0, 0.0],
  "fraud": false,
  "fraud_score": 0.0,
  "sca_required": true,
  "sca_required_score": 0.0,
  "treatment_strategy": "none",
  "FDZ_TRIGGERED_ACTIONS": "[]",
  "FDZ_OUTCOMES": "",
  "FDZ_RESULTS": "[]",
  ...
}
```

Copy

Please note that the section of the payload regarding metrics will change frequently, for instance, each time a new metric is added to the risk strategy.

Extra fieldsâ

Please find the list of extra fields that are sent in the messages payload.

Unique identifier (`FDZ_UUID`)â

This field is a unique identifier for every event extracted.

```
"FDZ_UUID": "524f8eff-2bf7-4eea-ad0f-f40f5710bf8c"
```

Copy

Actions triggered (`FDZ_TRIGGERED_ACTIONS`)â

This field is a **stringified** list of JSON Objects that contains all the triggered actions. Every JSON Object has a field `name` with the name of the triggered action.


```
"FDZ_TRIGGERED_ACTIONS": [
    {
        "arguments": {},
        "name": "Alert",
        "nestedSourceId": "o271PYG0ipWMJ4BgR2xINw",
        "sourceId": "34a95d87-6934-4a99-a0a0-0ddb7e485fc2"
    },
    {
        "arguments": {},
        "name": "Alert",
        "nestedSourceId": "t0vgfqUux9meytwbnj9Bxw",
        "sourceId": "e112ef90-899a-4d9f-854c-4d4307f2d2a6"
    }
]
```


Copy

Rules triggered and explanations (`FDZ_RESULTS`)â

This field is a **stringified** list of JSON Objects that contains all the rules that were triggered by the extracted event and their explanation.

Every **triggered rule** has two fields:

- **ruleId**: the triggered rule id
- **reason**: the explanation of why the rule was triggered


```
"FDZ_RESULTS": [
    {
        "explanations": [
            {
                "@type": "rule",
                "reason": "Customer name is Fraudulent Joe"
            }
        ],
        "result": {
            "@type": "clarule",
            "originalRuleId": "mI_NayHmjUkeQiG8ByNK1g",
            "outcome": "true",
            "ruleId": "o271PYG0ipWMJ4BgR2xINw",
            "scores": [],
            "targetOutcome": "fraud",
            "targetOutcomeLabel": "fraud"
        },
        "sourceId": "34a95d87-6934-4a99-a0a0-0ddb7e485fc2",
        "weight": 0
    }
]
```


Copy

Outcomes (`FDZ_OUTCOMES`)â

This field is a **stringified** JSON object that contains all the configured **outcomes** (e.g. `fraud` , `decision`).

Every **outcome** has two fields:

- **outcomeDecision**: the value chosen for the outcome

- **score**: the score of the outcome. If the outcome is a multi-classifier (e.g. `decision`), it will be a list of scores for each of the possible values

```
"FDZ_OUTCOMES": {
    "fraud": {
        "@type": "bincla",
        "outcomeDecision": true,
        "score": 0.0
    },
    "decision": {
        "@type": "mulcla",
        "outcomeDecision": "approve",
        "score": [
            0.0, # approve
            0.0, # decline
            0.0  # review
        ]
    }
}
```

Copy

Metrics/Plan operatorsâ

For every metric and plan operator there will be a field with the name of the metric/operator, e.g. `count_benefname_90d`.

Every field will have the metric value that was fetched when the event was consumed by Pulse:

```
"count_benefname_90d": 10
```

Copy

Feedback Updatesâ

If enabled, DTP will also extract feedback updates which are messages that refer to feedback events.

The feedback update extraction contains the following key fields:

- **feedbackLabel**: The feedback label for which the update refers
- **feedback.realOutcome**: The actual outcome value provided for the feedback label

```
{
    "@type": "insert",
    "appId": "599912dbf381eb06",
    "eventId": "ce3ffbc5-b388-4e68-a8fd-b2732102235e",
    "feedback": {
        "@type": "bincla",
        "metadata": {
            "inject": false,
            "lifecycle_id": "b99e215f-6823-4cf0-b82e-afd857bacfa9",
            "__eventTimestamp__": [
                "java.time.Instant",
                1731081106.106000000
            ]
        },
        "realOutcome": true,
        "reportInstant": 1731081106106
    },
    "feedbackLabel": "fraud",
    "timestamp": 1731081106.117000000,
    "workflowId": "daef64964ee3b9b9"
}
```

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Migrating from ETL to DTP

This page explains the key differences between **ETL** and **DTP** extractions, with a focus on the additional fields each tool retrieves beyond the standard schema fields.

For detailed schema information, see:

- DTP Extractions Schema
- ETL Extractions Schema

ETLâ

Below you can find a CSV example of an extraction, take into account that it may not represent the exact structure since the fields are obtained from the current Pulse workflow:

```
lifecycle_id,bulk_id,payment_type,payment_sub_type,product_code,payment_status,created_at,cutoff_at,sender_initiated_at,requested_execution_at,number_of_payments,recurring_payment,sender_transaction_amount,sender_transaction_currency,sender_transaction_bank_account_number,sender_transaction_notes,sender_bank_name,sender_bank_postal_code,sender_bank_address,sender_bank_city,sender_bank_country_code,receiver_bank_bic,receiver_bank_name,receiver_bank_postal_code,receiver_bank_address,receiver_bank_city,receiver_bank_country_code,receiver_transaction_amount,receiver_transaction_currency,receiver_transaction_bank_account_number,receiver_city,receiver_country_code,device_type,device_longitude,device_latitude,event_type,event_external_id,info_type,dispute_start_date,dispute_reason,dispute_notes,dispute_submitted_by,dispute_end_date,dispute_feedback_notes,dispute_refunded,oob_start_date,oob_comm_type,oob_notes,oob_end_date,oob_feedback_notes,oob_response_type,feedback_type,feedback_label,sender_name,sender_email,sender_phone,sender_zip,sender_dob,sender_occupation_status,sender_account_bic,sender_account_type,sender_account_currency,sender_account_balance,receiver_id,receiver_name,receiver_email,receiver_phone,receiver_street_addr,receiver_zip,receiver_region,receiver_dob,receiver_account_type,receiver_account_currency,direction,device_id,channel_type,sender_bank_sort_code,sender_bank_branch_id,receiver_bank_sort_code,receiver_bank_routing_number,event_occurred_at,segment_channel_type,auth_type,device_ip_country_code,session_id,device_language,sender_alias_type,text_for_receiver,device_user_agent,device_fingerprint,instant_transfers_review_outcome,device_browser,device_ip_region,feedback_value,sender_account_zip,sender_bank_type,condor_rule_code,receiver_cop_name,receiver_cop_match_name,office_id,recurring_payment_schdl_info,sender_account_ownership,payment_sca_proc_code,customer_market_segment,sender_transaction_type,sender_transaction_auth_type,payment_message_type,payment_revision_code,receiver_account_bank_id,receiver_bank_type,tpp_name_alt,tpp_payee_payer,number_of_payments_count,sender_phone_chg_date,customer_home_branch,recurring_payment_ref,payment_remittance_info,sender_transaction_ref,phone_session_id,user_guid,customer_onboarding_method,sender_transaction_cat_code,sender_orig,payment_related_ref,sender_transaction_narrative,sender_transaction_cop_name,sender_transaction_cop_matname,sender_bank_status,sender_account_number_external,receiver_email_chg_date,sender_email_chg_date,receiver_bank_status,receiver_merchant_category,receiver_details,tpp_name,payment_saf,customer_total_balance,customer_reported_income,customer_citizen_id_number,customer_preferred_language,customer_work_phone1_chg_date,customer_work_phone2_chg_date,customer_home_phone1_chg_date,customer_home_phone2_chg_date,customer_mob_phone1_chg_date,customer_mob_phone2_chg_date,customer_account_type,lifecycle_id_alt,customer_number_of_accounts,receiver_bank_addr_city,sender_transaction_original_amount,sender_transaction_original_currency,receiver_transaction_original_amount,receiver_transaction_original_currency,receiver_transaction_product,sender_transaction_reason,sender_transaction_product,sender_transaction_amount_usd,payment_message_source,sender_transaction_cop_result,browser_timezone,tpp_id,device_datetime,device_platform_type,customer_portfolio_region,customer_portfolio_product,customer_portfolio_channel,customer_portfolio_country,customer_portfolio_type,device_tdl,customer_change_new,customer_change_old,consent_id,device_cpu_class,device_fingerprint_type,device_banking_type,device_ip_address_v6,website_name,channel_user_id,pinpoint_risk_score,source,sender_transaction_exp_date,sender_transaction_code,auth_cam_level,customer_number,sender_transaction_sign_status,device_platform_type_reason,account_type,auth_method,request_type,customer_portfolio,customer_id_global,customer_id_external,customer_address_chg_
```

```
date,customer_phone_chg_date,customer_email_chg_date,sender_transaction_sole_ind,customer_accoun
t_number,customer_account_currency,customer_account_zip,customer_account_country,customer_accoun
t_nr_external,user_id_type,channel_daily_limit_currency,channel_name,staff_id,workstation_id,dev
ice_token_serial_no,device_mac_address,payment_entity,sender_transaction_details_ind,sender_tran
saction_category,sender_transaction_ref_nr,sender_transaction_ref_nr_alt,sender_transaction_stat,
sender_transaction_stat_reason,sender_transaction_effect_date,customer_change_reason_alt,custome
r_change_delivery,user_indicator,user_string,user_amount,customer_card_number,entity_id,entity_t
ype,idv_session_id,event_code,device_geo_ip_city,device_geo_ip_country_code,device_geo_ip_isp,de
vice_geo_longitude,device_geo_latitude,website_refferer_url,device_geo_timezone,device_gps_enabl
ed,device_data_enabled,device_wifi_bssid_connected,device_wifi_in_range,sanction_check,device_bl
uetooth_enabled,device_bluetooth_devices_conn,device_bluetooth_detected_list,device_imsi,device_
mcc,device_untrusted_keyboard,device_untrusted_screen_reader,device_hooking_frameworkds,devide_n
ative_codehooks,device_application_repackaged,device_debugger_attached,device_javascript_malform
ed,device_print,device_raw_jsc,device_is_desktop,device_is_gameconsole,device_rooted,device_risk_
score,device_timezone,device_ip_city,device_ip_latitude,device_ip_longitude,event_status,custome
r_country_code,customer_zip_code,customer_segment,customer_name,customer_dob,customer_address,cu
stomer_phone_number,customer_email,customer_branch_id,customer_id,account_id,device_emulator,dev
ice_java_script_enabled,sender_transaction_held,payment_restricted_template,sender_transaction_n
ew_bene,payment_type_direct_debit,payment_type_self_acc,device_wifi_status,sender_urgency,device_
timezone_offset,sender_initiated_at_local,sender_account_avg_balance,channel_daily_limit,event_s
erver_validated_at,event_updated_at,user_id,receiver_account_bank_name,payment_ref,device_os,cus
tomer_change_data_new,payment_type_sub_type,device_es_id,txn_type,device_dev_id,external_status_
reason,external_preliminary_decision,device_isp,is_scoring,billing_currency_code,change_event_de
livery,customer_change_data_old,device_ip_address,overnight_acc_ledger_bal,receiver_transaction_
amount_dbl,sender_account_balance_dbl,sender_account_open_date,sender_transaction_amount_dbl,var_
recurring_payment,receiver_creation_date,sender_registration_date,event_received_at,event_create
d_at,customer_fiv_marker,customer_fiv_marker_last_updated_at,customer_gender_code,customer_earni
ngs_amt_snapshot_dt,current_account_monthly_debit_amount,phone_bank_use_code,customer_current_ac
count_balance,customer_salary_amount,customer_forname,customer_surname,customer_home_street_addr,
customer_home_city,customer_home_zip,customer_home_zip_area_code,customer_home_region,customer_h
ome_country_code,customer_home_address_last_update_at,customer_corresp_street_addr,customer_corr
esp_city,customer_corresp_zip,customer_corresp_zip_area_code,customer_corresp_region,customer_co
rresp_country_code,customer_corresp_address_last_update_at,customer_email_last_update_at,custome
r_home_phone,customer_mobile_phone,customer_daytime_work_phone,customer_business_phone,customer_
hmphone_last_update_at,customer_mbphone_last_update_at,customer_wrkphone_last_update_at,customer_
businessphone_last_update_at,customer_registration_date,customer_citizenship,customer_portfolio_
class,is_historic,saf_ind,customer_fiv_markers,customer_pb_marker,customer_portfolio_segment,cus
tomer_voice_id,customer_voice_id_last_change_date,customer_annual_turnover,customer_business_type
,customer_chinese_comm_code,customer_rm_info,customer_marital_status,invalid_email_indicator,sen
der_account_channel_limit_double,sub_channel_name,customer_id_number,gift_indicator_code,custome
r_status,customer_rel_status,customer_type,customer_id_hashed,model_risk_score,bank_processing_a
pplication,bank_processing_type,reversal_indicator,check_book_request_date,cash_form,check_number
,thailand_fps_ind,gpb_manual_payment_ind,scameter_warning,sender_billing_amount,new_customer_flag
,customer_id_type,new_receiver_flag,receiver_creation_date_customer,customer_occupation_start_da
te,count_months_beneficiary_received_funds,avg_total_daily_transfer_out,avg_monthly_transfer_in,
browser_os_version,costumer_id_global,customer_portfolio_phys_type,customer_portfolio_remote_type
,ds_transaction_id,payment_message_source_d,sender_transaction_origin_amount,sender_transaction_
origin_currency,sender_transaction_reason_d,sender_transaction_sign_stts_d,customer_change_reason
,sender_account_channel_limit,delegate_id,delegate_initiated,approver_id,transaction_id,delegate_
of,delegates,sender_transaction_extra_1,sender_transaction_extra_2,sender_transaction_extra_3,cu
stomer_additional_info_1,customer_additional_info_2,customer_additional_info_3,customer_addition
al_info_4,account_holder_customer_id,payment_above_limit,customer_alias,ecw_question_and_answer_
details,ecw_final_action,ecw_answers_changed,ecw_duration,ecw_payment_details_changed,ecw_purpos
e_of_payment,ecw_country_payment_description,ecw_pop_cpd_mapped,ecw_response_time,payment_execut
ion_type,request_correlation_id,sfr_moneylock_hold_bal,sfr_total_hold_bal,delegate_without_produ
ct_indicator,sender_product_type,beneficiary_proxy_account_number,vocalink,tmx,fdz_channel,bioc,
FDZ_UUID,Customer average payment amount last
day,rules_triggered,formatted_results,outcomes_and_scores
f9ae8116-812d-4e0c-b658-dc3de1e31f92,b398a9e1-3900-4753-
ad45-8310ea5837234,sepa,credit_transfer,,entered,
1554198919000,1554198919000,1554196919000,1554996919000,1,true,
155000.0,270,5d2cf69d-2e0f-468c-981b-2e80f7419b4c,Notes on the transaction.,Gringotts,WC2N
5DU,Diagon Alley 33,London,440,4ad3fe88-317b-4b31-ad9b-9136424c17ad,Receiver Bank,WC7N 7EW,Some
Avenue 77,Manchester,440,155000.0,270,9a08f316-df1f-4096-9a2d-a904a82bb65e,receiver_city,440,web
```

```
browser,-31.3643,-77.4773,transfer_initiation,,out_of_band,1554196919000,test,Very
fraudulent.,test,1554196919000,,,1554196919000,test,Contacted John Doe to confirm.,
1535477895438,The client called to confirm it was not him making that
transaction.,call,out_of_band,,Sender Joe Doe,[email protected],001122334455,,19900101,looking
for a new project,PTCDG,CREDIT,270,1110000.0,receiver_id,Receiver Joe Doe,[email protected],
001122334455,receiver_street_addr,receiver_zip,receiver_region,19900101,CREDIT,270,outgoing,
7823d84a-6acc-43e6-84b2-dddc3122a248,branch,,,,,1764178132876,,password,440,53fcc910-
f92c-4cd5-84b6-4d437da309ca,be-440,,,"Mozilla/5.0 (Macintosh; U; Intel Mac OS X 10_8_6)
AppleWebKit/532.0.2 (KHTML, like Gecko) Chrome/14.0.855.0 Safari/
532.0.2",,false,Brave,OT,true,,,,,,,,,,,,,,,,,
1758901473241,0001-01-01,,,,,,,,,,,,,,0001-01-01,0001-01-01,,,,,,,,,,,,,,,,483,,395.75,270,,,
,,,155000.0,,,,,1758901473241,,,,,feedzai,,,,,,,,,,,,,,
0001-01-01,,,,,,,,,,,,0001-01-01,0001-01-01,0001-01-01,,,270,,,,,270,,,,,,,,,,,,0001-01-01,,,"["
"Eulah.Hammes24""]","[""Ludie_Denesik""]",[771.0],,,,,,,
440,,-140.6989,2.337,,,false,true,,false,true,false,,,,,true,false,false,true,true,true,false,,,
true,true,true,21,-07:00,New Dejaport,-48.0299,104.4205,success,
440,66720-331,commercial_banking,Fraudulent Joe,0001-01-01,224 Benjamin Flats,836-342-5605,
[email protected],,customer_1,d68107b9-0d6c-47e9-
a547-5e6ac5103e67,false,true,true,false,true,false,false,false,,,,
973.0,441.0,1758901473241,,,,,ioss,,,,,,,,,,
270,55551222,,161.223.207.161,100.5,400.75,1200.75,1668683369590,400.75,0,1758364512397,135439891
9000,1764178132988,1758901473241,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,true,13800.0,BS,
0HKHSXXXXXX796256,964,Insane,,,
2d18c716ba930802d187654c99fd097f5530f1479065428caa1a3ee1q7421,0.89,,,,1716369476000,,,,,,1.234523
3222E8,D,V,,,,,,,,,,,,,10,270,,,,,,false,,,,,Sender Transaction Additional Info 1,Sender
Transaction Additional Info 2,Sender Transaction Additional Info 3,Customer Additional Info
1,Customer Additional Info 2,Customer Additional Info 3,Customer Additional Info
4,account_holder_customer_id,false,,SQ1.SA2.SQ4.SA3.SQ8.SA1.W3,CON,Y,00.00.44.08,Y,RENT,RENT,Y,
03.08,FDP,VRWER3452BVQR2,1225.98,1225.98,true,UFAAE,beneficiary_proxy_account_number,"{""vocalin
k_score"":20.0,""mastercard_id"":"""",""alert_id"":""5aa89b5b-9fc0-4faf-bfaf-
c0454611d129"",""source_account_id"":""ada0e7c4-5705-4b76-8be8-6b3ccf8dafa0"",""destination_acco
unt_id"":""21837091-a0d2-4e16-a26c-c6cdd67a66fb"",""transaction_id"":""f211956c-4485-44e1-
a4f5-505dffbbcb50"",""transaction_value"":123.0,""score_generated_at"":
1554198919000,""transaction_currency"":""270""}","{""fuzzy_device_id"":"""",""ip_city"":"""",""i
p_country"":"""",""ip_longitude"":10.1339,""ip_latitude"":50.3646,""chall_policy_score"":
780.0,""ip_address"":"""",""digital_id"":"""",""device_isp"":"""",""summary_reason_code"":"""",""
policy_score"":21.0,""risk_score"":
21.0,""device_dev_id"":"""",""request_id"":"""",""true_isp"":null,""proxy_isp"":null,""dns_isp"":
null}",TRANSFERS,"{""is_bot"":false,""is_emulator"":true,""voice_scam_score"":
911,""is_malware"":false,""is_rat"":false,""is_recent_rat"":false,""custom_risk"":
21.0,""custom_risk_aliases"":""risk_alias"",""push_score"":null,""is_mobile_rat"":null,""pinpoin
t_risk_score"":null,""risk_score"":21.0,""chall_policy_score"":null}",02348085-801c-45ff-
bdc3-80204ad5e789,,lCyKop_QjP45br_tEjlNcg;lUQqjQhs2CSSxrRspC9NeQ;mGBzsBs9IOyaQn4hbg5E7g,"[{""exp
lanation"":""Customer name is Fraudulent
Joe"",""ruleId"":""lCyKop_QjP45br_tEjlNcg""}]","{""scam"":{""outcomeDecision"":false,""score"":
0.0},""fraud"":{""outcomeDecision"":true,""score"":0.0},""decision"":
{""outcomeDecision"":""approve"",""score"":[0.0,0.0,0.0]},""risk_level"":
{""outcomeDecision"":""low"",""score"":[0.0,0.0,0.0]},""shadow_scam"":
{""outcomeDecision"":false,""score"":0.0},""sca_required"":{""outcomeDecision"":true,""score"":
0.0},""shadow_fraud"":{""outcomeDecision"":false,""score"":0.0},""treatment_strategy"":
{""outcomeDecision"":""mobile_push_notification"",""score"":
[0.0,0.0,0.0,0.0]},""self_serve_strategy"":{""outcomeDecision"":""ivr_self_no"",""score"":
[0.0,0.0,0.0,0.0,0.0]}}"
```

Copy

Unique identifier (`FDZ_UUID`)â

This field is a unique identifier for every event extracted.

```
 lifecycle_id,...,FDZ_UUID,...
f9ae8116-812d-4e0c-b658-dc3de1e31f92,...,02348085-801c-45ff-bdc3-80204ad5e789,...
```

Copy

Scheduled metricsâ

For every scheduled metric there will be a field with the name of the metric, e.g. `count_benefname_90d`.

```
lifecycle_id,...,count_benefname_90d,...
f9ae8116-812d-4e0c-b658-dc3de1e31f92,...,10,...
```

Copy

Actions triggered (`action_X`)â

When this option is enabled the configured actions on the workflow are extracted as multiple fields, there's a field for every action.

Every field is a concatenation of `action_` with the name of the configured action, e.g.: `action_alert`. Then, for every field, it is said if the action was triggered or not (true/false).

```
lifecycle_id,...,action_alert,...
f9ae8116-812d-4e0c-b658-dc3de1e31f92,...,true,...
```

Copy

Rules triggered (`rules_triggered`)â

This field includes a semicolon-separated list of rule ids that were triggered by the extracted event.

```
lifecycle_id,...,rules_triggered,...
f9ae8116-812d-4e0c-b658-
dc3de1e31f92,...,lCyKop_QjP45br_tEjlNcg;lUQqjQhs2CSSxrRspC9NeQ;mGBzsBs9IOyaQn4hbg5E7g,...
```

Copy

Rules explanations (`formatted_results`)â

This field is a list of triggered rules with an explanation of why it was triggered.

```
lifecycle_id,...,formatted_results,...
f9ae8116-812d-4e0c-b658-dc3de1e31f92,...,[{"explanation":"Customer name is Fraudulent
Joe","ruleId":"lCyKop_QjP45br_tEjlNcg"}],...
```

Copy

Outcomes (`outcomes_and_scores`)â

This field is a JSON object that contains all the configured **outcomes** (e.g.: `fraud` , `decision`).

```
lifecycle_id,...,outcomes_and_scores,...
f9ae8116-812d-4e0c-b658-dc3de1e31f92,...,{"scam":{"outcomeDecision":false,"score":0.0},"fraud":
{"outcomeDecision":true,"score":0.0},"decision":{"outcomeDecision":"approve","score":
[0.0,0.0,0.0]},"risk_level":{"outcomeDecision":"low","score":[0.0,0.0,0.0]},"shadow_scam":
{"outcomeDecision":false,"score":0.0},"sca_required":{"outcomeDecision":true,"score":
0.0},"shadow_fraud":{"outcomeDecision":false,"score":0.0},"treatment_strategy":
{"outcomeDecision":"mobile_push_notification","score":[0.0,0.0,0.0,0.0]},"self_serve_strategy":
{"outcomeDecision":"ivr_self_no","score":[0.0,0.0,0.0,0.0,0.0]}},...
```

Copy

DTPâ

The DTP extractions are stored in parquet format, to have a CSV the output needs to be converted.

Below you can find a CSV example of an extraction (converted from parquet), take into account that it may not represent the exact structure since the fields are obtained from the current Pulse workflow:

```
FDZ_PARTITION_INDEX,FDZ_TENANT_KEY,FDZ_UUID,FDZ_timestamp,FDZ_SCHEMA_UID,FDZ_RESULTS,FDZ_OUTCOMES
,FDZ_TRIGGERED_ACTIONS,FDZ_SOURCE_ID,FDZ_WKFL_STATE,FDZ_INTERNAL_STATE,FDZ_DIFFS,FDZ_EXPOSED_SCH
EMA_DIFFB,account_holder_customer_id,account_id,account_type,approver_id,auth_cam_level,auth_met
hod,auth_type,avg_monthly_transfer_in,avg_total_daily_transfer_out,bank_processing_application,b
ank_processing_type,beneficiary_proxy_account_number,billing_currency_code,bioc,browser_os_versi
on,browser_timezone,bulk_id,cash_form,change_event_delivery,channel_daily_limit,channel_daily_li
mit_currency,channel_name,channel_type,channel_user_id,check_book_request_date,check_number,cmd_
auth_cam_level,cmd_auth_method,cmd_channel_type,cmd_event_description,cmd_payment_status,cmd_pay
ment_type,cmd_payment_type_sub_type,cmd_receiver_bank_address,cmd_receiver_bank_name,cmd_receive
r_bank_post_code,cmd_sender_account_type,cmd_sender_bank_address,cmd_sender_bank_name,cmd_sender_
bank_post_code,cmd_sender_transaction_cop_result,condor_rule_code,consent_id,costumer_id_global,
count_months_beneficiary_received_funds,created_at,current_account_monthly_debit_amount,Customer
averagepaymentamountlastday,customer_account_country,customer_account_currency,customer_account_
nr_external,customer_account_number,customer_account_type,customer_account_zip,customer_addition
al_info_1,customer_additional_info_2,customer_additional_info_3,customer_additional_info_4,custo
mer_address,customer_address_chg_date,customer_alias,customer_annual_turnover,customer_branch_id,
customer_business_phone,customer_business_type,customer_businessphone_last_update_at,customer_ca
rd_number,customer_change_data_new,customer_change_data_old,customer_change_delivery,customer_ch
ange_new,customer_change_old,customer_change_reason,customer_change_reason_alt,customer_chinese_
comm_code,customer_citizen_id_number,customer_citizenship,customer_corresp_address_last_update_at
,customer_corresp_city,customer_corresp_country_code,customer_corresp_region,customer_corresp_st
reet_addr,customer_corresp_zip,customer_corresp_zip_area_code,customer_country_code,customer_cur
rent_account_balance,customer_daytime_work_phone,customer_dob,customer_earnings_amt_snapshot_dt,
customer_email,customer_email_chg_date,customer_email_last_update_at,customer_fiv_marker,custome
r_fiv_marker_last_updated_at,customer_fiv_markers,customer_forname,customer_gender_code,customer_
hmphone_last_update_at,customer_home_address_last_update_at,customer_home_branch,customer_home_c
ity,customer_home_country_code,customer_home_phone,customer_home_phone1_chg_date,customer_home_p
hone2_chg_date,customer_home_region,customer_home_street_addr,customer_home_zip,customer_home_zi
p_area_code,customer_id,customer_id_external,customer_id_global,customer_id_hashed,customer_id_n
umber,customer_id_type,customer_marital_status,customer_market_segment,customer_mbphone_last_upd
ate_at,customer_mob_phone1_chg_date,customer_mob_phone2_chg_date,customer_mobile_phone,customer_
name,customer_number,customer_number_of_accounts,customer_occupation_start_date,customer_onboard
ing_method,customer_pb_marker,customer_phone_chg_date,customer_phone_number,customer_portfolio,c
ustomer_portfolio_channel,customer_portfolio_class,customer_portfolio_country,customer_portfolio_
phys_type,customer_portfolio_product,customer_portfolio_region,customer_portfolio_remote_type,cu
stomer_portfolio_segment,customer_portfolio_type,customer_preferred_language,customer_registrati
on_date,customer_rel_status,customer_reported_income,customer_rm_info,customer_salary_amount,cus
tomer_segment,customer_status,customer_surname,customer_total_balance,customer_type,customer_voi
ce_id,customer_voice_id_last_change_date,customer_work_phone1_chg_date,customer_work_phone2_chg_
date,customer_wrkphone_last_update_at,customer_zip_code,cutoff_at,decision,decision_scores,deleg
ate_id,delegate_initiated,delegate_of,delegate_without_product_indicator,delegates,device_applic
ation_repackaged,device_banking_type,device_bluetooth_detected_list,device_bluetooth_devices_conn
,device_bluetooth_enabled,device_browser,device_cpu_class,device_data_enabled,device_datetime,de
vice_debugger_attached,device_dev_id,device_emulator,device_es_id,device_fingerprint,device_fing
erprint_type,device_geo_ip_city,device_geo_ip_country_code,device_geo_ip_isp,device_geo_latitude,
device_geo_longitude,device_geo_timezone,device_gps_enabled,device_hooking_frameworkds,device_id,
device_imsi,device_ip_address,device_ip_address_v6,device_ip_city,device_ip_country_code,device_
ip_latitude,device_ip_longitude,device_ip_region,device_is_desktop,device_is_gameconsole,device_
isp,device_java_script_enabled,device_javascript_malformed,device_language,device_latitude,devic
e_longitude,device_mac_address,device_mcc,device_os,device_platform_type,device_platform_type_re
ason,device_print,device_raw_jsc,device_risk_score,device_rooted,device_tdl,device_timezone,devi
ce_timezone_offset,device_token_serial_no,device_type,device_untrusted_keyboard,device_untrusted_
screen_reader,device_user_agent,device_wifi_bssid_connected,device_wifi_in_range,device_wifi_sta
tus,devide_native_codehooks,direction,dispute_end_date,dispute_feedback_notes,dispute_notes,disp
ute_reason,dispute_refunded,dispute_start_date,dispute_submitted_by,ds_transaction_id,ecw_answer
s_changed,ecw_country_payment_description,ecw_duration,ecw_final_action,ecw_payment_details_chan
ged,ecw_pop_cpd_mapped,ecw_purpose_of_payment,ecw_question_and_answer_details,ecw_response_time,
entity_id,entity_type,event_code,event_created_at,event_external_id,event_occurred_at,event_rece
ived_at,event_server_validated_at,event_status,event_type,event_updated_at,external_preliminary_
decision,external_status_reason,fdz_channel,feedback_label,feedback_type,feedback_value,fraud,fr
aud_score,gift_indicator_code,gpb_manual_payment_ind,idv_session_id,info_type,instant_transfers_
review_outcome,internal_event_source,internal_payment_type,invalid_email_indicator,is_historic,i
s_scoring,isJade,lifecycle_id,lifecycle_id_alt,model_risk_score,new_customer_flag,new_receiver_f
lag,number_of_payments,number_of_payments_count,office_id,oob_comm_type,oob_end_date,oob_feedbac
```

```
k_notes,oob_notes,oob_response_type,oob_start_date,overnight_acc_ledger_bal,payment_above_limit,
payment_entity,payment_execution_type,payment_message_source,payment_message_source_d,payment_me
ssage_type,payment_ref,payment_related_ref,payment_remittance_info,payment_restricted_template,p
ayment_revision_code,payment_saf,payment_sca_proc_code,payment_status,payment_sub_type,payment_t
ype,payment_type_direct_debit,payment_type_self_acc,payment_type_sub_type,phone_bank_use_code,ph
one_session_id,pib_suspended,pinpoint_risk_score,product_code,receiver_account_bank_id,receiver_
account_bank_name,receiver_account_currency,receiver_account_type,receiver_bank_addr_city,receiv
er_bank_address,receiver_bank_bic,receiver_bank_city,receiver_bank_country_code,receiver_bank_na
me,receiver_bank_postal_code,receiver_bank_routing_number,receiver_bank_sort_code,receiver_bank_
status,receiver_bank_type,receiver_city,receiver_cop_match_name,receiver_cop_name,receiver_count
ry_code,receiver_creation_date,receiver_creation_date_customer,receiver_details,receiver_dob,rec
eiver_email,receiver_email_chg_date,receiver_id,receiver_merchant_category,receiver_name,receive
r_phone,receiver_region,receiver_street_addr,receiver_transaction_amount,receiver_transaction_am
ount_dbl,receiver_transaction_bank_account_number,receiver_transaction_currency,receiver_transac
tion_original_amount,receiver_transaction_original_currency,receiver_transaction_product,receive
r_zip,recurring_payment,recurring_payment_ref,recurring_payment_schdl_info,request_correlation_id
,request_type,requested_execution_at,reversal_indicator,risk_level,risk_level_scores,saf_ind,san
ction_check,sca_required,sca_required_score,scam,scam_score,scameter_warning,segment_channel_type
,self_serve_strategy,self_serve_strategy_scores,sender_account_avg_balance,sender_account_balance
,sender_account_balance_dbl,sender_account_bic,sender_account_channel_limit,sender_account_chann
el_limit_double,sender_account_currency,sender_account_number_external,sender_account_open_date,
sender_account_ownership,sender_account_type,sender_account_zip,sender_alias_type,sender_bank_ad
dress,sender_bank_branch_id,sender_bank_city,sender_bank_country_code,sender_bank_name,sender_ba
nk_postal_code,sender_bank_sort_code,sender_bank_status,sender_bank_type,sender_billing_amount,s
ender_dob,sender_email,sender_email_chg_date,sender_initiated_at,sender_initiated_at_local,sende
r_name,sender_occupation_status,sender_orig,sender_phone,sender_phone_chg_date,sender_product_ty
pe,sender_registration_date,sender_transaction_amount,sender_transaction_amount_dbl,sender_trans
action_amount_usd,sender_transaction_auth_type,sender_transaction_bank_account_number,sender_tra
nsaction_cat_code,sender_transaction_category,sender_transaction_code,sender_transaction_cop_mat
name,sender_transaction_cop_name,sender_transaction_cop_result,sender_transaction_currency,sende
r_transaction_details_ind,sender_transaction_effect_date,sender_transaction_exp_date,sender_tran
saction_extra_1,sender_transaction_extra_2,sender_transaction_extra_3,sender_transaction_held,se
nder_transaction_narrative,sender_transaction_new_bene,sender_transaction_notes,sender_transacti
on_origin_amount,sender_transaction_origin_currency,sender_transaction_original_amount,sender_tr
ansaction_original_currency,sender_transaction_product,sender_transaction_reason,sender_transact
ion_reason_d,sender_transaction_ref,sender_transaction_ref_nr,sender_transaction_ref_nr_alt,send
er_transaction_sign_status,sender_transaction_sign_stts_d,sender_transaction_sole_ind,sender_tra
nsaction_stat,sender_transaction_stat_reason,sender_transaction_type,sender_urgency,sender_zip,s
ession_id,sfr_moneylock_hold_bal,sfr_total_hold_bal,shadow_fraud,shadow_fraud_score,shadow_scam,
shadow_scam_score,source,staff_id,sub_channel_name,text_for_receiver,thailand_fps_ind,timestamp,
tmx,tpp_id,tpp_name,tpp_name_alt,tpp_payee_payer,transaction_id,treatment_strategy,treatment_str
ategy_scores,txn_type,user_amount,user_guid,user_id,user_id_type,user_indicator,user_string,uuid,
var_recurring_payment,vocalink,website_name,website_refferer_url,workstation_id
0,feedzai,02348085-801c-45ff-
bdc3-80204ad5e789,1764178132876,pP0Nb2qKXTKLXH1Bkx5HIA,"[{""explanations"":
[{""@type"":""rule"",""reason"":""Customer name is Fraudulent Joe""}],""result"":
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[],""targetOutcome"":""fraud"",""targetOutcomeLabel"":""fraud""},""sourceId"":""34a95d87-6934-4a9
9-a0a0-0ddb7e485fc2"",""weight"":0},{""explanations"":[],""result"":
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""ruleId"":""mGBzsBs9IOyaQn4hbg5E7g"",""scores"":
[],""targetOutcome"":""self_serve_strategy"",""targetOutcomeLabel"":""self_serve_strategy""},""s
ourceId"":""e1fa9a8a-f9f3-436a-9493-2ecf55df48fc"",""weight"":0}]","{""treatment_strategy"":
{""@type"":""mulcla"",""outcomeDecision"":""mobile_push_notification"",""score"":
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4a99-a0a0-0ddb7e485fc2""},{""arguments"":
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f9f3-436a-9493-2ecf55df48fc""}]",cdb5fcfe-c429-436e-879b-1bf570bcb1eb,"{""data"":
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{""@class"":""com.feedzai.pulse.api.types.MessagePath"",""endCause"":""END"",""path"":
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e709-4665-ab10-bdef3f92ac7c"":""2166c942-a3da-43ed-9f59-a404772cad99"",""2166c942-a3da-43ed-9f59-
a404772cad99"":""e1fa9a8a-f9f3-436a-9493-2ecf55df48fc"",""cdb5fcfe-
c429-436e-879b-1bf570bcb1eb"":""5342a480-e9d9-4ec2-b8b2-
b80bd868537b"",""e112ef90-899a-4d9f-854c-4d4307f2d2a6"":""4af67db2-e709-4665-ab10-
bdef3f92ac7c"",""5342a480-e9d9-4ec2-b8b2-b80bd868537b"":""ada6320a-561a-423d-86c4-
c93ca9a79022"",""ada6320a-561a-423d-86c4-c93ca9a79022"":""a2324ae4-c5a2-4d46-ac85-
a9c798d58e1a"",""a2324ae4-c5a2-4d46-ac85-a9c798d58e1a"":""34a95d87-6934-4a99-
a0a0-0ddb7e485fc2""}},""_FDZ_WF_CHAN_ID"":""lFP84JvtvfsabvhkkhRM-w""}}","{""globalState"":
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ab10-bdef3f92ac7c"":{""data"":{""NORMAL_MSG_CTRL_FLOW"":
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{""NORMAL_MSG_CTRL_FLOW"":[""com.feedzai.pulse.api.apps.enums.RuleControlFlow"",0]}},""5342a480-
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{""@class"":""com.feedzai.pulse.datascience.rte.runtime.MetricsWrapper"",""values"":
[null]},""_EXEC_ID"":""jUMq48XQ4sBvNqpiEkVGuQ""}},""e1fa9a8a-f9f3-436a-9493-2ecf55df48fc"":
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0]}}}}","{""5342a480-e9d9-4ec2-b8b2-b80bd868537b"":{""data"":{}}}","{""data"":
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a547-5e6ac5103e67,"","","","",password,,,"","",beneficiary_proxy_account_number,270,"{'is_bot':
false, 'is_emulator': true, 'voice_scam_score': 911, 'is_malware': false, 'is_rat': false,
'is_recent_rat': false, 'custom_risk': 21.0, 'custom_risk_aliases': risk_alias, 'push_score':
NULL, 'is_mobile_rat': NULL, 'pinpoint_risk_score': NULL, 'risk_score': 21.0,
'chall_policy_score': NULL}","","",b398a9e1-3900-4753-ad45-8310ea5837234,"",
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77,Receiver Bank,WC7N 7EW,CREDIT,Diagon Alley 33,Gringotts,WC2N 5DU,"","","","",,
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Additional Info 3,Customer Additional Info 4,224 Benjamin Flats,
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0001-01-01,,,,,,,,"",,,,"","",,,,,customer_1,"","",
2d18c716ba930802d187654c99fd097f5530f1479065428caa1a3ee1q7421,0HKHSXXXXXX796256,V,,"",,"","",,Fr
audulent Joe,"",483,,"",,
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","",,66720-331,1554198919000,approve,"[0.0, 0.0,
0.0]","",false,,true,,true,"","","",false,Brave,"",true,1758901473241,true,"",false,"",,"","",
440,"",2.337,-140.6989,"",false,false,7823d84a-6acc-43e6-84b2-dddc3122a248,"",
161.223.207.161,"",New Dejaport,
440,-48.0299,104.4205,OT,true,true,,true,false,be-440,-77.4773,-31.3643,"","",ioss,"","","","",
21,true,"",-07:00,"","",web browser,true,false,"Mozilla/5.0 (Macintosh; U; Intel Mac OS X
10_8_6) AppleWebKit/532.0.2 (KHTML, like Gecko) Chrome/14.0.855.0 Safari/
532.0.2","",false,false,true,outgoing,1554196919000,"",Very fraudulent.,test,"",
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0.89,D,"",1,1758901473241,"",test,1535477895438,The client called to confirm it was not him
making that transaction.,Contacted John Doe to confirm.,call,
1554196919000,100.5,false,"",FDP,"","","",,"","",false,"","","",entered,credit_transfer,sepa,fal
se,false,"",,"",YESD,"",,"",,270,CREDIT,,Some Avenue 77,4ad3fe88-317b-4b31-

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440,1758364512397,"","",19900101,[email protected],0001-01-01,receiver_id,"",Receiver Joe Doe,
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a904a82bb65e,270,,,"",receiver_zip,true,"","",VRWER3452BVQR2,"",1554996919000,"",low,"[0.0, 0.0,
0.0]","",true,true,0.0,false,0.0,"","",ivr_self_no,"[0.0, 0.0, 0.0, 0.0, 0.0]",
973.0,1110000.0,1200.75,PTCDG,,13800.0,270,"",1668683369590,"",CREDIT,"","",Diagon Alley
33,"",London,440,Gringotts,WC2N 5DU,"","","",123452332.22,19900101,[email protected],
0001-01-01,1554196919000,"",Sender Joe Doe,looking for a new project,"",
001122334455,0001-01-01,UFAAE,1354398919000,155000.0,400.75,155000.0,"",
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Transaction Additional Info 1,Sender Transaction Additional Info 2,Sender Transaction Additional
Info 3,true,"",true,Notes on the transaction.,
10,270,395.75,270,"","","","","","","","","","","","","",,53fcc910-f92c-4cd5-84b6-4d437da309ca,
1225.98,1225.98,false,0.0,false,0.0,"","",BS,"","",1764178132876,"{'fuzzy_device_id': '',
'ip_city': '', 'ip_country': '', 'ip_longitude': 10.1339, 'ip_latitude': 50.3646,
'chall_policy_score': 780.0, 'ip_address': '', 'digital_id': '', 'device_isp': '',
'summary_reason_code': '', 'policy_score': 21.0, 'risk_score': 21.0, 'device_dev_id': '',
'request_id': '', 'true_isp': NULL, 'proxy_isp': NULL, 'dns_isp':
NULL}","","","","","",mobile_push_notification,"[0.0, 0.0, 0.0, 0.0]","",[771.0],"","","",
[Eulah.Hammes24],[Ludie_Denesik],02348085-801c-45ff-bdc3-80204ad5e789,0,"{'vocalink_score': 20.0,
'mastercard_id': '', 'alert_id': 5aa89b5b-9fc0-4faf-bfaf-c0454611d129, 'source_account_id':
ada0e7c4-5705-4b76-8be8-6b3ccf8dafa0, 'destination_account_id': 21837091-a0d2-4e16-a26c-
c6cdd67a66fb, 'transaction_id': f211956c-4485-44e1-a4f5-505dffbbcb50, 'transaction_value': 123.0,
'score_generated_at': 1554198919000, 'transaction_currency': 270}","","",""
```


Copy

Unique identifier (`FDZ_UUID`)â

This field is a unique identifier for every event extracted.

```
lifecycle_id,...,FDZ_UUID,...
f9ae8116-812d-4e0c-b658-dc3de1e31f92,...,02348085-801c-45ff-bdc3-80204ad5e789,...
```

Copy

Actions triggered (`FDZ_TRIGGERED_ACTIONS`)â

This field is a list of JSON Objects that contains all the triggered actions.

```
lifecycle_id,...,FDZ_TRIGGERED_ACTIONS,...
f9ae8116-812d-4e0c-b658-dc3de1e31f92,...,[{"arguments":
{},"name":"Alert","nestedSourceId":"lCyKop_QjP45br_tEjlNcg","sourceId":"34a95d87-6934-4a99-
a0a0-0ddb7e485fc2"},{"arguments":
{},"name":"Alert","nestedSourceId":"lUQqjQhs2CSSxrRspC9NeQ","sourceId":"e112ef90-899a-4d9f-854c-4
d4307f2d2a6"},{"arguments":
{},"name":"Alert","nestedSourceId":"mGBzsBs9IOyaQn4hbg5E7g","sourceId":"e1fa9a8a-
f9f3-436a-9493-2ecf55df48fc"}],...
```

Copy

Rules triggered and explanations (`FDZ_RESULTS`)â

This field is a list of JSON Objects that contains all the rules that were triggered by the extracted event and their explanation.

```
lifecycle_id,...,FDZ_RESULTS,...
f9ae8116-812d-4e0c-b658-dc3de1e31f92,...,[{"explanations":[{"@type":"rule","reason":"Customer
name is Fraudulent Joe"}],"result":
{"@type":"clarule","originalRuleId":"mI_NayHmjUkeQiG8ByNK1g","outcome":"true","ruleId":"lCyKop_Q
jP45br_tEjlNcg","scores":
[],"targetOutcome":"fraud","targetOutcomeLabel":"fraud"},"sourceId":"34a95d87-6934-4a99-
a0a0-0ddb7e485fc2","weight":0},{"explanations":[],"result":
{"@type":"clarule","originalRuleId":"lWdQy83xgSh81tiMYg9EDA","outcome":"mobile_push_notification"
```

```
,"ruleId":"lUQqjQhs2CSSxrRspC9NeQ","scores":
[],"targetOutcome":"treatment_strategy","targetOutcomeLabel":"treatment_strategy"},"sourceId":"e1
12ef90-899a-4d9f-854c-4d4307f2d2a6","weight":0},{"explanations":[],"result":
{"@type":"clarule","originalRuleId":"lP7bp45ioG9Og22W9YBJMw","outcome":"ivr_self_no","ruleId":"m
GBzsBs9IOyaQn4hbg5E7g","scores":
[],"targetOutcome":"self_serve_strategy","targetOutcomeLabel":"self_serve_strategy"},"sourceId":"
e1fa9a8a-f9f3-436a-9493-2ecf55df48fc","weight":0}],...
```


Copy

Outcomes (`FDZ_OUTCOMES`)â

This field is a JSON object that contains all the configured **outcomes** (e.g. `fraud` , `decision`) and their scores.


```
lifecycle_id,...,FDZ_OUTCOMES,...
f9ae8116-812d-4e0c-b658-dc3de1e31f92,...,{"treatment_strategy":
{"@type":"mulcla","outcomeDecision":"mobile_push_notification","score":
[0.0,0.0,0.0,0.0]},"shadow_scam":{"@type":"bincla","outcomeDecision":false,"score":
0.0},"risk_level":{"@type":"mulcla","outcomeDecision":"low","score":[0.0,0.0,0.0]},"scam":
{"@type":"bincla","outcomeDecision":false,"score":0.0},"decision":
{"@type":"mulcla","outcomeDecision":"approve","score":[0.0,0.0,0.0]},"self_serve_strategy":
{"@type":"mulcla","outcomeDecision":"ivr_self_no","score":[0.0,0.0,0.0,0.0,0.0]},"sca_required":
{"@type":"bincla","outcomeDecision":true,"score":0.0},"shadow_fraud":
{"@type":"bincla","outcomeDecision":false,"score":0.0},"fraud":
{"@type":"bincla","outcomeDecision":true,"score":0.0}},...
```


Copy

Metricsâ

For every metric (realtime and scheduled) there will be a field with the name of the metric, e.g. `count_benefname_90d` .

```
lifecycle_id,...,count_benefname_90d,...
f9ae8116-812d-4e0c-b658-dc3de1e31f92,...,10,...
```

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ETL vs DTPâ

In this section we describe the differences between ETL and DTP extractions for non-schema fields.

Unique identifierâ

There's no change in the unique identifier field.

Tool	Field	Example
ETL	FDZ_UUID	02348085-801c-45ff-bdc3-80204ad5e789
DTP	FDZ_UUID	02348085-801c-45ff-bdc3-80204ad5e789

Metricsâ

In DTP **both** realtime and scheduled metrics are exported instead of only scheduled metrics as in the ETL.

Scheduled metricsâ

In DTP scheduled metrics are extracted just like in ETL.

Tool	Field	Example
ETL	metric_A	10
DTP	metric_A	10

Realtime metricsâ

Realtime metrics are only available in DTP.

Tool	Field	Example
ETL	N/A	N/A
DTP	metric_A	10

Actions triggeredâ

In DTP there's only a field with the triggered actions instead of a field for every action (and the value whether it was triggered or not).

Tool	Field	Example
ETL	action_alert	true
DTP	FDZ_TRIGGERED_ACTIONS	`[ { "name": "Alert" } ]` Copy

Rules triggered and explanationsâ

In DTP rules triggered and explanations are available in a single field `FDZ_RESULTS`.

Rules triggeredâ

Tool	Field	Example
ETL	rules_triggered	lCyKop_QjP45br_tEjlNcg;lUQqjQhs2CSSxrRspC9NeQ;mGBzsBs9IOyaQn4hbg5E7g
DTP	FDZ_RESULTS	`[ { "result": { "ruleId": "lCyKop_QjP45br_tEjlNcg" } }, { "result": { "ruleId": "lUQqjQhs2CSSxrRspC9NeQ" } }, { "result": { "ruleId": "mGBzsBs9IOyaQn4hbg5E7g" } } ]` Copy

Rules explanationsâ

<table>
<tr><th>Tool</th><th>Field</th><th>Example</th></tr>
<tr><td>ETL</td><td>formatted_results</td><td><pre>
[
  {
    "explanation": "Customer name is Fraudulent Joe",
    "ruleId": "lCyKop_QjP45br_tEjlNcg"
  }
]
</pre>
Copy</td></tr>
<tr><td>DTP</td><td>FDZ_RESULTS</td><td><pre>
[
  {
    "explanations": [
      {
        "reason": "Customer name is Fraudulent Joe"
      }
    ],
    "result": {
      "ruleId": "lCyKop_QjP45br_tEjlNcg"
    }
  }
]
</pre>
Copy</td></tr>
</table>

Outcomesâ

Outcomes are similar for ETL and DTP extractions. The DTP extractions only have an additional field `@type` .

<table>
<tr><th>Tool</th><th>Field</th><th>Example</th></tr>
<tr><td>ETL</td><td>outcomes_and_scores</td><td><pre>
{
  "scam": {
    "outcomeDecision": false,
    "score": 0.0
  },
  "fraud": {
    "outcomeDecision": true,
    "score": 0.0
  },
  "decision": {
    "outcomeDecision": "approve",
    "score": [0.0, 0.0, 0.0]
  },
  "risk_level": {
    "outcomeDecision": "low",
    "score": [0.0, 0.0, 0.0]
  }
}
</pre>
Copy</td></tr>
<tr><td>DTP</td><td>FDZ_OUTCOMES</td><td><pre>
{
  "scam": {
    "@type": "bincla",
    "outcomeDecision": false,
    "score": 0.0
</pre></td></tr>
</table>

Tool	Field	Example

		``
		  },
		  "fraud": {
		    "@type": "bincla",
		    "outcomeDecision": true,
		    "score": 0.0
		  },
		  "decision": {
		    "@type": "mulcla",
		    "outcomeDecision": "approve",
		    "score": [0.0, 0.0, 0.0]
		  },
		  "risk_level": {
		    "@type": "mulcla",
		    "outcomeDecision": "low",
		    "score": [0.0, 0.0, 0.0]
		  }
		}
		``
		Copy

On this page

GCP Setup

In this page you can find information about the necessary resource setup in Google Cloud Platform (GCP) for Data Transfer Pipeline (DTP).

Pub/Sub Setupâ

The DTP uses Pub/Sub topics as a vehicle to share messages from Pulse to the Dataflow pipeline and also for the Inactivity Service. It is possible to let DTP automatically create the Pub/Sub topics or it can be configured to connect to topics previously created.

For a complete reference of the configurations exposed by the DTP please refer to the DTP Configuration page.

Automatically create topicsâ

As a low-maintenance option, the DTP can be configured to automatically create the Pub/Sub topics it uses. These will have default configurations applied to them and it is not possible to tweak them.

To enable automatic topic creation, the following Ansible variables should be configured:

- `dtp_pubsub_topic_enable_creation` should be set to `true`.
- `dtp_pubsub_topic_name_prefix` should be set a string prefix to be used in the topic id.
  - The final topic name will be *"projects/{project}/topics/{prefix}-{app-id}"*.
- `dtp_inactivity_topic_enable_creation` should be set to `true`
- `dtp_inactivity_topic_id` should be set to the topic id for the DTP to create.

By design, each application will automatically have its own topic as the *app id* is used as part of the topic id.

Use existing topicsâ

If custom configuration for the topics is required, the DTP also supports connecting to existing topics. To enable this mode of operation, the follow Ansible variables should be configured:

- `dtp_pubsub_topic_enable_creation` should be set to `false`.
- `dtp_pubsub_topic_id` should be set to the topic id to use.
  - The topic name will be *"projects/{project}/topics/{dtp_pubsub_topic_id}"*.
- `dtp_inactivity_topic_enable_creation` should be set to `false`
- `dtp_inactivity_topic_id` should be set to the topic id for the DTP to connect to.

It is **crucial for the main topic id to be unique for each app**. Unlike the automatic topic creation mode, this approach does not inject the *app id* in the topic id so it up this Ansible variable to have distinct values for each application. That can be done using the `<app_slug>_dtp_pubsub_topic_id` syntax for the Ansible variable.

GCP Permissionsâ

In the following table you can find the minimal set of GCP resource permissions required by the service account attached to the machine running the Data Transfer Pipeline (DTP).

Permission	Description
dataflow.jobs.cancel	Allows the cancellation of Dataflow jobs.
dataflow.jobs.create	Allows the creation of Dataflow jobs.
dataflow.jobs.get	Allows the fetching of Dataflow jobs.

Permission	Description
dataflow.jobs.list	Allows listing Dataflow jobs of a project.
iam.serviceAccounts.actAs	Allows the service account to act as another service account (e.g. the Dataflow service account).
pubsub.schemas.create	Allows the creation of Pub/Sub schemas.
pubsub.schemas.delete	Allows the deletion of Pub/Sub schemas.
pubsub.schemas.get	Allows the fetching of Pub/Sub schemas.
pubsub.schemas.list	Allows listing Pub/Sub schemas of a project.
pubsub.subscriptions.consume	Allows the consumption of Pub/Sub topic messages by a subscription.
pubsub.subscriptions.create	Allows the creation of a Pub/Sub subscription.
pubsub.topics.attachSubscription	Allows attaching a subscription to a Pub/Sub topic.
pubsub.topics.create	Allows the creation of Pub/Sub topics.
pubsub.topics.publish	Allows the publish of messages to a Pub/Sub topic.
storage.buckets.get	Allows reading metadata of GCS buckets.
storage.objects.create	Allows the creation of objects in GCS buckets.
storage.objects.get	Allows the fetching of objects in GCS buckets.

Dataflow worker service accountâ

The DTP uses a Dataflow pipeline to process and store the data. When creating the job, the service account attached to the machine running DTP will impersonate the Dataflow service account. However, Dataflow uses two service accounts to manage security and permissions:

- **The Dataflow service account:** The Dataflow service uses the Dataflow service account as part of the job creation request, and during job execution to manage it.
- **The worker service account:** Worker instances use this service account to access input and output resources after the job is submitted.

By default, workers use the Compute Engine default service account of the project as the worker service account. This service account has the following email:

```
[email protected]
```

Copy

Although this service account can be used as the worker service account, it's recommended for a dedicated Dataflow worker service account to be created and used. The `dtp_dataflow_worker_service_account` variable can be used to set the desired service account to use as the Dataflow worker service account.

In the following table you can find the minimal set of GCP permissions required by the Dataflow **worker service account**.

Permission	Description
autoscaling.sites.readRecommendations	Allows reading autoscaling recommendations for a site (e.g., suggested capacity or scaling adjustments).
autoscaling.sites.writeMetrics	Allows writing custom metrics used by the autoscaler to make scaling decisions.
autoscaling.sites.writeState	Allows updating the autoscaling state of a site (e.g., reporting health or capacity state).
compute.instanceGroupManagers.update	Allows updating a managed instance group (MIG), such as changing instance templates, size, or update policies.
compute.instances.delete	Allows deleting virtual machine (VM) instances.
compute.instances.setDiskAutoDelete	Allows configuring whether attached disks are automatically deleted when the VM instance is deleted.

Permission	Description
dataflow.jobs.get	Allows the fetching of Dataflow jobs.
dataflow.shuffle.read	Allows reading intermediate shuffled data produced during a Dataflow pipeline execution.
dataflow.shuffle.write	Allows writing intermediate shuffled data during a Dataflow pipeline execution.
dataflow.streamingWorkItems.ImportState	Allows importing state data for streaming work items (used during streaming pipeline recovery or state restoration).
dataflow.streamingWorkItems.commitWork	Allows committing completed streaming work items and associated state changes.
dataflow.streamingWorkItems.getData	Allows retrieving data associated with a streaming work item.
dataflow.streamingWorkItems.getWork	Allows retrieving streaming work items assigned to a worker.
dataflow.streamingWorkItems.getWorkerMetadata	Allows retrieving metadata about a streaming worker (e.g., configuration or assignment details).
dataflow.workItems.lease	Allows leasing (claiming) batch work items for processing.
dataflow.workItems.sendMessage	Allows sending messages between Dataflow service components or workers.
dataflow.workItems.update	Allows updating the status or progress of a batch work item.
logging.logEntries.create	Allows writing log entries to Cloud Logging.
logging.logEntries.route	Allows routing log entries through log sinks to supported destinations.
monitoring.timeSeries.create	Allows writing custom metric data points to Cloud Monitoring time series.
pubsub.subscriptions.consume	Allows the consumption of Pub/Sub topic messages by a subscription.
pubsub.subscriptions.create	Allows the creation of a Pub/Sub subscription.
pubsub.topics.attachSubscription	Allows attaching a subscription to a Pub/Sub topic.
pubsub.topics.get	Allows the fetching of Pub/Sub topics.
storage.buckets.get	Allows reading metadata of GCS buckets.
storage.objects.create	Allows the creation of objects in GCS buckets.
storage.objects.delete	Allows the deletion of objects in GCS buckets.
storage.objects.get	Allows the fetching of objects in GCS buckets.

For more details regarding the service accounts that Dataflow uses and how, refer to the following documentation:

- Security and permissions for pipelines on Google Cloud
- Dataflow service account
- Worker service account

GCP Permission required for using CMEKâ

The `Cloud KMS CryptoKey Encrypter/Decrypter` role must be granted to the Service Account associated with the following services to enabled CMEK integration:

- Dataflow: `service-<PROJECT_NUMBER>@dataflow-service-producer-prod.iam.gserviceaccount.com`
- Compute Engine: `service-<PROJECT_NUMBER>@compute-system.iam.gserviceaccount.com`
- Pub/Sub: `service-<PROJECT_NUMBER>@gcp-sa-pubsub.iam.gserviceaccount.com`

Here, `<PROJECT_NUMBER>` refers to the project number (not the project ID) of your Google Cloud project.

For more details, refer to the following documentation:

- Dataflow CMEK integration
- Pub/Sub CMEK roles and permissions

Networkâ

By default all Dataflow jobs will be configured to use the `default` network. Given this network usage is usually more restricted in production environments, a different network and/or subnetwork may need to be configured.

This can be achieved by using the following deployment variables:

- `dtp_dataflow_network` : sets the Dataflow job network. If omitted along with `dataflow.subNetwork` the jobs will run on the `default` network
- `dtp_dataflow_subnetwork` : sets the Dataflow job subnetwork. If specified, GCP will determine the network and `dataflow.network` can be omitted. This configuration has to be set in the format `regions/REGION_NAME/subnetworks/SUBNETWORK_NAME`

For more details on Dataflow networking refer to Specify a network and subnetwork.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/installation-guide/dtp/service-dtp

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DTP Service

The Data Transfer Pipeline service (DTP) aims to be a more robust and scalable solution in comparison with ETL, by solving the following limitations:

- Better support for workflow schema change through the use of a workflow service that has information about the last published schema
- Events are streamed as soon they are processed in Pulse and sent to an upstream cloud native service (GCP Pub/Sub)
- Historical events extracted as they reach end of the Pulse workflow - it does not matter when they happened
- All the event fields that reach the end of the workflow will be sent to the Data Transfer Pipeline
- Data is sent in near real time to DSE and the extractions contain the metric values so the data sources can be immediately used (if there are not new metrics in the plan under test)

In this page one can find information about the service architecture, usage and monitoring.

Architectureâ

The DTP service uses the following GCP resources:

- Pub/Sub topics
- Pub/Sub schema registry
- Dataflow pipelines
- GCS buckets

Please refer to the DTP permissions page for the permissions required by the service account attached to the machine running the Data Transfer Pipeline (DTP).

The service architecture and the GCP resources utilization is represented in the following diagram:

Data Generationâ

An instance of the service runs in each Pulse workflow, streaming data generated in the Pulse workflow to GCP Pub/Sub.

Data Processingâ

The near real time messages streamed from the service are consumed by `1 + N` subscribers:

- DTP subscriber that consumes and transforms the data and makes it available to Pulse DS
    - This subscriber uses a Dataflow pipeline to process and store the data in a GCS bucket
- Any other subscribers developed by HSBC to consume and process messages. For instance, there can be a subscriber that processes and sends messages to FDR

Data Consumptionâ

Data generated from the DTP subscriber is available to be used in Pulse DS for risk strategy tests and upgrades. Any data consumed/generated by other subscribers can be used in other system (e.g. FDR).

Usageâ

As previously stated, DTP is a Pulse service and therefore has to be added to a Pulse application workflow.

By default it should appear in the Custom Services section in Pulse UI by going into the `Data Configuration` tab of an application. If it does not appear it can be manually added according to the following configuration:

```
Name: Data Transfer Pipeline
Type: Workflow
Binary: Provided
Binary Path: data-transfer-pipeline-service.jar
```

Copy

If the service is loaded correctly a set of configurations will then appear in the `Configuration` section. Please refer to [DTP configuration reference page] to understand how to configure the service.

After the service is configured it needs to be added to the workflow. To do so:

- Go the application workflow page
- Select the workflow
- Add a new `Custom Workflow Element` to the workflow
- Add a name (e.g. DTP), leave the `Target Outcome` empty and select `Data Transfer Pipeline` in the `Custom Service` dropdown
- Save the workflow and publish

Publishing the application workflow with the DTP service enabled will trigger the creation/update of the associated GCP resources.

info

The DTP service doesn't automatically remove any GCP resources, except the dataflow jobs marked as inactive by the Inactivity service.

Inactivity Serviceâ

The DTP service has an embedded inactivity service that detects and cancels inactive Dataflow jobs.

When enabled:

- Each Dataflow job pipeline includes an extra step that monitors the job inactivity
- If inactivity is detected, the service sends a message to the designated inactivity Pub/Sub topic
- The pipeline is then canceled, provided it is not the only active pipeline for its associated Pulse application and cluster color

Please refer to the Inactivity service section in the configuration page to understand how to configure the service.

Data Securityâ

The DTP service allows configuring security measures related to data encryption and regional restrictions.

Data Encryptionâ

It's possible to specify a customer-managed encryption key (CMEK) to encrypt data handled by Pub/Sub topics and Dataflow pipeline resources using the following property:

- `security.kmsKeyName`

The value should follow the format described in Configure a topic with CMEK.

If omitted, Google-managed encryption keys will be used by default.

Regional Restrictionsâ

It's possible to control where Pub/Sub is allowed to store and transmit data using the following properties:

- `security.allowedRegions` : A comma-separated list of allowed Google Cloud regions where Pub/Sub may store data. Example: "europe-west1,asia-east2".

```
# Ansible variable
dtp_security_allowed_regions:
  - europe-west1
  - asia-east2
```

Copy

- `security.enforceInTransit` : A boolean flag that, when enabled, enforces in-transit data restrictions to the same regions defined in `security.allowedRegions` .

```
# Ansible variable
dtp_security_enforce_in_transit: true
```

Copy

If regional restrictions are active, the following property becomes **mandatory**:

- `security.regionalEndpoint` : The regional Pub/Sub gRPC service endpoint to use when regional restrictions are enforced. Example: "europe-west1-pubsub.googleapis.com:443". The full list of available endpoints can be found [here](#).

```
# Ansible variable
dtp_security_regional_endpoint: "europe-west1-pubsub.googleapis.com:443"
```

Copy

info

When no regional restrictions are in place, the global Pub/Sub endpoint is used. This endpoint automatically routes traffic to the closes region to minimize latency.

Therefore, when using a regional endpoint, it's important to choose the appropriate one - preferably in the same region as other GCP resources and Pulse. If the regional endpoint is in a different region, performance can be affected, as cross-region requests are typically slower.

Logging and Monitoringâ

By default, the service has an `INFO` log level. This value can be changed by using the following Ansible variable:

```
data_transfer_pipeline_log_level: info
```

Copy

To build a monitoring solution for the DTP service please refer to the DTP Metrics Guide.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/installation-guide/etl/configuration-etl

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ETL Configurations

Ansible variablesâ

Variable	type	Mandatory	Default Value	Description
stop_on_any_error	BOOLEAN		true	boolean flag to stop extraction if any error happened
num_extractions_threads	NUMBER		1	Number of threads assigne to extractions
num_publishing_threads	NUMBER		1	Number of threads assigne to publishing
max_events_per_second	NUMBER		5000	Max number of events that can be loaded per second
event_storage_splits	NUMBER		1	The number of event storage splits.
split_size	NUMBER		134217728	The size of the event storage splits.
papp_file_path	STRING		"{{ etl_home_path }}/var/tfrb.papp"	The path where the tfrb.papp is set
database_pulse_eventstorage_hostname	STRING	X		the host of the Events Storage Database
database_pulse_eventstorage_port	STRING		5432	The port of the Events Storage Database
database_pulse_eventstorage_schema	STRING		eventstorage	The schema of the Event Storage Database
database_pulse_eventstorage_username	STRING		"{{ lookup('gcp_secret_manager', secret='database_pulse_eventstorage_username') }}"	Pulse Event Storage username
database_pulse_eventstorage_password	STRING		"{{ lookup('gcp_secret_manager', secret='database_pulse_eventstorage_password') }}"	Pulse Event Storage password
storage_bucket_eventstorage_path	STRING	X		Path of the folder on bucke where the events will be stored, e.g: 'gs:// BUCKET_NAME/FOLDER
storage_bucket_eventstorage_folder_prefix	STRING		event_store_	Prefix of the bucket folder where the events will be stored, e.g: 'gs:// BUCKET_NAME/FOLDER event_store_'
cron_jobs	LIST[CRON]		`daily extraction of last day`	List of cron jobs to run
etl_extraction_interval	[INTERVAL]			Interval to be used by the extraction. **Note:** Only considered if `cron_jobs` is not defined or empty
etl_java_opts	STRING		"-Xms8g -Xmx8g"	

Variable	type	Mandatory	Default Value	Description
				ETL Java options. This property can also be used override the metrics defaul port by setting it with `-Dcom.feedzai.commons letmeknow.reporter.r est.DropwizardRestRe porterFactory.port=< port>`
pulse_connection_mode	STRING	X	"PAPP"	ETL Connection mode. `PAPP` indicates that a local application will be used whereas `RUNTIME` indicates that the applicatio will be loaded from Pulse when the ETL process starts.
pulse_cluster_name	STRING			**Mandatory for `RUNTIME` mode**. Pulse cluster name in Zookeeper, usually set t `pulse{{ deployment_ olor }}`
zookeeper_nodes	STRING			**Mandatory for `RUNTIME` mode**. Zookeeper contact points (e.g. Zookeeper LB IP).
etl_include_result_tracking	BOOLEAN		False if ETL configured for Historical Data Loading, true otherwise	Whether the result tracking should be exported or not
etl_include_formatted_result_tracking	BOOLEAN		False if ETL configured for Historical Data Loading, true otherwise	Whether the formatted resu tracking should be exporte or not
etl_include_outcomes_extraction	BOOLEAN		False if ETL configured for Historical Data Loading, true otherwise	Whether the outcomes should be exported or not
etl_outcomes_in_single_field	BOOLEAN		False if ETL configured for Historical Data Loading, true otherwise	Whether the outcomes should be exported in a single field (JSON) or not (default - one column per outcome). Only works if extraction.includeOutcome or extraction.includeFeedbac are enabled
etl_include_internal_state	BOOLEAN		False if ETL configured for Historical Data Loading, true otherwise	Whether or not inernal stat (scheduled metrics) will be included during extractions
etl_include_triggered_actions	BOOLEAN		False if ETL configured for Historical Data Loading, true otherwise	Whether the triggered actions should be exported or not.
etl_project_all_fields	BOOLEAN		True if ETL configured for Historical Data Loading, true otherwise	The fields to be projected when `projectAllFields` is set to false. The ord of the fields in this list will match the order in which they are projected to the targetAdapter.
etl_include_feedbacks	BOOLEAN			

Variable	type	Mandatory	Default Value	Description
			False if ETL configured for Historical Data Loading, false otherwise	Whether the exported events should include feedback or not. The defau is false.
etl_include_workflow_state	BOOLEAN		False if ETL configured for Historical Data Loading, false otherwise	Whether the workflow state should be exported or not.
etl_include_event_diffs	BOOLEAN		False if ETL configured for Historical Data Loading, false otherwise	Whether the event diffs should be exported or not.
get_latest_published_app_version	BOOLEAN		false	**Only applies to `RUNTIME` mode**. If set to `true`, ET will try to load the publishe workflow schema to extrac events. If set to `false` (c if the current published schema is not found), ETL will load the current workflow schema.
etl_uuid_field	STRING		customer_id if ETL configured for Historical Data Loading, else use lifecycle_id	Indicates the CSV column where the event UUID is defined. When running ETI for Historical Data Loading this property is set to `cus omer_id` to prevent inserting errors due to non valid IDs.
etl_scheduled_metrics_manager	STRING		ALL_PLAN_EXECUTIONS	The type of the scheduled metrics manager to use when computing metrics. Can be one of `ALL_PLAN EXECUTIONS`, `CURRENT_ WORKFLOW`, `EMPTY` or `Z ERO`.
use_rsync	BOOLEAN	X	true	Configures the deploymen pipeline to use `rsync` instead of `copy`. Note tha this variable should be set to `false` in RHEL8+ installations.
etl_cm_rabbitmq_queue_type	STRING		CLASSIC	Sets the RabbitMQ queue type (possible values are CLASSIC and QUORUM). Note that this impacts only Pulse to Case Manager recovery, and is ignored in normal data science extractions to GCS buckets

Cron typeâ

Variable	type	Mandatory	Default Value	Description
name	STRING	X		Name of the cron job
user	STRING	X		The user which will run the cron job
job_arguments	STRING	X		Arguments for the ETL tool Cron Job
hour	STRING		*	The cron expression for `hour`

Variable	type	Mandatory	Default Value	Description
minute	STRING		*	The cron expression for `minute`
day	STRING		*	The cron expression for `day`
weekday	STRING		*	The cron expression for `weekday`
month	STRING		*	The cron expression for `month`

Default cron jobsâ

```
cron_jobs:
  - name: "Daily Extractions job of runtime data to Data Science environment"
    user: "{{ etl_unix_user }}"
    job_arguments: "--extraction -cf {{ etl_symlink_path }}/conf/event-storage-etl.properties -rei -1d --secret {{ etl_secret_path }}"
    hour: 0
    minute: 0
```

Copy

The `-rei` configuration follows [JTimeWindow Expressions](#).

In our example `-1d` means "the last day".

Other examples:

```
 - Last 2 days:                                   -2d/today
- Last week:                                      -1w
- From 3 months ago until 1 month ago: -3M/-1M
- From 30 minutes ago until now:          -30m/now
```

Copy

Interval typeâ

Variable	type	Mandatory	Default Value	Description
start_event_timestamp	TIMESTAMP	X		Unix timestamp of the extraction interval start timestamp
end_event_timestamp	TIMESTAMP	X		Unix timestamp of the extraction interval end timestamp

Runtime mode notesâ

ETL Runtime mode enables ETL to fetch the current schema from Pulse before it starts exporting events, keeping the schema up to date without requiring a new release.

Please find below some notes regarding this mode.

Since ETL will connect through RMI to Zookeeper to fetch Pulse configuration and then Pulse to fetch the schema, the VM running ETL requires the following firewall rules to be set:

- TCP connection to Pulse port 1099 (this port is configured in Pulse property `pulse.manager.rmi.localregistryport`)
- TCP connection to Pulse port 1199 (this port is configured in Pulse property `pulse.manager.rmi.objectport`)
- TCP connection to Zookeeper port 2181

Important noteâ

From 2.4.0 forward the below notice does no longer apply. By setting `get_latest_published_app_version` to `true` , ETL will extract events based on the published workflow schema, meaning that all columns the tool will try to fetch will already be present in the database.

> ETL will fetch the schema even if the application is not published. The consequence of this is that the workflow schema can not have any changes because the database will not have the new column(s)

created. This consequence is not problematic since the schema is only changed when the API is also changed (with a new release). Since this process is done in a B/G deployment, ETL will always pick up the current schema.

Scheduled metrics managerâ

ETL Runtime mode also supports schedule metrics extractions. This option can be enabled by setting `etl_include_internal_state` to `true`.

When extracting events from event storage to CSV a new configuration can be used to determine how ETL handles scheduled metrics extraction. In previous versions, when an event's scheduled metrics belonged to a plan that was no longer referenced in the workflow, extraction of the event would fail.

ETL now handles such cases gracefully and can also be configured to handle them in different ways. Find below the configuration options and how they influence schedule metrics extraction:

- `ALL_PLAN_EXECUTIONS` : ETL will search for the metrics definition in all existing plan executions
- `CURRENT_WORKFLOW` : ETL will search for metrics definitions in plan executions that are referenced in the workflow
- `EMPTY` : ETL will always return empty metrics
- `ZERO` : ETL will always return metrics with a value of zero

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Deleting Event Storage Records

Playbook to ensure that *Event Storage* won't have duplicate entries caused by multiple executions of Event Storage ETL for Historical Data Load over the same time range (e.g.: retries).

Configurationâ

Variable	type	Mandatory	Default Value	Description
database_pulse_eventstorage_host	String	X	The host of the events storage DB	
database_pulse_eventstorage_schema	String	X	The Schema of the events storage DB	
events_interval_start	Long	X	The start date of the interval that is required to delete it records in **Unix time** format	it has to be lower than the events_interval_end
events_interval_end	Long	X	The end date of the interval that is required to delete it records in **Unix time** format	it has to be greater than the events_interval_start

Running the Playbookâ

```
python3.6 ansible-playbook \
    -i project-default \
    -i project-hsbc \
    -i gcp \
    -i project-<env> \
    events-storage-records-deletion.yml
```

Copy

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ETL process from Pulse RealTime (RT) to DataScience Environment (DSE)

The goal of DS Data Migration is to make available the events processed in RT to the DSE environment.

This interaction is represented in the following diagram:

Usageâ

To run the event storage ETL processor we provide two ansible playbooks, `etl.yml` and `etl-single-run.yml` .

See Feedzai Ansible to understand how to run these playbooks and the playbook configuration reference.

info

Note that for `etl-single-run` playbook the `tag` is still `etl.install` .

In truth, the playbook is simply an abstraction to empty the `cron_jobs` variable (`cron_jobs: []`).

As an alternative, the variable can be emptied "manually" on the customer defined variables, before running the `etl` playbook.

ETL tool should be installed in a dedicated GCE (Google Cloud Engine) instance using the `etl.yml` playbook.

This playbook sets up a cron job to recurrently migrate a relative extraction interval of past data.

More precisely, the default configuration sets up a daily extraction of the data from the previous day.

The GCE instance should have the following permissions:

- Read RT Event Storage (Google Cloud SQL)
- Write to DSE data bucket (Google Cloud Storage bucket)

These accesses can be controlled via both firewall rules and service account permissions. Details on those topics are outside of the scope of this document.

Extraction Intervalâ

Relative extraction intervalâ

As explained before, since most commonly ETL runs in a scheduled manner, relative intervals are necessary to actually migrate different data.

This can be achieved with the `-rei` or `--relativeextractioninterval` flags that computes the timestamps following JTimeWindow Expressions.

You can use the default configuration as reference on how/where to use it.

Fixed extraction intervalâ

It is also possible to run ETL for a specific time interval, which can be defined using the following ansible variable / structure:

```
etl_extraction_interval:
  start_event_timestamp: <unix-timestamp>
  end_event_timestamp: <unix-timestamp>
```

Copy

warning

When recovering ETL extractions, make sure to follow the start and end timestamp restrictions described in the ETL Recovery Procedure

warning

The previous variables will only be considered if `cron_jobs` is not defined or an empty list.

Logging and testingâ

The default `logger.xml` is provided in `conf/logger.xml`. By default the execution logs will be redirected to:

```
<installation_dir>/log/etl.log
```

Copy

In the previous log file you will be able to find periodic âreportsâabout the status of the execution of the data migration about the number of events extracted, and published.

When the extraction finishes a log with the migration result is displayed, similar to the following one:

```
2023-01-23 19:45:17,897 [main]   ExecutionMetricFactory.report:330           Report:
[extracted: 8640000/0 @ 4880tps, filtered: 0, published: 8640000/0 @ 4880tps, queued: 0, elapsed
time: 1770532155051 ns]
2023-01-23 19:45:17,898 [main]   EventStorageEtlProcessor.runProcess:251       Finished execution
of the ETL process. Time elapsed: 1770532 milliseconds
```

Copy

Exceptions on the data migration will also be written to the previous log file.

Validations:

- After the execution of the tool the events extracted should be equal to the number of the events in the event storage for the same period
- The number of events extracted and published should be the same
- The log file should not have any exceptions
- The file written to the configured bucket should have the number of events published reported (and consequently the same number of events in the event storage for the extraction period)

To check the number of events in event storage:

```
SELECT count(*) FROM eventstorage."7b336ee2c9d8ac105d742c45dbde75"  where âFDZ_timestampâ >=
<start_timestamp_ms> and âFDZ_timestampâ <  <end_timestamp_ms>;
```

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On this page

Recover events from Pulse to Case Manager

The goal of Case Manager event recovery is to sync Pulse Event Storage with the Case Manager database if they become out of sync.

To support this process we have created a new playbook `etl-pulse-to-cm-single-run.yml` that is meant to run on demand. This playbook can be executed either with or without a time range but our suggestion is to use a fixed interval to prevent overflowing RabbitMQ and Case Manager and also to avoid duplication of events.

Executionâ

The recovery process can be executed with the following command:

```
python3 ansible-playbook \
    -i project-default \
    -i project-hsbc \
    -i gcp \
    -i project-<env> \
    etl-pulse-to-cm-single-run.yml
```

Copy

info

The recovery process behavior depends on the chosen routing strategy. For the DIRECT strategy, recovery only supports one channel per execution. For the MULTICHANNEL strategy, the process can handle multiple event types in a single run, routing them to their respective queues as defined in the configuration.

Execute for a fixed extraction intervalâ

Our recommendation is to run recovery only for a small time interval to prevent RabbitMQ and Case Manager to overflow with messages. This can be achieved by running the process with the following Ansible variables:

```
etl_extraction_interval:
  start_event_timestamp: <unix-timestamp>
  end_event_timestamp: <unix-timestamp>
```

Copy

Routing strategiesâ

The ETL offers two routing strategies to send events to Case Manager, configured via the `etl_cm_route_strategy` variable.

- `DIRECT` (Default): This strategy sends all recovered events to a single RabbitMQ queue. This is the preferred method for recovering events of workflows that process a single channel (e.g. Inbound Payments or Cards).
- `MULTICHANNEL` : This strategy routes events to different RabbitMQ queues based on an event field. This is the preferred method to recover events from a single workflow that should be routed to different Case Manager channel (e.g. Outbound Payments or Digital Activity).

For a `DIRECT` strategy, simply define the strategy and the queue:

```
etl_cm_route_strategy: DIRECT
etl_cm_rabbitmq_event_queue: "<event queue>"
```

Copy

For the `MULTICHANNEL` routing strategy the channel mappings are required. For example, to recover Outbound events (Transfers and Digital-Activity):

```
etl_cm_route_strategy: MULTICHANNEL
etl_cm_route_fieldName: fdz_channel
etl_cm_channel_queue_mappings:
  DIGITAL_ACTIVITY: "<digital activity queue>"
  TRANSFERS: "<transfers queue>"
```

Copy

To recover Cards events (Cards and Non-Mon) the channel mappings would be:

```
etl_cm_channel_queue_mappings:
  CARDS: "<cards queue>"
  NON_MON: "<non mon queue>"
```

Copy

Configurationâ

Please find below the available configuration options.

Variable	type	Mandatory	Default Value	Description
etl_cm_route_strategy	STRING		DIRECT	The routing strategy to use. Can be DIRECT (all events to a single queue) or MULTICHANNEL (events routed based on channel).
etl_rabbitmq_cm_brokers	STRING	X		The broker address and port in the format `host:port` (e.g. `rabbitmq:5672`).
etl_rabbitmq_cm_username	STRING	X		The username to be used when authenticating in RabbitMQ.
etl_rabbitmq_cm_password	STRING	X		The password to be used when authenticating in RabbitMQ.
etl_rabbitmq_cm_vhost	STRING	X	/	The vhost value that Pulse should use when connecting to RabbitMQ.
etl_cm_rabbitmq_event_queue	STRING	X	EVENT_QUEUE	The name of the targeted topic(e.g. `EVENT_QUEUE_CARDS`).
etl_cm_route_fieldName	STRING	X (for the MULTICHANNEL strategy)	"fdz_channel"	The field from the event payload that will be used to identify the channel. (e.g. `fdz_channel`).
etl_cm_channel_queue_mappings	DICTIONARY	X (for the MULTICHANNEL strategy)	{ 'TRANSFERS': 'EVENT_QUEUE_TRANSFERS', 'DIGITAL_ACTIVITY': 'EVENT_QUEUE_DIGITAL-ACTIVITY' }	For MULTICHANNEL strategy, this maps the channel type to a specific queue. Possible values: { 'TRANSFERS': 'EVENT_QUEUE_TRANSFERS', 'DIGITAL_ACTIVITY': 'EVENT_QUEUE_DIGITAL-ACTIVITY', 'CARDS': 'EVENT_QUEUE_CARDS', 'NON_MON': 'EVENT_QUEUE_NON-MON' }

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/installation-guide/etl/etl-sas-efm-to-pulse-rt

On this page

ETL process from SAS-EFM to Pulse Run Time

ETL process from SAS-EFM to Pulse Run Timeâ

The goal of Historical Data Load is to make available to Pulse RT event information received in the past before the environment to receive the events in real time is available.

This data will later be used to create / update the profile store - through the execution of the Scheduled Metics Jobs.

ETL will load the Historical Data from the F17 Datasource, which would be provided through CSV files to be loaded to the Events storage database. The diagram below represents the flow to load the historical data load.

Usageâ

The Historical Data ETL will be configured with the source of the CSV file stored in the storage buckets and the target will be the Pulse events storage database.

The ETL tool for historical data loading is the same tool that we use for the DS Data Migration, however it uses a different configuration:

```
 eventstorageetl.numExtractionThreads=25
eventstorageetl.numPublishingThreads=1
eventstorageetl.maxEventsPerSecond=1000
eventstorageetl.pulse.appId=APP_ID
eventstorageetl.pulse.workflowId=WORKFLOW_ID
eventstorageetl.pulse.connectionMode=PAPP
eventstorageetl.pulse.papp=var/tfrb.papp
eventstorageetl.storage=DATABASE
eventstorageetl.database.engine=com.feedzai.commons.sql.abstraction.engine.impl.PostgreSqlEngine
eventstorageetl.database.jdbc=JDBC_CONNECTION
eventstorageetl.database.schema=SCHEMA
eventstorageetl.database.username=USERNAME
eventstorageetl.database.password=ENCRYPTED_PASSWORD
eventstorageetl.transformers=com.feedzai.commons.eventstorage.etl.transform.filter.groovy.UuidKe
yTransformere
eventstorageetl.uuidkey.groovy.fieldName=FDZ_UUID
eventstorageetl.uuidkey.groovy.expression=UUID.randomUUID().toString()
eventstorageetl.csv.serializerConfig.schemaDetails.uuidField=lifecycle_id
eventstorageetl.csv.serializerConfig.schemaDetails.timestampField=event_occurred_at
eventstorageetl.csv.serializerConfig.workflowDetails.inputAdapter=INPUT_SERVICE
eventstorageetl.csv.uri=CSV_PATH
```

Copy

Please note the above is kept for historical reasons, but as of Feedzai release 2.4.0, the ETL tool no longer supports PAPP as the connectionMode, nor does it specify a .papp file

Installation with Ansibleâ

Historical ETL is installed as any other tool installed with the ansible toolkit, with playbook `historical-data-loading.yml` .

Executionâ

The way the tool will be executed in Production is still under review and it will be detailed later. Probably it will be provided an ansible task to launch the tool based on a list of GCS buckets provided via configuration.

The goal of this section is to provide raw information about how the tool is executed and under the hood how it will be called automatically later.

After the installation with ansible, to run the ETL to load Historical Data follow the following steps:

1. go to " `/opt/feedzai/historical-etl/etl` ".
2. run that command

```
JAVA_OPTS="-Xms8g -Xmx8g ./bin/event-storage-etl -cf ./conf/historical.properties --secret ./conf/secret
```

Copy

Logging and testingâ

The default logger.xml is provided in `conf/logger.xml` . By default the execution logs will be redirected to:

```
<installation_dir>/log/etl.log
```

Copy

In the previous log file you will be able to find periodic âreportsâabout the status of the execution of the data migration about the number of events extracted, and published.

When the extraction finishes a log with the migration result is displayed, similar to the following one:

```

2023-01-23 19:45:17,897 [main]   ExecutionMetricFactory.report:330            Report: [extracted: 8640000/0 @ 4880tps, filtered: 0, published: 8640000/0 @ 4880tps, queued: 0, elapsed time: 1770532155051 ns]
2023-01-23 19:45:17,898 [main]   EventStorageEtlProcessor.runProcess:251       Finished execution of the ETL process. Time elapsed: 1770532 milliseconds
```

Copy

Exceptions on the historical data load will also be written to the previous log file.

Validations:

- After the execution of the tool the events load to the event storage should be equal to the number of the events in bucket that is being loaded
    - Note the event storage may already have other events, so please check the number before and after the execution to know the number of events loaded
- The number of events extracted and published should be the same (tool logs)
- The log file should not have any exceptions

To check the number of events in event storage:

```
SELECT count(*) FROM <schema>."7b336ee2c9d8ac105d742c45dbde75"  where âFDZ_timestampâ >= <start_timestamp_ms> and âFDZ_timestampâ <  <end_timestamp_ms>;
```

Copy

Recommendationsâ

Configurationâ

To achieve the max speed for extraction and publishing we recommend the following configuration:

```
eventstorageetl.numPublishingThreads=25
eventstorageetl.maxEventsPerSecond=5000
eventstorageetl.databaseBatchSize=50
```

Copy

This configuration will ensure the maximum insertion rate while maintaining the stability of the system (processing transactions in real time).

From our tests, with that configuration the ETL should publish the records by rate of "4800 tps", and we run our tests on Compute Engine Instance of type n2-highmem-16, to take advantage of its CPUs capacity.

Historical data ingestionâ

Taking into account the large dataset expected, in order to reduce as much as possible the impact of possible failures (e.g.: invalid rows in the dataset, or even possible connection issues...) we recommend **against** running ETL in a single execution over the whole data.

Instead, it is better to partition the data and perform multiple executions of ETL with smaller datasets for specific known ranges of time (eg. 1 file per day).

This way, any failure will only affect the known partition and fixing it will be both easier and faster - thereâs no need to rollback and re-execute the whole dataset.

What to do in Case of Failure during middle of execution?â

Execution scenario:

Performing the historical data load for the known range **"9th of September 2022 7 AM till 9th of September 2022 11 AM"**.

Failure scenarios:

1. Connection issues
2. Due to an/many invalid row(s) in the CSV file.

In case 2., the invalid row(s) in the Dataset will need to be fixed.

In both cases we need to retry the same **ETL** execution. Before retrying we need to **delete all** records for that time range, otherwise it could end up with duplicates for rows already inserted previously.

How to delete these records?â

We have provided a python Script that we can run by ansible and is parameterized by the time range, so for example if we need to delete all records from **"9th of September 2022 7 AM till 9th of September 2022 11 AM"**.

We can run the ansible role for the time range of the equivalent unix timestamps to be from **1662703200000 to 1662717600000**.

Example for the ansible properties configuration would be like that:

```
events_interval_start: 1662703200000
events_interval_end: 1662717600000
```

Copy

Event Storage ETL

Welcome to the Event Storage ETL Guide.

In this section you will find a set of pages with information about the Event Storage ETL process:

- ETL process from Pulse RealTime (RT) to DataScience Environment (DSE)
- ETL configuration guide
- ETL extractions schema

On this page

ETL Extractions Schema

The ETL extractions contain the schema fields obtained directly from the Pulse Workflow of the live Pulse cluster and may include a set of additional fields depending on the configuration.

Below you can find a CSV example of an extraction, taking into account that it may not represent the exact structure since the fields are obtained from the current Pulse workflow:

```
lifecycle_id,bulk_id,payment_type,payment_sub_type,product_code,payment_status,created_at,cutoff_
at,sender_initiated_at,requested_execution_at,number_of_payments,recurring_payment,sender_transa
ction_amount,sender_transaction_currency,sender_transaction_bank_account_number,sender_transacti
on_notes,sender_bank_name,sender_bank_postal_code,sender_bank_address,sender_bank_city,sender_ba
nk_country_code,receiver_bank_bic,receiver_bank_name,receiver_bank_postal_code,receiver_bank_add
ress,receiver_bank_city,receiver_bank_country_code,receiver_transaction_amount,receiver_transact
ion_currency,receiver_transaction_bank_account_number,receiver_city,receiver_country_code,device_
type,device_longitude,device_latitude,event_type,event_external_id,info_type,dispute_start_date,
dispute_reason,dispute_notes,dispute_submitted_by,dispute_end_date,dispute_feedback_notes,disput
e_refunded,oob_start_date,oob_comm_type,oob_notes,oob_end_date,oob_feedback_notes,oob_response_t
ype,feedback_type,feedback_label,sender_name,sender_email,sender_phone,sender_zip,sender_dob,sen
der_occupation_status,sender_account_bic,sender_account_type,sender_account_currency,sender_acco
unt_balance,receiver_id,receiver_name,receiver_email,receiver_phone,receiver_street_addr,receive
r_zip,receiver_region,receiver_dob,receiver_account_type,receiver_account_currency,direction,dev
ice_id,channel_type,sender_bank_sort_code,sender_bank_branch_id,receiver_bank_sort_code,receiver_
bank_routing_number,event_occurred_at,segment_channel_type,auth_type,device_ip_country_code,sess
ion_id,device_language,sender_alias_type,text_for_receiver,device_user_agent,device_fingerprint,
instant_transfers_review_outcome,device_browser,device_ip_region,feedback_value,sender_account_z
ip,sender_bank_type,condor_rule_code,receiver_cop_name,receiver_cop_match_name,office_id,recurri
ng_payment_schdl_info,sender_account_ownership,payment_sca_proc_code,customer_market_segment,sen
der_transaction_type,sender_transaction_auth_type,payment_message_type,payment_revision_code,rec
eiver_account_bank_id,receiver_bank_type,tpp_name_alt,tpp_payee_payer,number_of_payments_count,s
ender_phone_chg_date,customer_home_branch,recurring_payment_ref,payment_remittance_info,sender_t
ransaction_ref,phone_session_id,user_guid,customer_onboarding_method,sender_transaction_cat_code,
sender_orig,payment_related_ref,sender_transaction_narrative,sender_transaction_cop_name,sender_
transaction_cop_matname,sender_bank_status,sender_account_number_external,receiver_email_chg_date
,sender_email_chg_date,receiver_bank_status,receiver_merchant_category,receiver_details,tpp_name,
payment_saf,customer_total_balance,customer_reported_income,customer_citizen_id_number,customer_
preferred_language,customer_work_phone1_chg_date,customer_work_phone2_chg_date,customer_home_pho
ne1_chg_date,customer_home_phone2_chg_date,customer_mob_phone1_chg_date,customer_mob_phone2_chg_
date,customer_account_type,lifecycle_id_alt,customer_number_of_accounts,receiver_bank_addr_city,
sender_transaction_original_amount,sender_transaction_original_currency,receiver_transaction_ori
ginal_amount,receiver_transaction_original_currency,receiver_transaction_product,sender_transact
ion_reason,sender_transaction_product,sender_transaction_amount_usd,payment_message_source,sende
r_transaction_cop_result,browser_timezone,tpp_id,device_datetime,device_platform_type,customer_p
ortfolio_region,customer_portfolio_product,customer_portfolio_channel,customer_portfolio_country,
customer_portfolio_type,device_tdl,customer_change_new,customer_change_old,consent_id,device_cpu_
class,device_fingerprint_type,device_banking_type,device_ip_address_v6,website_name,channel_user_
id,pinpoint_risk_score,source,sender_transaction_exp_date,sender_transaction_code,auth_cam_level,
customer_number,sender_transaction_sign_status,device_platform_type_reason,account_type,auth_met
hod,request_type,customer_portfolio,customer_id_global,customer_id_external,customer_address_chg_
date,customer_phone_chg_date,customer_email_chg_date,sender_transaction_sole_ind,customer_accoun
t_number,customer_account_currency,customer_account_zip,customer_account_country,customer_accoun
t_nr_external,user_id_type,channel_daily_limit_currency,channel_name,staff_id,workstation_id,dev
ice_token_serial_no,device_mac_address,payment_entity,sender_transaction_details_ind,sender_tran
saction_category,sender_transaction_ref_nr,sender_transaction_ref_nr_alt,sender_transaction_stat,
sender_transaction_stat_reason,sender_transaction_effect_date,customer_change_reason_alt,custome
r_change_delivery,user_indicator,user_string,user_amount,customer_card_number,entity_id,entity_t
```

```
ype,idv_session_id,event_code,device_geo_ip_city,device_geo_ip_country_code,device_geo_ip_isp,de
vice_geo_longitude,device_geo_latitude,website_refferer_url,device_geo_timezone,device_gps_enabl
ed,device_data_enabled,device_wifi_bssid_connected,device_wifi_in_range,sanction_check,device_bl
uetooth_enabled,device_bluetooth_devices_conn,device_bluetooth_detected_list,device_imsi,device_
mcc,device_untrusted_keyboard,device_untrusted_screen_reader,device_hooking_frameworkds,devide_n
ative_codehooks,device_application_repackaged,device_debugger_attached,device_javascript_malform
ed,device_print,device_raw_jsc,device_is_desktop,device_is_gameconsole,device_rooted,device_risk_
score,device_timezone,device_ip_city,device_ip_latitude,device_ip_longitude,event_status,custome
r_country_code,customer_zip_code,customer_segment,customer_name,customer_dob,customer_address,cu
stomer_phone_number,customer_email,customer_branch_id,customer_id,account_id,device_emulator,dev
ice_java_script_enabled,fdz_channel,sender_transaction_held,payment_restricted_template,sender_t
ransaction_new_bene,payment_type_direct_debit,payment_type_self_acc,device_wifi_status,sender_ur
gency,device_timezone_offset,sender_initiated_at_local,sender_account_avg_balance,channel_daily_
limit,event_server_validated_at,event_updated_at,user_id,receiver_account_bank_name,payment_ref,
device_os,customer_change_data_new,payment_type_sub_type,device_es_id,txn_type,device_dev_id,ext
ernal_status_reason,external_preliminary_decision,device_isp,is_scoring,billing_currency_code,ch
ange_event_delivery,customer_change_data_old,device_ip_address,overnight_acc_ledger_bal,receiver_
transaction_amount_dbl,sender_account_balance_dbl,sender_account_open_date,sender_transaction_am
ount_dbl,var_recurring_payment,vocalink,receiver_creation_date,sender_registration_date,event_re
ceived_at,event_created_at,customer_fiv_marker,customer_fiv_marker_last_updated_at,customer_gend
er_code,customer_earnings_amt_snapshot_dt,current_account_monthly_debit_amount,phone_bank_use_co
de,customer_current_account_balance,customer_salary_amount,customer_forname,customer_surname,cus
tomer_home_street_addr,customer_home_city,customer_home_zip,customer_home_zip_area_code,customer_
home_region,customer_home_country_code,customer_home_address_last_update_at,customer_corresp_str
eet_addr,customer_corresp_city,customer_corresp_zip,customer_corresp_zip_area_code,customer_corr
esp_region,customer_corresp_country_code,customer_corresp_address_last_update_at,customer_email_
last_update_at,customer_home_phone,customer_mobile_phone,customer_daytime_work_phone,customer_bu
siness_phone,customer_hmphone_last_update_at,customer_mbphone_last_update_at,customer_wrkphone_l
ast_update_at,customer_businessphone_last_update_at,customer_registration_date,customer_citizens
hip,customer_portfolio_class,is_historic,saf_ind,customer_fiv_markers,customer_pb_marker,custome
r_portfolio_segment,customer_voice_id,customer_voice_id_last_change_date,customer_annual_turnover
,customer_business_type,customer_chinese_comm_code,customer_rm_info,customer_marital_status,inva
lid_email_indicator,sender_account_channel_limit_double,sub_channel_name,customer_id_number,gift_
indicator_code,customer_status,customer_rel_status,customer_type,tmx,customer_id_hashed,bioc,mod
el_risk_score,bank_processing_application,bank_processing_type,reversal_indicator,check_book_req
uest_date,cash_form,check_number,thailand_fps_ind,gpb_manual_payment_ind,scameter_warning,sender_
billing_amount,new_customer_flag,customer_id_type,new_receiver_flag,receiver_creation_date_custo
mer,customer_occupation_start_date,count_months_beneficiary_received_funds,avg_total_daily_trans
fer_out,avg_monthly_transfer_in,browser_os_version,costumer_id_global,customer_portfolio_phys_ty
pe,customer_portfolio_remote_type,ds_transaction_id,payment_message_source_d,sender_transaction_
origin_amount,sender_transaction_origin_currency,sender_transaction_reason_d,sender_transaction_
sign_stts_d,customer_change_reason,sender_account_channel_limit,delegate_id,delegate_initiated,a
pprover_id,transaction_id,delegate_of,delegates,sender_transaction_extra_1,sender_transaction_ex
tra_2,sender_transaction_extra_3,customer_additional_info_1,customer_additional_info_2,customer_
additional_info_3,customer_additional_info_4,account_holder_customer_id,payment_above_limit,FDZ_
UUID,action_alert,Customer average payment amount last 
day,rules_triggered,formatted_results,outcomes_and_scores
f415b143-1a03-4bb7-a47d-230c478c7931,b398a9e1-3900-4753-
ad45-8310ea583724,sepa,credit_transfer,,entered,
1554198919000,1554198919000,1554196919000,1554996919000,1,true,155000.0,GPB,
7b37bd8c-0936-46aa-83cc-196255497783,Notes on the transaction.,Gringotts,WC2N 5DU,Diagon Alley 
33,London,UK,4ad3fe88-317b-4b31-ad9b-9136424c17ad,Receiver Bank,WC7N 7EW,Some Avenue 
77,Manchester,UK,155000.0,826,9a08f316-df1f-4096-9a2d-a904a82bb65e,receiver_city,PT,smartphone,
43.98,-19.56,transfer_initiation,,out_of_band,1554196919000,test,Very fraudulent.,test,
1554196919000,,,1554196919000,test,Contacted John Doe to confirm.,1535477895438,The client called 
to confirm it was not him making that transaction.,call,out_of_band,,Sender Joe Doe,[email prote
cted],001122334455,sender_zip,19900101,looking for a new project,PTCDG,CREDIT,USD,
1110000.0,receiver_id,Receiver Joe Doe,[email protected],
001122334455,receiver_street_addr,receiver_zip,receiver_region,
19900101,CREDIT,USD,outgoing,a8d92e56-a6c8-45ea-8db4-427a9337e68e,mobile,,,,,
1743690403724,,pin,GB,02fb50a1-56ce-42d9-97db-a5209094cf3a,he-RW,,,Mozilla/5.0 (Windows NT 6.1; 
Trident/7.0; rv:11.0) like Gecko,,false,Edge,IX,false,,,,,,,,,,,,,,,,,
1742989536163,0001-01-01,,,,,,,,,,,,,,,0001-01-01,0001-01-01,,,,,,,,,,,,,,,,,916,,395.75,978,,,
,,,155000.0,,,,,1742989536163,,,,,feedzai,,,,,,,,,,,,,
```

```
0001-01-01,,,,,,,,,,,,0001-01-01,0001-01-01,0001-01-01,,,,,,,,,,,,,,,,,,,,,0001-01-01,,"[""Bria
n78""]","[""Elenor.Olson""]",
[828.0],,,,,,,,,-90.1443,-16.7286,,,true,false,,false,false,true,,,,,true,true,false,true,true,t
rue,true,,,true,false,false,21,-01:00,Jonesborough,-40.5059,114.5007,fail,JE,
13615-680,commercial_banking,Dr. Jenna Satterfield,0001-01-01,7526 Layla Parkways,208-474-1633,[
email protected],,cutomer_1,d41b6343-c375-463e-
a204-8cc266b0c1ed,false,true,TRANSFERS,false,true,false,true,false,true,,,,
130.0,312.0,1742989536163,,,,,Android,,,,,,,,,,
826,55551222,,165.44.6.29,100.5,400.75,1200.75,1668683369590,400.75,0,"{""vocalink_score"":
20.0,""mastercard_id"":""""}",
1736206933068,1354398919000,1743690403890,1742989536163,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,
,,,,,,,true,13800.0,BS,
0HKHSXXXXXX796256,964,Insane,,,"{""fuzzy_device_id"":"""",""ip_city"":"""",""ip_country"":"""",""
ip_longitude"":152.2752,""ip_latitude"":-84.8857,""chall_policy_score"":
716.0,""ip_address"":"""",""digital_id"":"""",""device_isp"":"""",""summary_reason_code"":"""",""
policy_score"":21.0,""risk_score"":21.0,""device_dev_id"":"""",""request_id"":""""}",
2d18c716ba930802d187654c99fd097f5530f1479065428caa1a3ee1q7421,"{""is_bot"":true,""is_emulator"":
false,""voice_scam_score"":
486,""is_malware"":false,""is_rat"":true,""is_recent_rat"":true,""custom_risk"":
21.0,""risk_score"":
21.0,""custom_risk_aliases"":""risk_alias"",""push_score"":null,""is_mobile_rat"":null}",
0.89,,,,1716369476000,,,,,,1.2345233222E8,D,V,,,,,,,,,,,,,10,,,,,,,false,,,,,Sender Transaction
Additional Info 1,Sender Transaction Additional Info 2,Sender Transaction Additional Info
3,Customer Additional Info 1,Customer Additional Info 2,Customer Additional Info 3,Customer
Additional Info 4,account_holder_customer_id,false,dac5d527-15b1-4ba0-89e4-6c35eb9dfb8b,true,,,
[],"{""scam"":{""outcomeDecision"":false,""score"":0.0},""fraud"":
{""outcomeDecision"":false,""score"":0.0},""decision"":
{""outcomeDecision"":""approve"",""score"":[0.0,0.0,0.0]},""risk_level"":
{""outcomeDecision"":""low"",""score"":[0.0,0.0,0.0]},""shadow_scam"":
{""outcomeDecision"":false,""score"":0.0},""sca_required"":{""outcomeDecision"":true,""score"":
0.0},""shadow_fraud"":{""outcomeDecision"":false,""score"":0.0},""treatment_strategy"":
{""outcomeDecision"":""none"",""score"":[0.0,0.0,0.0,0.0]},""self_serve_strategy"":
{""outcomeDecision"":""none"",""score"":[0.0,0.0,0.0,0.0,0.0]}}"
```


Copy

ETL Data Extraction and Loading Processâ

The ETL tool extracts data from either the pulse event storage or a CSV file.

During the extraction process, the tool can include various configurations, such as whether to include scheduled metrics, internal state, or triggered actions. Scheduled metrics are specifically included as part of the internal state.

After transformation, the data is loaded into a target format, typically a CSV file in a Google Cloud Storage bucket. This allows the analytics team to access the data for further analysis and model training.

The ETL tool is configured to run periodically (e.g., every hour) to extract data from the previous hour, ensuring that the analytics team has access to relatively up-to-date information.

Additionally, the ETL tool can be utilized to load historical data into the event storage, which is essential for initializing scheduled metrics when launching new markets or features.

Extra fieldsâ

The following fields are added by enabling some properties in the tool configuration file.

Internal State (scheduled metrics)â

The internal state field contains, along with other information, the schedule metrics that enriched each event when it was processed by Pulse.

It can be enabled by setting the Ansible variable `etl_include_internal_state` to `true`.

This field is a JSON object contains various information including the metrics that were fetched for the event, identified by the `_METRICS` key as shown in the example below:


```
{
  "states": {
    "34a95d87-6934-4a99-a0a0-0ddb7e485fc2": {
      "data": {
        "NORMAL_MSG_CTRL_FLOW": [
          "com.feedzai.pulse.api.apps.enums.RuleControlFlow",
          0
        ]
      }
    },
    "5342a480-e9d9-4ec2-b8b2-b80bd868537b": {
      "data": {
        "_EXEC_ID": "mzJuEfUgOaLhbu60d2JM2Q",
        "_METRICS": {
          "@class": "com.feedzai.pulse.datascience.rte.runtime.MetricsWrapper",
          "values": [155000.0]
        }
      }
    }
  },
  "globalState": { "data": { "ALL_SCHEDULED_METRICS_FETCHES_HANDLED": true } }
}
```


Copy

For the above example let's assume the metric was created by `Metrics Plan A` with the name `Customer average payment amount last day` and its value for a given customer is `155000.0`.

During extraction ETL will look for each of the `_METRICS` key and create a new entry in the CSV file with the name and value of the metric. In this case ETL will look for the Plan snapshot given by `_EXEC_ID`, fetch the name of the metric and value from the `values` array.

The end result is a new column in the CSV file with the name `Customer average payment amount last day` and the value `155000.0`:

```
lifecycle_id,...,Customer average payment amount last day,...,
f415b143-1a03-4bb7-a47d-230c478c7931,...,155000.0,...
```

Copy

If `Plan A` had more than one metric, let's say for example `Customer max number of payments in the last 30 days` with a value of `20`, the CSV would look like the following (not necessarily in this order):

```
lifecycle_id,...,Customer average payment amount last day,Customer max number of payments in the last 30 days,...,
f415b143-1a03-4bb7-a47d-230c478c7931,...,155000.0,20,...
```

Copy

Unique identifier (`FDZ_UUID`)â

This field is a unique identifier for every event extracted.

The name of the field is equal to the attribute in the configuration (e.g.: `FDZ_UUID`).

It is enabled with the following property configuration: `eventstorageetl.csv.serializerConfig.schemaDetails.uuidField=FDZ_UUID`

```
524f8eff-2bf7-4eea-ad0f-f40f5710bf8c
```

Copy

Actions triggered (`action_X`)â

When this option is enabled the configured actions on the workflow are extracted as multiple fields, there's a field for every action.

Every field is a concatenation of `action_` with the name of the configured action, e.g.: `action_alert`. Then, for every field, it is said if the action was triggered or not (true/false).

It can be enabled by setting `eventstorageetl.extraction.includeTriggeredActions` or the Ansible variable `etl_include_triggered_actions` to `true`.

Example:

```
lifecycle_id,...,action_alert,...,
f415b143-1a03-4bb7-a47d-230c478c7931,...,true,...
```

Copy

Rules triggered (`rules_triggered`)â

This field includes a semicolon-separated list of rule ids that were triggered by the extracted event.

It can be enabled by setting `eventstorageetl.extraction.includeResultTracking` or the Ansible variable `etl_include_result_tracking` to `true`.

Example:

```
lifecycle_id,...,rules_triggered,...,
f415b143-1a03-4bb7-a47d-230c478c7931,...,lB3aZuzDthoLQmFrIHxMtw;g7GdIzY6kF5txP1Qu4JMaQ,...
```

Copy

Rules explanations (`formatted_results`)â

This field is a list of triggered rules with an explanation of why it was triggered.

It can be enabled by setting `eventstorageetl.extraction.includeFormattedResultTracking` or the Ansible variable `etl_include_formatted_result_tracking` to `true`.

Every **triggered rule** has two fields:

- **ruleId**: the triggered rule id.
- **explanation**: the explanation of why the rule was triggered.

```
[
    {
        "explanation": "Customer name is Fraudulent Joe",
        "ruleId": "k5guIAH53oJMrBh2Jl1IfQ"
    }
]
```

Copy

Example:

```
lifecycle_id,...,formatted_results,...,
f415b143-1a03-4bb7-a47d-230c478c7931,...,{"explanation":"Customer name is Fraudulent Joe","ruleId":"k5guIAH53oJMrBh2Jl1IfQ"},...
```

Copy

Outcomes (`outcomes_and_scores`)â

This field is a JSON object that contains all the configured **outcomes** (e.g.: `fraud`, `decision`).

It can be enabled by setting `eventstorageetl.extraction.outcomesInSingleField` or the Ansible variable `etl_outcomes_in_single_field` to `true` and also the property `eventstorageetl.extraction.includeOutcomes` (or the Ansible variable `etl_include_outcomes_extraction`).

Every **outcome** has two fields:

- **outcomeDecision**: the value chosen for the outcome.
- **score**: the score of the outcome, if the outcome is a multi-classifier (e.g.: `decision`), it will be a list of scores for each of the possible values.

```
{
    "fraud": {
        "outcomeDecision": true,
        "score": 0.0
    },
    "decision": {
        "outcomeDecision": "approve",
        "score": [
            0.0, # approve
            0.0, # decline
            0.0  # review
        ]
    }
}
```

Copy

Example:

```
 lifecycle_id,...,outcomes_and_scores,...,
f415b143-1a03-4bb7-a47d-230c478c7931,...,{"fraud":{"outcomeDecision":true,"score":
0.0},"decision":{"outcomeDecision":"approve","score":[0.0,0.0,0.0]}},...
```

Copy

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Historical Data Loading with ETL

Historical ETL is installed as any other tool installed with the ansible toolkit, using the playbook `historical-data-loading.yml`.

For more information, see ETL Configuration Reference.

Recommended Setupâ

In our tests, we achieved better performance - nearly 5000/tps - with the pre-bundled configurations when running on GCE, with an instance type `e2âhighcpuâ16`.

Rollback inserted transactionsâ

There may be some scenarios where ETL could fail during execution, due to for example having invalid rows in **CSV** files.

In case of failures where rows were already inserted, to re run the ETL process do as following:

1. Fix the root issue that caused the failure
2. Delete the records inserted by the previous execution
3. Run the **Historical ETL** again

Configuration Recommendations

On this page

Network and Connectivity

info

This page is based on the architecture of Feedzai standard solutions. Specific clients or custom solutions may have additional requirements, so this is not a definitive guide.

Firewall Rulesâ

Inboundâ

note

Inbound â other services connect to this service.

Service	Port	Description
HTTP	TCP 8080	Case Manager UI HTTP
Keycloak	TCP 7600	Intra-cluster communication
SSH	TCP 22	SSH access (if available)

Outboundâ

note

Outbound â this service connects to external services.

Service	Port	Description
RBDMS (PostgreSQL)	TCP 5432	
Keycloak	UDP 1812	Keycloak plugin for Radius server (OIDC)
Pulse	8080	Communication with Pulse via HTTP
Elasticsearch	9300	Communication with Elasticsearch via HTTP
Case Manager	8080	Communication with Case Manager via HTTP

Requirements

tip

For a complete list of Keycloak requirements, please refer to the official product documentation.

On this page

Logstash Pulse Workflow Audit Pipeline Recovery Tool

Playbook to run the *pulse-workflow-audit* Logstash pipeline in recovery mode.

This playbook can be executed for a specific interval when there is any problem with the pipeline and the data is not exported as it should.

By running this playbook, the *pulse-workflow-audit* pipeline will execute once for the defined interval and export the results to the configured GCP bucket.

This pipeline also supports a cron entry that restarts the *pulse-workflow-audit* pipeline on a schedule, and exposes variables to control whether to install base dependencies needed to have the cron daemon `crond` installed for the scheduling, as well as a variable to control the location of the temporary trimmed pipeline config that it needs as part of its restart operation.

Configurationâ

All the following variables are **mandatory**.

Variable	Type	Default Value	Description
logstash_pipeline_recovery_mode	boolean	true	Defines if we are running the pipeline in recovery mode. Needs to be **true**
logstash_pipeline_recovery_name	string	pulse-workflow-audit	Name of the pipeline that we are running in recovery mode
logstash_audit_pipeline_since	number		The start date of the interval to run the pipeline in **Unix time** format
logstash_audit_pipeline_until	number		The end date of the interval to run the pipeline in **Unix time** format
pulse.host	string		Pulse host name - in the format `http(s)://<host>:<port>`
pulse.user	string		Pulse user name
pulse.password	string		Pulse password
gcs_bucket_name	string		Bucket to where the pipeline will export the data
gcs_tmp_directory	string	/tmp/logstash/	Directory where temporary files are stored
max_file_size_kbytes	number	1024	Sets max file size in kbytes. 0 disable max file check
uploader_interval_secs	number	60	Uploader interval when uploading new files to GCS.
restart_pipeline_base_dependencies	list	[]	Dependencies to ensure the cron daemon is started for pipeline restarts
start_crond_service	boolean	false	Controls whether to install dependencies (restart_pipeline_base_dependencies)
temp_trimmed_pipeline_config_name	string	/tmp/trimmed_pipeline.conf	The location of the temporary trimmed pipeline needed by the pipeline restarter

Running the Playbookâ

```
python3.6 ansible-playbook \
    -i project-default \
```

```
    -i project-hsbc \
    -i gcp \
    -i project-<env> \
    execute-logstash-audit-pipeline.yml
```

Copy

On this page

Azul Platform Prime

Azul Platform Prime is a highly optimized JVM and elastic runtime that breaks traditional Java scale barriers and provides applications with improvement in Java responsiveness, scale, and throughput.

Why Azul Platform Prime?â

Pulse (real-time) is a memory intensive application due the way the real-time metrics are stored / calculated. Metrics are kept in memory and for each transaction the metrics for the entities involved (eg. Customer) are updated (according to the windows definitions transactions need to be expired, windows updated and metrics calculated) - and this can be a memory and GC intensive task depending on the number of transactions, metrics, size of the windows, etc).

Azul prime optimizes these kinds of workloads where low-latency / high-throughput with high memory usage is needed mainly due to its pauseless garbage collector.

info

Feedzai recommends the usage of Azul Prime (in Pulse Real Time machines) for the most demanding use cases (in terms of volumes). In those cases Azul Prime is required to accomplish the latencies and the throughput required.

Azul Platform Prime Installationâ

info

Azul Platform Prime is only required for Pulse RT nodes.

Azul Platform Prime can be downloaded through the Azul website

Pulse supports the last stable build compatible with the JDK8 (at the time of this document was written 22.08.301.0).

Alternatively Feedzai is also providing the packages in nexus:

- zing rpm zip tar.gz
- zing-zst rpm zip tar.gz

note

As the links to download the previous binaries from the Azul website are behind an account required link, Feedzai is providing the installation packages in nexus - since it was previously approved and configured as the standard way to provide Feedzai software to HSBC.

Due security guardrails, HSBC was having problems downloading the `RPM` packages, it was requested if it is possible to provide the packages in `zip` and `tar.gz` formats. This is the reason why there are multiple links for the Azul installation files.

To install the software please follow the steps on this page.

Validate the installationâ

Execute `java -version` or `/opt/zing/zing-<jdk-version>/bin/java -version` (if you don't have the java command on path):

The output should be similar to the one below:

```
java version "1.8.0-zing_19.07.0.0"
Zing Runtime Environment for Java Applications (build 1.8.0-zing_19.07.0.0-b3)
Zing 64-Bit Tiered VM (build 1.8.0-zing_19.07.0.0-b4-product-linux-X86_64, mixed mode)
```

Copy

License installationâ

The license key is not necessary to run Azul Platform Prime. However, to comply with your internal documents or processes regarding contract compliance, it may be required to install an Azul license key.

To install the licence key on a machine, copy the license file to your target hostâs `/etc/zing/` directory. To use any other file location, you must add `-XX:ZingLicenseFile=<filename>` to your Azul Zulu Prime JVM instance launch command. Licenses allow Azul Zulu Prime JVM instance launches up to the end of the day they are due to expire within the local timezone specified in the license file.

To force Azul Prime (version 20.10 and newer) to verify the license, add the following arguments: `-XX:+UseCommercialFeatures -XX:+ValidateLicenseKey`.

Some applications depend on the *JAVA_HOME* environment variable being set. So it is recommended to define the *JAVA_HOME* environment variable to point to the Azul Platform Prime installation directory, e.g. `$ export JAVA_HOME=/opt/zing/zing-<jdk-version>`.

Configurationsâ

The Azul Zulu Prime JVM recognizes the standard and non-standard command line options for the HotSpot VM that is a core component of Java Platform, Standard Edition (Java SE). If the Azul Zulu Prime JVM detects a HotSpot VM native command line option that it does not support, its response can be set with the following option:

```
-Xnativevmflags:[ignore|error|warn]
[source, shell,opts="user,check"]
```

Copy

The default option is ignore.

This is important because it allows us to test the Azul Prime installation without changing the already defined JVM parameters.

tip

One initial test to check if it is everything ok can be simply change the JVM and run everything else as usual.

Azul Platform Prime Memory allocationâ

Azul Platform Prime uses the Azul Zulu Prime JVM to run Java applications, similar to a typical JVM running a Java application. The significant difference is how memory is handled. Azul Zulu Prime JVM uses the Azul Zulu Prime System Tools (ZST) for managing the memory of three components: Contingency memory, Pause Prevention memory, and Zing Memory, the memory used for the Azul Zulu Prime JVMâs Garbage Collector.

Using Azul Zulu Prime JVM with the Garbage Collector and Contingency memory means that your Java application is no longer limited to the maximum heap size specified on the Java command line when you started your Java application. As an additional resource, Azul Zulu Prime JVMâs Garbage Collector might temporarily use Pause Prevention memory to prevent pauses that are caused by lack of available clean pages.

Azul Platform Prime provides two different methods to configure/instruct ZVM to use the system memory:

- **Reserve-at-config**: memory for use by the JVMs running on the machine can be reserved at configuration
- **Reserve-at-launch**: the memory reservation for each JVM can be deferred until the launch of that JVM process

For more details about ZVM memory allocation please refer to [Azul zing documentation](#).

Feedzai recommends reserved-at-config. On a typical production installation, around 80 % of the total system memory is reserved for zing, leaving 20 % for the OS/Kernel and/or off-heap allocation. Out of 80%, 75 % is set for Reservable Memory where all the applications JVMs will run, and the remaining ~4% are divided equally between Pause Prevention Memory and Contingency Memory.

For more information about how to set up the memory please check this guide.

Monitoringâ

Pulse already reports information about the JVM status by showing for instance the heap usage, garbage collector periodicity and execution time, and number of threads via the exposed monitoring endpoints.

The same metrics can be used to monitor the system running on top of Azul Platform Prime.

For more information please check the Pulse section in the monitoring guide.

Additionally Azul also provides an official runtime monitoring and diagnostics tool, for more information please check this page.

On this page

Blue-Green deployments

The recommended approach for zero downtime deployments is a technique known as blue-green deployment.

Blue-green deployment uses two almost identical clusters in the following way:

- At the start, one of the clusters is in use. Let's call it the blue environment.
- The other one is on standby, created from a new release. Let's call this one the green environment.
- When the green cluster is deployed, traffic is redirect to this cluster and deployment becomes the stand-by environment. It is available in case of a necessary rollback. We recommended this step to be manual so that every validation is done prior to changing traffic to the new cluster.

warning

Prior to a Blue-Green deployment RabbitMQ should have colored vhosts set up (`pulseblue` and `pulsegreen`)

Using a control-node as Blue-Green deployment orchestratorâ

When SSH connection is available between the control-node and service machines (Pulse, RabbitMQ, etc) it can orchestrate the entire deployment, excluding only validation and switching load balanceres tasks.

The first step is to federate the necessary queues, which can be done as follows:

```
python3.6 ansible-playbook \
    -i project-default \
    -i project-hsbc \
    -i gcp \
    -i project-<env> \
    rabbitmq-setup-blue-green-federation.yml
```

Copy

Note: the above playbook should only run on a single RabbitMQ node.

Pulse machines are installed using the current production opposite color and by copying its metadata. This can be achieve by using Ansible variables `deployment_color=<opposite_color>` and `keep_metadata_between_deployments=true` :

```
python3.6 ansible-playbook \
    -i project-default \
    -i project-hsbc \
    -i gcp \
    -i <region>/project-<env> \
    -i <region>/project-<env>/<portfolio> \
    pulse.yml
    -e deployment_color=<opposite_color>
    -e keep_metadata_between_deployments=true
```

Copy

After Pulse installation ends and the new cluster is validated, load balancers can be changed to point to the new cluster.

Federate queues can also be disabled by running the following playbook:

```
python3.6 ansible-playbook \
    -i project-default \
```

```
    -i project-hsbc \
    -i gcp \
    -i project-<env> \
    rabbitmq-disable-blue-green-federation.yml
```

Copy

Note: the above playbook should only run on a single RabbitMQ node.

Temporary solution for joint sessionâ

warning

This approach is currently under discussion and may not represent the final solution

When SSH connection between the orchestrator (control-node) and Pulse images is not available, the deployment can be achieved by offloading the steps that are prone to race conditions to a different machine(s).

The pipeline is defined as follows:

1. Deployment color is set to the current production environment opposite color
2. Pulse images are created with the latest Feedzai release
3. RabbitMQ queues replication (federation) is set up to allow new cluster to receive updates from the production cluster. Refer to [RabbitMQ federation](#) for details
4. A machine is created to clean up the existing metadata for colored cluster and to copy the metadata from the production cluster
5. Pulse machines are configured but ignore metadata and patches set up
6. A machine is create (can be the same as step 4) to apply patches and publish the application
7. After validating the new cluster, load balancers are updated to redirect traffic to the new cluster
8. RabbitMQ replication queues are removed
9. Old cluster may be kept for a period of time to be later removed. This ensures production cluster can be swapped to the previous stable cluster in case anything is wrong

Comparing to [Using a control-node as a Blue-Green deployment orchestrator](#), steps 4 to 6 split up existing Ansible roles while running from different machines.

The machine used in *Steps 3 and 8* to create and remove replication queues should have `rabbitmqctl` installed. This can be done by running the following playbook:

```
python3.6 ansible-playbook \
    --connection=local \
    -i project-default \
    -i project-hsbc \
    -i gcp \
    -i project-<env> \
    thirdparty.yml
    --tags=rabbitmq.install,rabbitmq.440
```

Copy

Note: this machine should have TCP connection to RabbitMQ instances on ports `4369` and `25672`.

After the machine is provisioned replication queues can be created/destroyed by running `rabbitmq-setup-blue-green-federation.yml` and `rabbitmq-disable-blue-green-federation.yml` respectively.

```
python3.6 ansible-playbook \
    --connection=local \
    -i project-default \
    -i project-hsbc \
    -i gcp \
    -i project-<env> \
    rabbitmq-setup-blue-green-federation.yml # or rabbitmq-disable-blue-green-federation.yml to destroy replication queues in Step 8
```

Copy

In *Step 4* only tags responsible for extracting and setting up the required components to run metadata clean up and copy current application metadata should be included. It is also required to set `keep_metadata_between_deployments=true` to enable metadata copy from existing cluster to the new cluster.

```
python3.6 ansible-playbook \
    -i project-default \
    -i project-hsbc \
    -i gcp \
    -i <region>/project-<env> \
    -i <region>/project-<env>/<portfolio> \
    pulse.yml
    -e deployment_color=<opposite_color>
    -e keep_metadata_between_deployments=true
    --tags pulse.050,pulse.060,pulse.110,pulse.120,pulse.210,pulse.220,pulse.415,pulse.425,pulse.
426,pulse.440,pulse.455,pulse.710,pulse.730,pulse.731,pulse.750
```

Copy

In *Step 5* all tags should be included except tags that are responsible for metadata clean up, application patches and publishing. The first were already executed in the previous step and the latter will be executed afterwards.

```
python3.6 ansible-playbook \
    -i project-default \
    -i project-hsbc \
    -i gcp \
    -i <region>/project-<env> \
    -i <region>/project-<env>/<portfolio> \
    pulse.yml
    -e deployment_color=<opposite_color>
    --tags=pulse.configure
    --skip-tags pulse.730,pulse.731,pulse.750,pulse.932,pulse.940,pulse.941,pulse.970
```

Copy

In *Step 6* included tags depend whether it is running in the same machine as *Step 4* or in a new machine. In case it is running in the same machine the playbook can run including only tags `pulse.940` , `pulse.941` , and `pulse.970` .

```
python3.6 ansible-playbook \
    -i project-default \
    -i project-hsbc \
    -i gcp \
    -i <region>/project-<env> \
    -i <region>/project-<env>/<portfolio> \
    pulse.yml
    -e deployment_color=<opposite_color>
    --tags pulse.940,pulse.941,pulse.970
```

Copy

When running in a new machine tags that set up the machine to run patches also need to be included. These are the same as in *Step 4* excluding tags `pulse.730` , `pulse.731` and `pulse.750` .

```
python3.6 ansible-playbook \
    -i project-default \
    -i project-hsbc \
    -i gcp \
    -i <region>/project-<env> \
    -i <region>/project-<env>/<portfolio> \
    pulse.yml
    -e deployment_color=<opposite_color>
    --tags pulse.050,pulse.060,pulse.110,pulse.120,pulse.210,pulse.220,pulse.415,pulse.425,pulse.
426,pulse.440,pulse.455,pulse.710,pulse.940,pulse.941,pulse.970
```

Copy

warning

Pulse Data Science deployments **should not** include `pulse.455` since this task is only required for Pulse Runtime deployments. Additionally, Pulse Data Science deployments should override the `apps_enabled` Pulse ansible variable to be empty: `apps_enabled: {}`

warning

Take into account that with the introduction of the global rollout, images need to be created with dependencies for the applications that are to be deployed.

This applies to Pulse (RT), Case Manager and helper images for both components and can be controlled by the `apps_enabled` variable.

For example, in a market that should have `Outbound` and `Inbound`, all images should be generated with:

```
apps_enabled:
  - <market>_outbound_payments
  - <market>_reference_data
  - <market>_inbound_payments
```

Copy

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Configuration Recommendations

tip

For the full Pulse configuration refer to the official Configuration Reference and the recommended Production Setup.

Setting up the JVM DNS cache TTLâ

A couple of security properties should be set up. This can be achieved by:

1. Making sure that the `security.overridePropertiesFile` property is set to true in the `$JAVA_HOME/jre/lib/security/java.security` file.
2. Creating a new file (e.g. `/opt/feedzai/pulse/conf/java.security.properties) with the following content` :

```
# conf/java.security.properties
networkaddress.cache.ttl=1
networkaddress.cache.negative.ttl=3
```

Copy

1. Using the previously created file to set up the security properties on Pulse by adding `-Djava.security.properties=conf/java.security.properties` to the `JVM_OPTS` variable on `/opt/feedzai/pulse/bin/startup.sh` :

```
# /opt/feedzai/pulse/bin/startup.sh snippet
JVM_OPTS="(...) -Djava.security.properties=conf/java.security.properties"
```

Copy

1. Using the previously created file to set up the security properties on App engines by adding `-Djava.security.properties=conf/java.security.properties` to the `pulse.global.appengine.jvmopts` property on Pulse properties.

```
# /opt/feedzai/pulse/conf/pulse.properties snippet
pulse.global.appengine.jvmopts=(...) -Djava.security.properties=conf/java.security.properties
```

Copy

warning

Setting up these properties affects the cache behavior of all network address name resolutions for both Pulse server and App engine processes.

From the `java.security` file, next to the `networkaddress.cache.ttl` property on, for example, Zulu JDK:

> "NOTE: setting this to anything other than the default value can have serious security implications. Do not set it unless you are sure you are not exposed to DNS spoofing attack."

Ansibleâ

note

Available ansible variables for running the ansible playbook.

Variable	Default Value
ansible_user	feedzai
ansible_local_user	{{ ansible_user }}
pulse_install_path	"/opt/feedzai"
pulse_home_path	"{{ pulse_install_path }}/product-pulse-{{ pulse_version }}"
pulse_license_src_path	{{ basesoftware_path }}/pulse-licenses
pulse_sso_idp_cert	N/A
pulse_sso_private_key	N/A
pulse_sso_public_key	N/A
pulse_sso_force_https	false
pulse_api_url	"{{ jwk_service_host }}"
cm_sso_private_key	N/A
cm_sso_public_key	N/A
reference_data_servers	reference-data
zookeeper_nodes	zookeeper
cassandra_nodes	cassandra

Variable	Default Value
messaging_brokers	"rabbitmq:5672"
keycloak_url	"http://{{ keycloak_hostname }}:{{ keycloak_port }}"
keycloak_hostname	keycloak
keycloak_port	8080
cassandra_pulse_keyspace_replication	"'class':'NetworkTopologyStrategy', 'dc1': 1"
cassandra_cluster_name	"{{ inventory }} cluster"
cassandra_pulse_keyspace	"profiles"
cassandra_table_name	"profiles"
database_pulse_eventstorage_hostname	"postgres"
database_pulse_eventstorage_port	"5432"
database_pulse_eventstorage_database_name	"feedzai"
database_pulse_metadata_hostname	"postgres"
database_pulse_metadata_port	"5432"
database_pulse_metadata_database_name	"feedzai"
database_pulse_eventstorage_read_only_hostname	"{{ database_pulse_eventstorage_hostname }}"
database_pulse_eventstorage_read_only_port	"{{ database_pulse_eventstorage_port }}"
database_pulse_eventstorage_read_only_database_name	"{{ database_pulse_eventstorage_port }}"

Variable	Default Value
database_pulse_metadata_jdbc	"jdbc:postgresql://{{ database_pulse_metadata_hostname }}:{{ database_ currentSchema=pulse&cancelSignalTimeout=2&connectTimeout=2&load
database_pulse_eventstorage_jdbc	"jdbc:postgresql://{{ database_pulse_eventstorage_hostname }}:{{ databa currentSchema=eventstorage&cancelSignalTimeout=2&connectTimeout=
database_pulse_eventstorage_jdbc_read_only	"jdbc:postgresql://{{ database_pulse_eventstorage_read_only_hostname currentSchema=eventstorage&cancelSignalTimeout=2&connectTimeout=
database_pulse_outbound_payments_eventstorage_hostname	"{{ database_pulse_eventstorage_hostname }}"
database_pulse_outbound_payments_eventstorage_port	"{{ database_pulse_eventstorage_port }}"
database_pulse_outbound_payments_eventstorage_database_name	"{{ database_pulse_eventstorage_database_name }}"
database_pulse_outbound_payments_eventstorage_read_only_hostname	"{{ database_pulse_eventstorage_read_only_hostname }}"
database_pulse_outbound_payments_eventstorage_read_only_port	"{{ database_pulse_eventstorage_read_only_port }}"
database_pulse_outbound_payments_eventstorage_read_only_database_name	"{{ database_pulse_eventstorage_read_only_database_name }}"
database_pulse_outbound_payments_eventstorage_username	"{{ database_pulse_eventstorage_username }}"
database_pulse_outbound_payments_eventstorage_password	"{{ database_pulse_eventstorage_password }}"
database_pulse_outbound_payments_eventstorage_schema	"{{ database_pulse_eventstorage_schema }}"

Variable	Default Value
database_pulse_outbound_payments_eventstorage_jdbc	"jdbc:postgresql://{{ database_pulse_outbound_payments_eventstorage_ {{database_pulse_outbound_payments_eventstorage_database_name }}
database_pulse_outbound_payments_eventstorage_jdbc_read_only	"jdbc:postgresql://{{ database_pulse_outbound_payments_eventstorage_ {{database_pulse_outbound_payments_eventstorage_read_only_databas
database_pulse_inbound_payments_eventstorage_hostname	"{{ database_pulse_eventstorage_hostname }}"
database_pulse_inbound_payments_eventstorage_port	"{{ database_pulse_eventstorage_port }}"
database_pulse_inbound_payments_eventstorage_database_name	"{{ database_pulse_eventstorage_database_name }}"
database_pulse_inbound_payments_eventstorage_read_only_hostname	"{{ database_pulse_eventstorage_read_only_hostname }}"
database_pulse_inbound_payments_eventstorage_read_only_port	"{{ database_pulse_eventstorage_read_only_port }}"
database_pulse_inbound_payments_eventstorage_read_only_database_name	"{{ database_pulse_eventstorage_read_only_database_name }}"
database_pulse_inbound_payments_eventstorage_username	"{{ database_pulse_eventstorage_username }}"

Variable	Default Value
database_pulse_inbound_payments_eventstorage_password	"{{ database_pulse_eventstorage_password }}"
database_pulse_inbound_payments_eventstorage_schema	"{{ database_pulse_eventstorage_schema }}"
database_pulse_inbound_payments_eventstorage_jdbc	"jdbc:postgresql://{{ database_pulse_inbound_payments_eventstorage_h {{database_pulse_inbound_payments_eventstorage_database_name }}?
database_pulse_inbound_payments_eventstorage_jdbc_read_only	"jdbc:postgresql://{{ database_pulse_inbound_payments_eventstorage_re {{database_pulse_inbound_payments_eventstorage_read_only_database
database_pulse_cards_eventstorage_hostname	"{{ database_pulse_eventstorage_hostname }}"
database_pulse_cards_eventstorage_port	"{{ database_pulse_eventstorage_port }}"
database_pulse_cards_eventstorage_database_name	"{{ database_pulse_eventstorage_database_name }}"
database_pulse_cards_eventstorage_read_only_hostname	"{{ database_pulse_eventstorage_read_only_hostname }}"
database_pulse_cards_eventstorage_read_only_port	"{{ database_pulse_eventstorage_read_only_port }}"
database_pulse_cards_eventstorage_read_only_database_name	"{{ database_pulse_eventstorage_read_only_database_name }}"

Variable	Default Value
database_pulse_cards_eventstorage_username	"{{ database_pulse_eventstorage_username }}"
database_pulse_cards_eventstorage_password	"{{ database_pulse_eventstorage_password }}"
database_pulse_cards_eventstorage_schema	"{{ database_pulse_eventstorage_schema }}"
database_pulse_cards_eventstorage_jdbc	"jdbc:postgresql://{{ database_pulse_cards_eventstorage_hostname }}:{{ {{ database_pulse_cards_eventstorage_jdbc_opts | join('&') }}"
database_pulse_cards_eventstorage_jdbc_read_only	"jdbc:postgresql://{{ database_pulse_cards_eventstorage_read_only_hos {{ database_pulse_cards_eventstorage_jdbc_opts_read_only | join('&') }}"
database_pulse_reference_data_eventstorage_hostname	"{{ database_pulse_outbound_payments_eventstorage_hostname }}"
database_pulse_reference_data_eventstorage_port	"{{ database_pulse_outbound_payments_eventstorage_port }}"
database_pulse_reference_data_eventstorage_database_name	"{{ database_pulse_outbound_payments_eventstorage_database_name }
database_pulse_reference_data_eventstorage_read_only_hostname	"{{ database_pulse_outbound_payments_eventstorage_read_only_hostn
database_pulse_reference_data_eventstorage_read_only_port	"{{ database_pulse_outbound_payments_eventstorage_read_only_port }}
database_pulse_reference_data_eventstorage_read_only_database_name	"{{ database_pulse_outbound_payments_eventstorage_read_only_datab
database_pulse_reference_data_eventstorage_username	"{{ database_pulse_outbound_payments_eventstorage_username }}"

Variable	Default Value
database_pulse_reference_data_eventstorage_password	"{{ database_pulse_outbound_payments_eventstorage_password }}"
database_pulse_reference_data_eventstorage_schema	"{{ database_pulse_outbound_payments_eventstorage_schema }}"
database_pulse_reference_data_eventstorage_jdbc	"jdbc:postgresql://{{ database_pulse_reference_data_eventstorage_hostr {{ database_pulse_reference_data_eventstorage_jdbc_opts | join('&') }}"
database_pulse_reference_data_eventstorage_jdbc_read_only	"jdbc:postgresql://{{ database_pulse_reference_data_eventstorage_read_ {{database_pulse_reference_data_eventstorage_read_only_database_na
jwk_service_db_jdbc	jdbc:postgresql://postgres:5432/feedzai?currentSchema=jwkservice
jwk_service_db_schema	jwkservice
jwk_service_db_username	jwkservice
jwk_service_db_password	secret
jwk_service_host	"http://localhost:8080"
rabbitmq_brokers	"{{ messaging_brokers }}"
rabbitmq_pulse_username	"guest"
rabbitmq_pulse_password	"guest"
rabbitmq_pulse_vhost	"/"

Variable	Default Value
rabbitmq_feedback_storage_service_vhost	"feedback-storage"
rabbitmq_casemanager_brokers	"{{ rabbitmq_brokers }}"
rabbitmq_casemanager_username	"guest"
rabbitmq_casemanager_password	"guest"
rabbitmq_casemanager_vhost	"/"
database_pulse_metadata_username	"pulse"
database_pulse_metadata_password	"secret"
database_pulse_metadata_schema	"pulse"
database_pulse_eventstorage_username	"eventstorage"
database_pulse_eventstorage_password	"secret"
database_pulse_eventstorage_schema	"eventstorage"
database_pulse_app_metadata_username	"{{ database_pulse_metadata_username }}"

Variable	Default Value
database_pulse_app_metadata_password	"{{ database_pulse_metadata_password }}"
database_pulse_app_metadata_schema	"{{ database_pulse_metadata_schema }}"
pull_hadoop_files_from_bucket	false
hadoop_files_download_dir	"/tmp"
gcp_bucket_name	"pulse-dataproc-config"
gcp_hadoop_conf_file	"hadoop-files.tar.gz"
model_training_directory	"hdfs:///user/pulse/models"
model_storage_directory	"hdfs:///user/pulse/models"
model_hpt_directory	"hdfs:///user/pulse/hpt-models"
pulse_liquibase_environment	

Variable	Default Value
pulse_commands_environment	
pulse_setup_thread_dumps_cron	false
pulse_thread_dumps_cron_jobs	[]
number_threads_workflow_recovery	1
import_tmx_udf	true
import_fiv_markers_udf	true
cross_channel_metrics_enabled	false
cross_channel_metrics_rmi_based_lookup	true
cross_channel_metrics_fetch_timeout_ms	200
cross_channel_metrics_num_threads	1
cross_channel_metrics_client_consumer_threads	8

Variable	Default Value
cross_channel_metrics_server_consumer_threads	8
cross_channel_metrics_require_workflow_ready	true
cross_channel_metrics_publisher_pool_size	32
cross_channel_metrics_publisher_queue_size	1000
cross_channel_metrics_timed_futures_pool_size	100

Variable	Default Value
cross_channel_metrics_message_ttl_ms	1000
cross_channel_metrics_publish_timeout_ms	1000
cross_channel_metrics_batched_requests_enabled	false
cross_channel_metrics_batch_max_size	10
cross_channel_metrics_batch_max_time_ms	10
cross_channel_metrics_max_inflight_size	500
reference_data_tenant_strategy	STATIC

Variable	Default Value
inbound_payments_tenant_strategy	STATIC
transfers_tenant_strategy	STATIC
digital_activity_data_tenant_strategy	STATIC
cards_tenant_strategy	STATIC
inbound_payments_tenant_api_payload_field	customer_portfolio_country
outbound_payments_tenant_api_payload_field	customer_portfolio_country
digital_activity_tenant_api_payload_field	customer_portfolio_country
cards_tenant_api_payload_field	customer_portfolio_country

Variable	Default Value
reference_data_tenant_api_payload_field	customer_country_code
reference_data_tenant_provider_parameter_key	tenant
reference_data_tenant_provider_fallback	feedzai
pulse_spark_eventlog_enabled	false
pulse_spark_eventlog_dir	"hdfs:///var/log/spark/apps"
tars_enabled	false
node_app_pool	"{{ lookup('ansible.builtin.env', 'NODE_APP_POOL') }}"
pulse_app_topology	
feature_plan_log_level	Info

Variable	Default Value
watchdog_check_period_millis	10000
watchdog_latency_actuation_cycles	5
cards_customer_enrichment_key	customer_id
cards_account_enrichment_key	account_id
cards_card_enrichment_key	card_pan
cards_token_enrichment_key	ref_digital_wallet_token
numberOfPartitions	1

Variable	Default Value
cm_list_enrichment_enabled	true
cm_list_enrichment_prefix	"cmd_"
additional_fields_enricher_active	true
additional_fields_enricher_fields	"pib_suspended, internal_payment_type, internal_event_source, isJade"
outbound_payments_feedback_strategy	SINGLE
inbound_payments_feedback_strategy	SINGLE
cards_feedback_strategy	SINGLE
partitioned_realtime_metrics_enabled	false

Variable	Default Value
rds_max_parallel_calls	4
workflow_validation_log_level	DEBUG
pulse_loggers	{}
lists_use_legacy_csv_format	false

Variable	Default Value
spark_datascience_jdk_home	
spark_distributed_jdk_home	
use_rsync	true
minimize_shuffles_enabled	true

Variable	Default Value
api_port	Outbound Payments: 8081 Inbound Payments:8084 Cards:8085
rds_api_port	8083
inbound_slack_future	1800000
inbound_slack_past	1800000
cards_slack_future	1800000
cards_slack_past	1800000

Variable	Default Value
transfers_slack_future	1800000
transfers_slack_past	1800000
digital_activity_slack_future	1800000
digital_activity_slack_past	1800000

Variable	Default Value
cards_offset_days	{debit: 4 credit: 5}
rabbitmq_queue_type	CLASSIC

(*) The configuration files should stored on a bucket in tar.gz format. The list of files inside the tar.gz should be:

- core-site.xml
- hdfs-site.xml
- mapred-site.xml
- yarn-site.xml

The service account used to fetch the configuration files from the bucket should have read permissions for that bucket (`storage.buckets.get` and `storage.buckets.list`).

Cron typeâ

Variable	type	Mandatory	Default Value	Description
name	STRING	X		Name of the cron job
hour	STRING		*	The cron expression for `hour`
minute	STRING		*	The cron expression for `minute`
day	STRING		*	The cron expression for `day`
weekday	STRING		*	The cron expression for `weekday`
month	STRING		*	The cron expression for `month`

TARS application topologyâ

```
pulse_app_topology:
  <app_1_id_as_defined_in_apps_enabled>:
```

```
    # should be the amount of instances across all pools (for apps without partitioned workflows)
    replication_factor: <app_1_replication_factor>
    app_pools:
      # dictionary of pools with the number of instances per pool
      <app_1_pool_name>:
        number_of_instances: <app_1_number_of_instances_in_pool>
  <app_2_id_as_defined_in_apps_enabled>:
    # should be the amount of instances across all pools (for apps without partitioned workflows)
    replication_factor: <app_2_replication_factor>
    app_pools:
      # dictionary of pools with the number of instances per pool
      <app_2_pool_1_name>:
        number_of_instances: <app_2_number_of_instances_in_pool_1>
      <app_2_pool_2_name>:
        number_of_instances: <app_2_number_of_instances_in_pool_2>
```

Copy

Where:

- `pulse_app_topology` is a dictionary:
    - each key is the application key as defined in `apps_enabled` and contains:
    - `replication_factor` : the total number of application instances (the sum of `app_pools[*].number_of_instances`)
    - a dictionary (`app_pools`) to define in which pools an application is deployed:
        - the key is the pool name
        - `number_of_instances` : the number of applications to deploy in the `pool`

By default this variable is not configured. An example configuration would be:

```
 # Define topology structure for deployment with:
# -> One Pulse node running in the outbound pool
# -> Outbound and Reference Data applications running in the outbound node
pulse_app_topology:
  uk_outbound_payments:
    replication_factor: 1
    app_pools:
      outbound:
        number_of_instances: 1
  uk_reference_data:
    replication_factor: 1
    app_pools:
      outbound:
        number_of_instances: 1
```

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Configuration Recommendations for Data Science Environment

tip

For the full Pulse configuration refer to the official Configuration Reference, the recommended Production Setup and Data Science Setup.

Ansibleâ

note

Available ansible variables for running the Data Science ansible playbook. These variables are an addition to the variables described in Pulse Configuration Recommendations.

Variable	Default Value	Description
jupyterlab_docker_image	docker.feedzai.com/feedzai/jupyterlab:{{ jupyterlab_version }}	Feedzai JupyterLab Docker Image
hdfs_feedzai_user	pulse	User that will be the owner of the folders created on Hadoop master nodes
hadoop_master_node	"hadoop_master"	Hadoop master node hostname
hadoop_conf_path	"/etc/hadoop/conf"	Hadoop configuration folder in the master nodes
spark_libs_install_path	"{{ pulse_install_path }}/product-pulse-{{ pulse_version }}"	Source path to Spark libs on the Pulse node
pulse_jupyterlab_setup_docker	true	Flag to enable setup of JupyterLab
model_training_directory	"hdfs:///user/pulse/models"	The directory location where trained models should be saved. This directory can be a local or distributed (HDFS) filesystem.
model_storage_directory	"hdfs:///user/pulse/models"	The directory location where imported models through App Collaboration should be saved. This directory can be a local or distributed (HDFS) filesystem
model_hpt_directory	"hdfs:///user/pulse/hpt-models"	The directory location where the hyperparameter tuning models should be saved between model training and evaluation. This directory can be a local or distributed (HDFS) filesystem.
pulse_jupyterlab_container_max_memory	8589934592 (8GB)	The maximum memory (in KB) allowed for JupyterLab containers. Other units can be used instead, e.g. `512m` would set maximum memory to 512 MB.
pulse_loggers	{}	Controls extra loggers to be enabled in Pulse. Keys in this dictionary are the loggers and the values the log level for that logger.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/installation-guide/pulse/dataproc-2-2-support

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Google Dataproc 2.2

With Google Dataproc 2.0 reaching end of life it becomes crucial to support the latest Dataproc releases. The limitation with Dataproc version (2.2) is that it comes with Java 11 by default and jobs triggered in Pulse require Java 8.

Dataproc uses the [Adoptium Temurin JDK](#) which offers Java 8 packages for both types of OS distributions (Rocky/Debian), so in this guide Java 8 and Java 11 refer to packages offered by Adoptium.

In this page you can find a description on how to install Java 8 in Dataproc 2.2 and also on how to configure Pulse to use the correct version.

ⓘnote

Java 8 should be available in all cluster instances (master(s) and workers).

Set upâ

As stated above Dataproc 2.2 comes with Java 11 by default. However Dataproc clusters allow other packages to be installed during the cluster creation by using two different approaches: **custom images** or **initialization scripts**.

Below you can find details on how to install such packages using both approaches.

Custom Imagesâ

By using this approach we can generate a custom image (either fresh or based on another one) that includes all the packages we require as follows:

- Create a customization script that installs the necessary packages:

```
#! /usr/bin/bash
# Install other packages if needed

# Install Java 8
yum install temurin-8-jdk
```

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- Save the file locally
- Generate a new image, for example, using the following Python script provided by Google, passing the local path to the script in `--customization-script` :

```
python3 generate_custom_image.py \
    --image-name=CUSTOM_IMAGE_NAME \
    [--family=CUSTOM_IMAGE_FAMILY_NAME] \
    --dataproc-version=IMAGE_VERSION \
    --customization-script=LOCAL_PATH \
    --zone=ZONE \
    --gcs-bucket=gs://BUCKET_NAME
```

Copy

🔥warning

The installation script should only install the Java 8 JDK and not set `JAVA_HOME`, which should be set to the default Java version shipped with Dataproc 2.2.

Please find a completed description of the above steps in [Google's Create a Dataproc custom image guide](#).

Initialization scriptsâ

This approach leverages the initialization action that is available in Dataproc clusters to run a script when the cluster instances are created as follows:

- Create a script that installs the necessary JDK version (or reuse the one in [Custom Images](#))
- Upload the script to a GCS Bucket accessible by the cluster
- Set the script path while creating the cluster. For example, if the cluster is created with the `gcloud` client:

```
gcloud dataproc clusters create CLUSTER_NAME \
    --region=${REGION} \
    --initialization-actions=gs://my-bucket/cloud-sql-proxy/v1.0/cloud-sql-proxy.sh \
    ...other flags...
```

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For more information please refer to [Initialization Actions](#).

Pulse configurationâ

After a Dataproc cluster is created with the necessary Java dependencies, a new Pulse cluster will also need to be created. It should use the new Dataproc cluster and include two new Ansible variables that set the `JAVA_HOME` when creating jobs in the cluster:

- `spark_datascience_jdk_home` : sets the `JAVA_HOME` for Data Science jobs, e.g. datasource analysis
- `spark_distributed_jdk_home` : sets the `JAVA_HOME` for other jobs, e.g. scheduled Metrics Jobs

Each of these variables should point to the Java 8 path in the Dataproc cluster which is by default installed in `/usr/lib/jvm/temurin-8-jdk` .

After the Pulse cluster is created you will notice two new properties in the Pulse UI:

```
pulse.global.services.modules.distributed.jobs.spark.forward.spark.executorEnv.JAVA_HOME
pulse.global.services.modules.datascience.spark.forward.spark.executorEnv.JAVA_HOME
```

Copy

If correctly configured, when a new job is created the Spark configuration for that job will include the `spark.executorEnv.JAVA_HOME` in its execution context and force the job to run with the Java 8 JDK.

ⓘnote

In Pulse Data Science deployments these properties might need to be set manually via the UI. No restart is needed since both properties are hot reloadable.

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GCP Resources Creation

GCP Resourcesâ

Some GCP Resources needs to be created manually, as they are not created by Pulse.

1-Storage Bucket:

The purpose of the Storage Bucket is to archive the DS Events that are sent to to PUB/SUB topic, they will ve saved as avro files.

tip

The name of the Bucket must be **globally unique** so that it is recommended to create buckets name based on their Stage. For example for prod we can call it "DS-Events-PROD" and for QA "DS-Events-QA".

2-Dataflow Job:

This Dataflow job is the one that transfers data from the PUB/SUB Topic to the previously mentioned Storage Bucket.

From the **GCP Console navigate** to **Dataflow**-->**Jobs**, and click **"Create job from template"**, and do the following:

- for **Job Name**: assign a proper name, for example **"ds-stream-archive-job"**.
- for **Dataflow template**: choose the option **"Pub/Sub to Avro files on Cloud Storage"**.

After choosing the template, it will show 1 drop down list and n 3 text input areas to enter:

- for the **Input Pub/Sub Topic** choose the pulse configured topic where ds-events are published.
- for the **Output directory** create a new folder called **"output"** in the previously created Storage bucket and select it.
- for the **Temporary Avro write directory** create a new folder called **"tmp-avro"** in the previously created Storage bucket and select it.
- for the **Temporary location** create a new folder called **"tmp"** in the previously created Storage bucket and select it.

How to persist JupyterLab notebooks

warning

This page is only applicable to Pulse Data Science installations

This is Feedzai's proposed solution to persist JupyterLab notebooks when VM boot disks are deleted upon VM recreation.

A new JupyterLab docker container spawns for each user that accesses it, allowing management of notebooks per-user. Those Docker containers have a volume bound to the host - Pulse DS.

When Pulse DS bootable disk is persistent, user notebooks are kept on VM recreation. In cases where the host bootable disk is not persistent the notebooks will be lost on machine recreation. To mitigate this problem Pulse DS instances can be configured to have two different disks:

- A non-persistent bootable disk. Pulse can be installed / configured in this disk since those steps should be achievable by the original image.
- A persistent disk that is attached to Pulse VMs during VM creation

The persistent disk mount target should be set to the directory where Pulse DS stores user notebooks (`/opt/feedzai/pulse/jupyterlab`). This ensures that, when a docker container is spawn, Jupyter Lab working directory for that specific user will be mapped to the persistent disk and will load its notebooks, even if Pulse DS VM was recreated.

To create and mount non-boot disks on Linux VMs follow the steps in [Add a persistent disk to your VM]. After mounting the persistent disk, permissions on mount target should be set to the following:

- owner should be user `feedzai`
- group should be group `1450` which is the group id mapping to group `feedzai-jupyterlab` in the docker containers

> ð**Note:** Google recommends the non-boot to be formatted as `ext4` . Doing so will create a `lost+found` file which breaks JupyterLab when Pulse tries to create the docker containers.
>
> To prevent this from happening permissions for this file should be change by running the following command:
>
> `sudo chmod 6776 /opt/feedzai/pulse/jupyterLab/lost+found/`

After everything is set up, validate installation by opening JupyterLab from Pulse DS UI and create/edit notebooks.

> ð**Note:** If `Permission denied` pop up appears to the user when creating or saving a notebook, the error is likely due to permission changes during the persistent disk mount.
>
> If SSH to the machine is enabled one can verify this by using `ls -la /opt/feedzai/pulse/jupyterLab` . The results should have every file and directory (aside from the top directory) with owner `feedzai` and group `feedzai-jupyterlab` .
>
> To set proper permissions run the following command:
>
> `sudo chown feedzai:1450 -R /opt/feedzai/pulse/jupyterLab`

Cluster with same portfolio applications

With the multi-tenancy support added to Pulse the system supports multiple portfolios of the same type in the same cluster. Although this architecture works in most cases, business/regulatory restrictions require a different setup. This page focus on an architecture where the Pulse cluster is shared but there is a clear segregation on the application side, including the workflow, risk strategy, Event Storage and others.

Overviewâ

Architecturally, having a cluster with different applications of the same portfolio does not extensively differ from a cluster with applications of different markets with the same portfolios. It can be viewed has having different portfolio applications in a cluster but with restrictions on the API input ports and event storage.

In this page we describe the changes that were done to support this new architecture. For simplicity, we will focus on a setup with:

- Two markets (`Market 1` and `Market 2`)
- Two Outbound Payments Applications, one for each market (`market1_outbound_payments` and `market2_outbound_payments`). These two applications can not share the same workflow/risk strategy
- One Cards Application for market 1
- Two Case Manager clusters, one for each market
- Two Reference Data Service instances, one for each market

In the above example we can see that by having different Case Manager and Reference Data Service instances, all data between the two applications is segregated, meaning that Case Manager will only show data from the application that belongs to the same market and that Pulse will only enrich events using the corresponding Reference Data Service.

note

Having different Event Storage databases per market is an optional requirement as this architecture allows different applications to share the same database.

The Event Storage tables are segregated by application ID meaning that, even if different markets share the same database instance, all data will be stored in the corresponding application tables.

Configurationâ

The bare minimum changes for the example in this page relate to database and API port configurations.

For the example above we will assume market 1 is the default one which means that for market we would need to set the following for Pulse:

- `market2_outbound_payments_database_pulse_outbound_payments_eventstorage_hostname` pointing to the `Outbound 2 Database`
- `market2_outbound_payments_database_pulse_outbound_payments_eventstorage_jdbc` with the default value but with the previous variable in the `hostname` definition
- `market2_outbound_payments_api_port` to an available port, different from the default port (`8081`). Given all applications run in the same instance, ports need to differ to avoid collision
- `apps_enabled` with the applications to deploy
- `markets` with the available markets

The Pulse configuration would look like the following:

```
# project-<env>/group_vars/pulse/market1_outbound_payments
market1_outbound_payments_app_slug                         : "market1_outbound_payments"
market1_outbound_payments_market                           : "market1"
market1_outbound_payments_portfolio                        : "outbound_payments"
database_pulse_outbound_payments_eventstorage_hostname: "market1_outbound_payments_database_host
name"
reference_data_servers                                     : "market1_rds_hostname"

# project-<env>/group_vars/pulse/market2_outbound_payments
market2_outbound_payments_app_slug                                                        : "market2_outbo
und_payments"
market2_outbound_payments_market                                                          : "market2"
market2_outbound_payments_portfolio                                                       : "outbound_paym
ents"
market2_outbound_payments_database_pulse_outbound_payments_eventstorage_hostname: "market2_outbo
und_payments_database_hostname"
market2_outbound_payments_database_pulse_outbound_payments_eventstorage_jdbc    : "jdbc:postgres
ql://{{ market2_outbound_payments_database_pulse_outbound_payments_eventstorage_hostname }}:
{{ database_pulse_outbound_payments_eventstorage_port }}/
{{database_pulse_outbound_payments_eventstorage_database_name }}?
{{ database_pulse_outbound_payments_eventstorage_jdbc_opts | join('&') }}"
market2_outbound_payments_api_port                                                        : 8086
market2_reference_data_servers                                                            : "market2_rds_h
ostname"

# project-<env>/group_vars/pulse/variables.yml
apps_enabled:
  - market1_outbound_payments
  - market1_cards
  - market2_outbound_payments

markets:
  market1: Market 1
  market2: Market 2
```

Copy

For Case Manager we would need less changes, mainly related with Pulse and RDS endpoint configurations. This would require a different set of variables per deployment, one per cluster, as shown below:

```
# Market 1 installation
# project-<env>/group_vars/case-manager/variables.yml
apps_enabled:
  - market1_outbound_payments
  - market1_cards

reference_data_servers: "market1_rds_hostname"

# Market 2 installation
# project-<env>/group_vars/case-manager/variables.yml
apps_enabled:
  - market2_outbound_payments

reference_data_servers: "market2_rds_hostname"
```

Copy

Although not included in this example, the Reference Data Applications would require their own application configuration, similar to the Outbound applications, as show below:

```
# project-<env>/group_vars/pulse/market1_reference_data
market1_reference_data_app_slug : "market1_reference_data"
market1_reference_data_market    : "market2"
market1_reference_data_portfolio: "reference_data"
reference_data_servers           : "market1_rds_hostname"
```

```yaml
# project-<env>/group_vars/pulse/market2_reference_data
market2_reference_data_app_slug : "market2_reference_data"
market2_reference_data_market   : "market2"
market2_reference_data_portfolio: "reference_data"
market2_reference_data_servers  : "market2_rds_hostname"
market2_reference_data_port     : 8087

# Enable applications in Pulse
# project-<env>/group_vars/pulse/variables.yml
apps_enabled:
  - market1_outbound_payments
  - market1_cards
  - market1_reference_data
  - market2_outbound_payments
  - market2_reference_data
```

Copy

On this page

Network and Connectivity

info

This page is based on the architecture of Feedzai standard solutions. Specific clients or custom solutions may have additional requirements, so this is not a definitive guide.

Firewall Rulesâ

tip

For a complete list of the the Firewall Configurations required for Pulse, please refer to the product documentation.

Inboundâ

note

Inbound â other services connect to this service.

Service	Port	Description
HTTP	TCP 8080	Pulse Server UI (HTTP)
HTTP	TCP 8443	Pulse Server UI (HTTPS)
HTTP	TCP 8081-8091	API HTTP Server (Feedzai Connectors)
YARN (Spark, HDFS)	TCP *	Job management
Pulse	TCP *	Communication between Pulse nodes (RMI)
SSH	TCP 22	SSH access (if available)

Outboundâ

note

Outbound â this service connects to external services.

Service	Port	Description
RDBMS (PostgreSQL)	TCP 5432	Access to database
RabbitMQ	TCP 5671-5672	RabbitMQ AMQP (5671 for TLS)
YARN (Spark, HDFS)	TCP *	Job submission and management
Pulse	TCP *	Communication between Pulse nodes (RMI)
Reference Data Service	TCP 8443-8444	Communication with Reference Data Service
Profile Store (Cassandra)	9042	Fetch data from Profile Store
ZooKeeper	2181	Fetch data from Profile Store
Feedzai Enrichment Server (Public Network)	443	Connection to Feedzai Enrichment Service for specific Feedzai IP

Load Balancersâ

For high available environments Load Balancers must be set as follow:

Service	Description	Algorithm	Healthcheck URL	Response Codes	Policy Example
Pulse	Entry point used by end users to access the web UI and by Case Manager Processor to interact remotely with Pulse.	Follow the leader	`HTTPS:8443/api/client/cluster/pulse/leader/global` `HTTP:8080/api/client/cluster/pulse/leader/global`	200 - Node is global leader. 503 - Node is not global leader	Checks should be performed every 30 seconds with a timeout of 5 seconds. Nodes should be removed after the second failure.
Pulse Application API	Entry point used to send events for Pulse to score them.	Round robin	`HTTP:8081/api/health`	200 - Pulse received event and AppEngine workflow processed it successfully 500 - Some error occurred processing the event in the AppEngine workflow 503 - Input adapter is not available to receive the event	Checks should be performed every 6 seconds with a timeout of 5 seconds. Nodes should be removed after the second failure.

How To

Welcome to the Pulse How To.

In this section you will find the following set of tutorials with information on how to configure Pulse to support global rollout:

- Deploy new applications
- Use application specific databases

Pulse Installation Guide

Welcome to the Pulse Installation Guide.

In this section you will find a set of pages with information about the Pulse installation process:

On this page

Requirements

info

This guide is just a quick reference of the requirements to install Pulse. For a comprehensive guide, please refer to the official documentation.

Service Requirementsâ

This service requires the following software to run.

Java Environmentâ

OpenJDK SE Development Kit version 1.8, with the following vendors (with their latest hotfix versions):

- Azul Zulu rpm zip tar.gz
- Azul Platform Prime (formerly Zing)
- Amazon Corretto
- Oracle

info

Product tests are done with the **Azul Zulu JVM**.

The following versions are not supported, and are known to have problems:

- 1.8.0u242
- 1.8.0u121
- 1.8.0u60
- 1.8.0u45 and earlier

Non-listed releases are either not supported or haven't been put to extensive testing.

Setup Requirementsâ

ZooKeeperâ

Pulse uses ZooKeeper for coordinating all of its components (Pulse Servers and Application Engines) in a datacenter. With ZooKeeper Pulse knows what components available in a datacenter, be informed when one of them goes down or rejoins the datacenter, create distributed locks, elect a leader, as well as maintain information about each component.

To setup Pulse with ZooKeeper, you should follow these steps:

1. Install a ZooKeeper ensemble and make sure it is working properly.
2. Configure `pulse.manager.cluster.dc.{dc}.zookeeper.nodes` with the nodes, or a load balancer (recommended). The `{dc}` parameter refers to the name of the datacenter the nodes belong to. It is defined in the `node.properties` file.
3. Configure the `node.properties` with the ZooKeeper namespace wish to use. If you are preparing a coloured deployment, use the colour of the cluster of the nodes being configured. For example, you can use `pulseblue`.
4. Start Pulse. If the service is able to connect to ZooKeeper, a new namespace will be created automatically with the name you have specified in the previous step.

info

The current installation of Pulse has been tested with ZooKeeper 3.6.1.

Profile Storeâ

note

All configurations referenced in this section are prefixed by `pulse.app.{appSlug}.services.rte.workflow.plan.scheduledMetrics.profileStore` - this prefix is denoted in the examples below by `{PREFIX}`.

Pulse relies on a read-optimized NoSQL database system to store entity profile data (e.g. Customer-level aggregations), and be able to fetch in real-time without compromising latencies. The following systems are supported:

- Apache Cassandra 2.1 or higher - version 3.11.3 is the recommended version, and DSE is also supported

You can decide which store to use with the configuration `{PREFIX}.storage`. You must also specify the name of the table that will store the profile data. That is set by `{PREFIX}.table`.

Cassandraâ

When using Cassandra, Pulse will also automatically create the profiles table, considering the following:

1. A Cassandra cluster is available, and working properly. To connect to Cassandra, you must define the contact points in `{PREFIX}.cassandra.contactPoints` (for example, `cassandra-1,cassandra-2`).
2. A user has been created for Pulse (see how to create users in this guide). Configuring Pulse with the credentials to access the Cassandra cluster is done by setting `{PREFIX}.cassandra.user|password`. Pulse requires a user that has permissions to:
    - Create a keyspace
    - Create tables
    - Modify (Insert, Delete, Update, Truncate)
    - Select

warning

Cassandra user credentials are stored in the `system_auth` keyspace, which is not replicated by default. This may make users unable to login if certain nodes are down. Read more about this in the DataStax documentation.

tip

To know more about the recommended Cassandra setup, refer to the Pulse official documentation. To see which configurations can be set to connect to Cassandra see the official configuration reference.

RDBMSâ

Pulse uses PulseDB for database access, which allows it to run on top of multiple database systems:

- Oracle 19.3 (or higher)
- PostgreSQL 10.1 (or higher) - driver version 42.2.18 should be used.
- SQL Server 2017 GA (or higher)
- H2 1.4.200 (or higher) - only recommended for testing purposes.

info

Feedzai validates the product primarily against **H2** and **PostgresSQL**.

caution

Before selecting a database engine, make sure you understand how different database systems are supported by our PulseDB.

Pulse relies on a relational database not only to store the events ingested and scored (also known as the "Event Storage"), but also to store metadata about the cluster and the applications that it serves. For that reason, before starting Pulse you must:

1. Provision a database with the engine of your choice.
2. Create a new user for Pulse.
3. Create a new schema for Pulse. If you are preparing a coloured deployment (i.e. to support Blue-Green deployments), you must create two schemas. For example, `pulseblue` and `pulsegreen` .
4. Update the Pulse configurations with the JDBC connection string, the schema, and the credentials for the user you just created.

The database settings must be configured for the following prefixes:

- Pulse Server Metadata - `pulse.global.database`
- Pulse AppEngine Metadata - `pulse.app.{appSlug}.database`
- AppEngine Event Storage - `pulse.app.{appSlug}.services.modules.rte.workflowmanager.eventStorage.database`

To connect to the database you need to provide the `jdbc` , `schema` , `username` and `password` settings.

tip

To see which configurations can be set to connect to a database see the [official configuration reference](#).

RabbitMQâ

RabbitMQ is the messaging platform Pulse uses to communicate between machines and processes within the same datacenter.

The following RabbitMQ and Erlang versions are supported:

- RabbitMQ - 3.8.23 with Erlang - 23.3.4+
- RabbitMQ - 3.9.8 with Erlang - 23.3.4+
- RabbitMQ - 3.13.7 with Erlang - 26.2.5+
- RabbitMQ - 4.2.3 with Erlang - 27.3.4+

Before starting Pulse you must configure the messaging cluster:

1. Provision a RabbitMQ cluster and ensure there Pulse is able to establish a connection.
2. Create a new vhost for Pulse. If you are preparing a coloured deployment (i.e. to support Blue-Green deployments), you must create two vhosts. For example, `pulseblue` and `pulsegreen` .

```
rabbitmqctl rabbitmqctl add_vhost pulseblue
rabbitmqctl rabbitmqctl add_vhost pulsegreen
```

Copy

1. Create a new user for Pulse.

```
rabbitmqctl add_user pulse <password>
```

Copy

1. Update the vhost policies to grant all permissions to the Pulse user.

```
# set write and read permissions on every resource
rabbitmqctl set_permissions pulse -p pulseblue ".*" ".*" ".*"
rabbitmqctl set_permissions pulse -p pulsegreen ".*" ".*" ".*"
```

Copy

ⓘnote

On single-node and testing environments, you can opt by using ActiveMQ as a local, embedded message broker. Nonetheless, ActiveMQ is not recommended even for standalone Production setups (e.g., using a standalone Pulse for model training activities).

No-Downtime Deploymentsâ

Pulse uses a **Blue-Green** deployment strategy to implement no-downtime deployments. In order to do that, the architecture of the Feedzai platform needs to be prepared for the deployment switching.

Blue-green deployment uses two almost identical clusters in the following way:

1. At the start, one of the clusters is in use. Let's call it the blue environment.
2. The other one is on standby, in the final testing phase of a development project. Let's call this one the green environment.
3. When the new software version has been validated, the system gets redirected to the green deployment, and the blue deployment becomes the stand-by environment. It is available in case of a necessary rollback.
4. When the new version is verified in production, the blue deployment can be upgraded, making it identical to the green one. While still on stand-by, the blue environment is ready to receive a new development project.

To support the segregation between the blue and the green clusters, it is necessary to do the following:

- Create coloured vhosts in RabbitMQ (the messages from each cluster are isolated).
- Create coloured namespaces in ZooKeeper (each cluster then has its own cluster). Pulse will actually create it automatically if it doesn't exist, but it still needs to be configured differently for the two clusters.

When starting a new deployment, the cluster being deployed must be configured with the opposite colour.

/opt/feedzai/pulse/confg/node.properties

```
node.cluster=pulsegreen
```

Copy

/opt/feedzai/pulse/confg/config.properties

```
configuration.clusterTablesPrefix=green
```

Copy

/opt/feedzai/pulse/confg/pulseinit.properties

```
pulse.manager.cluster.dc.dc1.messaging.vhost=pulsegreen
```

Copy

While the database configuration remains the same, the metadata tables are prefixed automatically with the value set by the `clusterTablesPrefix` property.

tip

The colour of the deployment should be a template in the configuration files, so that it can be updated dynamically when a new service instance is provisioned. This is particularly relevant in setups where immutable VM images are created and the same one is reused across the multiple environments (usually the case on Cloud-based setups).

RabbitMQ Federationâ

During a deployment, the active cluster continues to process events and synchronising state between the cluster nodes. However, if that state is not shared with new cluster, scoring will be impaired after switching the traffic to it. For that reason, it is important to **federate the event replicas between between the two clusters**. This is done with the [RabbitMQ Federation]() plugin.

To connect the green cluster to receive updates from the blue one, you must create a federation policy, and then apply it to the the queues

```
rabbitmqctl set_parameter [-p vhost] federation-upstream blue-green '{"uri": ["amqp://<user>:<pass>@<node 1>/pulseblue", "amqp://<user>:<pass>@<node 2>/pulseblue", ...]}'

rabbitmqctl set_policy [-p vhost] federate-blue-green 'global-fdz-metadata-sync|global-fdz-apps-service|global-fdz-tenant-sync|global-fdz-tenant-group-sync|global-fdz-sync-topic-.*' '{"federation-upstream": "blue-green"}' --apply-to exchanges
```

Copy

caution

When doing a Blue-Green deployment, only the topics above should be federated, otherwise there could be unwanted control messages being propagated between the two clusters (e.g. stopping the application).

Data Science Studioâ

Even though Pulse is commonly referred to as an Risk Engine and used for monitoring fraud in real-time, it is in fact a Data Science platform where the data analytics tooling can be integrated with a runtime environment.

The following are the requirements for setting up Pulse for a Data Science environment:

- A database for metadata storage, ideally a Relational one, such as PostgreSQL
- 1 machine for Pulse to execute on (a single replica)
- Coordination processes, which can be dockerized and/or executing in the same machine as Pulse:
  - 1 RabbitMQ process
  - 1 ZooKeeper process
- A Spark cluster for distributed job execution. The recommended approach is to run Spark on YARN (Cloudera, AWS EMR, Azure HDInsight, Google Dataproc), which requires a compliant Hadoop File System to exist in the cluster.

warning

Feedzai **does not recommended** the use of embedded coordination processes (i.e., using **embedded ZooKeeper** and **ActiveMQ**), as well as databases that are not production-ready, like **H2**.

For more information about the Data Science setup, refer to the official documentation.

Integration with JupyterLabâ

Pulse supports integration with JupyterLab.

- To enable Setup of JupyterLab:

  1. Docker must be installed on the Pulse machine.
  2. Feedzai Jupyterlab Docker Image must exist on Pulse machine.

- If the Docker software is not available on the Pulse machine, so the flag ***"pulse_jupyterlab_setup_docker"*** must be set to false, so that Pulse doesn't configure Jupyterlab erroneously.

On this page

Topology-Aware Resource Scheduling

Topology-Aware Resource Scheduling, or TARS for short, is the ability of distributing applications on a per node pool basis where a node pool is a set of Pulse instances.

This approach improves resiliency and resource management by:

- Isolating application processes, meaning that an application will keep running even if different node pools are degraded
- Giving the possibility of scaling machines according to the workload of each application

In this section you can find how TARS can be configured.

Overviewâ

As explained above TARS gives the possibility of distributing an application/set of applications in different node pools. The Pulse Server (PS) cluster will have to context of all deployed applications and acts as a single point of configuration for all.

In the following illustration you can find that:

- There are two node pools, `outbound` and `inbound`
- There are three Pulse instances, where two belong to `outbound` and one to `inbound`
- The Outbound application (Transfers and Digital Activity) is configured to be deployed in the `outbound` node pool
- The Inbound application is configured to be deployed in the `inbound` node pool
- The UI Load Balancer will connect to the Pulse leader node where the whole runtime cluster can be managed including the applications that are not deployed in the leader node
- The API Load Balancer on the other hand will only redirect traffic to the nodes that serve a specific application

This however does not mean one node pool can only serve one application. If, for instance, there are two applications that have a small workload they can be deployed in the same node pool.

The example below illustrates this case:

- The `Outbound` is still deployed in two (unshared) nodes
- The `Cards` application was configured to be deployed in the `inbound` node pool, hence sharing the node with the `Inbound` application
- The API Load Balancer for `Inbound` will now also redirect `Cards` requests for the `Cards` application running in the `inbound` node pool

Configurationâ

TARS configuration is controlled by the following Ansible variables:

- `apps_enabled` : should have all the applications to deploy
- `tars_enabled` : `true` to enable TARS, `false` otherwise
- `node_app_pool` : pool where the node belongs to, read from an environment variable
- `pulse_app_topology` : application topology in the structure defined below

```
pulse_app_topology:
  <app_1_id_as_defined_in_apps_enabled>:
    # should be the amount of instances across all pools (for apps without partitioned workflows)
    replication_factor: <app_1_replication_factor>
    app_pools:
```

```
      # dictionary of pools with the number of instances per pool
      <app_1_pool_name>:
        number_of_instances: <app_1_number_of_instances_in_pool>
  <app_2_id_as_defined_in_apps_enabled>:
    # should be the amount of instances across all pools (for apps without partitioned
workflows)
    replication_factor: <app_2_replication_factor>
    app_pools:
      # dictionary of pools with the number of instances per pool
      <app_2_pool_1_name>:
        number_of_instances: <app_2_number_of_instances_in_pool_1>
      <app_2_pool_2_name>:
        number_of_instances: <app_2_number_of_instances_in_pool_2>
```

Copy

The first thing to do regarding configuration is to define the topology you want to achieve. For the sake of this guide we will assume the topology defined in the second example given above, where `Inbound` and `Cards` will share a node pool and `Outbound` will have its own pool in two different nodes.

This means that when creating the instance groups they will need to have an environment variable indicating the pool they belong to. For this example the instance group for `Outbound` should have an environment variable `NODE_APP_POOL=outbound` and the `Inbound` / `Cards` should have its own set to `NODE_APP_POOL=inbound`.

warning

Since `node_app_pool` reads the node environment variable during deployment, the environment variable of each node needs to be set before running the deployment otherwise the pool will be configured as an empty string.

At this point with all the instance groups created we can continue configuration:

- Define the `apps_enabled` variable to include the applications to deploy

```
apps_enabled:
  - uk_outbound_payments
  - uk_reference_data
  - uk_inbound_payments
  - uk_cards
```

Copy

- Enable TARS by setting `tars_enabled: true`
- Define `pulse_app_topology` as follows:

```
pulse_app_topology:
  uk_outbound_payments:
    replication_factor: 2
    app_pools:
      outbound:
        number_of_instances: 2
  uk_reference_data:
    replication_factor: 1
    app_pools:
      outbound:
        number_of_instances: 1
  uk_inbound_payments:
    replication_factor: 1
    app_pools:
      inbound:
        number_of_instances: 1
  uk_cards:
    replication_factor: 1
    app_pools:
      inbound:
        number_of_instances: 1
```

Copy

Some notes regarding the `pulse_app_topology` :

- `uk_outbound_payments` has a `replication_factor` and `number_of_instances` of `2` because we want to have two instances of the `Outbound` application running in two different `outbound` nodes
- `uk_reference_data` will only be deployed in one of the `outbound` nodes
- `uk_inbound_payments` and `uk_cards` will be deployed in one `inbound` node

warning

Please note this example assumes the default `pulse_cluster_datacenters` variable was not overridden. In case it was the override needs to be extended as follows:

```
pulse_cluster_datacenters:
  - name:                                             "dc1"
    numberofnodes:                                    "{{ pulse_cluster_number_of_nodes }}"
    replicationfactor:                                "{{ pulse_cluster_replication_factor }}"
    (...)
    app_topology:                                     "{{ pulse_app_topology if tars_enabled else [] }}"
```

Copy

After deployment you should be able to see how the applications were distributed in the Pulse UI by accessing the datacenter page in the administration tab.

Additional configurationâ

As stated before using TARS enables application distribution across different Pulse nodes while keeping the application configuration accessible via a singe Pulse UI.

This means that after a TARS deployment, the UI Load Balancer will need to be updated to include all the backends (Pulse instances) while choosing to where redirect traffic based on the leader node.

As for the API Load Balancer, given the example on this page, it will need to be configured as follows:

- Redirect `Transfers` and `Digital Activity` traffic to the `outbound` nodes
- Redirect `Inbound` and `Cards` traffic to the `inbound` nodes

info

Find a detailed description of all available configurations in [Pulse Configuration Recommendations]

On this page

Configuration Recommendations

Ansibleâ

note

Available ansible variables for running the ansible playbook.

Variable	Default Value	Description
reference_data_app_port	8080	Reference data connection port
reference_data_admin_port	8081	Reference data administration connection port
reference_data_cassandra.tableName	feedzai_reference_data	Cassandra table name
reference_data_cassandra.keyspaceName	hsbc	Cassandra keyspace name
reference_data_cassandra.username	cassandra	Cassandra username
reference_data_cassandra.password	cassandra	Cassandra password
reference_data_cassandra.cluster	rds cluster	Cassandra cluster name
reference_data_cassandra.contactPoints	["localhost:9042"]	Cassandra contact points
reference_data_cassandra.consistencyLevel	"LOCAL_ONE"	Cassandra consistency level
reference_data_cassandra.replicationStrategy	"'class':'NetworkTopologyStrategy', 'dc1': 1"	Cassandra replication strategy
reference_data_jwt_secret	"0jd40327td208347t02k3487t23k40t87320t"	JWT secret, needs to be set
reference_data_use_https	false	When `true` RDS will use https for connection
keyStorePath	"{{ reference_data_home_path }}/conf/keystore.jks"	Keystore path, needs to be set when reference_data_use_https is `true`
keyStorePassword	N/A	Keystore password, needs to be set when reference_data_use_https is `true`
keyStoreType	"JKS"	Key store type, needs to be set when reference_data_use_https is `true`
keyManagerPassword	N/A	Key manager password, needs to be set when reference_data_use_https is `true`
validateCerts	true	

Variable	Default Value	Description
		Set to validate certificates, needs to be set when reference_data_use_https is `true`
validatePeers	true	Set to validate peers, needs to be set when reference_data_use_https is `true`
supportedProtocols	[TLSv1.2]	Set supported protocols, needs to be set when reference_data_use_https is `true`
supportedCipherSuites	[TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256]	Set supported ciphers, needs to be set when reference_data_use_https is `true`
allowRenegotiation	true	Set to allow renegotiation, needs to be set when reference_data_use_https is `true`
reference_data_secret_value	N/A	Value used to encrypt sensitive properties
reference_data_properties_to_encrypt	[]	List of properties to encrypt. Encrypted properties can be used by adding the suffix `_encrypted` to the original property name, i.e `<PROPERTY_NAME>_encrypted`. Doesn't support dictionary variables
reference_data_grace_period_ms	30000	The number of milliseconds the service will stay available after a shutdown request is received
reference_data_log_level	INFO	Default log level for all components in RDS. Please note that changing this value may impact the amount of produced logs.
reference_data_loggers	{}	Controls extra loggers in RDS, allowing to override the default value per logger. Keys in this dictionary are the loggers and the values the log level for that logger.

info

There are no configuration recommendations for this service besides the ones already described in the guide.

On this page

Network and Connectivity

info

This page is based on the architecture of Feedzai standard solutions. Specific clients or custom solutions may have additional requirements, so this is not a definitive guide.

Firewall Rulesâ

tip

For a complete list of the the Firewall Configurations required for Reference Data Service, please refer to the product documentation.

Inboundâ

note

Inbound âother services connect to this service.

Service	Port	Description
HTTP	TCP 8443-8444	HTTP Requests from Reference Data Service client.
SSH	TCP 22	SSH access (if available)

Outboundâ

note

Outbound âthis service connects to external services.

Service	Port	Description
Database (Cassandra)	TCP 9042	Connection to the database

Load Balancersâ

For high available environments Load Balancers must be set as follow:

Service	Description	Algorithm	Healthcheck URL	Response Codes	Policy Example
Reference Data Service	Used to read/write reference data.	Round robin	`HTTPS:9443/health` `HTTP:9080/health`	200 - Node is available	Checks should be performed every 10 seconds with a timeout of 2 seconds. Nodes should be removed after the second failure.

RDS Installation Guide

Welcome to the RDS Installation Guide.

In this section you will find a set of pages with information about the RDS Installation process:

- Requirements
- Network and Connectivity
- Configuration Recommendations

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/installation-guide/refdata/requirements-refdata

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Requirements

Service Requirementsâ

This service requires the following software to run.

Java Environmentâ

OpenJDK SE Development Kit version 1.8, with the following vendors (with their latest hotfix versions):

- Azul Zulu rpm zip tar.gz
- Azul Platform Prime (formerly Zing)
- Amazon Corretto
- Oracle

tip

Product tests are done with the **Azul Zulu JVM**.

The following versions are not supported, and are known to have problems:

- 1.8.0u242
- 1.8.0u121
- 1.8.0u60
- 1.8.0u45 and earlier

Non-listed releases are either not supported or haven't been put to extensive testing.

Setup Requirementsâ

Databaseâ

The Reference Data Service is a service responsible for storing reference data that belongs to individual entities (e.g. Card, Customer, etc). The type of storage that betters suits this use-case are *key-value* stores.

The following systems are supported:

- Apache Cassandra 2.1 or higer - version 3.11.3 is the recommended version, and DSE is also supported

Cassandraâ

When using Cassandra, RDS will also automatically create the entities tables, considering the following:

1. A Cassandra cluster is available, and working properly. To connect to Cassandra, you must define the contact points in `cassandra.contactPoints` (for example, `cassandra-1,cassandra-2`).
2. A user has been created for RDS with the following permissions:

- Create a Keyspace
- Create tables
- Modify (Insert, Delete, Update, Truncate)
- Select

The credentials of the user must be configured in `cassandra.user|password`.

On this page

Drop database index

Playbook to drop index *idx_event_occurred_at* from the *Event Storage* created by the [Fix Pulse Clock] playbook.

Configurationâ

All the following variables are **mandatory**.

Variable	Type	Default Value	Description
schema_migrator_database_username	string	lookup('gcp_secret_manager', secret='schema_migrator_database_username')	Database username
schema_migrator_database_password	string	lookup('gcp_secret_manager', secret='schema_migrator_database_password')	Database password

Running the Playbookâ

```
python3.6 ansible-playbook \
    -i project-default \
    -i project-hsbc \
    -i gcp \
    -i project-<env> \
    drop-db-index.yml
```

Copy

Fix Pulse Clock

Playbook to fix *Pulse* clock to ensure that the *Event Storage* won't have events in the future/past.

Configurationâ

All the following variables are **mandatory**.

Variable	Type	Default Value	Description
schema_migrator_range_start	integer		Number specifying at which position to start
schema_migrator_range_end	integer		Number specifying at which position to stop
schema_migrator_range_step	integer		Number specifying the incrementation
schema_migrator_database_username	string	lookup('gcp_secret_manager', secret='schema_migrator_database_username')	Database username
schema_migrator_database_password	string	lookup('gcp_secret_manager', secret='schema_migrator_database_password')	Database password

Running the Playbookâ

```
python3.6 ansible-playbook \
    -i project-default \
    -i project-hsbc \
    -i gcp \
    -i project-<env> \
    fix-pulse-clock.yml
```

Copy

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Overview

Schema migrator is Feedzai's solution to migrating existing data on a running system. It is ideal due to its capability of splitting the data in time ranges to **avoid locking up the database and impairing incoming traffic**.

Ansible variablesâ

Variable	Type	Default Value	Description
schema_migrator_run	boolean	true	Whether the script should be executed or not
schema_migrator_range_start	integer		Number specifying at which position to start
schema_migrator_range_end	integer		Number specifying at which position to start
schema_migrator_range_step	integer		Number specifying the incrementation
schema_migrator_script_file_path	integer	specific script path	Path to the script file to be executed
schema_migrator_configuration_path	string	/opt/feedzai/schema-migrator/conf/config.properties	Path to the tool configurations properties file
schema_migrator_database_username	string	lookup('gcp_secret_manager', secret='schema_migrator_database_username')	Database username
schema_migrator_database_password	string	lookup('gcp_secret_manager', secret='schema_migrator_database_password')	Database password
schema_migrator_database_jdbc	string	jdbc:postgresql://postgres:5432/feedzai	Database connection
schema_migrator_database_engine	string	com.feedzai.commons.sql.abstraction.engine.impl.PostgreSqlEngine	Database engine
schema_migrator_database_schema	string	eventstorage	Database schema where the script will be executed

On this page

Firewall Rules for Cassandra

info

This page is based on the architecture of Feedzai standard solutions. Specific clients or custom solutions may have additional requirements, so this is not a definitive guide.

Inboundâ

note

Inbound â other services connect to this service.

Service	Port	Description
Cassandra	TCP 7000-7001	Cluster communication
Pulse	TCP 9042	Requests from Cassandra Driver
Reference Data Service	TCP 9042	Requests from Cassandra Driver
SSH	TCP 22	SSH access (if available)

Outboundâ

note

Outbound â this service connects to external services.

Service	Port	Description
Cassandra	TCP 7000-7001	Connection to the database

On this page

Firewall Rules for RabbitMQ

info

This page is based on the architecture of Feedzai standard solutions. Specific clients or custom solutions may have additional requirements, so this is not a definitive guide.

Inboundâ

note

Inbound â other services connect to this service.

Service	Port	Description
Erlang	TCP 25672	Erlang inter-node and CLI tool communication
Erlang	TCP 4369	Peer discovery service (Erlang)
Pulse	TCP 5671-5672	RabbitMQ AMQP (5671 for TLS)
Case Manager	TCP 5671-5672	RabbitMQ AMQP (5671 for TLS)
Logstash	TCP 5671-5672	RabbitMQ AMQP (5671 for TLS)
HTTP	TCP 15672	RabbitMQ Management UI
SSH	TCP 22	SSH access (if available)

Outboundâ

note

Outbound â this service connects to external services.

Service	Port	Description
Erlang	TCP 25672	Erlang inter-node and CLI tool communication
Erlang	TCP 4369	Peer discovery service (Erlang)

Network Requirements

Welcome to the Network Requirements Guide.

In this section you will find a set of pages with information about Network Requirements:

- Firewall rules for Cassandra
- Firewall rules for RabbitMQ
- Firewall rules for ZooKeeper

On this page

Firewall Rules for ZooKeeper

info

This page is based on the architecture of Feedzai standard solutions. Specific clients or custom solutions may have additional requirements, so this is not a definitive guide.

Inboundâ

note

Inbound â other services connect to this service.

Service	Port	Description
ZooKeeper	TCP 2888	Peer communication
ZooKeeper	TCP 3888	Peer communication
Pulse	TCP 2181	ZooKeeper Client communication
SSH	TCP 22	SSH access (if available)

Outboundâ

note

Outbound â this service connects to external services.

Service	Port	Description
ZooKeeper	TCP 2888	Peer communication
ZooKeeper	TCP 3888	Peer communication

On this page

Recommendations for Cassandra

info

The configurations listed below result from the experience Feedzai has running this system. Nonetheless, it is important to review the configurations in light of the workload requirements, and adjusted accordingly.

Cassandra Configurationâ

The `cassandra.yaml` file is the main configuration file for Cassandra.

Property	Recommended Value	Description
cluster_name	`Feedzai Cassandra Cluster`	This setting prevents nodes in one logical cluster from joining another. All nodes in a cluster must have the same value.
num_tokens	256	Set this property for virtual node token architecture. Determines the number of token ranges to assign to this [vnode].
endpoint_snitch	`GossipingPropertyFileSnitch` (see note below)	The implementation used informs Cassandra about the network topology to route requests efficiently, and allows Cassandra to spread replicas around your cluster to avoid correlated failures.
concurrent_reads	`8 * #cores`	
concurrent_writes	`8 * #cores`	
trickle_fsync	true	Causes fsync to force the operating system to flush the dirty buffers at the set interval `trickle_fsync_interval_in_kb`, to prevent sudden flushes from impacting read latencies.
trickle_fsync_interval_in_kb	4096	The size of the fsync in kilobytes.
concurrent_compactors	`#cores`	The number of concurrent compaction processes allowed to run simultaneously on a node.
compaction_throughput_mb_per_sec	0	Throttles compaction to the specified Mb/second across the instance. Setting the value to 0 disables compaction throttling.
disk_failure_policy	stop	On disk failures, shuts down gossip and client transports, leaving the node effectively dead, but it still can be inspected with JMX.
commit_failure_policy	stop	On commit disk failures, shuts down gossip and client transports, leaving the node effectively dead, but it still can be inspected with JMX.
commitlog_sync	periodic	Ack writes immediately and sync the CommitLog every `commitlog_sync_period_in_ms`
commitlog_sync_period_in_ms	10000	Frequency of the CommitLog sync when in `periodic` mode.
auto_bootstrap	true	It causes new (non-seed) nodes migrate the right data to themselves automatically.

info

Endpoint Snitch

With the `GossipingPropertyFileSnitch` the rack and datacenter for the local node must be defined in `cassandra-rackdc.properties`.

If you are installing Cassandra on AWS, the recommendation is to use `Ec2Snitch`. This will rely on the instances and the availability zones they are done to define a better and more reliable topology.

If you are installing Cassandra on GCP, there is a `GoogleCloudSnitch` available.

tip

Make sure `ntp` is installed, and all Cassandra servers are in-sync.

Javaâ

Java Native Access (JNA)â

Using JNA is one way to improve Cassandra memory usage, and, hence, performance. If JNA is not installed and enabled, Cassandra will not boot unless `cassandra.boot_without_jna` is set to true.

JVM Optsâ

It is advisable that the `-Xms` and `-Xmx` flags are set to the same value. Optionally, `-Xmn` could be set as well, according to the resources and workload.

Make sure you define the most appropriate GC algorithm according to the workload and the heap size.

On this page

Recommendations for RabbitMQ

info

The configurations listed below result from the experience Feedzai has running this system. Nonetheless, it is important to review the configurations in light of the workload requirements, and adjusted accordingly.

RabbitMQ Configurationâ

Highly Available Queuesâ

RabbitMQ clustering capabilities do not offer replicated message queues by default, so a message queue only resides in the node which created them. As such, Feedzai recommends configuring Highly Available Queues.

In contrast to exchanges and bindings, which can always be considered to be on all nodes, queues can optionally run mirrors (additional replicas) on other cluster nodes.

Each mirrored queue consists of one **leader replica** and one or more **mirrors** (replicas). The leader is hosted on one node commonly referred to as the leader node for that queue. Each queue has its own leader node. All operations for a given queue are first applied on the queue's leader node and then propagated to mirrors.

warning

Consumers are connected to the leader regardless of which node they connect to, with mirrors dropping messages that have been acknowledged at the leader. Queue mirroring therefore enhances availability, but does not distribute load across nodes (all participating nodes each do all the work).

To create policy that defines the mirroring conditions, use the command below.

```
rabbitmqctl set_policy replicate-queues "^(fdz-sq-)" '{"ha-mode":"all", "expires":7200000, "ha-sync-batch-size":1000, "ha-promote-on-shutdown":"always"}' --apply-to queues
```

Copy

tip

All queues created by Feedzai services have the prefix `fdz-sq` , and are suffixed with a UUID to ensure that new processes will always get new queues - for example, `fdz-sq-fdz-workflow-publish-responsec6e7cef5-3239-4a9f-963b-98ef108eb4c5` .

Queue TTLâ

Defining a TTL for Pulse queues ensures that queues **without subscribers or without consuming the messages** will be deleted after a certain time. Feedzai recommends it to be at least two times the configured client session timeout in ZooKeeper. The TTL must also be confortably higher than the Workflow Recovery time, as during recovery Pulse nodes already subscribed to replicas, but haven't started to consume them. This also means that, in case of an incident, there is a limited period to recover the system without losing any messages.

The queue TTL is set using the `expires` configuration, as shown in the example above.

info

To ensure data is not lost in case of an incident with Case Manager, Feedzai usually recommends against setting the TTL for the Case Manager vhost.

Recommendations for ZooKeeper

info

There are no configuration recommendations for this service besides the ones already described in the guide.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/operations-guide/cm-queue-cleanup/cleanup-alerts

On this page

Case Manager Alert Queue Cleanup

In this page you can find the Case Manager alert cleanup process definition, how to configure and run it.

Alert queue cleanupâ

The Case Manager queue cleanup is a process that can be used to close all Case Manager alerted events that match a specific criteria with the intended reason. This is particularly useful before a go live so that Case Manager alert pages do not include any alerts that may have been already acted upon on a different platform.

This process is essentially a Python script that loads a set of events based on a given criteria, removes the queues from those events and closes them with a given reason. It also supports a mode in which it simply closes open alerts that are not assigned to any queue.

Requirementsâ

The script execution has the following requirements:

- Connection between the machine where the script is being run and the Case Manager load balancer
- A service account credentials
- Python version 3.8+ installed
- `requests` Python library installed

warning

This script was tested against Python version 3.8. If this specific version is not available, the script can be tested against the available one but take into account it may be unsupported.

Resourcesâ

The cleanup script can be downloaded from cleanup-alerts-queues.txt.

The extension must be changed from `.txt` to `.py.` after downloading. The rest of the documentation below assumes the script has been renamed to `cleanup-alerts-queues.py`.

Configurationâ

In this section you can find the configuration reference for the script.

Variable	Mandatory	Default Value	Description
`-h, --help`			Show an help message.
`-c, --url`	X		The Case Manager load balancer URL.
`-e, --exec-mode`			Flag indicating if the script should make any change. If not specific it will do a test run.
`-sk, --skip-ssl`		`False`	Flag indicating if SSL certificate validation should be skipped when making HTTPS requests
`-u, --username`		`pulse`	Username of the service account.
`-p, --password`	X		Password of the service account (will prompt if not specified).

Variable	Mandatory	Default Value	Description
`-s, --status-reason`	X		Status reason name to add when closing an event that will be used to fetch the reason id as shown in the UI.
`-ch, --channel`		`transfers`	Channel to search for events.
`-mints, --min_timestamp`		`1688169600000`	The minimum timestamp to fetch events in milliseconds.
`-mints, --max_timestamp`		`1688169600000`	The maximum timestamp to fetch events in milliseconds.
`-wq, --without_queue`		`1688169600000`	Flag indicating if the script should close only alerts not assigned to any queue.
`-t, --threshold`		`100`	Threshold to apply to the search filter.
`-r, --retries`		`3`	Number of retries when making HTTP requests.
`-b, --backoff`		`1`	Backoff factor for the HTTP requests.
`-tk, --tenant-keys`			The tenant keys (one or more) used to filter events. Should be used without `-ats, --all-tenants`.
`-ats, --all-tenants`			Include all tenants when filtering events. Should be used without `-tk, --tenant-keys`.

Running the script with the help flag (`-h`) will output the script information and available configuration:

```
$ python3 cleanup-alerts-queues.py -h
usage: cleanup-alerts-queues.py [-h] -c CM_URL [-e] [-sk] [-u USERNAME] [-p PASSWORD] -s
STATUS_REASON
                                [-ch CHANNEL] [-mints MIN_TIMESTAMP] [-maxts MAX_TIMESTAMP] [-
wq] [-t THRESHOLD]
                                [-r NUMBER_RETRIES] [-b BACKOFF_FACTOR] (-tk TENANT_KEYS [TENANT_
KEYS ...] | -ats)

This script cleanups Case Manager queued alerts by closing them with the given status reason.

optional arguments:
  -h, --help            show this help message and exit
  -tk, --tenant-keys TENANT_KEYS [TENANT_KEYS ...]
                        The tenant keys (one or more) used to filter events. Should be used
without `-ats, --all-tenants`.
  -ats, --all-tenants   Include all tenants when filtering events. Should be used without `-tk,
--tenant-keys`.

  -c CM_URL, --url CM_URL
                        Case Manager URL
  -e, --exec-mode       Run in change mode. If not specific will do a dry run.
  -sk, --skip-ssl       Skip SSL validation when making HTTPS requests.
  -u USERNAME, --username USERNAME
                        Username of the service account (default: pulse)
  -p PASSWORD, --password PASSWORD
                        Password of the service account (will prompt if not specified)
  -s STATUS_REASON, --status-reason STATUS_REASON
                        Status reason name to add when closing an event that will be used to
fetch the reason id.
  -ch CHANNEL, --channel CHANNEL
                        Channel to search for events (default: transfers)
  -mints MIN_TIMESTAMP, --min_timestamp MIN_TIMESTAMP
                        The minimum timestamp to fetch events in milliseconds (default: 168816960
0000)
  -maxts MAX_TIMESTAMP, --max_timestamp MAX_TIMESTAMP
                        The maximum timestamp to fetch events in milliseconds (default: 168816960
0000)
  -wq, --without_queue  Close only alerts not assigned to any queue. If not specified, will close
```

```
alerts not assigned
                        to a queue, clearing all queues.
  -t THRESHOLD, --threshold THRESHOLD
                        Threshold to apply to the search filter (default: 100)
  -r NUMBER_RETRIES, --retries NUMBER_RETRIES
                        Number of retries when making HTTP requests (default: 3)
  -b BACKOFF_FACTOR, --backoff BACKOFF_FACTOR
                        Backoff factor for the HTTP requests (default: 1)
```

Copy

Executionâ

The bare minimum execution only requires the fields that are define as being mandatory. All other fields will use their default values but take note that they might need to be adjusted.

info

When targeting a HTTPS endpoint, SSL certificate validation can fail if the certificate is not available in the machine or if it is not correctly configured.

To skip SSL certificate validation the `-sk/--skip-ssl` flag should be passed to the script:

`$ python3 cleanup-alerts-queues.py -c https://localhost:8090/ -sk [Other Options]`

It is suggested to execute the script without the `-e` flag to understand if all parameters are correctly set. Running in this mode will output the number of events it would try to update.

```
$ python3 cleanup-alerts-queues.py -c http://localhost:8090/ -s "Alert not investigated"
Service account password:
Running in test mode...
Found reason Alert not investigated with id d84ad2d1-xxxx-xxxx-xxxx-xxxxxxxxxxxx
Would try to close 11 events.
Finished running cleanup.
```

Copy

Once validated in test mode the script can be run in execution mode by passing the `-e/--exec-mode` flag:

```
$ python3 cleanup-alerts-queues.py -c http://localhost:8090/ -s "Alert not investigated" -e
Service account password:
Running in change mode...
Found reason Alert not investigated with id d84ad2d1-xxxx-xxxx-xxxx-xxxxxxxxxxxx
Clean queue executed for events: [{'id': '55f3cf56-xxxx-xxxx-xxxx-xxxxxxxxxxxx', 'timestamp': 173
2533728533, 'channelId': 'transfers'}, {'id': '6ec36930-xxxx-xxxx-xxxx-xxxxxxxxxxxx',
'timestamp': 1732533729218, 'channelId': 'transfers'}, {'id': '9735dfbf-xxxx-xxxx-xxxx-
xxxxxxxxxxxx', 'timestamp': 1732533730087, 'channelId': 'transfers'}, {'id': '9ad7ab9e-xxxx-xxxx-
xxxx-xxxxxxxxxxxx', 'timestamp': 1732533730728, 'channelId': 'transfers'}, {'id': '9fdd9fb4-xxxx-
xxxx-xxxx-xxxxxxxxxxxx', 'timestamp': 1732537566308, 'channelId': 'transfers'}, {'id': '4d844905-
xxxx-xxxx-xxxx-xxxxxxxxxxxx', 'timestamp': 1732537672724, 'channelId': 'transfers'}, {'id': '9806
7255-xxxx-xxxx-xxxx-xxxxxxxxxxxx', 'timestamp': 1732537673556, 'channelId': 'transfers'}, {'id':
'2064b6c7-xxxx-xxxx-xxxx-xxxxxxxxxxxx', 'timestamp': 1732537674340, 'channelId': 'transfers'}, {'
id': 'a4df2a90-xxxx-xxxx-xxxx-xxxxxxxxxxxx', 'timestamp': 1732537700312, 'channelId':
'transfers'}, {'id': '756e260b-xxxx-xxxx-xxxx-xxxxxxxxxxxx', 'timestamp': 1732537701173, 'channe
lId': 'transfers'}, {'id': '6729e208-xxxx-xxxx-xxxx-xxxxxxxxxxxx', 'timestamp': 1732537701825, '
channelId': 'transfers'}]
Move to close executed for transactions: [{'id': '55f3cf56-xxxx-xxxx-xxxx-xxxxxxxxxxxx', 'timest
amp': 1732533728533, 'channelId': 'transfers'}, {'id': '6ec36930-xxxx-xxxx-xxxx-xxxxxxxxxxxx', '
timestamp': 1732533729218, 'channelId': 'transfers'}, {'id': '9735dfbf-xxxx-xxxx-xxxx-
xxxxxxxxxxxx', 'timestamp': 1732533730087, 'channelId': 'transfers'}, {'id': '9ad7ab9e-xxxx-xxxx-
xxxx-xxxxxxxxxxxx', 'timestamp': 1732533730728, 'channelId': 'transfers'}, {'id': '9fdd9fb4-xxxx-
xxxx-xxxx-xxxxxxxxxxxx', 'timestamp': 1732537566308, 'channelId': 'transfers'}, {'id': '4d844905-
xxxx-xxxx-xxxx-xxxxxxxxxxxx', 'timestamp': 1732537672724, 'channelId': 'transfers'}, {'id': '9806
7255-xxxx-xxxx-xxxx-xxxxxxxxxxxx', 'timestamp': 1732537673556, 'channelId': 'transfers'}, {'id':
```

```
'2064b6c7-xxxx-xxxx-xxxx-xxxxxxxxxxxx', 'timestamp': 1732537674340, 'channelId': 'transfers'}, {'id': 'a4df2a90-xxxx-xxxx-xxxx-xxxxxxxxxxxx', 'timestamp': 1732537700312, 'channelId': 'transfers'}, {'id': '756e260b-xxxx-xxxx-xxxx-xxxxxxxxxxxx', 'timestamp': 1732537701173, 'channelId': 'transfers'}, {'id': '6729e208-xxxx-xxxx-xxxx-xxxxxxxxxxxx', 'timestamp': 1732537701825, 'channelId': 'transfers'}]
There are no more events to update that matches the condition
Finished running cleanup.
```

Copy

As you can see, when running in execution mode, the script will output the ids of the events it was able to close, which can be used for validation purposes.

Filtering eventsâ

There are some filtering options that can be used to better determine the amount of events to close.

Queuesâ

To filter events by assigned queue the `-wq, --without_queue` flag can be used. If provided, the script will only search for events that are not assigned to any queue and therefore will not try to clear any queues.

If not provided, the script will search for all events that are assigned or not to queues. Take into consideration the script will remove the queue from events meaning that in case of a re-run the output may not be consistent.

Tenantsâ

In multi tenancy setups the options `-ats, --all-tenants` and `-tk, --tenant-keys` can be used to filter events based on all or only the provided tenants. The second option supports one or more tenant ids, where `tenant id` is the Tenant's Unique ID as shown in the Pulse configuration page.

As an example, consider a multi tenant setup where:

- Tenant 1 has the Unique ID `tenant_1`
- Tenant 2 has the Unique ID `tenant_2`
- Tenant 3 has the Unique ID `tenant_3`

And the following cases:

- Clear events from all tenants
- Clear events from tenants 1 and 2
- Clear events from tenant 3

This would require running the script as follows:

```
# Case 1
$ python3 cleanup-alerts-queues.py -c http://localhost:8090/ (...) -ats

# Case 2
$ python3 cleanup-alerts-queues.py -c http://localhost:8090/ (...) -tk "tenant_1" "tenant_2"

# Case 3
$ python3 cleanup-alerts-queues.py -c http://localhost:8090/ (...) -tk "tenant_3"
```

Copy

warning

Please note that these options are mutually exclusive but still required.

Also note that `-ats, --all-tenants` should be used carefully in multi tenancy setups to prevent closing events wrongfully. Consider running the script in test mode prior to execution mode.

Validationâ

Validation can also be done in the UI by searching for events that fall in the same criteria. If the script processed every event correctly the search should bear no results.

To validate a subset of events one can check the operation status, close reason and queue in the event details of the events, as illustrated below.

Every operation done by the script is audited and can be seen in the event details page as well.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/operations-guide/how-to/cassandra-profiles-migration

On this page

Migrate Cassandra Profile Store

This document provides a step-by-step guide for migrating the Cassandra Profile Store, which stores the scheduled metrics (SMJs) values.

Please note this page outlines one of the many approaches to migrate a keyspace/table. This migration is not mandatory but reduces the possibility of metric collisions between channels.

Migration Overviewâ

Migrating the Profile Store from a single keyspace/table to multiple keyspaces/tables requires several steps to ensure that the metrics are not unavailable during the migration process. The process includes the following steps:

1. Prepare the new Cassandra keyspace and table by creating all the necessary data structures
2. Backfill the new table with the existing data to avoid null/zero metrics
3. Update the Pulse properties to start using the new keyspace/table
4. Clean up the old keyspace/table

Migration Stepsâ

In this section you can find a description of the migration steps.

For this migration process explanation, we will assume that the objective is to migrate the table `profiles` in the keyspace `profiles` to a new table `uk_oub_profiles` in a new keyspace `uk_profiles`.

Prepare the new Cassandra keyspace and/or tableâ

The first step is to prepare the new keyspace and/or table, ensuring it has the same definition has the existing one.

This can be done by following the next steps:

1. Describe the existing keyspace/table

```
DESCRIBE profiles;
```

Copy

2. Copy the output from the previous command, e.g.

```
CREATE KEYSPACE profiles WITH replication = {'class': 'NetworkTopologyStrategy', 'dc1': '1'}  AND durable_writes = true;

CREATE TABLE profiles.profiles (
    key text,
    column1 text,
    value text,
    PRIMARY KEY (key, column1)
) WITH CLUSTERING ORDER BY (column1 ASC)
    AND additional_write_policy = '99p'
    AND bloom_filter_fp_chance = 0.1
    AND caching = {'keys': 'ALL', 'rows_per_partition': 'NONE'}
    AND cdc = false
    AND comment = ''
    AND compaction = {'class': 'org.apache.cassandra.db.compaction.LeveledCompactionStrategy', 'max_threshold': '32', 'min_threshold': '4'}
```

```
    AND compression = {'chunk_length_in_kb': '16', 'class': 'org.apache.cassandra.io.comp
ress.LZ4Compressor'}
    AND memtable = 'default'
    AND crc_check_chance = 1.0
    AND default_time_to_live = 0
    AND extensions = {}
    AND gc_grace_seconds = 864000
    AND max_index_interval = 2048
    AND memtable_flush_period_in_ms = 0
    AND min_index_interval = 128
    AND read_repair = 'BLOCKING'
    AND speculative_retry = '99p';
```

Copy

3. Rename the keyspace/table and create the new keyspace/table

```
CREATE KEYSPACE uk_profiles WITH replication = {'class': 'NetworkTopologyStrategy', 'dc1':
 '1'}  AND durable_writes = true;

CREATE TABLE uk_profiles.uk_oub_profiles (
    key text,
    column1 text,
    value text,
    PRIMARY KEY (key, column1)
) WITH CLUSTERING ORDER BY (column1 ASC)
    AND additional_write_policy = '99p'
    AND bloom_filter_fp_chance = 0.1
    AND caching = {'keys': 'ALL', 'rows_per_partition': 'NONE'}
    AND cdc = false
    AND comment = ''
    AND compaction = {'class': 'org.apache.cassandra.db.compaction.LeveledCompactionStrat
egy', 'max_threshold': '32', 'min_threshold': '4'}
    AND compression = {'chunk_length_in_kb': '16', 'class': 'org.apache.cassandra.io.comp
ress.LZ4Compressor'}
    AND memtable = 'default'
    AND crc_check_chance = 1.0
    AND default_time_to_live = 0
    AND extensions = {}
    AND gc_grace_seconds = 864000
    AND max_index_interval = 2048
    AND memtable_flush_period_in_ms = 0
    AND min_index_interval = 128
    AND read_repair = 'BLOCKING'
    AND speculative_retry = '99p';
```

Copy

warning

Please note the commands above have default configuration values (e.g. replication factor) which might have to be adjusted according to the cluster configuration.

Backfill dataâ

To ensure that we have consistent metrics values while migrating the profile store keyspace and/or table, we should backfill it with the current data. Backfilling is an essential step to guarantee that metrics are always available in the current/new Pulse cluster.

This can be done using the CQL `COPY` command to export the current data and import it into the new table. Example:

```
COPY profiles.profiles TO 'profiles.csv';
COPY uk_profiles.uk_oub_profiles FROM 'profiles.csv';
```

Copy

Please refer to the official Cassandra documentation for more details regarding the CQL `COPY` command:

- COPY TO
- COPY FROM

Update Pulse propertiesâ

To complete the migration, some Pulse properties need to be redirected to the new keyspace/table:

- Keyspace name (Ansible variable: `cassandra_pulse_keyspace`):

  ```
  pulse.app.
  {appSlug}.services.rte.workflow.plan.scheduledMetrics.profileStore.cassandra.keyspace.name
  ```

  Copy
- Table name (Ansible variable: `cassandra_table_name`):

  ```
  pulse.app.{appSlug}.services.rte.workflow.plan.scheduledMetrics.profileStore.table
  ```

  Copy

The properties can be updated through Pulse UI or by using Ansible variables (requires a deploy). When updated through the Pulse UI, a publish with restart is required but all changes will be lost once a Blue/Green deployment is performed.

The application variable override should be used to set different keyspaces/tables for different applications. For example, considering a single UK keyspace and different tables per application, the configuration would be:

```
cassandra_pulse_keyspace                 : "uk_profiles"
uk_outbound_payments_cassandra_table_name: "uk_oub_profiles"
uk_cards_cassandra_table_name            : "uk_cds_profiles"
uk_inbound_payments_cassandra_table_name : "uk_inb_profiles"
```

Copy

After these changes are done and all applications are published, the migration process is now complete.

Clean-upâ

Once validated that everything is working as expected, the old keyspace/table can be safely removed:

```
DROP TABLE profiles.profiles;
DROP KEYSPACE profiles;
```

Copy

On this page

Change Log Levels at Runtime

Architectureâ

Feedzai's product is a distributed system with multiple components and each of those components is typically replicated. This can make investigating issues challenging and having more logs enabled could help in those investigations.

Typically changing log levels can be achieved by editing the local `logger.xml` files present in each machine but as we do not have access to them, an alternative solution was required. We've developed an API that runs in the Pulse Server which is able to affect log level changes, during runtime, across in each Pulse Server, Pulse Application and Case Manager instance.

This API writes a message to a new exchange in RabbitMQ (`hsbc-logging-sync`) and each component knows to read relevant messages from that queue and change their local log levels. The routing key of the message is the component which should affected by the log change.

On the consumer side, each Pulse Server, Pulse Application and Case Manager instance will have their own queues bound to this exchange and listening to particular routing keys. The table below lists the queues and bindings that are created for the exchange.

Component	Queue	Routing Key
Pulse Server	fdz-sq-hsbc-logging-sync-pulse-server-{host}	pulse-server
Pulse Application	fdz-sq-hsbc-logging-sync-pulse-application-{app_slug}-{host}	pulse-application:{app-slug}
Pulse Application	fdz-sq-hsbc-logging-sync-pulse-application-{app_slug}-{host}	pulse-application:*
Case Manager	fdz-sq-hsbc-logging-sync-case-manager-{host}	case-manager

All queues have the host in the queue name, ensuring that each replica will receive the messages. Each Pulse Application has two routing keys bound to their queue: one that affects all Pulse Applications and another specific to that app's slug.

The image below depicts what happens when a request to change a log level for a Pulse Application with slug `outbound` is sent to the API. First, upon receiving the request, the API will write a message to RabbitMQ using the routing key `pulse-application:outbound` . RabbitMQ will then route the message to the queues that are expecting that routing key. In this case, those would be `pulse-app-outbound-10.0.0.30` and `pulse-app-outbound-10.0.0.47` (note that the queue names have been simplified in this example). Eventually the consumers in each Pulse Application listening to those queues will notice the new messages and handle them by changing the log levels locally, according to the message payload.

The API lives in the Pulse Server and therefore inherits the authentication and authorization mechanisms there present.

What if Case Manager is connected to a different RabbitMQ cluster?

As the API lives in the Pulse Server, it inherits its RabbitMQ connection. However, it is possible to configure Case Manager to use a different RabbitMQ cluster. The API detects that by reading Pulse's Alert Storage Service configuration via the following properties:

- `pulse.global.services.AlertStorageService.dc.dc1.*`

If the brokers set in the Alert Storage Service global configuration are different from the ones configured for Pulse messaging, the API will connect to both clusters and route messages accordingly:

- any Pulse related requests will be written to the exchange in Pulse's RabbitMQ cluster
- any Case Manager related requests will be written to the exchange in Case Manager's RabbitMQ cluster

Loggers APIâ

The entrypoint to interact with this functionality is the Loggers API, available in each Pulse Server.

URL: `/admin/api/loggers/{targetComponent}/{loggerName}/{loggerLevel}`

Method: `POST` or `GET` (which allows the functionality to be accessed directly from the browser, using the permissions of the logged in user)

Authentication: This endpoint uses standard Pulse authentication. When using a `POST` request, a cookie must be sent in the header (see the API Authentication guide). When using a `GET` request via the browser, the user must have previously logged in via the Pulse UI in the same browser window.

Authorization: This endpoint required the logged in user to have the `security:logging:update` permission.

Parameters:

Name	Description
`targetComponent`	The component that should have the logger changed (this is used as the routing key in RabbitMQ). Valid values are: `pulse-server`, `pulse-application:{slug}`, `pulse-application:*` or `case-manager`.
`loggerName`	The logger name to be changed, for example `com.feedzai.hsbc.SomeClass`
`loggerLevel`	The logger level to be set. Valid values are `trace`, `debug`, `info`, `warn`, `error`, or `off` (case-insensitive)

Responses:

Code	Content-type	Response
`200`	`application/json`	OK - JSON that echoes back the parameters sent in the request.
`400`	`application/json`	Bad Request - JSON with the error message.
`401`	`text/html`	Unauthorized - Standard Jetty error page for unauthorized requests.
`403`	`application/json`	Forbidden - The logged in user is missing the required permissions.

Example using cURL:

```
curl -X POST http://<PULSE_UI>/admin/api/loggers/pulse-server/com.feedzai.pulse.service.jobmanag
er.exec.ExecutionManagerImpl/error \
  -H "Cookie: SSOcookie=cee99ab5-xxxx-xxxx-xxxx-xxxxxxxxxxxx" \
  -H "X-Requested-With: XMLHttpRequest"
```

Copy

Example response

```
{
  "component": "pulse-application:*",
  "loggerName": "com.feedzai.pulse.service.jobmanager.exec.ExecutionManagerImpl",
  "loggerLevel": "error"
}
```

Copy

Enabling Probe loggingâ

As a particularly useful case, the logging levels API can also be used to enable Pulse and Case Manager probes logging by setting specific probes-related loggers to log at `trace` level. Probes logging provides detailed timing information for incoming events and various other information that can be used in fine-grained performance debugging and analysis.

As probes logging creates a very large throughput of logging, which results in increased infrastructure and storage costs, and can even impact application performance, they should be used sparingly and only enabled for short periods of time when performing analysis and debugging, after which they should be changed back to `info` level.

Enabling Pulse Application probes for application with app slug (URL alias) `tfrb` - cURL

```
curl -X POST http://<PULSE_UI>/admin/api/loggers/pulse-application:tfrb/com.feedzai.metrics.probes.TextAbstractProbe/trace \
  -H "Cookie: SSOcookie=cee99ab5-xxxx-xxxx-xxxx-xxxxxxxxxxxx" \
  -H "X-Requested-With: XMLHttpRequest"
```

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Disabling Pulse Application probes for application with app slug (URL alias) `tfrb` - cURL

```
curl -X POST http://<PULSE_UI>/admin/api/loggers/pulse-application:tfrb/com.feedzai.metrics.probes.TextAbstractProbe/off \
  -H "Cookie: SSOcookie=cee99ab5-xxxx-xxxx-xxxx-xxxxxxxxxxxx" \
  -H "X-Requested-With: XMLHttpRequest"
```

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Enabling Case Manager probes - cURL

```
curl -X POST http://<PULSE_UI>/admin/api/loggers/case-manager/AlertManagerProbe/trace \
  -H "Cookie: SSOcookie=cee99ab5-xxxx-xxxx-xxxx-xxxxxxxxxxxx" \
  -H "X-Requested-With: XMLHttpRequest"
```

Copy

Disabling Case Manager probes - cURL

```
curl -X POST http://<PULSE_UI>/admin/api/loggers/case-manager/AlertManagerProbe/off \
  -H "Cookie: SSOcookie=cee99ab5-xxxx-xxxx-xxxx-xxxxxxxxxxxx" \
  -H "X-Requested-With: XMLHttpRequest"
```

Copy

On this page

Export HAR files

In this page you can find the steps to export a HAR (HTTP Archive) file from different browsers (Chrome, Firefox and Edge). A HAR file contains detailed network requests made by the browser when loading a webpage. It is especially useful for troubleshooting issues like slow page loads, network errors, or failed requests. By generating and sharing the HAR file with the Feedzai team, they can analyze the issue more effectively.

Key Sections:â

Export HAR file in Google Chromeâ

Please find below the steps to export a HAR file in Chrome:

1. Open Chrome and go to the problematic page
   - Navigate to the page where the issue occurs.
2. Access Developer Tools
   - Click the **three dots (â®)** in the top right corner.
   - Go to **More Tools > Developer Tools** or press `Ctrl + Shift + I` (`Cmd + Option + I` on Mac).
   - This will open the Developer Tools panel.
3. Select the Network Tab
   - In the Developer Tools panel, click the **Network** tab.
4. Start Recording
   - Ensure the **record button** (a round button in the upper-left corner) is **red**. If itâs **grey**, click it to start recording.
   - Check the **Preserve log** box to ensure all requests are recorded even when the page is refreshed.
5. Reproduce the issue
   - Reload the page and replicate the issue so the network requests can be captured.
6. Export the HAR file
   - After reproducing the issue, click the Export button to download the HAR file. The file will be saved in `.har` format.

Export HAR file in Mozilla Firefoxâ

Please find below the steps to export a HAR file in Firefox:

1. Open Firefox and go to the problematic page
   - Navigate to the page with the issue.

2. Open the Network Monitor
    - Press `Ctrl + Shift + E` (`Cmd + Option + E` on Mac) to open the Network Monitor.
3. Reproduce the issue
    - Perform the actions that lead to the issue while the network requests are being logged.
4. Export HAR file
    - Right-click on any entry in the File column and select **Save All As HAR**.
    - Save the HAR file to your computer.

Export HAR file in Microsoft Edgeâ

Please find below the steps to export a HAR file in Edge:

1. Open Edge and go to the problematic page
    - Navigate to the page where the issue occurs.
2. Access Developer Tools
    - Press `F12` or **Right-click > Inspect** to open the Developer Tools.
3. Select the Network Tab
    - Click the **Network** tab.
4. Start Recording
    - Ensure the **record button** is **red**. If itâs **grey**, click it to start recording.
    - Check the **Preserve log** box to ensure all requests are recorded even when the page is refreshed.
5. Reproduce the issue
    - Perform the actions that lead to the error while the network requests are being logged.
6. Export HAR file
    - After reproducing the issue, click the Export button to download the HAR file. The file will be saved in `.har` format.

Example: exporting a HAR file for a failed Case Manager search in Chromeâ

In this section you can find an example of how to export a HAR file after a failed search in Case Manager.

1. Open Chrome and navigate to the page where the search functionality is located.
2. Open Developer Tools:
    - Click the **three vertical dots** in the top-right corner of Chrome.
    - Go to **More Tools > Developer Tools**.
3. Go to the Network Tab:
    - Click on the **Network** tab in Developer Tools.
    - Ensure the **Preserve log** checkbox is checked. This ensures that the network log is retained even after the page reloads.
    - Make sure the **Record button** (round red circle in the top-left corner) is active (if itâs grey, click it to start recording).

4. Clear existing Logs:

    ◦ Before triggering the search, click the **clear button** (a small grey circle with a line through it) to clear any old network requests from the log.

5. Perform the failing search:

    ◦ In the webpage, perform the search action that fails

    ◦ As the request is made, it will appear in the **Network** tab.

6. Check for error responses:

    ◦ Look for any requests that have failed. These might have:

        ▪ **4xx (Client Errors)**: Such as **404 Not Found** or **400 Bad Request**.

        ▪ **5xx (Server Errors)**: Such as **500 Internal Server** Error or **502 Bad Gateway**.

7. Export the HAR file

    ◦ In the **Network** tab, youâ□□ll see an export button, which looks like a downward arrow icon.

    ◦ Click this button to export the HAR file.

    ◦ Choose a location to save the HAR file on your computer that will be shared.

On this page

Export and Import Pulse applications

Pulse offers application import and export to facilitate interoperability between installations and enable precise application state backups.

Exporting an Applicationâ

Applications are exported in a single PAPP file that contains all the application resources like schemas, workflows and so on. In some situations there are security resources that are also exported with the applications; more on that later.

warning

Note that application exports may contain personally-identifiable information (PII) if this data is directly used in any application resources, for example if a rule or a plan map operator checks a specific customer name. Care should be taken to ensure this is not the case before sharing an exported application.

PII is more likely to be found in Pulse lists, and therefore Feedzai always recommends that applications are exported without list items.

Users can easily export an application by selecting the `Export Application` option within the application configuration menu, which initiates the download of the application's PAPP file. Please note, this method of exporting the application will also export the list items stored in the application.

Applications can also be exported using a REST API endpoint. The API exposes a single authenticated endpoint that can be used to export the PAPP file. Authentication details are available in [API Authentication](#).

Calling the endpoint will return a similar PAPP file as if it has downloaded via the UI, but there are additional parameters can be tweaked.

`GET` `/pulseviews/api/apps/{id}/export?excludeItems={excludeItems}`

Parametersâ

Name	Type	Data type	Description
`id`	required	string	The ID of the application to export.
`excludeItems`	optional	boolean	Whether lists should be excluded from the export. Defaults to `false` but it is recommend to set to `true`.

Responsesâ

Code	Content-type	Response
`200`	`text/plain`	`200 OK`
`400`	`text/plain`	`400 Bad Request`
`401`	`text/html;charset=iso-8859-1`	`Error 401 Unauthorized`

Example cURLâ

```
curl -X GET http://<COMPONENT_URL>/apps/123456789/export?excludeItems=true \
  -H "Cookie: SSOcookie=b7ea8372-2108-46c8-9cae-d16a0c153dfa" \
```

```
  -H "X-Requested-With: XMLHttpRequest" \
  --output test_app.papp
```

Copy

warning

When exporting applications with `excludeItems` set to `false` (the default), the lists are fully loaded into memory, which often results in memory issues. We recommend that application exports are done via API requests with `excludeItems=true`.

Importing an Applicationâ

Importing an application is usually as simple as exporting it. The user only has to click on the `Import Application` button present in the Applications tab and an informative wizard will guide the entire process.

The Import Application wizard can have one to three steps depending on the complexity of the application being imported. If the application has neither security Groups nor Multi-Tenant resources, only one step will be needed which allows the user to upload the application PAPP file.

Applications With Multi-Tenancy Enabledâ

Importing and exporting applications that contain Multi-Tenancy resources do not require any additional configuration steps. However, there are a few considerations worth mentioning regarding the Application import with Multi-Tenancy. Tenants stored in the PAPP file are merged automatically by Pulse using the following algorithm:

1. If the Tenant being imported has the same Unique Id (Key) of a Tenant in the current Pulse Installation, all resources of the Tenant being imported will be added/mapped to the Tenant in the current Pulse Installation.
2. If there is no Tenant with the the same Unique Id in the current Pulse Installation, a new Tenant will be created verifying before hand that there is no naming collision.
    - If there is a name collision, the Tenant Unique Id will be appended at the end of the name, e.g. "Tenant Name (Tenant Unique Id)".
3. All the tenants stored in the PAPP file will be imported to the current Pulse Installation. Even if they are not being explicitly used in the Application being imported. Additional details:
    - Even the Tenants without any explicit resource are used in runtime in case of a Rules of the type "Shared by All Tenants".
    - When Exporting an Application, all the Tenants in the Pulse installation are stored in the PAPP file.

On this page

API Authentication

In this page you can find an outline of the authentication process, necessary parameters, and the required flow to interact with the Pulse APIs, for example, in the context of authenticating a service account before using the Pulse and Case Manager servlets that allow log levels changes at runtime or export and import Pulse applications.

Authentication Processâ

Pulse and Case Manager APIs require authentication for accessing and performing actions. Specifically, Basic Authentication is used for authenticating service accounts through their credentials (username and password). These credentials need to be encoded in Base64 before being sent in the API requests. The authentication flow is as follows:

- Take note of a Pulse service account's username and password
- Encode the username and password using a Base64 encoder of your choosing
- Send a request to the authentication endpoint with the credentials in the request body
- If the request succeeds, Pulse will return a session cookie that can be used in subsequent API calls, e.g. a request to the logging API

You can find below the authentication diagram:

After receiving a successful response the cookie can be used to make authenticated requests to the Pulse API to, for example, change logging levels via the Logging API.

API definitionâ

In this section you can find the authentication API endpoint definition.

Endpoint URLâ

`POST` `/pulseviews/api/sessions` `(authenticates a service account)`

Parametersâ

None

Responsesâ

Code	Content-type	Response
`200`	`application/json`	`JSON object with user information`
`401`	`text/html;charset=iso-8859-1`	`Error 401 Unauthorized`

Example cURLâ

```
curl -X POST http://<PULSE_URL>/pulseviews/api/sessions \
  -H "Content-Type: application/json" \
  -H "X-Requested-With: XMLHttpRequest" \
  -d '{"username": "<base64_username>","password": "base64_password"}' -c -
```

Copy

Example responseâ

```
{"blockedState":"NONE","createdAt":1732196807880,...}
#HttpOnly_localhost FALSE   /   FALSE   0   SSOcookie   cee99ab5-xxxx-xxxx-xxxx-xxxxxxxxxxxx
```

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RabbitMQ Segregation for Pulse and Case Manager

This document provides a step-by-step guide for migrating from a single RabbitMQ cluster to two segregated clusters: one dedicated to Pulse and another to Case Manager. The goal is to ensure a transition with no message loss, supporting a zero-downtime deployment.

Migration Overviewâ

Migrating RabbitMQ from a single shared cluster to segregated clusters for Pulse and Case Manager requires several steps to ensure zero message loss and minimal downtime. The process includes the following steps:

1. Prepare the new RabbitMQ cluster by creating all the necessary exchanges and queues for Case Manager
2. Enable maintenance mode in Case Manager to ensure no messages are consumed during the process
3. Create a RabbitMQ federation policy to forward messages from the old to the new cluster, preventing data loss and consumption duplication
4. Deploy Case Manager to consume messages from the new cluster
5. Deploy Pulse to publish Case Manager messages to the new cluster
6. Clean up the old Case Manager cluster once migration is complete

The federation step is crucial for a seamless migration, as it ensures that messages published to the old cluster during the transition are not lost and are safely streamed to the new cluster. This is achieved by:

- Setting a one-way policy that replicates messages published to the old RabbitMQ cluster to the new one
- Setting the acknowledge mode to on-confirm, so messages are only consumed a single time

It is expected that during the migration, both clusters will hold messages based on the federation policy above, which should start to be consumed once maintenance mode is disabled.

The diagram below illustrates the expected architecture after the migration:

Migration Stepsâ

In this section you can find a description of the migration steps.

Prepare the new RabbitMQ clusterâ

The first step is to prepare the new cluster, ensuring it has all the necessary queues/exchanges that Case Manager requires. This can be done in several ways, for instance:

- Creating a temporary Case Manager cluster, configured to use the new cluster, and decommissioning it afterwards
- Creating all the resources manually using the RabbitMQ Management UI
- Using `rabbitmqctl` to list and create all the necessary resources

warning

To avoid message loss it is crucial that all queues/exchanges are created in the new RabbitMQ cluster before enabling the federation policy.

If the first option is used make sure to remove the temporary cluster before creating the federation policy to ensure no messages are consumed.

Enable maintenance mode in Case Managerâ

To ensure that no messages are processed during the migration, maintenance mode should be enabled in Case Manager. By enabling maintenance mode it is guaranteed that messages are only consumed by the new cluster once the migration is concluded.

Set Up Federationâ

Setting up a federation policy to replicate all Case Manager messages to the new cluster guarantees that no messages are lost in the process and that each is only consumed once. The process is described as follows:

- Enable the federation plugin on both clusters:

  ```
  rabbitmq-plugins enable rabbitmq_federation
  rabbitmq-plugins enable rabbitmq_federation_management
  ```

  Copy
- On the new cluster, configure a federation upstream policy to connect to the old cluster. The 'ack-mode' is set to 'on-confirm' on the configuration of the federation, ensuring messages are only removed from the old cluster after being safely received and consumed by the new cluster. This establishes a link so the new cluster can pull messages from the old cluster during migration:
  - `<new-vhost>` : the vhost in the new Case Manager RabbitMQ cluster. It can be the same vhost name as in the old cluster
  - `<user>/<pass>` : RabbitMQ credentials with access to the old cluster
  - `<old-cluster>` : hostname or IP of the old cluster
  - `<old-vhost>` : the vhost used by Case Manager on the old cluster

  ```
  rabbitmqctl set_parameter -p <new-vhost> federation-upstream cm-federation '{"uri":
  ["amqp://<user>:<pass>@<old-cluster>/<old-vhost>"], "ack-mode":"on-confirm"}'
  ```

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- Apply a federation policy on the new cluster to specify which exchanges should receive messages from the old cluster via federation. Federate only the exchanges used by Case Manager (`CM_UPDATES_CASES` , `CM_UPDATES_EVENTS` , and `EVENT_QUEUE_.*`):

  ```
  rabbitmqctl set_policy -p <new-vhost> --apply-to exchanges cm-federation-policy
  "^(CM_UPDATES_CASES|CM_UPDATES_EVENTS|EVENT_QUEUE_.*)$" '{"federation-upstream":"cm-
  federation"}'
  ```

  Copy

warning

Ensure that `<new-vhost>` matches the vhost configured via Case Manager Ansible variables for the new cluster. For Pulse (Alert Storage Service), use the vhost set by the corresponding Pulse Ansible variable.

The diagram below illustrates the federation setup between the old and new RabbitMQ clusters. In particular, this shows how the architecture will look when both Pulse and Case Manager are still connected to the old RabbitMQ clusters, but after the new RabbitMQ cluster has been created and the federation link has been set up

Deploy Pulse and Case Managerâ

Following the federation setup, both Pulse and Case Manager should be deployed. The deployment is required so Pulse publishes messages to the new RabbitMQ cluster and Case Manager consumes messages from the new cluster.

To achieve this, the Ansible variables file should be updated for both components:

- Update configuration variables:
  - `rabbitmq_casemanager_brokers` : set to the new RabbitMQ cluster brokers
  - `rabbitmq_casemanager_vhost` : set to the new RabbitMQ cluster vhost

- Deploy Case Manager to the new RabbitMQ cluster. It will start consuming messages from the new cluster and pulling any remaining messages from the old cluster via federation
  - Before disabling maintenance mode on the new Case Manager deployment, stop/decommission the old Case Manager cluster. This ensures only the new cluster is active and prevents message consumption conflicts
- Deploy Pulse which will start publishing messages to the new cluster
- With federation configured using `ack-mode: on-confirm`, the federation link acts as the consumer for the old cluster, safely ensuring messages are only removed from the old cluster after being safely received and consumed by the new cluster

Please refer to Case Manager Configuration Reference and Pulse Configuration Reference for more information.

Cutover and cleanâ

warning

Make sure all Case Manager queues in the old RabbitMQ cluster are drained before proceeding with the cleanup.

Once both components are deployed and using the correct RabbitMQ cluster, any leftover configurations/resources can be safely removed:

- Remove federation policies and upstreams from the new cluster:

```
rabbitmqctl clear_policy -p <new-vhost> cm-federation-policy
rabbitmqctl clear_parameter -p <new-vhost> federation-upstream cm-federation
```

Copy

- Clean the old cluster Case Manager exchanges and queues:

```
rabbitmqctl delete_exchange -p <old-vhost> CM_UPDATES_CASES
rabbitmqctl delete_exchange -p <old-vhost> CM_UPDATES_EVENTS
rabbitmqctl delete_queue -p <old-vhost> EVENT_QUEUE_.*
```

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Use Enrichers in Case Manager

This guide outlines the process for configuring enrichers within Case Manager. Enrichers allow you to map raw data values from an event to more user-friendly names, enhancing clarity and readability for analysts. While the initial setup of enrichers requires assistance from the project team, once configured in the application, you can independently apply them to any field within Case Manager using the Self-Service UI functionality.

How to use an enricherâ

Using the Self-Service UI functionality from Case Manager, custom enricher types can be configured. Follow these steps to configure a field to use an enricher:

1. Within the Self-Service UI mode, edit an existing field or create a new one.
2. Fill the *Label* and *Field* form elements as usual.
3. Click on *Advanced* to expand the advanced configuration text box where the field JSON can be modified.
4. Edit the `type` field in the JSON to use the intended enricher type.

The following screenshot depicts the configuration of the field `event.fields.receiver_transaction_currency` using the `currencyEnrich` enricher which converts a numeric currency code to a human-readable format.

cm-configure-enricher

info

Case Manager will underline the type in red and say *"The code snippet is invalid. It can, however, still be submitted."*. This is normal as Case Manager is not capable of validating custom enricher types.

The *Preview* section can be used to confirm that the enricher is working as expected. In the example image, we can see the currency being displayed as `GBP`, confirming the enricher is properly mapping the source value (`826` in this case).

Available enrichersâ

The following enrichers are available:

Enricher Type	Enricher Description
`authMethodEnrich`	Enricher for the Authentication Method.
`cardCustomerPresentDetailsEnrich`	Enricher for the Card customer present details.
`countryEnrich`	Enricher for the Countries sent as 3-digit codes (ISO 3166-1 numeric).
`currencyEnrich`	Enricher for the Currencies sent as 3-digit codes (ISO 4217 numeric).
`deviceBankingType`	Enricher for the Device Banking Type.
`digitalWalletTokenTypeEnrich`	Enricher for the Digital Wallet token type.
`eventDescription`	Enricher for the Event Description.
`inboundPaymentTypeEnrich`	Enricher for the Inbound Payment Type.
`isBatchEnrich`	Enricher for the Batch flag.
`paymentStatusEnrich`	Enricher for the Payment Status.
`paymentTypeEnrich`	Enricher for the Payment Type.

Enricher Type	Enricher Description
`posCardholderVerMethodEnrich`	Enricher for the Cardholder verification method (Ecomm).
`senderTransactionCodeEnrich`	Enricher for the Sender Transaction Code.
`sicEnrich`	Enricher for the POS Code (sic).
`transReversalIndEnrich`	Enricher for the Transaction reversal indicator.
`transactionStatusEnrich`	Enricher for the Transaction Status Enrich.

Mappingsâ

This section includes the full mapping for each enricher.

Show the mapping for `authMethodEnrich`

Source Value	Enriched Value
1	One Time Password/SecureID Token
L	Login Password Verification
P	Pin Verification
I	ID verification
A	Address verification
Q	Secret Q and A
N	Pass Phase/Memorable Name
D	Digital Certificate
O	Others
U	unknown
V	Voice Recognition
F	Facial Recognition
E	Iris Recognition
G	Fingerprint Recognition
B	Biometric verification

Show the mapping for `cardCustomerPresentDetailsEnrich`

Source Value	Enriched Value
0	Cardholder was present
1	Cardholder was not present, unspecified
2	Cardholder was not present, mail/facsimile order
3	Cardholder was not present, phone/ARU order
4	Cardholder was not present, standing order/recurring transaction
5	Electronic order

Show the mapping for `countryEnrich`

Source Value	Enriched Value
100	Bulgaria
104	Myanmar
108	Burundi
112	Belarus

Source Value	Enriched Value
116	Cambodia
120	Cameroon
124	Canada
132	Cabo Verde
136	Cayman Islands (the)
140	Central African Republic (the)
144	Sri Lanka
148	Chad
152	Chile
156	China
158	Taiwan (Province of China)
162	Christmas Island
166	Cocos (Keeling) Islands (the)
170	Colombia
174	Comoros (the)
175	Mayotte
178	Congo (the)
180	Congo (the Democratic Republic of the)
184	Cook Islands (the)
188	Costa Rica
191	Croatia
192	Cuba
196	Cyprus
203	Czechia
204	Benin
208	Denmark
212	Dominica
214	Dominican Republic (the)
218	Ecuador
222	El Salvador
226	Equatorial Guinea
231	Ethiopia
232	Eritrea
233	Estonia
234	Faroe Islands (the)
238	Falkland Islands (the) [Malvinas]
239	South Georgia and the South Sandwich Islands
242	Fiji
246	Finland

Source Value	Enriched Value
248	Ã□land Islands
250	France
254	French Guiana
258	French Polynesia
260	French Southern Territories (the)
262	Djibouti
266	Gabon
268	Georgia
270	Gambia (the)
275	Palestine, State of
276	Germany
288	Ghana
292	Gibraltar
296	Kiribati
300	Greece
304	Greenland
308	Grenada
312	Guadeloupe
316	Guam
320	Guatemala
324	Guinea
328	Guyana
332	Haiti
334	Heard Island and McDonald Islands
336	Holy See (the)
340	Honduras
344	Hong Kong
348	Hungary
352	Iceland
356	India
360	Indonesia
364	Iran (Islamic Republic of)
368	Iraq
372	Ireland
376	Israel
380	Italy
384	CÃ´te d'Ivoire
388	Jamaica
392	Japan

Source Value	Enriched Value
398	Kazakhstan
400	Jordan
404	Kenya
408	Korea (the Democratic People's Republic of)
410	Korea (the Republic of)
414	Kuwait
417	Kyrgyzstan
418	Lao People's Democratic Republic (the)
422	Lebanon
426	Lesotho
428	Latvia
430	Liberia
434	Libya
438	Liechtenstein
440	Lithuania
442	Luxembourg
446	Macao
450	Madagascar
454	Malawi
458	Malaysia
462	Maldives
466	Mali
470	Malta
474	Martinique
478	Mauritania
480	Mauritius
484	Mexico
492	Monaco
496	Mongolia
498	Moldova (the Republic of)
499	Montenegro
500	Montserrat
504	Morocco
508	Mozambique
512	Oman
516	Namibia
520	Nauru
524	Nepal
528	Netherlands (the)

Source Value	Enriched Value
531	CuraÃ§ao
533	Aruba
535	Bonaire, Sint Eustatius and Saba
540	New Caledonia
548	Vanuatu
554	New Zealand
558	Nicaragua
562	Niger (the)
566	Nigeria
570	Niue
574	Norfolk Island
578	Norway
580	Northern Mariana Islands (the)
581	United States Minor Outlying Islands (the)
583	Micronesia (Federated States of)
584	Marshall Islands (the)
585	Palau
586	Pakistan
591	Panama
598	Papua New Guinea
600	Paraguay
604	Peru
608	Philippines (the)
612	Pitcairn
616	Poland
620	Portugal
624	Guinea-Bissau
626	Timor-Leste
630	Puerto Rico
634	Qatar
638	RÃ©union
642	Romania
643	Russian Federation (the)
646	Rwanda
652	Saint BarthÃ©lemy
654	Saint Helena, Ascension and Tristan da Cunha
659	Saint Kitts and Nevis
660	Anguilla
662	Saint Lucia

Source Value	Enriched Value
663	Saint Martin (French part)
666	Saint Pierre and Miquelon
670	Saint Vincent and the Grenadines
674	San Marino
678	Sao Tome and Principe
682	Saudi Arabia
686	Senegal
688	Serbia
690	Seychelles
694	Sierra Leone
702	Singapore
703	Slovakia
704	Viet Nam
705	Slovenia
706	Somalia
710	South Africa
716	Zimbabwe
724	Spain
728	South Sudan
729	Sudan (the)
732	Western Sahara
740	Suriname
744	Svalbard and Jan Mayen
748	Eswatini
752	Sweden
756	Switzerland
760	Syrian Arab Republic
762	Tajikistan
764	Thailand
768	Togo
772	Tokelau
776	Tonga
780	Trinidad and Tobago
784	United Arab Emirates (the)
788	Tunisia
792	Turkey
795	Turkmenistan
796	Turks and Caicos Islands (the)
798	Tuvalu

Source Value	Enriched Value
800	Uganda
804	Ukraine
818	Egypt
826	United Kingdom of Great Britain and Northern Ireland (the)
831	Guernsey
832	Jersey
833	Isle of Man
834	Tanzania, United Republic of
840	United States of America
850	Virgin Islands (U.S.)
854	Burkina Faso
858	Uruguay
860	Uzbekistan
862	Venezuela (Bolivarian Republic of)
876	Wallis and Futuna
882	Samoa
887	Yemen
892	Virgin Islands (British)
894	Zambia
004	Afghanistan
008	Albania
012	Algeria
016	American Samoa
020	Andorra
024	Angola
010	Antarctica
028	Antigua and Barbuda
032	Argentina
051	Armenia
036	Australia
040	Austria
031	Azerbaijan
044	Bahamas (the)
048	Bahrain
050	Bangladesh
052	Barbados
056	Belgium
084	Belize
060	Bermuda

Source Value	Enriched Value
064	Bhutan
068	Bolivia (Plurinational State of)
070	Bosnia and Herzegovina
072	Botswana
074	Bouvet Island
076	Brazil
086	British Indian Ocean Territory (the)
096	Brunei Darussalam
090	Solomon Islands

Show the mapping for `currencyEnrich`

Source Value	Enriched Value
104	MMK
108	BIF
116	KHR
124	CAD
132	CVE
136	KYD
144	LKR
152	CLP
156	CNY
170	COP
174	KMF
188	CRC
191	HRK
192	CUP
203	CZK
208	DKK
214	DOP
222	SVC
230	ETB
232	ERN
238	FKP
242	FJD
262	DJF
270	GMD
292	GIP
320	GTQ
324	GNF
328	GYD

Source Value	Enriched Value
332	HTG
340	HNL
344	HKD
348	HUF
352	ISK
356	INR
360	IDR
364	IRR
368	IQD
376	ILS
388	JMD
392	JPY
398	KZT
400	JOD
404	KES
408	KPW
410	KRW
414	KWD
417	KGS
418	LAK
422	LBP
426	LSL
430	LRD
434	LYD
446	MOP
454	MWK
458	MYR
462	MVR
480	MUR
484	MXN
496	MNT
498	MDL
504	MAD
512	OMR
516	NAD
524	NPR
532	ANG
533	AWG
548	VUV

Source Value	Enriched Value
554	NZD
558	NIO
566	NGN
578	NOK
586	PKR
590	PAB
598	PGK
600	PYG
604	PEN
608	PHP
634	QAR
643	RUB
646	RWF
654	SHP
682	SAR
690	SCR
702	SGD
704	VND
706	SOS
710	ZAR
728	SSP
748	SZL
752	SEK
756	CHF
760	SYP
764	THB
776	TOP
780	TTD
784	AED
788	TND
800	UGX
807	MKD
818	EGP
826	GBP
834	TZS
840	USD
858	UYU
860	UZS
882	WST

Source Value	Enriched Value
886	YER
901	TWD
925	SLE
926	VED
929	MRU
930	STN
931	CUC
932	ZWL
933	BYN
934	TMT
936	GHS
937	VEF
938	SDG
940	UYI
941	RSD
943	MZN
944	AZN
946	RON
947	CHE
948	CHW
949	TRY
950	XAF
951	XCD
952	XOF
953	XPF
960	XDR
965	XUA
967	ZMW
968	SRD
969	MGA
970	COU
971	AFN
972	TJS
973	AOA
975	BGN
976	CDF
977	BAM
978	EUR
979	MXV

Source Value	Enriched Value
980	UAH
981	GEL
984	BOV
985	PLN
986	BRL
990	CLF
994	XSU
997	USN
008	ALL
012	DZD
032	ARS
036	AUD
044	BSD
048	BHD
050	BDT
051	AMD
052	BBD
060	BMD
064	BTN
068	BOB
072	BWP
084	BZD
090	SBD
096	BND

Show the mapping for `deviceBankingType`

Source Value	Enriched Value
A	Mobile Phone App
C	Computer
M	Mobile Phone Browser

Show the mapping for `digitalWalletTokenTypeEnrich`

Source Value	Enriched Value
0	Not a token transaction
1	Apple Pay
2	Samsung Pay
3	Google Pay
F	Card on file token

Show the mapping for `eventDescription`

Source Value	Enriched Value
NTB	NTB Customer via Digital Flow to become a Customer
RRC	Mobile Registration Confirmation

Source Value	Enriched Value
RNO	On Token Selection Page - Chosing No Token Option
RST	On Token Selection Page - Chosing Soft Token Option
RHT	On Token Selection Page - Chosing Hard Token Option
ROT	Migration from CAM30 to Soft Token DSK CAM40
ROB	On Token Selection Page - Chosing Hard Token Option
RTC	CAM 30 Credentials Set up
RNA	Register Account to new Business Profile
RSX	Removal of Device (Handset) / De-Provisioning
AHT	Activation of m2FA CAM40 Hard Token Code PIN Activated
ARD	Hard Token Activation - Activate your Secure Key
ARF	Failure During Replacement Security Device Journey
ADE	Token Deactivation
ASA	Start Soft Token Activation process - Mobile app
ARA	Resend Activation Code (BIB)
APE	Activation of m2FA CAM30 Pass+EMAIL OTP+PIN Activated
ACB	New browser identified for the customer session
LSC	Logon Blocked due to Sanction Location
LLA	CAM10 Successful Entry of UserName
LLG	Logon Process Successful - Can now view accounts
LFX	Failed Attempt at Logon - CAM20
LFC	Failed Attempt at Logon - CAM40
LFB	Failed Attempt at Logon-CAM40 (PIN or Fingerprint)
LFO	Failed and Locked Logon after multiple CAM20 Failed
LFN	Failed and Locked Logon after multiple CAM30 Failed
LFM	Failed and Permanent Locked CAM40 multiple failed OTP
LFD	Failed Attempt at Logon - CAM20 Already Locked
LFF	Failed Attempt at Logon - CAM30 Already Locked
LFE	Failed Attempt at Logon - CAM40 Already Locked
LUI	Forgot User ID
LOF	Mobile App Logoff
LPN	Logon PosNeg Check for SaaS > FraudNet
LPE	Customer has logged and selected Brokerage Account
PSS	Successful Step up From CAM30 to CAM40 - HARD Token
PSF	Failed Step up from CAM30 to CAM40 - HARD Token
PSX	Locked Step up from from CAM30 to CAM40 - HARD Token
PMS	Successful Step up From CAM30 to CAM40 - SOFT Token
PMF	Failed Step up from CAM30 to CAM40 - SOFT Token
PMX	Locked Step up from CAM30 to CAM40 - SOFT Token
PSU	Successful Step up From CAM30 to CAM40

Source Value	Enriched Value
PMU	Failed Step up From CAM30 to CAM40 - Unknown Token
COL	Failed OLR process prior to credential resets
CMQ	Reset Security Credentials - Reset Memorable Questions
CSQ	Reset Security Credentials - Reset Security Questions
CPW	Password Reset - Successful
CCF	Full Credentials Reset Successful
COF	Reset Security Credentials - Off-line
CDU	Reset Security Credentials Hard Token Unlock Code
CPS	Passcode Reset - Code passcode Reset successful
CTH	Switch Token Type - Soft Token to Hard Token
CTS	Switch Token Type - Hard Token to Soft Token
DMQ	Edit Details - Edit Memorable Questions (or EMQ)
DSQ	Edit Details - Edit Security Questions (or ESQ)
DCP	Edit Details - Edit Password (or ECP)
DNU	New CAM10 Username / Change CAM10 Username
BBB	Balances Page Load Post Logon - Successful
OBC	Open Banking - Concent for Information Access
OBD	Open Banking Consent for Payment Creation
OBF	Open Banking Consent for Domestic Payment Creation
OBH	Open Banking Consent - International Payment Creation
OBE	Open Banking - Consent for Enduring Payment Creation
OBI	Open Banking, Open APIs - Consent for CBPII
OBS	Open Banking - Stop Sharing with TPP Confirmation
OSP	Open Banking - Stop Sharing with TPP Confirmation
OTA	Open Banking, Open APIs - AISP Usage
OBP	Open Banking, Open APIs - Make Payment
PL	Open Banking, Open APIs - Make International Payment
OPD	Open Banking, Open APIs - Make Domestic Payment
OBR	Open Banking Open APIs - Revoke direct from TPP for AISP
OBX	Open Banking Open APIs - Revoke direct from TPP for CBPII
URS	High Risk Transaction - Re-Auth OTP Successful
URF	High Risk Transaction - Re-Auth OTP Failed
URL	High Risk Transaction - Re-Auth OTP Locked
PPC	P2P Registration Complete
PPD	P2P De-Register Complete
PPA	P2P Amend Sending Details / Proxy
PPR	P2P Amend Recipient Details / Proxy
PPB	P2P Register Complete for Un-Banked customer
GTP	GTFF / P2P Payment Created / Sent

Source Value	Enriched Value
GTR	GTFF / P2P Payment Requested / Request Money
GTA	GTFF / P2P Payment Receive Accepted by Beneficiary
GTD	GTFF / P2P Payment Receive Declined by Beneficiary
TGB	Add new GTFF payee (GTFF equivalent of TNB)
TGE	Payment to existing GTFF Payee
TGN	Payment to new GTFF Payee (GTFF equivalent of TNI)
OAX	Opening Submission of Application - Accept
OAI	ICO CurrentAccountOpening Submission of Application
OAO	Current Account Opening Submission of Application
OAS	Savings Account Opening Submission of Application
OCR	Credit Card Limit Increase
OAL	Loan Account Opening Submission of Application
OAM	Mortgage Account Opening Submission of Application
OVD	Overdraft Submission of Application (New Limit)
OVA	Overdraft Submission of Application (Amending Limit)
OCL	Credit Card Limit Increase - Accept
OCC	Credit Card Application - Accept
OMU	Update Term Deposit Maturity Instruction
THA	ECW 1st Risk Score Check
THB	ECW 2nd Risk Score Check
THN	PaymentCreationEventforDomitic payment New bene
THD	PaymentCreationEventforDomitic paymentexitsingbene
THI	Payment CreationEvent for Intl payment New bene
THE	Payment CreationEvent for Intl paymentexitsingbene
TNB	New Beneficiary Creation
TNC	New Beneficiary Creation - Company (Me to Company)
TNE	Edit Beneficiary
TMM	Me to Me Payment/Transfer between AccountsMe to Me
TMX	Me to Not Me Payment - Domestic -Unknown Payee
TMI	Me to Not Me Payment International -Unknown Payee
TEB	Payment to Existing Beneficiary DomesticMetoNot Me
TND	Payment to New Beneficiary Domestic (Me to Not Me)
TEI	Payment to Existing Bene InternationalMe to Not Me
TNI	Payment to New Beneficiary InternationalMetoNot Me
TCE	Payment to Existing Beneficiary - (Me to Company)
TCN	Payment to New Beneficiary- Company(Me to Company)
TGV	Me to Me International/Global View/Global Transfer
TSN	Standing Order / Regular Payment Creation
TSA	Standing Order / Regular Payment Amend

Source Value	Enriched Value
TPP	P2P Make a Payment
TTD	Direct Debit Create / Amend Bene
TTS	Direct Debit Transfer - Change Send Account
TPA	Payment Authorisation(2nd Level Auth Post Creation
TUT	Payments - Saved Transfers Update Saved Transfers
TBT	Payments - Batch Payment to Multiple Beneficiaries
TPC	Payments - Make Payment to Credit Card Account
TUM	Modify Mandate (e.g. India UPI)
TTO	Make a Transfer - Cashiers Order
TBB	Launch Bank2Bank Service
TNP	P2P Make a Payment - New Beneficiary
DCT	Edit Details - New/Edit Telephone Number
DDT	Edit Details - New/Edit Telephone Number(BUSINESS)
DRT	Edit Details-Delete alltelephone numforthecustomer
DCE	Edit Details - New/Edit Email Address
DDE	Edit Details - New/Edit Email Address (BUSINESS)
DCD	Edit Details - New Address / Address Change
DDD	Edit Details-NewAddress / AddressChange (BUSINESS)
DMC	Edit Details - Edit Mobile Access Code (CMB)
DLT	Edit DetailsAdd,Amend,Remove Mobileor SMSor Email
DCJ	Edit Details - Occupation / Job Title
DCS	Edit Details - Salary Change
DPB	Edit Details Phone Banking-New / Reset Credentials
DPN	Edit Details -Request New Mobile PIN / Credentials
DSD	Edit Details - Add Secondary User
DSA	Edit Details - Amend Secondary User
DPE	Edit Details - Update Employer Name
DPF	Edit Details - Home Fax
DPX	Edit Details - Office Fax
DAN	Edit Details - Amend Account 'Display As' Name
QCQ	Request - Cheque Book - New or Replacement
QVQ	Request - Void Cheque
QCC	Request - Credit Card - New Card Request
QCV	Request - Credit Card-New Card Request VCProvided
QCY	Request - Credit Card - Replacement Card
QCZ	Request - Credit Card - Replacement Card - Damaged
QDC	Request - Debit Card - New Card Request
QDY	Request - Debit Card - Replacement Card
QDZ	Request - Debit Card - Replacement Card - Damaged

Source Value	Enriched Value
QRG	Request - Credit Card Activation Confirmation
QRH	Request - Debit Card Activation Confirmation
QIP	Request for cardcredit available into instantcash
QIC	Request for Card credit available into InstantCash
QZZ	Request - View PIN Online (Not Change)
QVV	Request - View CVV / CVC Online
QDV	Request - Generate Dynamic CVV / CVC
QCN	Request - Credit Card PIN Request
QDN	Request - Debit Card PIN Request
QCS	Request - Credit Card PIN Change STP
QDS	Request - Debit Card PIN Change STP
QZX	Request - Emergency Cash
QTB	Request - Block on Credit Card
QTD	Request - Block on Debit Card
QTR	Request - Block Removed from Credit Card
QTT	Request - Block Removed from Debit Card
QTN	Request - Travel Notification / Intent to Travel
QPS	Request - Paperless Statements
QVA	Register for a Virtual Account Digital Wallet
QPU	Request - 3DS Pin Update
QPV	Request - 3DS Pin View
IRA	IRA distribution form online request (US)
MML	Move Money Limit Change Confirmation -Daily Limit
BPL	Bill Payment Limit Change Confirmation -Master
LIP	International Payment limit Change Confirmation
LBA	BACS Payment Limit Change Confirmation
LCH	CHAPS Payment Limit Change Confirmation
LOD	Change Overall Daily Payment Limit
LIT	Internal transfer limit change confirmation
LND	New beneficiary payee limit change confirmation
LDP	Existing beneficiary payeelimit changeconfirmation
LAP	Autopay limit change confirmation
WER	Successful Registration for Chat Service - WeChat
WPM	Successful Registration for PayMe
WVV	Successful Registration for Verified by VISA
WCS	Link and Launch eCS Journey
CCP	Reset the CAM30 credentials online successfully.
ODM	Open Banking
TRP	Rejected Payment Creation

Source Value	Enriched Value
ODN	Bill Payment - Single Bene - New
ODE	Bill Payment - Single Bene - Existing
OLN	VRP - Consent for single bene - New
OLE	VRP - Consent for single bene - Existing
OLM	Variable Recurring Payments Consentfor - Me to Me
ODU	ACS- Payment Authorization Required
OAU	MULTI BILL PAYMENT -Payment Authorization Required
OMM	Payment Request - Single bene - Me-To-Me-Transfer
OIN	International Payment Request - New bene
OIM	INTERNATIONAL PAYMENTS - Payment Request -Me to Me
OIE	INTERNATIONALPAYMENTS Payment Reques-Existing bene
OFN	International Schedule Payments -Request-New bene
OFM	International Schedule Payments - Request-Me to Me
OFE	International Schedule PaymentRequest Existingbene
mRRX	Mobile Registration Cancelled
mRRF	Mobile Registration Failed
mAAA	Activation (not used anymore)
mAAC	Activation (not used anymore)
mAAE	Activation (not used anymore)
mAAF	Activation (not used anymore)
mAPS	Activation of m2FA CAM 30 Password+SMS OTP+PIN
mAPC	Cust success provisioned using CAM30+SMS+Email OTP
mAST	Soft Token Activation Complete Successful -2 AC
mASO	Soft Token Activation Complete Successful - one AC
mASP	Soft Token Activation usingUsernamePass and HToken
mASB	Soft Token Activation Complete with Mobile Back up
mASF	Soft token Activation failed
mRMC	Registration Activation/Provision App PIN Created
mRSC	Provisioned device with CAM40+soft tokenOTP/QR+SMS
mRRM	Provisioned device with CAM40+soft tokenOTP viaAPP
mRRV	Customermovedfrom SaaS to mAuthCAM40 on samedevice
mRRO	Customer migrated from SaaS DSK to mAuth CAM40
mRSZ	Switch of Trusted Device
mROH	Migration fromCAM40 Profile withHard Token to soft
mLHN	DSP iAuth Logon Authentication Fraud Check
mDTP	PushNotification to Mobile Device- Accepted
mDTD	PushNotification to Mobile Device- Declined
mDTF	PushNotification to Mobile Device- Fraud
mDTS	PushNotification to Mobile Device- Accepted(Setup)

Source Value	Enriched Value
mDCF	PushNotification to Mobile Device- Declined(Setup)
mDTG	PushNotification to Mobile Device- Fraud(Setup)
mDTT	PushNotification to Mobile Device- Accepted(TDS)
mDTC	PushNotification to Mobile Device- Declined(TDS)
mDTH	PushNotification to Mobile Device- Fraud(TDS)
mCDS	Deactivate SoftToken - Lost/ Stolen /Change device
mCPR	Passcode Reset - Code locked (or mLPR)
mCPO	Recover / Reset mPIN using CAM 30 Password
mCHT	Recover / Reset mPIN using CAM 40 Token
mCSW	Switch Soft OTP / DSK to a new Device
mTIE	TouchID / FaceID / Biometric ID - Enabled in App
mTID	TouchID / FaceID / Biometric ID - Disabled in App
mTIS	TouchID / FaceID / Biometric ID - Enabled in App
mTIL	TouchID /FaceID/ Logon failed Temp Lock
mFLC	Launch Call Connection
mUCA	3DS Card Auth Attempt - Approved
mUCD	3DS Card Auth Attempt - Declined
mBPR	Balance Peek - Registration Complete
mBPD	Balance Peek - DeRegistration Complete
mBPX	Alexa Registration for Balance Peek
mBPA	Alexa Balance Peek (Balance Peek Demised)
mPPE	P2P Register Complete for Visitor
mTUN	Create New Mandate (e.g. India UPI)
mTZA	Pay By Bank App Payment / ZAPP
mTLP	LocalPay featureotherthat current standard payment
mTAC	Payment as part of the account closure process
mDLO	Edit Details - Enable Location (GPS) Data Pull
mDLF	Edit Details - Disable Location (GPS) Data Pull
mRDC	Remote Chq Deposit Capture
mQVC	Request - Credit Card Provision to Mobile Wallet
mQVD	Request - Debit Card Provision to Mobile Wallet
mQIB	Balance Transfer - Payment to new Beneficiary
mQMS	Request - MasterCard 3D Secure Register
mQCL	Request - Contactless Enabled on Card
mQCO	Request - Contactless Disabled on Card
mQAE	Request - ATM Withdrawals Enabled on Card
mQAO	Request - ATM Withdrawals Disabled on Card
mQGB	Request - Gambling Block Enabled on Card
mQGO	Request - Gambling Block Disabled on Card

Source Value	Enriched Value
mQXE	Request -Online Transaction (eComm)Enabled On Card
mQXO	Request OnlineTransaction (eComm) Disabled On Card
mQSE	Request - POS Transaction Enabled On Card
mQSO	Request - POS Transaction Disabled On Card
mSLI	Library Injection Enabled
mSAM	App Mirroring Enabled
mSSD (ption	Screenshot Detected
mSSR (ption	Screenreader Detected
mMTL	Move Money Limit Change Confirmation
mWRR	Successful Registration for Chat Service - RMChat
mWRS	Successful Logon to RMChat
mWRF	Failed Logon to RMChat
mWRL	Failed & Locked Logon to RMChat
mWZE	Successful Zelle Enrollment
mWZS	Link and Launch Zelle Journey (SSO)
mNTB	NTB Customer via Digital Flow to become a Customer
mRRC	Mobile Registration Confirmation
mRTC	CAM 30 Credentials Set up
mAPE	Activation of m2FA CAM30Pass+EMAILOTP+PIN Activated
mAHT	Activation of m2FACAM40 HardTokenCode PIN Activated
mRSX	Removal of Device (Handset) / De-Provisioning
mROT	Migration from CAM30 to Soft Token DSK CAM40
mLSC	Logon Blocked due to Sanction Location
mLLA	CAM10 Successful Entry of UserName
mLLG	Logon Process Successful - Can now view accounts
mLFX	Failed Attempt at Logon - CAM20
mLFC	Failed Attempt at Logon - CAM40
mLFB	Failed Attempt at Logon-CAM40 (PIN or Fingerprint)
mLFO	Failed and Locked Logon after multiple CAM20Failed
mLFN	Failed and Locked Logon after multiple CAM30Failed
mLFM	Failed andPermanentLocked CAM40multiple failedOTP
mLFD	Failed Attempt at Logon - CAM20 Already Locked
mLFF	Failed Attempt at Logon - CAM30 Already Locked
mLFE	Failed Attempt at Logon - CAM40 Already Locked
mLOF	Mobile App Logoff
mLUI	Forgot User ID
mPSU	Successful Step up From CAM30 to CAM40
mPMU	Failed Step up From CAM30 to CAM40 - Unknown Token
mPSS	Successful Step up From CAM30 to CAM40 -HARD Token

Source Value	Enriched Value
mPSF	Failed Step up from CAM30 to CAM40 - HARD Token
mPSX	Locked Step up from from CAM30 toCAM40 -HARD Token
mPMS	Successful Step up From CAM30 to CAM40 -SOFT Token
mPMF	Failed Step up from CAM30 to CAM40 - SOFT Token
mPMX	Locked Step up from CAM30 to CAM40 - SOFT Token
mCPS	Passcode Reset - Code passcode Reset successful
mCTH	Switch Token Type - Soft Token to Hard Token
mCTS	Switch Token Type - Hard Token to Soft Token
mCPW	Password Reset - Successful
mCCF	Full Credentials Reset Successful
mDNU	New CAM10 Username / Change CAM10 Username
mBBB	Balances Page Load Post Logon - Successful
mOAI	ICO CurrentAccountOpening Submission ofApplication
mOAO	Current Account Opening Submission of Application
mOAS	Savings Account Opening Submission of Application
mOAL	Loan Account Opening Submission of Application
mOAM	Mortgage Account Opening Submission of Application
mOVD	Overdraft Submission of Application (New Limit)
mOVA	OverdraftSubmission of Application(Amending Limit)
mOCL	Credit Card Limit Increase - Accept
mOCR	Credit Card Limit Increase
mOCC	Credit Card Application - Accept
mOMU	Update Term Deposit Maturity Instruction
mOBC	Open Banking-Concent for Information Access
mOBD	Open BankinConsent for Payment Creation
mOBE	Open Banking-Consent for Enduring Payment Creation
mOBI	Confirmation of funds authorization
mOBS	Open Banking-Stop Sharing with TPP Confirmation
mOSP	Open Banking-Stop Sharing with TPP Confirmation
mURS	High Risk Transaction - Re-Auth OTP Successful
mURF	High Risk Transaction - Re-Auth OTP Failed
mURL	High Risk Transaction - Re-Auth OTP Locked
mPPA	P2P Amend Sending Details / Proxy
mPPB	P2P Register Complete for Un-Banked customer
mPPC	P2P Registration Complete
mPPD	P2P De-Register Complete
mPPR	P2P Amend Recipient Details / Proxy
mGTP	GTFF / P2P Payment Created / Sent
mGTR	GTFF / P2P Payment Requested / Request Money

Source Value	Enriched Value
mGTA	GTFF / P2P Payment Receive Accepted by Beneficiary
mGTD	GTFF / P2P Payment Receive Declined by Beneficiary
mTHA	ECW 1st Risk Score Check
mTHB	ECW 2nd Risk Score Check
mTHN	PaymentCreationEventforDomitic payment New bene
mTHD	PaymentCreationEventforDomitic paymentexitsingbene
mTHI	Payment CreationEvent for Intl payment New bene
mTHE	Payment CreationEvent for Intl paymentexitsingbene
mTNB	New Beneficiary Creation
mTNC	New Beneficiary Creation - Company (Me to Company)
mTNE	Edit Beneficiary
mTUM	Modify Mandate (e.g. India UPI)
mTMM	Me to Me Payment/Transfer between AccountsMe to Me
mTEB	Payment to Existing Beneficiary DomesticMetoNot Me
mTND	Payment to New Beneficiary Domestic (Me to Not Me)
mTEI	Payment to Existing Bene InternationalMe to Not Me
mTNI	Payment to New Beneficiary InternationalMetoNot Me
mTCE	Payment to Existing Beneficiary - (Me to Company)
mTCN	Payment to New Beneficiary- Company(Me to Company)
mTGV	Me to Me International/Global View/Global Transfer
mTSN	Standing Order / Regular Payment Creation
mTSA	Standing Order / Regular Payment Amend
mTPP	P2P Make a Payment
mTMX	Me to Not Me Payment - Domestic -Unknown Payee
mTMI	Me to Not Me Payment International -Unknown Payee
mTTD	Direct Debit Create / Amend Bene
mTTS	Direct Debit Transfer - Change Send Account
mTPA	Payment Authorisation(2nd Level Auth Post Creation
mTUT	Payments - Saved Transfers Update Saved Transfers
mTBT	Payments - Batch Payment to Multiple Beneficiaries
mTPC	Payments - Make Payment to Credit Card Account
mTBB	Launch Bank2Bank Service
mDCT	Edit Details - New/Edit Telephone Number
mDDT	Edit Details - New/Edit Telephone Number(BUSINESS)
mDRT	Edit Details-Delete alltelephone numforthecustomer
mDCE	Edit Details - New/Edit Email Address
mDDE	Edit Details - New/Edit Email Address (BUSINESS)
mDCD	Edit Details - New Address / Address Change
mDDD	Edit Details-NewAddress / AddressChange (BUSINESS)

Source Value	Enriched Value
mDLT	Edit DetailsAdd,Amend,Remove Mobileor SMSor Email
mDCJ	Edit Details - Occupation / Job Title
mDCS	Edit Details - Salary Change
mDPE	Edit Details - Update Employer Name
mDPB	Edit Details Phone Banking-New / Reset Credentials
mDPN	Edit Details -Request New Mobile PIN / Credentials
mDSD	Edit Details - Add Secondary User
mDSA	Edit Details - Amend Secondary User
mDAN	Edit Details - Amend Account 'Display As' Name
mQCQ	Request - Cheque Book - New or Replacement
mQCC	Request - Credit Card - New Card Request
mQCV	Request - Credit Card-New Card Request VCProvided
mQCY	Request - Credit Card - Replacement Card
mQCZ	Request - Credit Card - Replacement Card - Damaged
mQDC	Request - Debit Card - New Card Request
mQDY	Request - Debit Card - Replacement Card
mQDZ	Request - Debit Card - Replacement Card - Damaged
mQRG	Request - Credit Card Activation Confirmation
mQRH	Request - Debit Card Activation Confirmation
mQZZ	Request - View PIN Online (Not Change)
mQVV	Request - View CVV / CVC Online
mQDV	Request - Generate Dynamic CVV / CVC
mQCN	Request - Credit Card PIN Request
mQDN	Request - Debit Card PIN Request
mQCS	Request - Credit Card PIN Change STP
mQDS	Request - Debit Card PIN Change STP
mQIP	Request for cardcredit available into instantcash
mQIC	Request for Card credit available into InstantCash
mQZX	Request - Emergency Cash
mQTB	Request - Block on Credit Card
mQTD	Request - Block on Debit Card
mQTR	Request - Block Removed from Credit Card
mQTT	Request - Block Removed from Debit Card
mQTN	Request - Travel Notification / Intent to Travel
mQPS	Request - Paperless Statements
mQVA	Register for a Virtual Account Digital Wallet
mQPU	Request - 3DS Pin Update
mQPV	Request - 3DS Pin View
mMML	Move Money Limit Change Confirmation -Daily Limit

Source Value	Enriched Value
mBPL	Bill Payment Limit Change Confirmation -Master
mWER	Successful Registration for Chat Service - WeChat
mWPM	Successful Registration for PayMe
mWCS	Link and Launch eCS Journey
mPCS	change SOTP Password
mTRP	Rejected Payment Creation
mTRB	Rejected New Beneficiary Creation
mODM	Open Banking
mODN	Bill Payment - Single Bene - New
mODE	Bill Payment - Single Bene - Existing
mOLN	VRP - Consent for single bene - New
mOLE	VRP - Consent for single bene - Existing
mOLU	VRP - Consent for multiple bene - New or existing
mOSN	Standing Order - 1SCA - Payment request - New bene
mOIN	International Payment Request - New bene
mOLM	Variable Recurring Payments Consentfor - Me to Me
mMAD	Trusted Device -MobileApprove DeclinedNotification
mMAF	Trusted DeviceMobileApprove FailedFraudNotificatio
mMAL	Trusted Device -MobileApprove - LockedNotification

Show the mapping for `inboundPaymentTypeEnrich`

Source Value	Enriched Value
10	Single Immediate Payment
20	Return
25	Scheme Return Payment
30	Standing Order
40	Forward Dated Payment
50	DCA Bulk Payment
74	FP RTP HSBC Connect Payment
75	ZAPP Bank Payment
76	Open Banking Bill Payment
77	Open Banking Deferred Bill Payment
79	Paym Realtime Faster Payment/ P2P payment
90	PODS Bill Payment
91	PODS Deferred Bill Payment
92	HSBC Net SIP
93	Swift SIP
94	Net/Connect Batch Payment
95	Standing Orders
97	RBP Payment

Source Value	Enriched Value
99	Internal Faster Payment with media required
01	HSBC Net IAT Payment

Show the mapping for `isBatchEnrich`

Source Value	Enriched Value
R	Realtime
B	Batch

Show the mapping for `paymentStatusEnrich`

Source Value	Enriched Value
A	Approved
R	Reviewed

Show the mapping for `paymentTypeEnrich`

Source Value	Enriched Value
B	Bill Payment
T	Fund Transfer
K	Bulk Fund Transfer
W	Wire Transfer
P	Peer-to-Peer Payment

Show the mapping for `posCardholderVerMethodEnrich`

Source Value	Enriched Value
1	Non-authenticated transaction at a 3D-secure/secure code merchant
2	Successful authentication
9	Not Applicable

Show the mapping for `senderTransactionCodeEnrich`

Source Value	Enriched Value
510	Mastercard Other Credit
511	Mastercard UK Purchase Refund
512	Mastercard Intl Purchase Rev
513	Mastercard Intl Purchase Refund
514	Mastercard Intl ATM Reversal
515	Mastercard Recur Txn Rev
516	Mastercard UK Orig Credit
517	Mastercard Intl Orig Credit
519	Mastercard Credit Invalid PAN
522	Mastercard Cntctls UK Pur Rfnd
525	Mastercard Cntctls Intl Pur Rv
526	Mastercard Cntctls Intl Pur Rf
530	Mastercard NSTF CR
532	Mastercard UK IAP Pur Refund
535	Mastercard Intl IAP Pur Rev
536	Mastercard Intl IAP Pur Refnd

Source Value	Enriched Value
542	Mastercard UK Mob Cntct Pur Rf
545	Mastercard Intl Mob CntctPurRv
546	Mastercard Intl Mob CntctPurRf
549	Mastercard Contra Other Credit
550	Mastercard Default Credit
573	Mastercard UK Purc Refund Auth
574	MC Intl Purchase Refund Auth
575	MC Ctctless UK Purc Ref Auth
576	MC Ctctless Int Pur Ref Auth
577	MC Mobctct UK Purch Ref Auth
578	MC Mobctct IntlPrch Ref Auth
579	MC Post Office ATM Cash Depost
580	Mastercard UK Original Credit
581	Mastercard Intl Orig Credit
582	Mastercard IntOrig CR Tran Aut
583	Mastercard UKOrig CR Tran Auth
584	Mastercard UK Reversal Auth
602	TT BOI ONLINE CREDIT
604	TT BOI BATCH CREDIT
633	Positive Contra Credit
667	Mortgage Int Tax Relief
668	Supplementary List Total
670	Bank Giro/BACS Credit
671	Other Credit
672	Credit Branch Supp
673	Composite credit
674	Transfer
675	Automatic/ATM Transfer
676	Dividend
677	TCS Bancs Credit
679	Charges/Interest Credit
680	Comp Adj A/c Cr (not trunc)
682	Bureau Adj Contra Credit
684	Reversal of BACS Debit
688	Comp Adj Transaction
700	ATM Initiated Transfer
701	Customer Automatic Transfer
702	Repayment Automatic Transfer
703	Loan Protection Payment

Source Value	Enriched Value
704	Internal Automatic Transfer
705	Account Closure Auto Transfer
707	Accounting Entry Transaction
708	Personal Loan Interest Account
709	Personal Loan Int Susp Account
710	Structured Loan Drawdown
711	Structured Loan Closure
712	Unapplied Transaction
713	Amendment to Cash Balance
714	DSC Charges Exempt Adjustment
715	Inter Account Transfer
716	Real-time Cash only Credit
717	Credit to Exchange Account
718	Cash Out of Till
719	Cash Transfer to Safe
720	Intertill Cash Credit
721	Cash Transfer from Safe
725	Real-time Reversal of a Debit
726	Bill Payment
727	Telephone Banking Transfer
728	MPS Credit
729	Mortgage Drawdown Reversal
730	Returned Payment
731	Returned Payment Write-off
732	M/gage Drawdown Rev Deletion
733	Insurance Protection Refund
734	Insurance Protection Premium
735	Life/Pension Credit
736	Credit Card Cash Transfer
737	Credit Card Bill Payment
738	Credit Card Balance Transfer
739	Incentive Payment
740	Unrelated Account Transfer
741	Sundries
742	Credit Card Bill Pymt (Bureau)
743	Credit Card Bill Pymt (BACS)
744	Term Deposit Credit Transfer
745	Money Market Credit
747	CD Closing Credit

Source Value	Enriched Value
748	Money Market Opening (fixed)
749	Money Market Opening (notice)
750	Debit Authorisation Reversal
751	Maestro UK Purchse Refund Auth
752	Maestro UK Purchase Adj Auth
753	Maestro UK & Cashback Adj Auth
754	Solo Refund Authorisation
755	Solo Adjustment Authorisation
756	Solo with Cashback Adj Auth
757	Maestro Intl Purchase Ref Auth
758	Maestro Intl Purchase Adj Auth
759	Maestro Intl Cashback Adj Auth
760	Visa UK Purchase Refund Auth
761	Visa UK Purchase Adj Auth
762	Visa UK & Cashback Adj Auth
763	Visa Intl Purchase Refund Auth
764	Visa Intl Purchase Adj Auth
765	Visa Intl Cashback Adj Auth
766	Visa UK Reversal Auth
770	HSBC Republic Credit
771	CD Maturity Credit
772	Real-Time PIM Cash Credit
773	GENERIC PAYMENT CREDIT
774	PAPS Transfer
775	FP Incoming Standing Order CR
776	FP Returned Credit
777	FP Incoming Bill Payment CR
778	Media Incoming Bill Pay CR
779	GLE CR for Investment Sweep
780	Media In BP CR -no media setup
781	FP Corporate Re-credit
783	Internal standing order CR
784	Media standing order CR
790	Visa Intl Purchase Refund
791	Visa Intl Purchase Reversal
792	Visa Intl ATM Reversal
793	Visa UK Purchase Refund
794	Recurring Transaction Reversal
795	Visa Other Credit

Source Value	Enriched Value
796	Visa Credit Invalid PAN
798	ONUS SETTLEMENT CREDIT FUNDING
800	Me2Me Transfer - Credit
801	Credit Card Balance Transfer
803	Real-time Mixed Credit Cash
804	Mixed Credit Suspense Credit
805	HMS GP Retailer Credit
806	Account Opening Transfer
820	GIC Credit
821	Reversal of an GIC Debit
823	PSD D+1 RPS suspense credit
824	Visa Contactless Intl Purch Rf
825	Visa Contactless Intl Purch Rv
826	Visa Contactless UK Purch Ref
827	Visa Ctctless UK Purc Ref Auth
828	Visa Ctctless UK Purc Adj Auth
829	Visa Ctctless Int Pur Ref Auth
830	Visa Ctctless Int Pur Adj Auth
831	PSD D+1 RPS Fraud susp credit
832	Device Cash Account Credit
833	Branch Cash Transfer Credit
850	GIC Sell Credit
851	GIC Purchase Fee Credit
852	GIC Sell Debit Reversal
853	GIC Dividend
854	GIC Dividend FP Media
855	GIC Interest Distribution
856	GIC Interest Dist FP Media
857	GIC Property Income Dist
858	GIC Prop Income Dist FP Media
859	GIC Corporate Action
860	GIC Corporate Action FP Media
861	GIC Market Settlement
862	GIC Market Settlement FP Media
863	GIC Tax Credit
864	GIC Tax Debit
865	VisIntl Mob Cntct Purch Rfnd
866	VisIntl Mob Cntct Purch Rev
867	VisUK Mob Cntct Purch Rfnd

Source Value	Enriched Value
868	Visa MobCtct UK Purch Ref Auth
869	Visa MobCtct UK Purch Adj Auth
870	Visa MobCtct IntlPrch Ref Auth
871	Visa MobCtct IntlPrch Adj Auth
872	GIC Account Fee CR
874	Structured Loan Closure
875	ET ISA Payment
876	Post Office ATM Cash Deposit
877	PO ATM Cash Deposit Suspense
878	Visa Int Personal Payment Auth
879	Visa UK Personal Payment Auth
880	Non-Sterling Transaction Fee
881	Paym Incoming FP Credit
882	Paym FP Returned Credit
883	Visa UK Personal Payment
884	Visa Intl Personal Payment
885	Paym Incoming Link credit
886	Paym Link Suspense Credit
887	Lifepen Recredit
888	Lifepen Credit
889	Lifepen Credit FP Media
891	FP DCA ADJ CREDIT
893	FP FIM CREDIT
895	FP DA SIP CREDIT
897	FP DA Return Credit
898	FP DA SO CREDIT
900	FP DA FDP CREDIT
902	FP DA DCA Bulk Credit
904	FP DA Scheme Return Credit
905	FP DA Reversal Credit
906	GPS HSBCNet IAT
907	Visa UK Original Credit
908	Visa Intl Original Credit
911	Visa Intl Purchase Refund
912	Visa Intl Purchase Reversal
913	Visa UK Purchase Refund
914	Visa UK Purchase Refund Auth
915	Visa UK Purchase Adj Auth
916	Visa Intl Purchase Refund Auth

Source Value	Enriched Value
917	Visa Intl Purchase Adj Auth
918	Vaultex Cash Credit
919	VA Credit
920	JLP Card BT
921	JLP Card BT with media
922	M&S Card BT with media
923	Credit Card BT with media
924	JLP Card Credits
925	JLP Card Credits with media
926	GLE Automatic Transfer
928	HSBC SmartSave Credit
929	CHI Cheque Re-Credit
930	CHI Cheque Info only Credit
931	CHI Cheque Credit
932	CHI Credit
933	JLL Loan Disb
934	JLL Loan Disb with media
935	CHI Composite Re-Credit
936	CHI Composite Credit
937	CHI Cheque Credit Mobile
938	Visa Int Orig Credit Tran Auth
939	Visa Uk Orig Credit Tran Auth
940	OPEN BANKING TRANSFER CREDIT
941	OPEN BANKING INCOMING FP CR
942	OPEN BANKING FP RETURNED CR
943	DEBIT CARD DISPUTE CR ONLINE
944	CHI Cheque Credit Mobile
945	Adhoc Payment Cr
946	PBBA Refund Credit
947	CHI Cheque ADJ Credit
948	CHI Cheque REJ Credit
949	CHI Composite ADJ Credit
951	PBBA payout Credit
952	DEBIT CARD DISPUTE CR BATCH
955	CHI Bank Giro Credit
956	CHI Bank Giro Credit Incleared
957	CHI Bank Giro Credit Rescan
958	CHI Cheque Credit
960	CHI BGC RDC

Source Value	Enriched Value
961	CHI BGC BULK
962	OPEN BANKING BILL PAYMENT
963	VA CREDIT - FPS
965	CPR CLAIM - Credit
966	Charges/Interest Credit
967	CH 3000 CREDIT
968	Vaultex BPI Overs Credit
969	AT Cr Saving Acct Negative Bal
970	Generic Posting - Credit
971	Generic Posting - Credit
972	GENERIC POSTING MWS-CREDIT
973	GM SYSTEM - AUTO FX
974	GM INT ATM WITHDRAWAL
975	GM VISA UK PURCHASE
976	GM VISA INT PURCHASE
977	GM VISA UK CNTCTLS PURCHASE
978	GM VISA INT CNTCTLS PURCHASE
981	GM VISA PURCHASE REFUND
983	GM VISA OCT
984	GM VISA UK ATM WITHDRAWAL
985	GM DISPUTE REVERSAL
986	HSBC Pay MDR Credit
987	HSBC Pay customer refund

Show the mapping for `sicEnrich`

Source Value	Enriched Value
79	Chip fell back to read Magnetic stripe - counterfeit
80	Chip cards fall back to magnetic stripe - Moto
81	Internet
90	Entire magnetic stripe read - Counterfeit
91	PAN auto-entry via contactless magnetic stripe - Counterfeit
95	Initiated by chip CVV and chip data unreliable
00	Unknown
01	Manual (key entered)
02	Magnetic stripe read
03	Bar Code read
04	Optical character recognition (OCR) coding read
05	Initiated by chip card verification value
06	Internet
07	PAN auto-entry via contactless

Source Value	Enriched Value
S	Internet
U	Internet
V	Internet
T	Internet

Show the mapping for `transReversalIndEnrich`

Source Value	Enriched Value
A	Adjustment
F	Refund
D	Reversal (Derogatory)
R	Reversal (Non-Derogatory or Unknown)
N	Normal (not a reversal)
O	Other Reversal/refund/adjustment with a net increase to the open-to-buy/avail bal
P	Other Reversal/refund/adjustment with a net decrease to open-to-buy/avail bal
G	Chargeback

Show the mapping for `transactionStatusEnrich`

Source Value	Enriched Value
A	Approved/Successful
B	Cancelled by Bank
C	Cancelled by Customer
D	Declined/Unsuccessful
P	Pending
R	Reversal

Alerting in Case Manager

In this page you can find a set of alerts that can be added to the monitoring solution in order to improve awareness over Case Manager stability.

info

Note that these alerts are solemnly suggestions that might require new dashboards or alert adjustments based on HSBC specific environments.

Alert	Configuration	Description
New Event Throughput	alert level: Critical for alert: 1m frequency: 2h threshold value: 0.0000003 (messages per second)	Monitors the throughput of event processing in Case Manager. If throughput falls below the threshold, an alert is triggered.
New Event Latency	alert level: High for alert: 10m frequency: 10m threshold value: 60 (seconds)	Monitors the maximum latency for processing "new event" requests, specifically at the 99th percentile.
Queue Occupancy	alert level: High for alert: 10m frequency: 10m threshold value: 0.85 (percentage)	Monitors the average occupancy of the event processing queue. If the average occupancy exceeds 85% over the specified time span, an alert is triggered.
Event Queue Deadletter	alert level: High for alert: 5m frequency: 1h threshold value: 10 (events)	Monitors the total number of event queue deadletter events. If the sum of deadletters exceeds the defined threshold of 10 over the last hour, an alert is triggered.
Event Storage Deadletter	alert level: High for alert: 5m frequency: 1h threshold value: 10 (events)	Monitors the total number of event storage deadletter events. If the sum of deadletters exceeds the defined threshold of 10 over the last hour, an alert is triggered.
Event Automations Throughput	alert level: Moderate for alert: 10m frequency: 10m threshold value: 0 (events)	Monitors the overall throughput count when executing automation rules. If errors exceed the defined threshold of 0 over the last 10 minutes, an alert is triggered.
Event Automations Errors	alert level: Moderate for alert: 10m frequency: 10m threshold value: 0 (events)	Monitors the overall error count when executing automation rules. If errors exceed the defined threshold of 0 over the last 10 minutes, an alert is triggered.
Pulse Requests Errors	alert level: Moderate for alert: 10m frequency: 1m threshold value: 0.01 (percentage)	Monitors the throughput of requests to Pulse instances and tracks the occurrence of errors.
Database Connection Pool	alert level: Low for alert: 10m frequency: 5m threshold value: 0.90 (percentage)	Monitors the usage of the database connection pool, ensuring it does not exceed the defined threshold.

Alert	Configuration	Description
Messaging Healthcheck	alert level: Critical for alert: 5m frequency: 1m threshold value: 1 (event)	Monitors the messaging server healthcheck from Case Manager.
Database Connection	alert level: Critical for alert: 5m frequency: 1m threshold value: 1 (event)	Monitors the database healthcheck for Case Manager.

ⓘ**note**

- alert level : The severity of the alert (Low, Moderate, High, Critical).
- for alert : The duration for which the alert condition must persist before an alert is triggered.
- frequency : The interval at which the monitoring system checks the specified metrics or conditions.
- threshold value : A predefined limit or value that, when exceeded, triggers an alert.

Alerting in Pulse

In this page you can find a set of alerts that can be added to the monitoring solution in order to improve awareness over Pulse stability.

info

Note that these alerts are solemnly suggestions that might require new dashboards or alert adjustments based on HSBC specific environments.

Alert	Configuration	Description
Workflow Throughput	alert Level: Critical for alert: 1m frequency: 1m threshold value: 0.00001 (messages per second)	Measures the overall application workflow throughput per message type(normal, recovery, replica).
Workflow Errors	alert Level: High for alert: 10m frequency: 10s threshold value: 0 (events)	Measures the overall application workflow errors, considering all message types(normal, recovery and replica).
Events Dumped to Disk	alert Level: Critical for alert: 1m frequency: 1m threshold value: 0 (events)	Measures the overall application events spilled to disk. An event is spilled to disk when Pulse cannot write it on the event storage.
Auto Recovery Job	alert Level: Moderate for alert: 0s frequency: 5m threshold value: 0 (events)	Measures the overall number of events that are present in the in-memory queue to be written to the dump file in the failed folder, application and workflow.
Profile Fetches Latencies	alert Level: High for alert: 10m frequency: 1m threshold value: 0.4 (seconds)	Measures the overall max latency to fetch the profiles at percentile 99, per application, workflow and plan.
Pulse Application States	alert Level: Critical for alert: 12h frequency: 30s threshold value: 6 (STARTED)	Monitors the state of applications across all Pulse instances. If the application state is not STARTED (state < 6), an alert is triggered.
Pulse JVM GC Time	alert Level: High for alert: 10m frequency: 5m threshold value: 1 (seconds)	Measures maximum JVM GC duration. Long garbage collection pause durations indicate that the heap size is not adjusted for the actual usage.
App JVM GC Time	alert Level: High for alert: 10m frequency: 15m threshold value: 1 (seconds)	Measures maximum JVM GC duration for young and old gen per application and host, indicating heap size adjustment needs.
Pulse JVM Heap Usage	alert Level: Critical for alert: 5m frequency: 1m threshold value: 0.95 (percentage)	Measures JVM heap usage per Pulse instance, indicating high memory utilization.

Alert	Configuration	Description
App JVM Heap Usage	alert Level: Critical for alert: 5m frequency: 1m threshold value: 0.99 (percentage)	Measures JVM heap usage per application and host, indicating high memory utilization.
Pulse Threads Count	alert Level: Moderate for alert: 30m frequency: 10s threshold value: 2500 (threads)	Measures the current number of active threads in the JVM per Pulse instance.
App Threads Count	alert Level: Moderate for alert: 30m frequency: 10s threshold value: 3000 (threads)	Measures the current number of active threads in the JVM per application and host.
Pulse Heap Usage After GC	alert Level: High for alert: 15m frequency: 10s threshold value: 0.90 (percentage)	Measures the JVM heap usage after garbage collection per Pulse instance.
App Heap Usage After GC	alert Level: High for alert: 15m frequency: 10s threshold value: 0.9 (percentage)	Measures the JVM heap usage after garbage collection per application and host.
Pulse JVM File Descriptors Ratio	alert Level: High for alert: 10m frequency: 10s threshold value: 0.85 (percentage)	Measures the JVM file descriptors usage ratio per Pulse instance.
App JVM File Descriptors Ratio	alert Level: High for alert: 10m frequency: 10s threshold value: 0.85 (percentage)	Measures the JVM file descriptors usage ratio per application and host.
Pulse JVM Code Cache Usage	alert Level: Critical for alert: 10m frequency: 10s threshold value: 0.90 (percentage)	Measures the JVM code cache usage per Pulse instance.
App JVM Code Cache Usage	alert Level: Critical for alert: 10m frequency: 10s threshold value: 0.95 (percentage)	Measures the JVM code cache usage per application and host.
Messaging Healthcheck	alert Level: High for alert: 5m frequency: 2m threshold value: 1 (event)	Monitors the Pulse healthcheck for messaging communication with the messaging server.
Zookeeper Healthcheck	alert Level: High for alert: 5m frequency: 2m threshold value: 1 (event)	Monitors the Pulse healthcheck for Zookeeper communication.
Metadata DB Connection Pool	alert Level: Low for alert: 30m frequency: 1m	Monitors the metadata DB connection pool usage.

Alert	Configuration	Description
	threshold value: 0.9 (percentage)	
Metadata DB Connection Pool Per App	alert Level: Low for alert: 30m frequency: 1m threshold value: 0.9 (percentage)	Monitors the metadata connection pools usage per application.

note

- alert Level : The severity of the alert (Low, Moderate, High, Critical).
- for alert : The duration for which the alert condition must persist before an alert is triggered.
- frequency : The interval at which the monitoring system checks the specified metrics or conditions.
- threshold value : A predefined limit or value that, when exceeded, triggers an alert.

On this page

Collect Metrics

Feedzai products and components expose monitoring metrics on a per request fashion. They are [Dropwizard metrics] formatted in JSON and available on the following HTTP endpoints. The metrics endpoint should be used for consumption purposes, typically by a third-party tool such as Grafana or Prometheus. A companion *"rich metrics"* endpoint is also available for some components, which includes descriptions for the metrics to help users understand their purpose. Refer to the following endpoints per component:

- Pulse:
  - `https://<host:port>/metrics`
  - `https://<host:port>/metrics/rich`
- Case Manager:
  - `https://<host:port>/api/metrics`
  - `https://<host:port>/api/metrics/rich`
- Reference Data Service:
  - `https://<host:management_port>/metrics` (note the use of the management port, typically 8081)
  - `https://<host:management_port>/metrics/rich` (note the use of the management port, typically 8081)
- ETL:
  - `http://<host:port>/metrics` (port is typically 5567)

info

Note that these endpoints donât require authentication.

warning

All metrics are published to a single node. It's the operator's responsibility to use an external monitoring system to aggregate them afterwards.

WARNING - ETL metrics

ETL metrics are only exposed while the ETL process is executing. The process stays running for 20 seconds after the extraction finishes to allow fetching the metrics.

Next stepsâ

Learn how to interpret the available metrics, and how to use the `/metrics/rich` endpoint to understand the meaning of each metric.

On this page

Case Manager Database Table Structure

Find in this page a detailed guide on the important tables used for Case Manager.

AM_EVENT_STORAGEâ

- This table contains the referential records to all historical transactions, where the event request is stored in EVENT_DATA in raw json format. This includes all events for a given lifecycle_id, not just a single record per lifecycle_id.

Column Name	Column Type	Description
EVENT_ID	PRIMARY KEY	The event's unique identifier. Randomly generated UUID, generated for each event.
EVENT_VERSION_ID	PRIMARY KEY	It is usually equal to EVENT_ID. New Id gets generated in case the event is an update to the initial transaction.
EVENT_DATA	JSON	The field stores event data

AM_EVENT_DIGITAL_ACTIVITYâ

- The records in this table correspond to the Digital Activity events (e.g. Credentials Update, Device Update, New Payee, Alias, Profile Update, etc).
- The AM_EVENT_DIGITAL_ACTIVITY table only holds one record for all events with the same lifecycle_id. Subsequent events for the same lifecycle_id will update the record as opposed to creating a new record.
- Use this extract to find all the unique Digital Activity transactions uploaded using Digital Activity API

Column Name	Column Type	Description
EVENT_ID	PRIMARY KEY	Internal field. The event's unique identifier. Randomly generated UUID, equal to the corresponding FDZ_UUID in the Pulse's event storage.
EVENT_ALERT	BOOLEAN	Boolean for whether it is an alert or not.
EVENT_TENANT	STRING	Internal field which stores the tenant key.
QUEUE_ID	STRING	The ID of the current queue.
USER_ID	STRING	The ID of the current assignee.
ACCESS_GROUP_ID	STRING	The ID of the access group.
channelId	STRING	Channel identifier.
event_type	STRING	An internal field that indicates the endpoint from which the event entered.
fraud_score	STRING	Calculated fraud score.
outcome_decision	STRING	Event decision outcome.
amount	INTEGER	Transaction amount performed by the sender, in the currency specified by the `currency` field.
channel_type	STRING	Specifies the transaction payment channel.
currency_code	STRING	The local currency of the business.
customer_dob	STRING	Customer date of birth (YYYYMMDD).
customer_email	STRING	Customer's email.

Column Name	Column Type	Description
customer_id	STRING	Customer ID generated by the issuing bank.
customer_name	STRING	Customer's name.
customer_phone_number	STRING	Customer phone number.

AM_EVENT_TRANSFERSâ□□

- The records in this table correspond to the Transfer events (e.g. Initiation, Settlement, Info, Feedback, etc).
- The AM_EVENT_TRANSFERS database table only holds one record for all events with the same lifecycle_id. Subsequent events for the same lifecycle_id will update the record as opposed to creating a new record.
- Use this extract to find all the unique transfers uploaded using Transfers API

Column Name	Column Type	Description
EVENT_ID	PRIMARY KEY	Internal field. The event's unique identifier. Randomly generated UUID, equal to the corresponding FDZ_UUID in the Pulse's event storage.
EVENT_ALERT	BOOLEAN	Boolean for whether it is an alert or not.
EVENT_TENANT	STRING	Internal field which stores the tenant key.
QUEUE_ID	STRING	The ID of the current queue.
USER_ID	STRING	The ID of the current assignee.
ACCESS_GROUP_ID	STRING	The ID of the access group.
channelId	STRING	Channel identifier.
event_type	STRING	An internal field that indicates the endpoint from which the event entered.
fraud_score	STRING	Calculated fraud score.
outcome_decision	STRING	Event decision outcome.
merchant_city	STRING	Card acceptor city.
dispute_end_date	STRING	Timestamp corresponding to the moment when the dispute was closed.
dispute_feedback_notes	STRING	Contextual information associated with the dispute communication.
dispute_refunded	STRING	Whether the funds were refunded to the cardholder.
oob_feedback_notes	STRING	Contextual information associated with the out-of-band communication.
feedback_type	STRING	Identifies the different types of feedback.
dispute_notes	STRING	Contextual information left by the analyst at the time of the dispute opening.
dispute_reason	STRING	Why the dispute was opened.
dispute_start_date	STRING	Timestamp that corresponds to the time when the dispute was opened.
dispute_submitted_by	STRING	Who submitted the dispute.

AM_EVENT_STATUSâ□□

- Holds the history on status changes for events - previous state, new state, when the state change happened and who did it.

Column Name	Column Type	Description
EVENT_STATUS_ID	PRIMARY KEY	Internal field, randomly generated identifier for each status change.
OLD_STATUS_ID	STRING	The ID of the old status.
NEW_STATUS_ID	STRING	The ID of the new status.

Column Name	Column Type	Description
TIMESTAMP	INTEGER	Unix timestamp at which the event note was added.
USER_ID	STRING	ID of the user that added the note.

AM_EVENT_EXPLANATIONâ

- This table contains the explanations of why the event triggered an alert, that is the Rules that triggered the alert. This contains the rules explanation for events in AM_EVENT table.

Column Name	Column Type	Description
EVENT_ID	STRING	The event's unique identifier. Randomly generated UUID, equal to the corresponding FDZ_UUID in the Pulse's event storage.
EVENT_TIMESTAMP	INTEGER	Corresponds to the FDZ_timestamp field in the Pulse's event storage for this particular EVENT_ID.
EVENT_VERSION_ID	STRING	It is equal to EVENT_ID.
EVENT_VERSION_TIMESTAMP	INTEGER	It is equal to EVENT_TIMESTAMP.
EVENT_EXPLANATION_GENERATOR_ID	STRING	Contains the internal rule ID. The primary key of this table is composed of the event ID and the rule ID.
EVENT_EXPLANATION_GEN_NAME	STRING	Contains the rule name.
EVENT_EXPLANATION_TYPE	STRING	The type of trigger that caused the alert.
EVENT_EXPLANATION_DESCRIPTION	STRING	The explanation provided by the risk engine for triggering the rule.

AM_EVENT_NOTEâ

- Notes added by analysts that may include personal information. The key is the EVENT_NOTE_ID column.

Column Name	Column Type	Description
EVENT_NOTE_ID	PRIMARY KEY	Randomly generated identifier for each note manually added by the analysts.
NOTE_MESSAGE	STRING	Note added by the analyst.
USER_ID	STRING	ID of the analyst that added the note.

AM_EVENT_STATUS_REASONâ

- This will provide the reason why the status of an event was changed as selected in the UI.
- The table is related to the AM_STATUS_REASON table via the column STATUS_REASON_ID as the foreign key. The possible values for STATUS_REASON_ID is a small set of mostly static values that correspond to a few options in a drop-down menu that a Case Manager user can choose from in the UI.

Column Name	Column Type	Description
EVENT_STATUS_ID	PRIMARY KEY	ID of the status associated with this reason change. Maps to the EVENT_STATUS_ID field in AM_EVENT_STATUS.
STATUS_REASON_ID	FOREIGN KEY	Foreign key to AM_STATUS_REASON table.
TIMESTAMP	PRIMARY KEY	Unix timestamp at which the event note was added.

AM_SLAâ

- The table stores the SLA definition or metadata that is created to automatically perform actions on events.

Column Name	Column Type	Description
SLA_ID	PRIMARY KEY	ID of the SLA.
SLA_NAME	PRIMARY KEY	Name of the SLA.
SLA_DESCRIPTION	STRING	Description of the SLA.
SLA_GOAL	STRING	Goal of the SLA.
SLA_CONDITIONS	STRING	Conditions of the SLA.
SLA_ACTIONS	STRING	SLA actions.

AM_EVENT_SLAâ

- The table links specific events to SLA's.

Column Name	Column Type	Description
EVENT_ID	PRIMARY KEY	Unique event id.
EVENT_SLA_ID	PRIMARY KEY	Event sla unique id.
SLA_ID	FOREIGN KEY	Foreign key to AM_SLA table.

AM_AUTOMATIONâ

- The table stores the Automation rules created to automatically perform actions events and cases based on conditions.

Column Name	Column Type	Description
AUTOMATION_ID	PRIMARY KEY	ID of the service level agreement.
AUTOMATION_POSITION	UNIQUE KEY	Numeric position.
AUTOMATION_NAME	STRING	Name of Automation.
AUTOMATION_DESCRIPTION	STRING	Automation description.
AUTOMATION_CONDITIONS	STRING	Automation conditions.
AUTOMATION_ACTIONS	STRING	Automation actions.

`AM_CASE_*` â

ⓘnote

`AM_CASE_*` - They are not used by HSBC

Dead Letters System

info

Although described in conjunction, Pulse dead letter system runs independently from Case Manager

When Pulse and Case Manager receive events but are unable to persist them, for instance due to a connection error with the database, a recovery system is used to avoid losing important information.

This recovery system consists in storing failed events in Pulse and CM filesystems (dead letters) and a recovery job that parses these files and retries persisting them to the database.

Dead letters are stored in a `retry` folder under `<installation-folder>/var/dead-letters`. When the recovery job triggers, it reads the existing dead letters and tries to persist the events to the database. On success the dead letter is deleted from the filesystem otherwise they are kept in the filesystem until the recovery job next execution (by default 1 hour).

The recovery job maximum retries is by default 3 meaning that there is a 3 hour window to identify and resolve the database issue. If this threshold is reached, Pulse and CM move the dead letters to a failed folder under `<installation-folder>/var/dead-letters`.

Available dead letters metrics for Case Manager are described in Event Queue Deadletter and for Pulse in Events Spilled to Disk.

Monitoring the Infrastructure

info

Note that the following list is simply a reference to the most important metrics, and should not be considered as the single source of monitoring the healthiness of the system.

While monitoring the application is important, it is not possible to have a useful monitoring system if the infrastructure and the operating system are not monitored as well. There are many metrics that could be listed here, but the truth is that some are more important than others, and that list may vary with the system. For example, systems that are prone to spill data to disk should be provisioned with more storage, and disk usage monitoring is an absolute must, as if the disk runs out of space, we may end up with corrupted data and a crashed service.

These are the metrics that should absolutely be monitored for all VMs or containers:

- CPU stats (user, system, iowait & idle percentages)
- Memory usage (used, buffered, cached & free percentages)
- Virtual Memory statistics (dirty page flushes, writeback volume)
- Disk I/O (operations & amount of data transferred per unit time, time to service operations)
- Free disk space on the mount used for data directory
- File descriptors used by `beam.smp` process vs. max system limit
- Network throughput (bytes received, bytes sent) & maximum network throughput
- Network latency (particularly between replicas from the same service)

Case Manager

In this section you will find the key metrics to monitor, a detailed guide to the important tables used and the relevant alerts in Case Manager.

In this guide you will find:

- Which metrics are especially relevant for Case Manager.
- Which tables are important in Case Manager.
- Which alerts are especially relevant for Case Manager.

On this page

Monitoring Cassandra

tip

For a complete reference of the metrics provided by Cassandra, refer to the official documentation. Make sure to refer to the version being used.

Collecting metricsâ

Metrics in Cassandra are managed using the Dropwizard Metrics library. The recommendation from the Cassandra maintainers is to query the metrics via JMX.

Feedzai recommends the usage of Jolokia, a JMX-HTTP bridge plugin that makes JMX-based systems more suitable to agent-based solutions. This allows agents like Telegraf to easily poll the metrics from individual nodes and send them to a centralised metric storage.

note

The recommendation from Telegraf for monitoring Cassandra is also to use a Jolokia plugin.

Top Metrics to Monitorâ

In this section you will find which metrics are important to monitor in **Cassandra**.

info

Note that the following list is simply a reference to the most important metrics, and should not be considered as the single source of monitoring the healthiness of the system.

Table Operation Latencyâ

The metric names are all appended with the specific `Keyspace` and `Table` name: `org.apache.cassandra.metrics.Table.<MetricName>.<Keyspace>.<Table>`

Metric Name	Description
`ReadLatency`	Local read latency for this table.
`WriteLatency`	Local write latency for this table.
`PendingFlushes`	Estimated number of flush tasks pending for this table.
`PendingCompactions`	Estimate of number of pending compactions for this table.
`TotalDiskSpaceUsed`	Total disk space used by SSTables belonging to this table, including obsolete ones waiting to be GCâd.

tip

Table-specific metrics should be monitored for all the relevant tables in the system. In particular, the profiles table maintained for Pulse (i.e. the Profile Store) and the entity table maintained by the Reference Data Service.

Commit Logâ

Metrics specific to the `CommitLog` : `org.apache.cassandra.metrics.CommitLog.<MetricName>` .

Metric Name	Description
`CompletedTasks`	Total number of commit log messages written since [re]start.
`PendingTasks`	Number of commit log messages written but yet to be fsyncâd.
`TotalCommitLogSize`	Current size, in bytes, used by all the commit log segments.
`WaitingOnCommit`	The time spent waiting on CL fsync; for Periodic this is only occurs when the sync is lagging its sync interval.

Compactionâ

Metrics specific to the `Compaction` : `org.apache.cassandra.metrics.Compaction.<MetricName>` .

Metric Name	Description
`PendingTasks`	Estimated number of compactions remaining to perform.
`CompletedTasks`	Number of completed compactions since server [re]start.
`TotalCompactionsCompleted`	Throughput of completed compactions since server [re]start.

ⓘ**note**

The pending compactions per table are also reported under `org.apache.cassandra.metrics.Table.<MetricName>.<Keyspace>.<Table>.PendingCompactions`

JVM metricsâ

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/operations-guide/monitoring/monitoring-gcp

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Monitoring in GCP

info

Note that this section is simply a reference to the monitoring service from Google Cloud Platform, and is not a guide on how to implement a monitoring solution leveraging that service. For more details about that refer to the official documentation.

Operations Suite (formerly Stackdriver)â

When designing a new system, one of the parts you will need to consider is your monitoring solution. That can be a daunting task with all the decisions that need to be taken technology and architecture-wise. However, if you are on Google Cloud Platform, the Operations Suite service provides an integrated option for monitoring and logging for any system running on GCP.

Ops Agentâ

There are some metrics that it is impossible for applications to report as they often only have visibility about their process. Likewise, there are metrics that the hypervisor managing the instances can report, such as the memory allocated to a particular instance, but it cannot report which portion of that memory is actually being used. The typical approach to address such observability limitations is to install another process in the same virtual machine, such as an *agent*. This is why Google created the Ops Agent.

The Ops Agent is the primary agent for collecting logs and metrics from individual VMs. This agent uses Fluent Bit and OpenTelemetry Collector for collecting logs and metrics, respectively. In addition to collecting Google Compute Engine metrics, this agent can also be configured for third-party applications, such as ZooKeeper and Cassandra.

tip

If Ops Agent is not compatible with all the systems you want to monitor, a great alternative is Telegraf, one of the best server-based agents for collecting metrics, and can be configured with a Google Cloud Monitoring output plugin.

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Monitoring Logstash

Collecting metricsâ

Logstash automatically captures runtime metrics that can be used to monitor the health and performance of Logstash deployment.

The metrics collected by Logstash include:

- Logstash node info like pipeline settings, OS info, and JVM info.
- Plugin info, including a list of installed plugins.
- Node stats, like JVM stats, process stats, event-related stats, and pipeline runtime stats.

We can use monitoring APIs provided by Logstash to retrieve these metrics. These APIs are available by default without requiring any extra configuration.

Top Metrics to Monitorâ

Implement normal dashboards for Java application/ JVM and dashboards that show the performance of the pipelines. Example:

- JVM Garbage Collector Time
- JVM Heap Usage
- Received Events Throughput per Pipeline
- Emitted Events Throughput per Pipeline
- Events Latency per Pipeline

Pulse

In this section you will find the key metrics to monitor and the relevant alerts in Pulse.

In this guide you will find:

- Which metrics are especially relevant for Pulse.
- Which alerts are especially relevant for Pulse.

On this page

Monitoring RabbitMQ

tip

For a complete reference of the monitoring recommendations for RabbitMQ, refer to the official documentation. Make sure to refer to the version being used.

info

The details in this page may change depending on how you have installing your RabbitMQ cluster. For example, if you are using Kubernetes, you should be using the RabbitMQ Kubernetes Operator, and different recommendations apply.

Collecting metricsâ

RabbitMQ exposes several application-specific metrics that are collected and reported by RabbitMQ nodes. Some of these are cluster-wide, and others are node-specific. It is important to consider this when collecting and storing them.

The recommended monitoring approach from RabbitMQ is using the Prometheus plugin. Feedzai has also used the management plugin to serve metrics, and also allowing the cluster to be managed via a UI or an API. When activated, the management plugin provides an HTTP API.

warning

There are some performance concerns with the management plugin, which is why the recommendation is to use the Prometheus plugin. If you just wish to monitor cluster-wide metrics (see below), having the plugin enabled on a single node would suffice. However, for redundancy and node-specific metrics, all nodes in the cluster should have it enabled.

note

Telegraf provides an input plugin that relies on the RabbitMQ Management Plugin.

Top Metrics to Monitorâ

In this section you will find which metrics are important to monitor in **RabbitMQ**.

info

Note that the following list is simply a reference to the most important metrics, and should not be considered as the single source of monitoring the healthiness of the system.

Cluster-Wide Metricsâ

Cluster-wide metrics provide a high level view of cluster state, including the interaction between cluster nodes. To access these metrics, use the endpoint `GET /api/overview` from the HTTP API

Metric Name	Description
`object_totals.connections`	Total number of connections
`object_totals.channels`	Total number of channels
`object_totals.queues`	Total number of queues

Metric Name	Description
`object_totals.consumers`	Total number of consumers
`queue_totals.messages_ready`	Number of messages ready for delivery
`queue_totals.messages_unacknowledged`	Number of unacknowledged messages
`message_stats.publish_details.rate`	Message delivery rate
`message_stats.deliver_get_details.rate`	Message publish rate

tip

Consider capturing message metrics individually for the most important queues with the `GET /api/queues/{vhost}/{qname}` endpoint.

Node Metricsâ

Most of the metrics represent point-in-time absolute values of individual nodes. To access these metrics, use the endpoint `GET /api/nodes/{node}` from the HTTP API (or `GET /api/nodes` if your collector supports an array).

Metric Name	Description
`proc_total`	Erlang process limit
`proc_used`	Erlang processes used
`mem_limit`	Memory usage high watermark
`mem_used`	Total amount of memory used
`queue_totals.messages_ready`	Number of messages ready for delivery
`queue_totals.messages_unacknowledged`	Number of unacknowledged messages

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RabbitMQ Queues

This section presents the messaging topics used in Pulse and Case Manager.

This section describes each topic/queue in detail. Each topic/queue is presented taking in account its general purpose and a list of consumers and producers. Each topic may have one or more queues and each queue one or more consumers.

Pulse Queuesâ

This section describes the topic/queue Pulse uses.

Note: topics are identified by an `*` in front on their name. They can have more than one queue associated with them which is identified by the routing key.

Topic/Queue Name	Description	Consumers	Producers	Notes
fdz-sq-digital-activity-non-scoring-*	When a new non-scoring digital event (e.g. async event) is received by Pulse.	Pulse	Pulse	
fdz-sq-transfers-non-scoring-*	When a new non-scoring transfer event (e.g. async event) is received by Pulse.	Pulse	Pulse	
fdz-sq-hsbc-payments-feedback-listenerhsbc-payments-feedback-listener	When a new feedback is received by Pulse.	Pulse	Pulse	
fdz-sq-global-feedbacks-notification-Feedback-Notifier-Migrator	When a feedback change is received by Pulse.	Pulse	Pulse	
fdz-sq-global-feedbacks-queue-appId-<app_id>	When a new Feedback event is sent to the Feedback Storage Service.	Pulse	Pulse	If this queue piles up over time, feedback updates to the Pulse event storage table will not be written (the events in Pulse will be missing the fraud feedback labels), which can have an impact on SMJs depending on the metrics defined, and fraud labels coming from feedbacks will not arrive in Case Manager, leading to missing fraud feedback labels.
fdz-sq-global-fdz-sync-topic-*	Used for workflow replicas to be notified of normal events arriving the original workflow dead letter.	Pulse	Pulse	The number of messages in this topic/queue may increase substantially during an App Publish since the node being published does not process replica messages sent by the other instances. When the App Publish finishes the instance should be able to process the pending messages.
		Pulse	Pulse	

Topic/Queue Name	Description	Consumers	Producers	Notes
fdz-sq-global-fdz-app-*	This is a topic per application, shared by all replicas of that application. It has two main uses: - Communication between the Pulse Server - And the App Engines for publishing / stopping apps.			
fdz-sq-global-fdz-apps-service-app-service-*	This topic is used for two purposes: - Pulse Server leader to communicate application life cycle requests to followers. For example, after creating an app through the web API (on the Pulse Server leader), the app should be bootstrapped in all pulse servers; in order to achieve this, the pulse server leader publishes a Bootstrap App message for the other Pulse servers to know that they must bootstrap the app. - To broadcast changes in model object values. Currently this is used only by the Managed List Service, where changes on list items made on the GUI take effect immediately, with no need to re-publish the app.	Pulse	Pulse	
fdz-sq-global-fdz-dc-heartbeat-*	Receive heartbeat messages sent from other cluster monitor implementations and the self Data Center Heartbeat. Also, listen for Data Center Heartbeat Request messages, requesting each data center to report that it is alive.	Pulse	Pulse	
fdz-sq-global-fdz-metadata-sync-job-manager-*	For App Engines to be notified of request to update a job metadata.	Pulse	Pulse	
fdz-sq-global-fdz-metadata-sync-member-*	For App Engines to be notified of requests to reload the Configuration.	Pulse	Pulse	
fdz-sq-global-fdz-metadata-sync-pulse-config-manager-*	For Pulse Servers to be notified of requests to reload the Configuration.	Pulse	Pulse	
fdz-sq-global-fdz-metadata-sync-pulse-security-*	For Pulse Servers to be notified of requests to reload security metadata.	Pulse	Pulse	
fdz-sq-global-jwk-service-updates-*	When a JWK is created or revoked, the leader node sends an update for other replicas	Pulse	Pulse	
fdz-sq-fdz-cluster-dc1-dc-manager-*	This is used to broadcast to all the cluster members of the local data center that a given data center became primary.	Pulse	Pulse	
fdz-sq-fdz-data-science-jobs-<app_id>*	This is used to keep data science job status updated between Pulse instances	Pulse	Pulse	
fdz-sq-fdz-force-flush-topic-*	Keep workflow clocks synchronized between replicas.	Pulse	Pulse	
fdz-sq-fdz-processes-*	This topic is used for communications between external jobs (e.g. SMJs running	Pulse/ External Job		

Topic/Queue Name	Description	Consumers	Producers	Notes
	on Dataproc) and their launcher, the job manager service.		Pulse/ External Job	
fdz-sq-fdz-pulse-mgmt*	This is used by the pulse-mgmt tool to broadcast requests for disabling and shutting down a data center to all members of the current data center, in fail overs.	Pulse	Pulse	
fdz-sq-fdz-workflow-publish-*	This is used to synchronize the workflow publish state between Pulse instances	Pulse	Pulse	

Case Manager Queuesâ

This section describes each topic/queue Case Manager uses.

Note: Since cases are not being used all queues regarding cases in the following table are merged in `CM_UPDATES_CASES_*`.

Topic/Queue Name	Description	Consumers	Producers	Notes
CM_UPDATES_CASES_*	Cases Queues. **Note:** These queues are not in use	Logstash	Case Manager	
CM_UPDATES_EVENTS-EVENT-external-notifications-queue	When a new event (or an event update) arrives in CM.	Logstash	Case Manager	If no consumer is supposed to connect to this queue, it should be pruned occasionally so messages do not pile up.
CM_UPDATES_EVENTS-QUEUE-external-notifications-queue	An event's queue changes.	Logstash	Case Manager	If no consumer is supposed to connect to this queue, it should be pruned occasionally so messages do not pile up.
CM_UPDATES_EVENTS-STATUS-external-notifications-queue	When the state of an event (on any state machine) changes.	Logstash	Case Manager	If no consumer is supposed to connect to this queue, it should be pruned occasionally so messages do not pile up.
CM_UPDATES_EVENTS-USER-external-notifications-queue	The user assigned to an event has changed.	Logstash	Case Manager	If no consumer is supposed to connect to this queue, it should be pruned occasionally so messages do not pile up.
CM_UPDATES_EVENTS-RELATED_TRANSACTIONS-external-notifications-queue	When related transactions arrive to CM.	Logstash	Case Manager	If no consumer is supposed to connect to this queue, it should be pruned occasionally so messages do not pile up.
CM_UPDATES_EVENTS-NOTES-external-notifications-queue	When a new note is added to event	Logstash	Case Manager	If no consumer is supposed to connect to this queue, it should be pruned occasionally so messages do not pile up.
fdz-sq-EVENT_QUEUE_CARDS	When a new Cards event arrives to CM.	Case Manager	Pulse	In case this queue starts to pile up it means CM is not able to process events at a higher rate than they are being produced. This is not critical since the consequence is that events will show in CM later than when they were inserted in the queue. However, if events keep piling up in the queue for longer periods of time, this is a sign that something is wrong in the

Topic/Queue Name	Description	Consumers	Producers	Notes
				system, possibly in Case Manager or the database.
fdz-sq-EVENT_QUEUE_NON-MON	When a new Non-Mon event arrives to CM.	Case Manager	Pulse	In case this queue starts to pile up it means CM is not able to process events at a higher rate than they are being produced. This is not critical since the consequence is that events will show in CM later than when they were inserted in the queue. However, if events keep piling up in the queue for longer periods of time, this is a sign that something is wrong in the system, possibly in Case Manager or the database.
fdz-sq-EVENT_QUEUE_DIGITAL-ACTIVITY	When a new Digital Activity event arrives to CM.	Case Manager	Pulse	In case this queue starts to pile up it means CM is not able to process events at a higher rate than they are being produced. This is not critical since the consequence is that events will show in CM later than when they were inserted in the queue. However, if events keep piling up in the queue for longer periods of time, this is a sign that something is wrong in the system, possibly in Case Manager or the database.
fdz-sq-EVENT_QUEUE_TRANSFERS	When a new Transfers event arrives to CM.	Case Manager	Pulse	In case this queue starts to pile up it means CM is not able to process events at a higher rate than they are being produced. This is not critical since the consequence is that events will show in CM later than when they were inserted in the queue. However, if events keep piling up in the queue for longer periods of time, this is a sign that something is wrong in the system, possibly in Case Manager or the database.
fdz-sq-EVENT_QUEUE_65b71a29808f2220	When a new Inbound Payments event arrives to CM.	Case Manager	Pulse	In case this queue starts to pile up it means CM is not able to process events at a higher rate than they are being produced. This is not critical since the consequence is that events will show in CM later than when they were inserted in the queue. However, if events keep piling up in the queue for longer periods of time, this is a sign that something is wrong in the system, possibly in Case Manager or the database.

On this page

Building a Monitoring Solution

Monitoring has become one of the most critical aspects of operating and maintaining a reliable system. There are numerous technologies that can be leveraged to implement real-time application and infrastructure monitoring according to the industry best practises. Different tools are more appropriate for different architectures, so *you must decide what fits best into your exising system*.

Regardless of the technologies you use or plan to adopt, make sure your monitoring solution covers the following aspects:

- **Real-time feedback**: Regardless of whether you have push or pull-based architecture, knowing the state of your system in near real-time is critical to respond to incidents, or even preempt them.
- **Application state**: The applications are becoming progressively more complex, and being able to observe how the state of the applications change over time is key to pinpoint other issues in the system. For example, in the application system that implements quorum-based consistency (e.g. ZooKeeper), it is important to track the quorum state, as it can have adverse consequences on systems that depend on it.
- **Performance**: With systems growing in size and complexity, the performance of a healthy system should be considered - i.e. how much usable work can be accomplished by the system within a given period. If it varies a lot for a given system component, or degrades over time, that effect can propagate to other components as well, and impact the whole system. For that reason, it is really important that you track the latency of operations that sit in the critical path of your workload.
- **Availability**: This is the most crucial metric to monitor - whether a system or component is accessible and operate the way it is intended to.
- **Resource usage**: On specific architectures and workloads, the performance and availability of your system is closely coupled with the resources that are made available to the system to perform its task. For that reason, it is important to monitor the available capacity of the resources provisioned to your workload.

tip

Feedzai uses the renowned TIG stack

One of the solutions Feedzai has implemented for monitoring is based on the TIG stack. This acronym refers to the following systems:

Telegraf: A server-based agent used for collecting and sending metrics or events to a data sink.

InfluxDB: A scalable time series data platform that is often used for real-time analytics through metrics or events.

Grafana: A data visualization and analytics platform that integrates a myriad of technologies to provide real-time data visualization and alerting.

This stack implements a pull-based architecture, which deploys a telegraph agent on all VMs, containers, etc. It periodically collects operation metrics (i.e. CPU, RAM, disk, etc.) as well as application metrics (i.e. health-checks, internal state, etc.), and propagates them to InfluxDB. Grafana pulls data from InfluxDB both to feed its visualizations and to automatically trigger alerts based on the values or trends of specific metrics.

Next stepsâ

Learn how to monitor Feedzai systems and how to integrate them in your preferred monitoring solution.

On this page

Monitoring ZooKeeper

tip

For a complete reference of the monitoring recommendations for ZooKeeper, refer to the official documentation. Make sure to refer to the version being used.

Collecting metricsâ

Since version 3.6.0 ZooKeeper has started to expose several metrics to monitor multiple parts of the system. The recommendation from the maintainers is to use Prometheus with Grafana for visualization, but there is also an alternative to use InfluxDB alongside the Telegraf plugin.

It is also possible to start ZooKeeper with JMX and then use Jolokia, a JMX-HTTP bridge plugin that makes JMX-based systems more suitable to agent-based solutions.

Feedzai also recommends the usage of Exhibitor, a supervisor system for ZooKeeper originally implemented by Netflix that also provides an API-based facility to manage the cluster.

Top Metrics to Monitorâ

In this section you will find which metrics are important to monitor in **ZooKeeper**.

info

Note that the following list is simply a reference to the most important metrics, and should not be considered as the single source of monitoring the healthiness of the system.

Metric Name	Description
`avg_latency`	The average latency of the requests executed against ZooKeeper
`znode_count`	The number of znodes (suggested to be keept below 1000000)
`open_file_descriptor_count`	Number of files opened (suggested to be keept below 300)
`election_time_count`	Used to determine if there was a new leader election recently
`followers`	Number of followers, could be used to computed the quorum based on total number of nodes (leader only)

Get Started With Monitoring

info

Important note

Feedzai systems are preferably deployed in Feedzai Cloud. In such deployments, monitoring is performed internally by Feedzai, and there is no need to provide clients with monitoring instructions. However, for some clients, Feedzai software may be deployed in a client-hosted environment, i.e. on-premise or in a public cloud).

This monitoring guide covers client-hosted deployments and, for that reason, it is not part of our standard client-facing documentation. The guide should only be provided to clients that require instructions on how to monitor their own systems.

The purpose of this guide is to empower users to monitor Feedzai's system performance and investigate statuses autonomously.

In this guide you will learn:

- How to collect metrics.
- What kind of metrics there are.
- Which metrics are especially relevant for Pulse.
- Which metrics and tables are especially relevant for Case Manager.
- Which metrics are especially relevant for Reference Data Service.

tip

Note that this guide only covers some Feedzai services, as an example. Nevertheless, the same principles may apply to other products and components.

Monitoring Third-Party and Open Source Components

info

Important note

While Feedzai can provide recommendations on how to monitor third-party and open source software based on experience of using that software, Feedzai is not accountable for the software or its observability facilities. For client-hosted installations, clients are expected to install and operate such software, and that also extends to monitoring.

This guide only provides a reference to some of the most important metrics to monitor on some of these systems, and does not constitute a definitive monitoring guide.

The purpose of this guide is to complement the recommendations for monitoring Feedzai components with recommendations for third-party dependencies.

This guide covers:

- Monitoring recommendations for ZooKeeper.
- Monitoring recommendations for Cassandra.
- Monitoring recommendations for RabbitMQ.
- Monitoring recommendations for Logstash.

On this page

Top Metrics to Monitor in Case Manager

In this section you will find which metrics are important to monitor in **Case Manager**.

info

Note that the following list is simply a reference to the most important metrics, and should not be considered as the single source of monitoring the healthiness of the system.

For a complete and comprehensive monitoring guide, please refer to the official [Case Manager documentation]

Please also refer to [Understand Available Metrics] for more information.

note

The following Case Manager standard application tags are available in every metric that references `Standard CM application tags`:

- `metricType`
- `appType`

Event Processing Metricsâ

New Event Throughputâ

Measures the event processing throughput when the event type is "new event", for the selected Case Manager Instance. An event is considered to be "new" if the identifier of that event has never been seen. Otherwise, it is seen as an update.

Metric	Description	Tags	Type
`cm.event.processing.latency`	The event processing throughput for "new events"	• `eventType` • Standard CM application tags	Counter

What to Monitorâ

This metric is useful to ensure that the rate at which events are being processed is as expected. If this rate is lower than what Pulse is processing, then it could mean that there is a problem in the system, and there will be backpressure on RabbitMQ.

New Event Latencyâ

Measures the event processing latencies per percentile (50th, 95th, 99th, and 99.9th) when the event type is "new event", for the selected Case Manager instance.

Metric	Description	Tags	Type
`cm.event.processing.latency`	The event processing latency for "new events" at various percentiles.	• `eventType` • Standard CM application tags	Timer

What to Monitorâ

This metric is useful together with the previous throughput metric. As it reports the latency of the new events processed, if the throughput starts to decrease, we can use this metric as a complement to understand what can be contributing to that. For example, the system might be reaching memory exhaustion (see JVM metrics), or it's taking too long to run all the automations configured to trigger on event arrival.

Event Processing Queue Occupancyâ

Measures the percentage of the event processing queue being used, for the selected Case Manager instance.

Metric	Description	Tags	Type
`cm.event.processing.queue.ratio`	The percentage of the event processing queue being used.	• Standard CM application tags	Gauge

What to Monitorâ

This metric shows whether events are being queued in Case Manager's internal processing queue or not.

Event Queue Deadletterâ

The Event Queue deadletter is the process by which events that could not be processed successfully are written to local storage to be re-attempted later on by an automatic process. The retry outcomes are also measured.

Metric	Description	Tags	Type
`cm.event.queues.dump.storage.dump.entries.failed.counter`	The total number of failed event queue entries dumped to storage.	• Standard CM application tags	Counter
`cm.event.queues.dump.storage.dump.entries.retry.counter`	The total number of retried event queue entries dumped to storage.	• Standard CM application tags	Counter

What to Monitorâ

Events are not expected to be spilled to disk, and hence, this metric should report zero. Otherwise, this indicates that some events are failing to be ingested from the upstream queue and processed. This could happen, for example, if there is an automation or custom extension being invoked on event arrival that failed exceptionally.

Event Storage Deadletterâ

The Event Storage deadletter is the process by which events were processed but could not be persisted successfully. These events are written to local storage to be re-attempted later on by an automatic process. The retry outcomes are also measured.

Metric	Description	Tags	Type
`cm.event.storage.dump.storage.dump.entries.failed.counter`	The total number of failed event storage entries dumped to storage.	• Standard CM application tags	Counter
`cm.event.storage.dump.storage.dump.entries.retry.counter`	The total number of retried event storage entries dumped to storage.	• Standard CM application tags	Counter

What to Monitorâ

Events are not expected to be spilled to disk, and hence, this metric should report zero. Otherwise, this indicates that some events are failing to be persisted to the database. This could happen, for example, if there is a connectivity issue with the database, or if the operation is failing exceptionally.

Automation Rules Metricsâ

Event Automations Throughputâ

Measures the throughput of the automation rules execution, for a given automation rule identified by the tag `ruleName`.

Metric	Description	Tags	Type
`cm.automation`	The throughput of automation rule execution.	• `ruleName` • Standard CM application tags	Counter

What to Monitorâ

This metric is useful to ensure that the rate at which specific automations are being triggered is as expected. If you have a flow that depends on specific rules that trigger on event updates and such automations are not triggering, then it can indicate a failure in some part of the integration.

Event Automations Latencyâ

Measures the latency per percentile (50th, 95th, 99th, and 99.9th) of the automation rules execution, for a given automation rule identified by the tag `ruleName`.

Metric	Description	Tags	Type
`cm.automation`	The latency of automation rule execution at various percentiles.	• `ruleName` • Standard CM application tags	Timer

What to Monitorâ

This metric is useful together with the previous throughput metric. As it reports the latency of the automations being executed, if the throughput starts to decrease, we can use this metric as a complement to understand what can be contributing to that. For example, specific actions of certain automations could be inflicting some overhead in the system.

Event Automations Errorsâ

Measures the errors observed during the execution of automation rules, for individual rules. The rule is identified by the tag `ruleName`.

Metric	Description	Tags	Type
`cm.automation.error`	The rate of errors returned by the automation execution.	• `ruleName` • Standard CM application tags	Meter

What to Monitorâ

This metric provides a simple way to understand if the automations execution is failing during its execution. When there are no errors, the value of this counter is zero. If this metric reports a different value, then you must look into the log files for additional error details.

Integration Metricsâ

Pulse Requests Errorsâ

Measures the errors returned by the Pulse Server API, for the endpoint described in the path tag. Case Manager engages with Pulse for several purposes, such as propagating fraud feedbacks, updating list items, validating user roles and permissions, etc. Each operation is done in a different endpoint.

Metric	Description	Tags	Type
`cm.request.pulse.api.error`	The rate of errors returned by the Pulse endpoint.	• `path` • Standard CM application tags	Meter

What to Monitorâ

This metric provides a way to monitor the integration between Case Manager and Pulse, where errors are not expected to occur. If the error count over time is greater than zero, then that means that either there is a network issue preventing Case Manager from contacting Pulse, or Pulse is facing problems. In any case, if there are errors it means that some operations have failed and may need to be retried (e.g., listing entities).

Infrastructure Metricsâ

Database Connection Poolâ

Each instance of Case Manager has a lazy connection pool to a database. One of the most important to monitor is the one used by the Apps Service. This service runs in Pulse Server and is responsible for handling all metadata management. The metrics for the DB pool are available under the `cm.db.connection.pool` prefix.

The metrics of the DB pool are available under the `cm.db.connection.pool` prefix.

Metric	Description	Tags	Type
`cm.db.connection.pool.size`	The current size of the connection pool (if pool is lazy, the value can grow over time).	• Standard CM application tags	Gauge
`cm.db.connection.pool.free`	The number of free database connections.	• Standard CM application tags	Gauge
`cm.db.connection.pool.used`	The number of used database connections.	• Standard CM application tags	Gauge
`cm.db.connection.pool.waiting`	The number of threads waiting for a database connection.	• Standard CM application tags	Gauge
`cm.db.connection.pool.usage`	The current usage of the database connection pool (used/size in percent).	• Standard CM application tags	Gauge

What to Monitorâ

These metrics detect underlying issues with connecting to the DB, or potential problems interacting with it. For example, if connections are not being released because operations are blocked, or if the database is becoming a bottleneck. If the pool usage is always at close to 100%, then there's likely an issue.

JVM Metricsâ

Each Pulse Server and Pulse Application instance lives in their own JVM process. Therefore, JVM metrics for each of these processes are available. JVM metrics are available as `letmeknow.jvm` . You'll find many different metrics under this namespace from GC to Memory.

What to Monitorâ

Look into at least GC and Memory metrics. These give you an idea of the healthiness of the process.

Other Monitoring Metricsâ

info

The following metrics are still being improved and they might change in the near future.

Metric	Description	Tags	Type
`hsbc.afis.responsemeter.httpstatuscode`	Operational results (using the count value).	N/A	Counter
`hsbc.afis.retry`	Number of requests retried (using the count value).	N/A	Counter
`hsbc.afis.fallback`	Measures the number of events dumped to disk (using the count value).	N/A	Counter
`hsbc.afis.requestlatencyHistogram`	Requests latency (using the available percentile values).	N/A	Counter

On this page

Top Metrics to Monitor in ETL

In this section, you will find which metrics are important to monitor in **ETL**.

WARNING - ETL metrics

ETL metrics are only exposed while the ETL process is executing. The process stays running for 20 seconds after the extraction finishes to allow fetching the metrics.

info

Note that the following list is simply a reference to the most important metrics, and should not be considered the single way of monitoring the system's health.

Execution metricsâ

Metrics about the ETL process execution.

How to measure Execution metricsâ

JVM metrics are available as `etl.execution`. You will find different metrics under this namespace, including number of events read and written, as well well errors.

What to monitor in Execution metricsâ

The following metrics are the most useful when monitoring the ETL process:

- `etl.execution.extract.success` : reports the amount of successful reads performed
- `etl.execution.extract.error` : reports the amount of reads that resulted in an error
- `etl.execution.publish.success` : reports the amount of successful writes performed
- `etl.execution.publish.error` : reports the amount of writes that resulted in an error
- `etl.execution.publish-queue` : reports the size of the ETL publish queue
- `etl.execution.last-execution` : reports the status of the last execution
- `etl.execution.last-duration` : reports the duration of the last execution

It is important to monitor any differences between successful reads and writes, as well as any errors that occur when reading or writing. Furthermore, alerts should be set up for when the last execution fails and the execution time should also be kept in check.

JVM metricsâ

The ETL is a JVM process so the typicall metrics are available.

How to measure JVM metricsâ

JVM metrics are available as `letmeknow.jvm`. Youâll find many different metrics under this namespace, from GC to Memory.

What to monitor in JVM metricsâ

Look into at least GC, Memory and Threads metrics. These give you an idea of the healthiness of the process.

Logback metricsâ

Metrics are available for the amount of logs produced by the ETL.

How to measure Logback metricsâ

Logback metrics are available as `letmeknow.logback`. Youâll find many different metrics under this namespace, for each of the log levels.

What to monitor in Logback metricsâ

Look into the number of log messages being generated, particularly spikes in ERROR logs.

On this page

Top Metrics to Monitor in Pulse

In this section you will find which metrics are important to monitor in **Pulse**.

info

Note that the following list is simply a reference to the most important metrics, and should not be considered as the single source of monitoring the healthiness of the system.

Please refer to Understand Available Metrics for more information.

note

The following Pulse standard application tags are available in every metric that references `Standard Pulse application tags` :

- `metricType`
- `appType`
- `appSlug`

warning

Some metrics will only be available in the metrics endpoint if the value is greater than zero, otherwise it might be expected for them to not appear at all.

Workflowâ

Workflow Throughputâ

Metric	Description	Tags	Type
`pulse.workflow.global.event.stats`	Measures the global throughput of the selected workflow belonging to the specified appSlug. It displays the throughput per message type (normal, recovery and replica).	• `workflowDesc` • `workflowId` • `messageType` • Standard Pulse application tags	Counter

Because this metric is captured for each type of message, the messageType tag allows you to distinguish between:

- **NORMAL** - Used for news messages that arrive at workflow.
- **RECOVERY** - Used for messages that are received when the system is doing recovery during a publish operation.
- **REPLICA** - Used to propagate the result of a NORMAL message, to all other Pulse instances.

What to monitorâ

This metric is useful to ensure that the rate at which events are being processed is as expected. If you are injecting events at 100 events per second into the workflow, this metric should report a similar value. If the rate being reported is consistently below the expected number, this might indicate that events are not being processed fast enough. This can indicate that there is a problem somewhere in the system.

Workflow Latenciesâ

Metric	Description	Tags	Type
`pulse.workflow.global.event.stats`	Measures the global latencies per percentile (50, 75, 95, 99 and 99.9), of the selected workflow belonging to the specified appSlug.	• `workflowDesc` • `workflowId` • `messageType` • Standard Pulse application tags	Timer

Because this metric is captured for each type of message, the `messageType` tag allows us to distinguish between **NORMAL**, **RECOVERY** and **REPLICA** messages.

What to monitorâ

This metric is useful together with the previous throughput metric. As it reports the latency of the events being processed, if the throughput starts to decrease, we can use this metric as a complement to understand what can be contributing to that -- for example, the system might be reaching memory exhaustion (see JVM metrics), or itâs taking to long to fetch historical profiles from storage (see Profile Fetch Latencies metrics).

Workflow Errorsâ

Metric	Description	Tags	Type
`pulse.workflow.eventErrors.count`	Measures the global workflow errors registered on the selected pulse instance, appSlug and appSlug workflow.	• `workflowDesc` • `workflowId` • Standard Pulse application tags	Counter

What to monitorâ

This metric provides a simple way to understand if there are any errors during event processing in a workflow.

When there are no errors, the value of this counter is zero. If this metric reports a different value, then you must look into the log files for additional error details.

Workflow Validationâ

Metric	Description	Tags	Type
`pulse.workflow.validation.concurrent`	Measures the number of concurrent workflow validations.	• `metricType` • `appType`	Gauge
`pulse.workflow.validation.e2e`	Measures end-to-end workflow validation time: since the request is received in Pulse Server until the answer is returned to the user.	• `workflowDesc` • `workflowId` • `metricType` • `appType`	Gauge
`pulse.workflow.validation.remote`	Measures time spent doing a remote call to the app engine for workflow validation.	• `workflowDesc` • `workflowId`	Gauge

Metric	Description	Tags	Type
		• `metricType` • `appType`	

What to monitorâ

These metrics refer to the work validations which are issued whenever a user opens/edits a workflow.

These validations may take from seconds to minutes depending on the complexity of a workflow.

Eventsâ

Event Pending Persistence to Event Storageâ

Metric	Description	Tags	Type
`pulse.workflow.eventstorage.eventsQueuedSize`	Measures the number of events that are pending to be persisted to the Event Storage (on DB).	• `workflowDesc` • `workflowId` • Standard Pulse application tags	Counter

What to monitorâ

Events are persisted in batches and at periodic intervals. Both the size of the batch and interval at which the batch persistence is triggered are configurable in the Pulse Administration page.

The Pulse properties are:

- `services.modules.rte.workflowmanager.eventstorage.databasebatch.size`
- `services.modules.rte.workflowmanager.eventstorage.databasebatch.timeout`

As such, the number reported by this metric shouldnât greatly exceed the size configured for the batch (which defaults to 10000).

Custom Alert Storage Service Additional Fieldsâ

Metric	Description	Tags	Type
`hsbc.workflow.ass.missingExpectedFields.count`	Number of additional fields configured to enrich events that are missing from the workflow schema.	N/A	Counter

Workflow Eventsâ

Metric	Description	Tags	Type
`workflow.events`	Measures the events being processed by each workflow belonging to the specified appSlug.	Check detailed description below	Meter

Unlike the Workflow Throughput metrics, this metric only takes normal messages into account.

The following tags are available to further breakdown the events being processed:

- `fdz_channel` : describes the channel of the event.
- `event_type` : describes the event type of the event.
- `decision` : describes the workflow decision outcome, for the event.
- `fraud` : describes the workflow fraud outcome, for the event.
- `Alert` : describes whether the event resulted in an alert.
- `appSlug` : describes the appSlug.

- `tenant` : describes the tenant.
- `workflowId` : describes the workflowId associated with the event.

What to monitorâ

This metric provides a comprehensive breakdown of the events being processed. Its uses include, but are no limited to:

- roughly counting the number of events flowing in the system (possibly by channel or even event type)
- roughly monitoring the decline rate of the system (possibly by channel or even event type)
- roughly monitoring the alert rate of the system (possibly by channel or even event type).

Events Dumped to Diskâ

Metric	Description	Tags	Type
`pulse.dump.storage.dump.entries.retry.counter`	The total number of events that are currently stored in the disk to be retried by the auto recovery job.	• `workflowDesc` • `workflowId` • Standard Pulse application tags	Counter
`pulse.dump.storage.dump.entries.failed.counter`	The total number of events that are currently stored in the disk but that already reached the maximum amounts of retry, therefore are no longer reprocessed.	• `workflowDesc` • `workflowId` • Standard Pulse application tags	Counter

Measures the number of events dumped to disk per host, appSlug, and workflow element. When events reach the end of the workflow they are persisted in the Event Storage (Database). If the Event Storage isn't available (e.g. The Database is offline, there's a network error, or some other unexpected error) the events are dumped to disk and are reprocessed later on by the Event Storage Dump component.

What to monitorâ

Events are not expected to be dumped to disk. Therefore, this metric should report zero. Otherwise, this indicates that some events are failing to be persisted to the DB, which suggests an issue in the system.

Events spilled to Diskâ

Metric	Description	Tags	Type
`pulse.eventStorage.spillToDisk.count`	The number of events that failed persistence to event storage and that were dumped into local disk.	• `workflowDesc` • `workflowId` • Standard Pulse application tags	Counter
`pulse.eventStorage.spillToDisk.pendingQueue`	The number of events that failed persistence to event storage and that are pending in-memory to be dumped into local disk.	• `workflowDesc` • `workflowId` • Standard Pulse application tags	Gauge

Measures the number of events spilled to disk per host, appSlug and workflow element. Events are spilled to disk when, after being processed by the workflow, they fail to be persisted to the Event Storage (on the DB). This might happen because the DB is down, thereâs a network error, a timeout occurred, or some other unexpected error.

What to monitorâ

Events are not expected to be dumped to disk. Therefore, this metric should report zero. Otherwise, this indicates that some events are failing to be persisted to the DB, which suggests an issue in the system.

Auto Recovery Jobâ

Metric	Description	Tags	Type
`pulse.dump.storage.retry.queue.size`	Number of in-memory entries waiting to be written to the retry folder.	• `workflowDesc` • `workflowId` • Standard Pulse application tags	Gauge
`pulse.dump.storage.failed.queue.size`	Number of in-memory entries waiting to be written to the failed folder.	• `workflowDesc` • `workflowId` • Standard Pulse application tags	Gauge
`pulse.dump.storage.recovery.job`	Whether the auto recovery job is running or not.	• `workflowDesc` • `workflowId` • Standard Pulse application tags	Gauge

These sets of metrics capture the current status of the auto recovery job used for persisting Events Dumped to Disk.

These metrics are available under the `pulse.dump` prefix. To capture a specific workflow, use the `workflowDesc` tag. As an example, to capture auto recovery job metrics for the `Payments` workflow, look for `workflowDesc=Payments` .

What to monitorâ

These metrics are important to know if the recovery job may have issues processing the spilled to disk events and for how long it is running. For instance, if the recovery job runs for several hours it may be due to database issues or because the Recovery Job is taking too much time to process events.

Jobsâ

Scheduled Jobsâ

These sets of metrics capture the current status of all scheduled jobs running in the system.

Metric	Description	Tags	Type
`pulse.job.executions.scheduled`	Number of jobs that are scheduled.	• `jobType` • Standard Pulse application tags	Counter
`pulse.job.executions.waiting`	Number of jobs that are waiting.	• `jobType` • Standard Pulse application tags	Counter
`pulse.job.executions.executing`	Number of jobs that are executing.	• `jobType` • Standard Pulse application tags	Counter
`pulse.job.executions.failed`	Number of jobs that failed.	• `jobType` • Standard Pulse application tags	Counter
`pulse.job.executions.crashed`	Number of jobs that crashed.	• `jobType`	Counter

Metric	Description	Tags	Type
		• Standard Pulse application tags	
`pulse.job.executions.aborted`	Number of jobs that were aborted.	• `jobType` • Standard Pulse application tags	Counter

What to monitorâ

These metrics are especially important to detect jobs that have either failed, crashed or been aborted, as this is a sign that something went wrong -- in which case the log files will provide more detailed information.

Profile Fetches Latenciesâ

Metric	Description	Tags	Type
`pulse.workflow.profileFetches.statistics`	Provides statistics on the latency of fetching these profiles from the external storage.	• `elementId` • `workflowDesc` • `workflowId` • `elementDesc` • Standard Pulse application tags	Timer

What to monitorâ

This depends on the use case. Nevertheless, make sure you understand what are the expected latencies, and look out for any continuous deviation from those values. If the latency starts to increase, there could be an underlying problem with the profile store in Cassandra. Look in the log files for more details.

Pulse Application Statesâ

Pulse Statusâ

As an application server, Pulse is responsible for managing the lifecycle of the applications it serves. This lifecycle varies with the state at which the application is. For example, if an application is healthy and running as expected, it is said to be in state `STARTED`. However, if someone publishes a change, it transitions to state `APPLYING_CHANGES`.

How to monitorâ

There are 10 possible application states, and these are being reported as an integer between 1 and 10. The states and their corresponding value are:

- **STOPPING(1)**: Application is stopping. For each AppEngine, this means that it is shutting down the JVM.
- **STOPPED(2)**: Application has stopped. Therefore, it should not have any JVM process.
- **BOOTSTRAPPING(3)**: Application is bootstrapping, i.e. services are being instantiated. This means that each AppEngine in this state is starting the JVM and creating the services, which haven't started yet.
- **BOOTSTRAPPED(4)**: Application has been bootstrapped. This means that for each AppEngine in this state, the respective JVM is up and running and all services have been created (but not started).
- **STARTING(5)**: Application is starting. This means that each AppEngine is starting its services since they were only instantiated in the previous state.
- **STARTED(6)**: Application has started. This means that, for each AppEngines, the services have started and the definitions that were meant to be published have been loaded. Both the AppEngine and app as a whole transition to this state from `APPLYING_CHANGES`.
- **CRASHED(7)**: Application has crashed. This means that it does not have any AppEngine alive. Each AppEngine never assigns itself this state. This is not possible since its JVMs are off.

- **APPLYING_CHANGES(8)**: Application is applying changes. This includes loading (or unloading) the definitions to be published. The AppEngine may have transitioned to this state from `STARTED` (in case it's re-publishing an already published app) or from `STARTING` (in case it's publishing an app that was previously in the `BOOTSTRAPPED` state). In either case, and if the AppEngine is working properly, it will transition to the `STARTED` state.
- **DEGRADED(9)**: Application is in degraded state, i.e., there may be partitions with no replicas online and/or there are replicas running with distinct versions. This means that some AppEngines lack JVMs, and therefore they can't assign themselves this state. The healthy AppEngines do not transition to this state either.
- **SELF_HEALING(10)**: This is the cluster's overall state and it means that the application is being self-healed by a watchdog correction. The individual AppEngines do not transition to this state. This is an overall app state only.

All AppEngines' states are available under the below metrics. To capture a specific AppEngine, use the `appSlug` tag.

Metric	Description	Tags	Type
`pulse.app.state`	The state of the Pulse Application.	• Standard Pulse application tags	Gauge
`pulse.server.state`	The state of the Pulse Server.	• Standard Pulse application tags	Gauge

Note that the state of each application is shown via the Pulse interface.

What to monitor in Pulse Application Statesâ

It is normal for applications to switch between states as teams promote new changes. However, it is not normal to see the application stop, crash, or degrade. Tracking the state of the application is crucial to identify silent problems that may require manual intervention, such as repairing a cluster that has replicas running with different versions of Risk Strategy.

App Lifecycle Operationsâ

These metrics allow for monitoring app engine bootstrap and upstart operations.

Metric	Description	Tags	Type
`pulse.app.publish.count`	The number of times the AppEngine got a publish request since startup.	• Standard Pulse application tags	Counter
`pulse.app.publish.skipRecovery`	Tracks the number of times recovery operations have been skipped.	• Standard Pulse application tags	Counter
`pulse.app.publish.applyingChanges.state`	Reports the fine-grained states during the `APPLYING_CHANGES` phase. For more details, please consult [Fine-grained Application States]().	• Standard Pulse application tags	Gauge
`pulse.app.publish.bootstrap.success`	Measures the full duration from app process start until the bootstrap work finishes successfully.	• Standard Pulse application tags	Timer
`pulse.app.publish.upstart.success`	Measures the end-to-end upstart time in success cases (i.e. from when the upstart message is received until it finishes successfully).	• Standard Pulse application tags	Timer
`pulse.app.publish.upstart.error`	Measures the end-to-end upstart time in error cases (i.e. from when the upstart message is received until it finishes with an error).	• Standard Pulse application tags	Timer
`pulse.app.publish.upstartBeforeWorkflows.success`	Measures the time from upstart message reception until workflows start being processed (for validation, recovery, etc.).	• Standard Pulse application tags	Timer
			Timer

Metric	Description	Tags	Type
`pulse.app.publish.loadAndValidateWorkflows.success`	Measures the workflow loading and validation time in success cases (i.e. when all workflows loading/validation succeed).	• Standard Pulse application tags	
`pulse.app.publish.loadAndValidateWorkflows.error`	Measures the workflow loading and validation time in error cases (i.e. when at least one workflow loading/validation fails).	• Standard Pulse application tags	Timer
`pulse.app.publish.recovery.success`	Measures the workflow recovery time in success cases (i.e. when all workflows recover successfully).	• Standard Pulse application tags	Timer
`pulse.app.publish.recovery.error`	Measures the workflow recovery time in error cases (i.e. when at least one workflow fails recovery).	• Standard Pulse application tags	Timer
`pulse.app.publish.startWorkflows.success`	Measures the workflow start time in success cases (i.e. when all workflows start successfully).	• Standard Pulse application tags	Timer
`pulse.app.publish.startWorkflows.error`	Measures the workflow start time in error cases (i.e. when at least one workflow fails to start).	• Standard Pulse application tags	Timer

Fine-grained Application Statesâ

There are 10 possible fine-grained application states during the `APPLYING_CHANGES` state, reported as an integer between 0 and 6. The states and their corresponding value are:

- **IDLE(0)**: State reported when `pulse.app.state` is not `APPLYING_CHANGES`.
- **UNLOADING(1)**: Application is unloading services/workflows.
- **LOADING_BEFORE_WORKFLOWS(2)**: Application is is loading services before reaching workflow code (validation/recovery/start).
- **LOADING_AND_VALIDATING_WORKFLOWS(3)**: Application is instantiating and validating workflows.
- **RECOVERING_WORKFLOWS(4)**: Application is recovering workflows.
- **STARTING_WORKFLOWS(5)**: Application is starting workflows.
- **FINALIZING(6)**: Application is doing any operation that might occur in the `APPLYING_CHANGES` state after `STARTING_WORKFLOWS` and moving to `IDLE` again.

Pulse Publishâ

Pulse's publish process can also be monitored with the following metrics.

Metric	Description	Tags	Type
`pulse.publish.e2e.success`	Measures end-to-end publish time that ended in success.	• Standard Pulse application tags	Timer
`pulse.publish.e2e.failure`	Measures end-to-end publish time that ended in failure.	• Standard Pulse application tags	Timer

Note that these timers also track the count of each occurrence.

What to monitorâ

These metrics may help identify bottlenecks in the publishing process.

JVM metricsâ

Each Pulse Server and Pulse Application instance live in their own JVM process. Therefore, JVM metrics for each of these processes are available.

How to measureâ

JVM metrics are available as `letmeknow.jvm`. Youâll find many different metrics under this namespace, from GC to Memory.

What to monitorâ

Look into at least GC and Memory metrics. These give you an idea of the healthiness of the process.

Apps Service DB Connection Poolâ

There are several DB connections used in Pulse. However, one of the most important to monitor is the one used by the Apps Service. This service runs in Pulse Server and is responsible for handling all metadata management.

How to measureâ

The metrics for all DB pools are available under the `pulse.dbPool` prefix. To capture a specific pool, use the `poolName` tag. In this case, for the Apps Service DB connection pool, look for `poolName=appsService`.

There are several metrics that can be captured:

- **Connections usage**: `pulse.dbPool.usage`
- **Connection errors and timeouts**: `pulse.dbPool.connectionErrors`
- **Connection creation throughput and latency**: `pulse.dbPool.connectionCreationTimeStats`

What to monitor in Apps Service DB Connection Poolâ

These metrics detect underlying issues with connecting to the DB. For example, if the time to create a new connection takes much more than the usual average, or if the pool usage is always at 100% capacity, then thereâs likely an issue.

Data Transfer Pipeline Serviceâ

To support the Data Transfer Pipeline, there is a workflow service that forwards workflow messages and feedback updates to GCP.

How to monitorâ

There are three main aspects that can be monitored on the DTP workflow service:

- Batch executor
- Dump storage
- Processing metrics

The metrics have further breakdown using tags for the channel and processing type. Expect to see the following combinations:

- `channel=MESSAGE,processingType=NORMAL` for workflow events being processed for the first time
- `channel=FEEDBACK,processingType=NORMAL` for feedback updates being processed for the first time
- `channel=MESSAGE,processingType=RECOVERY` for workflow events being retried
- `channel=FEEDBACK,processingType=RECOVERY` for feedback updates being retried

Batch Executorâ

The following metrics expose the internal batch executors used by the workflow service:

Metric	Description	Tags	Type
`pulse.workflow.dtp.batch.executor.pool.activeThreadsCount`	Number of active threads.	• `workflowId` • `channel` • `processingType`	Counter
`pulse.workflow.dtp.batch.executor.pool.maxPoolSize`	The maximum pool size.	• `workflowId` • `channel` • `processingType`	Gauge
`pulse.workflow.dtp.batch.executor.pool.threadsUsageRatio`	The current thread usage ratio.	• `workflowId` • `channel` • `processingType`	Gauge
`pulse.workflow.dtp.batch.executor.pool.taskQueueCount`	The current size of the task queue.	• `workflowId` • `channel` • `processingType`	Gauge
`pulse.workflow.dtp.batch.executor.pool.taskQueueMaximumSize`	The maximum size of the task queue.	• `workflowId` • `channel` • `processingType`	Gauge
`pulse.workflow.dtp.batch.executor.pool.taskQueueUsageRatio`	The current usage ratio of the task queue.	• `workflowId` • `channel` • `processingType`	Gauge
`pulse.workflow.dtp.batch.executor.service.completedFlushes`	The number of completed flushes.	• `workflowId` • `channel` • `processingType`	Counter
`pulse.workflow.dtp.batch.executor.service.deadletterEvents`	The number of deadletter events.	• `workflowId` • `channel` • `processingType`	Counter
`pulse.workflow.dtp.batch.executor.service.failedFlushes`	The number of failed flushes.	• `workflowId` • `channel` • `processingType`	Counter
`pulse.workflow.dtp.batch.executor.service.lastFlush`	The timestamp of the latest flush.	• `workflowId` • `channel` • `processingType`	Counter
`pulse.workflow.dtp.batch.executor.service.pendingQueueSize`	The size of the pending queue.	• `workflowId` • `channel`	Gauge

Metric	Description	Tags	Type
		• `processingType`	
`pulse.workflow.dtp.batch.executor.service.pendingQueueUsageRatio`	The usage ratio of the pending queue.	• `workflowId` • `channel` • `processingType`	Gauge

Dump Storageâ

When enabled, the Dump Storage will store failed messages and feedback updates in disk so they can be retried at a later time. Note that for these metrics, the processing type will always be `RECOVERY`. The following metrics are available:

Metric	Description	Tags	Type
`pulse.workflow.dtp.dump.storage.dump.entries.retry.counter`	The number of entries in disk that will be retried.	• `workflowId` • `channel` • `processingType`	Counter
`pulse.workflow.dtp.dump.storage.dump.entries.failed.counter`	The number of entries in disk that have exhausted all retries.	• `workflowId` • `channel` • `processingType`	Counter
`pulse.workflow.dtp.dump.storage.retry.queue.size`	Number of in-memory entries waiting to be written to the retry folder.	• `workflowId` • `channel` • `processingType`	Gauge
`pulse.workflow.dtp.dump.storage.failed.queue.size`	The number of in-memory entries waiting to be written to the failed folder.	• `workflowId` • `channel` • `processingType`	Gauge
`pulse.workflow.dtp.dump.storage.recovery.action.latency`	The latency of the recovery job.	• `workflowId` • `channel` • `processingType`	Gauge
`pulse.workflow.dtp.dump.storage.recovery.job`	Whether the auto recovery job is running or not.	• `workflowId` • `channel` • `processingType`	Boolean

Processing Metricsâ

These metrics provide a high level view of how many messages and feedback updates are being processed (successfully and otherwise) by the service:

Metric	Description	Tags	Type
`pulse.workflow.dtp.processing.numberOfReceivedMessages`	The number of messages received.	• `workflowId` • `channel` • `processingType`	Counter
`pulse.workflow.dtp.processing.numberOfSuccessfullyProcessedMessages`	The number of successfully processed messages.	• `workflowId` • `channel` • `processingType`	Counter
`pulse.workflow.dtp.processing.numberOfFailedRecoverableMessages`	The number of messages that failed to be processed but were sent for recovery.	• `workflowId` • `channel` • `processingType`	Counter
`pulse.workflow.dtp.processing.numberOfFailedNonRecoverableMessages`	The number of messages that failed to be processed and are not recoverable.	• `workflowId` • `channel` • `processingType`	Counter
`pulse.workflow.dtp.processing.messageLatency`	The latency of the processing.	• `workflowId` • `channel` • `processingType`	Timer

What to monitor in Data Transfer Pipeline Serviceâ

It's important to keep track of spikes in messages that failed to be processed. If they are recoverable this may suggest connectivity issues with GCP but if they are not recoverable they may indicate data or integration issues.

It's equally important to keep track if failed messages are being successfully retried or piling up in disk.

On this page

Top Metrics to Monitor in Reference Data Service

In this section, you will find which metrics are important to monitor in **Reference Data Service**.

info

Note that the following list is simply a reference to the most important metrics, and should not be considered the single way of monitoring the system's health.

Database Throughputâ

The database throughput measures the data operations' throughput for a given entity.

How to measure Database Throughputâ

The database throughput metric is available as `reference-data-service.entity-repository` using the `count` value. Since this metric is captured both by each operation and entity, the tags operation and entityType allow you to have a granular view for each one, respectively. For example, you can monitor the throughput of `upsert` operations for the `customer` entity.

The operations available are similar to the CRUD repository:

- **Get**: Fetch an entity's data by the primary key.
- **Delete**: Delete an entity by the primary key.
- **Upsert**: Update the entity's data or create a new entity if it doesn't exist yet.
- **List**: Retrieve a list of entities based on a set of filters.

What to monitor in Database Throughputâ

The database throughput metric is useful to ensure that the rate at which records are being updated by / fetched from each entity is as expected. If the number of upsert operations is below the expected, there may be a problem with the processing of reference data events. Likewise, if the number of `get` operations is not consistent with the event throughput, the system may not be enriching transactions with reference data as expected. In both scenarios, this can indicate that there is a problem somewhere in the system.

Database Latencyâ

The database latency metric measures the latencies per percentile (50, 95, 99 and 99.9), of the selected database operations for the specified entity.

How to measure Database Latencyâ

The database latency is available as `reference-data-service.entity-repository` and uses the available percentile values. Since this metric is captured both by each operation and entity, the tags `operation` and `entityType` allow you to have a granular view for each one, respectively.

The operations available are the same as mentioned above.

What to monitor in Database Latencyâ

The database latency metric is useful to monitor the latency of specific operations and can be very useful together with the previous throughput metric. If the throughput decreases, this metric can contribute to explain that, for example, if the latency of operations in the database degrades, there may be an issue with the database.

"Entity Not Found" Errorsâ

The "Entity Not Found" Errors metric measures the errors of "entity not found" type - i.e., lookups by entity key that fail because there is no record for such key.

How to measure "Entity Not Found" Errorsâ

The "Entity Not Found" Errors metric is available as `reference-data-service.entity-repository.entitynotfound`, using the `count` value.

What to monitor in "Entity Not Found" Errorsâ

The "Entity Not Found" Errors metric is useful to understand how many events miss the reference data enrichment and, when tracked over time, can hint at potential integration problems. Reference data updates are often done as a result of a Change Data Capture (CDC) process, or other data pipelines, and such processes are prone to silent failures. This metric helps identify such integration failures. The count over time is expected to be higher than zero, as there may be cases in which events from a given entity are sent before Feedzai gets any records from that same entity. Nevertheless, the the error count should be stable, and an increase is a strong indication of a process failure upstream.

Database Errorsâ

The Database Errors metric measures all database errors excluding the ones of type "entity not found", as previously described.

How to measure Database Errorsâ

The Database Errors metric is available as `reference-data-service.entity-repository.error`, using the `count` value.

What to monitor in Database Errorsâ

This metric shows whether there are operations failing due to issues in the database. For example, if the service receives an upsert request and the database is unresponsive, the operation fails, and this counter is incremented.

JVM metricsâ

Each Pulse Server and Pulse Application instance live in their own JVM process. Therefore, JVM metrics for each of these processes are available.

How to measure JVM metricsâ

JVM metrics are available as `letmeknow.jvm`. Youâll find many different metrics under this namespace, from GC to Memory.

What to monitor in JVM metricsâ

Look into at least GC and Memory metrics. These give you an idea of the healthiness of the process.

On this page

Understand Available Metrics

Access `https://<host:port>/metrics` on your browser to see which metrics are reported. Note that a large number of metrics is reported.

This section explains the different types of metrics being reported and how they are grouped.

Available types of metricsâ

Learn below about the different types of metrics, what kind of information they report, and how they are represented.

Gaugesâ

Gauges are the simplest metric type. They represent a single value.

Example:

```
"audit.writer.activeThreadsCount,metricType=APPLICATION,appType=PULSE_SERVER": {
  "value": 0
}
```

Copy

Countersâ

Counters are similar to gauges, in that they also report a single value. Conceptually, a counter is a simple incrementing and decrementing 64-bit integer.

Example:

```
"pulse.dbPool.pendingThreads,metricType=APPLICATION,appType=PULSE_SERVER,poolName=appsService": {
  "count": 0
}
```

Copy

Metersâ

A meter measures the rate at which a set of events occur. Meters report the following values:

- **count** - The current number of events.
- **m1_rate** - The median rate at which events occur over a 1-minute window.
- **m5_rate** - The median rate at which events occur over a 5-minute window.
- **m15_rate** - The median rate at which events occur over a 15-minute window.
- **mean_rate** - The median rate at which events occur since they have started being recorded.
- **units** - The units of the rate.

Example:

```
"audit.writer.droppedEntries,metricType=APPLICATION,appType=PULSE_SERVER": {
   count: 0,
   m15_rate: 0,
   m1_rate: 0,
   m5_rate: 0,
   mean_rate: 0,
```

```
  units: "events/second"
}
```

Copy

Timersâ

A timer is basically a histogram of the duration of a type of event and a meter of the rate of its occurrence.

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Note that the elapsed times for events are measured internally in nanoseconds, using Javaâs high-precision System.nanoTime() method. Its precision and accuracy vary depending on the operating system and hardware.

Timers report the following values:

- **count** - The current number of events.
- **max** - The maximum duration since this metric has started being reported.
- **mean** - The maximum duration since this metric has started being reported.
- **min** - The minimum duration since this metric has started being reported.
- **stddev** - The standard deviation of the duration since this metric has started being reported.
- **p50** - The duration in the quantile 50.
- **p75** - The duration in the quantile 75.
- **p95** - The duration in the quantile 95.
- **p98** - The duration in the quantile 98.
- **p99** - The duration in the quantile 99.
- **p999** - The duration in the quantile 99.9.
- **m1_rate** - The median rate at which events occur over a 1-minute window.
- **m5_rate** - The median rate at which events occur over a 5-minute window.
- **m15_rate** - The median rate at which events occur over a 15-minute window.
- **mean_rate** - The median rate at which events occur since they have started being recorded.
- **rate_units** - The units of the rate.
- **duration_units** - The units of the duration.

Example:

```
"pulse.dbPool.connectionCreationTimeStats,metricType=APPLICATION,appType=PULSE_SERVER,poolName=appsService": {
  count: 3,
  max: 0,
  mean: 0,
  min: 0,
  p50: 0,
  p75: 0,
  p95: 0,
  p98: 0,
  p99: 0,
  p999: 0,
  stddev: 0,
  m15_rate: 0.05330846541330047,
  m1_rate: 4.86993058876713e-10,
  m5_rate: 0.003787289909588294,
  mean_rate: 0.0024885250809603644,
  duration_units: "seconds",
  rate_units: "calls/second"
}
```

Copy

Tagsâ

Before moving on to how to make use of available metrics, itâs also important to understand some additional information that is included in some metrics, which provides information about their context.

Pulse Server and Pulse Applicationsâ

A Feedzai Pulse installation is composed of two main services: Pulse Server and Pulse Application. Each service lives in its own JVM process. A Feedzai Pulse installation has one Pulse Server JVM process per node and one additional JVM process for each application, also in the same node.

When you query the `/metrics` endpoint, the retrieved information corresponds to the relevant Pulse Server JVM process and to the JVM processes of each application. Tags are used to distinguish between metrics that correspond to the Pulse Server and to each of the existing Pulse Applications.

Example 1: Pulse Server

```
pulse.dbPool.pendingThreads,metricType=APPLICATION,appType=PULSE_SERVER,poolName=appsService
```

Copy

Notice the **appType** and **poolName** tags. The first one indicates that this metric corresponds to the Pulse Server instance, while the later is used to identify the name of the DB connection pool whose metric is being reported. In this case, the pool name is required as all DB connection pool metrics have the same `pulse.dbPool.pendingThreads` namespace.

Example 2: Pulse Application

```
pulse.dbPool.pendingThreads,appSlug=transactions,metricType=APPLICATION,creator=SimpleRulesTestingReplayModule,appType=PULSE_APPLICATION,poolName=simpleRulesTesting
```

Copy

This metric also corresponds to `pulse.dbPool.pendingThreads`. However, the **appType** tag indicates *PULSE_APPLICATION*, which means that this metric belongs to a Pulse Application. Additionally, the metric contains an *appSlug=transactions* tag. This **appSlug** tag allows you to distinguish between different Pulse Applications running in the same system.

To recap, when monitoring these metrics, consider the following hierarchy:

1. First per **appType**.

- *PULSE_SERVER* indicates that a metric corresponds to the Pulse Server instance.
- *PULSE_APPLICATION* indicates that a metric corresponds to a Pulse Application instance.

1. Then per **appSlug**.

- Where the value is the name of the Pulse Application. This tag is only available when *appType=PULSE_APPLICATION*.

Metric formatâ

Metrics are collected using two endpoints. These endpoints collect the same metrics but output different information each:

`/metrics` provides:

- Version
- Metric type
- Metric namespace
- Value

`/metrics/rich` provides:

- Definition
- Description

- Group namespace
- Thread pool name
- Metric name

See below examples of how this information is displayed.

`/metrics` endpoint informationâ

Access `https://<host:port>/metrics` on your browser to display the relevant output. To help explain the format of the reported metrics, consider the example below:

```
{
    "version": "3.1.3",
    "gauges":
    {
            "audit.writer.activeThreadsCount,metricType=APPLICATION,appType=PULSE_SERVER":
            {
                    "value": 0
            }
...
```

Copy

What you see above is the first reported metric. Letâs break down its relevant parts:

- *gauges* represents the type of metric being reported. Metrics are grouped by their type. In this case, itâs a gauge and returns a value.
- *audit.writer.activeThreadsCount* is the namespace of the metric. In this case, you are looking at the *activeThreadsCount* of the *audit.writer*.
- *value* is the actual value of the metric. In this case, itâs the number of active threads.

As this example shows, all metrics are represented by their namespaces. However, this information may not be enough to get the actual meaning of each metric. To get more information, you can use the `/metrics/rich` endpoint.

Letâs see how to do this by taking the previous example that used a metric with the *audit.writer.activeThreadsCount* namespace.

`/metrics/rich` endpoint informationâ

In the `/metrics/rich` endpoint, while all the same metrics are available, they are grouped differently. Instead of being grouped by type (e.g. the *gauge* above) and represented by their namespace (e.g. *audit.writer.activeThreadsCount* above), they are provided as follows:

```
{
    "definition":
    {
            "description": "The usage ratio of audit writer executor threads and task queue.",
            "evaluators": [ ],
            "name": "audit.writer",
            "tags":
            {
                    "metricType": "APPLICATION",
                    "appType": "PULSE_SERVER"
            },
            "unit": "n/a"
    },
    "metric":
    {
            "metrics":
            {
                    "activeThreadsCount":
                    {
                            "value": 0
```

```
                                  },
...
```

Copy

This is the exact metric as above but represented differently. Letâs break the relevant parts down as before:

- *definition* includes the definition of the group of metrics that monitor a specific module/action.
- *description* explains the group of metrics. In this case, this group reports metrics regarding the thread pool of the audit writer.
- *evaluators* can be ignored as they will not be populated.
- *name* is the groupâs namespace. This is used as the prefix for the namespace of all metrics within the group.
- *tags* allows you to distinguish metrics that have the same namespace from each other. They are used to identify the name of a thread pool.
- *metrics* represents the actual metrics for this group.
- *unit* can be ignored as they will not be populated.
- *activeThreadsCount* is the metric name.

To recap, metrics are represented by their namespace in `/metrics`, and namespaces are composed by `<metric group prefix>.<metric name>`. Letâs apply this logic to our example, where:

- *audit.writer.activeThreadsCount* is the namespace of the metric.
- *audit.writer* is the prefix of the metric group to which the metric belongs.
- *activeThreadsCount* is the name of the metric within the group.

tip

To locate the `audit.writer.activeThreadsCount` metric in `/metrics`, search its group prefix in `/metrics/rich`, and then search its name under that group *(audit.writer)* and *activeThreadsCount*, respectively.

On this page

Configuration Recommendations

Please find below the complete configuration reference for the Pulse Application (`PAPP`) cleanup pipeline.

Ansibleâ

note

Available ansible variables for running the Ansible playbook.

Variable	Default Value	Description
ansible_user	feedzai	User for ansible execution
ansible_local_user	{{ ansible_user }}	User that will be the owner of the unpacked Pulse assembly, should be the same as `ansible_user`
cleanup_pulse_install_path	"/opt/feedzai"	Cleanup install path
cleanup_pulse_home_path	"{{ cleanup_pulse_install_path }}/{{ cleanup_pulse_home_prefix }}{{ project_version }}""	Cleanup home path
cleanup_database_pulse_metadata_hostname	"postgres"	Postgres Cleanup Metadata host name
cleanup_database_pulse_metadata_port	"5432"	Postgres Cleanup Metadata port
cleanup_database_pulse_metadata_database_name	"feedzai"	Postgres Cleanup Metadata Database name
cleanup_database_pulse_metadata_jdbc	"jdbc:postgresql://{{ cleanup_database_pulse_metadata_hostname }}:{{ cleanup_database_pulse_metadata_port }}/{{ cleanup_database_pulse_metadata_database_name }}?{{ cleanup_database_pulse_metadata_jdbc_opts | join('&') }}"	Cleanup JDBC for metadata which should be the same metadata DB as Pulse.
cleanup_database_pulse_metadata_username	"pulse"	Pulse Metadata username

Variable	Default Value	Description
cleanup_database_pulse_metadata_password	"secret"	Pulse Metadata password
cleanup_database_pulse_metadata_schema	"pulse"	Pulse Metadata schema
whitelist_by_name		List of schemas names to keep.
keep_ui_schemas		`true` to keep all schemas that show on the UI, `false` otherwise.
cleanup_last_days	3	Number of days to retain. For example, if set to 30, resources that were not updated in the last 30 days are eligible for cleanup.
cleanup_recent	5	Number of most recent resources to keep.
cleanup_apps	cleanup_apps: Â Â - uk_outbound_payments	Applications to cleanup.
cleanup_pulse_commands_contexts		Controls which cleanup patches will run. Possible values are `cleanup-schemas`, `cleanup-rule-snapshots`, `cleanup-plan-executions`, `cleanup-workflow` and multiple can be provided separated by comma.

Pulse Application Cleanup

In Feedzai a Pulse Application, `PAPP` for short, refers to a modular component that manages data processing pipelines for various tasks such as real-time machine learning, fraud detection, and data analytics. Each `PAPP` is a combination of configurations, rules, machine learning models, and scripts that are specifically designed to handle high-throughput data streams. By deploying `PAPPs` , Feedzai enables businesses to efficiently process, analyze, and act on large volumes of data in real-time.

Overviewâ

Over time, as `PAPPs` are created, modified and deployed, un-used or outdated `PAPPs` can accumulate leading to issues or performance degradation. Therefore `PAPP` cleanup is essential for maintaining an efficient and optimized environment within the Feedzai platform by:

- **Optimizing resources:** un-used `PAPPs` consume valuable system resources such as CPU, memory, and storage. Cleaning up these `PAPPs` frees up resources for active applications and tasks, improving overall system performance
- **Reducing costs**: reducing resource consumption by eliminating unnecessary `PAPPs` helps in minimizing operational costs, such as cloud storage, compute instances, and other infrastructure expenses
- **Improving Security and Compliance:** retaining outdated or un-used `PAPPs` poses a security risk, as these applications may contain sensitive data or outdated configurations. Regular cleanup ensures that only the necessary and secure applications remain active
- **Improving System Maintainability:** maintaining a clutter-free environment reduces the complexity of managing applications, configurations, and resources, leading to easier troubleshooting, monitoring, and system maintenance

This process can be done via the Pulse UI although it may be difficult to identify and remove the resources can be cleaned. To automate it we have developed a new cleanup pipeline which you can find in the following sections.

On this page

Pipeline Definition

In this page you can find the cleanup pipeline definition and how to configured and run it.

Cleanup Pipelineâ

The cleanup pipeline is an Ansible pipeline that can be configured and executed independently of Pulse, and will run the following set of cleanup patches for each of the configured application:

- Plan executions
- Schemas
- Rule Snapshots
- Workflows

To also enabled automation of the process, the pipeline was designed to be a one shot executor, meaning that the machine were it runs can be removed after execution and allowing it to run any number of times with different configurations.

The pipeline follows the same strategy of the other components by having install and configure stages. This is useful to create an immutable image that can be used in all environments but configured individually for each. The install tasks are identified by the tag `cleanup.install` while the configuration tasks by `cleanup.configure`.

Although it runs independently of Pulse it will connect to a running Pulse cluster via Zookeeper and to the metadata database to fetch the data to clean. To determine to which Pulse and Database the pipeline will connect to the `deployment_color` and `database` variables should be used.

warning

Since the pipeline will connect to Pulse via Zookeeper, the VM running it should be able to connect to Zookeper's port 2181.

Configurationâ

info

Find a detailed description of all available configurations in [Configuration Recommendations]

Cleanup patch configuration depends on the patch type. The only configuration that is shared between all patches is `cleanup_apps`, which should contain the list of applications key to cleanup, where the key should match the configuration in `apps_enabled`.

Please find below each patch description and configuration.

Plan executionsâ

The plan executions cleanup patch removes the oldest configured plan executions while keeping the configured most recent ones:

- `cleanup_last_days`: number of days to retain. For example, if set to 30, resources that were not updated in the last 30 days are eligible for cleanup
- `cleanup_recent`: number of most recent resources to keep

Clean schemasâ

The clean schemas patch removes schemas that are not being used in workflows, plans, rules projects, datasources and models. Similarly to the plans patch, it can be configured to remove only the oldest schemas, although it can also be configured to whitelist a set of schemas or to keep all schemas that appear on the UI:

- `cleanup_last_days` : number of days to retain
- `whitelist_by_name` : list of schemas names to keep
- `keep_ui_schemas` : `true` to keep all schemas that show on the UI, `false` otherwise

Rule Snapshotsâ

Similar to the plan executions cleanup patch but to clean rules project snapshots:

- `cleanup_last_days` : number of days to retain
- `cleanup_recent` : number of most recent resources to keep

Workflowsâ

Removes all Workflow scoring elements that have an external datasource, for example, a datasource stored on a GCS bucket. This patch does not have any additional configuration.

Executionâ

Executing the pipeline is fairly straightforward and only requires the inclusion of the provided defaults and the environment specific `variables.yml` file, which can be configured in its own inventory (e.g. `project-qa/uk`).

As an example, to run a cleanup on all UK applications for the last 5 days while keeping the 10 most recent executions of a resource you would need to:

1. Add a new base inventory for the portfolio/environment you are configuring, e.g. `project-qa/uk/group_vars`
2. Add a new host inventory for Cleanup, e.g. `project-qa/uk/group_vars/cleanup`
3. Inside a new variables file (`variables.yml`), under the inventory created previously, add the overrides for the specific portfolio/environment

```
cleanup_apps:
  - uk_outbound_payments
  - uk_reference_data
  - uk_inbound_payments

cleanup_last_days: 5
cleanup_recent: 10
```

Copy

Then you would need to run the pipeline as shown below:

```
python3.6 ansible-playbook \
    --connection=local \
    -i project-default \
    -i project-hsbc \
    -i gcp \
    -i project-qa/uk \
    pulse-app-cleanup.yml
    -e deployment_color=<current_color>
```

Copy

info

All execution logs are stored in `/opt/feedzai/cleanup/tools/commands/`

Control patch executionâ

To provide greater flexibility and control over which cleanup patches are executed, app cleanup has a mechanism where cleanup patches are managed through distinct contexts instead of a single one.

To control which patches are executed the `cleanup_pulse_commands_contexts` Ansible variable can be used with the following context options:

- `cleanup-plan-executions` : run the clean plan execution patch
- `cleanup-schemas` : run the clean schema patch
- `cleanup-rule-snapshots` : run the clean rule snapshots patch
- `cleanup-workflow` : run the clean workflow patch

To specify multiple patch contexts each of the context should be separated by a comma. For instance, to run only schema and plan execution cleanup patches we would use `cleanup_pulse_commands_contexts: "cleanup-schemas,cleanup-plan-executions"` .

On this page

Permissions Migration Tool

In this page you can find the documentation on the permissions migration tool, how to configure and run it.

Overviewâ

The `fdz-permissions-migrator` tool is designed to export and import Pulse permissions and their associated roles. This process can be used to promote/migrate permissions and their associated roles between environments.

Requirementsâ

The tool execution has the following requirements:

- A `bash` shell
- `jq` installed in the machine where the script is being run (tested with version 1.7.1)
- `curl installed in the machine where the script is being run (tested with version 7.68.0)
- The file must have execution permissions (`chmod u+x fdz-permissions-migrator`)
- Connection between the machine where the script is being run and the Pulse load balancer
- A service account credentials

Resourcesâ

The tool can be downloaded here: fdz-permissions-migrator.

Configurationâ

Running `./fdz-permissions-migrator help` will show the help message for the tool. It lists the available commands and options:

```
âž¯ ./fdz-permissions-migrator help
fdz-permissions-migrator [v1.0.0] - commandline tool to export and import Pulse permissions and roles between environments

usage: fdz-permissions-migrator <command> <options>

command:
  help                              Show this help message
  export                            Export permissions to stdout, in JSON format
  import                            Import all permissions and all roles from stdin, in JSON format
  import-role <role>                Import all permissions and the specified role from stdin, in JSON format

options:
  -u, --username USERNAME           The username of the service account (default: pulse)
  -p, --password PASSWORD           The password of the service account (default: pulse)
  -h, --hostname HOSTNAME           The Pulse Load Balancer URL (default: http://127.0.0.1:8080)
  -k, --insecure                    Sets the --insecure flag in the underlying cURL requests
  -q, --quiet                       Limit the output of skipped operations
  -nc, --no-cleanup                 Preserve folder with temporary files generated during execution
```

```
Example:
  $ ./fdz-permissions-migrator export --username pulse --hostname http://localhost:8080
```

Copy

Executionâ

Running the tool will usually require the user to specify `--hostname`, `--username` and `--password` as the default values are only useful for development environments. Besides that, the user should select one of the available commands: `export`, `import` or `import-role`.

Assume the following example:

- Permissions and roles are to be migrated from `Environment A` to `Environment B`
- `Environment A` is running under URL `http://env_a:8080/` and `Environment B` under `http://env_b:8080/`
- The default service account will be used (i.e. the default username and password)

The first step would be to export the permissions from `Environment A`:

```
$ ./fdz-permissions-migrator export --hostname http://env_a:8080
[
  {
    "name": "security:tenants:read",
    "roles": [
      "admin",
      "Fraud Prevention Agents - L1"
    ]
  },
  {
    "name": "security:tenants:write",
    "roles": [
      "admin"
    ]
  }
  ...
]
```

Copy

Export will always output the result to stdout but it can instead be redirected to a file:

Using HTTPS

This tool uses cURL to make HTTP requests to the remote servers and accepts HTTPS URLs. If there are any problems with certificate validation, the `--insecure` flag can be passed

```
$ ./fdz-permissions-migrator export --hostname http://env_a:8080 > permissions.json
```

Copy

From this point it depends whether all extracted roles should be import or not.

If yes, we would simply import all roles into `Environment B` by running:

```
$ ./fdz-permissions-migrator import --hostname http://env_b:8080 < permissions.json
```

Copy

If instead we want to import a single role, we can use the `import-role` command:

```
$ ./fdz-permissions-migrator import-role "Fraud Prevention Agents - L1" --hostname http://env_b:8080 < permissions.json
```

Copy

This way all permissions will be added to Pulse, but only the "Fraud Prevention Agents - L1" role will have its permissions modified.

Tip: Make it more quiet!

When dealing with production environments, the amount of roles and permissions being imported can be quite big. The tool will output one line for each permission being imported, even the environment already has that permission.

If you want to avoid logging operations that were skipped because the role/permission already existed, add the `--quiet` flag.

Dealing with passwords

Given that during export the script can not output anything so that stdout redirection does not break, the service account password cannot be prompted.

To make the execution more secure, the password can be passed onto the script by prompting it without printing its value:

```
$ read -sp "Password: " PASSWORD
$ ./fdz-permissions-migrator export --password $PASSWORD
```

Copy

Validation can be done in the Pulse UI by going to `Administration -> Security` and checking if the roles have the correct permissions.

ETL Recovery Procedure

This page describes the recovery procedure for ETL, from detecting errors in the extraction to generating a new extraction.

Detecting failuresâ

In order to detect failures, we must rely on the monitoring capabilities provided by the ETL. After each extraction, the values for `etl.execution.extract.error` and `etl.execution.publish.error` should be used to detect if any error occurred.

Recovery may also be needed if events failed to be written to the Event Storage database and were successfully retried only after the ETL process executed for that period of time. The events spilled to disk metrics can help identify these situations.

WARNING - ETL metrics

ETL metrics are only exposed while the ETL process is executing. The process stays running for 20 seconds after the extraction finishes to allow fetching the metrics.

Generating new extractionsâ

See fixed extraction interval as a reference on how to run ETL for a specific interval.

For compliance with HSBC reports, all extractions must span a time interval of exactly one hour. If recovery needs to be performed over a period larger than one hour, then multiple executions should be performed. The start timestamp of each extraction must be the start of the hour and the end timestamp must be the last millisecond of that same hour. For example, if we wanted to recover the period between `01/01/2024 10:00:00` and `01/01/2024 12:00:00` then we would need two extractions:

1. The first extraction between `01/01/2024 10:00:00.000` (UNIX timestamp `1704103200000`) and `01/01/2024 10:59:59.999` (UNIX timestamp `1704106799999`)
2. The second extraction between `01/01/2024 11:00:00.000` (UNIX timestamp `1704106800000`) and `01/01/2024 11:59:59.999` (UNIX timestamp `1704110399999`)

Consuming the new extractionsâ

The previous step generates a new extraction that is published to a Google Cloud Storage bucket containing all events from a given period. It is crucial for downstream systems to ensure the events that were previously successfully extracted are not duplicated when ingesting the new extraction. If the downstream systems are idempotent, the `FDZ_UUID` column can as the idempotency key. Otherwise, we recommend removing all events that match the new extraction period before ingesting.

On this page

Pulse Recovery Procedure

This document outlines the available options to recover Pulse following a catastrophic event or outage that disrupts all Pulse nodes.
Recovery focuses on restoring the system to a state where Pulse can process and score real-time payments and digital events, ensuring both real-time and scheduled metrics are accurate so that the DS/rule/model performance is as expected.

warning

Please note, the timings in this guide are relative to the time it takes a Pulse node to publish. The figures used were accurate as of April 2024, but may vary over time.

However, while the exact timings may differ, the principles remain the same.

Pulse operates with a three-node setup, where a single node requires approximately **1h 45m** to complete a publish. The following recovery strategies detail how to handle such scenarios effectively.

Recovery Strategiesâ

Option 1: Abort Publish and Perform Manual Publishâ

Process:â

1. Access the Pulse UI and abort the publish manually.
2. Ensure all three nodes are part of the cluster.
3. Initiate a parallel (non-rolling) publish for all nodes simultaneously.

Recovery Timeline:â

- **First 1h 50m:** No processing or scoring occurs while the abort and re-publish steps are performed. The timeline includes an additional ~5 minutes required to abort and initiate the manual re-publish.
- **After 1h 50m:** Full service and redundancy are restored simultaneously.

  note

  Note that parallel publishing across all three nodes may take slightly longer if database performance becomes a bottleneck. Pre-production testing is recommended to validate this behavior.

Considerations:â

- Events processed by the GCP load-balancer during the outage will likely have incorrect scores due to:
  - **Real-time metrics** being absent or incorrect.
  - **Workflow clock misalignment**, which could negatively affect model scores and alert rates.
- Despite these issues, previous testing has shown that the impact on scores and alert rates is manageable.

Manually publish with recovery skipped, then republishâ

Process:â

1. Access the Pulse UI and abort the publish manually.
2. Ensure all 3 Pulse server nodes are part of the cluster, and perform a parallel (non-rolling) publish with "skip recovery" selected.

3. After the publish, save the workflow without changes.
4. Perform a rolling publish (with application restart selected, without **skip recovery**)

Recovery Timeline:â

- **First 5â10 minutes:** Pulse will not be processing or scoring the events.
- **After 1h 45m:**
    - Partial service and full redundancy are restored at the same time.
    - Pulse will score events, but it will consider all real-time metrics as having the value zero.
    - All scheduled metrics will be correctly updated.
    - Model scores/performance and alert rates will be impacted.
- **After another 1h 45m (total 1h 55m from the start):**
- Half the transactions will have totally correct scores, and the other half will continue to have incorrect real-time metrics.

  info

  This is because after the first node publishes, one of the remaining two will go down. This means there are two nodes scoring at this point - one that just finished publishing, fully correct, and one with correct SMJs but incorrect real-time metrics.

- **After another 1h 45m (total ~3h 40m from the start):**
- Full service recovery is complete with correct real-time and scheduled metrics.

  info

  Full redundancy has not yet been achieved as 2 of 3 nodes will be scoring, while the 3rd will go down to update itself.

- **After another 1h 45m (total ~5h 25m from the start):** Full redundancy is achieved.

  info

  All the nodes are up at this and correctly updated.

Considerations:â

- This option ensures quicker partial service restoration compared to Option 1, with limited impact on metrics during the recovery process.
- Transactions processed during the initial phase may have incomplete real-time metrics, but overall service availability is restored sooner.
- Data Science (DS) team changes cannot be published during the recovery process.

Recommendationsâ

Feedzai recommends Option 2. It is the standard approach followed in the Feedzai cloud. It tends to have the smallest negative impact overall

- The service is mostly restored (with just real-time metrics missing) very quickly
- Followed by full restoration in steps (half the nodes fully recovered after another 1h 45m, and full restoration after another 1h 45m)

On this page

What's Changed

Release `0.11.2` â

Add `project-<env>` inventory to commands in Blue/Green documentationâ

Docs

See up-to-date information at Blue-Green Deployments

Fix missing `/` on ETL cron job arguments (defaults and documentation example)â

Bug Fix
Docs

See up-to-date information at ETL Configuration

Replace previous `check-only` strategy by Feedzai Encryption Tool lookup pluginâ

Internal

This ensures no sensitive data like usernames and passwords are leaked in logs and doesn't rely in an additional task and double lockups with changed facts in between.

Add Artifacts list to installation pageâ

Docs

See Artifacts to get the list of artifacts, the expected version and the respective download links.

Update documentation with load balancing informationâ

Docs

See up-to-date information at Load Balancers.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-0.x/Release-0.12

On this page

What's Changed

Release `0.12.0` â

Analysts see information about channel and authenticationâ

Feature FTMT-348

Case Manager now shows the description in place of the "machine code" (`channel_type`) to simplify its usage.

Code	Description
C	Payment Card at Card Reader Terminal (including online purchase and ATM)
D	Payment Card or Number with Online Details and Device Fingerprint Information
E	Payment Card or Number with Online Details
O	Online Banking (internet, mobile phone)
W	Online Banking with device fingerprint information
P	Phone Banking
H	Self Bank Branch
M	Correspondence(for non-mon and check deposit)
B	Bank Processing(include bank initiated non-mon maintenance, ACH debit, EFT processing)
F	Financial Consultant
R	Other
S	Merchant - Acquirer Processing with Device Fingerprint
T	Merchant - Acquirer Processing
U	Unknown
N	Not Applicable

* This mapping is controlled by Feedzai Engineering team and changes to it require a new client requirement and release.

Fix JupyterLab unable to startâ

Bug Fix

Our default configurations had a duplicate by mistake:

```
pulse.global.services.modules.jupyterlab.docker.image=source activate feedzai
pulse.global.services.modules.jupyterlab.docker.image=docker.feedzai.com/feedzai/jupyterlab:
{{ pulse_version }}
```

Copy

This was effectively ignoring the first configuration, which was necessary but with the key:

```
pulse.global.services.modules.jupyterlab.preStartCmd
```

Copy

We fixed our default configurations. As long as DSF deployments clean the Database configuration such that they're reloaded from the release bundle this shouldn't require any change.

Improve credentials handling in ETL toolâ

Internal

Previously, the lookup plugin `fdz_encrypt_tool` was explicitly called for the `_encrypted` variable since the encryption task assumed ETL would run in the same host as Pulse and would have its binaries to run the encryption tool (which isn't the case).

Since then, we've actually overridden all encrypt tasks to use this plugin by default instead (with the goal of improving error logging without leaking actual credentials). As such, we can simply go back to using the standard task.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-0.x/Release-0.13

On this page

What's Changed

Release `0.13.1` â

Autonomous model deploymentâ

Feature FTMT-100

Configure Pulse to store *Models* in *HDFS* by default.

With this new default:

- Data Science Environment stores Trained Models remotely, ensuring their persistence when a node is replaced.
- Pulse runtime Environment is able to deploy a model when running in multiple nodes.

Runtime and Data Science Environment configurations were updated with the new configurations. Nonetheless, the default values shouldn't require any change.

See Pulse documentation on training a model and app collaboration loop for more references.

You might also want to check importing an external model.

Disable JupyterLab terminalâ

Rework

In order to reduce security risks, we've disabled the option to spawn terminal instances from JupyterLab as previously agreed.

Change Data Science Environment schema Policy so that Pulse creates its tables at startupâ

Bug Fix

In the latest release we've disabled Liquibase in Data Science Environment so that Liquibase wouldn't clean up application status.

Unfortunately this change was tested with the legacy DS pipelines and wasn't working with the newer pipeline - used by HSBC - since `schemaPolicy` was set as default (`none`) in that pipeline.

We've fixed this configuration and changed our testing script so it doesn't use the legacy pipeline anymore.

Increase default column size for configuration valuesâ

Bug Fix

Some configurations could have more than the 4000 character limit established previously. We've increased `valuecolumnsize` in order to accommodate these configurations.

Fix failure when importing Catboost models via App Collaboration to HDFSâ

Bug Fix FTMT-100

Fixed an issue in Catboost adapter that prevented users from importing models via App Collaboration when Pulse is configured to store them in HDFS.

This feature is necessary when importing a Model trained in the Data Science Environment to a Pulse runtime Environment with multiple nodes.

Fix "unable to create Spark session" when using JupyterLab on Dockerâ

Bug Fix

Changed default variables so that JupyterLab is able to connect to a remote Hadoop Cluster when running on Docker, leveraging the same configurations as Pulse.

There's also a new version of JupyterLab Docker Image - `docker.feedzai.com/feedzai/jupyterlab:24.1.1-2` - with an updated version of GCS connector.

You can find it in our artifacts page.

This version is now set by default, but can be configured, which is reflected in Data Science Environment Configurations.

Removes unnecessary space (` `) prefix from encrypted secretsâ

Bug Fix

Detected by the project team even though no issue was reported. Affected the following roles: `pulse` , `case-manager` , `etl` , `reference-data` , `job-profile-tool` .

Release `0.13.0` â

Add additional fields to Feedback APIâ

Feature   FTMT-1085

Two new fields were added to the Feedback API:

- `external_status_reason` : optional field which represents the status of the transaction as manually reviewed by a fraud analyst. If absent the event will be updated with the remaining payload.
- `external_preliminary_decision` : field which represents the preliminary decision made by SAS-EFM. Should be one of `Accept` , `Reject` , `Refer`

See additional information regarding this feature at Feedback API status reason.

See up-to-date API documentation and Swagger definition at API Endpoints.

Move DataScience Environment Applications configurations to Global scopeâ

Bug Fix

A few configurations in Data Science environment were scoped to the application, which effectively meant they were ignored since Data Science has a different application.

Datasource statistics path fix is the highlight since it caused an error during validation. It also fixes **UDFs properties** (`java.lang.Math` and `com.feedzai.commons.udfs.StringTools`) and removes unnecessary configurations.

Simplify DataScience startup by skipping Liquibaseâ

Bug Fix

In the DS environment, Liquibase is only used to clean up and copy the database (from blue to green). The former was downright harmful since it would clean the application status in every machine restart. The latter is completely unnecessary as we don't need to perform blue/green deployments in this environment.

In order to ensure that the latest fixes are applied, this version of DataScience Environment should be deployed in a clean environment.

On this page

What's Changed

Release `0.14.0` â

New Playbook for Historical data loading into Pulse Runtime through ETLâ

Feature

Release of a new playbook for Historical data loading.

With this playbook we're able to load historical data from a CSV to a Pulse Runtime environment.

On this page

What's Changed

Release `0.15.1` â

Allow configuration of JVM Opts per environmentâ

Rework

We need JVM Opts to be set on the per each environment - i.e.: we don't expect DEV environments to require as much memory as Prod environments.

Previously, the service environment variables were set in the `<component>.install` ansible tags only, and since tasks with this tag are used to generate environment agnostic base images, service environment variables were not overridden per environment.

Fix "Load Historical Data by ETL" task tagsâ

Bug Fix

The task that actually launched the ETL tool was mistakenly tagged as `etl.install`. This actually meant ETL started when building the base images through Packer, which naturally fails.

We've changed the tag to `etl.configure`, which is actually the tag that runs when an instance starts - allowing for environment specific configurations and to actually run our services.

Add Project ETL configurations artifact to documentationâ

Docs

Adds Project ETL Configurations to our artifacts page. Even though this artifact already existed it wasn't clearly listed as one of the installation dependencies.

Release `0.15.0` â

Allow Ansible Role to call RabbitMQ federation API without SSHâ

Feature

There was an outstanding item from our Blue-Green proposal that, while it wasn't a blocker, it would mean real-time metrics could become inaccurate in that operation.

There was already a playbook to setup federation for the blue-green but the playbook assumed an SSH connection to a RabbitMQ Server host and running `rabbitmqctl` there.

With this new change `rabbitmqctl` can be used from a remote instance without SSH access. Blue-Green Documentation has been updated accordingly.

Add guardrails to avoid submission of events too far in the past or futureâ

Feature

In order to avoid future occurrences of the current production issue, we're deploying guardrails so that our API rejects events too far in the past or future - in our case, 30 minutes.

We want to reject events too far in the past since those would effectively induce errors in the current metrics (e.g.: an event sent with timestamp of "2 days before" has to be rejected so that it doesn't increase the count of events in the last day).

We also want to reject events too far in the future to avoid the current production issue.

Opposite to what we initially thought, this can already be deployed. **Additionally** we recommend to make a publish without recovery after deploying this version so that Pulse clock is effectively reset to "current time".

Performance improvement on Pulse Event Storage table accessâ

Feature

From now on Pulse Event Storage table is partitioned by timestamp in order to improve performance on read access. For now, it's implemented as monthly partition.

Fix Reference Data Service ansible roles tagâ

Bug Fix

Before change:

```
    - reference_data.415
   - reference_data.configure
```

Copy

After change:

```
    - reference-data.415
   - reference-data.configure
```

Copy

Set JupyterLab memory limitâ

Rework

Set JupyterLab memory limit to 8GB (`--ResourceUseDisplay.mem_limit=$((8*1024*1024*1024))`). We find this limit helpful to avoid unbounded memory usage by a single user. It is set to a fairly high value since, in our experience, it is uncommon that multiple users reach this maximum at the same time.

Bound the JupyterLab memory instead of the container in order to allow the software to self-manage its memory and give it a chance to optimize accordingly.

Ensure past events are considered by new plans real time and scheduled metricsâ

Rework

By default a plan ignores past events that were already processed by the workflow. We've changed this default to account for our specific case, where we expect to deploy a plan soon and we want that plan to account for the data that is currently sent to Feedzai Pulse.

Fix duplicate feedback updatesâ

Bug Fix

Fixed an issue that caused multiple feedback updates to be created when marking an event as Fraud / Not Fraud in Case Manager.

Include `Pulse Validation Tool` â

Feature

The tool is able to validate that a Pulse Data Science Environment is configured as expected, by running common commands / jobs.

Run the tool using the `pulse-validation-tool.yml` playbook, present in `installer-assembly`. The playbook requires `Pulse Validation Tool`, present in artifacts page.

See Pulse Validation Tool Configurations for more information (later note: deprecated and removed in release `2.7.0`).

Ansible Roles Upgradeâ

Feature

Task "[921] create default users" has been moved to "[925] create default users", and task "[921] deploy security roles" has been added. Currently, this new task is not enabled.

Before:

- [921] create default users

After:

- [921] deploy security roles (currently skipped)
- [925] create default users

On this page

What's Changed

Release `0.17.0` â

Input and Workflow Schema Updateâ

Feature

We made changes on the input schema (for the input APIs) and in the workflow schema to support HSBC V9 data mapping. Find the changes introduced in the release below.

Added fields

Field	Datatype	Description
sender_account_open_date	long	Opening date of the account (in milliseconds since the Unix epoch).
billing_currency_code	string	Billing Currency Code
change_event_delivery	string	Change Event Delivery
customer_change_data_old	string	Previous data change
device_ip_address	string	IP address v4 [Presently all IP are on v4. Oman/ Bahrain are using it at present]. Internet address obtained via ajax application
overnight_acc_ledger_bal	double	Customer balance at the end of previous day. Different that available balance. Available in SASEFM for use in Payment Alert Decisioning
receiver_transaction_amount_dbl	double	Transaction amount received by the receiver.
sender_account_balance_dbl	double	Available balance (including overdraft limits) in billing currency and in scientific notation prior to the transaction [also includes Sum of Balances on Sender Accounts]
sender_transaction_amount_dbl	double	Transaction amount in transaction currency and in scientific notation. This should always be a positive number
var_recurring_payment	string	Variable recurring payment indicator only for open banking payments
vocalink.vocalink_score	double	Vocalink Score
vocalink.mastercard_id	string	Vocalink Unique Identifier

Removed fields

Field
browser_os_version
costumer_id_global
customer_change_reason
customer_portfolio_phys_type
ds_transaction_id
payment_message_source_d
sender_account_channel_limit
sender_transaction_reason_d

Field
sender_transaction_sign_stts_d
sender_transaction_origin_amount
sender_transaction_origin_currency

Note: these fields were removed from the workflow schema but were kept as part of the API schema so we don't introduce a breaking change in the API. This allows that this set of fields can still be sent without the event being rejected by Pulse.

Updated fields

Field	Datatype	Description
bioc.custom_risk_aliases	double	Additional Biocatch Risk Factors
tmx.device_dev_id	string	Fingerprint hash STRING

Tests

After applying the changes to the input schema we ran a series of integration tests to validate the changes.

The first test case was to issue a request to all input endpoints using the payload as defined in schema v8. All tests passed since we still allow fields from schema v8 to be sent to the API although they are not part of the workflow schema:

The second test case was to issue a request to all input endpoints using the using a payload with the fields added in schema v9. All tests executed with success:

On this page

What's Changed

Release `1.0.0` â

warning

Previously we've released `0.17.0` so that documentation was publicly accessible (not possible with Release Candidates). Extraordinarily, release `1.0.0` doesn't include the changes present in `0.17.0`, meaning there's no changes to the APIs.

Feedzai Platform Upgradesâ

Feature

We've upgrade our base Solution and Platforms to TFB 5.0.3, Pulse 25.0.6 and Case Manager 13.8.0. Specific details on these product upgrades will be sent by email.

We want to highlight a few points:

- Pulse Upgraded it's Spark / Hadoop versions, clearing out a big chunk of previously present CVE's.
- Case Manager
  - Analyst Notes are now part of External Notifications (to be exported to FDR)
  - SLAs can now be scheduled in 15 minutes intervals, for example, at 9.00 or 11.15 or 14.45 or 20.30.
  - SLAs can now skip weekends

Artifacts page has been updated accordingly.

Allow Declining of Genuine transactionsâ

Feature

Previously in Case Manager the fraud label was tightly coupled with the Case Manager decision - approve or decline. With this change, some of the genuine status reasons have the suffix `- Decline`, which allows us to set these individually in specific cases. e.g.: customer confirms non fraudulent transaction but wishes to cancel it anyway as it is no longer relevant.

Implement Outbound Notificationsâ

Feature

We've implemented 2 Outbound Notifications in Case Manager.

1. Payment fulfillment - Case Manager Core Extension that is called automatically when the transaction status changes in Case Manager.
2. Send Mobile Payment Push - Case Manager Extension that can be used in Automation Rules and SLAs.

Introduction of Logstash Pipelinesâ

Feature

Introduction of Pipelines to export Case Manager notifications and rules metadata to a Google Cloud Storage bucket to be consumed by HSBC systems, for example for reporting.

Inclusion of Flexible Treatment Strategyâ

Rework

Previously, the treatment strategy was implemented by creating a state machine in Case Manager that matches with the Pulse outcome. This was not flexible because every time a new outcome value is added to Pulse it needs to be added to case manager configuration - and this required a new release. From now on, it is possible to decouple the treatment strategy in pulse from the Case Manager state machine. This way HSBC can add all the treatment strategy values required without the need of a new case manager release.

Fix Bug on Hyperlinkâ

Bug Fix

Fixed a bug where clicking on the Customer ID hyperlink would redirect the user to the search page instead of the customer page.

Add new fields to Case Managerâ

Bug Fix

On a previous release some fields were missing from Case Manager, meaning it was not possible to use them in Automation Rules and SLAs.

We have added them according to the following specification:

Field	UI Name
customer_portfolio_product	Customer Portfolio Product
customer_portfolio_channel	Customer Portfolio Channel
customer_portfolio_type	Customer Portfolio Type
customer_portfolio_class	Customer Account Level Classification
customer_portfolio_country	Customer Portfolio Country
customer_portfolio_region	Customer Portfolio Region
payment_type	Transaction Type
payment_sub_type	Transaction Sub-Type
source	Source
payment_message_source	Payment Message Source*
channel_type	Channel
customer_type	Customer Type
office_id	RPS Queue Name
customer_account_type	Customer Account Type
sender_transaction_type	Sender Transaction Type
sender_transaction_cop_result	Sender Transaction COP Result
sender_transaction_auth_type	Sender Transaction Authentication Type

ⓘ**note**

*Although in the Transfers Event Details page `payment_message_source` is translated to `Source`, when used in Automation Rules and/or SLAs it is translated to `Payment Message Source` so the user is able to differentiate between `source` and `payment_message_source`.

Add Alert ID to Transfers and Digital Activity Event Details Pageâ

Feature

This field was added to both Transfers and Digital Activity Details Page in the Transaction and Event Summary widget respectively.

In Transfers Transaction widget it was added after the `Lifecycle ID`:

In Digital Activity Event Summary widget it was added after the `Lifecycle ID`:

For events that did not trigger an alert the `Alert ID` will show in the respective widget with a `N/A` value.

Add Joint Account CIN to Transfers Event Details Pageâ

Feature

This field was added to Transfers Event Details Page in the Originator Details widget.

Remove a hyperlink from event details pageâ

Rework

We have removed the phone number hyperlink from the Event Details Page while keeping the Customer ID hyperlink. As a result the phone numbers will no longer be clickable but the customer ID hyperlink will remain as a way to access the customer page.

Generate Alert ID and add it as a searchable field in Case Managerâ

Feature

This ID is generated for alerts only. Its goal is to serve as client friendly identifier to send via SMS / Mobile Push Notification instead of using the raw ID that uniquely identifies each transaction.

This field can then be used to search for transactions in Case Manager.

Change on the limit of reasons per alertâ

Rework

We changed the configuration to limit the number of reasons that are able to be selected when confirming an event as Fraud or Not Fraud, now only one reason per alert is possible.

Fix deployment of Automation Rulesâ

Bug Fix

The previous deployment strategy was failing as we were trying to add new Automation Rules to a fixed position. Since the environment could've custom Automation Rules already, we can't use an hardcoded position.

The new strategy will append Automation Rules to the list, meaning it will compute the automation position so that it is equal to the current max position + 1.

The changelog was fixed so that this works in upper environments where 0.16.1 hasn't been deployed yet. At the same time, it disables the checksum validation so that it is also deployable on environments that were already deployed with the previous strategy.

To test this, we can verify that, if the changelog already ran before (in the case the environment didn't contain any custom Automation Rule) it will simply be skipped OR if it hadn't run yet it will run for the first time and compute the automation_position correctly.

Fix on Transaction History Mapping Fieldsâ

Bug Fix

On a previous release, we mentioned there was a bug during the mapping of some fields and we were showing values sent without mapping, this release fixes this issue.

The fields fixed are the following:

- `Transaction Type (payment_type)`
- `Currency (sender_transaction_currency)`
- `Event Type (txn_type)`
- `Login Last Authentication (auth_method)`
- `Device Type (device_banking_type)`

Allow setting status reasons via SLAs and Automation Rulesâ

Feature

Implemented the following custom extensions, which allow setting a specific status reason via SLAs and Automation Rules.

The extensions added are the following:

- `Set Reason | Payment Reversal as Unable to contact - Decline`
- `Set Reason | Customer Confirmed Fraud via Mobile Messaging (2 WAY)`
- `Set Reason | Customer Confirmed Genuine via Mobile Messaging (2 WAY)`
- `Set Reason | Payment Reversal Crypto Limit Reached - Decline`

In order to implement the previous extensions the following reasons had to be created additionally to the previously shared specification:

- `Payment Reversal Crypto Limit Reached - Decline`

The following SLAs and ARs were updated to make use of the new custom extensions:

- SLA: `Wait for Customer Response` -> `Set Reason | Payment Reversal as Unable to contact`
- AR: `Customer Notification - Fraud` -> `Set Reason | Customer Confirmed Fraud via Mobile Messaging (2 WAY)`
- AR: `Customer Notification - Genuine` -> `Set Reason | Customer Confirmed Genuine via Mobile Messaging (2 WAY)`

On this page

What's Changed

Release `1.1.4` â

Fix Feedzai API documentationâ

Bug Fix

We have fixed an issue where the below fields data types were not correct in the API documentation

Field	Previous type	Fixed type
sender_account_balance	number	integer
sender_transaction_amount	number	integer
receiver_transaction_amount	number	integer
receiver_creation_date	string	long
sender_registration_date	string	long
sanction_check	string	boolean
device_bluetooth_enabled	string	boolean
pinpoint_risk_score	number	string

Release `1.1.3` â

Fix installation process for Logstashâ

Bug Fix

We have fixed an issue where the Logstash plugins installation required internet access. We also added a toggle to skip the installation of Logstash itself, allowing pre-installed images to be used.

Release `1.1.2` â

Fix undefined variable error during installationâ

Bug Fix

We have fixed the issue found during installation by removing the undefined variable (`pulse_internal_comm_rsa_modulus`) usage as it is not required in the project.

Release `1.1.1` â

This version introduces an hotfix for the installation issues.

Fix index name in the Payments workflow tableâ

Bug Fix

An issue was identified when running version `1.0.0` due to a conflicting index (`idx_es_transfers_lifecycle_id`) in the Payments workflow table.

We have fixed the issue by renaming the Payments index to `idx_es_payments_lifecycle_id` so it does not clash with the Transfers workflow table lifecycle_id index.

Release `1.1.0` â

This version updates our APIs according to the latest DMA.

warning

For the new fields to be available in the ETL extractions, the new ETL version must be installed.

Update on Inbound APIâ

Feature

In this release we have included the Input and Workflow changes that were introduced in Release Candidate 0.17.0.

All the changes are described in Input and Workflow Schema Update.

New fields to Customers' Reference Data APIâ

Feature

We've added as requested in Customer Demographics DMA (F24_DMA_UK-IN_CUST_DEMOG_v. 01.02_Alpha_40_40_2023.05.12) new fields to the Customers' Reference Data API.

The available fields can be consulted in the API documentation: Reference Data.

Events enriched with Customer Reference Dataâ

Feature

We've updated the schema to include the new reference data fields.

Also, the Reference Data enrichers were configured in order to use the customer demographic data to enrich events.

Include Alert ID in CM Data Extractâ

Feature

We added the Alert Id field to Case Manager external notifications so it is available to build the necessary reports.

Changes on Case Manager Fields in Customer Details Pageâ

Rework

We've made a few changes in some fields on Case Manager Details Page, please find below a list of changes.

Change	Field
Removed	Account Type
Removed	Contact Number
Removed	Customer Postcode
Removed	Region
Added	Mobile Phone Number
Added	Home Phone Number
Added	Daytime Work Phone Number
Added	Business Phone Number
Added	Home Postcode

Change	Field
Added	Correspondence Postcode
Added	Customer Home Country
Added	Customer Correspondence Country

Changes on Case Manager Fields in Originator Details Pageâ

We've made a few changes in some fields on Case Manager Originator Details Page, please find below a list of changes.

Rework

Change	Field
Removed	WorkPhone2 lastChangeDate
Removed	HomePhone2 lastChangeDate
Removed	MobilePhone2 lastChangeDate
Renamed to	Work Phone Last Change Date
Renamed to	Home Phone Last Change Date
Renamed to	Mobile Phone Last Change Date

Add folder name prefix configuration to ETL toolâ

Feature FTMT-8225

Added a new property (`storage_bucket_eventstorage_folder_prefix`) to allow adding a prefix to the folder where events will be stored when running the ETL event storage extraction.

As an example, if the prefix is configured to `event_store_` , the output of an execution will be written to `gs://bucket/event_store_<start_datetime>-until-<end_datetime>/` .

For more information see ETL Configuration Reference.

Fix Search Results and Transaction History populating irrelevant dataâ

Bug Fix

The following fields no longer display irrelevant data on the Search Results and Transaction History widgets:

- Customer / Customer ID
- Originator Name
- Originator Account
- Beneficiary Name
- Device Location
- Device IP

Fix Device Location mapping in Case Managerâ

Bug Fix

The Device Location field is now correctly mapped to `device_geo_ip_isp` .

Changes on Beneficiary Account Numberâ

Feature

Beneficiary Account number now shows a combination of two fields: Beneficiary's bank BIC and Beneficiary's account number.

We also added a new search filter for BIC on the search screen.

Add Search by Beneficiary Account and Rename of the Fieldsâ

Feature

Include the possibility to search by Beneficiary Account on the search screen menus.

Also renamed the fields `Originator` and `Beneficiary` to `Originator Name` and `Beneficiary Name` .

Fix on Transaction Type Searchâ

Bug Fix

There was a bug that when we clicked on the Transaction Type search screen, the dropdown was showing the Transaction Type code instead of the Transaction Type description, we have fixed that so from now on it displays the Transaction Type description values.

New Operational Status Availableâ

Feature

We've added, as requested, new Operational Status to be available to use in Automation Rules and SLA's, namely:

- `Pending BOT Closure`
- `Pending Manual Closure`
- `Pending Crypto Reversal SMS`

On this page

What's Changed

Release `1.2.0` â

Logstash pipelines - support for multiple bucketsâ

Feature

We have added support in logstash pipelines to configure different buckets (per pipeline).

Role based bulk assign optionâ

Feature

We have created a new permission cm:resource:history:assign-bulk:* that allows a user to bulk assign events from the Alerts and Search pages.

This permission was added to all roles except for the Fraud Prevention Agents L1 and L2, as defined in the requirement.

Note: bulk assign in the event transaction history is still available to users with the Fraud Prevention Agents L1 or L2 roles.

Add new JVM flag to Application JVM Optsâ

Bug Fix

With the latest product upgrade it is required to add a new flag (`-XX:-UseAESIntrinsics`) to the application JVM Opts (`pulse.app.tfrb.appengine.jvmopts`
property) so Pulse is able to copy the necessary libraries to Dataproc when running Schedule Metrics (SMJs).

Note: If `tfrb_app_jvmopts` is being overridden in the Ansible pipelines it is required to add this new flag.

Customer Phone Search Modificationsâ

Rework

The search for Customer Phone on Case Manager was changed as requested, now this search box is able to filter to two different keys: `sender_phone` and `customer_phone` .

This fixes the reported issue where searching for Customer Phone Number on the search screen was not returning any results

Add New Fields to the Case Managerâ

Feature

We've added 3 new Fields to the Case Manager schema and also made them available while creating Automation Rule and SLAs.

Field	Type	UI Name
internal_payment_type	String	Internal Payment Type
internal_event_source	String	Internal Event Source
isJade	Boolean	Is Jade

Remove access to Lists Management page from L2 roleâ

Rework

Removed permission from L2 role so that respective users can't go to the lists management tab in Case Manager. L2 users can still add specific entities to lists from the Event Details page.

Feedzai Platform Upgradesâ

Feature

We've upgraded to Pulse 25.0.12 and Case Manager 13.9.0.

Add support for historical eventsâ

Feature

We've added support for receiving historical Digital Activity and Fraud Tagging events.

These events won't be accounted for in real-time metrics but can be used in scheduled metrics. In the case you don't want to account for these specific events even in scheduled metrics, there's an helpful generated field - `is_historical` which can be used for that purpose.

Add FDZ_UUID to the ETL extractionsâ

Feature

We've added the field `FDZ_UUID` to the ETL extractions to have a unique record identifier in the data extracted.

Remove workaround to Datasource Analysis failure in GCPâ

Rework

There was an issue with the latest product releases that prevented Datasource Analysis in GCP environments and required a project workaround.

The fix was already applied upstream and the project workaround was therefore reverted.

Include Customer Segment fields in CM Data Extractâ

Feature

Added the following fields to the Case Manager external notifications payload.

Fields
customer_id
customer_portfolio_region
customer_portfolio_product
customer_portfolio_channel
customer_portfolio_country
customer_portfolio_type
customer_portfolio_class

Add Transformations and rename fields for Digital Activity event in Case Managerâ

Rework

We've added Transformations for field `True IP Geo` to show Country name when 3 Digit Code is received for Digital Activity event.

Fields updated are

Section	Field
Network	True IP Geo
Transaction History - Digital Activity	True IP Geo

We have renamed `Voice Score Type_New` to `Smart ID Score` in the widget and Transaction History.

We have added the new field `Payment Reference` to Transaction History that directly maps to `sender_transaction_notes`.

Add PIB suspension indicatorâ

Feature

We've updated the Case Manager transfers and digital activity schemas to include the PIB suspension indicator (`pib_suspended`).

Case Manager external notifications were also updated to include this field.

In order to send this field to Case Manager, we created a custom Alert Storage Service.

This service enriches the event with a new `additional_fields` field, this is enriched with information from the message that is being processed in Pulse.

To enable this enrichment the following property must have a value `true` :

- pulse.app.tfrb.com.feedzai.hsbc.pulse.services.customalertstorageservice.additionalfields.enricher.active

And the information fields to be added need to be included here (list of comma separated strings e.g.: pib_suspended,another_field):

- pulse.app.tfrb.com.feedzai.hsbc.pulse.services.customalertstorageservice.additionalfields.enricher.fields

Add network widget field mappingâ

Feature

We have added a new field called Device ISP and IP location is renamed to IP City

Change the mapping of Event Type to payment_type_sub_typeâ

Rework

We have changed the mapping for the field - Event Type which is in the following widgets :

Event: Digital Activity

Section: Event Summary, Transaction History, Search Screen Page and Alert Search Page

Field: Event Type

Old Mapping: txn_type

New Mapping: payment_type_sub_type

Fix issue with Case Manager reasonsâ

Bug Fix

We've fixed an issue where previously deleted reasons were being re-added to Case Manager.

Add Missing Resource for External Status Reason on Case Managerâ

Bug Fix

We've added missing resource for field `External Status Reason` which causes CM to show the field database name instead of its pretty name.

External Status Reason extension is enabledâ

Feature

We've enabled the External Status Reason extension, allowing the user to set a reason via Feedback API using the `external_status_reason` field.

To allow this we've removed the `transfer status` state machine, and moved its reasons to the `status` state machine.

We also removed the following automation rules that were used to sync the state between the two state machines:

- Handle Transfers Status change to Fraud (Feedbacks)
- Handle Transfers Status change to Not Fraud (Feedbacks)

Change on event date formatâ

Bug Fix

We have fixed an issue where events were showing in a 12 hour format without the AM/PM marker, instead of a 24 hour format.

Renamed Customer Account Level Classification field reference on Case Managerâ

Bug Fix

We've renamed all the references of the field `Customer Account Level Classification` to `Segment`, as initially intended.

Customer Name Search Modificationsâ

Rework

The search for Customer Name on Case Manager was changed as requested, now this search box is able to filter to two different keys: `sender_name` (Like in the Originator Name on the Advanced Search) and `customer_name`.

Change on Amount Fields Formatâ

Bug Fix

We've made some changes on how the amount values were being displayed, previously they were displaying like "$1200", and we were requested to change it to display like "1200 (USD)".

This change was applied to the following widgets:

Section: Transaction

Fields: Originator Amount & Amount(Bene)

Section: Originator Account Details

Fields: Balance

Section: Originator Details

Fields: Total Balance

Inclusion of Event Description fieldâ

Feature

We've added a new field called `Event Description` to Case Manager, this field is present for Digital Activity Events, in the Event Summary and Transaction History widget.

This new field is mapping to `sender_transaction_code`, it receives the specific code, and if there is a match with a valid code, a transformation is done, and the correspondent description is shown in Case Manager. Example: QPS -> Request - Paperless Statements.

As agreed, to not occupy too much space in Case Manager UI, we used the short version of the description

New Field Mapping in Network Section on Transfer eventsâ

Feature

We've made some changes on the mapping of the fields in the Network section, please find below the list of modifications.

Field	Section	Change	Feedzai Mapping	AFIS Mapping
Device ISP	Network	Added	device_geo_ip_isp	ipAddressIsp
Device Location	Transaction History + Search Screen	Rework	device_geo_ip_city	geoIpCity
IP City	Transaction History + Search Screen	Renamed	device_ip_city	ipAddressCity

On this page

What's Changed

Release `1.4.1` â

Fix Rules Triggered widget not showing in Case Manager for events with lifecycle_id length higher than 40 charactersâ

Bug Fix

Events sent with a `lifecycle_id` with more than 40 characters were showing an empty Rules Triggered widget in Case Manager due to a database limitation.

This issue was addressed and events will now show all rules that were triggered.

Fix Feedback historic eventsâ

Bug Fix

Transfers feedback events were being marked as historical based on a wrong configuration (`*.transfers.event.timestamp.slack.past`).

This is now fixed and feedback events will be marked as historical based on `*.transfers.event.timestamp.slack.historic` .

Release `1.4.0` â

ETL documentation improvementsâ

Rework

A few additional documents are now available in Feedzai documentation:

1. ETL process from Pulse RealTime (RT) to DataScience Environment (DSE)
2. Top Metrics ETL
3. ETL Recovery Procedure

Additionally, we've updated ETL Configurations and Collect Metrics now details how to collect the ETL metrics.

FIV Markersâ

Feature

Support for FIV Markers is now available.

See full context in Concepts | FIV Markers and Custom UDFs | FIV Markers to understand how they can be used in Pulse Rules.

Analyst lists configurationâ

Feature

We've added 8 new lists to Pulse, namely:

- UK Payments_Hot Beneficiary Sort Code and Account Numbers
- UK Payments_Vulnerable Customer List

- UK Payments_Hot Customer ID
- UK Payments_Crypto Beneficiary Name
- UK Payments_Crypto Payment Reference
- UK Payments_Low Risk Sort Code
- UK Payments_Low Risk Account
- UK Payments_Ex Jade Customers

These lists were also added to Case Manager, and can be consulted in the Lists Management.

Each one of them was mapped to a schema field, the mapping is as shown in the following table:

List name	CM field	CM schema field
UK Payments_Hot Beneficiary Sort Code and Account Numbers	Bene Account Number	rcv_transact_bank_acc_number
UK Payments_Vulnerable Customer List	Customer ID	customer_id
UK Payments_Hot Customer ID	Customer ID	customer_id
UK Payments_Crypto Beneficiary Name	Beneficiary Details - Name	rcv_name
UK Payments_Crypto Payment Reference	Transaction Notes	text_for_receiver
UK Payments_Low Risk Sort Code	Bank Identifier Code	receiver_bank_bic
UK Payments_Low Risk Account	Bene Account Number	rcv_transact_bank_acc_number
UK Payments_Ex Jade Customers	Customer ID	customer_id

All the OOB lists in Case Manager for **transfers** were removed.

The Update Lists` screen for transfers in Case Manager was also edited. As of now, there are two entities available to be configured:

- `Customer ID`, mapped to `UK Payments_Hot Customer ID` (should be removed ahead, configured due to a CM limitation)
- `Beneficiary Account`, mapped to `UK Payments_Hot Beneficiary Sort Code and Account Numbers`

As requested, we kept the OOB lists configured on CM for **Digital Activity**.

Field Mapping changes in Case Managerâ

Rework

We have implemented the requested field mapping changes in Case Manager. Please refer to the table below for a list of the applied changes.

Field	Field	Old mapping	New Mapping	Required Transformation
Transfers- Originator Account Details	Account Type	sender_bank_type	sender_account_type	Yes
Transfers - Originator Details	Phone Number	sender_phone	customer_mobile_phone	No
Transaction History All Channels	Customer Phone Number	customer_phone	customer_mobile_phone	No
Alert Search Page	Customer Phone Number	customer_phone	customer_mobile_phone	No
Search Screen Results + Search Filters	Customer Phone Number	customer_phone	customer_mobile_phone	No
Transfers - Originator Details	Phone Number Last Changed	sender_phone_chg_date	customer_mbphone_last_update_at	No
Transfers - Originator Details	Email Address	sender_email	customer_email	No

Field	Field	Old mapping	New Mapping	Required Transformation
Digital Activity - Event Summary	Adding Lifecycle ID	NA	lifecycle_id	No
Transfers - Transaction	Event Date	created_at	event_occurred_at	No
Transfers - Transaction	Payment Date	event_occurred_at	created_at	No

Feedzai Platform Upgradesâ

Feature

We've upgraded to Pulse 25.0.19.

Fix a Bug with SLAs not skipping weekendsâ

Bug Fix

With the Case Manager upgrade to version 13.10.2, we have successfully resolved an issue related to certain SLAs inappropriately triggering during weekends. This bug fix ensures that SLAs now accurately skip weekends as intended.

Fix Payment Fulfillment extensionâ

Bug Fix

We've identified issues when using the Payment fulfillment extension in the following scenarios:

1. When a user without admin permissions marks an event as Fraud/Not Fraud
2. When an event is marked as Fraud/Not Fraud via the Feedback API

The first issue was due to the missing permission `cm:state-machine:event:read:decision`, which was added to Pulse and to the following roles:

- developer
- Fraud Prevention Agents - L1
- Fraud Prevention Agents - L2
- Fraud Prevention Manager
- Fraud Strategist
- Fraud Strategist Manager
- Monitoring and Auditing

Note: any other role that was not created by the installation should be manually updated.

The second issue was addressed in the extension implementation.

Optimize Alert and Search Pagesâ

Rework

Alert and Search pages were taking several seconds to load due to event enrichment.

We've optimize enrichment and these pages should load significantly faster.

Deprecate custom extensions to set reason in Case Manager Automation Rules & SLAsâ

Rework

With the upgrade to Case Manager 13.10.0, Automation Rules & SLAs that set a specific status can now also set the respective reason.

With this, the the following Custom Extensions have been deprecated:

- `Set Reason | Payment Reversal as Unable to contact - Decline`

- `Set Reason | Customer Confirmed Fraud via Mobile Messaging (2 WAY)`
- `Set Reason | Customer Confirmed Genuine via Mobile Messaging (2 WAY)`
- `Set Reason | Payment Reversal Crypto Limit Reached - Decline`

warning

Migration has to be done manually.

As of now we've kept the functionality as is such that existing Automation Rules keep working as before.

The custom extensions will be removed in the next release.

New Customers' Reference Data API fieldsâ

Feature

We've added the following fields to Customer Reference Data API :

- customer_pb_marker
- customer_portfolio_segment
- customer_voice_id
- customer_voice_id_last_change_date

The available fields can be consulted in the API documentation: Reference Data.

Logstash pipelines recovery toolâ

Feature

We've created a recovery tool for the Logstash pipelines.

This allows the re-execution of the `pulse-workflow-audit` pipeline when an error occurs and the data is not exported to the GCP bucket.

Further documentation on this tool can be found here: Logstash Recovery Tool.

Update Reference Data authentication tokenâ

Bug Fix

We have noticed a high number of log entries since the existing Reference Data token was missing the expiration date property.

This property, set to expire in 2030, was added and a new token was generated. There's an ongoing effort to revamp authentication across all Feedzai components which is expected to be finished before this expiration date.

Update documentation so that Artifacts are displayed as a table instead of a list.â

Rework

This approach not only is more readable but also allows for potential future extensions (in new columns).

Include missing Digital Activity fields in CM Data Extractâ

Bug Fix

We have detected that `lifecycle_id` and `event_type` fields from Digital Activity events were not being sent in the notifications payload.

The missing fields were added to Case Manager external notifications payload.

Labeling modifications for customer_portfolio_class in ARs and SLAsâ

Rework

We have made refinements to the labeling of customer_portfolio_class. From now, it will be identified as Customer Portfolio Class within Automation Rules and SLAs.

However, we have retained the designation Segment for the Event Details page.

Empty Application for DSE Environmentsâ

Feature

We have added an artifact with an empty application for DSE. This application can be imported in DSE environments so that models exported from DSE can be imported via App Collaboration in Pulse Runtime without manually editing the exported files.

The application can be found in the Artifacts page.

Optimize Custom Alert Storage Serviceâ

Rework

We've changed the Custom Alert Storage Service developed in Add PIB suspension indicator to validate the additional fields that are present in the workflow schema during creation. With this change logs will have a single entry if any additional field is missing from the workflow schema instead of a log entry per event.

We've also added a new metric with the count of missing fields. Metric definition can be viewed in Custom Alert Storage Service Additional Fields.

On this page

What's Changed

Release `1.5.0` â

Fix `saf_ind` field not being available on Consent and Behaviour endpointsâ

Bug Fix

On a previous release, we added support for the `saf_ind` field on most endpoints. However, due to a bug, the field was not available on the Consent and Behaviour endpoints.

This release adds `saf_ind` as a valid field on the Consent and Behaviour endpoints.

Add new property to enable Pulse Authentication when behind a proxyâ

Feature

After changing Pulse Load Balancer to an Application Load Balancer we started facing issues in authentication via Keycloak due to a proxy redirection to an insecure (HTTP) Pulse URL.

To fix this we have added a new Ansible property (`pulse_sso_force_https`) that when set to `true` will force a redirection to the secure Pulse URL.

For the complete description and default variables values see Pulse Configuration Reference.

Add Case Manager Rolling Upgrades documentationâ

Feature

We have added documentation and two new Ansible playbooks to support Case Manager Rolling Upgrades.

Documentation can be found in Rolling Upgrades.

Fix ETL documentation field names inconsistenciesâ

Rework

We have fixed the ETL documentation field name inconsistencies. Please refer to the table below for the fixed mappings.

Field name in documentation	Field name in Extraction
sender_transaction_bank_accoun	sender_transaction_bank_account_number
receiver_transaction_bank_acco	receiver_transaction_bank_account_number
instant_transfers_review_outco	instant_transfers_review_outcome
sender_transaction_original_am	sender_transaction_original_amount
sender_transaction_original_cu	sender_transaction_original_currency
receiver_transaction_origin_am	receiver_transaction_original_amount
receiver_transaction_origin_cu	receiver_transaction_original_currency
receiver_transaction_amount_db	receiver_transaction_amount_dbl
customer_fiv_marker_last_updat	customer_fiv_marker_last_updated_at
customer_earnings_amt_snapshot	customer_earnings_amt_snapshot_dt

Field name in documentation	Field name in Extraction
currentaccount_monthly_debit	current_account_monthly_debit_amount
customer_current_account_balan	customer_current_account_balance
customer_home_address_last_upd	customer_home_address_last_update_at
customercorresp_address_last	customer_corresp_address_last_update_at
customer_hmphone_last_update_a	customer_hmphone_last_update_at
customer_mbphone_last_update_a	customer_mbphone_last_update_at
customerwrkphone_last_update	customer_wrkphone_last_update_at
customer_businessphone_last_up	customer_businessphone_last_update_at

On this page

What's Changed

Release `1.6.3` â

Change Pulse publish task to wait via RMIâ

Bug Fix

We identified issues running the application via REST API when the publish takes more than 10 minutes (in a Cloud environment).

We are changing ths approach by issuing the publish asynchronously via REST (so it can be auditable) and waiting for it to be published via RMI.

Release `1.6.2` â

Add `Tmx.request_id` to ETL extractionsâ

Bug Fix

We've fixed an issue where ETL extractions were not including the `Tmx.request_id` field, added in release `1.6.0` .

This is now fixed and the extractions will include the new field.

Release `1.6.1` â

Add sender_transaction_original_currency to Pulse APIâ

Feature

One new field was added to the Transfers API:

- `sender_transaction_original_currency` : The currency in which `sender_transaction_original_amount` was sent, as a `string` .

This field already existed in the Pulse workflow schema and was already available in Case Manager (although it was not being populated). As of this release, this field is now also available to the Pulse API.

Release `1.6.0` â

Enable/disable toggle for AFIS Case Manager extensionsâ

Feature

We have implemented the ability to enable or disable the calling of the AFIS Case Manager extensions via the following two ansible variables (which both default to true to preserve backward compatibility):

- `afis_payment_fulfillment_cm_extension_enabled`
- `afis_mobile_outbound_cm_extension_enabled`

This will help disable extensions that are not fully configured, to prevent log spam with large stack traces and make it easier to provide meaningful logs during Feedzai support issues.

Case Manager Enrichment via Pulse Listsâ

Rework

We've added a new feature where Case Manager enrichment can be achieved by using Pulse lists.

To support this feature we've also introduced the following changes:

- Added the `Case Manager Enrichment Plan` to Pulse with a predefined set of fields to enrich
- Added the following Pulse lists for enrichment:
    - Authentication method
    - Channel Type
    - Transaction Type
    - Transaction Status
    - Account Type
    - Sender Transaction Cop Result
    - CAM Level
    - Event Type
    - Event Description
    - Beneficiary Bank Details
- Added Case Mappings for the newly generated fields

Please check the Case Manager Event Enrichment Guide and Pulse Configuration Reference for more information.

Delete Reference Data Endpointâ

Feature

We have implemented endpoints to delete Reference Data. This endpoint allows deleting any of the Reference Data entities sent to Feedzai and performs a hard delete of the entity (i.e. it is not recoverable once deleted). The API documentation has been updated with the new endpoints.

API additionsâ

Feature

One new field was added to the Transfers API:

- `sender_transaction_original_amount` : The original amount of the transaction by the sender, in the sender's local currency. This value is a `double` (floating-point). Example: `4.23`

This field already existed in the Pulse workflow schema and was already available in Case Manager (although it was not being populated). As of this release, this field is now also available to the Pulse API.

Case Manager Version Upgraded to 13.10.5â

Feature

We've upgraded the Case Manager Version to 13.10.5.

This upgrade will allow to benefit some improvements in performance, mainly regarding inserts into the database.

It also fixes a bug where the use of reasons in Automation Rule conditions was being ignored and not providing the desired outcomes.

Optimize Transaction History widgetâ

Rework

Transaction history widget in Event details page was taking several seconds to load due to slow event enrichment.

While this may have to be changed in the future, for now, we've moved that enrichment to the client side which should result in that widget loading significantly faster without changing functionality.

Case Manager Search Page Guardrailsâ

Feature

We've implement a set of guardrails in Case Manager's Search Page to enhance search performance and limit long running queries.

Please find below the changes:

- Date filter (From/To) is now mandatory and by default is limited to one week
  - `From` will default to last week
  - `To` will default to the beginning of the next day
  - **Note:** these defaults will only be applied when the user does not have any previous search stored in cache
- `Get Next` will now search unreviewed events only up to one week, instead of searching without boundary. This value is configurable during installation (check [Case Manager Configuration] for more information)
- The maximum number of filters in the search page is now 4 instead of 6

Find below an example of the search page when a new user logs in:

Fix ETL extracting duplicate events on time boundariesâ

Bug Fix

We've fixed an issue where events would be extracted twice if their timestamp was precisely on the cron boundaries.

For example, if the ETL is configured to run every hour, prior to this fix an event with timestamp *11:00:00.000* would be extracted by:

1. the execution extracting events from *10:00:00.000* to *11:00:00.000*
2. the execution extracting events from *11:00:00.000* to *12:00:00.000*

After this fix, the upper bound timestamp is now exclusive and the event would only be extracted by the second execution.

Refactor quick filtersâ

Feature

We've refactored quick filters in searches to be aligned with the 6 months data retention period.

The refactoring includes the removal of the `All time`, `Last 5 years` and `Last year` options and the inclusion of `Last 6 months` option.

This change applies to the following pages:

- Alerts page
- Customer page (transaction history widget)
- Event details page (transaction history widget)

Please see below an example of the Alerts Page after the change.

Fix Originator Transaction Amount fieldâ

Bug Fix FTMT-13383

We've fixed a misconfiguration in CM that prevented the `Originator Transaction Amount` field from working as expected for the transfers' automation rules.

This field will now show as `Amount` in the Automation Rules conditions configuration.

Enable Case Manager Garbage Collection logsâ

Feature

We've detected that Case Manager garbage collection logs were not enabled.

These are now enabled by default and are generated in Case Manager `logs` folder with the format `cm_GC_%p_%t.log`.

Add support for runtime mode in ETLâ

Feature

In previous releases ETL required an updated version of Pulse application to be able to export events. This would mean that when a new project version with schema changes was released a new ETL release was also needed.

In order to configure this mode check our [runtime mode documentation reference](#).

Add request_id to TMXâ

Feature

We have made changes on the input schema (for the input APIs) and in the workflow schema. Find the changes introduced in the release below.

Field	Datatype	Description
tmx.request_id	string	Unique identifier for TMX calls

Case Manager Version upgraded to 13.10.4â

Feature

In order to benefit from some improvements in performance, mainly regarding inserts into the database, we've upgrade Case Manager Version to 13.10.4.

ETL Single Run fixes and improvementsâ

Bug Fix

Depending on the installation structure, some log configurations (e.g.: persist to file in `log` folder) were not applied. This has been fixed and it should now work independently of the installation structure.

Similarly it was reported that `papp_file_path` had to be configured as an absolute path for ETL single run to work. This is no longer necessary. Nonetheless, we recommend leaving it with the default (absolute) path.

Additionally, we've fixed a few inconsistencies on our documentation, namely:

- ETL Configuration reference now reports the actual default `papp_file_path` - which is the absolute path
- References to `min` / `max` ... `_event_timestamp` where updated to the actual variables `start` / `end` `_event_timestamp`

Add missing indices for search fields in Case Manager tablesâ

Bug Fix

Some fields used in Case Manager searches were missing database indices. This release adds indices for the following fields:

- `sender_phone`, in the Transfers events table
- `sender_email`, in the Transfers events table
- `lifecycle_id`, in the Digital Activity events table
- `amount`, in the Digital Activity events table
- `STATUS_ID`, in the Common events table
- `amount`, in the Common events table
- `score`, in the Common events table

Fixed empty JSONs being produced by workflow audit extractionsâ

Bug Fix

Fixed an issue where the Workflow Audit Pipeline would write an empty JSON object to the GCP bucket if there was an error with the HTTP requests to Pulse.

Fixed missing workflow information extractionsâ

Bug Fix

Fixed an issue that caused application publishes done via Pulse Commands (for example, at the end of a Blue/Green deployment) to be missing from the `pulse-workflow-audit` pipeline extractions.

This was fixed by changing the publish action from an RMI call to Pulse to a REST API call. As a result, a new property `pulse_api_url` is required for the Pulse Ansible role. More information can be found in the [configuration page].

Change Data Type to prevent Events not being persistedâ

Bug Fix

Some Events were failing to be inserted in the database due to three fields having more than the configured length in the database.

To prevent that these values are no longer an issue and to fix the events not being persisted to the database, we've changed the Data Type of the following fields to `TEXT` :

- device_user_agent
- website_refferer_url
- device_language

Feedzai Platform Upgradesâ

Feature

We've upgraded to Pulse 25.0.26.

This upgrade introduces a performance increase in the Auto Recovery Job, used to recover events that are spilled to disk.

We've also added a new section in Pulse metrics page with metrics that can be used to monitor the [Auto Recovery Job].

Customer phone number and Email not populating correct values in Search screenâ

Bug Fix

We've fixed a bug where Customer Phone Number and Customer Email were being incorrectly mapped, in the Transfers and Digital Activity tab, on the Search and Alert page.

They are now being mapped for the same fields as in the `All Channels` tab.

Allow setting environment variables for Liquibaseâ

Bug Fix

Previously Liquibase execution environment was not configurable, which would be useful for setting JVM options for instance.

We now allow this configuration via the following variables:

- `cm_liquibase_environment` : used in Case Manager pipeline
- `pulse_liquibase_environment` : used in Pulse pipeline for database changes
- `pulse_commands_environment` : used in Pulse pipeline for application changes

For more information regarding Pulse configuration options see [Pulse Configuration Recommendations].

For more information regarding Case Manager configuration options see [Case Manager Configuration Recommendations].

Allow setting environment variables for ETL scheduled runâ

Bug Fix

Previously we only allowed setting environment variables for ETL in the single run setup.

We now allow this also in the scheduled run by setting the `etl_java_opts` variable as required. For more information see ETL Configuration Reference.

Add a new page in Pulse that reports the versions being usedâ

Feature

A new page was added to Pulse which reports Pulse, Solution and project versions running on that environment.

The page can be accessed via a web browser, by navigating to `admin/api/version/pulse-details` in Pulse.

Make PaymentFulfillment extension more resilient to missing valuesâ

Bug Fix

Previously multiple attributes were accessed directly without checking if they were present or not, which potentially could result in a failure that wasn't clear in the logs.

Input and Workflow Schema Updateâ

Feature

We made changes on the input schema (for the input APIs) and in the workflow schema. Find the changes introduced in the release below.

Updated fields

Old:

Field	Datatype	Description
bioc.custom_risk_aliases	double	Additional Biocatch Risk Factors

New:

Field	Datatype	Description
bioc.custom_risk_aliases	string	Additional Biocatch Risk Factors

Add Customer Reference Data API fields to Workflow schemaâ

Rework

We've added the following fields from Customer Reference Data API to the schema and data extractions:

- customer_pb_marker
- customer_portfolio_segment
- customer_voice_id
- customer_voice_id_last_change_date

The available fields can be consulted in the API documentation: Reference Data.

warning

For the new fields to be available in the ETL extractions, the new ETL version must be installed.

Edit the Customer FIV Marker field in Customer detailsâ

Rework

We've edited the `Customer FIV Marker` mapping in the Customer Details page from `customer_fiv_marker` to `customer_fiv_markers`.

Fix Logstash extractions for CM events notesâ

Bug Fix

We've fixed an issue where the Logstash extractions had the wrong schema for the `case-manager-events-notes` pipeline.

On this page

What's Changed

Release `1.7.0` â

Fixes and improvements to periodic thread dumpsâ

Bug Fix

For the periodic thread dumps feature, the linux `logrotate` utility was using a location in `/var/lib` to which the user running Pulse and Case Manager would not necessarily have access, leading to the periodic thread dumps not being rotated, and the log file becoming large. This script now uses a location within the Pulse directory itself now to store its logrotate status (for Pulse `/opt/feedzai/pulse/scripts/logrotate-status/jstack_logrotate.status`).

Additionally, the thread dumps taken now support multiple processes in the case of running on Pulse (supports multiple applications, and the spark driver which controls SMJs and Pulse data science jobs).

Reference Data Logs are now compressedâ

Feature

During the Go Live there were alerts regarding the disk usage in the Reference Data cluster since the service logs were not being compressed.

To prevent similar warnings or errors in the future, Reference Data logs are now automatically compressed when rotated.

New variable added to RDS configuration recommendationsâ

Feature

In order to avoid increased timeouts fetching Reference Data for enrichment, a new variable named `reference_data_grace_period_ms` can be found here: [RDS configuration recommendations]

Add new indexes to optimize Case Manager SLAs and Feedbacksâ

Feature

We have added an index on `EVENT_SLA_ACTIVE` in the `AM_EVENT_SLA` table.

We also updated the `AM_EVENT_SLA` and `AM_EVENT_STORAGE` partitioning functions to include `EVENT_TIMESTAMP` in the partition indexes.

By having this option, every new partition will have the following set of indexes that were identified as points of optimization for SLA and Feedback processing:

- `(EVENT_SLA_TIMESTAMP_BREACH, EVENT_TIMESTAMP)` in `AM_EVENT_SLA`
- `(EVENT_SLA_ACTIVE, EVENT_TIMESTAMP)` in `AM_EVENT_SLA`
- `(EVENT_VERSION_ID, EVENT_TIMESTAMP)` in `AM_EVENT_STORAGE`

Add support for filtered indexes in Case Managerâ

Feature

In previous releases Case Manager partitions did not have filtered indexes propagated from the main table due to lack of support in the partitioning functions.

This support was added so that any filtered index that exists in the main table is propagated to new partitions.

In particular, an index on the `amount` field for `AM_EVENT_COMMON` and `AM_EVENT_DIGITAL_ACTIVITY` with a `not null` constraint was added using this filtered index functionality.

Enable periodic thread dumps for Case Manager and Pulseâ

Feature

We have implemented a new feature that will take periodic thread dumps into the local file system for both Pulse and Case Manager.

Check Pulse Configuration Reference and Case Manager Configuration Reference for more information.

Note: The generated files will be stored in `<component_installation_dir>/scripts/log` where `<component_installation_dir>` is Case Manager/Pulse installation directory.

Add new fields to the rules extracted by the Pulse Workflow Audit pipelineâ

Feature

We have added five fields to the Pulse Workflow Audit pipeline:

- `workflow.rules[].condition`
- `workflow.rules[].variables`
- `workflow.rules[].rulesProjectSnapshotDesc`
- `workflow.elements[type = rules].rulesProjectDesc`
- `workflow.elements[type = plan].planDesc`

These fields allow more fine-grained control over rule and plan changes. More details can be found on the Data Schema page.

Fix behaviour and Consent endpoints returning error codesâ

Bug Fix

We have fixed an issue where the `consent` and `behaviour` API endpoints were returning `500` error codes due to missing type definitions for the `numeric double` type used by the `sender_transaction_original_amount` field.

Fix Event Description enrichment list itemâ

Bug Fix

We have fixed a wrongly created list item in the Event Description enrichment list which had its key set to key plus description.

This is now set as follows:

- key: `mNTB`
- description: `NTB Customer via a Digital Flow to become a Customer. When Application Completed and Successful`

Note that this change does not require the plan to be updated in/outside the workflow.

Add documentation for Case Manager tablesâ

Docs

We have added documentation for Case Manager tables under Case Manager Tables.

Add documentation for RabbitMQ Queuesâ

Docs

We have added documentation describing RabbitMQ Queues used by Pulse and Case Manager.

Docs

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-2.x/Release-2.0

On this page

What's Changed

Release `2.0.0` â

New fields added for AMHâ

Feature

The following fields were added to Pulse Schemas and APIs:

Reference Data:

- `customer_annual_turnover`
- `customer_business_type`
- `customer_chinese_comm_code`
- `customer_rm_info`
- `customer_marital_status`
- `customer_rel_status`
- `customer_type`

Transfers:

- `invalid_email_indicator`
- `sub_channel_name`
- `customer_id_number`
- `sender_account_channel_limit_double`

Digital Activity:

- `customer_id_number`
- `gift_indicator_code`
- `customer_status`
- `sender_billing_amount`

Add new fields to the Payments workflow schemaâ

Feature

The following fields were added to the Payments workflow schema:

- `customer_total_balance`
- `customer_reported_income`
- `customer_citizen_id_number`
- `customer_preferred_language`

These fields were already part of the Customer Reference Data API and will now be forwarded to the Payments workflow when available.

Add new Customer reference data fields to Case Manager schemaâ

Feature

The following fields were added to the Case Manager schema and are now shown in the Customer Details page:

- `customer_business_phone`
- `customer_businessphone_last_update_at`

- `customer_citizen_id_number`
- `customer_citizenship`
- `customer_corresp_city`
- `customer_corresp_country_code`
- `customer_corresp_region`
- `customer_corresp_street_addr`
- `customer_corresp_zip`
- `customer_corresp_zip_area_code`
- `customer_email_last_update_at`
- `customer_fiv_marker_last_updated_at`
- `customer_forname`
- `customer_gender_code`
- `customer_hmphone_last_update_at`
- `customer_home_address_last_update_at`
- `customer_home_city`
- `customer_home_country_code`
- `customer_home_phone`
- `customer_home_region`
- `customer_home_street_addr`
- `customer_home_zip`
- `customer_home_zip_area_code`
- `customer_mbphone_last_update_at`
- `customer_mobile_phone`
- `customer_pb_marker`
- `customer_preferred_language`
- `customer_registration_date`
- `customer_reported_income`
- `customer_segment`
- `customer_surname`
- `customer_total_balance`

Fix Customer Search link in Case Managerâ

Bug Fix

Fixed an issue where clicking on the ID of a Customer when search in the Customer Search page would redirect to the incorrect page.

Customer Reference Data field `customer_total_balance` now accepts decimalsâ

Rework

The `customer_total_balance` field of the Customer Reference Data API now accepts decimal numbers. Previously it was limited to integer numbers.

The updated type and examples can be found in the API Reference.

Fix hardcoded hostnames in Pulse and Case Manager database connection stringsâ

Bug Fix

We have added the following database-related variables to the ansible variables for both Pulse and Case Manager deployment configuration, to allow for simpler and more fine-grained tuning of the JDBC connection strings in both components.

For Case Manager, the following properties are now available:

- `database_casemanager_hostname`
- `database_casemanager_port`
- `database_casemanager_database_name`
- `database_casemanager_ro_hostname`

- `database_casemanager_ro_port`
- `database_casemanager_ro_database_name`

For Pulse, the following properties are now available:

- `database_pulse_eventstorage_hostname`
- `database_pulse_eventstorage_port`
- `database_pulse_eventstorage_database_name`
- `database_pulse_metadata_hostname`
- `database_pulse_metadata_port`
- `database_pulse_metadata_database_name`
- `database_pulse_eventstorage_read_only_hostname`
- `database_pulse_eventstorage_read_only_port`
- `database_pulse_eventstorage_read_only_database_name`

Check Pulse Configuration Reference and Case Manager Configuration Reference for more information.

Add periodic restart of pulse-workflow-audit logstash pipelineâ

Feature

Configurable periodic restart functionality has been added to the *pulse-workflow-audit* logstash pipeline, based on a Linux `cron` entry that runs a script to do so. Further details on this feature may be found in Logstash Pulse Workflow Audit Pipeline Recovery Tool.

Update currency field descriptionsâ

Bug Fix

We have fixed the Payments schema currency fields descriptions from `Alpha-3` to `3-Digit Numeric Code` to match the value definition sent to Pulse API.

Case Manager Version Upgraded to 13.12.3â

Feature

We've upgraded the Case Manager Version to 13.12.3. This upgrade will fix conditions for reasons in SLA not working.

Fix the transaction history decision option not filtered based on permission.â

Bug Fix

We have fixed an issue in Case Manager where the decision options in the transaction history widget were not being filtered by permissions.

Set Case Manager default states without Automation Rulesâ

Feature

`Breach Status` and `Operational Status` will now be automatically set to `Not Breached` and `Unreviewed`, respectively, without the need of Automation Rules.

By having these default states we greatly improve the processing time of new events.

Notes:

The following Automation Rules can be disabled:

- `Set Operational Status to Unreviewed and Breach Status`
- `Set Operational Status to Unreviewed`

Also any `Event Arrival` Automation Rule output that sets either of these states to their default value can be removed.

Add missing Case Manager enrichmentâ

Bug Fix

We have fixed the issue where CAM Level mapping not displaying correctly in the Digital Activity event Details Page.

Add missing Case Manager translationsâ

Bug Fix

We have fixed the issue where some fields in case manager were not having any translations.

Add missing fields to Payments + Outcomes schemaâ

Bug Fix

We fixed an issue in which following fields had not been properly added to the `Payments + Outcomes` schema:

- `customer_fiv_marker`
- `customer_mbphone_last_update_at`
- `customer_fiv_marker_last_updated_at`
- `customer_gender_code`
- `customer_earnings_amt_snapshot_dt`
- `current_account_monthly_debit_amount`
- `phone_bank_use_code`
- `customer_current_account_balance`
- `customer_salary_amount`
- `customer_forname`
- `customer_surname`
- `customer_home_street_addr`
- `customer_home_city`
- `customer_home_zip`
- `customer_home_zip_area_code`
- `customer_home_region`
- `customer_home_country_code`
- `customer_home_address_last_update_at`
- `customer_corresp_street_addr`
- `customer_corresp_city`
- `customer_corresp_zip`
- `customer_corresp_zip_area_code`
- `customer_corresp_region`
- `customer_corresp_country_code`
- `customer_corresp_address_last_update_at`
- `customer_email_last_update_at`
- `customer_home_phone`
- `customer_mobile_phone`
- `customer_daytime_work_phone`
- `customer_business_phone`
- `customer_hmphone_last_update_at`
- `customer_wrkphone_last_update_at`
- `customer_businessphone_last_update_at`
- `customer_registration_date`
- `customer_citizenship`

We've added an application patch in this release that will ensure the fields are there during the deployment (this is expected to be a no-op as the fields were added manually to the schema during the 1.6 deployment process).

On this page

What's Changed

Release `2.10.1` â

Update Case Manager EVENT_ID field length in the `AM_EVENT_AUTOMATION` tableâ

Bug Fix FTMT-32347 HSBC-3708

In order to keep the consistency across tables, for events with big `lifecycle_id`, we've updated the field `event_id` length in the `AM_EVENT_AUTOMATION` table for the type `VARCHAR(1024)`.

Address CVE'sâ

Feature

In this release, we addressed multiple CVEs and security-related FTMT tickets.

Please refer to the table below for a detailed summary of the updates and component versions affected.

Component	FTMT Ticket	Feedzai Ticket	Old Version	Upgraded Version	CVE	Additional Notes
Pulse	FTMT-30590 FTMT-30225 FTMT-28076	HSBC-3567 HSBC-3630	25.0.91	25.1.0	CVE-2023-6378 CVE-2023-6481 CVE-2024-7254 CVE-2025-24970 CVE-2025-30065 CVE-2025-48924 CVE-2025-52999 CVE-2025-5115	Addresses a security vulnerability identified in `jackson-core` version Upgrade library versions for local project-components (that don't come directly from the core Feedzai products).
Pulse Commands	FTMT-30225 FTMT-28076	HSBC-3567 HSBC-3630	25.14.38	25.17.3	CVE-2023-6378 CVE-2023-6481 CVE-2024-7254 CVE-2025-24970 CVE-2025-30065 CVE-2025-48924 CVE-2025-52999	
ETL	FTMT-30224	HSBC-3565	18.0.0	18.1.0	CVE-2024-1597 CVE-2024-7254 CVE-2024-13009 CVE-2025-52999	
Reference Data	FTMT-30224	HSBC-3568	3.7.8	3.7.11	CVE-2024-13009	
	FTMT-30226	HSBC-3619	7.8.4	7.8.5	CVE-2025-47273	Prevents a security

Component	FTMT Ticket	Feedzai Ticket	Old Version	Upgraded Version	CVE	Additional Notes
Ansible Portable						vulnerability identified in the `setuptools` version, that was updated to 79.0.1.
External Notifications	FTMT-30590	HSBC-3621	1.1.2	2.1.1	CVE-2022-1471 CVE-2025-52999	Full details on the Upgrade Data Extraction Pipelines to version 2.1.1 release note.
JupyterLab	FTMT-30590 FTMT-30225	HSBC-3633 HSBC-3681	25.0.46	25.0.49-gcs-2.2.29	CVE-2025-52999 CVE-2023-44487 CVE-2022-1471 CVE-2024-13009	This upgrade addresses the vulnerabilities in Pulse and removes the vulnerable libraries included by the Hadoop dependency. Please find the new Docker image in the Installation Guide.
Case Manager	FTMT-30224	HSBC-3547	N/A	N/A	CVE-2023-5072 CVE-2023-6378 CVE-2023-6481 CVE-2024-1597 CVE-2024-13009 CVE-2025-52999	Upgrade library versions for local project-components (that don't come directly from the core Feedzai products).
Spark Distribution	FTMT-30590 FTMT-30225	HSBC-3657	3.3.2-hadoop-3.3.4	3.5.7-hadoop-3.3.5	CVE-2025-52999	Addresses a security vulnerability identified in `jackson-core` version by upgrading the Spark dependency. Please note that this upgrade does not address the CVE that is bundled with

Component	FTMT Ticket	Feedzai Ticket	Old Version	Upgraded Version	CVE	Additional Notes
						Hadoop 3.3.6 given there is no version that addresses it.
Ansible Roles	FTMT-30590 FTMT-30225	HSBC-3657	33.1.0	34.0.0	CVE-2025-52999	Pipeline changes to adopt new Spark distribution archive.
GCS Connector	FTMT-28985	HSBC-3682	1.9.4-hadoop3	hadoop3-2.2.29	CVE-2019-17571 CVE-2020-7692 CVE-2020-8908 CVE-2020-15250 CVE-2021-4104 CVE-2021-22573 CVE-2022-23302 CVE-2022-23305 CVE-2022-23307 CVE-2023-2976 CVE-2023-26464	The upgraded version is the last version that supports Java 8

warning

Please note that starting in this version the `Spark Jars` archive was deprecated in favor of `Spark Dist`.

Any non-Feedzai pipelines that depend on this artifact should be updated to the new one that can be found in the [Installation Guide](#).

warning

Starting in this version the JupyterLab will be having a different naming, including the GCS Connector version provided in it, making it clear which component versions the image includes.

This change also requires a change in the load process. Instead of loading the image as `docker.feedzai.com/feedzai/jupyterlab:<jupyterlabVersion>`, it will now have to be loaded as `docker.feedzai.com/feedzai/jupyterlab:<jupyterlabVersion>-gcp`.

Upgrade Data Extraction Pipelines to version 2.1.1â

Feature FTMT-26459 FTMT-30590 HSBC-3367 HSBC-3621 HSBC-3330

We've upgraded the Data Extraction Pipelines to version 2.1.1.

This upgrade includes the following changes:

- Support for Logstash 9.1.4 by upgrading the existing pipelines and service configuration files. It addresses `CVE-2022-1471` and `CVE-2025-52999`
- Logstash can now be executed with a keystore
- Fix an issue where the Logstash recovery process pipeline would get stuck, causing Ansible to wait indefinitely

warning

Please note this is a breaking change since the new Data Extraction Pipelines package requires Java 17 or 21.

For more information refer to Data Extraction Pipelines Requirements and also to Data Extraction Pipelines Configurations.

Add multi tenancy support in the Case Manager alert cleanup script

Feature FTMT-31365 HSBC-3599

We have added multi tenancy support in the Case Manager alert cleanup script.

This change adds two new script options that allows the script to filter events based on all tenants or a specific set of tenant ids.

For more information please refer to the new filtering section in the Case Manager alert queue cleanup page.

Enable the `External Status Reason Updater` extension for Inbound Payments

Feature FTMT-30475 HSBC-3579

We have updated the `External Status Reason Updater` extension to allow Inbound Payments events.

The extension will now set the event's status reason based on the `external_status_reason` field value sent in the feedback event if it matches any Case Manager configured reasons.

Release `2.10.0`

Upgrade Pulse Commands version to 25.14.38

Feature HSBC-3493

We have upgraded Pulse Commands to version 25.14.38. This is a technical change which prevents patch failures when workflow schemas are deleted or implicitly defined.

Revamp documentation for the Case Manager AFIS extensions

Docs HSBC-3480

We've updated the documentation for the Case Manager AFIS Extensions.

The new page combines the schemas and payload examples for all Case Manager AFIS extensions, replacing the previous two individual pages for *Mobile Outbound* and *Payment Fulfillment*. This new page distinguishes between the *Send Mobile Payment Push* and *Send 2 WAY / 1 WAY SMS*, which previously had the same schema and payload and now have distinct values for the `notification_type` field. It also contains the schema and a payload example for the new *Inbound Payment Notification* extension.

All anchors that previously linked to the old pages have been updated to link to the new page instead.

Change Transaction Type filter for Inbound Payments

Feature FTMT-30267 HSBC-3503

Previously, the `Transaction Type` filter for **Inbound Payments** only allowed the selection of predefined values. This has now been changed to a standard text box, allowing the search using any value.

Please note that for accurate results the search box value must exactly match the Transaction Type name. For example, for events with a Transaction Type of `Bulk Fund Transfer`, searching for `Bulk Fund` would not yield any results.

Add `fdz_channel` to Cards and Inbound Payments

Feature FTMT-30250 HSBC-3434

The `fdz_channel` field, which describes the channel of the event, was missing for the **Cards** and **Inbound Payments** channels.

The field will now be automatically enriched with the following values and also be included in ETL extractions:

- Cards: `CARDS`
- Inbound Payments: `INBOUND-PAYMENTS`

Change the `notification_type` value for the *"Send 2 WAY / 1 WAY SMS"* Case Manager extensionâ

Feature FTMT-29922 HSBC-2940

We've changed the value of the `notification_type` field in the *"Send 2 WAY / 1 WAY SMS"* Case Manager extension from `"payment_mobile_push"` to `"2_way_1_way_sms"`.

This change makes it possible to distinguish the *"Send 2 WAY / 1 WAY SMS"* and *"Send Mobile Payment Push"* notifications downstream.

Please refer to the AFIS Extensions reference for more details.

Add *Inbound Payment Notification* Case Manager extensionâ

Feature FTMT-29922 HSBC-3408 HSBC-3444

We've added a new Case Manager extension named *Inbound Payment Notification* that is able to send notifications to AFIS from Inbound Payments automation rules. This extension has a different schema compared to the Outbound Payments extensions and includes a `rule_explanations` field that lists the rule explanations triggered by the event.

By default the extension is enabled, but it can be disabled by setting `inbound_payment_notification_extension_enabled` to `false`. For more details please refer to the Case Manager Configuration Recommendations and the AFIS Extensions reference.

Add Ansible variable to control HSBCNet Advanced Search fieldsâ

Feature FTMT-30368 HSBC-3460

We've introduced a new Ansible variable `show_hsbcnet_search_fields` that allows showing or hiding the HSBCNet specific Advanced Search fields.

By default these fields are hidden but can be enabled in HSBCNet markets by setting the variable to `True` during deployment.

Add support for distinct RabbitMQ brokers for Pulse and Case Managerâ

Feature HSBC-3430

We've added support for distinct RabbitMQ brokers for Pulse and Case Manager. The following Ansible variables were changed:

Component	New Ansible Variable	Old Ansible Variable
Pulse	`rabbitmq_casemanager_brokers`	N/A
Case Manager	`rabbitmq_casemanager_brokers`	`rabbitmq_brokers`
Logstash	`rabbitmq_casemanager_brokers`	`rabbitmq_brokers`
Logstash	`rabbitmq_casemanager_username`	`rabbitmq_insights_logstash_username`
Logstash	`rabbitmq_casemanager_password`	`rabbitmq_insights_logstash_password`
Logstash	`rabbitmq_casemanager_vhost_blue`	`rabbitmq_pulse_vhost_blue`
Logstash	`rabbitmq_casemanager_vhost_green`	`rabbitmq_pulse_vhost_green`

Note that `rabbitmq_brokers` is still required in Pulse and defines the RabbitMQ brokers for Pulse internal communication.

For more details please refer to:

- Pulse Configuration Recommendations
- Case Manager Configuration Recommendations
- Data Extraction Pipelines Configurations

Default values for the new variables

The new Ansible variables default to the variables they are replacing (e.g. `rabbitmq_casemanager_brokers` defaults to `rabbitmq_brokers`, if not defined). This means that this change is backwards compatible and the default behaviour is to continue using a single RabbitMQ for all components.

For clarity, we recommend changing the variable names in the HSBC Ansible Variables files to use the new ones instead.

Make `minimizeShuffles` property configurable via Ansible

Feature FTMT-30345 HSBC-3447

We've added support to configure the property `services.modules.datascience.context.minimizeshuffles` via Ansible.

The property can now be configured using the following Ansible variable:

- `minimize_shuffles_enabled`: a boolean flag that determines if the group by fields can be re-sorted in order to minimize shuffles (defaults to *true*)

For more details please refer to the [Pulse Configuration](#) page.

Add `node.properties` template to Pulse Data Science Assembly

Feature HSBC-3461

We've added the `node.properties.j2` template to the Data Science assembly.

Previously Data Science installations defaulted to a template that only supported single node setups, meaning that in the Pulse UI the node would always show the localhost IP (`127.0.0.1`).

Add `lifecycle_id` index for Cards and Inbound Event Storage

Feature HSBC-3466

We've added the following index to the Inbound and Cards Event Storage tables to improve the feedback processing time:

- `idx_es_<eventstorage.tableName>_lifecycle_id`: index on column `lifecycle_id`

warning

This index will be automatically applied to the main (non-partitioned) table, the partitioning function, and all *new* partitions created after this update. Existing partition tables created prior to this change will *not* be affected and will require manual indexing if needed.

Add Decision Typology outcome to Inbound Payments

Feature FTMT-29925 HSBC-3411

We've added the `decision_typology` outcome to the **Inbound Payments** workflow. The field was also added to the **Inbound Payments + Outcomes** schema.

This outcome can have the following values:

- `approve_fraud`
- `approve_scam`
- `review_fraud`
- `review_scam`
- `decline_fraud`
- `decline_scam`
- `none`

ⓘnote

The values added to the outcome are the same that as the **Outbound Payments** baseline. It is possible that some values have since been added to the **Outbound Payments** outcome. If that's the case, any missing values should be added manually to the **Inbound Payments** outcome to ensure consistency between both workflows.

Add Payment Fulfillment Extension for Inbound Paymentâ

Feature FTMT-29917 HSBC-3409

We've updated the Payment Fulfillment extension to support Inbound Payments events in addition to Outbound.

The schema is the same as for Outbound events, as described in the AFIS Extensions reference.

Add Inbound Payments application to Data Science application artifactâ

Feature HSBC-3414

The Data Science applications assembly has been updated to include the **Inbound Payments** application.

This update ensures that the complete assembly can now be imported into HSBC's Data Science environments, guaranteeing compatibility of the initial schemas with Pulse runtime. The update assembly can be found in the Installation Guide.

Fix Case Manager partial updates on Digital Activity eventsâ

Bug Fix FTMT-31199 HSBC-3399

We've fixed an issue where Digital Activity feedback events where processed as full updates of the original event. This would cause some fields, e.g. `decision`, to be updated although that is not the intended behaviour.

Add new fields to Case Managerâ

Feature FTMT-28352 HSBC-3281

We've added new search fields and transaction history filter fields in Case Manager.

We've added the following fields to the Case Manager advanced search page:

fields	Channel
session_id	Transfers
device_id	Transfers
device_ip_address	Transfers/Digital Activity
device_ip_address_v6	Transfers/Digital Activity
user_id	Transfers/Digital Activity
sender_transaction_ref_nr_alt	Transfers/Digital Activity
device_user_agent	Transfers/Digital Activity

We've added the following fields to the Event Details `transaction history` dropdown filter:

fields	label
device_ip_address	IP Address
device_ip_address_v6	IP Address v6
user_id	User ID
sender_transaction_bank_account_number	Debit Account Number
alert_id	Alert ID

fields	label
sender_transaction_ref_nr_alt	Instruction Reference Number
receiver_transaction_bank_account_number	Beneficiary Account Number
receiver_bank_bic	Beneficiary Bank Sort Code
receiver_name	Beneficiary Name
device_user_agent	User Agent
payment_type	Payment type

Change deployment pipelines to support RHEL8+â

Feature FTMT-26831 HSBC-3086
Also In: 2.8.3 2.9.2

We've changed the Case Manager, Pulse and ETL deployment pipelines to support installations on RHEL8+ that were failing due to restrictions on the `rsync` module execution.

By default the pipelines will still have the previous behaviour. This can be changed during deployment by setting the `use_rsync` Ansible variable to `false`.

For more details please refer to:

- Case Manager Recommendations
- Pulse Configuration Recommendations
- ETL Configuration Recommendations

Improve currency display in Case Managerâ

Feature WPB-349833 HSBC-3128

We have updated some Case Manager widgets where a generic currency sign (Â¤) could be displayed.

The widgets now have custom logic to detect ISO 4217 currency codes (both the 3-letter and 3-digit codes) and format the currency amount according to the browser locale. When Case Manager cannot detect a valid currency it will now display the amount value without any currency symbols.

Upgrade JupyterLab version to 25.0.46â

Feature FTMT-27382 HSBC-3084 HSBC-3375

We have upgraded JupyterLab to version 25.0.46 to fix the missing `timestampField` dependencies in List operators when duplicating a plan using the `ds-api`, as well as including the SHAP Python package.

The new Docker image will have to be loaded in existing environments. Please find it in the Installation Guide.

Improve Case Manager and Pulse Monitoring guidesâ

Feature HSBC-3252 HSBC-3253

We've made some changes to Case Manager and Pulse monitoring guide for better readability and organization.

You can review the updated version at Top Metrics to Monitor in Case Manager and Top Metrics to Monitor in Pulse

Add back pulse.configure and case-manager.configure tags to Pulse and Case Manager 810 Ansible tasksâ

Bug Fix FTMT-29677 HSBC-3393
Also In: 2.8.2 2.9.1

We have added back the `pulse.configure` and `case-manager.configure` tags to Pulse and Case Manager Ansible tasks `810.1`, respectively, so that these task are not skipped during Pulse or Case Manager deployment. These tasks was being skipped in releases `2.8`, `2.8.1`, and `2.9`, leading to certain Pulse server or Case Manager Ansible variables being ignored (such as Pulse Java options, which includes the Pulse server heap memory size parameter).

Add `Originator Details` and `Beneficiary Details` tabs in Case Manager Event Details pageâ

Feature FTMT-10124 HSBC-3359

The event details page for both **Inbound and Outbound Payments** now includes `Originator Details` and `Beneficiary Details` tabs for quick access to the corresponding widgets.

Add Dead Letter Queueing system to Data Extraction Pipelinesâ

Feature HSBC-3255

We've upgraded the Data Extraction Pipelines to version 1.1.2, which includes:

- The possibility of enabling Logstash Dead Letter Queueing (DLQ) system to recover events that errored during processing
- A recovery pipeline for the `pulse-workflow-audit` events present in the DLQ

For more information refer to the Data Extraction Pipelines overview page.

Add new fields to Outbound Paymentsâ

Feature FTMT-27741 HSBC-3290

We've added the following fields to the Outbound Payments schema and the Mobile Outbound extensions:

Field Name	Datatype	Description
tmx.true_isp	String	The legitimate Internet Service Provider that assigned the original IP address used by an end-user to access the internet. This ISP is tied directly to the user's physical location or device - without any intermediate.
tmx.proxy_isp	String	It is the service provider offering proxy IP addresses that mask the user's actual IP address. These includes VPNs, residential proxies, datacenter proxies, and anonymizers. In essence, the ISP visible to the internet is that of the proxy provider, not the user's real ISP.
tmx.dns_isp	String	It refers to the ISP seen during Domain Name System (DNS) resolution - the process of converting human-readable domain names into IP addresses. This is typically the ISP of the DSN resolver the user is.

Please refer to the AFIS Extensions reference for more details.

Allow concurrent Pulse application publishesâ

Feature FTMT-29333 HSBC-3358

The number of concurrent application publishes in Pulse will now be set to the number of applications configured in the `apps_enabled` variable, so that for instance inbound and outbound can be published concurrently. Specifically, this change sets the `pulse.global.services.core.apps.operationThreadPoolNumberOfThreads` property in Pulse.

Add new endpoints for List Managementâ

Feature HSBC-3101

We've added a new Entity Tagging API which can be used to perform list management operations programmatically. There are four new endpoints:

- **GET Entity Tagging** âused to retrieve list items
- **DELETE Entity Tagging** âused to remove list items
- **POST Entity Tagging** âused to create new list items
- **PUT Entity Tagging** âused to upsert (create or update) list items

The new endpoints live in the Reference Data application Input Service, sharing the same port as the Reference Data API.

For a full specification check the [Entity Tagging API documentation]() and refer to the [Postman Collection]() for examples.

Complex Lists

When it comes to Complex Lists, these endpoints only support lists that have a single value which must be named `value`.

Add new fields to the Mobile Outbound extensionsâ

Bug Fix FTMT-28271 HSBC-3276

The following fields were added to Case Manager `2 Way/1 Way SMS` and `Mobile Payment Push` extensions for the Digital Activity channel:

Field Name	Datatype	Description
customer_id_global	String	Global Customer ID
user_id	String	Online Banking User ID

For the Digital Activity channel we have also changed the `payment_sub_type` and `payment_type` fields in Case Manager (which also applies to the fields in SLAs, automation rules, and the AFIS extension) to use the raw values received from Pulse, instead of the values enriched via self-service List enrichment.

This should have no impact on the existing solution, as these fields were not being used, but will allow the raw value to be used for these fields if used again in the future.

Please refer to the [AFIS Extensions]() reference for more details.

Fix color coding issues in Case Manager widgetsâ

Bug Fix FTMT-27709 HSBC-3215

We've fixed the color coding issues in several Case Manager widgets (e.g. `Alerted`, `System Decision`) caused by the removal of an internal package in a previous version.

When navigating to, for example, Case Manager Event Details page fields will display their correct colors:

Add new fields to Reference Dataâ

Feature FTMT-15088 HSBC-3304 HSBC-3474

The following fields were added to the Customer Reference Data API:

Field Name	Datatype
customer_portfolio_region	String
customer_portfolio_product	String
customer_portfolio_class	String
customer_portfolio_type	String
customer_portfolio_channel	String
customer_portfolio_country	string
customer_account_country_code	string

Add bulk actions, new columns, search event fields, and permissionsâ

Feature FTMT-10124 HSBC-3196

We've added customizations to Case Manager. These updates include:

- New Bulk Actions with new permissions: The following actions are available on bulk under the identified permissions:

Page/Widget	Description	Permission
Transaction history	Add or update reasons	cm:resource:history:reason-bulk:*
Transaction history	Change status (Move to Hold, In-progress, Close, or Queue)	cm:resource:history:status-bulk:*
Transaction history	Update lists	cm:resource:history:update-list-bulk:*
Alert page	Add or update reasons	cm:resource:alert:reason-bulk:*
Alert page	Change status (Move to Hold, In-progress, Close, or Queue)	cm:resource:alert:status-bulk:*
Alert page	Update lists	cm:resource:alert:update-list-bulk:*

- Search Capability: Users can now search for events based on their Operational Status.
- Linked Alerts: The 'Linked Alerts' widget has been updated to include the Beneficiary Name and Beneficiary Account Number.

Update Alert ID generationâ

Feature WPB-380307 HSBC-3131

Previously, Alert IDs were created by selecting the last 8 characters of the lifecycle ID.

We've now improved this logic. Alert IDs are generated by hashing the full lifecycle ID using SHA-256, encoding the hash with Base64, removing any non-alphanumeric characters, and then taking the last 8 characters of the result.

As a side effect, the restriction that the Alert ID will only be generated when the lifecycle ID has at least 8 characters has been lifted.

Add ETL playbook for Pulse ES to Case Manager recoveryâ

Feature FTMT-27051 FTMT-27144 HSBC-3232
Also In: 2.8.1 2.9.0

We've added a new ETL playbook to enable event recovery from Pulse event storage to Case Manager. It supports two routing strategies to send events to Case Manager, configured via the `etl_cm_route_strategy` variable.

Please refer to [Recover events from Pulse to Case Manager]() for more details.

Add support for externally created Pub/Sub subscriptions for DTPâ

Feature FTMT-27152 HSBC-3202
Also In: 2.8.1 2.9.0

We've added support for the DTP to connect to Pub/Sub subscriptions externally created in Google Cloud Platform. Three new Ansible variables control this behaviour:

- `dtp_pubsub_subscription_enable_creation` : a boolean flag that determines if the DTP should create the Pub/Sub subscription when starting (defaults to *false*)
- `dtp_pubsub_subscription_id` : the subscription id for the DTP to use
  - This value is used only when `dtp_pubsub_subscription_enable_creation` is set to `false`
- `inactivity.subscription.enableCreation` : a boolean flag that determines if the DTP should create the Inactivity Service Pub/Sub subscription when starting (defaults to *false*)

Please refer to the DTP Configuration page for more details.

🔥warning

The new boolean flags that determine if the DTP automatically creates the subscriptions default to `false`. This means that if not explicitly configured otherwise, the DTP will expect the subscriptions to already exist in GCP. This is a departure from the previous behaviour which always created the subscriptions on start.

Add new field to the Mobile Outbound extensionsâ

Bug Fix FTMT-28271 HSBC-3245
Also In: 2.8.1 2.9.0

The following field was added to Case Manager `2 Way/1 Way SMS` and `Mobile Payment Push` extensions:

Field Name	Datatype	Channels	Description
channel_user_id	String	Digital	Online/Internet Banking User ID.

Please refer to the AFIS Extensions reference for more details.

New Case Manager enrichers documentationâ

Docs HSBC-3233

We've added a new documentation page that describes how to use Case Manager enrichers and lists the available ones.

Add support for Dataproc 2.2 in Pulse pipelineâ

Feature FTMT-22240 HSBC-2978
Also In: 2.8.4 2.9.5

To fully support the Dataproc 2.2 migration described in Google Dataproc 2.2, we've added two new configuration option to set the Java version used by Pulse jobs (e.g. datasource analysis, scheduled metrics):

- `spark_datascience_jdk_home`: Sets the `JAVA_HOME` for Data Science jobs executed in Dataproc, e.g. datasource analysis
- `spark_distributed_jdk_home`: Sets the `JAVA_HOME` for other jobs executed in Dataproc, e.g. scheduled Metrics Jobs

For more details about the new variables and their default values, refer to Pulse Configuration Recommendations.

Add new fields to the Mobile Outbound extensionsâ

Feature FTMT-27710 HSBC-3210
Also In: 2.9.0

The following fields were added to Case Manager `2 Way/1 Way SMS` and `Mobile Payment Push` extensions for Inbound and Outbound Payments channels:

Field Name	Datatype	Description
receiver_account_bank_id	String	Receiver Account Bank ID
receiver_bank_bic	String	Beneficiary Bank Identification Code

Please refer to the AFIS Extensions reference for more details.

Fix error when adding Notes to Customersâ

Bug Fix FTMT-26842 HSBC-3226

We've resolved an issue that occurred when adding notes to a Customer from the Event Details page.

The bug was triggered for events where the `lifecycle_id` exceeded 40 characters, which caused the note creation to fail. To address this, we've increased the allowed length of the `lifecycle_id` field to support longer values.

Add support for defining allowed Pub/Sub regions in DTPâ

Feature FTMT-27702 HSBC-3212
Also In: 2.8.1 2.9.0

We've added support in the DTP for defining the allowed Pub/Sub regions where data may be stored and transmitted. This behaviour is controlled by three new Ansible variables:

- `dtp_security_allowed_regions` : A list of allowed regions where data may be stored. Example:

```
dtp_security_allowed_regions:
  - europe-west1
  - asia-east2
```

Copy

- `dtp_security_enforce_in_transit` : A boolean flag that enforces in-transit restrictions to the same regions defined for storage.
- `dtp_security_regional_endpoint` : The regional Pub/Sub gRPC service endpoint to use.

Please refer to the Data Security section in the DTP Configuration page for more details.

Add new performance configuration options and probe logs to Case Managerâ

Feature HSBC-3216
Also In: 2.4.10 2.8.1 2.9.0

We've added a new set of performance configuration options to the Case Manager deployment pipeline. These configurations were already part of Case Manager but were not exposed during deployment.

On top of these changes we also added a deployment option to enable Case Manager probe logs. These logs can be useful to pinpoint issues by checking in which actions Case Manager is taking the most time. Please note that enabling this feature will create a considerable amount of logs, so it should only be done in a controlled environment.

Please find a complete description of the changes in Case Manager Performance Configuration.

Add configurable usage of the legacy list formatâ

Feature HSBC-3151
Also In: 2.9.0

Weâve introduced support for configuring the list import/export format via the following Ansible variable:

- `lists_use_legacy_csv_format`

By default, the variable value is set to true, preserving the previous behavior. To enable the new format, set it to false. Please note that in case the variable is not defined, it will use the old import/export format.

This enables installations to choose between the legacy and new list formats based on their needs:

- Strategic Market installations can adopt the new list format, making use of tenant-specific exports.
- UK and AMH installations must continue using the legacy format to maintain compatibility.

Tne new list format provides some improvements, like:

- Introduction of data headers to the exported lists
- Use the tenant key or the tenant group name in the exported files
- Complex lists are also more intuitive, as each field is in a different column instead of in a json format

For more details about the new variables and their default values, refer to Pulse Configuration Recommendations.

Add new fields to the Mobile Outbound extensionsâ

Feature FTMT-27243 HSBC-3158
Also In: 2.9.0

The following fields were added to Case Manager `2 Way/1 Way SMS` and `Mobile Payment Push` extensions for Inbound and Outbound Payments channels:

Field Name	Datatype	Description
device_ip_address	String	IP address
tmx.ip_address	String	True IP Address
tmx.ip_city	String	True IP City
tmx.ip_country	String	True IP Country
device_banking_type	String	Online Banking Device Type
device_os	String	Banking device Operating System Version
device_user_agent	String	Accept User Agent HTTP Header

Please refer to the AFIS Extensions reference for more details.

Add new field to Inbound and Outbound Paymentsâ

Feature FTMT-27248 HSBC-3176
Also In: 2.9.0

We have added the below field to Pulse and Case Manager schemas, for Outbound and Inbound Payments:

Field Name	Datatype
beneficiary_proxy_account_number	String

Add new Case Manager enrichment fields for Inbound Paymentsâ

Feature FTMT-24520 HSBC-2696
Also In: 2.9.0

The following fields can now be enriched when displayed in Case Manager for Inbound Payments events:

- sender_transaction_code (senderTransactionCodeEnrich)
- sender_bank_country_code (countryEnrich)
- receiver_bank_country_code (countryEnrich)
- is_batch (isBatchEnrich)
- payment_type (inboundPaymentTypeEnrich)
- payment_status (paymentStatusEnrich)
- sender_transaction_original_currency (currencyEnrich)

With these changes, the field sent in the payload can now be transformed, as long as it is correctly mapped and matches a valid value.

Assuming this, if in an event payload `sender_transaction_code` is `510`, what will be displayed in Case Manager is `Mastercard Other Credit`

These new enriched fields can be added to Case Manager directly via the self-serviceable edit feature. This can be achieved by using the advanced configuration and a template similar to the following:

```
{
    "key": "event.fields.sender_transaction_original_currency",
    "type": "currencyEnrich",
    "title": "Sender Transaction Original Currency"
}
```

Copy

ⓘnote

The `type` value should correspond to what is inside the parentheses in the list of new fields above. If the type is not changed, what will be displayed is exactly the same value that is in the payload.

Add new fields to `vocalink` complex fieldâ

Feature FTMT-27190 HSBC-3110 HSBC-3204
Also In: 2.9.0

We've added new fields to the existing `vocalink` for Inbound and Outbound Payments API and workflow schema:

Field Name	Datatype
alert_id	String
source_account_id	String
destination_account_id	String
transaction_id	String
transaction_value	Double
score_generated_at	Long
transaction_currency	String

warning

Due to the complex nature of HSBCâs production workflow, which contains many elements with multiple copies of the schema, this change may cause a normal blue-green deployment to fail. A manual step will be required to update the objects in the Pulse UI.

Add GCP permissions documentation for the DTP Serviceâ

Docs FTMT-27152 HSBC-3136
Also In: 2.8.1 2.9.0

We've added a new page with the minimal set of GCP permissions needed for the DTP service.

Please refer to the DTP permissions page for more details.

Also, the subscription created by the DTP service for the Inactivity topic is now configured to never expire.

Add support for externally created Pub/Sub topics for DTPâ

Feature FTMT-27152 HSBC-3138
Also In: 2.8.1 2.9.0

We've added support for the DTP to connect to Pub/Sub topics externally created in Google Cloud Platform. Three new Ansible variables control this behaviour:

- `dtp_pubsub_topic_enable_creation` : a boolean flag that determines if the DTP should create the Pub/Sub topics when starting (defaults to *false*)
- `dtp_pubsub_topic_id` : the topic id for the DTP to use
    - This value is used only when the topic creation is **disabled**
- `dtp_inactivity_topic_enable_creation` : a boolean flag that determines if the DTP should create the Inactivity Service Pub/Sub topic when starting (defaults to *false*)

Please refer to the DTP Configuration page and the new GCP Setup page for more details.

warning

The new boolean flags that determine if the DTP automatically creates the topics default to `false`. This means that if not explicitly configured otherwise, the DTP will expect the topics to already exist in GCP. This is a departure from the previous behaviour which always created the topics on start.

Add performance-related Ansible variables for cross-channel metricsâ

Feature HSBC-3168
Also In: 2.4.10 2.5.7 2.8.1 2.9.0

We've added Ansible variables that can be used to control the performance of cross-channel metrics:

- `cross_channel_metrics_fetch_timeout_ms` controls the timeout when fetching cross-channel metrics (defaults to `200ms`)
- `cross_channel_metrics_num_threads` controls the number of threads used when fetching cross-channel metrics (defaults to `1`)

More information can be found in Pulse Configuration Recommendations.

Make RDS not copy logs to system logâ

Feature HSBC-3152
Also In: 2.8.3 2.9.2

The Reference Data service was copying its request logs into the system logs (`/var/log/messages`) by default. This was controlled by the undocumented `reference_data_log_console` variable, which was set to `true` by default. We've set this to `false` in the `project-default` Ansible configuration.

The request log will instead go to the `request.log` file in the normal reference data log folder, and will be rotated and compressed daily, with 5 compressed files. We do not expect that this file will need to be exported to stackdriver.

Add Workflow Monitoring Service to the Reference Data applicationâ

Feature HSBC-3140
Also In: 2.8.1 2.9.0

We've added the Workflow Monitoring Service to the Reference Data application.

This service will populate the `workflow.events` metric with the events that are processed by that workflow and can be used to create monitoring dashboards.

Compute Event Storage table names based on the application idâ

Feature HSBC-3094
Also In: 2.8.1 2.9.0

Up until this point Event Storage table names were fixed and thus not configurable. This meant that these tables would be created automatically but their name would in fact only match the default application ids.

To further support the global rollout we've changed this behaviour so that the Event Storage table name for a given application is calculated based on the configured application ids.

Please note this change will only affects new environments that do not use the default application ids.

Add tenancy information to rules extracted by the `Pulse Workflow Audit` pipelineâ

Feature FTMT-26643 HSBC-3066
Also In: 2.9.0

We've added three new fields to the `Pulse Workflow Audit` pipeline:

- `workflow.rules[].isMultiTenant`: indicates whether a rule snapshot belongs to a multi tenanted rules projects

- `workflow.rules[].tenantId` : indicates the tenant key of the rules snapshot if it belongs to a multi tenanted rules projects and `null` otherwise
- `workflow.elements[type = rules].isMultiTenant` : indicates if the workflow element belongs to a multi tenanted rules projects

Please note that `Duplicated for Tenant` rules projects will have `tenantId=null` but still show `isMultiTenant=true` .

Please refer to the Data Schema page for more details.

Add filtered index on `customer_citizen_id_number` to optimize Case Manager searchesâ

Feature HSBC-3050
Also In: 2.8.1 2.9.0

We've recreated the `customer_citizen_id_number` indexes to improve performance by removing non-null values from the index. This change is specially helpful during Case Manager searches in deployments where `customer_id` is used instead of `customer_citizen_id_number` .

Pulse Version Upgradeâ

Feature FTMT-26830 WPB-378442 HSBC-3076 HSBC-3141
Also In: 2.4.9 (Pulse Version 25.0.85) 2.5.7 (Pulse Version 25.0.85) 2.8.1 (Pulse Version 25.0.85) 2.9.0 (Pulse Version 25.0.85)

We've upgraded Pulse version to 25.0.90 for the 2.10.0 release.

This upgrade includes:

- Performance improvements for real-time cross-channel metrics
- A fix for issue with file descriptor leaks that were affecting the Pulse server.

Add Ansible variable to control Reference Data Service loggersâ

Feature HSBC-3093
Also In: 2.4.9 2.5.7 2.7.3 2.8.1 2.9.0

We've added Ansible variables that allow log levels to be configured for Reference Data Service:

- `reference_data_log_level` controls the log level for all components
- `reference_data_loggers` controls the log level for individual components

The `reference_data_loggers` variable can be configured as a dictionary of loggers as keys and log level as values, for example:

```
reference_data_loggers:
    com.example: debug
```

Copy

More information can be found in the Reference Data Service Configuration Reference.

New API to change log levelsâ

Feature HSBC-3058

We've added a new API that allows log levels to be modified during runtime. This API can be used to change the log levels in the Pulse Server, Pulse Applications and Case Manager. It uses RabbitMQ to make sure a single request to the API is propagated across all replicas of a given component, allowing each to change its log level.

The API can be used via authenticated `GET` or `POST` requests to the Pulse UI host and requires the logged in user to have the `security:logging:update` permission:

```
POST http(s)://<PULSE_UI>/admin/api/loggers/{targetComponent}/{loggerName}/{loggerLevel}
```

By using the `targetComponent` parameter, the API will propagate the log change request to either Pulse Server, Pulse Application or Case Manager. A more comprehensive architecture and API spec can be found in the [Change Log Levels at Runtime]() guide.

Improve DTP Dataflow pipeline dependencies file stagingâ

Feature HSBC-2602

Currently, every time a publish occurs with the DTP service enabled, the Dataflow pipeline dependencies file is uploaded to the configured staging bucket.

We have updated this behavior; from now on, the dependencies file will only be uploaded once per release.

The current active dependencies file version is in the Pulse property: `pulse.global.services.rte.workflow.dtp.dataflow.fileToStageVersion`.

Case Manager version upgraded to 13.22.0â

Feature FTMT-18187 FTMT-22596 FTMT-22865 FTMT-22986 FTMT-27682 WPB-239611 HSBC-2885 HSBC-2958 HSBC-3214

We've upgraded Case Manager to version 13.22.0 which includes the following:

- Fix an issue where the event note payload was not including the event payload (including the event tenant name)
- Fix an issue where refreshing Case Manager's search page would throw a warning if the previous search had a multi-criteria filter (e.g. `Customer ID`)
- A new *Smart Search* functionality described below
- A new feature where the Activity Log now shows when a change was triggered by an Automation Rule (more details below)

Smart Searchâ

With *Smart Search*, when searching for events using filters for channel-specific fields (Outbound, Digital Activity, Cards, or Inbound Payments), the search results will only display events that exactly match the selected filters.

Hereâs how it works:

- If a search is performed using only common filters, all tabs will be enabled, displaying the total number of filtered events.
- If a search includes a channel-specific filter (for example, *Beneficiary Name* in Inbound Payments), only the corresponding tab will be enabled, showing the total number of matching events. The tabs corresponding to the other use cases will not display any results.

ⓘ**note**

Since the From and To Date filter is mandatory in all searches, the All Channels tab will always display all events, regardless of other search filters.

Automation Rules in Activity Logâ

When there is a change in an event that is the result of an Automation Rule triggering, the Activity Log now shows the Automation Rule instead of the **system** user. Clicking the Automation Rule name will link to the respective Automation Rule (if the user has permissions to access it).

This feature has two important caveats:

1. It is **not retroactive**, meaning that only Automation Rules triggered in this version will be shown in the Activity Log.
2. If the **Automation Rule is deleted, Case Manager will not be able to show it** in the Activity Log and will revert back to the previous behaviour of showing the **system** user as the actor.
    - Note: this **does not affect disabled Automation Rules** âthose will show up in the Activity Log.

Add CMEK configuration for DTP resource creationâ

Feature FTMT-27701 HSBC-3137
Also In: 2.8.1 2.9.0

We've added Customer-managed encryption keys (CMEK) support for the DTP service.

The key name can be configured by using the `dtp_security_kms_key_name` Ansible variable. If omitted DTP resources will have the default behaviour and use Google-managed encryption keys.

The key configuration should follow the format described in [Configured a topic with CMEK](#).

Please refer to [DTP configuration](#) and [GCP permission required for using CMEK](#) for more information.

Add new fields to Cardsâ

Feature FTMT-25618 FTMT-29492 HSBC-3100 HSBC-3443
Also In: 2.9.0

We have added the below fields to the Cards API schema:

Field Name	Datatype	Also In 2.9.0
transaction_priority	String	X
global_customer_identifier	String	X
on_us_flag	String	X
auth_tran_type	String	X
cardholder_bill_currency_code	String	X
card_account_pre_tran_cash_bal	Double	
card_account_cash_limit	Double	
card_account_client_def_limit	Double	
card_delinquency_days	Integer	
card_currency_conv_rate	Double	
card_account_txn_count_limit	Long	
card_limit_currency_conv_rate	Double	
card_account_limit_type	String	
card_entity_usage	String	
card_pb_user_id_usage	String	
card_channel_level_count_limit	Long	
card_channel_currency_conv_rate	Double	
card_phone_banking_user_id	String	
card_previous_action_info	String	
card_action_until_timestamp	Long	
card_customer_type_indicator	String	
card_clnt_def_tran_exp_date	Long	
card_utc_indicator	String	
card_multi_events	String	
card_auth_cust_old_address	String	
card_hub_guid_customer_id	String	

Field Name	Datatype	Also In 2.9.0
digital_wallet_token_txn_new	String	
card_clring_cheque_mail_cntry	String	
card_unique_id	String	
card_account_sending_fund	String	
card_stip_reason_cde_changes	String	
card_expiry_date_val_result	String	
card_cnp_tran_not_thresh_amt	Double	
card_upstream_fraud_score	Double	
card_upstream_fraud_reason_code	String	
card_update_pass_wallet_date	Long	
card_activity_comp_detl_1	String	
card_activity_comp_detl_2	String	
card_activity_comp_detl_3	String	
card_client_transaction_type	String	
card_message_destination_name	String	
card_transaction_component_type	String	
card_instant_credit_indicator	String	
card_biometric_verify_ind	String	
card_customer_present_details	String	
card_message_expiry_date	Long	
card_termnl_pan_entry_capblty	String	
card_customer_type	String	
card_customer_language	String	
card_customer_level_limit_amt	Double	
card_customer_level_limit_cnt	Long	
card_customer_level_conv_rate	Double	
card_current_action_code	String	
card_travel_indicator	String	
card_cnp_tran_notification	String	
card_mobile_match	String	
card_atm_transaction_type	String	
card_auth_system_warning_code	String	
card_acc_last_txn_date	Long	
card_active_flag	String	
card_frst_use_flag	String	
card_pri_first_use_flag	String	
card_cnp_threshold_amount	Double	
card_cycles_over_limit	Integer	
card_dispatch_branch_sortcode	String	

Field Name	Datatype	Also In 2.9.0
card_last_auth_date	Long	
card_last_exp_cmf_date	Long	
card_ond_vec_value	Integer	
card_san_account_number	String	
card_address_home_pc	String	
card_customer_income	Double	
card_held_dob	String	
card_staff_member_indicator	String	
card_vip_indicator	String	
created_at	Long	
sender_initiated_at	Long	
sender_initiated_at_local	String	
event_created_at	Long	
event_received_at	Long	
local_datetime_alt	String	
card_latitude_longitude	String	
card_c_type_atm_debit	String	
card_account_change	String	
card_hcc_box	String	
card_last_expiry_date	Long	
card_ond_vec_status	String	
card_p_ac_t_atm_debit	String	
card_code_card_type	String	
card_account_new_status	String	
crypto_transaction_info	String	
card_transaction_ref_number	String	
lifecycle_id_alt	String	
saf_ind	String	

The `Cards` and `Cards + Outcomes` schemas were updated with all the above fields.

On this page

What's Changed

Release `2.11.0` â

Upgrade Pulse to version 25.1.2â

Bug Fix FTMT-30238 HSBC-3849

We've upgraded Pulse to version 25.1.2.

This change includes an upgrade of Apache Commons BeanUtils to address the CVE-2025-48734 vulnerability.

Upgrade Pulse to version 25.1.1â

Bug Fix HSBC-3790
Also In: 2.10.2

We've upgraded Pulse to version 25.1.1.

This upgrade fixes an issue where both the live and upgrade clusters would consume feedback messages, resulting in inconsistent data.

Fix issue with Pulse's Alert Storage Service not forwarding messages to Case Managerâ

Bug Fix HSBC-3754

We've fixed an issue that could occur with Pulse not forwarding messages to Case Manager when there was both a RabbitMQ connection loss forcing Pulse's Alert Storage Service to reconnect to RabbitMQ, and a manual Pulse UI configuration reload.

Add ETL and DTP extractions comparison documentationâ

Docs HSBC-3738

We have created a guide which explains the key data format differences between ETL and DTP extractions to help with the tool migration process. The guide focus on the additional fields each tool retrieves beyond the standard schema fields.

The guide is available in Migrating from ETL to DTP.

Fix API specification examplesâ

Bug Fix HSBC-3722

We have fixed an issue in the API specification where field types and examples differed in datatype. For example, `customer_annual_turnover` showed as `number` but the example would should as a `string` (`"15000000.00"`).

Please refer to the API specification page for more information.

Add Digital Activity channel to Mobile Outbound extensionsâ

Bug Fix FTMT-32724 HSBC-3709
Also In: 2.10.1 2.9.3

We have added the Digital Activity channel to the Mobile Outbound extensions. The base schema of the notifications is the same as for Outbound Payments. Please refer to the AFIS Extensions reference for more details.

Add new fields to Cardsâ

Feature FTMT-24023 FTMT-30076 FTMT-32738 FTMT-33019 HSBC-3666 HSBC-3755 HSBC-3749

We have added the below fields to the Cards schema and workflow:

New Fields

Field Name	Datatype
address_match_ind_3ds	STRING
buyer_daily_provisn_attempt_3ds	STRING
buyer_purchase_velocity_3ds	STRING
challenge_method_final_3ds	STRING
date_received_3ds	STRING
device_channel_3ds	STRING
digital_id_trust_score_3ds	DOUBLE
purchase_amount_3ds	DOUBLE
purchase_exponent_3ds	STRING
shipping_indicator_3ds	STRING
shipping_post_region_3ds	STRING
svc_acs_transaction_id_3ds	STRING
card_action_on_data_indicator	STRING
card_active_flg_last_change_date	STRING
card_change_data_new	STRING
card_change_data_old	STRING
card_cpp_exp_date_proxy	LONG
card_cpp_list_date	LONG
card_details_availability_ind	STRING
card_entity_id	STRING
card_entity_type	STRING
card_expiration_date_3ds	LONG
card_involving_entity	STRING
card_new_city_name	STRING
card_new_country_code	STRING
card_new_phn_country_code	STRING
card_new_phone_number	STRING
card_new_post_code	STRING
card_new_state_name	STRING
card_nm_effective_timestamp	LONG
card_old_city_name	STRING
card_old_country_code	STRING
card_old_phn_country_code	STRING
card_old_phone_number	STRING
card_old_post_code	STRING

Field Name	Datatype
card_old_state_name	STRING
card_onlinebanking_user_id	STRING
card_reissue_type_flag	STRING
card_temp_limit_chnage_indicator	STRING
card_temporary_lmt_exp_date	LONG
card_transfer_entity_from_key	STRING
card_transfer_entity_to_key	STRING
card_transfer_reason	STRING
card_txn_sub_type	STRING
card_transaction_ref_number_alt	STRING
card_website_name	STRING
customer_level_limit_type	STRING
digital_wallet_token_txn_new	STRING
new_card_activation_flag	STRING
old_card_activation_flag	STRING
transaction_status_3ds	STRING
user_amount	DOUBLE
user_indicator	STRING
user_string	STRING
observed_ip_address_3ds	STRING
network_risk_score_3ds	STRING
device_browser_language_3ds	STRING
device_region_3ds	STRING
issuer_name_3ds	STRING
arcot_merch_alert_score_3ds	STRING
arcot_tpd_provider_id_3ds	STRING
matched_policy_3ds	STRING
matched_regulatory_rule_3ds	STRING
policy_advice_id_3ds	STRING
policy_results_3ds	STRING
policy_advice_score_3ds	STRING
regulatory_advice_id_3ds	STRING
regulatory_rule_results_3ds	STRING
rule_score_source_3ds	STRING
rule_set_advice_id_3ds	STRING
rule_set_matched_rule_3ds	STRING
rule_set_result_3ds	STRING
rule_set_score_3ds	DOUBLE
sr_rules_advice_id_3ds	STRING

Field Name	Datatype
sr_rules_set_version_3ds	STRING
transaction_failure_reason_3ds	STRING
tran_origination_cntry_3ds	STRING
card_saf_ind	STRING
tran_id_3ds	STRING
ucaf_value_3ds	STRING
customer_id_personal	STRING
customer_id_business	STRING
tmx.digital_id_trust_score	DOUBLE
tmx.chall_reason_code	STRING

The `Cards` and `Cards + Outcomes` schemas were updated with all the above fields.

Update API documentationâ

Docs FTMT-30058 HSBC-3661

We've update the API resources documentation to include:

- Missing Cards and Inbound Payments endpoint information
- Non Mon endpoint documentation, including how the `event_type` field is generated
- Missing tag descriptions in the API specification page

For more information please refer to API Additional Resources.

Add Non-Mon Endpointâ

Feature FTMT-30076 HSBC-3637

We've created a new endpoint to allow the receiving and registering of non monetary events.

For a more in-depth explanation of this new endpoint please refer to the Non-Mon Documentation page.

Add new fields to Outbound Payments and Digital Activityâ

Feature FTMT-28937 HSBC-3591

We have added the below elements to the `bioc` tuple in Outbound Payments and Digital Activity API schemas:

Field Name	Datatype
pinpoint_risk_score	Integer
chall_policy_score	Integer

The `Payments` and `Payments + Outcomes` schemas were updated with all the above fields.

Fix Transaction History header scroll behaviorâ

Bug Fix FTMT-30567 HSBC-3476

Weâve fixed a bug that made the Transaction History header disappear while scrolling. It now stays visible at the top of the list.

Create RabbitMQ segregation guide for Pulse and Case Managerâ

Docs FTMT-476 HSBC-3431

We have created a guide on how to segregate a RabbitMQ cluster by component, focused on Pulse and Case Manager. The guide focus on a migration with no message loss and minimal downtime.

For more information please refer to the RabbitMQ Segregation for Pulse and Case Manager page.

Add new fields to Cardsâ

Feature FTMT-30565 HSBC-3627

We have added the below fields to the Cards API schema:

Field Name	Datatype
card_deposit_type	String
card_minimum_amount_due	Double
card_purchase_timestamp	Timestamp
credit_card_post_ref_number	String
debit_card_auth_ref_number	String
card_client_amount	Double
cardholder_zip	String
cards_purchase_utc_indicator	String

The `Cards` and `Cards + Outcomes` schemas were updated with all the above fields.

On this page

What's Changed

Release `2.2.0` â

Add functionality to remove null byte from the payloadâ

Bug Fix

We have implemented a functionality to remove null bytes from the payload to prevent errors when storing events in the Pulse and Case Manager databases.

Add placeholder inbound payments applicationâ

Feature

We have added a placeholder Pulse application to real-time Pulse for inbound payments, to begin supporting the inbound payments use case. The schema used for now is the one used by the outbound payments application. As each application can be independently deployed, this does not have an impact on existing installations, but unblocks a future inbound payments installation.

Input and Workflow Schema Updateâ

Feature

The following fields were added to the `Payments` and `Payments + Outcomes` schemas and Pulse APIs:

Field	Datatype	Description
bioc.push_score	double	Push API score as received from BioCatch
customer_id_hashed	string	Hashed version of puidUserId used to call Vendor Tech

These fields were also added in Case Manager as follows:

- `customer_id_hashed`
  - Transfers Originator Details widget
  - Digital Activity Event Summary widget
- `bioc.push_score`
  - Digital Activity Vendor Risk Data widget

Add channel based permissionsâ

Feature

We have created the below new permissions for inbound payments.

- `cm:channel:inbound:read-only`
- `cm:channel:inbound:read-write`
- `cm:state-machine:event-status:write:status_inbound_payments`
- `cm:state-machine:event:read:status_inbound_payments`
- `cm:state-machine:event-status:write:status_inbound_payments`
- `cm:state-machine:status-reason:delete:status_inbound_payments`
- `cm:state-machine:status-reason:read:status_inbound_payments`
- `cm:state-machine:status-reason:write:status_inbound_payments`

Add Self Serve Strategy outcomeâ

Feature

Please find below the changes regarding the new Serf Service Strategy outcome.

Pulse:

- We have added the Self Serve Strategy outcome to the Payments workflow
- The corresponding field was added to the `Payments + Outcomes` schema

Case Manager:

- A matching field was added to Case Manager so it can be used in Automation Rules and SLAs
- Transaction widget in Event Details page was updated to show the Serve Self Strategy field
- External notifications payload was updated to send both Serve Self Strategy and Treatment Strategy

ETL:

- The outcome will be extracted in the outcomes column

New Ansible task to revert Case Manager changes in Databaseâ

Feature

We've created a new Ansible task to facilitate reverting Case Manager database changes.

This means that if for some reason there are compatibility issues between Case Manager versions, we can quickly revert the changes and keep Case Manager running.

The `Rolling Upgrades` documentation was updated. Please check it at Rolling Upgrades

What's Changed

Release `2.3.4` â

Fix an issue with inbound permissions during deploymentâ

Bug Fix HSBC-1958

We've fixed an issue where, when inbound permissions were already present in the `admin` role, these were not getting added again, causing the deployment pipeline to fail. These permissions are no longer being added to the role, and in fact are now removed from the role as they are not strictly needed.

Release `2.3.3` â

Fix an issue with the ETL extractionâ

We've fixed an issue in the ETL tool extractions, in which the ETL tool was looking for a nonexistent schema field (and database column) used for testing.

Release `2.3.2` â

ETL Papp format fixâ

We have fixed an issue in the ETL tool, in which the papp file it was using to connect to Pulse was using the new default Pulse papp export format introduced recently. As the current ETL tool does not currently support this format, we have updated the papp file it uses to use the legacy file format, to restore the ETL tool's functionality.

Release `2.3.1` â

Add Inbound Payments to Case Manager Search Screenâ

Feature HSBC-2104

We've added the Inbound Payments channel to the Case Manager search screen. Following this change, an Inbound Payments tab will appear in search results as it does for other channels, and search criteria will also retrieve results for the Inbound Payments channel. Additionally, a new advanced search section for Inbound Payments has been added with the same search criteria as the Outbound Payments (Transfers) advanced search criteria section.

Add Inbound Payments Case Manager Event Details Pageâ

Feature HSBC-2102

We've added the Inbound Payments channel to Case Manager Event Details Page, defaulting to the same default displayed widgets as Outbound Payments. What widgets are displayed can be controlled, as for Outbound Payments and Digital Activity, by the self-service UI editing directly in Case Manager, so the added fields represent only a baseline that may be modified directly in the UI.

Pulse Version Upgraded to 25.0.52â

Feature FTMT-14325 HSBC-1981

We've upgraded Pulse to version 25.0.52.

This version of Pulse has improved support for cross-channel metrics. A new Ansible variable `cross_channel_metrics_enabled` was created to control whether cross-channel metrics are enabled. More information can be found in the [Pulse Configuration Reference](#).

Remove permissions with typoâ

Bug Fix

We've removed the `cm:action:reveal-schema:card_pan_plain_text` and `cm:resource:manager.*` permissions since these were created with a typo.

By removing them we guarantee the correct permissions are assigned to user roles.

Added `customer_id_number` to Consent and Behaviour endpointsâ

Bug Fix FTMT-17711 HSBC-2152

We fixed a bug where `customer_id_number` field was not available for Consent and Behaviour endpoints.

This issue is now resolved and the field can be used on the respective endpoints.

New field for Celebrus Model Scoreâ

Feature FTMT-14242 HSBC-2001

We've added a new field `model_risk_score` to the Pulse API for the Outbound Payments. We've also updated the Payments workflow schema to include this new field.

Reconcile Inbound Schema Differences for CMâ

Feature HSBC-2101

We have added below fields and indexes to Inbound Payments CM schema, making them available to use in SLAs and automation rules.

Please find below the newly added fields

Inbound Payments

Field Name	Datatype
dispute_end_date	TIMESTAMP
dispute_feedback_notes	STRING
dispute_refunded	STRING
oob_feedback_notes	STRING
feedback_type	STRING
oob_end_date	TIMESTAMP
oob_response_type	STRING
dispute_notes	STRING
dispute_reason	STRING
dispute_start_date	TIMESTAMP
dispute_submitted_by	STRING
oob_comm_type	STRING
oob_notes	STRING
oob_start_date	TIMESTAMP
bulk_id	STRING

Field Name	Datatype
channel_type	STRING
created_at	TIMESTAMP
customer_transaction_amount	TIMESTAMP
device_browser	STRING
device_browser_version	STRING
device_first_seen	TIMESTAMP
device_imei_number	STRING
device_ip_address	STRING
device_ip_address_location	STRING
device_ip_country_code	STRING
device_ip_region	STRING
device_isp	STRING
device_language	STRING
device_latitude	STRING
device_longitude	STRING
device_operating_system	STRING
device_os_version	STRING
device_type	STRING
device_user_agent	STRING
direction	STRING
event_external_id	STRING
event_occurred_at	TIMESTAMP
execution_at	TIMESTAMP
frequency	STRING
geographic_scope	STRING
last_sca_at	TIMESTAMP
malware_flag	BOOLEAN
number_of_payments	INTEGER
pass_reset	BOOLEAN
payment_status	STRING
payment_sub_type	STRING
payment_type	STRING
product_code	STRING
proxy_flag	BOOLEAN
receiver_account_id	STRING
receiver_bank_branch_id	STRING
recurring_payment	BOOLEAN
reference_number	STRING
requested_execution_at	TIMESTAMP

Field Name	Datatype
sca_risk_score	TIMESTAMP
sender_account_balance	TIMESTAMP
sender_account_bic	STRING
sender_account_credit_limit	TIMESTAMP
sender_account_currency	STRING
sender_account_debt_balance	TIMESTAMP
sender_account_id	STRING
sender_account_last_activity	TIMESTAMP
sender_account_number_of_cards	INTEGER
sender_account_open_date	TIMESTAMP
sender_account_status	STRING
sender_account_type	STRING
sender_alias	STRING
sender_alias_last_update_time	STRING
sender_alias_registration_time	STRING
sender_alias_status	STRING
sender_alias_type	STRING
sender_bank_branch_id	STRING
sender_citizen_id_number	STRING
sender_city	STRING
sender_country_code	STRING
sender_currency	STRING
sender_dob	STRING
sender_email_change_flag	BOOLEAN
sender_id	STRING
sender_initiated_at	TIMESTAMP
sender_kyc_lvl	STRING
sender_number_of_accounts	INTEGER
sender_number_of_cards	INTEGER
sender_occupation	STRING
sender_passport_number	STRING
sender_phone_change_flag	BOOLEAN
sender_region	STRING
sender_registration_date	TIMESTAMP
sender_risk_rating	STRING
sender_segment	STRING
sender_ssn	STRING
sender_street_addr	STRING
sender_tax_number	STRING

Field Name	Datatype
sender_total_balance	TIMESTAMP
sender_total_credit_limit	TIMESTAMP
sender_zip	STRING
session_register_date	TIMESTAMP
transaction_goal	STRING
transaction_method	STRING
user_email_address	STRING
user_phone_number	STRING
user_phone_number_country_code	STRING
channelId	STRING
customer_portfolio_product	STRING
customer_portfolio_channel	STRING
customer_portfolio_type	STRING
customer_portfolio_class	STRING
customer_portfolio_country	STRING
customer_portfolio_region	STRING
source	STRING
payment_message_source	STRING
customer_type	STRING
office_id	STRING
customer_account_type	STRING
sender_transaction_type	STRING
sender_transaction_cop_result	STRING
sender_transaction_auth_type	STRING
external_status_reason	STRING
receiver_bank_bic	STRING
internal_payment_type	STRING
internal_event_source	STRING
text_for_receiver	STRING
lifecycle_id_alt	STRING
sender_transaction_original_amount	DOUBLE
sender_transaction_original_currency	STRING
customer_id_number	STRING
rcv_name	STRING
receiver_transaction_bank_account_number	STRING
rcv_tnx_mm_susp_bank_acc_num	STRING
rcv_tnx_ops_susp_bank_acc_num	STRING
rcv_tnx_bank_redirect_acc_num	STRING
is_batch	STRING

Field Name	Datatype
reversal_code	INTEGER
reversal_reason	STRING
inhibit_code	INTEGER
stop_valid_code	INTEGER
product_valid_code	INTEGER
credit_valid_code	INTEGER
repayment_valid_code	INTEGER
account_not_found_code	INTEGER
account_transaction_valid_code	INTEGER
mm_payment_status	STRING

Please find below the newly added indexes

Inbound Payments

Column Name
payment_type
payment_sub_type
rcv_name
receiver_bank_bic
rcv_transact_bank_acc_number
sender_transaction_type

Inbound Payments Schema Changesâ

Feature HSBC-2043

We have made the requested adjustments to Inbound Payments and Reference Data schema.

Please find below the new added fields

Inbound Payments

Field Name	Datatype
receiver_transaction_mm_suspense_bank_account_number	String
receiver_transaction_ops_suspense_bank_account_number	String
receiver_transaction_bank_redirect_account_number	String
is_batch	String
reversal_code	Integer
reversal_reason	String
inhibit_code	Integer
stop_valid_code	Integer
product_valid_code	Integer
credit_valid_code	Integer
repayment_valid_code	Integer
account_not_found_code	Integer

Field Name	Datatype
account_transaction_valid_code	Integer
mm_payment_status	String

Reference Data - Customer

Field Name	Datatype
line_of_business	String
business_start_date	Long
employer_name	String
employer_start_date	Long
app_fraud_source	String
app_fraud_overall_decision	String
app_fraud_overall_decision_text	String
app_fraud_session_id	String
app_fraud_timestamp	Long
app_fraud_bio_score	Double
app_fraud_bio_risk_level	String
app_fraud_bio_is_rat	String
app_fraud_bio_is_bot	String
app_fraud_bio_is_emulator	String
app_fraud_bio_reason_code	String
app_fraud_tmx_policy_score	Double
app_fraud_tmx_risk_rating	String
app_fraud_tmx_summary_reason_codes	String
app_fraud_tmx_device_id	String
app_fraud_tmx_fuzzy_device_id	String
app_fraud_tmx_fuzzy_device_id_confidence	Integer
app_fraud_tmx_first_party_score	Double
app_fraud_tmx_true_ip	String
app_fraud_tmx_proxy_ip	String
app_fraud_tmx_digital_id_trust_score	Double

Reference Data - Account

Field Name	Datatype
product_type	String
account_closed_date	Long

Add Inbound Payments channel to Customer Details Pageâ

Feature HSBC-2103

We've added the Inbound Payments channel to Case Manager Customer Details page Transaction History.

Add Inbound Payments channel to Alerts Pageâ

Feature HSBC-2105

We've added the Inbound Payments channel to the Case Manager Alerts page.

Add Inbound Payments channel to Customer Details Pageâ

Feature HSBC-2103

We've added the Inbound Payments channel to Case Manager Customer Details page Transaction History.

AMH Outbound Payments Schema Changesâ

Feature FTMT-16494 FTMT-16649 HSBC-1992

We have made the requested adjustments to Outbound Payments schema.

Please find below the newly added fields

Outbound Payments - Transfers

Field Name	Datatype
bank_processing_application	String
bank_processing_type	String
reversal_indicator	String
check_book_request_date	Long
cash_form	Integer
check_number	String
thailand_fps_ind	Integer
gpb_manual_payment_ind	Integer
scameter_warning	Integer
sender_billing_amount	Double

Outbound Payments - Digital

Field Name	Datatype
new_customer_flag	String
scameter_warning	String

Release `2.3.0` â

Enable multi-tenancy in Case Managerâ

Feature HSBC-2071

We've enabled multi-tenancy in Case Manager. This feature can be enabled by using the Ansible variable `cm_multi_tenancy_enabled` . Please check [Case Manager Configuration Reference] for more information.

Add List Management Service to workflowâ

Feature FTMT-14956 FTMT-14957 FTMT-17203 HSBC-1970

We've added the List Management Service to the Payments and Inbound Payments workflows.

See [Configure List Management Service] for information on how to configure the service.

New search field for HKID/Passport numberâ

Feature FTMT-15538 HSBC-1908

We have modified the customer id field in the search to also consider the HKID or passport number in addition to the customer ID. Specifically, the search will now query both the fields `customer_citizen_id_number` and `customer_id`, returning results when either of these criteria match the value in the search box.

Add a new extension for Case Managerâ

Feature FTMT-16590 HSBC-1993

We have added new Extension `Send 2 WAY / 1 WAY SMS` for the Case Manager

- `Send Mobile Payment Push` - Case Manager Extension that can be used in Automation Rules and SLAs.

We have added new ansible variables to control this extension separately from the existing Send Mobile Payment Push, which are comprised of all the variables for the Send Mobile Payment Push extension, but with _send_sms appended at the end of the variable names. Full details may be found in the [Case Manager Configuration Reference]

We have added new Operational Status `Remind Customer`

Additional fields added to the CM extension are

- `customer_portfolio_class`
- `operational_status`
- `customer_id_number`
- `alert_id`
- `treatment_strategy`
- `self_serve_strategy`
- `customer_name`
- `customer_surname`
- `customer_mobile_phone`
- `customer_email`
- `customer_preferred_language`
- `sender_transaction_original_amount`
- `sender_transaction_original_currency`

Added the Inbound Payments APIâ

Feature HSBC-1987

We've added the Inbound Payments API to the workflow of Inbound Payments applications.

Refer to the API Reference section for a specification of the available endpoints.

Add missing permissions for read-only admin roleâ

Feature FTMT-17207 HSBC-1875

We've added a new set of permissions that will be required for the read-only admin role.

You can find the up to date information in the Permissions Guide.

Ensure Delete Reference Data Endpoint is included in workflowâ

Bug Fix FTMT-17231 HSBC-2033 HSBC-1951 HSBC-2088

Use the reference data delete endpoint bundled together with the other reference data endpoint classes, so that the existing workflow configuration works without needing to add the HSBC-specific one. This solves an issue that required a UI workaround during an AMH QA deployment.

To support this change under the hood, we've also removed the following Pulse services (these previously showed up in the Data Configuration tab in the Pulse UI) which shipped by default with the original Feedzai application, but were not being used and were simply cluttering the application. In doing so, we were also able to remove some underlying dependencies which make the deployment package slightly leaner and safer.

- FraudForce Enricher
- FraudForce Enrichment Service
- Revelock Enricher
- Revelock Enrichment Service
- List Management Service (this is an older default one, and in this release is being replaced by the `Workflow List Management Service`)
- Expired Metrics Service
- Outcome Priority Enforcer

Remove references to DynamoDBâ

Docs HSBC-1941

See up-to-date information at Pulse Installation Guide and Monitoring Services

Pulse Version Upgraded to 25.0.46â

Feature FTMT-14325 HSBC-1980

We've upgraded Pulse version to 25.0.46.

This version of Pulse has support for the cross-channel metrics.

Case Manager version Upgraded to 13.14.3â

Feature FTMT-14311 HSBC-1994

This upgrade includes the addition of the **AND** and **OR** logic operators in Automation Rules and SLAs.

Refer to Case Manager Documentation for more information.

Add Case Manager support for Global Rolloutâ

Feature HSBC-1957

We have added support for multi application deployments in Case Manager.

Please find a description of the changes in Global Rollout Documentation.

Input and Workflow Schema Updateâ

Feature FTMT-15663 HSBC-1881

The following field was added to the `Payments` and `Payments + Outcomes` schemas and Pulse APIs:

Field	Datatype	Description
bioc.is_mobile_rat	boolean	BioCatch Rat indicator for Mobile

This field was also added in Case Manager as follows:

- `bioc.is_mobile_rat`
    - Digital Activity Vendor Risk Data widget

Exclude outcomes and results from ETL when configured for Historical Data Loadingâ

Feature FTMT-14963 HSBC-1731

We have changed ETL to exclude outcomes and results when configured for Historical Data Loading. Please find the updated configuration reference in ETL Configurations.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-2.x/Release-2.4

On this page

What's Changed

Release `2.4.3` â

Change Outbound Payments Amount and Currency mappingsâ

Feature FTMT-21000 HSBC-2399

We've changed the `amount` and `currency` mappings to `sender_transaction_original_amount` and `sender_transaction_original_currency`, respectively, for Outbound Payments in Case Manager.

This change affects the `Common` (for `Transfers` events) and `Transfers` tabs in the following pages:

- Alerts page
- Search Page
- Event Details transaction history widget
- Customer Details transaction history widget

Change Inbound Payments Amount and Currency mappingsâ

Feature FTMT-21027 HSBC-2402

We've changed the `amount` and `currency` mappings to `receiver_transaction_amount_dbl` and `receiver_transaction_currency`, respectively, for Inbound Payments in Case Manager.

This change affects the `Common` (for `Inbound Payments` events) and `Inbound Payments` tabs in the following pages:

- Alerts page
- Search Page
- Event Details transaction history widget
- Customer Details transaction history widget

Release `2.4.2` â

Add ansible support for segregating outbound/inbound/cards channels in Case Managerâ

Bug Fix HSBC-2277

We've added the configuration to allow the setup of a dedicated databases to store the event storage of Case Manager's event channels. The following variable identifies the usage of the dedicated databases. The first channel of the list identifies the main database, which includes all the AM_* tables (with the exception of AM_EVENT_STORAGE and AM_EVENT__<channel>). Example with the available values:

```
pdb_active_channels:
  - transfers
  - cards
  - digital-activity
  - inbound-payments
```

Copy

Each of the following variables needs a suffix _<channel> that identifies the channel and the configuration of the dedicated database.

- database_casemanager_engine

- database_casemanager_jdbc
- database_casemanager_ro_jdbc
- database_casemanager_username
- database_casemanager_password
- database_casemanager_schema
- database_casemanager_schemapolicy
- database_casemanager_pool_size
- database_casemanager_pool_lazy
- database_casemanager_ro_pool_size
- database_casemanager_ro_username
- database_casemanager_ro_password
- database_casemanager_ro_schema
- database_casemanager_password_encrypted
- database_casemanager_reconnect_on_lost
- database_casemanager_max_number_retries
- database_casemanager_retry_interval
- database_casemanager_query_select_timeout
- database_casemanager_ro_password
- database_casemanager_ro_reconnect_on_lost
- database_casemanager_ro_max_number_retries
- database_casemanager_ro_retry_interval
- database_casemanager_ro_query_select_timeout

Release `2.4.1` â

Fix ETL not using new RUNTIME modeâ

Bug Fix HSBC-2166

We have fixed an issue where the ETL was not using a live connection to Pulse for ETL extractions. The live connection is needed for the inclusion of scheduled metrics in ETL extractions.

Add Decision Typology outcomeâ

Feature FTMT-17700 HSBC-2196

We have added a new `decision_typology` outcome to the Payments workflow in the Outbound Payments application. This outcome can have the following values:

- `approve_fraud`
- `approve_scam`
- `review_fraud`
- `review_scam`
- `decline_fraud`
- `decline_scam`
- `none` (default)

A `decision_typology` field has also been added to the `Payments + Outcomes` schema, allowing it to be used in the Risk Strategy.

Release `2.4.0` â

ETL Tool now extracts scheduled metricsâ

Feature HSBC-2149 HSBC-2166 HSBC-1988

The ETL tool now extracts scheduled metrics in its extractions.

In order to do so, the ETL tool now connects to the live application when determining the schema to use in its extractions. Depending on the value set in `get_latest_published_app_version` (default: `true`), the ETL tool will either extract events based on the published workflow schema or load the current (potentially-unpublished) workflow schema. It will always fall back to the latest unpublished version in the rare case when the application is not published (such as during a Pulse outage).

The previous `pulse_connection_mode` and `papp_file_path` ansible variables have been removed, and these properties will be ignored in any ansible inventories that were previously using them.

A new ansible variable was added to control whether the ETL tool includes metrics in its extraction, defaulting to `true`:

- `include_internal_state`

See [ETL configuration reference](#) for more information.

Reconciliation of Transfers and Digital Activity endpointsâ

Feature HSBC-2162

We have reconciled the API definition of all Transfers and Digital Activity endpoints.

Please find below the changes.

All Transfers Endpoints:

Field Name	Datatype
customer_status	String
gift_indicator_code	Long
new_customer_flag	String

All Transfers Endpoints except for Feedback:

Field Name	Datatype
external_preliminary_decision	String
external_status_reason	String

All Digital Activity Endpoints:

Field Name	Datatype
bank_processing_application	String
bank_processing_type	String
cash_form	String
check_book_request_date	Long
check_number	String
external_preliminary_decision	String
external_status_reason	String
gpb_manual_payment_ind	String
invalid_email_indicator	Boolean
reversal_indicator	String
sender_account_channel_limit_double	Double
sub_channel_name	String
thailand_fps_ind	String

Behaviour and Consent Endpoints:

Field Name	Datatype
customer_status	String
gift_indicator_code	Long
sender_billing_amount	Double

Reference Data Schema and Workflow moved to their own applicationâ

Feature HSBC-2187 HSBC-2200 HSBC-2201 HSBC-2202 HSBC-2203

Since Reference Data is intended to be common across multiple applications, we have created a new, separate app specifically for Reference Data.

As a result, we've made some changes to the way the apps are organized:

- In the Outbound Payments app, we removed the Reference Data schema and workflow.
- We now have a new Reference Data application that includes the schema and workflow previously present in the Outbound Payments application.

Additionally, it is important to note that the Reference Data App (e.g. `uk_reference_data`) must be added to the `apps_enabled` variable.

Also, tags used in monitoring metrics (for example those used in Grafana) will change automatically with these modifications. Therefore, to ensure everything continues to function as expected, please review the monitoring metrics endpoints (pulse `/metrics` endpoint) for any resource using these tags (e.g., appSlug, workflowId, etc.).

Update database retention scripts to include inbound tablesâ

Feature HSBC-1979

We've updated the database retention scripts to include the inbound Event Storage and Case Manager tables..

See Retention Scripts for more information.

Change Outbound Payments Amount mappingâ

Feature FTMT-17872 FTMT-18154 HSBC-2206 HSBC-2212 HSBC-1947

Consequence of the recent change to send a transaction `amount` in the `sender_transaction_amount_dbl` field instead of `sender_transaction_amount` , Case Manager may not show the correct value in searches or show `N/A` .

We've changed the current mapping to a new field `amount_dbl` based on `sender_transaction_amount_dbl` . This change affects the following pages:

- Alerts page
- Search Page
- Event Details transaction history widget
- Customer Details transaction history widget

Note: Apart from the last page, this change only applies to the `Transfers` tab. In a future release we will deprecate the `amount` field completely, remap all channel tabs `Amount` columns to the `amount_dbl` field and also change the search filter.

The new field is available to be used in Automation Rules and SLAs with the name `Amount (Double)` .

We've also extended the Mobile Outbound extension to include the new field (`sender_transaction_amount_dbl`) in the request payload. Please see the updated payload in the AFIS Extensions reference.

Add support to configure multi tenancy using an API fieldâ

Feature HSBC-2108

To further support the multi tenancy `REQUEST` strategy introduced in [Configure Reference Data Service to use multi tenancy], we've extended the deployment pipeline to make it possible to configured the payload field from which Pulse will read the tenant value.

Configuration is based on the channel/application, using the following Ansible variables:

- `inbound_payments_tenant_api_payload_field`
- `outbound_payments_tenant_api_payload_field`
- `digital_activity_tenant_api_payload_field`
- `reference_data_tenant_api_payload_field`

More information can be found in the Pulse Configuration Reference.

Add support for Reference Data Service multi tenancyâ

Feature HSBC-2171

We've added support to configure Reference Data Service multi tenancy during deployment.

This configuration can be done using the following Ansible variables:

- `reference_data_tenant_strategy`
- `inbound_payments_tenant_strategy`
- `transfers_tenant_strategy`
- `digital_activity_data_tenant_strategy`

Setting these to `REQUEST` will enable RDS to use an API field to retrieve the tenant, instead of the default `feedzai` tenant.

More information can be found in the Pulse Configuration Reference.

Case Manager version upgraded to 13.15.0â

Feature FTMT-17140 HSBC-1983

We've upgraded Case Manager to version 13.15.0. With this upgrade we enabled a feature that allows queues to be exported and imported, improving the process of migrating queues between environments.

Access to this feature is controlled by adding the following two permissions to a role:

- `cm:resource:settings:import-export:*`
- `cm:action:settings:import-export:*`

These permissions have also been included in the Permissions Guide.

Add functionality to remove null byte from the Inbound Payments payloadâ

Feature HSBC-2167

We have made the transformation that strips null bytes from string fields in payloads work for inbound payments (it previously already existed for outbound payments).

Add support for Customer multi-tenancy in Case Managerâ

Feature HSBC-2172 HSBC-2177

We've added support for multi-tenanted customers in Case Manager. The Customer Search page now has a form field to specify the tenant to be searched. Leaving the field empty will continue to use the default tenant, keeping the previous behaviour. Additionally, all links to the customer page now include the tenant in the URL, allowing multi-tenanted deployments to correctly link to customers.

Add missing Reference Data fields to API documentationâ

Bug Fix HSBC-2188

Weâ□□ve updated the API documentation since it was outdated and missing several fields, for example `product_type` and `account_closed_date`.

Add `external_status_reason` to Feedback events from Case Managerâ□□

Feature FTMT-16108 HSBC-1909

We've added functionality to populate the `external_status_reason` field in Feedback workflow events when they are triggered by status changes in Case Manager. The value will be the status reason selected when marking the event as Fraud or Not Fraud.

Inbound Payments Schema Changesâ□□

Feature FTMT-17618 HSBC-2121

We have made the requested schema changes to Inbound Payments and Reference Data.

Please find below the newly added fields

Inbound Payments

Field Name	Datatype
rfb_flag	Boolean

Reference Data - Customer

Field Name	Datatype
customer_business_type_last_update_date	Long
customer_reported_income_last_update_date	Long
customer_annual_turnover_last_update_date	Long
customer_portfolio_segment_last_update_date	Long
employer_name_last_update_date	Long
customer_occupation_status_last_update_at	Long

Support Alert ID Transformation for Inbound Payments Alert Storage Serviceâ□□

Feature HSBC-2170

We have added Alert ID Transformation to the inbound payments app/workflow so that Alert ID is available in CM for inbound payments.

On this page

What's Changed

Release `2.5.2` â

Add decision outcome columns to Digital Activity Case Manager tablesâ

Feature FTMT-20451 HSBC-2425

Add `DECISION_ID` and `EVENT_DECISION_ID` columns to the `AM_EVENT_DIGITAL_ACTIVITY` table in order to resolve an issue appearing in search following the Mobile Outbound extension support for Digital Activity added in version `2.5.1`.

Add deployment guides for Pulse and Case Managerâ

Docs HSBC-2315

We've added a new `How To` section with a set of deployment guides for Pulse and Case Manager.

These guides cover the following subjects:

- How to deploy new applications/channels
- Use application/channel specific databases

Please find the Pulse guides in Pulse How To and the Case Manager guides in Case Manager How To.

Release `2.5.1` â

Add Digital Activity support for Case Manager Mobile Outbound extensionâ

Feature FTMT-21346 HSBC-2425

We've added the `decision` state machine to Digital Activity. Enabling the execution of the Mobile Outbound extension to Digital Activity events.

warning

The feature is available in version 2.5 and will be available again in version >= 2.7.X. It will **not** be included in minor version 2.6.

Release `2.5.0` â

Add Documentation for Multi tenancyâ

Feature HSBC-2086

Find a descriptive multi tenancy guide in Multi-Tenancy Concepts.

Add Fields to Payment Fulfillment Extensionâ

Feature FTMT-18681 HSBC-2268

We've added the below fields in the Payment Fulfillment API for the Outbound API call from Case Manager :

Field Name	Channel
user_id	Transfers
customer_id_global	Transfers
sender_initiated_at_local	Transfers
lifecycle_id_alt	Transfers
customer_id_number	Transfers
payment_sub_type	Transfers
receiver_transaction_bank_account_number	Transfers
sender_transaction_ref_nr_alt	Transfers
sender_transaction_original_amount	Transfers
sender_transaction_original_currency	Transfers
operational_status	Transfers
oob_notes	Transfers
treatment_strategy	Transfers
self_serve_strategy	Transfers

Please refer to the AFIS Extensions reference for full details.

Case Manager version upgraded to 13.16.2â

Feature FTMT-14311 FTMT-17140 HSBC-1997

We've upgraded Case Manager to version 13.16.2.

With this upgrade we enabled a feature that allows SLAs and Automation rules to be exported and imported, improving the process of migrating them between environments. This functionality uses the same permissions of the queues import/ export feature enabled on a previous version.

We have also enabled a feature that provides the count of alerts in each queue, broken down by state machine state. This can be found in Queues settings page in Case Manager. Note that a time filter must be selected for the counts to appear.

Remove Data Migration Serviceâ

Feature HSBC-2210

We've removed all existing data migration service properties from the Pulse configuration, as they weren't being used.

Add PosNeg lists for Inbound Payments Appâ

Feature HSBC-2173 HSBC-2261

We've added 8 new lists to Pulse and Case Manager for Inbound Payments App, namely:

- Payments_Hot Beneficiary Sort Code and Account Numbers
- Payments_Low Risk Account Numbers
- Payments_Vulnerable Customer List
- Payments_Hot Customer ID
- Payments_Ex Jade Customers
- Payments_Crypto Beneficiary Name
- Payments_Crypto Payment Reference
- Payments_Low Risk Sort Code

Change Case Manager Amount mappingsâ

Feature FTMT-18233 HSBC-2256

We've changed the `amount` column mapping, in the remaining pages and channels, to the new field `amount_dbl` based on `sender_transaction_amount_dbl`. This change affects the following pages:

- Alerts page
- Search Page
- Event Details transaction history widget
- Customer Details transaction history widget

Add Fields to Mobile Outbound extensionsâ

Feature FTMT-18136 HSBC-2238

We've added the below 4 fields to be included in the 2Way SMS / 1 Way SMS for Outbound API call from Case Manager :

Field Name	Channel
payment_message_source	Transfers
payment_sub_type	Transfers
thailand_fps_ind	Transfers
customer_alias	Transfers

Please refer to the AFIS Extensions reference for full details.

Add customer ID and customer ID number to Case Manager external notificationsâ

Feature FTMT-18381 HSBC-2244

We've added the `customer_id` and `customer_id_number` fields to Case Manager external notifications payload.

Please find below the changes included in external notifications payload.

Field Name	Channel
customer_id	Inbound Payments
customer_id_number	Inbound Payments
customer_id_number	Transfers

Fix ETL hostname parameterâ

Bug Fix FTMT-16570 HSBC-2003

We have fixed an issue where ETL had problem connecting to database. As part of resolution we have changed the hostname for ETL from `database_pulse_eventstorage_host` to `database_pulse_eventstorage_hostname`.

Enable Spark History Server integrationâ

Feature FTMT-15172 HSBC-1814

We've enabled the Spark History Server integration with Pulse. Configuration uses the following two new Ansible variables:

- `pulse_spark_eventlog_enabled`
- `pulse_spark_eventlog_dir`

By default, the Spark logs are disabled (except for DSE).

More information can be found in the Pulse Configuration Reference.

ETL extraction for HDL fails if the lifecycle_id is not a valid UUIDâ

Bug Fix HSBC-2197

We have updated the ETL configuration for historical data loading so that it no longer fails when the lifecycle_id is an invalid UUID. Please find the updated configuration reference in ETL Configurations.

Add support for Reference Data Service multi tenancy with the tenant as a query parameterâ

Feature HSBC-2220

We've added support to collect the tenant as a query parameter for Reference Data Service multi tenancy during deployment.

This configuration can be done using the following Ansible variables:

- `reference_data_tenant_provider_parameter_key` - Configures the parameter key for the query parameter strategy (the default value is tenant)
- `reference_data_tenant_provider_fallback` - Configures the tenant value if no tenant in the request is found (the default value is feedzai)

On this page

What's Changed

Release `2.6.0` â

Workflow Monitoring Service added to Inbound Payments Workflowâ

Feature HSBC-2365

We've added the Workflow Monitoring Service to the Inbound Payments Workflow.

This service will populate the `workflow.events` metric with the events that are processed by that workflow and can be used to create monitoring dashboards.

Upgrade JupyterLab version to 25.0.28â

Feature HSBC-2279

We have upgraded JupyterLab to version 25.0.28.

The new docker image will have to be loaded in existing environments. Please find the newest version in the [Installation Guide].

Add Voice Biometrics (Nuance) field to Pulseâ

Feature FTMT-14249 FTMT-17979 HSBC-2207

We've added a new complex field `voice_biometrics_nuance` to Outbound and Digital Activity APIs and to the `Payments` and `Payments + Outcomes` schemas.

Please find below the field definition:

Field Name	Datatype
is_in_watchlist	Boolean
session_auth_decision	String
session_risk_level	Integer
text_decision	String
biometric_decision	String
pbf_score	Double
pbf_decision	String
watchlist_score	Double
watchlist_decision	String
fraudster_score	Double
text_score	Double
highest_fraud_score	Double
highest_fd_norm_score	Double
identified_voiceprint_id	String
session_auth_score	Double

Field Name	Datatype
session_veredict	Integer
expected_fa_rate	Double
session_risk_score	Double
synth_score	Double
pbc_score	Double

Add customer_ssn field to Pulseâ

Feature FTMT-19691 HSBC-2347

We have added the `customer_ssn` field to the `Payments` and `Payments + Outcomes` schemas.

Please find below the field definition:

Field Name	Datatype
customer_ssn	String

Add premier marker to Pulse and Case Managerâ

Feature FTMT-18234 HSBC-2293

We've added the `premier_marker` field to the `Payments` and `Payments + Outcomes` schemas.

The field was also added to Case Manager and is available to be used in Automation Rules and SLAs.

Field Name	Datatype
premier_marker	String

Change Inbound Transfer Initiation mandatory fieldsâ

Bug Fix FTMT-17618 HSBC-2121 HSBC-2319

The following fields were made optional in the Inbound Transfer Initiation endpoint:

- device_id
- receiver_bank_bic
- receiver_bank_country_code
- receiver_name
- receiver_transaction_bank_account_number

Add account_pw_change_flag field to Pulseâ

Feature FTMT-19381 HSBC-2305

We have added a new field `account_pw_change_flag` to Outbound and Digital Activity APIs and to the `Payments` and `Payments + Outcomes` schemas. Please find below the field definition:

Field Name	Datatype
account_pw_change_flag	Boolean

Add new schema field to Inbound Paymentsâ

Feature FTMT-18883 HSBC-2284

We have added a new field `customer_reported_income_dbl` to the Customer Reference Data API and to the `Inbound Payments` and `Inbound Payments + Outcomes` schemas.

Please find below the field definition:

Field Name	Datatype
customer_reported_income_dbl	Double

Update Mobile Outbound Data Schema Documentationâ

Feature HSBC-2304

We've updated the documentation for the AFIS Extensions reference.

Add partitioning to dedicated channel databasesâ

Feature HSBC-2302

We've added the execution of the Case Manager database changelog patches to the dedicated channel databases, so that the partitioning and index trigger functions that already exist for the single common database will also be automatically added for the channel-specific databases.

The following variables are used to setup the properties for each channel (each of the following variables needs a specific channel defined in its suffix _<channel>, which can be one of `transfers`, `digital_activity`, `inbound_payments`, and `cards`).

- `database_casemanager_jdbc_<channel>`
- `database_casemanager_username_<channel>`
- `database_casemanager_schema_<channel>`

Add Biocatch Push field to Pulseâ

Feature FTMT-19304 HSBC-2294

We have added a new complex field `biopush` to Outbound and Digital Activity APIs and to the `Payments` and `Payments + Outcomes` schemas.

Please find below the field definition:

Field Name	Datatype
bio_push_score	Double
bio_push_id	Double
bio_push_name	String
bio_push_platform_name	String
bio_push_context	String
bio_push_mule_account_risk_level	String

We've also changed the `Push Score` mapping in Case Manager's Digital Activity Event Details to the new field.

On this page

What's Changed

Release `2.7.0` â

Add property to configure JupyterLab container memoryâ

Feature HSBC-2634

We've added a new property to control the maximum memory used by JupyterLab Docker containers.

By default it will limit memory usage to 8GB per container but that value can be changed by using the following Ansible property:

- `pulse_jupyterlab_container_max_memory`

More information can be found in Pulse DSF Configuration Reference.

Add missing permissions to Pulseâ

Bug Fix HSBC-2644

We have added the following permissions to Pulse:

- `application:managedlistitems:revision:read`
- `application:managedlistitems:revision:update`
- `application:managedlistitems:revision:create`
- `cm:section:event:read:*`
- `cm:section:event:create:*`
- `cm:section:event:delete:*`
- `cm:section:event:update:*`
- `admin:configuration:read`
- `admin:configuration:update`
- `admin:configuration:delete`

For more details on the existing permissions please refer to the Permissions Guide page.

Addition of new fieldsâ

Feature FTMT-22488 HSBC-2492

We have added the below fields to the Inbound Payments API:

Field Name	Datatype
fps_customer_id_type	Integer
customer_id_type	Long
receiver_account_ownership_details	String

The Customer Reference Data API was also updated with the following fields:

Field Name	Datatype
customer_id_type	String
customer_occupation_start_date	String

Field Name	Datatype
count_months_beneficiary_received_funds	Integer
avg_total_daily_transfer_out	Double
avg_monthly_transfer_in	Double

The Outbound Payments API was also updated with the following fields:

Field Name	Datatype
customer_id_type	String
new_receiver_flag	String
receiver_creation_date_customer	String

The Digital Activity API was also updated with the following fields:

Field Name	Datatype
customer_id_type	String

The following schemas were updated accordingly:

- `Inbound Payments`
- `Inbound Payments + Outcomes`
- `Payments`
- `Payments + Outcomes`
- `Cards

Add new fields to the Mobile Outbound extensionsâ

Feature FTMT-21172 HSBC-2417

We've added the below two fields to Case Manager `2 Way/1 Way SMS` and `Mobile Payment Push` extensions:

Field Name	Channel
user_id	Transfers
customer_id_global	Transfers

Please refer to the AFIS Extensions reference for more details.

Enable Cards in Case Managerâ

Feature HSBC-2614

We've enabled the Cards channel in Case Manager.

Although the `Cards` tab was already previously visible in search and alert listings, it would not actually have shown any cards events.

In order to enable this functionality, an entry for the cards application in the Case Manager `apps_enabled` must be added.

For more details on how to enable channels in Case Manager please refer to the Deploy a new channel page.

Enable control of Cleanup Patches with Context-Specific Executionâ

Feature FTMT-9287 HSBC-2391

To provide greater flexibility and control over which cleanup patches are executed, we have introduced a mechanism where cleanup patches are managed through distinct contexts rather than a single one.

The Ansible variable `cleanup_pulse_commands_contexts` defines the patches that will be executed. The available values are:

- `cleanup-schemas`
- `cleanup-rule-snapshots`
- `cleanup-plan-executions`
- `cleanup-workflow`

We can specify which contexts to run by including the desired values in the above Ansible variable.

Please refer to the [cleanup configuration]() page for more configuration details.

Add Case Manager alert queue cleanup documentationâ

Docs HSBC-2607

We've added a new page with the cleanup process to remove any pending alerts in Case Manager. You can find the page in [Case Manager alert queue cleanup]() which also includes the required Python script and its configuration.

Add account and customer reference data enrichment in the Cards applicationâ

Feature HSBC-2590

We've added account and customer reference data enrichment in the Cards application.

By default, enrichment for account will use the `account_id` field and for customer the `customer_id` field. These can be changed during deployment by setting the following Ansible variables:

- `cards_account_enrichment_key`
- `cards_customer_enrichment_key`

More information can be found in [Pulse Configuration Reference]().

Remove `tagname` property from Liquibase configuration filesâ

Rework HSBC-2582

We have removed the `tagName` parameter from the Liquibase configuration files since it was not being used.

Add functionality to remove null byte from the Cards payloadâ

Feature HSBC-1969

We have made the transformation that strips null bytes from string fields in payloads for card events (it previously already existed for inbound and outbound payments).

Add Pulse recovery procedure documentationâ

Docs HSBC-1996

We've added new page about Pulse Recovery Procedure and is now available in Feedzai documentation here [Pulse Recovery Procedure]()

This guide was previously shared in April 2024, but was not included in the Feedzai documentation. For completeness, we have included this guide in the documentation.

New fields added to Inbound Paymentsâ

Feature HSBC-2367

We've added new fields to the Inbound Payments Workflow Schemas, please find below the added fields:

Field Name	Datatype
customer_number_of_cards	Integer

Field Name	Datatype
customer_total_credit_limit	Double
is_domestic	Boolean
receiver_account_avg_balance	Double
receiver_account_balance_dbl	Double
receiver_account_credit_limit	Double
receiver_account_last_activity	Long
receiver_acc_last_payment_amount	Double
receiver_acc_last_payment_date	Long
receiver_account_number_of_cards	Integer
receiver_account_open_date	Long
receiver_account_status	String
receiver_bank_branch_id	String
receiver_registration_date	Long
reversal_flag	String
sca_risk_flag	Boolean
sca_risk_score	Double
sender_city	String
sender_country_code	String
sender_street_addr	String

Add Logging API Authentication Documentationâ

Docs HSBC-2591

We've added documentation on how to authenticate in Pulse via API to complement the logging levels changes API flow.

Case Manager version upgraded to 13.17.4â

Feature FTMT-21578 FTMT-21582 FTMT-22456 HSBC-2469 HSBC-2456 HSBC-2416

We've upgraded Case Manager to version 13.17.4.

This version addresses the following issues:

- When deleting notes and files from cases, the Activity log would always incorrectly show the user who created them as the owner of the action
- Support for expandable and collapsible Activity Log

It also adds support for features that:

- Allow entity page filtered by tenant
- Allow duplicated schema fields in Case Manager

Add Alerts section to the Monitoring Guideâ

Docs HSBC-2474

We've created a new section under the Monitoring Guide for Pulse and Case Manager pages with a suggestion of alerts.

Note that these are solemnly suggestions that might require new dashboards or alert adjustments based on HSBC specific environments.

Fix the account mapping key for inbound and outbound endpointsâ

Bug Fix HSBC-2486

We've fixed the account enricher keys for both Inbound and Outbound endpoints. The `sender_account_*` fields will be enriched with Account Reference Data based on the `sender_transaction_bank_account_number` and that `receiver_account_*` will use `receiver_transaction_bank_account_number`. Ensure that the `*_bank_account_number` fields contain the same IDs used when sending the account reference data for successful enrichment.

Fix cross channel metrics configuration for secondary channelsâ

Bug Fix HSBC-2475

We have fixed the issue with cross-channel metrics where the required properties were not enabled for the secondary channels. The variable `cross_channel_metrics_enabled` now enables the cross-channel metrics for all apps in the `apps_enabled` variable.

Case Manager hides Reasons based on the channelâ

Feature FTMT-22215 HSBC-2503
Also In: 2.4.6

We've implemented a tactical solution where Case Manager now hides Outbound Payments reasons when marking *Fraud* / *Not Fraud* for Inbound Payments, and vice-versa.

This is a temporary solution where the Case Manager UI will hide those values when showing the dialog to the analysts, despite it still being available in the backend. The solution requires all Inbound Payments reasons' to have `Inbound` as their prefix, so they can be detected and properly hidden at the correct moment.

Extend Cards applicationâ

Feature HSBC-1978 HSBC-2478 HSBC-2499 HSBC-2549

We've extended the Cards application by adding the following services to the Cards workflow:

- HTTP Input Service (Cards API)
- Monitoring Service
- List Management Service
- Historic Events Service (Historic Events Concepts Page)
- Alert Storage Service

Refer to the API Reference section for a specification of the available endpoints.

Fix hyperlinks to the Customer Details page in single tenanted Case Manager deploymentsâ

Bug Fix FTMT-22659 HSBC-2509
Also In: 2.4.6

We have fixed an issue where hyperlinks for the Customer Details page (e.g. in the Event Details page) were using the event tenant instead of the Customer's default tenant in single-tenanted Case Manager deployments.

Remove Pulse Validation Toolâ

Rework HSBC-2483

A security vulnerability was identified in the `pulse-environment-validation-tool`.

Since this is an internal tool with limited usage and its not intended to be released, we have decided to remove it from the project packages and associated documentation to mitigate any potential risks.

Add extra logging to application symbolic link creationâ

Feature HSBC-2415

We've added extra logging to the deployment task that creates the symbolic links for applications.

Pulse 455 Ansible task will now output a set of information that may be useful to identify potential issues during the creation of Pulse immutable images.

Please find below a snippet of the output:

```
TASK [[455.1] Log task variables] *******************************************
Wednesday 16 October 2024  09:21:55 +0000 (0:00:00.060)       0:01:17.043 *****
ok: [pulse] => (item=uk_outbound_payments) => {}

MSG:

- pulse_unix_user: feedzai
- pulse_unix_group: feedzai
- pulse_home_path: /opt/feedzai/product-pulse-25.0.52
- apps_enabled: ['uk_outbound_payments', 'uk_inbound_payments', 'uk_cards', 'uk_reference_data']

...

TASK [pulse : [455.4.1] Get directory stats] **********************************
Wednesday 16 October 2024  09:21:56 +0000 (0:00:00.331)       0:01:17.703 *****
ok: [pulse] => (item=uk_outbound_payments, app_ext_path=/opt/feedzai/product-pulse-25.0.52/ext/
tfrb)
ok: [pulse] => (item=uk_inbound_payments, app_ext_path=/opt/feedzai/product-pulse-25.0.52/ext/
uk_inbound_payments)
ok: [pulse] => (item=uk_cards, app_ext_path=/opt/feedzai/product-pulse-25.0.52/ext/cards)
ok: [pulse] => (item=uk_reference_data, app_ext_path=/opt/feedzai/product-pulse-25.0.52/ext/
uk_reference_data)

TASK [pulse : [455.4.2] Debug application symbolic links] **********************
Wednesday 16 October 2024  09:21:56 +0000 (0:00:00.252)       0:01:17.956 *****
ok: [pulse] => (item=/opt/feedzai/product-pulse-25.0.52/ext/tfrb) => {}

MSG:

- Is symbolic link: True
- Is directory: False
- Link source: /opt/feedzai/product-pulse-25.0.52/ext/tfrb
- Link target: /opt/feedzai/product-pulse-25.0.52/ext/base_outbound_payments

...
```

Copy

Upgrade liquibase-core version to 4.30.0â

Internal FTMT-14099 HSBC-2482 HSBC-2484

We've upgraded the bundled version of liquibase from 4.19.0 to 4.30.0.

This internal upgrade of the liquibase core library also upgrades its transient dependencies, some of which might have had open CVEs, such as SnakeYAML.

Have Garbage Collection logs per applicationâ

Bug Fix HSBC-2095

We've fixed a bug where the various application Garbage Collection (GC) logs were saved in the same file. From now on each application will have its own GC log identified by the application id, e.g. `app-tfrb_GC_pid2155_2024-08-20_02-32-41.log.0.current`.

Note that GC logs are rotated over the `log` extension, meaning they will either end with:

- A number indicating the rotation

- The `current` extension, meaning it's the current GC log

This is specially important if log export to GCP Cloud Logging is based on the extension.

Add customer_occupation_status to Inbound Workflow Schemaâ

Bug Fix FTMT-22435 HSBC-2476

We've fixed an issue where `customer_occupation_status` was not being enriched by adding it to the Inbound Workflow Schema.

Remove deprecated Spark propertiesâ

Feature HSBC-2446

We've changed the below deprecated configuration properties in favor of the corresponding new ones:

- `spark.forward.spark.yarn.executor.memoryOverhead` to `spark.executor.memoryOverhead`
- Spark master `yarn-client` to `master=yarn` and `deployMode=client`

Fix Case Manager read only database properties set with non read-only Ansible variablesâ

Bug Fix HSBC-2382

We've fixed an issue where some Case Manager read only database properties where being set with non read-only Ansible variables by creating the following new properties and adjusting the respective configuration templates:

- database_casemanager_ro_reconnect_on_lost
- database_casemanager_ro_max_number_retries
- database_casemanager_ro_retry_interval
- database_casemanager_ro_pool_lazy
- database_casemanager_ro_schemapolicy

To configure per channel databases a corresponding channel variable was created for each of the above, which can be identified by the channel suffix (e.g. `database_casemanager_ro_reconnect_on_lost_transfers`).

Please find more information in the CaseManager Configuration Reference.

Add `Also In` badge to documentationâ

Docs HSBC-2455

We've added a new badge to the Release Notes pages indicating in which versions a task was also included.

Add Topology-Aware Resource Scheduling (TARS) supportâ

Feature HSBC-2340
Also In: 2.4.5 2.5.4

We've added Topology-Aware Resource Scheduling, TARS for short, support to the deployment pipelines.

Please find a deployment guide in Topology-Aware Resource Scheduling and the complete configuration reference in Pulse Configuration Recommendations.

Add missing UDF importsâ

Bug Fix FTMT-21005 HSBC-2454

We've fixed an issue where the UDF imports listed below were missing. This will be managed by ansible variable `import_solutions_udfs`

- pulse.app.tfrb.services.modules.fraudmodel.import.Revelock
- pulse.app.tfrb.services.modules.fraudrules.import.Revelock
- pulse.app.tfrb.services.modules.datascience.context.import.Revelock

- pulse.app.tfrb.services.modules.datascience.context.rules.import.Revelock

Fix Null Fields in Notifications Sent to CCMâ

Bug Fix FTMT-21442 HSBC-2447
Also In: 2.4.5 2.5.4

We've fixed a bug where certain fields in the notification sent to CCM, such as `sender_initiated_at`, were being sent as `null`

Remove the duplicated mapping for `sender_transaction_original_amount` in Case Managerâ

Bug Fix FTMT-21582 HSBC-2441
Also In: 2.4.5 2.5.4

We've removed the duplicated mapping for `sender_transaction_original_amount` in Case Manager for the Outbound Payments schema.

This was causing multiple issues, notably the inability to access old alerts in Case Manager.

We also added `sender_transaction_amount_dbl` to the Case Manager schema as this was needed to avoid API changes in the outbound extension and ensure the correct value is sent in the extension payload.

Pulse Version Upgrade to 25.0.73â

Feature FTMT-14099 HSBC-2484 HSBC-2414

We've upgraded Pulse version to 25.0.73.

This version addresses the following issues:

- Upgraded SnakeYAML to a later version to address CVE-2022-1471
- Added support for Postgres 14

Enable logger configuration that tracks metrics creation in Feature Plansâ

Feature FTMT-20290 HSBC-2397

We've enabled the configuration of the logger that tracks metrics creation in Feature Plans.

The logger level can be configured by setting the `feature_plan_log_level` Ansible variable.

More information can be found in the Pulse Configuration Reference.

Add missing Case Manager translationâ

Bug Fix HSBC-2437

We've added the missing Case Manager translation string for below:

- quick-preview.eventReadOnly.inbound-payments

Pulse Application cleanup processâ

Feature FTMT-9287 HSBC-2325

We've created a new process to clean up the following Pulse Application resources:

- Plan executions
- Schemas
- Rule Snapshots
- Workflows

Please find the detailed instructions and configuration reference in Pulse Application Cleanup.

Add Case Manager enrichers to Inbound Paymentsâ

Feature HSBC-2373

We've added the following enrichers for the Inbound Payments channel in Case Manager.

Field	Enricher	Target Field
event.fields.customer_country_code	country-iso-3166-1-numeric	customer_country_code
event.fields.sender_transaction_currency	currency-alpha-3-code	sender_transaction_currency
event.fields.receiver_country_code	country-iso-3166-1-numeric	receiver_country_code
event.fields.receiver_transaction_currency	currency-alpha-3-code	receiver_transaction_currency
event.fields.sender_account_currency	currency-alpha-3-code	sender_account_currency
event.fields.receiver_account_currency	currency-alpha-3-code	receiver_account_currency
event.fields.payment_revision_code	hsbc_so_mandate_status	payment_revision_code

Add Historic Event Service to Inbound Payments Workflowâ

Feature HSBC-2366

We've added the Historic Event Service to the Inbound Payments Workflow.

The service uses the same logic as Outbound Payments (Transfers and Digital Activity), described in the Historic Events Concepts Page.

Add Customer ID Number to Digital Activity external notifications from Case Managerâ

Feature HSBC-2383

We've added the `customer_id_number` field to the external notifications send by Case Manager for Digital Activity events. This is a feature parity change as the field was already being sent in external notifications for the Outbound Payments and Inbound Payments channels.

Add support for distinct Pulse event storage databasesâ

Feature HSBC-2338

We've added support for Inbound Payments and Cards to have a different Event Storage database from the Outbound Payments one.

This is achieved by having a new set of Ansible variables that segregate Event Storage configurations per portfolio:

- `database_pulse_outbound_payments_eventstorage_*` variables configure the Outbound Payments Event Storage
- `database_pulse_inbound_payments_eventstorage_*` variables configure the Inbound Payments Event Storage
- `database_pulse_cards_eventstorage_*` variables configure the Cards Event Storage

Their default values are being mapped to the corresponding `database_pulse_eventstorage_*` variable, meaning that by default both will continue to use the same event storage database that was previously configured. For a comprehensive list of the new database variables and their default values refer to the Pulse Configuration Reference.

Add new fields to Inbound Paymentsâ

Feature FTMT-19912 HSBC-2346

We have added the below fields to the Inbound Payments API:

Field Name	Datatype
customer_number_of_cards	Integer

Field Name	Datatype
overnight_acc_ledger_bal_update_date	Long
transaction_type	String

The Customer Reference Data API was also updated with the following fields:

Field Name	Datatype
customer_business_segment	String
customer_business_segment_last_update_at	Long

The `Inbound Payments` and `Inbound Payments + Outcomes` schemas were updated with all the above fields.

Add configurable Watchdog Properties for Enhanced Recovery Managementâ

Feature HSBC-2554

We've added support for configuring watchdog properties through the following Ansible variables:

- `watchdog_check_period_millis`
- `watchdog_latency_actuation_cycles`

By increasing these values, the system is better equipped to handle transient issues, such as temporary queue buildup in RabbitMQ, by allowing additional time before initiating the recovery process. This enhancement improves system stability and reduces unnecessary recovery actions in high-load scenarios. For more details about the new variables and their default values, refer to Pulse Configuration Recommendations.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-2.x/Release-2.8

On this page

What's Changed

Release `2.8.0` â

Make ETL hardcoded values configurableâ

Feature FTMT-26243 HSBC-3052

We've extended the ETL configuration to allow `includeTriggeredActions` to be set to `false` for historical data loading (leaving it as `true` by default for data extraction) and also allowing it to be set via an Ansible variable. Also, a few variables that were previously hardcoded are now configurable.

Please find below the variable definition, including their default and default for HDL:

Variable	Datatype	Default Value	Default for HDL
include_triggered_actions	Boolean	true	false
project_all_fields	Boolean	true	true
include_feedbacks	Boolean	false	false
include_workflow_state	Boolean	false	false
include_event_diffs	Boolean	false	false

This is a breaking change for the `includeTriggeredActions` variable in the CSV schema for historical data loading. Moving forward, users no longer need to manually add action columns, as they are now handled automatically.

Please refer to [ETL Configuration Reference] for more information.

Add Ansible variable to control Pulse and Case Manager loggersâ

Feature HSBC-3042
Also In: 2.4.9 2.5.7 2.7.3

We've added Ansible variables that allow log levels to be configured for Pulse and Case Manager.

- `pulse_loggers` controls loggers for Pulse
- `cm_loggers` controls loggers for Case Manager

Each variable can be configured as a dictionary of loggers as keys and log level as values, for example:

```
pulse_loggers:
    com.feedzai.hsbc: debug
    com.feedzai.pulse: trace
```

Copy

More information can be found in the configuration reference for Pulse and Case Manager.

Add documentation on Pulse applications export and importâ

Docs HSBC-3030

We've added a new documentation page to describe the process of exporting and importing Pulse applications.

Please refer to Export and Import Pulse applications for more information.

Data Transfer Pipeline (DTP)â

Feature

The Data Transfer Pipeline (DTP) is a new tool intended to replace the ETL.

The DTP goal is to make event data processed in Pulse RT available to the Data Science environment.

Compared to the ETL, the DTP provides better support for workflow schema changes by using a workflow service that has access to the last published schema.

Events and all their fields that reach the end of the workflow are streamed to an upstream cloud native service (GCP Pub/Sub) as soon as they are processed in Pulse.

For detailed information about DTP, its functionality, and usage, refer to the provided documentation here.

Add Outbound Event Storage indices for `event_occurred_at` and `event_received_at` â

Feature FTMT-24568 HSBC-3014

We've added the following indices to the Outbound Event Storage table:

- `idx_event_occurred_at` , a `btree` index on column `event_occurred_at`
- `idx_event_received_at` , a `btree` index on column `event_received_at`

These indices are not required nor used by the Feedzai product and are intended to boost performance of HSBC internal processes.

warning

These indices will be automatically applied to the main (non-partitioned) table, the partitioning function, and all *new* partitions created after this update. Existing partition tables created prior to this change will *not* be affected and will require manual indexing if needed.

Add new field to the Payment Fulfillment extensionâ

Feature FTMT-25899 HSBC-3012

The following field was added to Case Manager Payment Fulfillment extension:

Field Name	Datatype	Channels	Description
channel_name	string	transfers	Channel Name

Please refer to the AFIS Extensions reference for full details.

Add Case Manager assign and lock alert condition for cardsâ

Feature HSBC-3013

We've implemented the `assign and lock` feature for Cards in Case Manager.

The working logic is identical to the one present in Outbound Payments.

From now on, only Alerted events will be displayed inside `Linked Alerts` section with "Assignee".

Extend Case Manager to also include the Cards channel when hiding reasonsâ

Feature HSBC-2977

We've extended the existing tactical solution where Case Manager hides reasons for channels other than the one of the current event, allowing it to also hide Cards reasons along with Outbound Payments and Inbound Payments when marking *Fraud* / *Not Fraud* decisions.

The solution requires all Cards reasons' to have `Cards` as their prefix, so they can be detected and properly hidden at the correct moment.

Remove the deployment dependency on TFB Pulse and Case Manager configurationsâ

Internal HSBC-2770

We've removed the dependency on TFB Pulse and Case Manager configuration assemblies. As a result, we've removed those entries from the Installation Guide list of artifacts. This means that HSBC is no longer required to download those artifacts when installing new versions.

This is not a breaking change as the presence of those artifacts does not hinder the installation. In that case they would simply be ignored. It also does not does not affect any available functionality or configuration in use.

Add configuration for RDS `maxParallelCalls` propertyâ

Feature HSBC-2968

We've added a configuration to control the maximum number of parallel calls that the RDS client used by Pulse to connect to RDS can make. It will default to the previous value that Pulse was using (4). This can be configured using the following Ansible variable (and can be configured per-app by prefixing the variable with `<app_slug>_`:

Variable	Datatype	Default Value
rds_max_parallel_calls	Integer	4

Please refer to Pulse Configuration Reference for more information.

Update Data Extraction Pipelines Configuration Documentationâ

Docs HSBC-2971

We've update the documentation of some Ansible variables described in the Data Extraction Pipelines Configuration page. This is a documentation-only change and does not affect the variables used by the installation pipelines.

Previously the documentation was referencing internal variables when it should have been referencing our external variables that have a nomenclature more similar to other components. Please find below a summary of the changes. Note that some variables that were previously documented as colour-specific now use the same variable for both colours.

Previous Ansible Variable	New Ansible Variable
`rabbitmq_blue.host`	`rabbitmq_brokers`
`rabbitmq_green.host`	`rabbitmq_brokers`
`rabbitmq_blue.vhost`	`rabbitmq_pulse_vhost_blue`
`rabbitmq_green.vhost`	`rabbitmq_pulse_vhost_green`
`rabbitmq_blue.user`	`rabbitmq_insights_logstash_username`
`rabbitmq_green.user`	`rabbitmq_insights_logstash_username`
`rabbitmq_blue.password`	`rabbitmq_insights_logstash_password`
`rabbitmq_green.password`	`rabbitmq_insights_logstash_password`
`pulse.host`	`pulse_elb`
`pulse.app_id`	`pulse_app_id`
`pulse.workflow_id`	`pulse_workflow_id`
`pulse.user`	`pulse_insights_logstash_username`
`pulse.password`	`pulse_insights_logstash_password`

Enable debug-level logs for Pulse workflow validationâ

Feature HSBC-2982

We have enabled extra `DEBUG` logs that can be used to track how long workflow validation is taking. Among others, the new messages include:

- *"Starting remote inbound_payments workflow validation."* which can be used to track the start of the workflow validation
- *"Finished inbound_payments workflow validation."* which can be used to track the end of the validation

The time between both log messages will determine how long validation took. These log messages are present in the Pulse Server log file (usually `pulse.log`) and not in the application logs.

The logging level for workflow validation can be controlled by changing the `workflow_validation_log_level` ansible variable. It will default to `DEBUG`, whereas previously when it was not configurable, it was set (hardcoded) to `INFO`.

More information can be found in the [Pulse Configuration Reference](#).

Private Banking Delegateâ

Feature WPB-350942 WPB-365306 HSBC-2735 HSBC-2736 HSBC-2737 HSBC-2738 HSBC-2739 HSBC-2740 HSBC-2741 H

We've added the necessary features to enable the PB Delegate use case.

Please find below a list of changes.

Pulse

New Fieldsâ

The following fields were added to Pulse API:

Field Name	Datatype	Endpoints
delegate_id	String	Transfers & Digital Activity
delegate_initiated	Boolean	Transfers & Digital Activity
approver_id	String	Transfers & Digital Activity
transaction_id	String	Transfers & Digital Activity
account_holder_customer_id	String	Transfers & Digital Activity
delegate_of	List of Objects	Customer Reference Data
delegates	List of Objects	Customer Reference Data

Each entry of the above list of objects has the following fields:

Field Name	Datatype
customer_id	String
date_added	Long

The following schemas were updated accordingly:

- `Payments`
- `Payments + Outcomes`

Delegate Watchlist for Outbound Paymentsâ

We have created the `Delegate Watchlist` list in the Outbound Payments application to support the Personal Banking Delegate use case. This list can be used, for instance, to store delegate ID numbers of customers that were flagged after risky transactions.

See Configure List Management Service for information on how to configure the service.

The list was also added to Case Manager to allow an analyst to update it when, for example, marking an event as Fraud or Not Fraud.

Case Manager

Schemaâ

The following fields were added to the Case Manager schema:

Field Name	Datatype	Channels	Description
delegate_id	String	Transfers/Digital Activity	The CIN from the delegate that is currently active on the PB customer account.
delegate_initiated	Boolean	Transfers	Whether the payment is being made by a delegate.
approver_id	String	Digital Activity	The CIN from the customer that approves a pending payment.
transaction_id	String	Transfers/Digital Activity	Identifier to correlate initiation and approval events of a transaction.

Event Details Pageâ

`By Delegate` is now present in both Transfers and Digital Activity events, mapping to the `delegate_id` field

Also in Case Manager Transaction History widget, in the Transfers and Digital Activity channels there is a new column and filter for `Delegate CIN` that is indeed mapping to the `delegate_id` field.

Customer Details Pageâ

There is a new widget called `Transaction History Delegate` in the Customer Details Page.

This widget filters all the Transfers and Digital Activity events for a given `delegate_id`.

As a result, both the `Transaction History` and `Transaction History Delegate` have a new column for `Delegate CIN`, that is mapping to the `delegate_id` field.

This new widget is controlled by a new ansible variable, `show_delegate_transaction_history_widget`, when set to True, that is the default value, the Transaction History Delegate widget and the Delegate CIN column will appear in Case Manager.

This new variable was added to Case Manager Configuration Recommendations.

We've added two new widgets to the Customer Details Page, `Delegates` and `Delegates of`:

- `Delegate Of` displays a list of PB customers for whom the customer is a delegate
- `Delegates` displays a list of customers who have delegated on the customer

Please refer to the PB Delegate Documentation for a more detailed documentation about the PB Delegate use case.

Add outcome fields to Cardsâ

Feature FTMT-24022 HSBC-2970

We've added the following outcomes to the Cards workflow to align with Outbound:

- `shadow_fraud`
- `scam`
- `shadow_scam`
- `self_serve_strategy`
- `decision_typology`

Add tenant field to Case Manager external notificationsâ

Feature FTMT-25381 HSBC-2907

We've added the `tenant` field to Case Manager external notifications payload for all channels.

New operational status statesâ

Feature FTMT-24511 HSBC-2918

We've added new states to the Operational Status state machine in Case Manager:

- Pending Mule Review
- Mule Review In Progress
- Customer Contact Awaited
- Pending Mule Investigation Closure
- Mule Investigation Completed

These new states can be used in Automation Rules to better support mule investigations in Case Manager.

Update Case Manager EVENT_ID field lengthâ

Bug Fix HSBC-1820

In order to keep the consistency across tables, and to prevent some features to stop working for events with big `lifecycle_id`, we've updated the field `event_id` length in the following tables:

- `AM_CASE_EVENT`
- `AM_EVENT`
- `AM_EVENT_ACCESS_GROUP`
- `AM_EVENT_CASE`
- `AM_EVENT_FILE`
- `AM_EVENT_IN_REVIEW`
- `AM_EVENT_NOTE`
- `AM_EVENT_QUEUE`
- `AM_EVENT_SLA`
- `AM_EVENT_STATUS`
- `AM_EVENT_STORAGE`
- `AM_EVENT_USER`
- `AM_EXTENSION_LOG`
- `AM_LIST_UPDATE`

The `transaction_id` field of the `AM_EVENT_COMMON` table was also updated since this column is computed from `EVENT_ID`.

Add new payment above limit field to Pulseâ

Feature WPB-356058 HSBC-2923

We've added a new field to the Outbound Payments API and Digital Activity API:

Field Name	Datatype
payment_above_limit	Boolean

The following schemas were updated accordingly:

- `Payments`
- `Payments + Outcomes`

Add new fields to Case Manager external notificationsâ

Feature FTMT-24847 HSBC-2908

We've added the below fields to the Case Manager external notifications payload.

Field Name	Channel
treatment_strategy	Inbound Payments
self_serve_strategy	Inbound Payments
pib_suspended	Inbound Payments

Add new generic fields to Pulseâ

Feature WPB-334507 HSBC-2598

We have added the below fields to the Digital Activity and Transfers API:

Field Name	Datatype
sender_transaction_extra_1	String
sender_transaction_extra_2	String
sender_transaction_extra_3	String
customer_additional_info_1	String
customer_additional_info_2	String
customer_additional_info_3	String
customer_additional_info_4	String

The Customer Reference Data API was also updated with the following fields:

Field Name	Datatype
customer_additional_info_1	String
customer_additional_info_2	String
customer_additional_info_3	String
customer_additional_info_4	String

The following schemas were updated accordingly:

- `Payments`
- `Payments + Outcomes`

All fields were also made available in Case Manager to be used in Automation Rules and SLAs.

Add configurable Real-time PKernel Partitioningâ

Feature HSBC-2641

We've added a new Pulse configuration property `services.rte.workflow.plan.partitionedRealtimeMetrics` to control real-time PKernel partitioning by application:

- When set to `true`, there will be multiple PKernel Engines (one per workflow partition) generating instances
- When `false` (the default), there will be only one PKernel Engine generating instances

With a partitioned PKernel setup each node will be responsible for only a part of the calculated real-time metrics, with other nodes still having replicas. Partitioning the PKernel in this way can lead to dramatic speed-ups in workflow recovery times, which typically accounts for the vast majority of the time it takes for a node to publish or to recover after it crashes.

Please note, as of this `2.8` release, Pulse does not support cross-channel metrics with a partitioned PKernel.

More information can be found in the Pulse Configuration Reference.

Add new mortgage fields to Pulseâ

Feature FTMT-24519 HSBC-2919

We've added two new mortgage fields to the Customer Reference Data API:

Field Name	Datatype
customer_mortgage_flag	Boolean
customer_number_of_mortgage_accounts	Integer

The following schemas were updated accordingly:

- `Inbound Payments`
- `Inbound Payments + Outcomes`

Disable Self Service Rules Testing storageâ

Feature HSBC-2615

We've changed the following variables to disable the Self Service Rules Testing storage:

- `pulse.global.services.rte.workflow.rules.forwardToTestingStorage`
- `pulse.global.services.fraud.simplerulestestingservice.enable`

This prevents data related to this feature from being stored in the database `ssrt_events` table, optimizing disk usage.

Fix Ansible pipeline in Pulse assuming eth0 as the network interface nameâ

Bug Fix HSBC-2912
Also In: 2.7.2

We have added a workaround to the technical issue in which the `eth0` interface was being assumed in the ansible pipelines.

Add receiver_product_type field to Pulse and Case Managerâ

Feature FTMT-23881 HSBC-2778

We have added the below field to the Inbound Payments API:

Field Name	Datatype
receiver_product_type	String

The following schemas were updated accordingly:

- `Inbound Payments`
- `Inbound Payments + Outcomes`

The new field was also added to the Case Manager schema.

Fix Cards amount column in Case Managerâ

Feature HSBC-2807

We've changed the Case Manager mapping for the Cards field `schema.amount_dbl` to use the `amount` field from the API request. The `schema.amount` field still exists in Case Manager but will be empty from now on.

All Case Manager widgets and tables were changed to always use `schema.amount_dbl` when displaying the Amount for Cards events.

Please note that Cards Automation Rules and SLAs should use the `Amount (Double)` field.

Add new field to the Mobile Outbound extensionsâ

Feature FTMT-24874 HSBC-2791
Also In: 2.7.2

The following field was added to Case Manager `2 Way/1 Way SMS` and `Mobile Payment Push` extensions:

Field Name	Datatype	Channels	Description
channel_user_id	String	Transfers	Online/Internet Banking User ID.

Please refer to the AFIS Extensions reference for more details.

Create `Cards + Outcomes` schema and `Treatment Strategy Priority` list in Cards applicationâ

Feature HSBC-2767

We have created `Cards + Outcomes` schema and `Treatment Strategy Priority` list to support the Alert Prioritization in the `Get Next` feature.

The list addition is non-destructive: if it list already exists in a given environment, it will not be added (nor will the list items of the existing list be modified).

Likewise, the workflow will not be updated to prevent any issues in installations that are already using an existing list.

Add new feedback update strategy to Outbound paymentsâ

Feature FTMT-20994 HSBC-2722
Also In: 2.5.6 2.7.2

In previous versions, when a feedback event reached Pulse, all events related to the feedback would be updated and reinjected into the workflow. This would generate unnecessary load in Case Manager since it only requires a single feedback event.

To circumvent this issue we've introduced a new feedback update strategy to control which payment related events are to be updated by a feedback event. This can be configured in the following properties:

- transfers.endpoints.puttransferfeedback.feedbackUpdateStrategy
- transfers.feedbacklistener.feedbackUpdateStrategy

The strategy applied to both properties can be configured using the following Ansible variable:

Variable	Datatype	Default Value
outbound_payments_feedback_strategy	String	SINGLE

The value of the property can be one of:

- `ALL` : Updates all events with the same lifecycle id (corresponds to the old behaviour)
- `SINGLE` : Updates each Case Manager event with a given lifecycle id

To better understand the different behaviours please take the following example:

- A transfer initiation reaches Pulse
- Two info requests are made to update the initiation
- A feedback event is made

Having the previous scenarios with the configuration set to `ALL` would mean three feedback events would be reinjected into the workflow and later processed by Case Manager.

On the other hand setting it to `SINGLE` would mean only one event would be reinjected into the workflow. The default strategy is now `SINGLE` and it is our suggestion to keep this strategy.

Please refer to Pulse Configuration Reference and the Feedback Update Strategy concepts page for more details.

Add new feedback update strategy to Cards and Inbound Paymentsâ

Feature HSBC-2903

Following the changes done in Add new feedback update strategy to Outbound payments, we've extended the feedback update strategy to Cards and Inbound Payments.

Functionality wise the behaviour is the same as described for Outbound Payments, only the configuration properties change:

- Inbound:
    - inboundpayments.endpoints.putinboundtransferfeedback.feedbackUpdateStrategy
    - inboundpayments.feedbacklistener.feedbackUpdateStrategy
- Cards:
    - cards.endpoints.feedback.feedbackUpdateStrategy
    - cards.feedback.listener.feedbackUpdateStrategy

The strategy applied to both properties can be configured using the following Ansible variables:

Variable	Datatype	Default Value
inbound_payments_feedback_strategy	String	SINGLE
cards_feedback_strategy	String	SINGLE

Change Cards Event Details pageâ

Feature HSBC-2576

We have made several changes to the default configuration of the Event Details page for Card events.

Previously, some fields were unused and displayed `N/A` .

This issue has been resolved, and now all fields on the page are designed to display meaningful data.

Upgrade ETL to version 17.0.0â

Feature HSBC-2594

We've upgraded ETL to version 17.0.0.

This version introduces a new ETL mode to address extraction of event scheduled metrics when the metrics were computed by a plan that is no longer referenced by the workflow.

The new mode is configured to search metrics definition in all existing plan executions and to not fail extraction if the plan execution no longer exists in Pulse.

For more information refer to the ETL configuration reference.

Upgrade Reference Data Service to version 3.7.8â

Feature HSBC-2721

This change includes an upgrade of SnakeYAML to address the CVE-2022-1471 vulnerability.

Revert liquibase-core 4.30 upgradeâ

Internal HSBC-2703

We've reverted the prior `liquibase-core` version bump from 4.29.2 to 4.30.0, to avoid a bug that sometimes causes it to execute slowly.

Remove unused Case Manager columnsâ□□

Feature HSBC-2723

We have removed `TREATMENT_STRATEGY_ID` and `EVENT_TREATMENT_STRATEGY_ID` from Case Manager columns as they are no longer used.

In environments in which the columns do not exist, this change is a no-op and will not fail.

ⓘnote

For completeness and to match earlier branches (such as the `2.4.8` release), these columns were added in this release in environments that don't exist, but here they are once again removed.

Add support to filter events without status reason to Payment Fulfillment extensionâ□□

Feature FTMT-20994 HSBC-2701
Also In: 2.5.6 2.7.2

We've added a filtering option to the Payment Fulfillment extension that will filter out events when a status reason is not present. The filter can be enabled or disabled using the following Ansible variable:

Variable	Datatype	Default Value
payment_fulfillment_exclude_events_without_status_reason	Boolean	true

Please refer to [Case Manager Configuration Reference]() for more information.

Add new fields to the Inbound Payments and Digital channels in Case Managerâ□□

Feature FTMT-24516 HSBC-2464

We've made available a new set of fields to Automation Rules and SLAs for the Inbound Payments and Digital channels:

Field Name	Inbound Payments	Digital Activity
sender_transaction_code	â□□¸□	â□□¸□
sender_transaction_narrative	â□□¸□	â□□¸□
sender_transaction_notes	â□□¸□	â□□¸□
sender_bank_country_code	â□□¸□	â□□¸□
receiver_account_currency	â□□¸□	â□□¸□
receiver_bank_country_code	â□□¸□	â□□¸□
receiver_transaction_currency	â□□¸□	â□□¸□
receiver_account_bank_id	â□□¸□	â□□¸□
overnight_acc_ledger_bal	â□□¸□	â□□¸□
sender_transaction_ref	â□□¸□	â□□¸□
customer_fiv_markers	â□□¸□	â□□¸□
receiver_account_type	â□□¸□	â□□¸□
customer_business_type	â□□¸□	â□□¸□
customer_portfolio_segment	â□□¸□	â□□¸□
customer_annual_turnover	â□□¸□	â□□¸□
customer_reported_income	â□□¸□	â□□¸□
rfb_flag	â□□¸□	
receiver_transaction_amount_dbl	â□□¸□	

Field Name	Inbound Payments	Digital Activity
line_of_business	â□□¸□	
business_start_date	â□□¸□	
customer_business_type_last_update_date	â□□¸□	
customer_reported_income_last_update_date	â□□¸□	
customer_annual_turnover_last_update_date	â□□¸□	
customer_portfolio_segment_last_update_date	â□□¸□	
employer_name	â□□¸□	
employer_name_last_update_date	â□□¸□	
employer_start_date	â□□¸□	
customer_occupation_status	â□□¸□	
customer_occupation_status_last_update_at	â□□¸□	
receiver_account_open_date	â□□¸□	

Additionally the following plan enriched fields were also made available to Inbound Payments and Digital Activity:

Field Name	Inbound Payments	Digital Activity
cmd_payment_type	â□□¸□	â□□¸□
cmd_payment_sub_type	â□□¸□	â□□¸□
cmd_event_description (cmd_sender_transaction_code in Inbound)	â□□¸□	â□□¸□
cmd_sender_bank_name	â□□¸□	â□□¸□
cmd_receiver_bank_name	â□□¸□	â□□¸□

Please note that the enrichment plan to support this feature for Inbound Payments is not available by default and requires manual creation. Please refer to [Case Manager event enrichment guide] to learn how to configure it.

Add new Pulse list for Inbound Paymentsâ□□

Feature HSBC-2653

We have created a new Pulse list for treatment strategy in the Inbound Payments application in order to support the Alert Prioritisation in the `Get Next` feature, so that the baseline Feedzai inbound deployment more closely matches the outbound deployment.

The new list is called `Treatment Strategy Priority` and the addition is non-destructive: if this list already exists in a given environment, it will not be added (nor will the list items of the existing list be modified).

Likewise, the workflow will not be updated, so that the runtime Pulse of environments that are already using an existing list will not be affected.

The content of the list has the same outcomes that were initially present in the Outbound Payments list that existed before Outbound Payments was deployed and HSBC took ownership of the list. Specifically, these default outcomes are:

Treatment Strategy	Priority
BIBACH	4
CHAPS	10
FASTER	1
GBLMONEY	10
GBLWALLET	10

Treatment Strategy	Priority
GFX	10
INTERNAL	2
SO	3

Add new fields to Pulseâ

Feature HSBC-2409

We have added the below fields to the `Payments` and `Payments + Outcomes` schemas in order to more closely match the fields comprising the Outbound Payments API:

Field Name	Datatype
browser_os_version	String
costumer_id_global	String
customer_portfolio_phys_type	String
customer_portfolio_remote_type	String
ds_transaction_id	String
payment_message_source_d	String
sender_transaction_origin_amount	Long
sender_transaction_origin_currency	String
sender_transaction_reason_d	String
sender_transaction_sign_stts_d	String
customer_change_reason	String
sender_account_channel_limit	Double

We have added the below field to the `Inbound Payments` and `Inbound Payments + Outcomes` schemas in order to more closely match the fields comprising the Inbound Payments API:

Field Name	Datatype
sender_account_channel_limit	Double

Enable advanced search for Cardsâ

Feature HSBC-2578

We've enabled the advanced search for Cards in the search page.

Also the alerts page now has the Cards channel fully working and filtering only the corresponding events.

Add treatment strategy columns to Case Manager databaseâ

Bug Fix HSBC-2710
Also In: 2.4.8 2.5.6 2.7.2

We've added two columns to `AM_EVENT_INBOUND_PAYMENTS` that are needed for Case Manager to start:

- `TREATMENT_STRATEGY_ID`
- `EVENT_TREATMENT_STRATEGY_ID`

These columns must exist (and must match the columns from `AM_EVENT_COMMON` and `AM_EVENT_TRANSFERS`) for legacy reasons.

Add new Inbound fields to Case Manager external notificationsâ

Feature FTMT-23796 FTMT-24496 HSBC-2672

We've added the below fields to the Case Manager external notifications payload.

Field Name	Channel
event_type	Inbound Payments
lifecycle_id	Inbound Payments
alert_id	Inbound Payments

We've created the Notifications Schema page with the notifications schema and payload examples for the various channels. Please refer to it for more details.

Configure Cards alert id transformationâ

Feature HSBC-2652

We've added a new `alert_id` field to the Cards workflow schema to enable the alert id transformation.

This transformation generates a new `alert_id` field from the event's `lifecycle_id` and endpoint name, which will then be used as the event identifier in Case Manager.

In practice this transformation will make Case Manager merge authorization and info events that have the same `lifecycle_id`. Clearings will always generate a new event in Case Manager, for example:

- Send `autorization` with `lifecycle_id=lifecycle_1` : new event in Case Manager
- Send `info` with `lifecycle_id=lifecycle_1` : updates existing event in Case Manager
- Send `clearing` with `lifecycle_id=lifecycle_1` : new event in Case Manager

Please note that this transformation is different from the Inbound/Outbound alert id transformation which takes the configured last n characters from the `lifecycle_id`.

Cards schema changesâ

Feature HSBC-2581

In order to make the CM and Pulse schema for the Cards application as much as identical as possible, we've done several changes to some fields.

The following fields were removed from Case Manager schema:

```
 - acceptor_city
- acceptor_dba_name
- acceptor_mastercard_risk
- acceptor_open_date
- acceptor_risk
- acceptor_risky_flag
- acceptor_street_addr
- acceptor_visa_risk
- acceptor_zip
- account_available_credit
- account_credit_limit
- account_last_activity
- atc1
- atc2
- card_last4
- last_sca_at
- pos_card_capture_cap
- terminal_latitude
- terminal_longitude
- terminal_poc_flag
- terminal_street_addr
```

```
- dispute_end_date
- dispute_feedback_notes
- dispute_refunded
- feedback_type
- oob_end_date
- oob_feedback_notes
- oob_response_type
- dispute_notes
- dispute_reason
- dispute_start_date
- dispute_submitted_by
- info_type
- oob_comm_type
- oob_notes
- oob_start_date
- customer_portfolio_producte
- customer_portfolio_region
- payment_message_source
- customer_type
- office_id
- customer_account_type
- sender_transaction_type
- sender_transaction_cop_result
- sender_transaction_auth_type
- payment_type
- payment_sub_type
```

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The following fields were added to the Pulse Schema:

```
 - customer_phone
- customer_business_phone
- customer_businessphone_last_update_at
- customer_citizen_id_number
- customer_citizenship
- customer_corresp_city
- customer_corresp_country_code
- customer_corresp_region
- customer_corresp_street_addr
- customer_corresp_zip
- customer_corresp_zip_area_code
- customer_email_last_update_at
- customer_fiv_marker_last_updated_at
- customer_gender_code
- customer_hmphone_last_update_at
- customer_home_address_last_update_at
- customer_home_city
- customer_home_country_code
- customer_home_phone
- customer_home_region
- customer_home_street_addr
- customer_home_zip
- customer_home_zip_area_code
- customer_mbphone_last_update_at
- customer_pb_marker
- customer_preferred_language
- customer_registration_date
- customer_surname
- customer_total_balance
- sender_email
- sender_name
- sender_phone
- customer_forname
- customer_reported_income
```

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Fix Case Manager enrichment errorâ

Bug Fix FTMT-23776 FTMT-23777 FTMT-23780 HSBC-2664

We've fixed an issue in Case Manager where some enrichers were not properly loaded due to a malformed Javascript function.

This would lead to Case Manager showing the enricher name (e.g. `paymentTypeEnrich`) instead of the actual enriched value.

Add Alert Storage Service transformations to the Inbound Payments and Cards applicationsâ

Feature FTMT-23716 HSBC-2532

The Alert Storage Service for Outbound Payments benefited from a number of transformations that were not available in the other applications. These transformations include, among others, the ability to populate an `additional_fields` field in Case Manager with values generated during the workflow.

As we ported these functionalities, we merged what were previously two independent functionalities:

- The ability to have `additional_fields` based on a prefix (which is being used to import `cmd_*` fields into Case Manager)
    - Please note that the enrichment plan and lookup lists were not created as part of this task. Please refer to [Case Manager event enrichment guide]() to learn how to create the necessary resources to support this feature
- The ability to have `additional_fields` based on exact name matching (which by default is making the following fields available under `additional_fields`: `pib_suspended, internal_payment_type, internal_event_source, isJade`):
    - These fields are controlled via the `additional_fields_enricher_fields` Ansible variable

warning

This refactor made it so the configuration for the exact name matching enrichment was ported into the same configuration namespace as the prefix based enrichment.

This results in some Pulse configuration properties changing:

- `services.alertstorageservice.alerttransformations.cmenrichment.enrichedFields`: accepts a list of additional fields to enrich
    - Previously was `com.feedzai.hsbc.pulse.services.customalertstorageservice.additionalfields.enricher.fields`
- `services.alertstorageservice.alerttransformations.cmenrichment.enabled`: now acts as a toggle for both the prefix based enrichment and exact name matching enrichment
    - Previously only controlled the prefix based enrichment

Unless manual overrides were done, this requires no further action from HSBC as this release automatically uses the new properties.

Add Cards use caseâ

Feature HSBC-2575

We've added the Cards use case, along with the application in Pulse and the channel in Case Manager. While technically, an old baseline for Cards existed from the project outset, this was not used at HSBC. We have expanded on it so that the Cards use case is fully implemented and ready to use.

There are several release notes related to the Cards use case and changes that were done in this release notes page which can be consulted for the details on each feature that was implemented.

ⓘnote

To enable the Cards application, the following Ansible variables must be set for both Pulse and Case Manager:

```
apps_enabled:
  - <other apps>
  - <market>_cards

...

<market>_cards_app_slug: <market>_cards
<market>_cards_market: <market>
<market>_cards_portfolio: cards
```

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For example for the UK, the above variables would be:

```
apps_enabled:
  - <other UK apps>
  - uk_cards

...

uk_cards_app_slug: uk_cards
uk_cards_market: uk
uk_cards_portfolio: cards
```

Copy

For full details, please see the existing documentation for how to deploy new apps/channels, which applies to all applications/channels, including cards:

- Pulse - How to deploy new applications
- Pulse Configuration Reference and default variables
- Case Manager - How to deploy new channels
- Global Rollout Documentation

Case Manager version upgraded to 13.17.7â

Feature FTMT-24041 HSBC-2658
Also In: 2.4.7 2.7.1

We've upgraded Case Manager to version 13.17.7.

This version addresses the following issues in multi database setups:

- Automation rules intended to set the event queue were not working properly
- SLAs were not updating the operational status correctly after breaching

Add Python 3 support in the Ansible deployment pipelinesâ

Feature FTMT-22381 HSBC-2453
Also In: 2.7.2

We've updated the deployment pipelines to support setting the interpreter Python version.

By default it will now use Python 3 (the version that is linked to the `/usr/bin/python3` path) and can be changed by setting the Ansible built-in variable `ansible_python_interpreter`.

warning

In version 2.7.1 we introduced a `python_ansible_interpreter` variable which is no longer in use. The Ansible built-in variable `ansible_python_interpreter` is now the only way to change the Python interpreter.

On this page

What's Changed

Release 2.9.0 â

Add new product type field to Paymentsâ

Feature FTMT-27619 HSBC-3191

We've added a new field to the Payments schema:

Field Name	Datatype
sender_product_type	String

The `Payments` and `Payments + Outcomes` Pulse schemas were updated with the new fields. The fields were also added to the Case Manager schema for the Transfers and Digital Activity channels.

Add new PB Delegate field to Paymentsâ

Feature WPB-376255 HSBC-3177

We've added a new field to the Payments schema:

Field Name	Datatype
delegate_without_product_indicator	Boolean

The `Payments` and `Payments + Outcomes` Pulse schemas were updated with the new fields. The fields were also added to the Case Manager schema for the Transfers and Digital Activity channels.

Add new SRF fields to Paymentsâ

Feature FTMT-27191 HSBC-3175

We've added new fields to the Payments schema:

Field Name	Datatype
sfr_moneylock_hold_bal	Double
sfr_total_hold_bal	Double

The `Payments` and `Payments + Outcomes` Pulse schemas were updated with the new fields. The fields were also added to the Case Manager schema for the Transfers and Digital Activity channels.

Add new CMB ECW fields to Paymentsâ

Feature FTMT-27244 HSBC-3160

We've added new fields to the Payments schema:

Field Name	Datatype
ecw_question_and_answer_details	String
ecw_final_action	String

Field Name	Datatype
ecw_answers_changed	String
ecw_duration	String
ecw_payment_details_changed	String
ecw_purpose_of_payment	String
ecw_country_payment_description	String
ecw_pop_cpd_mapped	String
ecw_response_time	String
payment_execution_type	String
request_correlation_id	String

The `Payments` and `Payments + Outcomes` Pulse schemas were updated with the new fields. The fields were also added to the Case Manager schema for the Transfers and Digital Activity channels.

Generate HAR Export Documentationâ

Docs HSBC-3035

We've added a new page with the process to export a HAR (HTTP Archive).

Please refer to Export HAR files for more details.

Add new fields to Cardsâ

Feature FTMT-25998 HSBC-3044

We have added the below fields to the Cards API schema:

Field Name	Datatype
customer_portfolio_region	String
customer_portfolio_country	String
customer_portfolio_product	String
customer_portfolio_class	String
customer_portfolio_type	String
customer_portfolio_channel	String
account_portfolio_hier_2	String
merchant_id	String
scheme_fraud_score	String

The `Cards` and `Cards + Outcomes` schemas were updated with all the above fields.

note

With the addition of `customer_portfolio_country`, we've changed the default `tenant` field mapping in Cards accordingly to align with the Outbound and Inbound Payments behavior.

On this page

What's Changed

Release `3.0.2` â

Add support for adding IPv6 addresses to `PosNeg Listed IP Addresses` from Case Managerâ

Feature FTMT-36830 HSBC-4427

We've enabled the ability to add IPv6 addresses (`device_ip_address_v6`) entries to the `PosNeg Listed IP Addresses` list directly from the event details page in Case Manager. This is the same list already used by IPv4 addresses (`device_ip_address`).

Add columns to Digital Activity Transaction Historyâ

Feature FTMT-36748 HSBC-4426

We've added the following columns to the Transaction History widget in the Event Details page, for the Digital Activity channel:

Column Name	Field Name
IP Country	device_ip_country_code
IP/Device ISP	device_geo_ip_isp

Additionally, the following columns specific to HSBCNet were added:

Column Name	Field Name
IP Address v6	device_ip_address_v6
Device User-Agent	device_user_agent
Malware Score	pinpoint_risk_score

Add columns to Transfers Transaction Historyâ

Feature FTMT-36746 HSBC-4425

We've added the following HSBCNet columns to the Transaction History widget in the Event Details page, for the Transfers channel:

Column Name	Field Name
User ID	user_id
Customer Country Code	customer_country_code
Sender Bank Country Code	sender_bank_country_code
Receiver Bank Name	receiver_bank_name
Receiver Bank Country Code	receiver_bank_country_code
Receiver Account Bank ID	receiver_account_bank_id
Payment Held Indicator	sender_transaction_held
Transaction Sign Status	sender_transaction_sign_status

Column Name	Field Name
Payment Message Source	payment_message_source
Transaction Notes	text_for_receiver
Customer Business Type	customer_business_type

Release `3.0.1` â

Fix block code expiration value not computing with offsetâ

Bug Fix FTMT-37370 HSBC-4407

We've fixed an issue where the fields `cm_transaction_block_date` and `cm_transaction_block_expiration` in Case Manager extension payloads were not computed using the configured offset days (default `5`).

Revert `bioc` nested fields orderâ

Bug Fix HSBC-4382
Also In: 2.11.3

We've fixed an issue where the `bioc` nested fields order introduced in version 2.11 caused replica ingestion failures during cluster upgrades.

The field order has been reverted to its pre-2.11 state to prevent similar issues in future upgrades.

info

This issue only affected Outbound, as the field order did not change for other channels.

Fix Reasons dropdown options becoming unavailableâ

Bug Fix FTMT-36949 HSBC-4366

We've fixed an issue which, under certain conditions, caused all options to disappear from the Reasons dropdown when trying to set an event's Block Code or Operational Status.

Fix database error preventing SLAs from breachingâ

Bug Fix FTMT-36820 HSBC-4358
Also In: 2.11.3

We've fixed an inconsistency between database column sizes that prevented SLAs from breaching when Case Manager was deployed on machines with very long hostnames (70 or more characters).

Fix `Move to Unresolved` option visibility for Cards channelâ

Bug Fix HSBC-4354

The `Move to Unresolved` action in the Cards channel's dropdown menu was previously restricted by the `cm:resource:alert:status-bulk:*` permission. We have removed this permission requirement, allowing users to see the "Move to Unresolved" option regardless of their permissions.

Add network configuration for DTPâ

Feature HSBC-4353

We've added support to configure the network and subnetwork in which DTP Dataflow worker nodes are created. This can be configured during deployment using the following Ansible variables:

- `dtp_dataflow_network` : sets the Dataflow job network
- `dtp_dataflow_subnetwork` : sets the Dataflow job subnetwork

warning

Note that the two parameters are not required to be set at the same time. For example, when setting `subnetwork` the `network` can be omitted and GCP will determine the network automatically.

For more details on Dataflow networking refer to Specify a network and subnetwork.

We've also added a new property to configure if worker pool machines should be created with public IP addresses. This is available via the `dtp_dataflow_use_public_ips` variable and defaults to `false`.

Please refer to DTP Network Configuration and to the DTP Configuration Reference for more information.

Release `3.0.0` â

Upgrade Case Manager to version 13.31.0â

Feature FTMT-27426 HSBC-4280

We've upgraded Case Manager version to 13.31.0.

Case Manager now supports RabbitMQ 4.2. As part of this upgrade, it is now possible to configure Case Manage to use RabbitMQ `QUORUM` queues.

This is configurable using the `rabbitmq_casemanager_use_quorum_queues` Ansible variable. To ensure backwards compatibility, the default value of the variable is `False`.

For more information please refer to the Case Manager Configuration Page.

note

Although classic queues are still supported in RabbitMQ 4.2 please note they aren't mirrored which removes high availability. The default value is only meant as an upgrade point from 3.X versions.

Upgrade Data Extraction Pipelines to version 2.2.0â

Feature FTMT-27426 HSBC-4280

We've upgraded the Data Extraction Pipelines to version 2.2.0. As part of this upgrade, it is now possible to configure the Data Extraction Pipelines to use RabbitMQ `QUORUM` queues.

This is configurable using the following Ansible variables:

- `rabbitmq_casemanager_queue_type_green`
- `rabbitmq_casemanager_queue_type_blue`

To ensure backwards compatibility, the default value of both variables is `CLASSIC`.

For more information please refer to the Data Extraction Pipelines Configuration Page.

Upgrade ETL Tool to version 18.3.0â

Feature FTMT-27426 HSBC-4280

We've upgraded the ETL extraction tool to version `18.3.0`. As part of this upgrade, it is now possible to configure the ETL tool to use RabbitMQ `QUORUM` queues. Please note, this does not have an impact on the regular ETL extractions used to extract events to GCS buckets for analytics. This will only be applicable to the recovery of events from Pulse to Case Manager.

This is configurable using the `etl_cm_rabbitmq_queue_type` Ansible variable.

To ensure backwards compatibility, the default value of both variables is `CLASSIC`.

For more information please refer to the ETL Configuration Guide.

Upgrade Pulse to version 25.1.9â

We've upgraded Pulse to version 25.1.9.

This upgrade include the following changes:

HSBC-3866
Bug Fix HSBC-3866
Also In: 2.11.2

Fixes an issue where SMJs did not retrieve values from the secondary channel when different event storages were configured.

HSBC-3998
Bug Fix HSBC-3998
Also In: 2.11.2

Fixes an issue where SMJs did not retrieve values from the secondary channel when the workflow partition keys are incompatible or the second channel has a different number of partitions.

This upgrade also adds new monitoring metrics. For more information, please consult Workflow Validation metrics, App Lifecycle Operations metrics, and Pulse Publish metrics.

HSBC-4156
Feature HSBC-4156
Also In: 2.11.2

Includes additional logging to help identify the root cause of an error that occurs when deleting datasources in the Data Science environment.

For these logs to be available, the following classes must be set to the `DEBUG` level:

- com.feedzai.pulse.service.apps.manager.datascience.AppDataScienceDataSourceManagerImpl
- com.feedzai.pulse.service.apps.manager.AbstractModelManager
- com.feedzai.pulse.service.apps.manager.AppSchemaManagerImpl

This can be done during deployment by using the `pulse_loggers` Ansible variable.

warning

After replicating the issue with the above log levels enabled, the loggers should be disabled to avoid excessive logging.

FTMT-26214 / HSBC-4218
Feature FTMT-26214 HSBC-4218
Also In: 2.11.2

Includes extended support for real-time cross-channel metrics. Cross-channel metrics now have a messaging-based implementation which adds support for partitioned PKernel and workflow partitioning. The new implementation can be enabled by setting the following Ansible variable to `false`:

- `cross_channel_metrics_rmi_based_lookup` (default: `true`)

Please check Pulse Configuration Reference for more information.

HSBC-4259
Feature HSBC-4259
Also In: 2.11.2

Upgrades pac4j from version `4.5.7` to `4.5.9` to address critical CVE-2026-29000.

FTMT-27426 / HSBC-4126
Feature FTMT-27426 HSBC-4126

Pulse now supports RabbitMQ `4.2`. As part of this upgrade, it is now possible to configure Pulse to use RabbitMQ `QUORUM` queues.

This is configurable using the `rabbitmq_queue_type` Ansible variable. To ensure backwards compatibility, the default value of the variable is `CLASSIC`.

For more information please refer to the [Pulse Configuration Page].

ⓘ**note**

Although classic queues are still supported in RabbitMQ 4.2 please note they aren't mirrored which removes high availability. The default value is only meant as an upgrade point from 3.X versions.

Upgrade Reference Data Service to version 3.7.12â

Feature HSBC-4244
Also In: 2.11.2

We've upgraded the Reference Data Service to version 3.7.12, which addresses the `pac4j` CVE-2026-29000 vulnerability.

Add Transaction Block Code to Cardsâ

Feature FTMT-30641 FTMT-34449 HSBC-3667
Feature FTMT-34504 FTMT-34504 HSBC-3984 HSBC-4044
Feature FTMT-32330 HSBC-4031
Feature FTMT-29506 FTMT-34942 HSBC-4120
Feature FTMT-32330 HSBC-4143
Feature FTMT-27481 HSBC-4212

We've added a set of changes for both Pulse and Case Manager to introduce the Transaction Block Code feature, for Cards event.

Given this work has been done in multiple tasks, we've outlined the major changes in [Transaction Block Page].

Fix Digital Activity status and reasons bulk actionsâ

Bug Fix FTMT-34940 HSBC-4223

We've fixed an issue with the bulk actions for Digital Activity events in the Transaction History widget, which did not update the correct state machine when labelling alerts as Fraud or Not Fraud.

Labelling Digital Activity events through bulk actions now correctly triggers Automated Rules based on the Digital Activity State Machine and its Reasons can be updated.

Add `alert_number` to Cards and Non-Monâ

Feature FTMT-35849 HSBC-4206

We've added a new `alert_number` field to Cards and Non-Mon, containing the Alert ID for these channels. The Alert ID generation follows the same approach used for the other channels: it's generated by hashing the full lifecycle ID using SHA-256, encoding the hash with Base64, removing any non-alphanumeric characters, and then taking the last 8 characters of the result.

We've also extended the Alert ID search box in Case Manager to search for `alert_id` and `alert_number` and added both fields to the Cards/Non-Mon Event Details page baseline.

Add `customer_country_code` to Cards, Token and Account Reference Data entitiesâ

Feature FTMT-34393 FTMT-34969 HSBC-4205

We've added the `customer_country_code` field to the Cards, Token and Account Reference Data entities.

This changes ensures that all entities have a common field where the tenant can be specified and allow the enrichment of events in multi-tenanted environments.

The default value of the variable `reference_data_tenant_api_payload_field` has been changed to `customer_country_code`.

For more information, refer to the Pulse Configuration Recommendations.

Update Case Manager to version 13.30.0â

Feature FTMT-35919 HSBC-4201

We've upgraded Case Manager to version 13.30.0.

The Case Manager's SLA scheduler now performs a graceful shutdown, which ensures its resources are safely and consistently released when the system stops. This fixes an issue where SLAs would not trigger due to a database lock.

Add fdz_channel to the Payment Fulfillment extensionâ

Feature FTMT-33848 HSBC-4178

The following field was added to Case Manager Payment Fulfillment extension:

Field Name	Datatype	Channels
fdz_channel	String	Transfers, Inbound Payments

Please refer to the AFIS Extensions for more details.

Fix `Transaction Sub-Type` field inconsistent mappings in Case Managerâ

Bug Fix FTMT-32085 HSBC-4177

We've fixed inconsistent mappings in Case Manager regarding the `Transaction Sub-Type` field. Previously, the column in the transaction history widget was getting its value from `sender_transaction_code`. We have changed it to get its value from `cmd_payment_sub_type` to keep it consistent with the Search and Alerts pages.

Add new fields to Cards and Non-Monâ

Feature FTMT-35019 HSBC-4175

We've added the following fields to the Cards and Non-Mon endpoints and workflow schemas:

Field Name	Datatype
digital_wallet_tar_ref_num	String
digital_wallet_tcn_ref_num	String
account_other_balances_2	Object
account_other_balances_2.amount	Number
account_other_balances_2.type	String
card_ond_vec_value_new	String

Add support for RabbitMQ 3.13.7â

Feature FTMT-27426 HSBC-4154

Feedzai components now support and have been tested against RabbitMQ `3.13.7`. The RabbitMQ `3.13` branch is still under extended commercial support through December 2027.

Add Session ID column to Transaction History and Search Pagesâ

Feature FTMT-34971 HSBC-4152

We've added the `Session ID` column (mapped to `session_id) in the following pages:

- Event Search Page: Transfers tab, between `Beneficiary Account` and `Assignee`
- Alerts Page: Transfers tab, between `Beneficiary Account` and `Assignee`
- Event Details Transaction History: Transfers tab, between `Device ID` and `Assignee`
- Customer Details Transaction History: Transfers tab, between `Beneficiary Account` and `Assignee`

Add previous transactions to Cards `Send 2-WAY SMS` extensionâ

Feature FTMT-34477 HSBC-4146

We've added the `previous_transactions` field to the Cards `Send 2-WAY SMS` extension. This field will be populated by listing the last transaction that fall under the following criteria:

- Same `card_pan` value
- Same `customer_id` value
- Happened in the last 24 hours (calculated from the time the extension was triggered)
- Is not the event that triggered the extension

For more details please refer to AFIS Extensions.

Add Breach Status option to Cards and Non-Mon SLAsâ

Feature FTMT-34982 HSBC-4145

We've enabled the `Set Breach Status` and `Set Initial Breach Status` options in Breach Actions for SLAs in the Cards and Non-Mon channels.

Change fields required by the Cards Info endpointâ

Feature FTMT-34893 HSBC-4133

We've changed the mandatory fields of the Cards Info endpoint. When calling this endpoint, only the following fields are now required:

- `card_pan`
- `lifecycle_id`
- `event_occurred_at`
- `customer_id`

For more information, please refer to the API reference.

Add new fields to Cards and Non-Monâ

Feature FTMT-34892 HSBC-4045

We've added the following fields to the Cards and Non-Mon endpoints and workflow schemas:

Field Name	Datatype
customer_id_number	String
customer_id_global	String
dw_auth_mthd	String

Add new enrichers for Cards and Non-Mon eventsâ

Feature FTMT-28255 HSBC-4034

We've added the following enrichers to be used in Cards and Non-Mon events:

Field Name	Enricher Name
sic	sicEnrich

Field Name	Enricher Name
trans_reversal_ind	transReversalIndEnrich
digital_wallet_token_type	digitalWalletTokenTypeEnrich
card_customer_present_details	cardCustomerPresentDetailsEnrich
pos_cardholder_ver_method	posCardholderVerMethodEnrich

For more details on the available values and configuration please refer to the Case Manager Enrichers page.

Add new fields to Outbound Payments and Digital Activityâ

Feature FTMT-32720 FTMT-34654 HSBC-4030

We have added the below elements to the `bioc` tuple in Outbound Payments and Digital Activity API schemas:

Field Name	Datatype
trust_score	Integer
mule_account_score	Integer
max_30_day_score	Integer
chall_risk_score	Double
chall_voice_scam_score	Double

The `Payments` and `Payments + Outcomes` schemas were updated with all the above fields.

Add Reference Data fields to the Outbound workflow schemasâ

Feature FTMT-34648 HSBC-4029

We've added the following fields to Transfers and Digital Activity to be enriched with Account Reference Data information:

New Transfers and Digital Activity Fields

Field Name	Datatype	Entity
sender_business_start_date	Long	Accounts
sender_customer_occupation_status	String	Accounts
sender_customer_reported_income_dbl	Double	Accounts
sender_line_of_business	String	Accounts
sender_product_type	String	Accounts
receiver_business_start_date	Long	Accounts
receiver_customer_occupation_status	String	Accounts
receiver_customer_reported_income_dbl	Double	Accounts
receiver_line_of_business	String	Accounts
receiver_product_type	String	Accounts

We've added the following fields to the Account Reference Data API:

New Fields

Field Name	Datatype	Entity
business_start_date	Long	Accounts
customer_occupation_status	String	Accounts
customer_reported_income_dbl	Double	Accounts

Field Name	Datatype	Entity
line_of_business	String	Accounts
product_type	String	Accounts

Add new fields to the Mobile Outbound extensionâ

Feature FTMT-34647 HSBC-4028

The following fields were added to Case Manager `Mobile Payment Push` extensions for the Outbound Payments channel:

Field Name	Datatype	Description
sender_account_type	String	Type of sender account
receiver_account_type	String	Type of receiver account
customer_account_type	String	Type of customer account
account_type	String	Account type
product_code	String	The code of the product offered by the bank that was used to initiate the transfer
sender_product_type	String	Type of product held

Please refer to the AFIS Extensions reference for more details.

Add new schema fields to Outbound and Inbound Paymentsâ

Feature FTMT-32720 HSBC-4027

We've added the following fields to the Outbound (Transfers and Digital) API and workflow schemas:

Field Name	Datatype	Description
tmx.digital_id_trust_score	Number	Digital ID Trust Score
tmx.chall_reason_code	String	Challenger Risk Reason Code

The following field was also added to Outbound and Inbound Payments:

Field Name	Datatype	Description
vocalink.vocalink_chall_score	Number	Challenge Vocalink Score

Add new Token reference data entityâ

Feature FTMT-34393 FTMT-34969 HSBC-4024

We've added a new Token reference data entity to the Reference Data API. The entity key is `ref_digital_wallet_token` and its schema is the following:

New Token fields

Field Name	Datatype
tar_unique_refernce_number	String
tcn_unique_refernce_number	String
ref_customer_id	String
ref_card_pan	String
ref_tar_local_datetime	String
ref_tcn_local_datetime	String
ref_tar_local_datetime_alt	String

Field Name	Datatype
ref_tcn_local_datetime_alt	String
ref_digital_wallet_secu_elem_id	String
ref_digital_wallet_eligible_path	String
ref_digital_wallet_locale_lang	String
ref_card_mobile_match	String
ref_digital_wallet_source_pan	String
ref_digital_wallet_token_device_type	String
ref_digital_wallet_token	String
ref_digital_wallet_provider_reason	String
ref_digital_wallet_provider_risk	String
ref_digital_wallet_token_source	String
ref_digital_wallet_pan_source	String
ref_digital_wallet_provider_acc_score	String
ref_digital_wallet_provider_device_score	String
ref_avs_code	String

We've also added support for Cards enrichment in the Cards workflow by adding all of the Token entity fields to the Cards API and workflow schemas. By default Cards enrichment will be done using `ref_digital_wallet_token` but the field can be changed during deployment using the Ansible variable `cards_token_enrichment_key` .

Please refer to the Pulse Configuration Reference and to the Reference Data Tokens API for more details.

Make DTP Dataflow worker service account configurableâ

Feature HSBC-3997

We've made the service account for Data Transfer Pipeline Dataflow worker instances configurable during deployment using the `dtp_dataflow_worker_service_account` Ansible variable.

We've also updated the documentation to include the specific permissions this account requires. Please refer to the GCP Permissions page for more details.

Add Card Pan transaction grouping option in Case Managerâ

Feature FTMT-34471 HSBC-3978

We've added the option of grouping events by Card Pan (`card_pan` field) in the transaction history. When selected, the transaction history is filtered to show all transactions that have taken place on the card with the same number.

We've also fixed some inconsistencies on how `card_pan` was showing in the Case Manager UI. The `Card Pan` label will now show the exact value sent in `card_pan` on Event Details and tables, instead of being computed from `card_bin` and `card_last4` .

Add new extensions for the Cards and Non-Mon channelsâ

Feature FTMT-34467 FTMT-34477 FTMT-34482 HSBC-3977 HSBC-4147

We've created the following extensions to be used for the Cards channel via automation rules and SLA rules:

Extension Name	Notification Type
`Send 2-WAY SMS`	`cards_2_way_sms`
`Send 1-WAY SMS`	`cards_1_way_sms`

Extension Name	Notification Type
`Send Reminder SMS`	`cards_reminder_sms`

The extensions `Send 1-WAY SMS` and `Send Reminder SMS` have also been enabled for the Non-Mon channel, with the same notification types.

Additionally, `Send Mobile Payment Push` has been enabled for Cards to send notifications with the type `cards_mobile_push`.

These extensions are enabled by default, but can be disabled by setting the following to `false`:

- `send_2_way_sms_extension_enabled`
- `send_1_way_sms_extension_enabled`
- `send_reminder_sms_extension_enabled`

For more details please refer to the Case Manager Configuration Recommendations and the AFIS Extensions reference.

Add new fields to Cards and Non-Monâ

Feature FTMT-34453 HSBC-3975

We've added the following fields to the Cards and Non-Mon endpoints and workflow schemas:

Field Name	Datatype
info_type	String
dispute_start_date	Timestamp
dispute_reason	String
dispute_notes	String
dispute_submitted_by	String
dispute_end_date	Timestamp
dispute_feedback_notes	String
dispute_refunded	String
oob_start_date	Timestamp
oob_comm_type	String
oob_notes	String
oob_end_date	Timestamp
oob_feedback_notes	String
oob_response_type	String
feedback_type	String
feedback_label	String

Document slack Ansible variablesâ

Docs FTMT-34140 HSBC-3920

We have documented all the past and future slack Ansible variables.

Please refer to Pulse Configuration Recommendations for more details.

Fix mismatch in field formats in Cardsâ

Bug Fix FTMT-34154 HSBC-3918
Also In: 2.11.1

We've fixed a mismatch in the Cards API and Workflow schemas where the following fields had a different data type from the requirement.

Field Name	Previous Datatype	New Datatype
user_amount	Double	List<Double>
user_indicator	String	List<String>
user_string	String	List<String>

The field types have been corrected in the API specification, Cards and Cards + Outcomes schemas, and database.

warning

The database columns corresponding to the updated fields will be dropped during the deployment pipeline. Until the deployment completes and the column is re-added, any Cards events that contain these fields will fail to be stored in event storage and won't be recovered.

Add new Cards outcome to API specificationâ

Docs FTMT-33585 HSBC-3911

We've updated the API specification to include the `wallet_decision` outcome in the Cards response.

We've also updated the API titles to be more readable.

Please refer to Non-Mon and Cards Authorization for more details.

Add new fields to Cardsâ

Feature FTMT-33679 HSBC-3884

We have added the below fields to the Pulse Cards API schema:

Field Name	Datatype
card_relay_attk_code	String
dw_token_msg_rsn_cde_des	String
card_customer_instruction	String
card_customer_instruction_desc	String

The `Cards` and `Cards + Outcomes` schemas were updated with all the above fields.

Add *Card Status Update* Case Manager extensionâ

Feature FTMT-33602 HSBC-3883
Feature FTMT-36284 HSBC-4249

We've added a new Case Manager extension named *Card Status Update* that is able to send notifications to AFIS from Cards (and Non-mon) status update automation rules.

The behaviour of this extension is identical to the Mobile/SMS extensions for Outbound Payments, and to the Inbound Payment Notification extension for Inbound Payments, but with a much simpler schema in the API call made to AFIS.

An example of the schema sent for this extension is as below:

```
{
  "notification_type" : "card_status_update",
  "account_number" : "1005103001044120",
  "operational_status" : "closed",
  "fdz_channel" : "cards",
  "source" : "MSC",
  "trans_message_type" : "TypeA",
```

```
  "customer_portfolio_channel" : "portfoliochannel",
  "customer_portfolio_country" : "feedzai",
  "event_occurred_at" : "1773684342216",
  "event_type" : "feedback",
  "customer_portfolio_type" : "portfoliotype",
  "treatment_strategy" : "none",
  "alert_id" : "6e684d92-49ec-309a-9112-4ebc70e8d3ad",
  "customer_portfolio" : "EU_PC_CR_MS_GNRL",
  "customer_portfolio_class" : "elite",
  "self_serve_strategy" : "none",
  "customer_id_global" : "HK",
  "card_transaction_ref_number" : "card_1",
  "card_pan" : "card_1",
  "lifecycle_id" : "ad3c8300-8d7d-40ac-8c2a-9db15e1feef5",
  "customer_portfolio_region" : "US-EAST",
  "system_decision" : "approve",
  "lifecycle_id_alt" : "9e00eeba-daa7-48ed-9c92-c2125108d94e",
  "account_id" : "c0e68e4e-215f-47cb-b3cd-0d67a9ad881f",
  "card_transaction_component_type" : "TRX",
  "smh_source" : "",
  "customer_id" : "cutomer_1",
  "wallet_decision" : "none",
  "card_txn_sub_type" : "TypeB",
  "customer_id_number" : "0HKHSXXXXXX796256",
  "customer_portfolio_product" : "product"
}
```


Copy

By default the extension is enabled, but it can be disabled by setting `card_status_update_extension_enabled` to `false`. For more details please refer to the Case Manager Configuration Recommendations and the AFIS Extensions reference.

Add operational status filter to the Cards Fulfillment extensionâ

Feature FTMT-33601 HSBC-3882

We've extended the Fulfillment extension to accept the `operational_status` state machine in addition to the `status` state machine.

We have also updated the filter logic to support specific statuses. By default, the extension executes for all `status` changes, but limits `operational_status` changes to `On Hold` or `Unresolved`. This behavior can be configured during deployment using the Ansible variable `fulfillment_channels_state_machines`. Please refer to Case Manager Configuration Reference for more information.

Add new fields to the Case Manager Cards/Non-Mon external notificationsâ

Feature FTMT-27481 FTMT-34765 HSBC-3880 HSBC-4032

We've added the following fields to the Case Manager Cards and Non-Mon external notifications payload:

New Fields

Field Name	Datatype
lifecycle_id_alt	String
card_transaction_ref_number	String
card_transaction_component_type	String
source	String
customer_portfolio_region	String
customer_portfolio_country	String
customer_portfolio_channel	String

Field Name	Datatype
customer_portfolio_product	String
customer_portfolio_type	String
customer_portfolio_class	String
customer_portfolio	String
account_id	String
account_number	String
card_phone_banking_user_id	String
customer_id	String
customer_email	String
customer_name	String
customer_id_personal	String
customer_id_business	String
customer_id_number	String
customer_id_global	String
card_txn_sub_type	String
trans_message_type	String
acquirer_id	String
amount_dbl	Number
trans_model_amt	Number
cardholder_bill_amount	Number
cardholder_bill_currency_code	String
trans_currency_code	String
currency_code	String
treatment_strategy	String
oob_comm_type	String
oob_notes	String
oob_feedback_notes	String
oob_response_type	String
feedback_type	String
feedback_label	String
external_status_reason	String
channel_type	String
card_pan	String
saf_ind	String
sender_initiated_at	Timestamp
sender_initiated_at_local	String
local_datetime	String
event_occurred_at	Timestamp
crypto_transaction_info	String

Field Name	Datatype
card_type	String
outcome_decision	String
wallet_decision	String

For more details please refer to the Notifications Schema page.

Add scheduled metrics Cassandra table migration guideâ

Docs HSBC-3860

We've added a guide for migrating scheduled metrics Cassandra keyspaces/tables. This can be used to decouple the existing setup and reduce the possibility of metrics collisions in multi-channel deployments.

For more information please refer to Cassandra Profiles Migration.

Add Spanish (Mexico) language to Case Managerâ

Feature FTMT-29151 HSBC-3858

Spanish (Mexico) is now available in Case Manager, enabling users to work in Mexican Spanish across the application.

All translations are available in the Manage Messages page, introduced in Update Case Manager to version 13.28.0.

Upgrade Ansible Roles to version 34.1.0â

Feature HSBC-3850

We've upgraded Ansible Roles to version 34.1.0. This version updates the Case Manager pipeline to accept the `SELF_HEALING` Pulse application state, in addition to `STARTED` and `STARTING`.

We also fixed an issue where the pipeline waited for an application that did not always exist. Case Manager will now correctly identify the available applications and wait for them to reach their desired states.

Fix Case Manager logger configuration via Ansibleâ

Bug Fix FTMT-27146 HSBC-3842
Also In: 2.10.3 2.11.1 2.8.6 2.9.7

We have fixed an issue in which some logging levels were not correctly being updated via the `cm_loggers` Ansible variable due to a logger XML formatting issue.

Add new search field to Case Managerâ

Feature FTMT-28258 HSBC-3806

We've added the `Card Number` search field to the common filters in Case Manager's search page. The new search field maps to the `card_pan` event field.

Add Wallet Decision outcome to Cards and Non-Monâ

Feature FTMT-33585 HSBC-3804

We've added the `wallet_decision` outcome to the **Cards** workflow. The field was also added to the **Cards + Outcomes** schema and is included in the real-time response for requests to the Authorization and Non-Mon endpoints.

This outcome can have the following values by default:

- `RED`
- `YLW`
- `GRN`
- `none`

Enable the `External Status Reason Updater` extension for Cardsâ

Feature FTMT-28948 HSBC-3802

We have updated the `External Status Reason Updater` extension to allow Cards events.

The extension will now set the event's status reason based on the `external_status_reason` field value sent in the feedback event if it matches any Case Manager configured reasons.

Add the Non-Monetary channel to Pulse and Case Managerâ

Feature FTMT-30076 HSBC-3689
Feature FTMT-27453 FTMT-29259 FTMT-30076 HSBC-3690
Feature FTMT-28765 HSBC-3805
Feature FTMT-27479 HSBC-3803
Feature FTMT-27453 FTMT-30076 HSBC-3798

We've added the Non-Monetary (Non-Mon) channel to Pulse and Case Manager. Given this work has been done in multiple tasks, we've outlined the major changes in Non-Mon Channel.

Update database retention scriptsâ

Feature HSBC-3789

We've updated the database retention scripts to include the following tables:

- AM_EVENT_AUTOMATION
- AM_CASE_FILE
- AM_CASE_SOURCE_TYPE
- AM_EVENT_FILE
- AM_EXTENSION_LOG
- AM_FILE
- AM_EVENT_ACCESS_GROUP

See Retention Scripts for more information.

Add support for Zookeeper version 3.9â

Feature FTMT-30224 HSBC-3786

This release officially adds support for Zookeeper 3.9.

Update Case Manager to version 13.28.0â

Feature FTMT-29151 HSBC-3777

We've upgraded Case Manager to version 13.28.0.

This release introduces the Manage Messages feature, giving HSBC direct control over message translations in Case Manager.

This change also introduces a new set of permissions to control which users can access and manage Case Manager messages:

Messages Permission	Description
`cm:section:settings-message:read:*`	Lets the user see the Manager Messages page in the Settings section.
`cm:resource:message:read:*`	Lets the user read messages.
`cm:action:settings-message:create:*`	Lets the user create messages in the Settings section.
`cm:resource:message:create:*`	Lets the user create messages.
`cm:action:settings-message:update:*`	Lets the user update messages in the Settings section.

Messages Permission	Description
`cm:resource:message:update:*`	Lets the user update messages.
`cm:action:settings-message:delete:*`	Lets the user delete messages in the Settings section.
`cm:resource:message:delete:*`	Lets the user delete messages.

Please refer to [Case Manager Documentation] for more information and to the [Permissions Guide] for the full permissions list.

Add Reference Data fields to the Cards workflow schemasâ

Feature FTMT-32500 HSBC-3745
Feature FTMT-34761 HSBC-3993

We've added the following fields to the `Cards` and `Cards + Outcomes` schemas:

New Fields

Field Name	Datatype	Entity
account_closed_date	Long	Accounts
account_customers	List<String>	Accounts
product_type	String	Accounts
account_iban	String	Accounts
app_fraud_bio_is_bot	String	Customers
app_fraud_bio_is_emulator	String	Customers
app_fraud_bio_is_rat	String	Customers
app_fraud_bio_reason_code	String	Customers
app_fraud_bio_risk_level	String	Customers
app_fraud_bio_score	Double	Customers
app_fraud_overall_decision	String	Customers
app_fraud_overall_decision_text	String	Customers
app_fraud_session_id	String	Customers
app_fraud_source	String	Customers
app_fraud_timestamp	Long	Customers
app_fraud_tmx_device_id	String	Customers
app_fraud_tmx_digital_id_trust_score	Double	Customers
app_fraud_tmx_first_party_score	Double	Customers
app_fraud_tmx_fuzzy_device_id	String	Customers
app_fraud_tmx_fuzzy_device_id_confidence	Integer	Customers
app_fraud_tmx_policy_score	Double	Customers
app_fraud_tmx_proxy_ip	String	Customers
app_fraud_tmx_risk_rating	String	Customers
app_fraud_tmx_summary_reason_codes	String	Customers
app_fraud_tmx_true_ip	String	Customers
business_start_date	Long	Customers
customer_age	String	Customers
customer_alias_last_update_time	Long	Customers

Field Name	Datatype	Entity
customer_annual_turnover	Double	Customers
customer_annual_turnover_last_update_date	Timestamp	Customers
customer_branch_id	String	Customers
customer_business_type	String	Customers
customer_business_type_last_update_date	Timestamp	Customers
customer_businessphone_last_update_at	Timestamp	Customers
customer_corresp_address_last_update_at	Timestamp	Customers
customer_daytime_work_phone	String	Customers
customer_fiv_marker_last_updated_at	Timestamp	Customers
customer_fiv_markers	String	Customers
customer_hmphone_last_update_at	Timestamp	Customers
customer_home_address_last_update_at	Timestamp	Customers
customer_mbphone_last_update_at	Timestamp	Customers
customer_mobile_phone	String	Customers
customer_number_of_accounts	Integer	Customers
customer_occupation_status	String	Customers
customer_occupation_status_last_update_at	Timestamp	Customers
customer_portfolio_segment	String	Customers
customer_portfolio_segment_last_update_date	Timestamp	Customers
customer_reported_income_last_update_date	Timestamp	Customers
customer_voice_id	String	Customers
customer_voice_id_last_change_date	Timestamp	Customers
customer_wrkphone_last_update_at	Timestamp	Customers
delivery	String	Customers
employer_name	String	Customers
employer_name_last_update_date	Long	Customers
employer_start_date	Long	Customers
line_of_business	String	Customers

Note also that the following fields are new to the Customer Reference Data API:

- `customer_age`
- `delivery`

These can now be used to enrich Cards events via Reference Data.

Fix an issue with the ETL extraction MULTICHANNEL strategyâ

Bug Fix HSBC-3696

We've fixed an issue with the ETL extraction tool using the MULTICHANNEL routing strategy, where the extraction to Case Manager would fail if it found events whose channel was not mapped in `etl_cm_channel_queue_mappings`.

The MULTICHANNEL strategy can now be used to recover the events of a single channel from a multichannel workflow (e.g. recover only Digital Activity events).

We've also upgraded ETL to version 18.2.2 which reduces the amount of logging when a channel-queue mapping is missing.

Add model information to Cards and Non Mon responseâ

Feature FTMT-29505 HSBC-3668

We've updated the response object from the Cards authorization and Non Mon endpoint, to include the model information that was used to reach an outcome. The `model` field includes the following structured information:

```
    "model": [
        {
            "modelId": "6f41c74c7d491cd8",
            "modelVersionId": "53d00a6cc43347f7",
            "modelName": "dummy_model_test",
            "outcome": "isfraud",
            "score": [
                0.9340874776977786,
                0.0659125223022215
            ]
        }
    ]
```

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See up-to-date API documentation and Swagger definition at [Cards Authorization Endpoints]() and [Non Mon]().

Add Operational Status 'Unresolved'â

Feature FTMT-31854 HSBC-3663

We've added a new Operational status `Unresolved`, with the same functionality as `On-Hold`. It can be selected by clicking on the drop-down option `Move to Unresolved`, which is available in Alert Details and as a bulk action in Transaction History.

Improve the Case Manager `External Status Reason Updater` extensionâ

Feature FTMT-27146 FTMT-27219 HSBC-3396 HSBC-3620

We've improved the Case Manager `External Status Reason Updater` extension by eliminating the need for a database call to load the event's context. This improvement ensures that the extension runs with the exact event context that triggered it.

Extend improved currency to all amount and currency columnsâ

Feature WPB-349833 HSBC-3412

We've standardized the event tables across Case Manager using a single Amount column to represent currency amounts along with the respective currency symbol. This change uses the improved currency support added in [Improve currency display in Case Manager]().

warning

Please note that the Currency table column no longer exists in any event tables.

Extend the PaymentFulfillment extension to work for Cards and Non-Mon channelsâ

Feature FTMT-29506 HSBC-3391 HSBC-3999

We've updated the PaymentFulfillment extension to support Cards and Non-Mon channel events. On the back of this support we've also configured the fulfillment Cards and Non-Mon schemas, which can be found in [AFIS Extensions]().

Additionally, the state machines that may trigger PaymentFulfillment are now configurable during deployment by using the Ansible variable `fulfillment_channels_state_machines`. For more information regarding this feature please refer to the [Case Manager Configuration Reference](#).

Add `customer_id_number` to the Case Manager `Customer ID` search filterâ

Feature WPB-349829 HSBC-3132

We've added the `customer_id_number` field to the Case Manager `Customer ID` search filter, which now includes this field plus `customer_id` and `customer_citizen_id_number`.

Add support for multiple applications of the same typeâ

Feature HSBC-3095

We've added support for deployments with multiple applications of the same type.

The changes address two limitations:

- Allow overriding Liquibase database configurations: Liquibase now supports database configuration overrides per application, similar to what is supported for other configurations. For example, to use a different database for an equal portfolio application, the database override can be set as `<app_slug>_database_pulse_outbound_payments_eventstorage_hostname`
- Allow overriding the API ports for Outbound and Reference Data Service (RDS): this change introduces two new Ansible variables to configure in which port the Pulse and RDS APIs will listen for requests:
  - `api_port`: configure the API port for Outbound Payments
  - `rds_api_port`: configure the API port for RDS

Please refer to [Cluster with same portfolio applications](#) for a detailed explanation of the changes and to the [Pulse Configuration Reference](#) for deployment information.

Remove unused Pulse binariesâ

Rework HSBC-2122

We've removed a set of Pulse binaries related to the following unused services:

- workflow-list-manager-workflow-service
- expired-metrics-pulse-service
- fraudforce-workflow-service
- revelock-workflow-service
- revelock-udfs
- data-migration-service
- outcome-priority-enforcer-service
- enrichment-framework-lib

Along with the removal of these binaries, we have also removed the `import_solutions_udfs` Ansible variable given the binaries it imported are no longer available.

This change should not impact upgrades unless the current deployed version has a very early version installed (e.g. 1.6). In these versions the services were still part of the applications and Pulse required the binaries to be available before the more recent version could remove their reference.

If any deployment from the outlined early versions is planned, there are some steps that are required prior to the update:

- Keep only the following services from each application and removing all remaining using the Pulse UI:
  - Alert Storage Service
  - Data Transfer Pipeline Service
  - HTTP Input Service
  - Historic Events Service
  - Logging Admin Service
  - Workflow List Manager Service
  - Workflow Monitoring Service

- Remove the following properties:
    - `*.services.modules.fraudmodel.import.ListedKeywords`
    - `*.services.modules.datascience.context.import.ListedKeywords`
    - `*.services.modules.datascience.context.rules.import.ListedKeywords`
    - `*.services.modules.fraudrules.import.ListedKeywords`
    - `*.services.modules.fraudmodel.import.ValueFormatTools`
    - `*.services.modules.datascience.context.import.ValueFormatTools`
    - `*.services.modules.datascience.context.rules.import.ValueFormatTools`
    - `*.services.modules.fraudrules.import.ValueFormatTools`
    - `*.services.modules.fraudmodel.import.DateConversionTools`
    - `*.services.modules.datascience.context.import.DateConversionTools`
    - `*.services.modules.datascience.context.rules.import.DateConversionTools`
    - `*.services.modules.fraudrules.import.DateConversionTools`
    - `*.services.modules.fraudmodel.import.Revelock`
    - `*.services.modules.fraudrules.import.Revelock`
    - `*.services.modules.datascience.context.import.Revelock`
    - `*.services.modules.datascience.context.rules.import.Revelock`

Asynchronous Operation

The Cards Connector supports asynchronous operation. This page covers the foundational aspects to help you with the integration process.

Non-get endpoints support an `async` query parameter which, if `true`, will trigger the asynchronous processing of the request. If the `async` query parameter is set to false, the system will assure the synchronous operation. The default value for the `async` query parameter can be defined per endpoint using configuration.

When invoking an endpoint using the asynchronous operation mode, request validations such as authentication, authorization and payload specific validations (e.g. schema validations and timestamp validations) are executed immediately. Any invalid request will fail immediately as it would in the synchronous operation mode.

Once validations are successfully passed, the request is sent to a message broker (such as RabbitMQ) where it waits to be processed by the system at a controlled rate. Once the request is successfully persisted in the message broker, the asynchronous request will terminate with an HTTP 202 Accepted response, denoting that the request will be eventually processed. Since the request was not fully processed yet, any response fields that result from processing the request (most notably the score, decision and triggered actions on the Authorization endpoint will not be present in this response).

Requests are consumed from the message broker at a configurable rate and execute the same operations as they would if they were synchronous. Once the request is completed, a response will be produced and stored in a message broker. This response is the same as the one that would have been produced in synchronous operation mode. You can use the Responses Get endpoint to retrieve the responses from the message broker.

The responses obtained from the Responses Get endpoint can be correlated to the original request using the `lifecycle_id` response field, which will have the same value that was sent in the homonym request field.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/api-reference/cards/additional-resources/generated-fields

Generated Fields

The Cards API generates fields that will be sent to the Pulse Workflow alongside the ones received in the API.

event_typeâ

The `event_type` will be set according to the endpoint used to receive the request:

Endpoint	event_type
Authorization	authorization
Info	info
Clearing	clearing
Feedback	feedback

Check the Non-Mon Endpoint documentation for more information on how this field is handled in the Non-Mon Endpoint.

event_received_atâ

The `event_received_at` represents the time the request reached the Cards/Non-Mon API as a timestamp, in milliseconds.

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Non-Mon Endpoint

As of version 2.11.0 we have introduced a new Non-Mon (Non-Monetary) endpoint to allow Feedzai to receive and register non-monetary events.

In this page one can find the schema and event type specificity of these endpoints.

Schemaâ

The Non-Mon endpoint utilizes the same schema that is used for Cards events.

Given the endpoint is deployed in the Cards application and to keep the same consistency as all other applications (mainly Outbound), the schema of the endpoint is the same of all Cards endpoints.

Please find the API specification in Non-Mon API documentation.

Generated fieldsâ

The Non-Mon API generates fields that will be sent to the Pulse Workflow alongside the ones received in the API.

event_typeâ

Unlike all other existing endpoints, the non-mon endpoint has a different way of generating the `event_type` field. When receiving an event the endpoint will look for an existing `event_type` value in the payload. If the field does not exist or if it is an empty string then the endpoint will use the default value of `nonmon`.

This field can then be used to differentiate between various categories of non-monetary events (such as 3DS or Digital Wallet) directly in the event storage tables or in ETL extractions.

Please find below some examples:

Upstream `event_type`	Event actual `event_type` value
`empty` or `null`	`nonmon`
`nonmon`	`nonmon`
`digitalwallet`	`digitalwallet`
`3ds`	`3ds`

note

In the Case Manager UI the original type of the non-monetary event is always available as `schema.event_type`, while `event.fields.event_type` can be updated to `Info` or `Feedback` if the event receives an Info or a Feedback respectively.

fdz_channelâ

The `fdz_channel` field for Non-Mon requests will be set to `NON-MON`, which adds support for:

- Forwarding the requests to their own Case Manager channel
- Differentiating channel data in ETL exports
- Recovering messages from Pulse to Case Manager

Asynchronous Operation

The Digital Activity Connector is configured with synchronous operation out of the box. However, it also supports asynchronous operation. This page covers the foundational aspects to help you with the integration process.

Non-get endpoints support an `async` query parameter which, if `true`, will trigger the request's asynchronous processing. If the `async` query parameter is set to false, the system will assure the synchronous operation. The default value for the `async` query parameter can be defined per endpoint using the configuration.

When invoking an endpoint that is using the asynchronous operation mode, request validations such as authentication (e.g. schema validations and timestamp validations) are executed immediately. Any invalid request will fail immediately as it would in the synchronous operation mode.

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Authentication Response Scenarios

This section presents several scenarios that might occur in the Digital Activity's Authentication response, depending on the rules that are triggered in the Pulse workflow. Check the Authentication endpoint for more information about detailed request and response examples.

When no rules are triggered, the following fields and default values are shown in the Authentication endpoint's response:

- `"alert": false` This field is changed to 'true' when any rule with the action 'alert' is triggered. See examples of possible scenarios below.
- `"action_codes": []` This field is populated with the actions of the first triggered rule, other than the 'alert' action. Note that the 'alert' action is represented by the `alert` field.
- `"secondary_action_codes":[]` This field is populated with all the non-'alert' actions of every triggered rule, except the first triggered rule.

When rules are triggered, they match the following scenarios:

Scenario 1â

- **The first triggered rule:**
  - Action field - Alert
- **Excerpt from the Digital Activity's Authentication response:**

```
{
"alert": true,
"action_codes" = [],
"secondary_action_codes" = []
}
```

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Scenario 2â

- **The first triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Digital Activity's Authentication response:**

```
{
"alert": true,
"action_codes" = ["Action 1", "Action 2", "Action 3"],
"secondary_action_codes" = []
}
```

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Scenario 3â

- **The first triggered rule:**
  - Action field - Alert
- **The second triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Digital Activity's Authentication response:**

```
{
"alert": true,
"action_codes" = [],
"secondary_action_codes" = ["Action 1", "Action 2", "Action 3"]
}
```

Copy

Scenario 4â

- **The first triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **The second triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Digital Activity's Authentication response:**

```
{
"alert": true,
"action_codes" = ["Action 1", "Action 2", "Action 3"],
"secondary_action_codes" = ["Action 1", "Action 2", "Action 3"]
}
```

Copy

Scenario 5â

- **The first triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **The second triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **The third triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 4
- **Excerpt from the Digital Activity's Authentication response:**

```
{
"alert": true,
"action_codes" = ["Action 1", "Action 2", "Action 3"],
"secondary_action_codes" = ["Action 1", "Action 2", "Action 3", "Action 4"]
}
```

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Scenario 6â

- **The first triggered rule:**
  - This rule doesn't have actions.
- **The second triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Digital Activity's Authentication response:**

```
{
"alert": true,
"action_codes" = [],
"secondary_action_codes" = ["Action 1", "Action 2", "Action 3"]
}
```

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Generated Fields

The Digital Activity API generates fields that will be sent to the Pulse Workflow alongside the ones received in the API.

event_typeâ

The `event_type` will be set according to the endpoint used to receive the request:

Endpoint	event_type
Authentication	authentication
Credentials Update	credentials_update
Device	device
New Payee	new_payee
Profile Update	profile_update
Alias	alias

event_received_atâ

The `event_received_at` represents the time the request reached the Digital Activity API as a timestamp, in milliseconds.

Asynchronous Operation

The Inbound Payments Connector supports asynchronous operation. This page covers the foundational aspects to help you with the integration process.

Non-get endpoints support an `async` query parameter which, if `true`, will trigger the asynchronous processing of the request. If the `async` query parameter is set to false, the system will assure the synchronous operation. The default value for the `async` query parameter can be defined per endpoint using configuration.

When invoking an endpoint using the asynchronous operation mode, request validations such as authentication, authorization and payload specific validations (e.g. schema validations and timestamp validations) are executed immediately. Any invalid request will fail immediately as it would in the synchronous operation mode.

Once validations are successfully passed, the request is sent to a message broker (such as RabbitMQ) where it waits to be processed by the system at a controlled rate. Once the request is successfully persisted in the message broker, the asynchronous request will terminate with an HTTP 202 Accepted response, denoting that the request will be eventually processed. Since the request was not fully processed yet, any response fields that result from processing the request (most notably the score, decision and triggered actions on the Transfer Initiation endpoint will not be present in this response).

Requests are consumed from the message broker at a configurable rate and execute the same operations as they would if they were synchronous. Once the request is completed, a response will be produced and stored in a message broker. This response is the same as the one that would have been produced in synchronous operation mode. You can use the Responses Get endpoint to retrieve the responses from the message broker.

The responses obtained from the Responses Get endpoint can be correlated to the original request using the `lifecycle_id` response field, which will have the same value that was sent in the homonym request field.

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Generated Fields

The Inbound Payments API generates some fields which will be sent to the Pulse Workflow alongside the ones received in the API.

event_typeâ

The `event_type` will be set according to the endpoint used to receive the request:

Endpoint	event_type
Initiation	inbound_payment_initiation
Info	info
Settlement	settlement
Feedback	feedback

event_received_atâ

The `event_received_at` represents the time the request reached the Inbound Payments API as a timestamp, in milliseconds.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/api-reference/inbound-payments/additional-resources/inbound-initiation-response-scenarios

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Inbound Initiation Response Scenarios

This section presents several scenarios that might occur in the Inbound Payments Initiation response, depending on the rules that are triggered in the Pulse workflow. Check the Inbound Payments Initiation Endpoint for more information about detailed request and response examples.

When no rules are triggered, the following fields and default values are shown in the Inbound Payments Initiation response:

- `"alert": false` This field is changed to `true` when any rule with the action `alert` is triggered. See examples of possible scenarios below
- `"action_codes": []` This field is populated with the actions of the first triggered rule, other than the `alert` action. Note that the `alert` action is represented by the `alert` field
- `"secondary_action_codes":[]` This field is populated with all the non- `alert` actions of every triggered rule, except the first triggered rule

When rules are triggered, they match the following scenarios:

Scenario 1â

- **The first triggered rule:**
  - Action field - Alert
- **Excerpt from the Inbound Initiation response:**

```
{
    "alert": true,
    "action_codes": [],
    "secondary_action_codes": []
}
```

Copy

Scenario 2â

- **The first triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Inbound Initiation response:**

```
{
    "alert": true,
    "action_codes": ["Action 1", "Action 2", "Action 3"],
    "secondary_action_codes": []
}
```

Copy

Scenario 3â

- **The first triggered rule:**
  - Action field - Alert
- **The second triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Inbound Initiation response:**

```
{
    "alert": true,
    "action_codes": [],
    "secondary_action_codes": ["Action 1", "Action 2", "Action 3"]
}
```

Copy

Scenario 4â

- **The first triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **The second triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Inbound Initiation response:**

```
{
    "alert": true,
    "action_codes": ["Action 1", "Action 2", "Action 3"],
    "secondary_action_codes": ["Action 1", "Action 2", "Action 3"]
}
```

Copy

Scenario 5â

- **The first triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **The second triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **The third triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 4
- **Excerpt from the Inbound Initiation response:**

```
{
    "alert": true,
    "action_codes": ["Action 1", "Action 2", "Action 3"],
    "secondary_action_codes": ["Action 1", "Action 2", "Action 3", "Action 4"]
}
```

Copy

Scenario 6â

- **The first triggered rule:**
  - This rule doesn't have actions.
- **The second triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Inbound Initiation response:**

```
{
    "alert": true,
    "action_codes": [],
    "secondary_action_codes": ["Action 1", "Action 2", "Action 3"]
}
```

Copy

Asynchronous Operation

The Transfers Connector supports asynchronous operation. This page covers the foundational aspects to help you with the integration process.

Non-get endpoints support an `async` query parameter which, if `true`, will trigger the asynchronous processing of the request. If the `async` query parameter is set to false, the system will assure the synchronous operation. The default value for the `async` query parameter can be defined per endpoint using configuration.

When invoking an endpoint using the asynchronous operation mode, request validations such as authentication, authorization and payload specific validations (e.g. schema validations and timestamp validations) are executed immediately. Any invalid request will fail immediately as it would in the synchronous operation mode.

Once validations are successfully passed, the request is sent to a message broker (such as RabbitMQ) where it waits to be processed by the system at a controlled rate. Once the request is successfully persisted in the message broker, the asynchronous request will terminate with an HTTP 202 Accepted response, denoting that the request will be eventually processed. Since the request was not fully processed yet, any response fields that result from processing the request (most notably the score, decision and triggered actions on the Transfer Initiation endpoint will not be present in this response.

Requests are consumed from the message broker at a configurable rate and execute the same operations as they would if they were synchronous. Once the request is completed, a response will be produced and stored in a message broker. This response is the same as the one that would have been produced in synchronous operation mode. You can use the Responses Get endpoint to retrieve the responses from the message broker.

The responses obtained from the Responses Get endpoint can be correlated to the original request using the `lifecycle_id` response field, which will have the same value that was sent in the homonym request field.

On this page

Generated Fields

The Transfers API generates some fields which will be sent to the Pulse Workflow alongside the ones received in the API.

event_typeâ

The `event_type` will be set according to the endpoint used to receive the request:

Endpoint	event_type
Initiation	transfer_initiation
Info	info
Settlement	settlement
Feedback	feedback

event_received_atâ

The `event_received_at` represents the time the request reached the Transfers API as a timestamp, in milliseconds.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/api-reference/transfers/additional-resources/transfer-initiation-response-scenarios

On this page

Transfer Initiation Response Scenarios

This section presents several scenarios that might occur in the Transfer Initiation response, depending on the rules that are triggered in the Pulse workflow. Check the Transfer Initiation Endpoint for more information about detailed request and response examples.

When no rules are triggered, the following fields and default values are shown in the Transfer Initiation response:

- `"alert": false` This field is changed to `true` when any rule with the action `alert` is triggered. See examples of possible scenarios below
- `"action_codes": []` This field is populated with the actions of the first triggered rule, other than the `alert` action. Note that the `alert` action is represented by the `alert` field
- `"secondary_action_codes":[]` This field is populated with all the non-`alert` actions of every triggered rule, except the first triggered rule

When rules are triggered, they match the following scenarios:

Scenario 1â

- **The first triggered rule:**
    - Action field - Alert
- **Excerpt from the Transfer Initiation response:**

```
{
    "alert": true,
    "action_codes": [],
    "secondary_action_codes": []
}
```

Copy

Scenario 2â

- **The first triggered rule:**
    - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Transfer Initiation response:**

```
{
    "alert": true,
    "action_codes": ["Action 1", "Action 2", "Action 3"],
    "secondary_action_codes": []
}
```

Copy

Scenario 3â

- **The first triggered rule:**
    - Action field - Alert
- **The second triggered rule:**
    - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Transfer Initiation response:**

```
{
    "alert": true,
    "action_codes": [],
    "secondary_action_codes": ["Action 1", "Action 2", "Action 3"]
}
```

Copy

Scenario 4â

- **The first triggered rule:**
    - Action field - Alert + Action 1 + Action 2 + Action 3
- **The second triggered rule:**
    - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Transfer Initiation response:**

```
{
    "alert": true,
    "action_codes": ["Action 1", "Action 2", "Action 3"],
    "secondary_action_codes": ["Action 1", "Action 2", "Action 3"]
}
```

Copy

Scenario 5â

- **The first triggered rule:**
    - Action field - Alert + Action 1 + Action 2 + Action 3
- **The second triggered rule:**
    - Action field - Alert + Action 1 + Action 2 + Action 3
- **The third triggered rule:**
    - Action field - Alert + Action 1 + Action 2 + Action 4
- **Excerpt from the Transfer Initiation response:**

```
{
    "alert": true,
    "action_codes": ["Action 1", "Action 2", "Action 3"],
    "secondary_action_codes": ["Action 1", "Action 2", "Action 3", "Action 4"]
}
```

Copy

Scenario 6â

- **The first triggered rule:**
    - This rule doesn't have actions.
- **The second triggered rule:**
    - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Transfer Initiation response:**

```
{
    "alert": true,
    "action_codes": [],
    "secondary_action_codes": ["Action 1", "Action 2", "Action 3"]
}
```

Copy

On this page

Deploy a new channel

In Case Manager the entire process of controlling which channels are deployed is controlled during the deployment phase. In this guide you can find the necessary configurations and commons issues when deploying new channels.

Case Manager channel relation to Pulse applicationsâ

Channel configuration in Case Manager is tightly coupled with the applications deployed in Pulse.

While Pulse does not require that Case Manager deploys all channels related to existing Pulse applications, the other way around is different. Any Case Manager channel must have a matching Pulse application otherwise might fail to fetch metadata from the configured application during startup.

Configuring channelsâ

note

By default all variables for the entire UK market are already provided

Channel configuration in Case Manager is done using the following variables:

- `apps_enabled`
- `app_slug`
- `app_portfolio`

By default `apps_enabled` in Case Manager is configured to deploy the UK outbound channels which includes Outbound Payments (Transfers) and Digital Activity:

```
apps_enabled:
  - uk_outbound_payments
```

Copy

Please find below two examples on how to deploy a new channel for the UK market and for a new market.

Deploying a new channel in the UKâ

As stated previously, all UK channel configurations are supported by default, being the only requirement to override `apps_enabled`. This can be done by overriding the default variable, preferably in its own inventory, as follows:

1. Add a new base inventory for the portfolio/environment you are configuring, e.g. `project-qa/uk/group_vars`
2. Add a new host inventory for Case Manager, e.g. `project-qa/uk/group_vars/case-manager`
3. Inside a new variables file (`variables.yml`), under the inventory created previously, add the overrides for the specific portfolio/environment

```
apps_enabled:
  - uk_outbound_payments
  - uk_inbound_payments
# Other overrides
```

Copy

Deploying in a new marketâ□□

Aside from `apps_enabled` any additional market needs to also configure `app_slug` and `app_portfolio`. In the following example we describe how to deploy `outbound` and `inbound` for the HK market:

1. Add a new base inventory as described in the first point of [Deploying a new channel in the UK]() for HK (`project-qa/hk/group_vars`)
2. Add a new inventory for all hosts, e.g. `project-qa/hk/group_vars/all`. This step is optional if this was already created to deploy an application in Pulse
3. Inside the previous folder create two files
    1. `hk_outbound_payments.yml` which holds the configuration for the HK outbound application

    ```
     hk_outbound_payments_app_slug: hk_outbound_payments # should match the app_slug in Pulse
    hk_outbound_payments_market: hk
    hk_outbound_payments_portfolio: outbound_payments
    ```

    Copy

    2. `hk_inbound_payments.yml` which holds the configuration for the HK inbound application

    ```
     hk_inbound_payments_app_slug: hk_inbound_payments # should match the app_slug in Pulse
    hk_inbound_payments_market: hk
    hk_inbound_payments_portfolio: inbound_payments
    ```

    Copy

4. In the end you should have a structure as follows

```
.
â□□â□□â□□ project-default
â□□   â□□â□□â□□ group_vars
â□□       â□□â□□â□□ ...
â□□â□□â□□ project-hsbc
â□□   â□□â□□â□□ hosts.yml
â□□â□□â□□ gcp
â□□   â□□â□□â□□ group_vars
â□□       â□□â□□â□□ ...
â□□â□□â□□ project-qa # being qa what was defined previously
    â□□â□□â□□ hk
        â□□â□□â□□ group_vars
            â□□â□□â□□ all
            â□□   â□□â□□â□□ hk_outbound_payments.yml
            â□□   â□□â□□â□□ hk_inbound_payments.yml
            â□□â□□â□□ case-manager
                â□□â□□â□□ variables.yml
```

Copy

5. In the `project-qa/hk/group_vars/case-manager/variables.yml` set the channels to deploy

```
apps_enabled:
  - hk_outbound_payments
  - hk_inbound_payments
# Other overrides
```

Copy

6. Finally during deployment pass your newly created inventory

```
python3.6 ansible-playbook \
    -i project-default \
    -i project-hsbc \
    -i gcp \
    -i project-qa/hk \
    case-manager.yml
```

Copy

ⓘnote

The structure described in point 4 is optional. In this example we have segregated by environment and market but another approach could be applied, for example by segregating first by market and then by environment:

```
.
â â â  hk
    â â â  project-qa
    â      â â â  ...
    â â â  project-preprod
        â â â  ...
```

Copy

Common Issuesâ

Please find below a description of common deployment issues.

Deployments fails due to missing variablesâ

This issue may occur if `app_slug` and `app_portfolio` are undefined as illustrated below.

```
AnsibleError: An unhandled exception occurred while templating '...' original message: An
unhandled exception occurred while templating '{{ dict(app_slugs | zip(app_portfolios)) |
dict2items(key_name='app_slug', value_name='app_portfolio') | list }}
'. Error was a <class 'ansible.errors.AnsibleError'>, original message: An unhandled exception
occurred while templating '{{ apps_enabled | map('regex_replace', '^(.*)$', '\1_app_slug') |
map('extract', vars) | list }}
'. Error was a <class 'KeyError'>, original message: 'hk_inbound_payments_app_slug'
```

Copy

As you can see, Ansible is telling us that `hk_inbound_payments_app_slug` is undefined but it is required to compute the channel configuration.

While this should not occur for the UK market, since the project assembly includes are UK configurations, it can happen for other markets which in this case is the HK market.

The fix in this case would be to create the file structure to support this market, as described in [Deploying in a new market](#).

Case Manager fails to startâ

During startup Case Manager can fail with the below error although the cause can be two different errors.

```
2024-08-22 16:02:30,490 [main] ERROR c.f.a.w.v.DefaultValidationStrategy - Validation errors
detected.
com.feedzai.am.api.metadata.ValidationException: The following errors were found while
initializing Case Manager:

State Machines (4 errors)
1) There was an error connecting to pulse to retrieve the configured outcomes.
```

Copy

This error indicates that at least one channel in Case Manager was unable to fetch the required information from Pulse. The root cause for this issue can be:

1. The application is not running/deployment in Pulse

This issue can be fixed by deploying the application in Pulse and recreating/restarting Case Manager

1. The application id is wrongly configured and does not match any application running in Pulse

For the second issue, see the following example:

- The `app_slug` for a Pulse inbound application is `hk_inbound_payments`
- In Case Manager we configured the `hk_inbound_payments_app_slug` to be `hk_inbound`

As you can see the `app_slugs` do not match. During deployment Case Manager calculates the application id according to the configured `<market>_<channel>_app_slug` but in this case it will not match any application running in Pulse.

To fix this issue you would have to change `hk_inbound_payments_app_slug` from `hk_inbound` to `hk_inbound_payments` and redeploy Case Manager.

See [Deploying in a new market] for more information.

Best Practicesâ

To ensure a smooth deployment see the following set of best practices.

Double-check consistencyâ

Before deploying always verify that:

- Case Manager `apps_enabled` matches the deployed applications in Pulse
- The channel configuration (`app_slug` and `app_portfolio`) match the corresponding application configuration in Pulse

Use environment specific configurationsâ

Follow a inventory structure segregated by market and environment, for instance, as described in [Deploying in a new market]. With this structure you guarantee that a component has the correct configuration for its specific market and environment.

info

Find a detailed description of all available configurations in [Case Manager Configuration Recommendations]

On this page

Use channel specific databases

Case Manager supports using separate databases for channel-specific tables, optimizing performance by distributing queries and storage across multiple databases. This setup provides effective isolation between channels and improves overall system efficiency.

Overviewâ

Database segregation is done in a main-dedicated relation, meaning the main database will hold all `AM_*` tables which include system tables and the `AM_EVENT` and `AM_EVENT_STORAGE` for the channels that are configured to use this database. On the other hand the dedicated database will store only the `AM_EVENT` and `AM_EVENT_STORAGE` tables for the channels that are configured to use it.

In the following sections we will demonstrate how to configure a dedicated database for one channel while keeping the others running against the main database.

Configuring channel-specific databasesâ

warning

Ensure that the required databases are created with sufficient CPU resources, storage, and the necessary permissions replicated from the base database. Confirm that firewall rules are configured to allow Case Manager to connect to each specific database.

Also ensure that each database has adequate maximum connections available. While channel-specific databases typically require fewer connections, it is recommended to set the connection limits similar to those of the base database to ensure smooth operation.

In this section we demonstrate how to configure Case Manager with three different channels where two will use the main database and the other a dedicated database.

As an example consider the following:

- We have deployed two databases which we will consider as main and dedicated. Each database is using a proxy running locally where port `3307` connects to the transfers database and port `3308` to the inbound one
- Outbound (transfers) and Digital Activity channels will use the main database
- Inbound will use the dedicated database
- For simplicity we will consider there is no read only database in the setup

The first thing to do is to configure `pdb_active_channels` as follows:

```
pdb_active_channels:
  - transfers
  - inbound-payments
```

Copy

With this configuration the deployment will:

- Use the `transfers` configured database as the main database. Since `transfers` and `digital-activity` are usually tied together, by omitting the latter from the configuration the channel will use the `transfers` database
- Configure `inbound-payments` to use a dedicated database

warning

The item order in `pdb_active_channels` matters. Having `transfers` as the first item is not the same as having it in any other order since the first item's configured database will be considered as the main database.

The next step is to set the variables that configure the connection and connection options for each channel. These variables extend the following ones by adding the channel suffix (`_<channel>`) to them:

- `database_casemanager_jdbc`
- `database_casemanager_username`
- `database_casemanager_password`
- `database_casemanager_schema`
- `database_casemanager_schemapolicy`
- `database_casemanager_pool_size`
- `database_casemanager_pool_lazy`
- `database_casemanager_reconnect_on_lost`
- `database_casemanager_max_number_retries`
- `database_casemanager_retry_interval`
- `database_casemanager_query_select_timeout`

By default the suffixed variables use the value from the database variables defined in [Case Manager Configuration Recommendations](#).

info

While in this example we are omitting the read only database, the same variable extensibility applies in the form of `database_casemanager_ro_*_<channel>` .

For simplicity let us assume all database variables are the same except for the connection string. At this point we already have configured them according to the Case Manager configuration recommendations and we are only missing setting the channel specific connection strings:

```
 # Configure transfers connection string
# Note that database_casemanager_jdbc_opts can also be suffixed
database_casemanager_jdbc_transfers: "jdbc:postgresql://localhost:3307/feedzai?
{{ database_casemanager_jdbc_opts | join('&') }}"

# Configure inbound payments connection string
# Note that database_casemanager_jdbc_opts can also be suffixed
database_casemanager_jdbc_inbound_payments: "jdbc:postgresql://localhost:3308/feedzai?
{{ database_casemanager_jdbc_opts | join('&') }}"
```

Copy

Deploying Case Manager will now set the `transfers` database as main, `digital-activity` will use the same database and `inbound-payments` will use its own database to store event information.

On this page

Deploy new applications

In Pulse the entire process of controlling which applications are deployed is controlled during the deployment phase. In this guide you can find the necessary configurations and commons issues when deploying new channels.

Pulse application relation to Case Manager channelsâ

Pulse does not need to know which channels are deployed in Case Manager since applications just publish events to their channel specific queue for Case Manager to consume. If Case Manager (or if a channel for a given application) is not deployed, events will still be published to the queues but will be left unconsumed.

Deploying applicationsâ

note

By default all variables for the entire UK market are already provided

Application configuration in Pulse is done using the following variables:

- `apps_enabled`
- `app_slug`
- `<app>_market`
- `<app>_portfolio`

By default `apps_enabled` in Pulse is configured to deploy the UK outbound application which includes Outbound Payments (Transfers) and Digital Activity, and also the Reference Data application:

```
apps_enabled:
  - uk_outbound_payments
  - uk_reference_data
```

Copy

Please find below three examples on how to deploy a new application for the UK market, one for a new market (HK) and one final one for a new market that is not configured by default.

Deploying a new application in the UKâ

As stated previously all UK applications are supported by default, being the only requirement to override `apps_enabled`. This can be done by overriding the default variable, preferably in its own inventory, as follows:

1. Add a new base inventory for the portfolio/environment you are configuring, e.g. `project-qa/uk/group_vars`
2. Add a new host inventory for Pulse, e.g. `project-qa/uk/group_vars/pulse`
3. Inside a new variables file (`variables.yml`), under the inventory created previously, add the overrides for the specific portfolio/environment

   ```
   apps_enabled:
     - uk_outbound_payments
     - uk_reference_data
     - uk_inbound_payments
   # Other overrides
   ```

   Copy

warning

`apps_enabled` is incremental. This means that after an application is deployed, new applications should be added to the variable but the existing ones shouldn't be deleted.

This applies to all deployments including Topology-Aware Resource Scheduling.

Deploying in a new market (HK)â

The HK market is configured by default except for the applications that should be deployed in it.

Aside from `apps_enabled` every applications requires to have the following variables:

- `app_slug`
- `app_market`
- `app_portfolio`

In the example we describe how to deploy `outbound`, `reference data` and `inbound` for the HK market:

1. Add a new base inventory as described in the first point of Deploying a new application in the UK for HK (`project-qa/hk/group_vars`)
2. Add a new inventory for all hosts, e.g. `project-qa/hk/group_vars/all`. This step is optional if this was already created to deploy a channel in Case Manager
3. Inside the previous folder create three files
    1. `hk_outbound_payments.yml` which holds the configuration for the HK outbound application

    ```
    hk_outbound_payments_app_slug: hk_outbound_payments
    hk_outbound_payments_market: hk
    hk_outbound_payments_portfolio: outbound_payments
    ```

    Copy
    2. `hk_reference_data.yml` which holds the configuration for the HK reference data application

    ```
    hk_reference_data_app_slug: hk_reference_data
    hk_reference_data_market: hk
    hk_reference_data_portfolio: reference_data
    ```

    Copy
    3. `hk_inbound_payments.yml` which holds the configuration for the HK inbound application

    ```
    hk_inbound_payments_app_slug: hk_inbound_payments
    hk_inbound_payments_market: hk
    hk_inbound_payments_portfolio: inbound_payments
    ```

    Copy
4. In the end you should have a structure as follows

```
.
âââ project-default
â   âââ group_vars
â       âââ ...
âââ project-hsbc
â   âââ hosts.yml
âââ gcp
â   âââ group_vars
â       âââ ...
âââ project-qa # being qa what was defined previously
    âââ hk
        âââ group_vars
            âââ all
            â   âââ hk_outbound_payments.yml
            â   âââ hk_reference_data.yml
            â   âââ hk_inbound_payments.yml
```

```
        â□□â□□â□□ pulse
            â□□â□□â□□ variables.yml
```

Copy

5. In the `project-qa/hk/group_vars/pulse/variables.yml` set the applications to deploy

```
apps_enabled:
  - hk_outbound_payments
  - hk_reference_data
  - hk_inbound_payments
# Other overrides
```

Copy

6. Finally during deployment pass your newly created inventory

```
python3.6 ansible-playbook \
    -i project-default \
    -i project-hsbc \
    -i gcp \
    -i project-qa/hk \
    pulse.yml
    -e deployment_color=<color_to_deploy>
```

Copy

ⓘnote

The structure described in point 4 is optional. In this example we have segregated by environment and market but another approach could be applied, for example by segregating first by market and then by environment:

```
.
â□□â□□â□□ hk
    â□□â□□â□□ project-qa
    â□□    â□□â□□â□□ ...
    â□□â□□â□□ project-preprod
        â□□â□□â□□ ...
```

Copy

Deploying in a new, non-default marketâ□□

Deployments for new markets require the same steps described Deploying in a new market (HK) with a small addition. For the sake of this example we assume the new market to be New Market (NM).

If you see the `markets` variable that is included in the assembly, you will notice there are two markets defined:

```
markets:
  uk: UK
  hk: Hong Kong
```

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For any additional market deployment this variable will need to be overridden as follows:

1. Follow step one described in Deploying a new application in the UK for the new market (project-qa/nm/group_vars)
2. Add a new host inventory for Pulse, e.g. `project-qa/nm/group_vars/pulse`
3. Inside a new variables file (`variables.yml`), under the inventory created previously, add the overrides for the specific portfolio/environment

```
markets:
  uk: UK
  hk: Hong Kong
```

```
  nm: New Market
# Other overrides
```

Copy

tip

To make this variable accessible in all deployments, it could be overridden in `project-qa/group_vars/all/variables.yml` instead

After setting up the new market follow the remaining steps described in Deploying a new application in the UK to set up the applications to deploy.

Application templates use the following structure:

```
 <market>_<appName>_app_slug: <market>_<appName>
<market>_<appName>_market: <market>
<market>_<appName>_portfolio: <appName>
```

Copy

Where `market` is the market where the application belongs to (as defined in the `markets` variable) and `appName` the application name in Snake Case.

As an example, the configuration for the NM Outbound application would be as follows:

```
 # nm_outbound_payments.yml
nm_outbound_payments_app_slug: nm_outbound_payments
nm_outbound_payments_market: nm
nm_outbound_payments_portfolio: outbound_payments
```

Copy

Correlating with the `apps_enabled` by the `<app>_app_slug` configuration:

```
 apps_enabled:
   - nm_outbound_payments
# Other overrides
```

Copy

To deploy the new application we would include the new inventory in the deployment command:

```
 python3.6 ansible-playbook \
     -i project-default \
     -i project-hsbc \
     -i gcp \
     -i project-qa/nm \
     pulse.yml
     -e deployment_color=<color_to_deploy>
```

Copy

info

Find a detailed description of all available configurations in Pulse Configuration Recommendations

On this page

Use application specific databases

Pulse supports using separate databases for application-specific tables, optimizing performance by distributing queries and storage across multiple databases. This setup provides effective isolation between applications and improves overall system efficiency.

Overviewâ

In this guide we will do an overview on how to configure application specific databases with the assumptions:

- The deployment includes Outbound and Inbound Payments with a database for each (`DB1` for Outbound and `DB2` for Inbound)
- The Outbound Payments database was chosen to hold all applications' metadata

In practice, the result of these assumptions means `DB2` will hold only the Event Storage table, where all events Inbound events will be stored. On the other hand `DB1` will store application metadata for Outbound and Inbound Payments' applications and also the Outbound Payments Event Storage.

In the following section we will demonstrate how to configure a dedicated database for each application.

Configuring application-specific databasesâ

warning

Ensure that the required databases are created with sufficient CPU resources, storage, and the necessary permissions replicated from the base database. Confirm that firewall rules are configured to allow Pulse to connect to each specific database.

Also ensure that each database has adequate maximum connections available. While application-specific databases typically require fewer connections, it is recommended to set the connection limits similar to those of the base database to ensure smooth operation.

As an example consider the following:

- We have deployed two databases to support the two applications to deploy. Each database is using a proxy running locally where port `3307` connects to the outbound/metadata (`DB1`) database and port `3308` to the inbound one (`DB2`)
- For simplicity we will consider there is no read only database in the setup
- Both applications will be deployed in the same deployment process
- Both applications are for the UK market

The first step is to create a new inventory for the deployment as described in [Deploying a new application in the UK](), where any specific variables for the deployment will be defined.

Since both applications are to be deployed in the same deployment, we need to add application specific variables. Without it both applications would use the same values to define their database connections. To achieve this we will change `project-qa/pulse/group_vars/pulse/variables.yml` as follows:

```
# Define applications to deploy
apps_enabled:
  - uk_outbound_payments
  - uk_reference_data
  - uk_inbound_payments
```

```
# Set metadata jdbc configuration
database_pulse_metadata_hostname: "localhost"
database_pulse_metadata_port:     "3307"

# Set outbound jdbc configuration
# Take the same value as the metadata configuration since they will share the same database
database_pulse_eventstorage_hostname: "{{ database_pulse_metadata_hostname }}"
database_pulse_eventstorage_port:     "{{ database_pulse_metadata_port }}"

# Set inbound jdbc configuration
database_pulse_inbound_payments_eventstorage_hostname: "localhost"
database_pulse_inbound_payments_eventstorage_port:     "3308"
```

Copy

Notice two things in the example above:

- By default channel based variables point to the default (without channel) variables
- Since we want to defined a different Inbound Payments database connection string we need to override its respective variables. These follow the syntax `database_pulse_<channel>_eventstorage_<variable_name>` as you can see above

warning

For versions earlier than 2.7.0 channel based variables are defined in the format `<app_name>_<variable_name>`, where `<app_name>` is the application name as defined in `apps_enabled`.

In the context of this guide, we would configure the connection string as follows:

```
 # Set inbound jdbc configuration
uk_inbound_payments_database_pulse_eventstorage_hostname: "localhost"
uk_inbound_payments_database_pulse_eventstorage_port:     "3308"
uk_inbound_payments_database_pulse_eventstorage_jdbc:     "jdbc:postgresql://
{{ uk_inbound_payments_database_pulse_eventstorage_hostname }}:
{{ uk_inbound_payments_database_pulse_eventstorage_port }}/
{{database_pulse_eventstorage_database_name }}?{{ database_pulse_eventstorage_jdbc_opts |
join('&') }}"
```

Copy

The deployment would continue by issuing the installation command:

```
 python3.6 ansible-playbook \
    -i project-default \
    -i project-hsbc \
    -i gcp \
    -i project-qa/uk \
    pulse.yml
    -e deployment_color=<color_to_deploy>
```

Copy

Assuming the deployment ran without issues, you should be able to see the configuration in the Pulse configuration page as illustrated below.

note

We would not need to define the Inbound variables as `<app>_<variable_name>` if Inbound was deployed on a different Pulse node. We could instead have the default variables defined (as we did for Outbound) and pass a application-specific inventory during deployment

info

Find a detailed description of all available configurations in Pulse Configuration Recommendations

On this page

Case Manager Setup

In this page you can find a set of suggestions for Case Manager monitoring using Grafana.

Suggestionsâ

info

There are VM resource related metrics (CPU, Memory, IO, etc) that can also be added to Grafana. These were left out on purpose Google already has out of the box monitoring support for these resources.

These suggestions can be either new dashboards, changes to existing dashboards or completely new dashboard pages. Please find the metric description in Case Manager Top Metrics.

Cluster pageâ

Currently there is only one page where we can filter metrics by each of the instances. Although useful, having a cluster view monitoring page gives a view of how the overall cluster health. Given this, our suggestion is to add a new page with the following metrics:

Row Title	Title	Description	Query Details
Healthchecks - Case Manager	Database	The database health check relies on the shared CM's connection pool to execute the statement SELECT 1. If the connection is successfully retrieved and the statement finishes, the health check succeeds, if any error occurs in any of the steps the check fails. A timeout of 5 seconds was given when retrieving a connection to avoid hanging the health check response. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'Database' , $timeFilter **Group By:** time($__interval), "host"
Healthchecks - Case Manager	Messaging	The Messaging Server Health Check is achieved by trying to initialize a new connection with the messaging server. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'Messaging' , $timeFilter **Group By:** time($__interval), "host"
Healthchecks - Case Manager	Pulse	The Pulse Health Check when triggered tries to connect to pulse and tries to login. The outcome is a bool,	**Query A** **Field:** "result" **From:** "healthchecks"

Row Title	Title	Description	Query Details
		meaning that it was successfully executed when 1, otherwise is 0.	**Where:** "healthcheck" = 'Pulse' , $timeFilter **Group By:** time($__interval), "host"
Healthchecks - Case Manager	Tokenizer	The tokenizer health checks validates if there is a tokenizer instance reachable and if the authorization configured is valid. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'Tokenizer' , $timeFilter **Group By:** time($__interval), "host"
Healthchecks - Case Manager	RabbitMQ Consumers per Event Queue	Measures the total number of consumers per event queue. Having a number of consumers less then the number of nodes represent a problem in the sense that only one/some of the casemanager(s) is/are consuming events, not distributing the load.	**Query A** **Field:** "consumers" **From:** "rabbitmq_queue" **Where:** queue =~ /*.EVENT_QUEUE.*/ , vhost =~ / hsbcasemanager/ , $timeFilter **Group By:** time($__interval), "queue", "vhost"
Case Manager End-to-End Check	End-to-End Check	Measures the execution of a operation set, constituted by 5 operations: - login - get alert list - load first alert list - get all queues - load first queue The operations are executed sequentially, and if one fails the value is immediately returned, meaning that, the set is successfully executed when the outcome is 5. It means also, that, when value is 1 for instance, the login was successfully done, but if fails getting the alert list.	**Query A** **Field:** "operations" **From:** "casemanager_monitoring" **Where:** $timeFilter **Group By:** time($__interval)
Case Manager End-to-End Check	End-to-End Check - Elapsed Time	Measures the elapsed time to execute a set of operations, per case manager instance. This graph is coupled with the "Operation Set Execution", once it shares the same operations The operations are executed sequentially, and if one fails the elapsed time will be the the time when the operation failed minus the start time.	**Query A** **Field:** "elapsed_time" **From:** "casemanager_monitoring" **Where:** $timeFilter **Group By:** time($__interval)
Event Processing - New Event	New Event - Throughput	Measures the overall throughput for the event processing when the event type is: "new event".	**Query A** **Field:** "count" **From:** "cm.event.processing.latency" **Where:** eventType='NEW_EVENT' , $timeFilter

Row Title	Title	Description	Query Details
			Group By: time($__interval), "host" **Query C** **Field:** "uptime" **From:** "system" **Where:** $timeFilter **Group By:** time($interval), "host"
Event Processing - New Event	New Event - Latency per Percentile	Measures the overall cluster event processing max latency, when the event type is "new event". It displays the higher latencies for percentile 50, 90, 99, 99.9 and also the maximum.	**Query A** **Field:** "p50", "p95", "p99", "p999" **From:** "cm.event.processing.latency" **Where:** eventType='NEW_EVENT' , $timeFilter **Group By:** time($__interval), "host" **Query B** **Field:** "max" **From:** "cm.event.processing.latency" **Where:** eventType='NEW_EVENT' , $timeFilter **Group By:** time($__interval)
Event Processing - Status Change	Status Change - Throughput	Measures the overall throughput for the event processing when the event type is: "status change".	**Query A** **Field:** "count" **From:** "cm.event.processing.latency" **Where:** eventType='STATUS_CHANGE' , $timeFilter **Group By:** time($__interval), "host"
Event Processing - Status Change	Status Change - Latency per Percentile	Measures the overall cluster event processing max latency, when the event type is "status change". It displays the higher latencies for percentile 50, 90, 99, 99.9 and also the maximum.	**Query A** **Field:** "p50", "p95", "p99", "p999" **From:** "cm.event.processing.latency" **Where:** eventType='STATUS_CHANGE' , $timeFilter **Group By:** time($__interval), "host" **Query B** **Field:** "max" **From:** "cm.event.processing.latency" **Where:** eventType='STATUS_CHANGE' , $timeFilter **Group By:** time($__interval)
Automation Rules	Automation Rules - Throughput and Errors	Measures the overall throughput and errors when executing automation rules.	**Query A** **Field:** "count" **From:** "cm.automation" **Where:** $timeFilter **Group By:** time($__interval), "host", "ruleName"
Automation Rules	Automation Rules - Latency per Percentile	Measures the overall cluster latency at percentile 99 to execute an automation rule. It displays highest automation rule latency at percentile 99.9 per instance.	**Query A** **Field:** "p999" **From:** "cm.automation" **Where:** $timeFilter **Group By:** time($__interval), "host", "ruleName"
Pulse Requests	Pulse Requests - Throughput and Errors	Measures the overall throughput and the errors in the communication with pulse instances.	**Query A** **Field:** "count" **From:** "cm.request.pulse.api" **Where:** $timeFilter **Group By:** time($__interval), "host", "path"
Pulse Requests	Pulse HTTP Endpoints and Latencies	Measures the latency in the communication with pulse instances.	**Query A** **Field:** "count", "p99" **From:** "cm.request.pulse.api" **Where:** $timeFilter **Group By:** "host", "path"
Event Processing	Event Processing Queue Occupancy	Measures the event processing queue occupancy	**Query A** **Field:** "value"

Row Title	Title	Description	Query Details
		(internal cm queue), per case manager instance.	**From:** "cm.event.processing.queue.ratio" **Where:** $timeFilter **Group By:** time($__interval), "host"
Event Processing	Event Processing Batch audit-batch Queue Occupancy	Measures the audit batch queue occupancy (internal cm queue), per case manager instance.	**Query A** **Field:** "value" **From:** "cm.event.processing.queue.occupancy.audit-batch" **Where:** $timeFilter **Group By:** time($__interval), "host"
Event Processing	Event Processing Batch audit-batch Flush Latency at p99	Measures the audit batch queue latency (internal cm queue), per case manager instance.	**Query A** **Field:** "p99" **From:** "cm.event.processing.batch.flush.latency.audit-batch" **Where:** $timeFilter **Group By:** time($__interval), "host"
Event Processing	Event Processing Batch explanations-batch Queue Occupancy	Measures the explanations batch queue occupancy (internal cm queue), per case manager instance.	**Query A** **Field:** "value" **From:** "cm.event.processing.queue.occupancy.explanations-batch" **Where:** $timeFilter **Group By:** time($__interval), "host"
Event Processing	Event Processing Batch explanations-batch Flush Latency at p99	Measures the explanations batch queue latency (internal cm queue), per case manager instance.	**Query A** **Field:** "p99" **From:** "cm.event.processing.batch.flush.latency.explanations-batch" **Where:** $timeFilter **Group By:** time($__interval), "host"
Event Processing	Event Processing Batch event-storage-batch Queue Occupancy	Measures the event storage batch queue occupancy (internal cm queue), per case manager instance.	**Query A** **Field:** "value" **From:** "cm.event.processing.queue.occupancy.event-storage-batch" **Where:** $timeFilter **Group By:** time($__interval), "host"
Event Processing	Event Processing Batch event-storage-batch Flush Latency at p99	Measures the event storage batch queue latency (internal cm queue), per case manager instance.	**Query A** **Field:** "p99" **From:** "cm.event.processing.batch.flush.latency.event-storage-batch" **Where:** $timeFilter **Group By:** time($__interval), "host"
Event Processing	Event Processing Batch event-common-batch Queue Occupancy	Measures the event common batch queue occupancy (internal cm queue), per case manager instance.	**Query A** **Field:** "value" **From:** "cm.event.processing.queue.occupancy.event-common-batch" **Where:** $timeFilter **Group By:** time($__interval), "host"
Event Processing	Event Processing Batch event-common-batch Flush Latency at p99	Measures the event common batch queue latency (internal cm queue), per case manager instance.	**Query A** **Field:** "p99" **From:** "cm.event.processing.batch.flush.latency.event-common-batch" **Where:** $timeFilter **Group By:** time($__interval), "host"
Event Processing	Event Processing Batch event-queue-automations-batch Queue Occupancy	Measures the event queue automations batch queue occupancy (internal cm	**Query A** **Field:** "value" **From:** "cm.event.processing.queue.occupancy.event-queue-automations-batch"

Row Title	Title	Description	Query Details
		queue), per case manager instance.	**Where:** $timeFilter **Group By:** time($__interval), "host"
Event Processing	Event Processing Batch event-queue-automations-batch Flush Latency at p99	Measures the event queue automations batch queue latency (internal cm queue), per case manager instance.	**Query A** **Field:** "p99" **From:** "cm.event.processing.batch.flush.latency.event-queue-automations-batch" **Where:** $timeFilter **Group By:** time($__interval), "host"
Deadletters	Deadletter Events	Measures the total number of deadletters.	**Query A** **Field:** "count" **From:** "cm.event.processing.deadletter.count" **Where:** $timeFilter **Group By:** time($__interval), "host"
DB Connection Pool	DB Connection Pool usage	Measures the overall db connection pool usage.	**Query A** **Field:** "value" **From:** "cm.db.connection.pool.usage" **Where:** $timeFilter **Group By:** time($__interval), "host"
DB Connection Pool	DB Connection Pool Waiting Threads	Measures the overall db connection pool waiting threads.	**Query A** **Field:** "count" **From:** "cm.db.connection.pool.waiting" **Where:** $timeFilter **Group By:** time($__interval), "host"
Case Manager HTTP Endpoints and Latencies	Case Manager HTTP Endpoints and Latencies	Measures the latency of Case Manager HTTP endpoints.	**Query A** **Field:** "count", "p99" **From:** "cm.request.api" **Where:** $timeFilter **Group By:** "host", "path"
Login Errors	Login Error Percent(10 Minute Window)	Measures the number of logins and its error percentage over a 10 minute window.	**Query A** **Field:** "value" **From:** "cm.request.login" **Where:** $timeFilter **Group By:** time($__interval), "host"
JVM	JVM Heap Usage	Measures the jvm heap usage per component instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.heap.usage" **Where:** $timeFilter **Group By:** time($__interval), "host"
JVM	JVM Heap Usage After GC	Measures the jvm heap usage after gc per instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.heap.after-gc.usage" **Where:** $timeFilter **Group By:** time($__interval), "host"
JVM	JVM Garbage Collector Time	Measures maximum JVM GC duration. Long garbage collection pause durations indicate that the heap size is not adjusted for the actual usage.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.gc.G1-Old-Generation.time" **Where:** $timeFilter **Group By:** "host"
JVM	JVM Active Threads Count	Measures the current number of active threads in the JVM per instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.threads.count" **Where:** $timeFilter **Group By:** time($__interval), "host"
JVM			

Row Title	Title	Description	Query Details
	JVM File Descriptors Ratio	Measures the jvm file descriptors usage ratio per instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.filedescriptor" **Where:** $timeFilter **Group By:** time($__interval), "host"

The `Row Title` column refers to a Grafana feature to group dashboards by type. This is only a suggestion and thus not mandatory.

Node pageâ

These suggestions relate to the existing Case Manager dashboard page. You may find some new dashboards or changes to existing ones.

Our suggestions are to add the dashboards in the following table with a filter by each of the available nodes and applications:

Row Title	Title	Description	Query Details
Case Manager - Healthchecks	Database	The database health check relies on the shared CM's connection pool to execute the statement SELECT 1. If the connection is successfully retrieved and the statement finishes, the health check succeeds, if any error occurs in any of the steps the check fails. A timeout of 5 seconds was given when retrieving a connection to avoid hanging the health check response. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0. This healthcheck corresponds to the case manager selected instance.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'Database' , "host" = '$host' , $timeFilter **Group By:** time($__interval)
Case Manager - Healthchecks	Messaging	The Messaging Server Health Check is achieved by trying to initialize a new connection with the messaging server. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0. This healthcheck corresponds to the case manager selected instance.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'Messaging' , "host" = '$host' , $timeFilter **Group By:** time($__interval)
Case Manager - Healthchecks	Pulse	The Pulse Health Check when triggered tries to connect to pulse and tries to login, if it succeeds the check will also succeed. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0. This healthcheck corresponds to the case manager selected instance.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'Pulse' , "host" = '$host' , $timeFilter **Group By:** time($__interval)
	New Event - Throughput	Measures the event processing throughput when the event type	**Query A** **Field:** "count"

Row Title	Title	Description	Query Details
Event Processing - New Event		is "new event", for the selected case manager instance.	**From:** "cm.event.processing.latency" **Where:** "host" = '$host' , eventType='NEW_EVENT' , $timeFilter **Group By:** time($__interval)
Event Processing - New Event	New Event - Latency per Percentile	Measures the event processing latencies when the event type is "new event", for the selected case manager instance.	**Query A** **Field:** "p50", "p95", "p99", "p999", "max" **From:** "cm.event.processing.latency" **Where:** "host" = '$host' , eventType='NEW_EVENT' , $timeFilter **Group By:** time($__interval)
Event Processing - Status Change	Status Change - Throughput	Measures the event processing throughput when the event type is "status change", for the selected case manager instance.	**Query A** **Field:** "count" **From:** "cm.event.processing.latency" **Where:** "host" = '$host' , eventType='STATUS_CHANGE' , $timeFilter **Group By:** time($__interval)
Event Processing - Status Change	Status Change - Latency per Percentile	Measures the event processing latencies when the event type is "status change", for the selected case manager instance.	**Query A** **Field:** "p50", "p95", "p99", "p999", "max" **From:** "cm.event.processing.latency" **Where:** "host" = '$host' , eventType='STATUS_CHANGE' , $timeFilter **Group By:** time($__interval)
Automation Rules	Automation Rule $automationRule - Throughput and Error Percent	For the selected case manager instance, measures the throughput to execute the selected automation rule and the associated error percentage.	**Query A** **Field:** "count" **From:** "cm.automation" **Where:** "host" = '$host' , "ruleName" = '$automationRule' , $timeFilter **Group By:** time($__interval)
Automation Rules	Automation Rule $automationRule - Latency per Percentile	For the selected case manager instance, measures the latency per percentile (50, 95, 99, 99.9 and 100) to execute the selected automation rule.	**Query A** **Field:** "p50", "p95", "p99", "p999", "max" **From:** "cm.automation" **Where:** "host" = '$host' ruleName = '$automationRule' , $timeFilter **Group By:** time($__interval)
Pulse Requests	Pulse Requests - Throughput and Error Percent	For the selected case manager instance, measures the throughput of the communications with pulse and the error percent inherent to this.	**Query A** **Field:** "count" **From:** "cm.request.pulse.api" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval), "path"
Pulse Requests	Pulse HTTP Endpoints and Latencies	Measures the latency in the communication with pulse instances.	**Query A** **Field:** "count", "p99" **From:** "cm.request.pulse.api" **Where:** "host" = '$host' , $timeFilter **Group By:** "path"
Event Processing	Event Processing Queue Occupancy	Measures the percentage of the event processing queue being used, for the selected case manager instance.	**Query A** **Field:** "value" **From:** "cm.event.processing.queue.ratio" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)
Event Processing	Events Waiting in Queue	Measures the number of events waiting to be processed, for the selected case manager instance.	**Query A** **Field:** "value" **From:** "cm.event.processing.queue.used.count" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)

Row Title	Title	Description	Query Details
Deadletters	Deadletter Events	Measures the number of events that were selected case manager deadletter.	**Query A** **Field:** "count" **From:** "cm.event.processing.deadletter.count" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)
DB Connection Pool	DB Connection Pool Usage	Measures the db connection pool usage, for the selected case manager instance.	**Query A** **Field:** "value" **From:** "cm.db.connection.pool.usage" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)
DB Connection Pool	DB Connection Pool Waiting Threads	Measures the db connection pool waiting threads, for the selected case manager instance.	**Query A** **Field:** "count" **From:** "cm.db.connection.pool.waiting" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)
Case Manager Executor Thread Pool and Task Queue	Executor Thread Pool Usage	Measures the thread pool usage, for the selected case manager instance.	**Query A** **Field:** "value" **From:** "cm.executor.pool.threadsUsageRatio" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)
Case Manager Executor Thread Pool and Task Queue	Tasks on Executor's Queue	Measures the the number of tasks within the executor's queue, on the selected case manager instance.	**Query A** **Field:** "value" **From:** "cm.executor.pool.taskQueueEventsCount" **Where:** "host" = '$host' $timeFilter **Group By:** time($__interval)
Case Manager HTTP Endpoints and Latencies	Case Manager HTTP Endpoints and Latencies	Measures the latency of Case Manager HTTP endpoints.	**Query A** **Field:** "count", "p99" **From:** "cm.request.api" **Where:** "host" = '$host' , $timeFilter **Group By:** "path"
Login Errors	Login Error Percent	Measures the login error percent on the selected case manager instance.	**Query A** **Field:** "value" **From:** "cm.request.login.error.percent" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)
JVM	JVM Heap	Measures the jvm heap usage of the selected instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.heap.used" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval) **Query B** **Field:** "value" **From:** "letmeknow.jvm.memory.heap.max" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval), "host"
JVM	JVM Heap Used (By Gen)	For the selected instance, considering the heap used, displays the usage of each generation.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.pools.G1-Old-Gen.used" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)
JVM	JVM Heap Used After GC (By Gen)	For the selected instance, considering the heap used, displays the usage of each generation after gc.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.pools.G1-Old-Gen.used-after-gc"

Row Title	Title	Description	Query Details
			Where: "host" = '$host' , $timeFilter **Group By:** time($__interval)
JVM	JVM Heap Usage After GC	Measures the jvm heap usage after gc on the selected instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.heap.after-gc.usage" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)
JVM	JVM Garbage Collector Time	Measures the JVM GC duration on the selected instance. Long garbage collection pause durations indicate that the heap size is not adjusted for the actual usage.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.gc.G1-Old-Generation.time" **Where:** "host" = '$host' , $timeFilter **Group By:** N/A
JVM	JVM Active Threads Count	Measures the current number of active threads in the JVM on the selected instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.threads.count" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)
JVM	JVM File Descriptors Ratio	Measures the jvm file descriptors usage ratio of the selected instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.filedescriptor" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)

On this page

Grafana

In this page you can find a description of the recommend setup for Grafana.

Overviewâ

For some components (e.g. Pulse and Case Manager) it is recommended to have two dashboards:

- Cluster: holds the cluster view containing data for all component nodes. Ideally this page shouldn't have a filter, and dashboards should aggregated values of all nodes
- Node: holds dashboards with metrics breakdown per node. Note that some of the dashboards in the page can monitor the same metrics as the cluster page

This distinction is important because it may help pinpointing an issue with a specific node while keeping a view of the overall cluster healthiness.

info

The following example assumes the datasource is InfluxDB

To better understand the differences, imagine we want to monitor Case Manager latency when processing new events (`New Event Latency`). We would need the following pages:

- Cluster page:
  - No filters
  - `New Event Latency` dashboard for all percentiles, where:
    - We first select the `mean` value of the various metric values (`p50` , `p95` , etc), grouped by `host` and filtered by `eventType='NEW_EVENT'`
    - We select the `max` value of each metrics values without grouping by hosts
- Node page:
  - A filter by node (name, IP or any other metadata that identifies the node and can be used to filter metric queries)
  - `New Event Latency` dashboard for all percentiles, where:
    - We select the `mean` value of the various metric values (`p50` , `p95` , etc) filtered by `eventType='NEW_EVENT'` and `"host" = '$host'` . Note that the filter `$host` is based on [Grafana templating feature]

We can see that the overall new event processing reached around 400ms maximum. If we compare this latency value with the node example we can see that the node maximum processing time was around 300ms which means this node was not the one with the highest processing time.

info

This example refers to Case Manager `cm.event.processing.latency` metric (or `cm_event_processing_latency_*` in Prometheus)

Metrics in Prometheusâ

Feedzai products expose various metric fields in a single measurement. For instance `cm.event.processing.latency` will have, among others, `p50` and `p99` .

While in some solutions these are exported (and can be queried) in the same way, in Prometheus each measurement field is exposed as a single metric. In the above example, `p50` for the `cm.event.processing.latency` metric would be `cm_event_processing_latency_p50`.

Comparing to InfluxDB, a single query that aggregates various fields of a measurement would have to be split into `n` queries, where `n` is the amount of fields in the measurement. Considering the example in [Expectation](), in Prometheus it would look like following:

```
# Query A
cm_event_processing_latency_p50 {
    duration_units = "seconds",
    eventType = "NEW_EVENT",
    metricType = "APPLICATION",
    metric_type = "timer",
    rate_units = "calls/second"
}

# Query B
cm_event_processing_latency_p99 {
    duration_units = "seconds",
    eventType = "NEW_EVENT",
    metricType = "APPLICATION",
    metric_type = "timer",
    rate_units = "calls/second"
}

...
```

Copy

Component metricsâ

Please find below the list of Grafana suggestions for each of the available components:

- Pulse Setup
- Case Manager Setup
- Reference data Setup

On this page

Pulse Setup

In this page you can find a set of suggestions for Pulse monitoring using Grafana.

Suggestionsâ

info

These suggestions are based on the shared resources.

info

There are VM resource related metrics (CPU, Memory, IO, etc) that can also be added to Grafana. These were left out on purpose Google already has out of the box monitoring support for these resources.

These suggestions can be either new dashboards, changes to existing dashboards or completely new dashboard pages. Please find the metric description in Pulse Top Metrics.

Cluster pageâ

Currently there is only one page where we can filter metrics by each of the instances. Although useful, having a cluster view monitoring page gives a view of how the overall cluster health. Given this, our suggestion is to add a new page with the following metrics:

Row Title	Title	Description	Query Details
Healthchecks - Pulse	Login	Displays the login healthcheck. This healthcheck is carried out by pulse in order to validate the login operation. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'login' , "source" = 'pulse' , $timeFilter **Group By:** time($__interval), "host"
Healthchecks - Pulse	Messaging	Displays the messaging healthcheck. This healthcheck is carried out by pulse in order to validate the communication with messaging server. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'messaging' , "source" = 'pulse' , $timeFilter **Group By:** time($__interval), "host"
Healthchecks - Pulse	Metadata DB	Displays the metadata db healthcheck. This healthcheck is carried out by pulse in order to validate the communication with metadata db. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'metadata' , "source" = 'pulse' , $timeFilter **Group By:** time($__interval), "host"
Healthchecks - Pulse	Zookeeper	Displays the zookeeper healthcheck. This healthcheck is carried out by pulse in order to validate the communication with zookeeper. The outcome is a bool, meaning that it	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'zookeeper' , "source" =

Row Title	Title	Description	Query Details
		was successfully executed when 1, otherwise is 0.	'pulse' , $timeFilter **Group By:** time($__interval), "host"
Healthchecks - Pulse	Leadership	This healthcheck indicates the leadership status of each Pulse instance, represented by 1	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck"='leadership' , "source"='pulse' , $timeFilter **Group By:** time($__interval), "host" **Query B** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck"='leadership' , "source"='pulse' , $timeFilter **Group By:** time($__interval), "host"
Healthchecks - Pulse	Jobs Status	This healthcheck verifies that jobs can be submitted and ran in the Dataproc cluster. This healthcheck is based on a dummy job that is submitted every X configured time and reported in its metrics consumed by Telegraf. The outcome is a boolean, meaning that it was successfully executed when 1, otherwise is 0.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'job-processing-cluster' , $timeFilter **Group By:** time($__interval), "host", "source" **Query B** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'job-processing-cluster' , $timeFilter **Group By:** time($__interval), "host", "source"
Healthchecks - Apps	Login	Displays the appSlug login healthcheck. This healthcheck is carried out by the appSlug in order to validate the login operation. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck"='login' , source != 'pulse' , $timeFilter **Group By:** time($__interval), "host", "source"
Healthchecks - Apps	Leadership	This healthcheck indicates the app's leadership status, represented by 1 and 0 respectively.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck"='leadership' , "source"!='pulse' , $timeFilter **Group By:** time($__interval), "source", "host" **Query B** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck"='leadership' , "source"!='pulse' , $timeFilter **Group By:** time($__interval), "host", "source"
States	Pulse Instances State	Displays the pulse instances states: - 1: INITIAL - 2: BOOTSTRAPPING - 3: RUNNING - 4: STOPPING	**Query A** **Field:** "value" **From:** "pulse.server.state" **Where:** $timeFilter **Group By:** time($__interval), "host"
States	Pulse Apps State	Displays the pulse appSlug states: - 1: INITIAL - 3: BOOTSTRAPPING - 4: BOOTSTRAPPED	**Query A** **Field:** "value" **From:** "pulse.app.state" **Where:** $timeFilter

Row Title	Title	Description	Query Details
		- 5: STARTING - 6: STARTED - 8: APPLYING_CHANGES - 10: SELF_HEALING	**Group By:** time($__interval), "host", "appSlug" **Query B** **Field:** "value" **From:** "pulse.app.state" **Where:** $timeFilter **Group By:** time($__interval), "host", "appSlug"
Leadership Changes	Pulse Leadership Changes	Measures the number of pulse leadership changes.	**Query A** **Field:** "count" **From:** "pulse.server.leadership.count" **Where:** $timeFilter **Group By:** "host"
Leadership Changes	Pulse Apps Leadership Changes	Measures the number of appSlug leadership changes.	**Query A** **Field:** "count" **From:** "pulse.app.leadership.count" **Where:** $timeFilter **Group By:** "host", "appSlug"
Watchdog Activations	Watchdog Activations	Measures the number watchdog activation per appSlug. Watchdog activation is triggered when there is a problem with an appSlug and subsequently, needs to be recovered.	**Query A** **Field:** "count" **From:** "pulse.server.watchdog.trigger" **Where:** $timeFilter **Group By:** "host", "appSlug", "operation"
Connection Pools	Metadata DB Connection Pools Usage	Measures the metadata db connection pool usage. High values implies waiting time to access to metadata db which may degrade the component performance.	**Query A** **Field:** "value" **From:** "pulse.dbPool.usage" **Where:** appSlug = '' , $timeFilter **Group By:** time($__interval), "host", "poolName" **Query B** **Field:** "value" **From:** "pulse.dbPool.usage" **Where:** appSlug = '' , $timeFilter **Group By:** time($__interval), "host", "poolName" **Query C** **Field:** "value" **From:** "pulse.dbPool.maxPoolSize" **Where:** appSlug = '' , $timeFilter **Group By:** time($__interval), "host", "poolName"
Connection Pools	Connection Pools per App Usage	Measures the metadata connection pools usage per appSlug.	**Query A** **Field:** "value" **From:** "pulse.dbPool.usage" **Where:** appSlug != '' , $timeFilter **Group By:** time($__interval), "host", "appSlug", "poolName" **Query B** **Field:** "value" **From:** "pulse.dbPool.maxPoolSize" **Where:** appSlug != '' , $timeFilter **Group By:** time($__interval), "host", "appSlug", "poolName"
Connection Pools	Metadata DB Connection Errors and Timeouts	Measures the number of errors and timeouts occurred when trying to get a metadata db connection.	**Query A** **Field:** "count" **From:** "pulse.dbPool.connectionErrors" **Where:** appSlug = '' , $timeFilter **Group By:** time($__interval), "host", "poolName"

Row Title	Title	Description	Query Details
			Query B **Field:** "count" **From:** "pulse.dbPool.connectionObtainTimeout" **Where:** appSlug = '' , $timeFilter **Group By:** time($__interval), "host", "poolName" **Query C** **Field:** "count" **From:** "pulse.dbPool.connectionErrors" **Where:** appSlug = '' , $timeFilter **Group By:** time($__interval), "host", "poolName" **Query D** **Field:** "count" **From:** "pulse.dbPool.connectionObtainTimeout" **Where:** appSlug = '' , $timeFilter **Group By:** time($__interval), "host", "poolName"
Connection Pools	DB Connection Errors and Timeouts per App	Measures the number of errors and timeouts occurred when the appSlug tries to get a connection .	**Query A** **Field:** "count" **From:** "pulse.dbPool.connectionErrors" **Where:** appSlug != '' , $timeFilter **Group By:** time($__interval), "host", "poolName", "appSlug" **Query B** **Field:** "count" **From:** "pulse.dbPool.connectionObtainTimeout" **Where:** appSlug != '' , $timeFilter **Group By:** time($__interval), "host", "poolName", "appSlug" **Query C** **Field:** "count" **From:** "pulse.dbPool.connectionErrors" **Where:** appSlug != '' , $timeFilter **Group By:** time($__interval), "host", "poolName" **Query D** **Field:** "count" **From:** "pulse.dbPool.connectionObtainTimeout" **Where:** appSlug != '' , $timeFilter **Group By:** time($__interval), "host", "poolName"
Jobs	Jobs Scheduled	Measures the number of jobs scheduled for execution broken-down by job type; i.e., with state SCHEDULED (1).	**Query A** **Field:** "count" **From:** "pulse.job.executions.scheduled" **Where:** jobType = 'Workflow' , $timeFilter **Group By:** time($__interval), *
Jobs	Jobs Waiting	Measures the number of jobs that are waiting for an agent to be free broken-down by job type; i.e., with state WAITING_AGENT (8).	**Query A** **Field:** "count" **From:** "pulse.job.executions.waiting" **Where:** $timeFilter **Group By:** time($__interval), *
Jobs	Jobs Executing	Measures the number of jobs that are currently in state EXECUTING (2) broken-down by job type.	**Query A** **Field:** "count" **From:** "pulse.job.executions.running" **Where:** jobType = 'Workflow' , $timeFilter **Group By:** time($__interval), *
Jobs	Jobs Failed	Measures the number of jobs that failed broken-down by both job type and	**Query A** **Field:** "count" **From:** "pulse.job.executions.failed"

Row Title	Title	Description	Query Details
		state: ERROR (5), CRASHED (6), ABORTED (7).	**Where:** jobType = 'Workflow' , $timeFilter **Group By:** time($__interval), ***Query B*** ***Field:*** *"count"* ***From:*** *"pulse.job.executions.failed"* ***Where:*** *jobType = 'Workflow' , $timeFilter* ***Group By:*** *time($__interval),*
Jobs	Jobs Finished	Measures the number of jobs that completed successfully broken-down by job type; i.e., with state FINIDHED (3).	**Query A** **Field:** "count" **From:** "pulse.job.executions.success" **Where:** jobType = 'Workflow' , $timeFilter **Group By:** time($__interval), * **Query B** **Field:** "count" **From:** "pulse.job.executions.success" **Where:** jobType = 'Workflow' , $timeFilter **Group By:** time($__interval), "host", "appSlug"
Workflow	Global Workflow Throughput per Message Type	Measures the overall appSlug workflow throughput per message type(normal, recovery, replica).	**Query A** **Field:** "count" **From:** "pulse.workflow.global.event.stats" **Where:** $timeFilter **Group By:** time($__interval), "host", "appSlug", "workflowDesc", "messageType"
Workflow	Global Workflow Errors	Measures the overall appSlug workflow errors, considering all message types(normal, recovery and replica).	**Query A** **Field:** "count" **From:** "pulse.workflow.eventErrors.count" **Where:** $timeFilter **Group By:** time($__interval), "host", "appSlug", "workflowDesc"
Workflow	Global Workflow latency per Message Type at Percentile 99	Measures the overall workflow latency at percentile 99 per pulse instance, appSlug, workflow and message type.	**Query A** **Field:** "p99" **From:** "pulse.workflow.global.event.stats" **Where:** "messageType" != 'RECOVERY' , $timeFilter **Group By:** time($__interval), "host", "appSlug", "workflowDesc", "messageType" **Query B** **Field:** "p99" **From:** "pulse.workflow.global.event.stats" **Where:** "messageType" = 'RECOVERY' , $timeFilter **Group By:** time($__interval), "host", "appSlug", "workflowDesc" **Query C** **Field:** "p99" **From:** "pulse.workflow.global.event.stats" **Where:** "messageType" = 'NORMAL' , $timeFilter **Group By:** time($__interval), "host", "appSlug", "workflowDesc"
Workflow	Events Spilled to Disk	Measures the overall appSlug events spilled to disk. An event is spilled to disk when pulse cannot write it on the event storage. This scenario must be avoided, otherwise require human action to move the events from the disk to the event storage.	**Query A** **Field:** "count" **From:** "pulse.eventStorage.spillToDisk.count" **Where:** $timeFilter **Group By:** time($__interval), "host", "appSlug", "workflowDesc"

Row Title	Title	Description	Query Details
Workflow	Global Workflow Throughput by API endpoint	Measures the overral appSlug workflow throughput of normal messages by Endpoint (initiation, info, etc).	**Query B** **Field:** "count" **From:** "workflow.events" **Where:** $timeFilter **Group By:** time($__interval), "workflowDesc", "elementDesc", "host", "appSlug", "event_type"
Event Storage Dumps	Events Storage Dump Failed and Retry Events	Measures the overall number of events that were dumped to disk and are either pending to be retried or reached to the max amount of retries and need to be re-injected manually.	**Query A** **Field:** "count" **From:** "pulse.dump.storage.dump.entries.retry.counter" **Where:** $timeFilter **Group By:** time($__interval), "host", "appSlug", "workflowDesc" **Query B** **Field:** "count" **From:** "pulse.dump.storage.dump.entries.failed.counter" **Where:** $timeFilter **Group By:** time($__interval), "host", "appSlug", "workflowDesc"
Event Storage Dumps	Events Storage Dump Failed and Retry Queue Size	Measures the overall number of events that are present in the in-memory queue to be written to the dump file in the failed folder, appSlug and workflow.	**Query A** **Field:** "value" **From:** "pulse.dump.storage.failed.queue.size" **Where:** $timeFilter **Group By:** time($__interval), "host", "appSlug", "workflowDesc" **Query B** **Field:** "value" **From:** "pulse.dump.storage.retry.queue.size" **Where:** $timeFilter **Group By:** time($__interval), "host", "appSlug", "workflowDesc"
Event Pending Dumps	Event Pending Persistence to Event Storage	Measures the overall number of events that are pending to be persisted to the databased.	**Query A** **Field:** "count" **From:** "pulse.workflow.eventstorage.eventsQueuedSize" **Where:** $timeFilter **Group By:** time($__interval), "host", "appSlug", "workflowDesc"
Profile Fetches	Profile Fetches Latency at Percentile 99 (Excluding Timeouts)	Measures the overall max latency to fetch the profiles at percentile 99, per appSlug, workflow and plan.	**Query A** **Field:** "p99" **From:** "pulse.workflow.profileFetches.statistics" **Where:** $timeFilter **Group By:** time($__interval), "host", "appSlug", "workflowDesc", "elementDesc"
Profile Fetches	Profile Fetches Timeouts	Measures the number of profile fetches that were timed out, per appSlug, workflow and plan.	**Query A** **Field:** "count" **From:** "pulse.workflow.profileFetches.timeouts" **Where:** $timeFilter **Group By:** time($__interval), "host", "appSlug", "workflowDesc", "elementDesc"
JVM	JVM Heap Usage (Pulse Server)	Measures the jvm heap usage per pulse instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.heap.usage" **Where:** appSlug = '' , $timeFilter **Group By:** time($__interval), "host"

Row Title	Title	Description	Query Details
			Query B **Field:** "value" **From:** "letmeknow.jvm.memory.heap.usage" **Where:** appSlug = '' , $timeFilter **Group By:** time($__interval), "host"
JVM	JVM Heap Usage (per App)	Measures the jvm heap usage per application and host.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.heap.usage" **Where:** appSlug != '' , $timeFilter **Group By:** time($__interval), "host", "appSlug" **Query B** **Field:** "value" **From:** "letmeknow.jvm.memory.heap.usage" **Where:** appSlug != '' , $timeFilter **Group By:** time($__interval), "host", "appSlug" **Query C** **Field:** "value" **From:** "letmeknow.jvm.memory.heap.usage" **Where:** appSlug != '' , $timeFilter **Group By:** time($__interval), "host", "appSlug"
JVM	JVM Heap Usage After GC (Pulse Server)	Measures the jvm heap usage after gc per pulse instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.heap.after-gc.usage" **Where:** appSlug = '' , $timeFilter **Group By:** time($__interval), "host"
JVM	JVM Heap Usage After GC (per App)	Measures the jvm heap usage after gc per application and host.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.heap.after-gc.usage" **Where:** appSlug != '' , $timeFilter **Group By:** time($__interval), "host", "appSlug"
JVM	JVM Code Cache Usage (Pulse Server)	Measures the jvm code cache usage per pulse instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.pools.CodeCache.usage" **Where:** appSlug = '' , $timeFilter **Group By:** time($__interval), "host"
JVM	JVM Code Cache Usage (per App)	Measures the jvm code cache usage per application and host.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.pools.CodeCache.usage" **Where:** appSlug != '' , $timeFilter **Group By:** time($__interval), "host", "appSlug"
JVM	JVM Garbage Collector Time (Pulse Server)	Measures maximum JVM GC duration. Long garbage collection pause durations indicate that the heap size is not adjusted for the actual usage.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.gc.G1-Old-Generation.time" **Where:** appSlug = '' , $timeFilter **Group By:** "host" **Query B** **Field:** "value" **From:** "letmeknow.jvm.gc.G1-Young-Generation.time" **Where:** appSlug = '' , $timeFilter **Group By:** "host"
JVM	JVM Garbage Collector Time (per App)	Measures maximum JVM GC duration for young and old gen, per application and host. Long garbage collection	**Query A** **Field:** "value" **From:** "letmeknow.jvm.gc.G1-Old-Generation.time"

Row Title	Title	Description	Query Details
		pause durations indicate that the heap size is not adjusted for the actual usage.	**Where:** appSlug != '' , $timeFilter **Group By:** "host", "appSlug" **Query B** **Field:** "value" **From:** "letmeknow.jvm.gc.G1-Young-Generation.time" **Where:** appSlug != '' , $timeFilter **Group By:** "host", "appSlug"
JVM	JVM Active Threads Count (Pulse Server)	Measures the current number of active threads in the JVM per pulse instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.threads.count" **Where:** appSlug = '' , $timeFilter **Group By:** time($__interval), "host"
JVM	JVM Active Threads Count (per App)	Measures the current number of active threads in the JVM per application and host.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.threads.count" **Where:** appSlug != '' , $timeFilter **Group By:** time($__interval), "host", "appSlug"
JVM	JVM File Descriptors Ratio (Pulse Server)	Measures the jvm file descriptors usage ratio per instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.filedescriptor" **Where:** appSlug = '' , $timeFilter **Group By:** time($__interval), "host"
JVM	JVM File Descriptors Ratio (per App)	Measures the jvm file descriptors usage ratio per application and host.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.filedescriptor" **Where:** appSlug != '' , $timeFilter **Group By:** time($__interval), "host", "appSlug"

Note that in the cluster page some dashboards are separated. For instance, instead of having the `Jobs` dashboards separated by job state, we can have a dashboard that gathers all states. Given that in cluster view we are not filtering by any node, it is recommend to have disassociated dashboards so they do not get cluttered with information.

Also note the `Row Title` column which refers to a Grafana feature to group dashboards by type. This is only a suggestion and thus not mandatory.

Node pageâ

These suggestions relate to the existing Pulse dashboard page. You may find some new dashboards or changes to existing ones.

Our suggestions are to add the dashboards in the following table with a filter by each of the available nodes and applications:

Row Title	Title	Description	Query Details
Healthchecks - Pulse	Login	Displays the login healthcheck on the selected instance. This healthcheck is carried out by pulse in order to validate the login operation. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "source" = 'pulse' , "healthcheck" = 'login' , "host" = '$host' , $timeFilter **Group By:** time($__interval)

Row Title	Title	Description	Query Details
Healthchecks - Pulse	Messaging	Displays the messaging healthcheck on the selected instance. This healthcheck is carried out by pulse in order to validate the communication with rabbitmq. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "source" = 'pulse' , "healthcheck" = 'messaging' , "host" = '$host' , $timeFilter **Group By:** time($__interval)
Healthchecks - Pulse	Metadata DB	Displays the metadata db healthcheck on the selected instance. This healthcheck is carried out by pulse in order to validate the communication with metadata db. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "source" = 'pulse' , "healthcheck" = 'metadata' , "host" = '$host' , $timeFilter **Group By:** time($__interval)
Healthchecks - Pulse	Zookeeper	Displays the zookeeper healthcheck on the selected instance. This healthcheck is carried out by pulse in order to validate the communication with zookeeper. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "source" = 'pulse' , "healthcheck" = 'zookeeper' , "host" = '$host' , $timeFilter **Group By:** time($__interval)
Pulse State and Leadership Changes	Pulse Leadership Changes	Displays pulse leadership changes, for the selected pulse instance.	**Query A** **Field:** "count" **From:** "pulse.server.leadership.count" **Where:** "host" = '$host' , $timeFilter **Group By:** N/A
Healthchecks - Apps	Login	Displays the login healthcheck for the specified app on the selected instance. This healthcheck is carried out by the appSlug in order to validate the login operation. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck"='login' , source =~ /^app.$appSlug.*/ , "host" = '$host' , $timeFilter **Group By:** time($__interval)
Healthchecks - Apps	Leadership	Displays the leadership healthcheck for the specified app on the selected instance. This healthcheck is carried out by the appSlug in order to validate the login operation. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck"='leadership' , source =~ /^app.$appSlug.*/ , "host" = '$host' , $timeFilter **Group By:** time($__interval)
Watchdog Activations	Watchdog Activations	Measures the number of watchdog activations triggered for the selected appSlug and the specified pulse instance.	**Query A** **Field:** "count" **From:** "pulse.server.watchdog.trigger" **Where:** "host" = '$host' , "appSlug" = '$appSlug' , $timeFilter **Group By:** "operation"
Jobs	Jobs Scheduled	Measures the number of jobs scheduled for execution broken-down by job type; i.e., with state SCHEDULED (1).	**Query A** **Field:** "count" **From:** "pulse.job.executions.scheduled" **Where:** jobType = 'Workflow' , "host" = '$host' , "appSlug" = '$appSlug' , $timeFilter **Group By:** time($__interval), "host", "appSlug", "jobType"
Jobs	Jobs Waiting	Measures the number of jobs that are waiting for an agent to be free broken-down by job type; i.e., with state WAITING_AGENT (8).	**Query A** **Field:** "count" **From:** "pulse.job.executions.waiting" **Where:** jobType = 'Workflow' , "host" = '$host' ,

Row Title	Title	Description	Query Details
			"appSlug" = '$appSlug' , $timeFilter **Group By:** time($__interval), "host", "appSlug", "jobType"
Jobs	Jobs Executing	Measures the number of jobs that are currently in state EXECUTING (2) broken-down by job type.	**Query A** **Field:** "count" **From:** "pulse.job.executions.running" **Where:** jobType = 'Workflow' , "host" = '$host' , "appSlug" = '$appSlug' , $timeFilter **Group By:** time($__interval), "host", "appSlug", "jobType"
Jobs	Jobs Failed	Measures the number of jobs that failed broken-down by both job type and state: ERROR (5), CRASHED (6), ABORTED (7).	**Query A** **Field:** "count" **From:** "pulse.job.executions.failed" **Where:** jobType = 'Workflow' , "host" = '$host' , "appSlug" = '$appSlug' , $timeFilter **Group By:** time($__interval), "host", "appSlug", "jobType"
Jobs	Jobs Finished	Measures the number of jobs that completed successfully broken-down by job type; i.e., with state FINIDHED (3).	**Query A** **Field:** "count" **From:** "pulse.job.executions.success" **Where:** jobType = 'Workflow' , "host" = '$host' , "appSlug" = '$appSlug' , $timeFilter **Group By:** time($__interval), "host", "appSlug", "jobType"

On this page

Reference Data Setup

In this page you can find a set of suggestions for Reference Data monitoring using Grafana.

Suggestionsâ

info

There are VM resource related metrics (CPU, Memory, IO, etc) that can also be added to Grafana. These were left out on purpose Google already has out of the box monitoring support for these resources.

These suggestions can be either new dashboards, changes to existing dashboards or completely new dashboard pages. Please find the metric description in Reference Data Top Metrics.

Cluster pageâ

Currently there is only one page where we can filter metrics by each of the instances. Although useful, having a cluster view monitoring page gives a view of how the overall cluster health. Given this, our suggestion is to add a new page with the following metrics:

Row Title	Title	Description	Query Details
Healthchecks - RDS	Reference Data Service	Displays reference data server healthcheck. The outcome is a bool, meaning that it assumes the value one when everything is okay with the service itself, otherwise is 0.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'reference-data-server' , $timeFilter **Group By:** time($__interval), "host"
Healthchecks - RDS	Deadlocks	Displays deadlocks healthcheck. This healthcheck is carried out by reference data service in order to validate if some of its threads entered in deadlock. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'deadlocks' , $timeFilter **Group By:** time($__interval), "host"
Healthchecks - RDS	Entity Repository	Displays entity repository healthcheck. This healthcheck is carried out by reference data service in order to validate if it is able to reach entity repository, which is an abstraction for a non-sql db. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'entity-repository' , $timeFilter **Group By:** time($__interval), "host"
HTTP Requests	HTTP Throughput	Measures reference-data HTTP throughput.	**Query A** **Field:** "count"

Row Title	Title	Description	Query Details
			From: "io.dropwizard.jetty.MutableServletContextHandler.requests" **Where:** $timeFilter **Group By:** time($interval), "host"
HTTP Requests	HTTP Throughput per Method	Measures reference-data HTTP throughput per method.	**Query A** **Field:** "count" **From:** "io.dropwizard.jetty.MutableServletContextHandler.get-requests" **Where:** $timeFilter **Group By:** time($interval), "host"
HTTP Requests	2XX Requests Rate	Measures reference-data HTTP throughput with return code falls into 2xx.	**Query A** **Field:** "count" **From:** "io.dropwizard.jetty.MutableServletContextHandler.2xx-responses" **Where:** $timeFilter **Group By:** time($interval), "host"
HTTP Requests	4XX Errors	Measures the number of 4xx errors returned by reference-data.	**Query A** **Field:** "count" **From:** "io.dropwizard.jetty.MutableServletContextHandler.4xx-responses" **Where:** $timeFilter **Group By:** time($interval), "host"
HTTP Requests	5XX Errors	Measures the number of 5xx errors returned by reference-data.	**Query A** **Field:** "count" **From:** "io.dropwizard.jetty.MutableServletContextHandler.5xx-responses" **Where:** $timeFilter **Group By:** time($interval), "host"
Global Database Requests	Global Database Throughput	Measures the overall database throughput, considering all operations executed by reference-data. It also shows a break down per operation.	**Query A** **Field:** "count" **From:** "reference-data-service.entity-repository" **Where:** $timeFilter **Group By:** time($interval), "host", "operation"
Global Database Requests	Global Database Latency at Percentile 99	Measures global database latency at percentile 99 considering all operations executed by reference-data.	**Query A** **Field:** "p99" **From:** "reference-data-service.entity-repository" **Where:** $timeFilter **Group By:** time($__interval), "host", "operation"
Global Database Requests	Global Database Errors	Measures the total number of errors registered while performing some operations onto the database.	**Query A** **Field:** "count" **From:** "reference-data-service.entity-repository.error" **Where:** $timeFilter **Group By:** time($__interval), "host"
Global Database Requests	Global Database Errors - Entities Not Found	Measures the total number of errors caused by entities not found, registered while performing some operations onto the database.	**Query A** **Field:** "count" **From:** "reference-data-service.entity-repository.entitynotfound" **Where:** $timeFilter **Group By:** time($__interval), "host"
Database Operations	Database Throughput Per Entity	Measures reference-data throughput per entity.	**Query A** **Field:** "count" **From:** "reference-data-service.entity-repository" **Where:** $timeFilter **Group By:** time($interval), "host", "entityType", "operation"
	Global Database Latency at	Measures database latency at percentile 99 per entity,	**Query A** **Field:** "p99"

Row Title	Title	Description	Query Details
Database Operations	Percentile 99 per Entity	considering all operations executed by reference-data.	**From:** "reference-data-service.entity-repository" **Where:** $timeFilter **Group By:** time($__interval), "host", "entityType", "operation"
JVM	JVM Heap Usage	Measures the jvm heap usage per component instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.heap.usage" **Where:** $timeFilter **Group By:** time($__interval), "host"
JVM	JVM Heap Usage After GC	Measures the jvm heap usage after gc per instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.heap.after-gc.usage" **Where:** $timeFilter **Group By:** time($__interval), "host"
JVM	JVM Garbage Collector Time	Measures maximum JVM GC duration. Long garbage collection pause durations indicate that the heap size is not adjusted for the actual usage.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.gc.G1-Old-Generation.time" **Where:** $timeFilter **Group By:** "host"
JVM	JVM Active Threads Count	Measures the current number of active threads in the JVM per instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.threads.count" **Where:** $timeFilter **Group By:** time($__interval), "host"
JVM	JVM File Descriptors Ratio	Measures the jvm file descriptors usage ratio per instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.filedescriptor" **Where:** $timeFilter **Group By:** time($__interval), "host"

The `Row Title` column refers to a Grafana feature to group dashboards by type. This is only a suggestion and thus not mandatory.

Node pageâ

These suggestions relate to the existing Reference Data dashboard page. You may find some new dashboards or changes to existing ones.

Our suggestions are to add the dashboards in the following table with a filter by each of the available nodes and applications:

Row Title	Title	Description	Query Details
Healthchecks - RDS	Reference Data Service	Displays reference data server healthcheck. The outcome is a bool, meaning that it assumes the value one when everything is okay with the service itself, otherwise is 0.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'reference-data-server' , "host" = '$host' , $timeFilter **Group By:** time($__interval)
Healthchecks - RDS	Deadlocks	Displays deadlocks healthcheck. This healthcheck is carried out by reference data service in order to validate if some of its threads entered in deadlock. The outcome is a bool, meaning that it was	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'deadlocks' , "host" = '$host' , $timeFilter **Group By:** time($__interval)

Row Title	Title	Description	Query Details
		successfully executed when 1, otherwise is 0.	
Healthchecks - RDS	Entity Repository	Displays entity repository healthcheck. This healthcheck is carried out by reference data service in order to validate if it is able to reach entity repository, which is an abstraction for a non-sql db. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'entity-repository' , "host" = '$host' , $timeFilter **Group By:** time($__interval)
HTTP Requests	HTTP Throughput	Measures reference-data service HTTP throughput on the selected instance.	**Query A** **Field:** "count" **From:** "io.dropwizard.jetty.MutableServletContextHandler.requests" **Where:** "host" = '$host' , $timeFilter **Group By:** time($interval), "host"
HTTP Requests	HTTP Throughput per Method	Measures reference-data service HTTP throughput per method on the selected instance.	**Query A** **Field:** "count" **From:** "io.dropwizard.jetty.MutableServletContextHandler.get-requests" **Where:** "host" = '$host' , $timeFilter **Group By:** time($interval), "host"
HTTP Requests	2XX Requests Rate	Measures reference-data service HTTP throughput, on the selected instance, with return code falls into 2xx.	**Query A** **Field:** "count" **From:** "io.dropwizard.jetty.MutableServletContextHandler.2xx-responses" **Where:** "host" = '$host' , $timeFilter **Group By:** time($interval), "host"
HTTP Requests	4XX Errors	Measures the number of 4xx errors returned by the selected instance.	**Query A** **Field:** "count" **From:** "io.dropwizard.jetty.MutableServletContextHandler.4xx-responses" **Where:** "host" = '$host' , $timeFilter **Group By:** time($interval), "host"
HTTP Requests	5XX Errors	Measures the number of 5xx errors returned by the selected instance.	**Query A** **Field:** "count" **From:** "io.dropwizard.jetty.MutableServletContextHandler.5xx-responses" **Where:** "host" = '$host' , $timeFilter **Group By:** time($interval), "host"fill(none)
Global Database Requests	Global Database Throughput	Measures the overall database throughput, considering all operations executed by the selected instance. It also shows a break down per operation.	**Query A** **Field:** "count" **From:** "reference-data-service.entity-repository" **Where:** "host" = '$host' , $timeFilter **Group By:** time($interval), "host", "operation"
Global Database Requests	Global Database Errors	Measures the total number of errors registered, on the selected instance, while performing some operations onto the database.	**Query A** **Field:** "count" **From:** "reference-data-service.entity-repository.error" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval), "host"

Row Title	Title	Description	Query Details
Global Database Requests	Global Database Errors - Entities Not Found	Measures the total number of errors caused by entities not found, registered on the selected instance while performing some operations onto the database.	**Query A** **Field:** "count" **From:** "reference-data-service.entity-repository.entitynotfound" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval), "host"fill(none)
Global Database Requests	Global Database Latency per Percentile for Operation	Measures global database latency at percentile 99 considering all operations executed by the selected instance.	**Query A** **Field:** "p50", "p95", "p99", "p999", "max" **From:** "reference-data-service.entity-repository" **Where:** "host" = '$host' , operation='$operation' , $timeFilter **Group By:** time($__interval), "host"
Database Operation	Operation throughput for Entity	Measures the database throughput, considering the operations executed by the selected instance using the selected entity type.	**Query A** **Field:** "count" **From:** "reference-data-service.entity-repository" **Where:** "host" = '$host' , operation='$operation' , entityType = '$entity' , $timeFilter **Group By:** time($__interval), "host", "operation"
Database Operation	Operation for Entity Latency Per Percentile	Measures the latency per percentile to execute the specified operation on the selected entity type using the selected instance.	**Query A** **Field:** "p50", "p95", "p99", "p999", "max" **From:** "reference-data-service.entity-repository" **Where:** "host" = '$host' , operation='$operation' , entityType = '$entity' , $timeFilter **Group By:** time($__interval), "host", "operation"
JVM	JVM Heap	Measures the jvm heap usage of the selected instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.heap.used" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval) **Query B** **Field:** "value" **From:** "letmeknow.jvm.memory.heap.max" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval), "host"
JVM	JVM Heap Used (By Gen)	For the selected instance, considering the heap used, displays the usage of each generation.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.pools.G1-Old-Gen.used" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)
JVM	JVM Heap Used After GC (By Gen)	For the selected instance, considering the heap used, displays the usage of each generation after gc.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.pools.G1-Old-Gen.used-after-gc" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)
JVM	JVM Heap Usage After GC	Measures the jvm heap usage after gc on the selected instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.heap.after-gc.usage" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)
JVM	JVM Garbage Collector Time	Measures the JVM GC duration on the selected instance. Long garbage collection pause durations indicate that the heap size is not adjusted for the actual usage.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.gc.G1-Old-Generation.time" **Where:** "host" = '$host' , $timeFilter **Group By:** N/A

Row Title	Title	Description	Query Details
JVM	JVM Active Threads Count	Measures the current number of active threads in the JVM on the selected instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.threads.count" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)
JVM	JVM File Descriptors Ratio	Measures the jvm file descriptors usage ratio of the selected instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.filedescriptor" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-0.x/release-0.16/whats-changed

On this page

What's Changed

Release `0.16.1` â

Add Release Notesâ

Docs

We've now added `0.16.0` Release Notes as the release was previously issued without them.

Standardize amount fieldsâ

Bug Fix

We've noticed a slight discrepancy on the amount fields being shown in Case Manager User Interface.

In Transfers - Transaction, we were already displaying the decimal amount, which isn't the one used in this schema version.

As of 0.16.1, we standardized all amount fields to look the same across all screens. For release 0.17.0 we plan to use the new fields on the schema.

Release `0.16.0` â

Case Manager Workflow MVPâ

Rework

We have customized the CM Workflow according with the MVP specification. See Case Manager Workflow MVP for a comprehensive list of changes.

Allow setting status reason via APIâ

Feature FTMT-1085

Previously, tagging via feedback API wouldn't set any status reason in Case Manager. This can now be set via the `external_status_reason` field in the feedback API.

Case Manager UI MVPâ

Rework

We have customized the CM UI according with the MVP specification. See Case Manager UI MVP for a comprehensive list of changes.

Add Linked Events widget to Case Manager event details pageâ

Feature

Adds a widget to the event details page that shows all linked events.

linked-alerts

Update commons UDFs to support UK postal codesâ

Feature

Adds support to zip Code with less than 3 characters.

Add Workflow Monitoring Service to workflowâ

Feature

These new metrics are separated by channel (Transfers and Digital Activity). The Workflow Events section was updated with the new metric.

Set correct Zulu version in documentationâ

Docs

We've updated our requirements to link to the correct Zulu version. Example: Case Manager requirements - Java Environment

Add dead letter system documentationâ

Docs

Dead Letter System.

Integrate Liquibase Sessionlock extensionâ

Feature

Change Liquibase lock mechanism from "database entry" to a native RDBMS lock. With this change, when our pipelines terminate unexpectedly, the lock will automatically release soon after. Previously, the database entry would remain there until it was manually dropped, causing a crashloop.

Add lifecycle_id index in Event Storageâ

Bug Fix

Adds a DB index on the `lifecycle_id` column, avoiding `504 gateway timeouts` when calling the Feedback API.

warning

Creating the index is a concurrent operation that won't affect significantly the running application but it will require a long deployment window that should be tested in Pre-Production against relevant data.

Create UDF and list for TMX risk scoreâ

Feature

Created UDF `Tmx.risk` which receives the chain of codes and returns the respective risk. This is documented in Custom UDFs.

On this page

What's Changed

Release `1.3.1` â

Add new "Pulse Workflow Audit" pipeline to Logstashâ

Feature

A new Logstash pipeline was added which extracts workflow information (including rules and workflow elements) whenever there is a publish.

The documentation was updated with the data schema for this pipeline.

Add new Field to Event Notificationâ

Bug Fix

New field `sender_transaction_bank_account_number` added to Payment Fulfillment outbound call from Case Manager to AFIS.

It will appear now in the Payload coming from Event notification.

Release `1.3.0` â

Case Managerâ

Changes to Case Manager fields and positionsâ

Rework

We have customized the CM UI accordingly with the new requirements.

See Case Manager Transfers UI changes for a comprehensive list of Transfers Event UI page changes.

See Case Manager Digital Activity UI changes for a comprehensive list of Digital Activity Event UI page changes.

See Search Page UI changes for a comprehensive list of Search Page UI page changes.

See Search Result UI changes for a comprehensive list of Search Result UI page changes.

See Customer Details UI changes for a comprehensive list of Search Result UI page changes.

Add Saf Indicator to Case Managerâ

Feature

Created Saf Indicator field in Payments schema, with the objective of replacing the payment_saf field.

Saf Indicator is available in Case Manager automation rules and SLAs.

Fix scores with decimal numbers numbers in some Case Manager tablesâ

Bug Fix

We're now rounding scores so that they're always displayed as whole numbers.

Bulk actions made available in transaction historyâ

Rework

As requested, we've made the bulk actions (fraud/not fraud) available in the transaction history.

Note that with this change, analysts will be able to label transactions without being assigned to them beforehand.

Fix status reason not getting updated via custom extensionâ

Bug Fix

Fixed an issue that prevented the status reason to be updated by the `Set Reason | Payment Reversal as Unable to Contact - Decline` custom extension.

Achieved by changing the corresponding status reason `Payment Reversal as Unable to Contact - Decline` to match the reason configured in the custom extension.

warning

The reason shouldn't be edited for the extension to work.

Fix transactions not opening in Case Managerâ

Bug Fix

As part of a previous feature, we've removed the `status_transfers` state machine.

This caused the transactions in one of these states to not open in Case Manager.

Because of that, we've re-enabled the state machine but it's marked as **deprecated** and will no longer be updatable.

The transactions will use one of the remaining state machines.

Search field "Customer Email" now searches both `sender_email` and `customer_email` .â

Rework

The search for Customer Email on Case Manager was changed as requested, now this search box is able to filter to two different keys: `sender_email` and `customer_email` .

This fixes the reported issue where searching for Customer Email on the search screen was not returning any results

Update Get Next so it no longer fetches non-alerted events.â

Bug Fix

Updated Assign and Lock configurations so that the Get Next no longer fetches non-alerted events.

From now on the Linked Alerts widget will only show alerted events.

Event Storage ETLâ

Add `formatted_results` to ETL extractionsâ

Feature

We've added a new column `formatted_results` to the ETL extractions to store rules explanations in the following format:

```
[
    {
        "ruleId": "rule_id_1",
        "explanation": "Customer name is Fraudulent Joe"
    },
    {
```

```
        "ruleId": "rule_id_2",
        "explanation": "This rule triggered because customer name is Joe. Customer name: Fraudulent Joe."
    }
]
```

Copy

A new property has been added to enable/disable this new column: `eventstorageetl.extraction.includeFormattedResultTracking`.

By default, we've enabled this new property.

ETL tool support for dynamic outcomes and scores.â

Feature

This column holds all the outcomes and scores in the following format:

```
{
    "fraud": {
        "outcomeDecision": true,
        "score": 0.0
    },
    "decision": {
        "outcomeDecision": "approve",
        "score": [
            0.0,
            0.0,
            0.0
        ]
    }
}
```

Copy

There's a new property `eventstorageetl.extraction.outcomesInSingleField` that when enabled exports a new column `outcomes_and_scores`.

By default, we've enabled this new property.

For this to work `eventstorageetl.extraction.includeOutcomes` (`include_outcomes` ansible variable) needs to be enabled as well.

Add option to disable outcomes extraction in ETLâ

Feature

The ansible variable `include_outcomes` can be changed to `false` to disable the outcomes extraction.

Miscâ

Feedzai Platform Upgradesâ

Feature

We've upgrade Reference Data Service to version 3.6.7. This upgrade resolves an issue that caused some PATCH and PUT reference data requests to fail to persist data.

Artifacts page has been updated accordingly.

Update logstash pipelinesâ

Feature

We've updated the logstash pipelines, the new version includes:

- added a ansible configuration variable to limit the extractions max file size.
- enabled "Add Note" pipeline

The external notifications installation documentation was also updated according to the previous changes.

Increase Application JVM heap size according to performance testsâ

Improvement

warning

In order to have these configured by default in both *production* and *pre-production* you should confirm that `project-<env>` inventories are included when running Feedzai ansible-toolkit.

In practice that's `-i project-prod` and `-i project-pre-prod` respectively.

Fix Case Manager partitioning scriptsâ

Bug Fix

We've identified an issue in Case Manager partitioning scripts where project field indexes were not being created in the partitioned tables which led to performance degradation.

This version includes a fix to the partitioning scripts to add project field indexes. We've also included `EVENT_TIMESTAMP` to increase the database performance regarding Case Manager searches.

Improvements on External Notifications resiliency and observabilityâ

Improvement

The monitoring documentation as been updated accordingly.

Add a new page in Case Manager that reports the versions being usedâ

Feature

A new page was added to Case Manager which reports the Case Manager version, Solution version and project version running on that environment.

The page can be accessed via a web browser, by navigating to `/ext/version.json` in Case Manager.

Add `Shadow Fraud` , `Scam` and `Shadow Scam` outcomesâ

Feature

Scam Score is now available to be used in Case Manager, for example, in Automation Rules.

The shadow counterparts haven't been added to Case Manager core schema since shadow models are "non-live" and therefore not usually shown in Case Manager. Monitoring these models performance is a very different task that leverages different tools.

Allow JWK service configuration in High Available environmentsâ

Rework FTMT-9531

We've made JWK service endpoints configurable to support API requests across all nodes in High Available environments by introducing a new variable in the Ansible pipelines:

- jwk_service_host

For the complete description and default variables values see Pulse Configuration Reference.

Note: In environments with only one Pulse node this new variable doesn't need to be overridden.

On this page

Global Rollout

In this section, you will find comprehensive details regarding the main changes in the structure of our project to accommodate the global rollout changes.

This solution currently preserves backwards compatibility with the existing UK Outbound Payments solution, such that no changes to the deployment process or ansible variables should be explicitly needed there.

Please find below the list of changes introduced below.

General improvementsâ

Merging of default inventoriesâ

We have merged inventories that were essentially being used for the same purpose.

From this point forward, `dsf` and `default` inventories should be dropped in favor of `project-ds` and `project-default`, respectively.

The playbook examples in `Installation Guide` were updated accordingly.

Pulseâ

Cards applicationâ

DSF environment can now automatically create both the outbound payments application (on an initial deployment, when an existing payments application does not already exist) and the cards application (on an initial deployment, when an existing cards application does not already exist), using patches.

The `apps_enabled` variable controls which apps are created in Pulse DSF, the same as Pulse Runtime. This variable is documented in this [section](#).

We also added a patch to create a default Cards application in a Pulse Runtime Deployment, this does not run by default, but is controlled by the `pulse_commands_context` variable in the Pulse variables.yml file.

For example, setting this variable to have the value "cards" on a Pulse deployment, will run the patches related with the Cards Application.

The schema for the Cards application was also created based on the mappings that were accorded in the DMAs for the endpoints, namely Authorization, Authentication 3DS and Non Monetary Events.

Changes on variable directory structureâ

To allow support for global rollout initiatives, we've made significant changes to our folder structure organization.

These changes provide a more dynamic variable directory structure, using nested Ansible inventories for better flexibility and customization.

Specifically, inventories containing variables defined per-region or per-market should always be specified *after* the default ones. For example, to include `uk/project-prod/cards` variables for the UK region that take precedence over the `prod` environment variables (`project-prod` inventory), we might have something like the following:

```
python3.6 ansible-playbook \
    -i project-default \
```

```
    -i project-prod \
    -i ... \
    -i uk/project-prod/cards \
    pulse.yml ...
```

Copy

Please consult the updated Installing Feedzai with Ansible documentation for more details. The Blue-Green Documentation has also been updated accordingly.

Support for running patches for multiple appsâ

This is done by using a new Ansible variable named `apps_enabled` .

The `apps_enabled` variable is used to define which of the available apps will actually be deployed.

By doing so, patches will also run for each application, namely once per app declared in `apps_enabled` .

Each app has portfolios as well, ensuring we only run patches that are specific to a particular app type. (e.g. HSBC could deploy UK Cards and HSBC Cards under the same context, as they will have different app slugs it still guarantees different patch executions).

Case Managerâ

Support for multi regional channelsâ

Similarly to Pulse, we have introduced a new Ansible variable (`apps_enabled`) to control which channels are deployed in Case Manager.

My setting this variable with the applications that are deployed in Pulse, Case Manager deployments will configure the various channels to connect to the correct application and workflow.

If a channel's application/workflow is not yet deployed in Pulse, it will point to the outbound payments application but it won't process any events since Pulse will not publish anything to the channel respective queue.

You can find a detailed description of the deployment configuration in Case Manager Configuration Recommendations.

On this page

What's Changed

Release `2.1.0` â

New Ansible task to create symbolic links for multiple appsâ

Feature

In order to support multiple regions and market deployments, we have created a new Ansible task that creates filesystem symbolic links for each app slug to the directories that store the various libraries for the papp.

Case Manager Version Upgraded to 13.13.2â

Feature

We've upgraded Case Manager Version to 13.13.2.

Improve Case Manager extension configuration cachingâ

Rework

Case Manager was reading the configuration file to obtain the custom action (mobile outbound, payment fulfillment) on every trigger of the extension (via automation rules, for instance) along with the secret to decrypt the configuration. This could lead to rare occurrences in which too many files were being opened at the operating system level. This configuration is now only being read once on the first trigger of the automation rule for each extension type, and reusing this for all subsequent triggers of the extension.

Enable pkernel eager compilationâ

Feature

We have enabled eager compilation of the realtime metrics backend during Pulse application publishes. Previously, this compilation was performed on the first event processed after a publish, which would result in longer Pulse response times and timeouts on each node in the first seconds immediately following a publish.

New variable added to enable multithreaded workflow recoveryâ

Feature

A new ansible variable `number_threads_workflow_recovery` has added to set the number of threads used by Pulse to recover windows for real-time metrics during a full rolling publish. Previously, Pulse was using only a single thread, and this will continue to be the default. However, this number is now configurable.

Control custom UDFs importâ

Feature

We have added the possibility of controlling if custom UDFs (TMX and FivMarkers) are imported during deployment with the following variables:

- `import_tmx_udf`
- `import_fiv_markers_udf`

For more information regarding Pulse configuration options see [Pulse Configuration Recommendations](#).

Global Rollout Documentationâ

Docs

Initial support for multiple regions and multiple portfolios within individual regions has been added.

Please check the detailed documentation in Global Rollout Documentation.

Change Mobile Outbound Payment to send double amount field instead of integerâ

Feature

We've changed the Mobile Outbound notification to have the amount as a decimal instead of an integer. For example, an amount of Â£5 would previously be shown as 500 (in pence) and is now shown as 5.0 (in pounds).

Customer Details

Before:

After:

Field Modifications

Notes:

Upon investigating the original request, we have concluded that allowing multiple phone numbers to be added to a customer under a single field could potentially pose a risk and cause issues with metrics. Therefore, we recommend a different approach to this field.

One possible solution is to create separate fields for additional phone numbers, such as a "2nd Phone Number" field. This will allow for more flexibility while avoiding potential issues with metrics.

Change	Field
Fixed	Date of Birth
Fixed	Customer Since
Fixed	Customer Segment
Fixed	Email
Fixed	Contact Number
Fixed	Region
Fixed	Country
Renamed	Zip Code -> Customer Postcode
Renamed	Originator ID -> Customer ID
Renamed	Customer Segment -> Account Type
Removed	Driver's License
Removed	Citizen ID Number
Removed	Occupation
Removed	Street address
Removed	Passport Number
Removed	City
Removed	Tax Number
Removed	Annual Income
Removed	Country

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-0.x/release-0.16/case-manager-ui-mvp/digital-activity-change-data-details

Digital Activity - Change Data Details

This is a new widget, below is an example of what it will look like:

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-0.x/release-0.16/case-manager-ui-mvp/digital-activity-cop-details

Digital Activity - COP Details

New widget:

Digital Activity - Device

Before:

After:

Field Modifications:

Change	Field
Added	Device ID Type
Added	Fuzzy Device ID
Added	Device TDL
Added	HTTP User Agent
Added	Server Date time
Added	Browser Date
Added	Browser Timezone
Fixed	Device Channel
Fixed	Device Type
Fixed	Operating System
Fixed	Device Language
Removed	Model & Brand
Removed	Registration Date
Removed	Browser
Removed	Mobile App
Removed	Device Location
Removed	Device Screen Height
Removed	Device Screen Width
Removed	Registration Date
Removed	Device Phone Number
Removed	Device Email
Removed	Device Risk Rating
Removed	Device Risk Code
Removed	Device Risk Reason

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-0.x/release-0.16/case-manager-ui-mvp/digital-activity-event-summary

Digital Activity - Event Summary

Before:

After:

Field Modifications

Change	Field
Added	Source
Added	User ID
Added	Name of website
Enriched	Event Type
Fixed	Date
Fixed	Event Sub Type
Fixed	Customer Date of Birth
Removed	Customer Name
Removed	Event Status
Removed	Fail Reason

Digital Activity - Network

Before:

After:

Field Modifications:

Change	Field
Added	True IP City
Added	True IP Geo
Added	IP Country
Fixed	IP Location
Removed	Proxy

Digital Activity - Session

Before:

After:

Field Modifications:

Change	Field
Added	CAM Level
Added	Login Last Authentication
Fixed	BionicID
Removed	Account ID
Removed	Session First Seen

Digital Activity - Vendor Risk Data

New widget:

As agreed, for fields that are booleans we will use the Yes(True)/No(False) annotation instead of 1/0, as it will be more user friendly to analysts.

Transaction History - Digital Activity

Widget after modifications:

Field Modifications

Notes: We have identified an issue on our side that is currently blocking the transformation of values in 3 fields within this widget. The affected fields are: `Event Type`, `Login Last Authentication`, and `Device Type`. We are currently conducting an investigation to address this issue and plan to include a fix in a later release.

As a result, these fields will be displayed without any transformation in the current version

Change	Field
Added	Data source
Added	CAM Level
Added	Login Last Authentication
Added	Fuzzy Device ID
Added	True IP City
Added	True IP Geo
Added	TDL
Added	Device Language
Added	Server Date time
Added	TMX score
Added	Voice Scam score
Added	Biocatch ATO score
Added	Is Rat
Added	Is Recent Rat
Added	Is Bot
Added	Is Emulator
Added	Is Malware
Added	Change Data Reason
Added	Change Data Previous
Added	Change Data Delivery
Added	Change Data New
Added	Voice Score Type New
Added	Receiver Name
Added	Payment COP Result
Added	Receiver created date
Fixed	Date
Fixed	Event Type
Fixed	Event Sub type

Change	Field
Fixed	Device ID
Fixed	Device Type
Removed	Account ID
Removed	Customer Email
Removed	Customer Phone Number
Removed	Customer Address
Removed	Device Email
Removed	Device Phone Number
Removed	Jailbroken Check
Removed	Most Restrictive Rule ID

Transaction History - Transfers

Below you can find screenshots of how this widget will look like after all the required modifications done:

Field Modifications

Notes:

Initially, the "amount" field was intended to be a concatenation of the currency and amount fields. However, during the development process, we found a limitation that prevented us from implementing this mapping as intended. As a workaround, we have implemented two separate fields for the currency and amount.

Similarly, we have also encountered limitations when mapping the 3-digit currency code to the intended code, and when mapping the Transaction Type to the corresponding description. As of now, we will display the values sent without mapping, while we continue to work on finding alternatives to resolve these issues and implement them correctly.

Change	Field
Added	Currency
Fixed	Amount
Fixed	Transaction Type
Fixed	Transaction Sub-type
Fixed	Device ID
Removed	Direction

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-0.x/release-0.16/case-manager-ui-mvp/transfers-beneficiary-account-details

Transfers - Beneficiary Account Details

Before:

After:

Field Modifications

Change	Field
Enriched	Currency
Fixed	Type
Fixed	BIC
Renamed	Beneficiary Bank Account Number -> Bene Account Number
Removed	ID
Removed	IBAN
Removed	Open Date
Removed	Balance
Removed	Credit Limit
Removed	Debt Balance
Removed	Status
Removed	Last Activity
Removed	Number of Cards

Transfers - Beneficiary Bank

Below you can find screenshots of how this widget will look like after all the required modifications:

Field Modifications

Change	Field
Fixed	Name
Fixed	Beneficiary Bank Address
Fixed	Bank City

Transfers - Beneficiary Details

Before:

After:

FIelds Modification:

Change	Fields
Enriched	Currency
Enriched	Country
Renamed	Zip Code -> Bene Postcode
Fixed	Name
Fixed	Registration Date
Removed	Beneficiary ID
Removed	Segment
Removed	Date of Birth
Removed	Citizen ID Number
Removed	Passport Number
Removed	Social Security Number
Removed	Tax Number
Removed	Email Address
Removed	Phone Number
Removed	Address
Removed	City
Removed	Region
Removed	Number of Cards
Removed	Number of Accounts
Removed	Total Balance
Removed	Total Credit Limit
Removed	KYC Level
Removed	Risk Rating
Removed	Occupation
Removed	Beneficiary Alias
Removed	Beneficiary Alias Type
Removed	Beneficiary Alias Status
Removed	Beneficiary Alias Registration Time
Removed	Beneficiary Alias Last Update Time

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-0.x/release-0.16/case-manager-ui-mvp/transfers-device

Transfers - Device

`Session Alerts` and `Session` widgets were removed, whose fields have been merged with the Device widget.

Before:

After:

Field Modifications

Change	Field
Fixed	Device ID
Fixed	Device Type
Fixed	Device OS
Fixed	Device Language
Fixed	Device User-Agent
Fixed	Associated Email
Fixed	Email Changed
Fixed	Associated Phone Number
Fixed	Phone Number Changed
Fixed	Type
Fixed	Session ID
Delayed	Malware Infected
Delayed	Device First Seen
Removed	BionicIDâ¢
Removed	Session First Seen
Removed	Latitude
Removed	Longitude
Removed	Device IMEI
Removed	Password Reset Requested
Removed	View

Transfers - Network

Before:

After:

Field Modifications

Change	Field
Fixed	IP Address
Renamed	Device IP Address Location -> IP City
Removed	Device ISP
Removed	Proxy

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-0.x/release-0.16/case-manager-ui-mvp/transfers-originator-account-details

Transfers - Originator Account Details

Before:

After:

Field Modifications

Change	Field
Added	Segment
Enriched	Currency
Fixed	Balance
Renamed	Originator Bank Account Number -> Customer Bank Account Number
Renamed	Type -> Account type
Removed	ID
Removed	IBAN
Removed	BIC
Removed	Open Date
Removed	Credit Limit
Removed	Debt Balance
Removed	Status
Removed	Last Activity
Removed	Number of Cards

Transfers - Originator Bank

Below you can find screenshots of how this widget will look like after all the required modifications:

Field Modifications

Change	Field
Fixed	Name
Fixed	Originator Bank Address
Removed	Bank City

Transfers - Originator Details

Before:

After:

Field Modifications

Notes:

Upon investigating the original request, we have concluded that allowing multiple phone numbers to be added to a customer under a single field could potentially pose a risk and cause issues with metrics. Therefore, we recommend a different approach to this field.

One possible solution is to create separate fields for additional phone numbers, such as a "2nd Phone Number" field. This will allow for more flexibility while avoiding potential issues with metrics.

Change	Field
Added	Phone Number Last Changed
Added	Work Phone 1 Last Change Date
Added	Work Phone 2 Last Change Date
Added	Home Phone 1 Last Change Date
Added	Home Phone 2 Last Change Date
Added	Mobile Phone 1 Last Change Date
Added	Mobile Phone 2 Last Change Date
Enriched	Currency
Enriched	Country
Fixed	Date of Birth
Fixed	Email Address
Fixed	Phone Number
Fixed	Address
Fixed	Region
Fixed	Total Balance
Renamed	Originator ID -> Customer ID
Renamed	Zip Code -> Customer Postcode
Removed	Joint Account CIN
Removed	Segment
Removed	Registration Date
Removed	Citizen ID Number
Removed	Passport Number
Removed	Social Security Number
Removed	Tax Number
Removed	City

Change	Field
Removed	Number of Cards
Removed	Number of Accounts
Removed	Total Credit Limit
Removed	KYC Level
Removed	Risk Rating
Removed	Occupation
Removed	Originator Alias
Removed	Originator Alias Type
Removed	Originator Alias Status
Removed	Originator Alias Registration Time
Removed	Originator Alias Last Update Time

Transfers - Transaction

Before:

After:

Field Modifications

Notes:

~~Currently, the Amt (Bene) field will display the amount along with the currency. However, we are currently awaiting the addition of a new field in the schema that will allow us to display the amount in decimal format. Until this issue is resolved, it is expected that this field will display "N/A".~~

As of 0.16.1 we standardized all amount fields to look the same across all screens. For release 0.17.0 we plan to use the new fields on the schema.

The VRP_INDICATOR field isn't present on the schema yet, so for now it is expected that this field will display "N/A".

Change	Field
Added	RPS Queue Name
Added	Amt (Bene)
Added	FX Deal ID
Added	Payment Date local timestamp
Added	VRP_INDICATOR
Added	Transfer to self account indicator
Added	New Beneficiary indicator flag
Added	New Beneficiary create date
Added	Payee Cop Name
Added	CoP Description
Added	SO MandateID
Added	SO Mandate Status ID
Added	SO MandateID+Ref
Enriched	Transaction Type
Enriched	Channel
Enriched	CoP Description
Fixed	Source
Fixed	Original Amount
Fixed	Payment Value Date
Fixed	Frequency
Fixed	Recurring Payment
Fixed	Beneficiary Bank Branch ID
Fixed	Reference Number
Fixed	Transaction Notes

Change	Field
Fixed	Transaction Method
Renamed	Date -> Event Date
Removed	Payment Value Date
Removed	Product Code
Removed	Bulk ID
Removed	Geographic Scope
Removed	Originator Urgency
Removed	Number of Payments
Removed	SCA Result
Removed	Last SCA
Removed	Transaction Goal
Removed	Good Beneficiary Flag
Removed	Bad Beneficiary Flag
Removed	Total Cust for Bene(Customers paid for Beneficiary)
Removed	Beneficiary First Paid Date
Removed	Total Num Payment Bene
Removed	Max Value Paymt Bene
Removed	Bene Cust Status
Removed	Cust First for Bene
Removed	SO Ammend Date
Removed	SO Ammend Time
Removed	SO Action Type
Removed	SO Start Date
Removed	Signed Status Code
Removed	SO Fequency Info
Removed	SO Schedule Ref Num
Removed	SO referrence
Removed	Most Restrictive Rule ID

Case Manager UI MVP

Please find below a comprehensive list of the changes that have been applied to all widgets:

- Transfers - Transaction
- Transfers - Originator Details
- Transfers - Originator Account Details
- Transfers - Beneficiary Details
- Transfers - Beneficiary Account Details
- Transfers - Originator Bank
- Transfers - Beneficiary Bank
- Transfers - Network
- Transfers - Device
- Customer Details
- Digital Activity - Event Summary
- Digital Activity - Network
- Digital Activity - Session
- Digital Activity - Device
- Digital Activity - COP Details
- Digital Activity - Vendor Risk Data
- Digital Activity - Change Data Details
- Transaction History - Transfers
- Transaction History - Digital Activity

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-0.x/release-0.16/case-manager-workflow-mvp/workflow-changes

On this page

Case Manager Workflow MVP

Add new operational status statesâ

Rework

Operational status now includes 2 new states:

- Wait to contact Customer
- Contact Customer

These 2 new states are core to the Case Manager Workflow MVP.

Update Alerts Page and Get Nextâ

Rework

Updated Alerts Page and Get Next so that only "Unreviewed" alerts are considered.

Update Assign and Lock Criteriaâ

Rework

Change on Assign and lock transfers schema in order to make the linking criteria to be the customer id like requested

Implement integration between Pulse and Case Manager regarding treatment strategyâ

Feature

Add new *outcome* to Workflow - `treatment_strategy` - defaulting to `none` (manual). This outcome is core to the Case Manager Workflow MVP.

Replace Status Reasonsâ

Rework

Changed existing Case Manager Status Reasons according to HSBC request. Note that, since there's no Status Reason for Digital Activity and one is mandatory to mark as Fraud / Not Fraud, currently it's not possible to mark digital activity events.

With that in mind, a *Fraud Prevention Manager* can add new Status Reasons to allow the previous behavior.

Implement core SLA's and Automation Rulesâ

Rework

We've implemented the core SLA's and Automation Rules to implement Case Manager Workflow MVP.

Digital Activity - Confirmation of Payee Details

Event: Digital Activity

Widget: Confirmation of Payee Details

Before:

After:

Action	Field	UI
Rename Section	COP Details	Confirmation of Payee Details
Rename	Result of the Payment COP	Result of the Payment CoP
Rename	Payment Bene Create Date	Payment Beneficiary Created Date

Digital Activity - Device

Event: Digital Activity

Widget: Device

Before:

After:

Action	Field	UI
Remove	Device Channel	

Digital Activity - Event Summary

Event: Digital Activity

Widget: Event Summary

Before:

After:

Action	Field	UI
Rename	Channel	Device Channel

Digital Activity - Vendor Risk Data

Event: Digital Activity

Widget: Vendor Risk Data

Before:

After:

Action	Field	UI
Moved from Change Data Details	Smart ID Score	

Case Manager Digital Activity UI Page Changes

Please find below a comprehensive list of the changes that have been applied to all widgets:

- Digital Activity - Event Summary
- Digital Activity - Device
- Digital Activity - Confirmation of Payee Details
- Digital Activity - Vendor Risk Data

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-1.x/release-1.3/case-manager-search-page/search-digital-activity-rework

Search Page - Digital Activity

Widget: Search Page Digital Activity Advanced Search

Before:

After:

Action	Field	UI
Remove	Account ID	
Remove	Device Email	
Remove	Device Phone Number	
Rename	IP Location	IP City

Search Page - Top Section

Widget: Search Page Top Section

Before:

After:

Action	Field	UI
Remove	Rule ID	

Search Page - Transfers

Widget: Search Page Transfers Advanced Search

Before:

After:

Action	Field	UI
Remove	Direction	
Remove	Originator Name	
Rename	BIC	Beneficiary Bank Identifier Code
Remove	Device ID	
Remove	Citizen ID Number	

Case Manager Search UI Page Changes

Please find below a comprehensive list of the changes that have been applied to all widgets:

- Search Page - Top Section
- Search Page - Transfers
- Search Page - Digital Activity

Search Result - Digital Activity

Widget: Search Result Digital Activity

Action	Field	UI
Remove	Account ID	
Remove	Device Phone Number	
Rename	IP Location	IP City

Search Result - Transfers

Widget: Search Result Transfers

Before:

After:

Action	Field	UI
Remove	Direction	
Rename	Originator Name	Customer Name
Rename	Originator Phone Number	Customer Phone Number
Rename	Originator Email	Customer Email
Remove	Beneficiary Email	
Remove	Device ID	

Case Manager Search Result UI Page Changes

Please find below a comprehensive list of the changes that have been applied to all widgets:

- Search Result - Transfers
- Search Result - Digital Activity

Customer Details

Widget: Customer Details

Before:

After:

Action	Field	UI
Add	Customer FIV Marker	

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-1.x/release-1.3/case-manager-transfers-ui-changes/transfers-beneficiary-account-details-rework

Transfers - Beneficiary Account Details

Event: Transfer

Widget: Beneficiary Account Details

Before:

After:

Action	Field	UI
Rename	BIC	Bank Identifier Code

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-1.x/release-1.3/case-manager-transfers-ui-changes/transfers-beneficiary-bank-rework

Transfers - Beneficiary Bank

Event: Transfer

Widget: Beneficiary Bank

Grid is removed and fields moved to Beneficiary Account Details

Before:

After:

Action	Field	UI
Rename	Name	Bank Name
Rename	Beneficiary Bank Address	Bank Address
Move	Bank City	

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-1.x/release-1.3/case-manager-transfers-ui-changes/transfers-beneficiary-details-rework

Transfers - Beneficiary Details

Event: Transfer

Widget: Beneficiary Details

Before:

After:

Action	Field	UI
Rename	Bene Postcode	Postcode

Transfers - Confirmation of Payee Details

Event: Transfer

Widget: Confirmation of Payee check

Before:

After:

Action	Field	UI
Rename section	Confirmation of Payee check	Confirmation of Payee Details
Moved from Transaction Grid and Renamed	Payee Cop Name	Payee CoP Name
Rename	COP Payment Match Name	Payee CoP Match Name
Moved from Transaction Grid and Renamed	CoP Description	Payment CoP Description
Rename	Beneficiary Name Response - COP	Beneficiary Name Response
Rename	Beneficiary Match Name Response - COP	Beneficiary Match Name Response

Transfers - Device

Event: Transfer

Widget: Device

Device Grid has been removed

Action	Field	UI
Moved to Transaction Grid	Session ID	

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-1.x/release-1.3/case-manager-transfers-ui-changes/transfers-network-rework

Transfers - Network

Event: Transfer

Widget: Network

Network Grid has been removed

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-1.x/release-1.3/case-manager-transfers-ui-changes/transfers-originator-account-details-rework

Transfers - Originator Account Details

Event: Transfer

Widget: Originator Account Details

Before:

After:

Transfers - Originator Bank

Event: Transfer

Widget: Originator Bank

Grid is removed and fields moved to Originator Account Details

Before:

After:

Action	Field	UI
Rename	Name	Bank Name
Rename	Originator Bank Address	Bank Address

Transfers - Originator Details

Event: Transfer

Widget: Originator Details

Before:

After:

Action	Field	UI
Rename	Customer Postcode	Postcode

Transfers - Transaction

Event: Transfer

Widget: Transaction

Before:

After:

Action	Field	UI
Rename	VRP_INDICATOR	Variable Recurring Payment Indicator
Rename	Transfer to self account indicator	Transfer to self account Indicator
Rename	New Beneficiary indicator flag	New Beneficiary Indicator
Rename	New Beneficiary create date	New Beneficiary Created Date
Rename	Payee Cop Name	Payee CoP Name
Rename	SO MandateID	SO Mandate ID
Remove	SO MandateID+Ref	
Add	SO Mandate ID + Schedule Number	
Add	Treatment Strategy	
Add	Treatment Strategy Priority	
Moved to Confirmation of Payee Details Grid	Payee Cop Name	
Moved to Confirmation of Payee Details Grid	CoP Description	
Moved from Device Grid	Session ID	

Transaction Status field Transformations

Input	Transformation
A	Approved/Successful
B	Cancelled by Bank
C	Cancelled by Customer
D	Declined/Unsuccessful
P	Pending
R	Reversal

SO Mandate Status ID field Transformation

Input	Transformation
O	Original
C	Changed
D	Deleted

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-1.x/release-1.3/case-manager-transfers-ui-changes/ui-changes-transfers

Case Manager Transfers UI Page Changes

Please find below a comprehensive list of the changes that have been applied to all widgets:

- Transfers - Transaction
- Transfers - Originator Details
- Transfers - Originator Account Details
- Transfers - Confirmation of Payee Details
- Transfers - Beneficiary Details
- Transfers - Beneficiary Account Details
- Transfers - Originator Bank
- Transfers - Beneficiary Bank
- Transfers - Network
- Transfers - Device

On this page

HSBC-2602-Improve-DTP-Dataflow-pipeline-dependencies-file-staging

Improve DTP Dataflow pipeline dependencies file stagingâ

Feature HSBC-2602

Currently, every time a publish occurs with the DTP service enabled, the Dataflow pipeline dependencies file is uploaded to the configured staging bucket.

We have updated this behavior; from now on, the dependencies file will only be uploaded once per release.

The current active dependencies file version is in the Pulse property: `pulse.global.services.rte.workflow.dtp.dataflow.fileToStageVersion`.

On this page

HSBC-2696-Add-new-Case-Manager-enrichment-fields-for-Inbound-Payments

Add new Case Manager enrichment fields for Inbound Paymentsâ

Feature FTMT-24520 HSBC-2696
Also In: 2.9.0

The following fields can now be enriched when displayed in Case Manager for Inbound Payments events:

- sender_transaction_code (senderTransactionCodeEnrich)
- sender_bank_country_code (countryEnrich)
- receiver_bank_country_code (countryEnrich)
- is_batch (isBatchEnrich)
- payment_type (inboundPaymentTypeEnrich)
- payment_status (paymentStatusEnrich)
- sender_transaction_original_currency (currencyEnrich)

With these changes, the field sent in the payload can now be transformed, as long as it is correctly mapped and matches a valid value.

Assuming this, if in an event payload `sender_transaction_code` is `510`, what will be displayed in Case Manager is `Mastercard Other Credit`

These new enriched fields can be added to Case Manager directly via the self-serviceable edit feature. This can be achieved by using the advanced configuration and a template similar to the following:

```
{
    "key": "event.fields.sender_transaction_original_currency",
    "type": "currencyEnrich",
    "title": "Sender Transaction Original Currency"
}
```

Copy

note

The `type` value should correspond to what is inside the parentheses in the list of new fields above. If the type is not changed, what will be displayed is exactly the same value that is in the payload.

HSBC-2885-HSBC-2958-HSBC-3214-HSBC-3376-HSBC-3482-Case-Manager-version-upgraded-to-13-22-0

Case Manager version upgraded to 13.22.0â

Feature FTMT-18187 FTMT-22596 FTMT-22865 FTMT-22986 FTMT-27682 WPB-239611 HSBC-2885 HSBC-2958 HSBC-3214

We've upgraded Case Manager to version 13.22.0 which includes the following:

- Fix an issue where the event note payload was not including the event payload (including the event tenant name)
- Fix an issue where refreshing Case Manager's search page would throw a warning if the previous search had a multi-criteria filter (e.g. `Customer ID`)
- A new *Smart Search* functionality described below
- A new feature where the Activity Log now shows when a change was triggered by an Automation Rule (more details below)

Smart Searchâ

With *Smart Search*, when searching for events using filters for channel-specific fields (Outbound, Digital Activity, Cards, or Inbound Payments), the search results will only display events that exactly match the selected filters.

Hereâs how it works:

- If a search is performed using only common filters, all tabs will be enabled, displaying the total number of filtered events.
- If a search includes a channel-specific filter (for example, *Beneficiary Name* in Inbound Payments), only the corresponding tab will be enabled, showing the total number of matching events. The tabs corresponding to the other use cases will not display any results.

note

Since the From and To Date filter is mandatory in all searches, the All Channels tab will always display all events, regardless of other search filters.

Automation Rules in Activity Logâ

When there is a change in an event that is the result of an Automation Rule triggering, the Activity Log now shows the Automation Rule instead of the **system** user. Clicking the Automation Rule name will link to the respective Automation Rule (if the user has permissions to access it).

This feature has two important caveats:

1. It is **not retroactive**, meaning that only Automation Rules triggered in this version will be shown in the Activity Log.
2. If the **Automation Rule is deleted, Case Manager will not be able to show it** in the Activity Log and will revert back to the previous behaviour of showing the **system** user as the actor.
   - Note: this **does not affect disabled Automation Rules** âthose will show up in the Activity Log.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-2.x/release-2.10.0/individual-release-notes/HSBC-2940-Change-the--notification-type--value-for-the-Send-2-WAY--1-WAY-SMS-Case-Manager-extension

On this page

HSBC-2940-Change-the--notification-type--value-for-the-Send-2-WAY--1-WAY-SMS-Case-Manager-extension

Change the `notification_type` value for the *"Send 2 WAY / 1 WAY SMS"* Case Manager extensionâ

Feature FTMT-29922 HSBC-2940

We've changed the value of the `notification_type` field in the *"Send 2 WAY / 1 WAY SMS"* Case Manager extension from `"payment_mobile_push"` to `"2_way_1_way_sms"`.

This change makes it possible to distinguish the *"Send 2 WAY / 1 WAY SMS"* and *"Send Mobile Payment Push"* notifications downstream.

Please refer to the AFIS Extensions reference for more details.

HSBC-2978-Add-support-for-Dataproc-2-2-in-Pulse-pipeline

Add support for Dataproc 2.2 in Pulse pipelineâ

Feature FTMT-22240 HSBC-2978
Also In: 2.8.4 2.9.5

To fully support the Dataproc 2.2 migration described in Google Dataproc 2.2, we've added two new configuration option to set the Java version used by Pulse jobs (e.g. datasource analysis, scheduled metrics):

- `spark_datascience_jdk_home` : Sets the `JAVA_HOME` for Data Science jobs executed in Dataproc, e.g. datasource analysis
- `spark_distributed_jdk_home` : Sets the `JAVA_HOME` for other jobs executed in Dataproc, e.g. scheduled Metrics Jobs

For more details about the new variables and their default values, refer to Pulse Configuration Recommendations.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-2.x/release-2.10.0/individual-release-notes/HSBC-3050-Add-filtered-index-on--customer-citizen-id-number--to-optimize-Case-Manager-searches

On this page

HSBC-3050-Add-filtered-index-on--customer-citizen-id-number--to-optimize-Case-Manager-searches

Add filtered index on `customer_citizen_id_number` to optimize Case Manager searchesâ

Feature HSBC-3050
Also In: 2.8.1 2.9.0

We've recreated the `customer_citizen_id_number` indexes to improve performance by removing non-null values from the index. This change is specially helpful during Case Manager searches in deployments where `customer_id` is used instead of `customer_citizen_id_number`.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-2.x/release-2.10.0/individual-release-notes/HSBC-3058-New-API-to-change-log-levels

On this page

HSBC-3058-New-API-to-change-log-levels

New API to change log levelsâ

Feature HSBC-3058

We've added a new API that allows log levels to be modified during runtime. This API can be used to change the log levels in the Pulse Server, Pulse Applications and Case Manager. It uses RabbitMQ to make sure a single request to the API is propagated across all replicas of a given component, allowing each to change its log level.

The API can be used via authenticated `GET` or `POST` requests to the Pulse UI host and requires the logged in user to have the `security:logging:update` permission:

```
POST http(s)://<PULSE_UI>/admin/api/loggers/{targetComponent}/{loggerName}/{loggerLevel}
```

By using the `targetComponent` parameter, the API will propagate the log change request to either Pulse Server, Pulse Application or Case Manager. A more comprehensive architecture and API spec can be found in the Change Log Levels at Runtime guide.

On this page

HSBC-3066-Add-tenancy-information-to-rules-extracted-by-the--Pulse-Workflow-Audit--pipeline

Add tenancy information to rules extracted by the `Pulse Workflow Audit` pipelineâ

Feature FTMT-26643 HSBC-3066
Also In: 2.9.0

We've added three new fields to the `Pulse Workflow Audit` pipeline:

- `workflow.rules[].isMultiTenant` : indicates whether a rule snapshot belongs to a multi tenanted rules projects
- `workflow.rules[].tenantId` : indicates the tenant key of the rules snapshot if it belongs to a multi tenanted rules projects and `null` otherwise
- `workflow.elements[type = rules].isMultiTenant` : indicates if the workflow element belongs to a multi tenanted rules projects

Please note that `Duplicated for Tenant` rules projects will have `tenantId=null` but still show `isMultiTenant=true` .

Please refer to the Data Schema page for more details.

On this page

HSBC-3076-HSBC-3141-Pulse-Version-Upgrade

Pulse Version Upgradeâ

Feature FTMT-26830 WPB-378442 HSBC-3076 HSBC-3141
Also In: 2.4.9 (Pulse Version 25.0.85) 2.5.7 (Pulse Version 25.0.85) 2.8.1 (Pulse Version 25.0.85) 2.9.0 (Pulse Version 25.0.85)

We've upgraded Pulse version to 25.0.90 for the 2.10.0 release.

This upgrade includes:

- Performance improvements for real-time cross-channel metrics
- A fix for issue with file descriptor leaks that were affecting the Pulse server.

HSBC-3084-HSBC-3375-Upgrade-JupyterLab-version-to-25-0-46

Upgrade JupyterLab version to 25.0.46â

Feature FTMT-27382 HSBC-3084 HSBC-3375

We have upgraded JupyterLab to version 25.0.46 to fix the missing `timestampField` dependencies in List operators when duplicating a plan using the `ds-api`, as well as including the SHAP Python package.

The new Docker image will have to be loaded in existing environments. Please find it in the Installation Guide.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-2.x/release-2.10.0/individual-release-notes/HSBC-3086-Change-deployment-pipelines-to-support-RHEL8-

On this page

HSBC-3086-Change-deployment-pipelines-to-support-RHEL8-

Change deployment pipelines to support RHEL8+â

Feature FTMT-26831 HSBC-3086
Also In: 2.8.3 2.9.2

We've changed the Case Manager, Pulse and ETL deployment pipelines to support installations on RHEL8+ that were failing due to restrictions on the `rsync` module execution.

By default the pipelines will still have the previous behaviour. This can be changed during deployment by setting the `use_rsync` Ansible variable to `false`.

For more details please refer to:

- Case Manager Recommendations
- Pulse Configuration Recommendations
- ETL Configuration Recommendations

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-2.x/release-2.10.0/individual-release-notes/HSBC-3093-Add-Ansible-variable-to-control-Reference-Data-Service-loggers

On this page

HSBC-3093-Add-Ansible-variable-to-control-Reference-Data-Service-loggers

Add Ansible variable to control Reference Data Service loggersâ

Feature HSBC-3093
Also In: 2.4.9 2.5.7 2.7.3 2.8.1 2.9.0

We've added Ansible variables that allow log levels to be configured for Reference Data Service:

- `reference_data_log_level` controls the log level for all components
- `reference_data_loggers` controls the log level for individual components

The `reference_data_loggers` variable can be configured as a dictionary of loggers as keys and log level as values, for example:

```
reference_data_loggers:
    com.example: debug
```

Copy

More information can be found in the [Reference Data Service Configuration Reference](#).

On this page

HSBC-3094-Compute-Event-Storage-table-names-based-on-the-application-id

Compute Event Storage table names based on the application idâ

Feature HSBC-3094
Also In: 2.8.1 2.9.0

Up until this point Event Storage table names were fixed and thus not configurable. This meant that these tables would be created automatically but their name would in fact only match the default application ids.

To further support the global rollout we've changed this behaviour so that the Event Storage table name for a given application is calculated based on the configured application ids.

Please note this change will only affects new environments that do not use the default application ids.

On this page

HSBC-3100-HSBC-3443-Add-new-fields-to-Cards

Add new fields to Cardsâ

Feature FTMT-25618 FTMT-29492 HSBC-3100 HSBC-3443
Also In: 2.9.0

We have added the below fields to the Cards API schema:

Field Name	Datatype	Also In 2.9.0
transaction_priority	String	X
global_customer_identifier	String	X
on_us_flag	String	X
auth_tran_type	String	X
cardholder_bill_currency_code	String	X
card_account_pre_tran_cash_bal	Double	
card_account_cash_limit	Double	
card_account_client_def_limit	Double	
card_delinquency_days	Integer	
card_currency_conv_rate	Double	
card_account_txn_count_limit	Long	
card_limit_currency_conv_rate	Double	
card_account_limit_type	String	
card_entity_usage	String	
card_pb_user_id_usage	String	
card_channel_level_count_limit	Long	
card_channel_currency_conv_rate	Double	
card_phone_banking_user_id	String	
card_previous_action_info	String	
card_action_until_timestamp	Long	
card_customer_type_indicator	String	
card_clnt_def_tran_exp_date	Long	
card_utc_indicator	String	
card_multi_events	String	
card_auth_cust_old_address	String	
card_hub_guid_customer_id	String	
digital_wallet_token_txn_new	String	
card_clring_cheque_mail_cntry	String	

Field Name	Datatype	Also In 2.9.0
card_unique_id	String	
card_account_sending_fund	String	
card_stip_reason_cde_changes	String	
card_expiry_date_val_result	String	
card_cnp_tran_not_thresh_amt	Double	
card_upstream_fraud_score	Double	
card_upstream_fraud_reason_code	String	
card_update_pass_wallet_date	Long	
card_activity_comp_detl_1	String	
card_activity_comp_detl_2	String	
card_activity_comp_detl_3	String	
card_client_transaction_type	String	
card_message_destination_name	String	
card_transaction_component_type	String	
card_instant_credit_indicator	String	
card_biometric_verify_ind	String	
card_customer_present_details	String	
card_message_expiry_date	Long	
card_termnl_pan_entry_capblty	String	
card_customer_type	String	
card_customer_language	String	
card_customer_level_limit_amt	Double	
card_customer_level_limit_cnt	Long	
card_customer_level_conv_rate	Double	
card_current_action_code	String	
card_travel_indicator	String	
card_cnp_tran_notification	String	
card_mobile_match	String	
card_atm_transaction_type	String	
card_auth_system_warning_code	String	
card_acc_last_txn_date	Long	
card_active_flag	String	
card_frst_use_flag	String	
card_pri_first_use_flag	String	
card_cnp_threshold_amount	Double	
card_cycles_over_limit	Integer	
card_dispatch_branch_sortcode	String	
card_last_auth_date	Long	
card_last_exp_cmf_date	Long	

Field Name	Datatype	Also In 2.9.0
card_ond_vec_value	Integer	
card_san_account_number	String	
card_address_home_pc	String	
card_customer_income	Double	
card_held_dob	String	
card_staff_member_indicator	String	
card_vip_indicator	String	
created_at	Long	
sender_initiated_at	Long	
sender_initiated_at_local	String	
event_created_at	Long	
event_received_at	Long	
local_datetime_alt	String	
card_latitude_longitude	String	
card_c_type_atm_debit	String	
card_account_change	String	
card_hcc_box	String	
card_last_expiry_date	Long	
card_ond_vec_status	String	
card_p_ac_t_atm_debit	String	
card_code_card_type	String	
card_account_new_status	String	
crypto_transaction_info	String	
card_transaction_ref_number	String	
lifecycle_id_alt	String	
saf_ind	String	

The `Cards` and `Cards + Outcomes` schemas were updated with all the above fields.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-2.x/release-2.10.0/individual-release-notes/HSBC-3101-Add-new-endpoints-for-List-Management

On this page

HSBC-3101-Add-new-endpoints-for-List-Management

Add new endpoints for List Managementâ

Feature HSBC-3101

We've added a new Entity Tagging API which can be used to perform list management operations programmatically. There are four new endpoints:

- **GET Entity Tagging** âused to retrieve list items
- **DELETE Entity Tagging** âused to remove list items
- **POST Entity Tagging** âused to create new list items
- **PUT Entity Tagging** âused to upsert (create or update) list items

The new endpoints live in the Reference Data application Input Service, sharing the same port as the Reference Data API.

For a full specification check the Entity Tagging API documentation and refer to the Postman Collection for examples.

Complex Lists

When it comes to Complex Lists, these endpoints only support lists that have a single value which must be named `value`.

On this page

HSBC-3110-HSBC-3204-Add-new-fields-to--vocalink--complex-field

Add new fields to `vocalink` complex fieldâ

Feature FTMT-27190 HSBC-3110 HSBC-3204
Also In: 2.9.0

We've added new fields to the existing `vocalink` for Inbound and Outbound Payments API and workflow schema:

Field Name	Datatype
alert_id	String
source_account_id	String
destination_account_id	String
transaction_id	String
transaction_value	Double
score_generated_at	Long
transaction_currency	String

warning

Due to the complex nature of HSBCâs production workflow, which contains many elements with multiple copies of the schema, this change may cause a normal blue-green deployment to fail. A manual step will be required to update the objects in the Pulse UI.

On this page

HSBC-3128-Improve-currency-display-in-Case-Manager

Improve currency display in Case Managerâ

Feature WPB-349833 HSBC-3128

We have updated some Case Manager widgets where a generic currency sign (Â¤) could be displayed.

The widgets now have custom logic to detect ISO 4217 currency codes (both the 3-letter and 3-digit codes) and format the currency amount according to the browser locale. When Case Manager cannot detect a valid currency it will now display the amount value without any currency symbols.

On this page

HSBC-3131-Update-Alert-ID-generation

Update Alert ID generationâ

Feature WPB-380307 HSBC-3131

Previously, Alert IDs were created by selecting the last 8 characters of the lifecycle ID.

We've now improved this logic. Alert IDs are generated by hashing the full lifecycle ID using SHA-256, encoding the hash with Base64, removing any non-alphanumeric characters, and then taking the last 8 characters of the result.

As a side effect, the restriction that the Alert ID will only be generated when the lifecycle ID has at least 8 characters has been lifted.

On this page

HSBC-3136-Add-GCP-permissions-documentation-for-the-DTP-Service

Add GCP permissions documentation for the DTP Serviceâ

Docs FTMT-27152 HSBC-3136
Also In: 2.8.1 2.9.0

We've added a new page with the minimal set of GCP permissions needed for the DTP service.

Please refer to the DTP permissions page for more details.

Also, the subscription created by the DTP service for the Inactivity topic is now configured to never expire.

On this page

HSBC-3137-Add-CMEK-configuration-for-DTP-resource-creation

Add CMEK configuration for DTP resource creationâ

Feature FTMT-27701 HSBC-3137
Also In: 2.8.1 2.9.0

We've added Customer-managed encryption keys (CMEK) support for the DTP service.

The key name can be configured by using the `dtp_security_kms_key_name` Ansible variable. If omitted DTP resources will have the default behaviour and use Google-managed encryption keys.

The key configuration should follow the format described in Configured a topic with CMEK.

Please refer to DTP configuration and GCP permission required for using CMEK for more information.

On this page

HSBC-3138-Add-support-for-externally-created-PubSub-topics-for-DTP

Add support for externally created Pub/Sub topics for DTPâ

Feature FTMT-27152 HSBC-3138
Also In: 2.8.1 2.9.0

We've added support for the DTP to connect to Pub/Sub topics externally created in Google Cloud Platform. Three new Ansible variables control this behaviour:

- `dtp_pubsub_topic_enable_creation` : a boolean flag that determines if the DTP should create the Pub/Sub topics when starting (defaults to *false*)
- `dtp_pubsub_topic_id` : the topic id for the DTP to use
    - This value is used only when the topic creation is **disabled**
- `dtp_inactivity_topic_enable_creation` : a boolean flag that determines if the DTP should create the Inactivity Service Pub/Sub topic when starting (defaults to *false*)

Please refer to the DTP Configuration page and the new GCP Setup page for more details.

warning

The new boolean flags that determine if the DTP automatically creates the topics default to `false`. This means that if not explicitly configured otherwise, the DTP will expect the topics to already exist in GCP. This is a departure from the previous behaviour which always created the topics on start.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-2.x/release-2.10.0/individual-release-notes/HSBC-3140-Add-Workflow-Monitoring-Service-to-the-Reference-Data-application

On this page

HSBC-3140-Add-Workflow-Monitoring-Service-to-the-Reference-Data-application

Add Workflow Monitoring Service to the Reference Data applicationâ

Feature HSBC-3140
Also In: 2.8.1 2.9.0

We've added the Workflow Monitoring Service to the Reference Data application.

This service will populate the `workflow.events` metric with the events that are processed by that workflow and can be used to create monitoring dashboards.

On this page

HSBC-3151-Add-configurable-usage-of-the-legacy-list-format

Add configurable usage of the legacy list formatâ

Feature HSBC-3151
Also In: 2.9.0

Weâve introduced support for configuring the list import/export format via the following Ansible variable:

- `lists_use_legacy_csv_format`

By default, the variable value is set to true, preserving the previous behavior. To enable the new format, set it to false. Please note that in case the variable is not defined, it will use the old import/export format.

This enables installations to choose between the legacy and new list formats based on their needs:

- Strategic Market installations can adopt the new list format, making use of tenant-specific exports.
- UK and AMH installations must continue using the legacy format to maintain compatibility.

Tne new list format provides some improvements, like:

- Introduction of data headers to the exported lists
- Use the tenant key or the tenant group name in the exported files
- Complex lists are also more intuitive, as each field is in a different column instead of in a json format

For more details about the new variables and their default values, refer to Pulse Configuration Recommendations.

On this page

HSBC-3152-Make-RDS-not-copy-logs-to-system-log

Make RDS not copy logs to system logâ

Feature HSBC-3152
Also In: 2.8.3 2.9.2

The Reference Data service was copying its request logs into the system logs (`/var/log/messages`) by default. This was controlled by the undocumented `reference_data_log_console` variable, which was set to `true` by default. We've set this to `false` in the `project-default` Ansible configuration.

The request log will instead go to the `request.log` file in the normal reference data log folder, and will be rotated and compressed daily, with 5 compressed files. We do not expect that this file will need to be exported to stackdriver.

On this page

HSBC-3158-Add-new-fields-to-the-Mobile-Outbound-extensions

Add new fields to the Mobile Outbound extensionsâ

Feature FTMT-27243 HSBC-3158
Also In: 2.9.0

The following fields were added to Case Manager `2 Way/1 Way SMS` and `Mobile Payment Push` extensions for Inbound and Outbound Payments channels:

Field Name	Datatype	Description
device_ip_address	String	IP address
tmx.ip_address	String	True IP Address
tmx.ip_city	String	True IP City
tmx.ip_country	String	True IP Country
device_banking_type	String	Online Banking Device Type
device_os	String	Banking device Operating System Version
device_user_agent	String	Accept User Agent HTTP Header

Please refer to the AFIS Extensions reference for more details.

On this page

HSBC-3168-Add-performance-related-Ansible-variables-for-cross-channel-metrics

Add performance-related Ansible variables for cross-channel metricsâ

Feature HSBC-3168
Also In: 2.4.10 2.5.7 2.8.1 2.9.0

We've added Ansible variables that can be used to control the performance of cross-channel metrics:

- `cross_channel_metrics_fetch_timeout_ms` controls the timeout when fetching cross-channel metrics (defaults to `200ms`)
- `cross_channel_metrics_num_threads` controls the number of threads used when fetching cross-channel metrics (defaults to `1`)

More information can be found in Pulse Configuration Recommendations.

On this page

HSBC-3176-Add-new-field-to-Inbound-and-Outbound-Payments

Add new field to Inbound and Outbound Paymentsâ

Feature FTMT-27248 HSBC-3176
Also In: 2.9.0

We have added the below field to Pulse and Case Manager schemas, for Outbound and Inbound Payments:

Field Name	Datatype
beneficiary_proxy_account_number	String

On this page

HSBC-3196-Add-bulk-actions--new-columns--search-event-fields--and-permissions

Add bulk actions, new columns, search event fields, and permissionsâ

Feature FTMT-10124 HSBC-3196

We've added customizations to Case Manager. These updates include:

- New Bulk Actions with new permissions: The following actions are available on bulk under the identified permissions:

Page/Widget	Description	Permission
Transaction history	Add or update reasons	cm:resource:history:reason-bulk:*
Transaction history	Change status (Move to Hold, In-progress, Close, or Queue)	cm:resource:history:status-bulk:*
Transaction history	Update lists	cm:resource:history:update-list-bulk:*
Alert page	Add or update reasons	cm:resource:alert:reason-bulk:*
Alert page	Change status (Move to Hold, In-progress, Close, or Queue)	cm:resource:alert:status-bulk:*
Alert page	Update lists	cm:resource:alert:update-list-bulk:*

- Search Capability: Users can now search for events based on their Operational Status.
- Linked Alerts: The 'Linked Alerts' widget has been updated to include the Beneficiary Name and Beneficiary Account Number.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-2.x/release-2.10.0/individual-release-notes/HSBC-3202-Add-support-for-externally-created-PubSub-subscriptions-for-DTP

On this page

HSBC-3202-Add-support-for-externally-created-PubSub-subscriptions-for-DTP

Add support for externally created Pub/Sub subscriptions for DTPâ

Feature FTMT-27152 HSBC-3202
Also In: 2.8.1 2.9.0

We've added support for the DTP to connect to Pub/Sub subscriptions externally created in Google Cloud Platform. Three new Ansible variables control this behaviour:

- `dtp_pubsub_subscription_enable_creation` : a boolean flag that determines if the DTP should create the Pub/Sub subscription when starting (defaults to *false*)
- `dtp_pubsub_subscription_id` : the subscription id for the DTP to use
    - This value is used only when `dtp_pubsub_subscription_enable_creation` is set to `false`
- `inactivity.subscription.enableCreation` : a boolean flag that determines if the DTP should create the Inactivity Service Pub/Sub subscription when starting (defaults to *false*)

Please refer to the DTP Configuration page for more details.

warning

The new boolean flags that determine if the DTP automatically creates the subscriptions default to `false` . This means that if not explicitly configured otherwise, the DTP will expect the subscriptions to already exist in GCP. This is a departure from the previous behaviour which always created the subscriptions on start.

On this page

HSBC-3210-Add-new-fields-to-the-Mobile-Outbound-extensions

Add new fields to the Mobile Outbound extensionsâ

Feature FTMT-27710 HSBC-3210
Also In: 2.9.0

The following fields were added to Case Manager `2 Way/1 Way SMS` and `Mobile Payment Push` extensions for Inbound and Outbound Payments channels:

Field Name	Datatype	Description
receiver_account_bank_id	String	Receiver Account Bank ID
receiver_bank_bic	String	Beneficiary Bank Identification Code

Please refer to the AFIS Extensions reference for more details.

On this page

HSBC-3212-Add-support-for-defining-allowed-PubSub-regions-in-DTP

Add support for defining allowed Pub/Sub regions in DTPâ

Feature FTMT-27702 HSBC-3212
Also In: 2.8.1 2.9.0

We've added support in the DTP for defining the allowed Pub/Sub regions where data may be stored and transmitted. This behaviour is controlled by three new Ansible variables:

- `dtp_security_allowed_regions` : A list of allowed regions where data may be stored. Example:

```
dtp_security_allowed_regions:
  - europe-west1
  - asia-east2
```

Copy

- `dtp_security_enforce_in_transit` : A boolean flag that enforces in-transit restrictions to the same regions defined for storage.
- `dtp_security_regional_endpoint` : The regional Pub/Sub gRPC service endpoint to use.

Please refer to the Data Security section in the DTP Configuration page for more details.

HSBC-3215-Fix-color-coding-issues-in-Case-Manager-widgets

Fix color coding issues in Case Manager widgetsâ

Bug Fix FTMT-27709 HSBC-3215

We've fixed the color coding issues in several Case Manager widgets (e.g. `Alerted` , `System Decision`) caused by the removal of an internal package in a previous version.

When navigating to, for example, Case Manager Event Details page fields will display their correct colors:

On this page

HSBC-3216-Add-new-performance-configuration-options-and-probe-logs-to-Case-Manager

Add new performance configuration options and probe logs to Case Managerâ

Feature HSBC-3216
Also In: 2.4.10 2.8.1 2.9.0

We've added a new set of performance configuration options to the Case Manager deployment pipeline. These configurations were already part of Case Manager but were not exposed during deployment.

On top of these changes we also added a deployment option to enable Case Manager probe logs. These logs can be useful to pinpoint issues by checking in which actions Case Manager is taking the most time. Please note that enabling this feature will create a considerable amount of logs, so it should only be done in a controlled environment.

Please find a complete description of the changes in Case Manager Performance Configuration.

HSBC-3226-Fix-error-when-adding-Notes-to-Customers

Fix error when adding Notes to Customersâ

Bug Fix FTMT-26842 HSBC-3226

We've resolved an issue that occurred when adding notes to a Customer from the Event Details page.

The bug was triggered for events where the `lifecycle_id` exceeded 40 characters, which caused the note creation to fail. To address this, we've increased the allowed length of the `lifecycle_id` field to support longer values.

On this page

HSBC-3232-Add-ETL-playbook-for-Pulse-ES-to-Case-Manager-recovery

Add ETL playbook for Pulse ES to Case Manager recoveryâ

Feature FTMT-27051 FTMT-27144 HSBC-3232
Also In: 2.8.1 2.9.0

We've added a new ETL playbook to enable event recovery from Pulse event storage to Case Manager. It supports two routing strategies to send events to Case Manager, configured via the `etl_cm_route_strategy` variable.

Please refer to Recover events from Pulse to Case Manager for more details.

On this page

HSBC-3233-New-Case-Manager-enrichers-documentation

New Case Manager enrichers documentationâ

Docs HSBC-3233

We've added a new documentation page that describes how to use Case Manager enrichers and lists the available ones.

On this page

HSBC-3245-Add-new-field-to-the-Mobile-Outbound-extensions

Add new field to the Mobile Outbound extensionsâ

Bug Fix FTMT-28271 HSBC-3245
Also In: 2.8.1 2.9.0

The following field was added to Case Manager `2 Way/1 Way SMS` and `Mobile Payment Push` extensions:

Field Name	Datatype	Channels	Description
channel_user_id	String	Digital	Online/Internet Banking User ID.

Please refer to the AFIS Extensions reference for more details.

On this page

HSBC-3252-HSBC-3253-Improve-Case-Manager-and-Pulse-Monitoring-guides

Improve Case Manager and Pulse Monitoring guidesâ

Feature HSBC-3252 HSBC-3253

We've made some changes to Case Manager and Pulse monitoring guide for better readability and organization.

You can review the updated version at Top Metrics to Monitor in Case Manager and Top Metrics to Monitor in Pulse

On this page

HSBC-3255-Add-Dead-Letter-Queueing-system-to-Data-Extraction-Pipelines

Add Dead Letter Queueing system to Data Extraction Pipelinesâ

Feature HSBC-3255

We've upgraded the Data Extraction Pipelines to version 1.1.2, which includes:

- The possibility of enabling Logstash Dead Letter Queueing (DLQ) system to recover events that errored during processing
- A recovery pipeline for the `pulse-workflow-audit` events present in the DLQ

For more information refer to the Data Extraction Pipelines overview page.

On this page

HSBC-3276-Add-new-fields-to-the-Mobile-Outbound-extensions

Add new fields to the Mobile Outbound extensionsâ

Bug Fix FTMT-28271 HSBC-3276

The following fields were added to Case Manager `2 Way/1 Way SMS` and `Mobile Payment Push` extensions for the Digital Activity channel:

Field Name	Datatype	Description
customer_id_global	String	Global Customer ID
user_id	String	Online Banking User ID

For the Digital Activity channel we have also changed the `payment_sub_type` and `payment_type` fields in Case Manager (which also applies to the fields in SLAs, automation rules, and the AFIS extension) to use the raw values received from Pulse, instead of the values enriched via self-service List enrichment.

This should have no impact on the existing solution, as these fields were not being used, but will allow the raw value to be used for these fields if used again in the future.

Please refer to the AFIS Extensions reference for more details.

On this page

HSBC-3281-Add-new-fields-to-Case-Manager

Add new fields to Case Managerâ

Feature FTMT-28352 HSBC-3281

We've added new search fields and transaction history filter fields in Case Manager.

We've added the following fields to the Case Manager advanced search page:

fields	Channel
session_id	Transfers
device_id	Transfers
device_ip_address	Transfers/Digital Activity
device_ip_address_v6	Transfers/Digital Activity
user_id	Transfers/Digital Activity
sender_transaction_ref_nr_alt	Transfers/Digital Activity
device_user_agent	Transfers/Digital Activity

We've added the following fields to the Event Details `transaction history` dropdown filter:

fields	label
device_ip_address	IP Address
device_ip_address_v6	IP Address v6
user_id	User ID
sender_transaction_bank_account_number	Debit Account Number
alert_id	Alert ID
sender_transaction_ref_nr_alt	Instruction Reference Number
receiver_transaction_bank_account_number	Beneficiary Account Number
receiver_bank_bic	Beneficiary Bank Sort Code
receiver_name	Beneficiary Name
device_user_agent	User Agent
payment_type	Payment type

On this page

HSBC-3290-Add-new-fields-to-Outbound-Payments

Add new fields to Outbound Paymentsâ

Feature FTMT-27741 HSBC-3290

We've added the following fields to the Outbound Payments schema and the Mobile Outbound extensions:

Field Name	Datatype	Description
tmx.true_isp	String	The legitimate Internet Service Provider that assigned the original IP address used by an end-user to access the internet. This ISP is tied directly to the user's physical location or device - without any intermediate.
tmx.proxy_isp	String	It is the service provider offering proxy IP addresses that mask the user's actual IP address. These includes VPNs, residential proxies, datacenter proxies, and anonymizers. In essence, the ISP visible to the internet is that of the proxy provider, not the user's real ISP.
tmx.dns_isp	String	It refers to the ISP seen during Domain Name System (DNS) resolution - the process of converting human-readable domain names into IP addresses. This is typically the ISP of the DSN resolver the user is.

Please refer to the AFIS Extensions reference for more details.

On this page

HSBC-3304-HSBC-3474-Add-new-fields-to-Reference-Data

Add new fields to Reference Dataâ

Feature FTMT-15088 HSBC-3304 HSBC-3474

The following fields were added to the Customer Reference Data API:

Field Name	Datatype
customer_portfolio_region	String
customer_portfolio_product	String
customer_portfolio_class	String
customer_portfolio_type	String
customer_portfolio_channel	String
customer_portfolio_country	string
customer_account_country_code	string

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-2.x/release-2.10.0/individual-release-notes/HSBC-3358-Allow-concurrent-Pulse-application-publishes

On this page

HSBC-3358-Allow-concurrent-Pulse-application-publishes

Allow concurrent Pulse application publishesâ

Feature FTMT-29333 HSBC-3358

The number of concurrent application publishes in Pulse will now be set to the number of applications configured in the `apps_enabled` variable, so that for instance inbound and outbound can be published concurrently. Specifically, this change sets the `pulse.global.services.core.apps.operationThreadPoolNumberOfThreads` property in Pulse.

HSBC-3359-Add--Originator-Details--and--Beneficiary-Details--tabs-in-Case-Manager-Event-Details-page

Add `Originator Details` and `Beneficiary Details` tabs in Case Manager Event Details pageâ

Feature FTMT-10124 HSBC-3359

The event details page for both **Inbound and Outbound Payments** now includes `Originator Details` and `Beneficiary Details` tabs for quick access to the corresponding widgets.

On this page

HSBC-3393-Add-back-pulse-configure-and-case-manager-configure-tags-to-Pulse-and-Case-Manager-810-Ansible-tasks

Add back pulse.configure and case-manager.configure tags to Pulse and Case Manager 810 Ansible tasksâ

Bug Fix FTMT-29677 HSBC-3393
Also In: 2.8.2 2.9.1

We have added back the `pulse.configure` and `case-manager.configure` tags to Pulse and Case Manager Ansible tasks `810.1`, respectively, so that these task are not skipped during Pulse or Case Manager deployment. These tasks was being skipped in releases `2.8`, `2.8.1`, and `2.9`, leading to certain Pulse server or Case Manager Ansible variables being ignored (such as Pulse Java options, which includes the Pulse server heap memory size parameter).

On this page

HSBC-3399-Fix-Case-Manager-partial-updates-on-Digital-Activity-events

Fix Case Manager partial updates on Digital Activity eventsâ

Bug Fix FTMT-31199 HSBC-3399

We've fixed an issue where Digital Activity feedback events where processed as full updates of the original event. This would cause some fields, e.g. `decision`, to be updated although that is not the intended behaviour.

On this page

HSBC-3408-HSBC-3444-Add-Inbound-Payment-Notification-Case-Manager-extension

Add *Inbound Payment Notification* Case Manager extensionâ

Feature FTMT-29922 HSBC-3408 HSBC-3444

We've added a new Case Manager extension named *Inbound Payment Notification* that is able to send notifications to AFIS from Inbound Payments automation rules. This extension has a different schema compared to the Outbound Payments extensions and includes a `rule_explanations` field that lists the rule explanations triggered by the event.

By default the extension is enabled, but it can be disabled by setting `inbound_payment_notification_extension_enabled` to `false`. For more details please refer to the Case Manager Configuration Recommendations and the AFIS Extensions reference.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-2.x/release-2.10.0/individual-release-notes/HSBC-3409-Add-Payment-Fulfillment-Extension-for-Inbound-Payment

On this page

HSBC-3409-Add-Payment-Fulfillment-Extension-for-Inbound-Payment

Add Payment Fulfillment Extension for Inbound Paymentâ

Feature FTMT-29917 HSBC-3409

We've updated the Payment Fulfillment extension to support Inbound Payments events in addition to Outbound.

The schema is the same as for Outbound events, as described in the AFIS Extensions reference.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-2.x/release-2.10.0/individual-release-notes/HSBC-3411-Add-Decision-Typology-outcome-to-Inbound-Payments

On this page

HSBC-3411-Add-Decision-Typology-outcome-to-Inbound-Payments

Add Decision Typology outcome to Inbound Paymentsâ

Feature FTMT-29925 HSBC-3411

We've added the `decision_typology` outcome to the **Inbound Payments** workflow. The field was also added to the **Inbound Payments + Outcomes** schema.

This outcome can have the following values:

- `approve_fraud`
- `approve_scam`
- `review_fraud`
- `review_scam`
- `decline_fraud`
- `decline_scam`
- `none`

note

The values added to the outcome are the same that as the **Outbound Payments** baseline. It is possible that some values have since been added to the **Outbound Payments** outcome. If that's the case, any missing values should be added manually to the **Inbound Payments** outcome to ensure consistency between both workflows.

HSBC-3414-Add-Inbound-Payments-application-to-Data-Science-application-artifact

Add Inbound Payments application to Data Science application artifactâ

Feature HSBC-3414

The Data Science applications assembly has been updated to include the **Inbound Payments** application.

This update ensures that the complete assembly can now be imported into HSBC's Data Science environments, guaranteeing compatibility of the initial schemas with Pulse runtime. The update assembly can be found in the Installation Guide.

On this page

HSBC-3430-Add-support-for-distinct-RabbitMQ-brokers-for-Pulse-and-Case-Manager

Add support for distinct RabbitMQ brokers for Pulse and Case Managerâ

Feature HSBC-3430

We've added support for distinct RabbitMQ brokers for Pulse and Case Manager. The following Ansible variables were changed:

Component	New Ansible Variable	Old Ansible Variable
Pulse	`rabbitmq_casemanager_brokers`	N/A
Case Manager	`rabbitmq_casemanager_brokers`	`rabbitmq_brokers`
Logstash	`rabbitmq_casemanager_brokers`	`rabbitmq_brokers`
Logstash	`rabbitmq_casemanager_username`	`rabbitmq_insights_logstash_username`
Logstash	`rabbitmq_casemanager_password`	`rabbitmq_insights_logstash_password`
Logstash	`rabbitmq_casemanager_vhost_blue`	`rabbitmq_pulse_vhost_blue`
Logstash	`rabbitmq_casemanager_vhost_green`	`rabbitmq_pulse_vhost_green`

Note that `rabbitmq_brokers` is still required in Pulse and defines the RabbitMQ brokers for Pulse internal communication.

For more details please refer to:

- Pulse Configuration Recommendations
- Case Manager Configuration Recommendations
- Data Extraction Pipelines Configurations

Default values for the new variables

The new Ansible variables default to the variables they are replacing (e.g. `rabbitmq_casemanager_brokers` defaults to `rabbitmq_brokers`, if not defined). This means that this change is backwards compatible and the default behaviour is to continue using a single RabbitMQ for all components.

For clarity, we recommend changing the variable names in the HSBC Ansible Variables files to use the new ones instead.

On this page

HSBC-3434-Add--fdz-channel--to-Cards-and-Inbound-Payments

Add `fdz_channel` to Cards and Inbound Paymentsâ

Feature FTMT-30250 HSBC-3434

The `fdz_channel` field, which describes the channel of the event, was missing for the **Cards** and **Inbound Payments** channels.

The field will now be automatically enriched with the following values and also be included in ETL extractions:

- Cards: `CARDS`
- Inbound Payments: `INBOUND-PAYMENTS`

On this page

HSBC-3447-Make--minimizeShuffles--property-configurable-via-Ansible

Make `minimizeShuffles` property configurable via Ansibleâ

Feature FTMT-30345 HSBC-3447

We've added support to configure the property `services.modules.datascience.context.minimizeshuffles` via Ansible.

The property can now be configured using the following Ansible variable:

- `minimize_shuffles_enabled` : a boolean flag that determines if the group by fields can be re-sorted in order to minimize shuffles (defaults to *true*)

For more details please refer to the [Pulse Configuration](#) page.

HSBC-3460-Add-Ansible-variable-to-control-HSBCNet-Advanced-Search-fields

Add Ansible variable to control HSBCNet Advanced Search fieldsâ

Feature FTMT-30368 HSBC-3460

We've introduced a new Ansible variable `show_hsbcnet_search_fields` that allows showing or hiding the HSBCNet specific Advanced Search fields.

By default these fields are hidden but can be enabled in HSBCNet markets by setting the variable to `True` during deployment.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-2.x/release-2.10.0/individual-release-notes/HSBC-3461-Add--node-properties--template-to-Pulse-Data-Science-Assembly

On this page

HSBC-3461-Add--node-properties--template-to-Pulse-Data-Science-Assembly

Add `node.properties` template to Pulse Data Science Assemblyâ

Feature HSBC-3461

We've added the `node.properties.j2` template to the Data Science assembly.

Previously Data Science installations defaulted to a template that only supported single node setups, meaning that in the Pulse UI the node would always show the localhost IP (`127.0.0.1`).

On this page

HSBC-3466-Add--lifecycle-id--index-for-Cards-and-Inbound-Event-Storage

Add `lifecycle_id` index for Cards and Inbound Event Storageâ

Feature HSBC-3466

We've added the following index to the Inbound and Cards Event Storage tables to improve the feedback processing time:

- `idx_es_<eventstorage.tableName>_lifecycle_id` : index on column `lifecycle_id`

warning

This index will be automatically applied to the main (non-partitioned) table, the partitioning function, and all *new* partitions created after this update. Existing partition tables created prior to this change will *not* be affected and will require manual indexing if needed.

On this page

HSBC-3480-Revamp-documentation-for-the-Case-Manager-AFIS-extensions

Revamp documentation for the Case Manager AFIS extensionsâ

Docs HSBC-3480

We've updated the documentation for the Case Manager AFIS Extensions.

The new page combines the schemas and payload examples for all Case Manager AFIS extensions, replacing the previous two individual pages for *Mobile Outbound* and *Payment Fulfillment*. This new page distinguishes between the *Send Mobile Payment Push* and *Send 2 WAY / 1 WAY SMS*, which previously had the same schema and payload and now have distinct values for the `notification_type` field. It also contains the schema and a payload example for the new *Inbound Payment Notification* extension.

All anchors that previously linked to the old pages have been updated to link to the new page instead.

On this page

HSBC-3493-Upgrade-Pulse-Commands-version-to-25-14-38

Upgrade Pulse Commands version to 25.14.38â

Feature HSBC-3493

We have upgraded Pulse Commands to version 25.14.38. This is a technical change which prevents patch failures when workflow schemas are deleted or implicitly defined.

On this page

HSBC-3503-Change-Transaction-Type-filter-for-Inbound-Payments

Change Transaction Type filter for Inbound Paymentsâ

Feature FTMT-30267 HSBC-3503

Previously, the `Transaction Type` filter for **Inbound Payments** only allowed the selection of predefined values. This has now been changed to a standard text box, allowing the search using any value.

Please note that for accurate results the search box value must exactly match the Transaction Type name. For example, for events with a Transaction Type of `Bulk Fund Transfer`, searching for `Bulk Fund` would not yield any results.

On this page

HSBC-3579

Enable the `External Status Reason Updater` extension for Inbound Paymentsâ

Feature FTMT-30475 HSBC-3579

We have updated the `External Status Reason Updater` extension to allow Inbound Payments events.

The extension will now set the event's status reason based on the `external_status_reason` field value sent in the feedback event if it matches any Case Manager configured reasons.

On this page

HSBC-3599

Add multi tenancy support in the Case Manager alert cleanup scriptâ

Feature FTMT-31365 HSBC-3599

We have added multi tenancy support in the Case Manager alert cleanup script.

This change adds two new script options that allows the script to filter events based on all tenants or a specific set of tenant ids.

For more information please refer to the new filtering section in the Case Manager alert queue cleanup page.

HSBC-3621

Upgrade Data Extraction Pipelines to version 2.1.1â

Feature FTMT-26459 FTMT-30590 HSBC-3367 HSBC-3621 HSBC-3330

We've upgraded the Data Extraction Pipelines to version 2.1.1.

This upgrade includes the following changes:

- Support for Logstash 9.1.4 by upgrading the existing pipelines and service configuration files. It addresses `CVE-2022-1471` and `CVE-2025-52999`
- Logstash can now be executed with a keystore
- Fix an issue where the Logstash recovery process pipeline would get stuck, causing Ansible to wait indefinitely

warning

Please note this is a breaking change since the new Data Extraction Pipelines package requires Java 17 or 21.

For more information refer to Data Extraction Pipelines Requirements and also to Data Extraction Pipelines Configurations.

On this page

HSBC-3632

Address CVE'sâ

Feature

In this release, we addressed multiple CVEs and security-related FTMT tickets.

Please refer to the table below for a detailed summary of the updates and component versions affected.

Component	FTMT Ticket	Feedzai Ticket	Old Version	Upgraded Version	CVE	Additional Notes
Pulse	FTMT-30590 FTMT-30225 FTMT-28076	HSBC-3567 HSBC-3630	25.0.91	25.1.0	CVE-2023-6378 CVE-2023-6481 CVE-2024-7254 CVE-2025-24970 CVE-2025-30065 CVE-2025-48924 CVE-2025-52999 CVE-2025-5115	Addresses a security vulnerability identified in `jackson-core` version Upgrade library versions for local project-components (that don't come directly from the core Feedzai products).
Pulse Commands	FTMT-30225 FTMT-28076	HSBC-3567 HSBC-3630	25.14.38	25.17.3	CVE-2023-6378 CVE-2023-6481 CVE-2024-7254 CVE-2025-24970 CVE-2025-30065 CVE-2025-48924 CVE-2025-52999	
ETL	FTMT-30224	HSBC-3565	18.0.0	18.1.0	CVE-2024-1597 CVE-2024-7254 CVE-2024-13009 CVE-2025-52999	
Reference Data	FTMT-30224	HSBC-3568	3.7.8	3.7.11	CVE-2024-13009	
Ansible Portable	FTMT-30226	HSBC-3619	7.8.4	7.8.5	CVE-2025-47273	Prevents a security vulnerability identified in the `setuptools` version, that was updated to 79.0.1.
	FTMT-30590	HSBC-3621	1.1.2	2.1.1		Full details on the

Component	FTMT Ticket	Feedzai Ticket	Old Version	Upgraded Version	CVE	Additional Notes
External Notifications					CVE-2022-1471 CVE-2025-52999	Upgrade Data Extraction Pipelines to version 2.1.1 release note.
JupyterLab	FTMT-30590 FTMT-30225	HSBC-3633 HSBC-3681	25.0.46	25.0.49-gcs-2.2.29	CVE-2025-52999 CVE-2023-44487 CVE-2022-1471 CVE-2024-13009	This upgrade addresses the vulnerabilities in Pulse and removes the vulnerable libraries included by the Hadoop dependency. Please find the new Docker image in the Installation Guide.
Case Manager	FTMT-30224	HSBC-3547	N/A	N/A	CVE-2023-5072 CVE-2023-6378 CVE-2023-6481 CVE-2024-1597 CVE-2024-13009 CVE-2025-52999	Upgrade library versions for local project-components (that don't come directly from the core Feedzai products).
Spark Distribution	FTMT-30590 FTMT-30225	HSBC-3657	3.3.2-hadoop-3.3.4	3.5.7-hadoop-3.3.5	CVE-2025-52999	Addresses a security vulnerability identified in `jackson-core` version by upgrading the Spark dependency. Please note that this upgrade does not address the CVE that is bundled with Hadoop 3.3.6 given there is no version that addresses it.
Ansible Roles	FTMT-30590 FTMT-30225	HSBC-3657	33.1.0	34.0.0	CVE-2025-52999	Pipeline changes to adopt new Spark

Component	FTMT Ticket	Feedzai Ticket	Old Version	Upgraded Version	CVE	Additional Notes
						distribution archive.
GCS Connector	FTMT-28985	HSBC-3682	1.9.4-hadoop3	hadoop3-2.2.29	CVE-2019-17571 CVE-2020-7692 CVE-2020-8908 CVE-2020-15250 CVE-2021-4104 CVE-2021-22573 CVE-2022-23302 CVE-2022-23305 CVE-2022-23307 CVE-2023-2976 CVE-2023-26464	The upgraded version is the last version that supports Java 8

warning

Please note that starting in this version the `Spark Jars` archive was deprecated in favor of `Spark Dist`.

Any non-Feedzai pipelines that depend on this artifact should be updated to the new one that can be found in the Installation Guide.

warning

Starting in this version the JupyterLab will be having a different naming, including the GCS Connector version provided in it, making it clear which component versions the image includes.

This change also requires a change in the load process. Instead of loading the image as `docker.feedzai.com/feedzai/jupyterlab:<jupyterlabVersion>`, it will now have to be loaded as `docker.feedzai.com/feedzai/jupyterlab:<jupyterlabVersion>-gcp`.

HSBC-3708

Update Case Manager EVENT_ID field length in the `AM_EVENT_AUTOMATION` tableâ

Bug Fix FTMT-32347 HSBC-3708

In order to keep the consistency across tables, for events with big `lifecycle_id`, we've updated the field `event_id` length in the `AM_EVENT_AUTOMATION` table for the type `VARCHAR(1024)`.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-2.x/release-2.11.0/individual-release-notes/HSBC-3267

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HSBC-3267

Add new fields to Cardsâ

Feature FTMT-30565 HSBC-3627

We have added the below fields to the Cards API schema:

Field Name	Datatype
card_deposit_type	String
card_minimum_amount_due	Double
card_purchase_timestamp	Timestamp
credit_card_post_ref_number	String
debit_card_auth_ref_number	String
card_client_amount	Double
cardholder_zip	String
cards_purchase_utc_indicator	String

The `Cards` and `Cards + Outcomes` schemas were updated with all the above fields.

On this page

HSBC-3431

Create RabbitMQ segregation guide for Pulse and Case Managerâ

Docs FTMT-476 HSBC-3431

We have created a guide on how to segregate a RabbitMQ cluster by component, focused on Pulse and Case Manager. The guide focus on a migration with no message loss and minimal downtime.

For more information please refer to the RabbitMQ Segregation for Pulse and Case Manager page.

On this page

HSBC-3476

Fix Transaction History header scroll behaviorâ

Bug Fix FTMT-30567 HSBC-3476

Weâve fixed a bug that made the Transaction History header disappear while scrolling. It now stays visible at the top of the list.

On this page

HSBC-3591

Add new fields to Outbound Payments and Digital Activityâ

Feature FTMT-28937 HSBC-3591

We have added the below elements to the `bioc` tuple in Outbound Payments and Digital Activity API schemas:

Field Name	Datatype
pinpoint_risk_score	Integer
chall_policy_score	Integer

The `Payments` and `Payments + Outcomes` schemas were updated with all the above fields.

On this page

HSBC-3637

Add Non-Mon Endpointâ

Feature FTMT-30076 HSBC-3637

We've created a new endpoint to allow the receiving and registering of non monetary events.

For a more in-depth explanation of this new endpoint please refer to the Non-Mon Documentation page.

On this page

HSBC-3661

Update API documentationâ

Docs FTMT-30058 HSBC-3661

We've update the API resources documentation to include:

- Missing Cards and Inbound Payments endpoint information
- Non Mon endpoint documentation, including how the `event_type` field is generated
- Missing tag descriptions in the API specification page

For more information please refer to API Additional Resources.

On this page

HSBC-3666

Add new fields to Cardsâ

Feature FTMT-24023 FTMT-30076 FTMT-32738 FTMT-33019 HSBC-3666 HSBC-3755 HSBC-3749

We have added the below fields to the Cards schema and workflow:

New Fields

Field Name	Datatype
address_match_ind_3ds	STRING
buyer_daily_provisn_attempt_3ds	STRING
buyer_purchase_velocity_3ds	STRING
challenge_method_final_3ds	STRING
date_received_3ds	STRING
device_channel_3ds	STRING
digital_id_trust_score_3ds	DOUBLE
purchase_amount_3ds	DOUBLE
purchase_exponent_3ds	STRING
shipping_indicator_3ds	STRING
shipping_post_region_3ds	STRING
svc_acs_transaction_id_3ds	STRING
card_action_on_data_indicator	STRING
card_active_flg_last_change_date	STRING
card_change_data_new	STRING
card_change_data_old	STRING
card_cpp_exp_date_proxy	LONG
card_cpp_list_date	LONG
card_details_availability_ind	STRING
card_entity_id	STRING
card_entity_type	STRING
card_expiration_date_3ds	LONG
card_involving_entity	STRING
card_new_city_name	STRING
card_new_country_code	STRING
card_new_phn_country_code	STRING
card_new_phone_number	STRING
card_new_post_code	STRING

Field Name	Datatype
card_new_state_name	STRING
card_nm_effective_timestamp	LONG
card_old_city_name	STRING
card_old_country_code	STRING
card_old_phn_country_code	STRING
card_old_phone_number	STRING
card_old_post_code	STRING
card_old_state_name	STRING
card_onlinebanking_user_id	STRING
card_reissue_type_flag	STRING
card_temp_limit_chnage_indicator	STRING
card_temporary_lmt_exp_date	LONG
card_transfer_entity_from_key	STRING
card_transfer_entity_to_key	STRING
card_transfer_reason	STRING
card_txn_sub_type	STRING
card_transaction_ref_number_alt	STRING
card_website_name	STRING
customer_level_limit_type	STRING
digital_wallet_token_txn_new	STRING
new_card_activation_flag	STRING
old_card_activation_flag	STRING
transaction_status_3ds	STRING
user_amount	DOUBLE
user_indicator	STRING
user_string	STRING
observed_ip_address_3ds	STRING
network_risk_score_3ds	STRING
device_browser_language_3ds	STRING
device_region_3ds	STRING
issuer_name_3ds	STRING
arcot_merch_alert_score_3ds	STRING
arcot_tpd_provider_id_3ds	STRING
matched_policy_3ds	STRING
matched_regulatory_rule_3ds	STRING
policy_advice_id_3ds	STRING
policy_results_3ds	STRING
policy_advice_score_3ds	STRING
regulatory_advice_id_3ds	STRING

Field Name	Datatype
regulatory_rule_results_3ds	STRING
rule_score_source_3ds	STRING
rule_set_advice_id_3ds	STRING
rule_set_matched_rule_3ds	STRING
rule_set_result_3ds	STRING
rule_set_score_3ds	DOUBLE
sr_rules_advice_id_3ds	STRING
sr_rules_set_version_3ds	STRING
transaction_failure_reason_3ds	STRING
tran_origination_cntry_3ds	STRING
card_saf_ind	STRING
tran_id_3ds	STRING
ucaf_value_3ds	STRING
customer_id_personal	STRING
customer_id_business	STRING
tmx.digital_id_trust_score	DOUBLE
tmx.chall_reason_code	STRING

The `Cards` and `Cards + Outcomes` schemas were updated with all the above fields.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-2.x/release-2.11.0/individual-release-notes/HSBC-3709

On this page

HSBC-3709

Add Digital Activity channel to Mobile Outbound extensionsâ

Bug Fix FTMT-32724 HSBC-3709
Also In: 2.10.1 2.9.3

We have added the Digital Activity channel to the Mobile Outbound extensions. The base schema of the notifications is the same as for Outbound Payments. Please refer to the AFIS Extensions reference for more details.

On this page

HSBC-3722

Fix API specification examplesâ

Bug Fix HSBC-3722

We have fixed an issue in the API specification where field types and examples differed in datatype. For example, `customer_annual_turnover` showed as `number` but the example would should as a `string` (`"15000000.00"`).

Please refer to the API specification page for more information.

On this page

HSBC-3738

Add ETL and DTP extractions comparison documentationâ

Docs HSBC-3738

We have created a guide which explains the key data format differences between ETL and DTP extractions to help with the tool migration process. The guide focus on the additional fields each tool retrieves beyond the standard schema fields.

The guide is available in Migrating from ETL to DTP.

On this page

HSBC-3754

Fix issue with Pulse's Alert Storage Service not forwarding messages to Case Managerâ

Bug Fix HSBC-3754

We've fixed an issue that could occur with Pulse not forwarding messages to Case Manager when there was both a RabbitMQ connection loss forcing Pulse's Alert Storage Service to reconnect to RabbitMQ, and a manual Pulse UI configuration reload.

On this page

HSBC-3790

Upgrade Pulse to version 25.1.1â

Bug Fix HSBC-3790
Also In: 2.10.2

We've upgraded Pulse to version 25.1.1.

This upgrade fixes an issue where both the live and upgrade clusters would consume feedback messages, resulting in inconsistent data.

On this page

HSBC-3849

Upgrade Pulse to version 25.1.2â

Bug Fix FTMT-30238 HSBC-3849

We've upgraded Pulse to version 25.1.2.

This change includes an upgrade of Apache Commons BeanUtils to address the CVE-2025-48734 vulnerability.

On this page

HSBC-2122

Remove unused Pulse binariesâ

Rework HSBC-2122

We've removed a set of Pulse binaries related to the following unused services:

- workflow-list-manager-workflow-service
- expired-metrics-pulse-service
- fraudforce-workflow-service
- revelock-workflow-service
- revelock-udfs
- data-migration-service
- outcome-priority-enforcer-service
- enrichment-framework-lib

Along with the removal of these binaries, we have also removed the `import_solutions_udfs` Ansible variable given the binaries it imported are no longer available.

This change should not impact upgrades unless the current deployed version has a very early version installed (e.g. 1.6). In these versions the services were still part of the applications and Pulse required the binaries to be available before the more recent version could remove their reference.

If any deployment from the outlined early versions is planned, there are some steps that are required prior to the update:

- Keep only the following services from each application and removing all remaining using the Pulse UI:
  - Alert Storage Service
  - Data Transfer Pipeline Service
  - HTTP Input Service
  - Historic Events Service
  - Logging Admin Service
  - Workflow List Manager Service
  - Workflow Monitoring Service
- Remove the following properties:
  - `*.services.modules.fraudmodel.import.ListedKeywords`
  - `*.services.modules.datascience.context.import.ListedKeywords`
  - `*.services.modules.datascience.context.rules.import.ListedKeywords`
  - `*.services.modules.fraudrules.import.ListedKeywords`
  - `*.services.modules.fraudmodel.import.ValueFormatTools`
  - `*.services.modules.datascience.context.import.ValueFormatTools`
  - `*.services.modules.datascience.context.rules.import.ValueFormatTools`
  - `*.services.modules.fraudrules.import.ValueFormatTools`
  - `*.services.modules.fraudmodel.import.DateConversionTools`
  - `*.services.modules.datascience.context.import.DateConversionTools`
  - `*.services.modules.datascience.context.rules.import.DateConversionTools`
  - `*.services.modules.fraudrules.import.DateConversionTools`
  - `*.services.modules.fraudmodel.import.Revelock`
  - `*.services.modules.fraudrules.import.Revelock`
  - `*.services.modules.datascience.context.import.Revelock`
  - `*.services.modules.datascience.context.rules.import.Revelock`

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HSBC-3095

Add support for multiple applications of the same typeâ

Feature HSBC-3095

We've added support for deployments with multiple applications of the same type.

The changes address two limitations:

- Allow overriding Liquibase database configurations: Liquibase now supports database configuration overrides per application, similar to what is supported for other configurations. For example, to use a different database for an equal portfolio application, the database override can be set as `<app_slug>_database_pulse_outbound_payments_eventstorage_hostname`
- Allow overriding the API ports for Outbound and Reference Data Service (RDS): this change introduces two new Ansible variables to configure in which port the Pulse and RDS APIs will listen for requests:
    - `api_port` : configure the API port for Outbound Payments
    - `rds_api_port` : configure the API port for RDS

Please refer to Cluster with same portfolio applications for a detailed explanation of the changes and to the Pulse Configuration Reference for deployment information.

On this page

HSBC-3132

Add `customer_id_number` to the Case Manager `Customer ID` search filterâ

Feature WPB-349829 HSBC-3132

We've added the `customer_id_number` field to the Case Manager `Customer ID` search filter, which now includes this field plus `customer_id` and `customer_citizen_id_number`.

On this page

HSBC-3391

Extend the PaymentFulfillment extension to work for Cards and Non-Mon channelsâ

Feature FTMT-29506 HSBC-3391 HSBC-3999

We've updated the PaymentFulfillment extension to support Cards and Non-Mon channel events. On the back of this support we've also configured the fulfillment Cards and Non-Mon schemas, which can be found in AFIS Extensions.

Additionally, the state machines that may trigger PaymentFulfillment are now configurable during deployment by using the Ansible variable `fulfillment_channels_state_machines`. For more information regarding this feature please refer to the Case Manager Configuration Reference.

On this page

HSBC-3412

Extend improved currency to all amount and currency columnsâ

Feature WPB-349833 HSBC-3412

We've standardized the event tables across Case Manager using a single Amount column to represent currency amounts along with the respective currency symbol. This change uses the improved currency support added in Improve currency display in Case Manager.

warning

Please note that the Currency table column no longer exists in any event tables.

On this page

HSBC-3620

Improve the Case Manager `External Status Reason Updater` extensionâ

Feature FTMT-27146 FTMT-27219 HSBC-3396 HSBC-3620

We've improved the Case Manager `External Status Reason Updater` extension by eliminating the need for a database call to load the event's context. This improvement ensures that the extension runs with the exact event context that triggered it.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-3.x/release-3.0.0/individual-release-notes/HSBC-3663

On this page

HSBC-3663

Add Operational Status 'Unresolved'â

Feature FTMT-31854 HSBC-3663

We've added a new Operational status `Unresolved`, with the same functionality as `On-Hold`. It can be selected by clicking on the drop-down option `Move to Unresolved`, which is available in Alert Details and as a bulk action in Transaction History.

On this page

HSBC-3668

Add model information to Cards and Non Mon responseâ

Feature FTMT-29505 HSBC-3668

We've updated the response object from the Cards authorization and Non Mon endpoint, to include the model information that was used to reach an outcome. The `model` field includes the following structured information:

```
    "model": [
        {
            "modelId": "6f41c74c7d491cd8",
            "modelVersionId": "53d00a6cc43347f7",
            "modelName": "dummy_model_test",
            "outcome": "isfraud",
            "score": [
                0.9340874776977786,
                0.0659125223022215
            ]
        }
    ]
```

Copy

See up-to-date API documentation and Swagger definition at [Cards Authorization Endpoints] and [Non Mon].

On this page

HSBC-3696

Fix an issue with the ETL extraction MULTICHANNEL strategyâ

Bug Fix HSBC-3696

We've fixed an issue with the ETL extraction tool using the MULTICHANNEL routing strategy, where the extraction to Case Manager would fail if it found events whose channel was not mapped in `etl_cm_channel_queue_mappings`.

The MULTICHANNEL strategy can now be used to recover the events of a single channel from a multichannel workflow (e.g. recover only Digital Activity events).

We've also upgraded ETL to version 18.2.2 which reduces the amount of logging when a channel-queue mapping is missing.

On this page

HSBC-3745

Add Reference Data fields to the Cards workflow schemasâ

Feature FTMT-32500 HSBC-3745
Feature FTMT-34761 HSBC-3993

We've added the following fields to the `Cards` and `Cards + Outcomes` schemas:

New Fields

Field Name	Datatype	Entity
account_closed_date	Long	Accounts
account_customers	List<String>	Accounts
product_type	String	Accounts
account_iban	String	Accounts
app_fraud_bio_is_bot	String	Customers
app_fraud_bio_is_emulator	String	Customers
app_fraud_bio_is_rat	String	Customers
app_fraud_bio_reason_code	String	Customers
app_fraud_bio_risk_level	String	Customers
app_fraud_bio_score	Double	Customers
app_fraud_overall_decision	String	Customers
app_fraud_overall_decision_text	String	Customers
app_fraud_session_id	String	Customers
app_fraud_source	String	Customers
app_fraud_timestamp	Long	Customers
app_fraud_tmx_device_id	String	Customers
app_fraud_tmx_digital_id_trust_score	Double	Customers
app_fraud_tmx_first_party_score	Double	Customers
app_fraud_tmx_fuzzy_device_id	String	Customers
app_fraud_tmx_fuzzy_device_id_confidence	Integer	Customers
app_fraud_tmx_policy_score	Double	Customers
app_fraud_tmx_proxy_ip	String	Customers
app_fraud_tmx_risk_rating	String	Customers
app_fraud_tmx_summary_reason_codes	String	Customers
app_fraud_tmx_true_ip	String	Customers
business_start_date	Long	Customers
customer_age	String	Customers
customer_alias_last_update_time	Long	Customers

Field Name	Datatype	Entity
customer_annual_turnover	Double	Customers
customer_annual_turnover_last_update_date	Timestamp	Customers
customer_branch_id	String	Customers
customer_business_type	String	Customers
customer_business_type_last_update_date	Timestamp	Customers
customer_businessphone_last_update_at	Timestamp	Customers
customer_corresp_address_last_update_at	Timestamp	Customers
customer_daytime_work_phone	String	Customers
customer_fiv_marker_last_updated_at	Timestamp	Customers
customer_fiv_markers	String	Customers
customer_hmphone_last_update_at	Timestamp	Customers
customer_home_address_last_update_at	Timestamp	Customers
customer_mbphone_last_update_at	Timestamp	Customers
customer_mobile_phone	String	Customers
customer_number_of_accounts	Integer	Customers
customer_occupation_status	String	Customers
customer_occupation_status_last_update_at	Timestamp	Customers
customer_portfolio_segment	String	Customers
customer_portfolio_segment_last_update_date	Timestamp	Customers
customer_reported_income_last_update_date	Timestamp	Customers
customer_voice_id	String	Customers
customer_voice_id_last_change_date	Timestamp	Customers
customer_wrkphone_last_update_at	Timestamp	Customers
delivery	String	Customers
employer_name	String	Customers
employer_name_last_update_date	Long	Customers
employer_start_date	Long	Customers
line_of_business	String	Customers

Note also that the following fields are new to the Customer Reference Data API:

- `customer_age`
- `delivery`

These can now be used to enrich Cards events via Reference Data.

On this page

HSBC-3777

Update Case Manager to version 13.28.0â

Feature FTMT-29151 HSBC-3777

We've upgraded Case Manager to version 13.28.0.

This release introduces the Manage Messages feature, giving HSBC direct control over message translations in Case Manager.

This change also introduces a new set of permissions to control which users can access and manage Case Manager messages:

Messages Permission	Description
`cm:section:settings-message:read:*`	Lets the user see the Manager Messages page in the Settings section.
`cm:resource:message:read:*`	Lets the user read messages.
`cm:action:settings-message:create:*`	Lets the user create messages in the Settings section.
`cm:resource:message:create:*`	Lets the user create messages.
`cm:action:settings-message:update:*`	Lets the user update messages in the Settings section.
`cm:resource:message:update:*`	Lets the user update messages.
`cm:action:settings-message:delete:*`	Lets the user delete messages in the Settings section.
`cm:resource:message:delete:*`	Lets the user delete messages.

Please refer to Case Manager Documentation for more information and to the Permissions Guide for the full permissions list.

On this page

HSBC-3786

Add support for Zookeeper version 3.9â

Feature FTMT-30224 HSBC-3786

This release officially adds support for Zookeeper 3.9.

On this page

HSBC-3789

Update database retention scriptsâ

Feature HSBC-3789

We've updated the database retention scripts to include the following tables:

- AM_EVENT_AUTOMATION
- AM_CASE_FILE
- AM_CASE_SOURCE_TYPE
- AM_EVENT_FILE
- AM_EXTENSION_LOG
- AM_FILE
- AM_EVENT_ACCESS_GROUP

See Retention Scripts for more information.

On this page

HSBC-3798

Add the Non-Monetary channel to Pulse and Case Managerâ

Feature	FTMT-30076	HSBC-3689		
Feature	FTMT-27453	FTMT-29259	FTMT-30076	HSBC-3690
Feature	FTMT-28765	HSBC-3805		
Feature	FTMT-27479	HSBC-3803		
Feature	FTMT-27453	FTMT-30076	HSBC-3798	

We've added the Non-Monetary (Non-Mon) channel to Pulse and Case Manager. Given this work has been done in multiple tasks, we've outlined the major changes in Non-Mon Channel.

On this page

HSBC-3802

Enable the `External Status Reason Updater` extension for Cardsâ

Feature FTMT-28948 HSBC-3802

We have updated the `External Status Reason Updater` extension to allow Cards events.

The extension will now set the event's status reason based on the `external_status_reason` field value sent in the feedback event if it matches any Case Manager configured reasons.

On this page

HSBC-3804

Add Wallet Decision outcome to Cards and Non-Monâ

Feature FTMT-33585 HSBC-3804

We've added the `wallet_decision` outcome to the **Cards** workflow. The field was also added to the **Cards + Outcomes** schema and is included in the real-time response for requests to the Authorization and Non-Mon endpoints.

This outcome can have the following values by default:

- `RED`
- `YLW`
- `GRN`
- `none`

On this page

HSBC-3806

Add new search field to Case Managerâ

Feature FTMT-28258 HSBC-3806

We've added the `Card Number` search field to the common filters in Case Manager's search page. The new search field maps to the `card_pan` event field.

On this page

HSBC-3842

Fix Case Manager logger configuration via Ansibleâ

Bug Fix FTMT-27146 HSBC-3842
Also In: 2.10.3 2.11.1 2.8.6 2.9.7

We have fixed an issue in which some logging levels were not correctly being updated via the `cm_loggers` Ansible variable due to a logger XML formatting issue.

On this page

HSBC-3850

Upgrade Ansible Roles to version 34.1.0â

Feature HSBC-3850

We've upgraded Ansible Roles to version 34.1.0. This version updates the Case Manager pipeline to accept the `SELF_HEALING` Pulse application state, in addition to `STARTED` and `STARTING`.

We also fixed an issue where the pipeline waited for an application that did not always exist. Case Manager will now correctly identify the available applications and wait for them to reach their desired states.

On this page

HSBC-3858

Add Spanish (Mexico) language to Case Managerâ

Feature FTMT-29151 HSBC-3858

Spanish (Mexico) is now available in Case Manager, enabling users to work in Mexican Spanish across the application.

All translations are available in the Manage Messages page, introduced in Update Case Manager to version 13.28.0.

On this page

HSBC-3860

Add scheduled metrics Cassandra table migration guideâ

Docs HSBC-3860

We've added a guide for migrating scheduled metrics Cassandra keyspaces/tables. This can be used to decouple the existing setup and reduce the possibility of metrics collisions in multi-channel deployments.

For more information please refer to Cassandra Profiles Migration.

On this page

HSBC-3880

Add new fields to the Case Manager Cards/Non-Mon external notificationsâ

Feature FTMT-27481 FTMT-34765 HSBC-3880 HSBC-4032

We've added the following fields to the Case Manager Cards and Non-Mon external notifications payload:

New Fields

Field Name	Datatype
lifecycle_id_alt	String
card_transaction_ref_number	String
card_transaction_component_type	String
source	String
customer_portfolio_region	String
customer_portfolio_country	String
customer_portfolio_channel	String
customer_portfolio_product	String
customer_portfolio_type	String
customer_portfolio_class	String
customer_portfolio	String
account_id	String
account_number	String
card_phone_banking_user_id	String
customer_id	String
customer_email	String
customer_name	String
customer_id_personal	String
customer_id_business	String
customer_id_number	String
customer_id_global	String
card_txn_sub_type	String
trans_message_type	String
acquirer_id	String
amount_dbl	Number
trans_model_amt	Number
cardholder_bill_amount	Number
cardholder_bill_currency_code	String

Field Name	Datatype
trans_currency_code	String
currency_code	String
treatment_strategy	String
oob_comm_type	String
oob_notes	String
oob_feedback_notes	String
oob_response_type	String
feedback_type	String
feedback_label	String
external_status_reason	String
channel_type	String
card_pan	String
saf_ind	String
sender_initiated_at	Timestamp
sender_initiated_at_local	String
local_datetime	String
event_occurred_at	Timestamp
crypto_transaction_info	String
card_type	String
outcome_decision	String
wallet_decision	String

For more details please refer to the Notifications Schema page.

On this page

HSBC-3882

Add operational status filter to the Cards Fulfillment extensionâ

Feature FTMT-33601 HSBC-3882

We've extended the Fulfillment extension to accept the `operational_status` state machine in addition to the `status` state machine.

We have also updated the filter logic to support specific statuses. By default, the extension executes for all `status` changes, but limits `operational_status` changes to `On Hold` or `Unresolved`. This behavior can be configured during deployment using the Ansible variable `fulfillment_channels_state_machines`. Please refer to Case Manager Configuration Reference for more information.

On this page

HSBC-3883

Add *Card Status Update* Case Manager extensionâ

Feature	FTMT-33602	HSBC-3883
Feature	FTMT-36284	HSBC-4249

We've added a new Case Manager extension named *Card Status Update* that is able to send notifications to AFIS from Cards (and Non-mon) status update automation rules.

The behaviour of this extension is identical to the Mobile/SMS extensions for Outbound Payments, and to the Inbound Payment Notification extension for Inbound Payments, but with a much simpler schema in the API call made to AFIS.

An example of the schema sent for this extension is as below:


```
{
  "notification_type" : "card_status_update",
  "account_number" : "1005103001044120",
  "operational_status" : "closed",
  "fdz_channel" : "cards",
  "source" : "MSC",
  "trans_message_type" : "TypeA",
  "customer_portfolio_channel" : "portfoliochannel",
  "customer_portfolio_country" : "feedzai",
  "event_occurred_at" : "1773684342216",
  "event_type" : "feedback",
  "customer_portfolio_type" : "portfoliotype",
  "treatment_strategy" : "none",
  "alert_id" : "6e684d92-49ec-309a-9112-4ebc70e8d3ad",
  "customer_portfolio" : "EU_PC_CR_MS_GNRL",
  "customer_portfolio_class" : "elite",
  "self_serve_strategy" : "none",
  "customer_id_global" : "HK",
  "card_transaction_ref_number" : "card_1",
  "card_pan" : "card_1",
  "lifecycle_id" : "ad3c8300-8d7d-40ac-8c2a-9db15e1feef5",
  "customer_portfolio_region" : "US-EAST",
  "system_decision" : "approve",
  "lifecycle_id_alt" : "9e00eeba-daa7-48ed-9c92-c2125108d94e",
  "account_id" : "c0e68e4e-215f-47cb-b3cd-0d67a9ad881f",
  "card_transaction_component_type" : "TRX",
  "smh_source" : "",
  "customer_id" : "cutomer_1",
  "wallet_decision" : "none",
  "card_txn_sub_type" : "TypeB",
  "customer_id_number" : "0HKHSXXXXXX796256",
  "customer_portfolio_product" : "product"
}
```


Copy

By default the extension is enabled, but it can be disabled by setting `card_status_update_extension_enabled` to `false`. For more details please refer to the [Case Manager Configuration Recommendations] and the [AFIS Extensions] reference.

On this page

HSBC-3884

Add new fields to Cardsâ

Feature FTMT-33679 HSBC-3884

We have added the below fields to the Pulse Cards API schema:

Field Name	Datatype
card_relay_attk_code	String
dw_token_msg_rsn_cde_des	String
card_customer_instruction	String
card_customer_instruction_desc	String

The `Cards` and `Cards + Outcomes` schemas were updated with all the above fields.

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HSBC-3911

Add new Cards outcome to API specificationâ

Docs FTMT-33585 HSBC-3911

We've updated the API specification to include the `wallet_decision` outcome in the Cards response.

We've also updated the API titles to be more readable.

Please refer to Non-Mon and Cards Authorization for more details.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-3.x/release-3.0.0/individual-release-notes/HSBC-3918

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HSBC-3918

Fix mismatch in field formats in Cardsâ

Bug Fix FTMT-34154 HSBC-3918
Also In: 2.11.1

We've fixed a mismatch in the Cards API and Workflow schemas where the following fields had a different data type from the requirement.

Field Name	Previous Datatype	New Datatype
user_amount	Double	List<Double>
user_indicator	String	List<String>
user_string	String	List<String>

The field types have been corrected in the API specification, Cards and Cards + Outcomes schemas, and database.

warning

The database columns corresponding to the updated fields will be dropped during the deployment pipeline. Until the deployment completes and the column is re-added, any Cards events that contain these fields will fail to be stored in event storage and won't be recovered.

On this page

HSBC-3920

Document slack Ansible variablesâ

Docs FTMT-34140 HSBC-3920

We have documented all the past and future slack Ansible variables.

Please refer to Pulse Configuration Recommendations for more details.

On this page

HSBC-3975

Add new fields to Cards and Non-Monâ

Feature FTMT-34453 HSBC-3975

We've added the following fields to the Cards and Non-Mon endpoints and workflow schemas:

Field Name	Datatype
info_type	String
dispute_start_date	Timestamp
dispute_reason	String
dispute_notes	String
dispute_submitted_by	String
dispute_end_date	Timestamp
dispute_feedback_notes	String
dispute_refunded	String
oob_start_date	Timestamp
oob_comm_type	String
oob_notes	String
oob_end_date	Timestamp
oob_feedback_notes	String
oob_response_type	String
feedback_type	String
feedback_label	String

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HSBC-3977

Add new extensions for the Cards and Non-Mon channelsâ

Feature FTMT-34467 FTMT-34477 FTMT-34482 HSBC-3977 HSBC-4147

We've created the following extensions to be used for the Cards channel via automation rules and SLA rules:

Extension Name	Notification Type
`Send 2-WAY SMS`	`cards_2_way_sms`
`Send 1-WAY SMS`	`cards_1_way_sms`
`Send Reminder SMS`	`cards_reminder_sms`

The extensions `Send 1-WAY SMS` and `Send Reminder SMS` have also been enabled for the Non-Mon channel, with the same notification types.

Additionally, `Send Mobile Payment Push` has been enabled for Cards to send notifications with the type `cards_mobile_push`.

These extensions are enabled by default, but can be disabled by setting the following to `false`:

- `send_2_way_sms_extension_enabled`
- `send_1_way_sms_extension_enabled`
- `send_reminder_sms_extension_enabled`

For more details please refer to the Case Manager Configuration Recommendations and the AFIS Extensions reference.

On this page

HSBC-3978

Add Card Pan transaction grouping option in Case Managerâ

Feature FTMT-34471 HSBC-3978

We've added the option of grouping events by Card Pan (`card_pan` field) in the transaction history. When selected, the transaction history is filtered to show all transactions that have taken place on the card with the same number.

We've also fixed some inconsistencies on how `card_pan` was showing in the Case Manager UI. The `Card Pan` label will now show the exact value sent in `card_pan` on Event Details and tables, instead of being computed from `card_bin` and `card_last4` .

On this page

HSBC-3997

Make DTP Dataflow worker service account configurableâ

Feature HSBC-3997

We've made the service account for Data Transfer Pipeline Dataflow worker instances configurable during deployment using the `dtp_dataflow_worker_service_account` Ansible variable.

We've also updated the documentation to include the specific permissions this account requires. Please refer to the GCP Permissions page for more details.

On this page

HSBC-4024

Add new Token reference data entityâ

Feature FTMT-34393 FTMT-34969 HSBC-4024

We've added a new Token reference data entity to the Reference Data API. The entity key is `ref_digital_wallet_token` and its schema is the following:

New Token fields

Field Name	Datatype
tar_unique_refernce_number	String
tcn_unique_refernce_number	String
ref_customer_id	String
ref_card_pan	String
ref_tar_local_datetime	String
ref_tcn_local_datetime	String
ref_tar_local_datetime_alt	String
ref_tcn_local_datetime_alt	String
ref_digital_wallet_secu_elem_id	String
ref_digital_wallet_eligible_path	String
ref_digital_wallet_locale_lang	String
ref_card_mobile_match	String
ref_digital_wallet_source_pan	String
ref_digital_wallet_token_device_type	String
ref_digital_wallet_token	String
ref_digital_wallet_provider_reason	String
ref_digital_wallet_provider_risk	String
ref_digital_wallet_token_source	String
ref_digital_wallet_pan_source	String
ref_digital_wallet_provider_acc_score	String
ref_digital_wallet_provider_device_score	String
ref_avs_code	String

We've also added support for Cards enrichment in the Cards workflow by adding all of the Token entity fields to the Cards API and workflow schemas. By default Cards enrichment will be done using `ref_digital_wallet_token` but the field can be changed during deployment using the Ansible variable `cards_token_enrichment_key` .

Please refer to the Pulse Configuration Reference and to the Reference Data Tokens API for more details.

On this page

HSBC-4027

Add new schema fields to Outbound and Inbound Paymentsâ

Feature FTMT-32720 HSBC-4027

We've added the following fields to the Outbound (Transfers and Digital) API and workflow schemas:

Field Name	Datatype	Description
tmx.digital_id_trust_score	Number	Digital ID Trust Score
tmx.chall_reason_code	String	Challenger Risk Reason Code

The following field was also added to Outbound and Inbound Payments:

Field Name	Datatype	Description
vocalink.vocalink_chall_score	Number	Challenge Vocalink Score

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HSBC-4028

Add new fields to the Mobile Outbound extensionâ

Feature FTMT-34647 HSBC-4028

The following fields were added to Case Manager `Mobile Payment Push` extensions for the Outbound Payments channel:

Field Name	Datatype	Description
sender_account_type	String	Type of sender account
receiver_account_type	String	Type of receiver account
customer_account_type	String	Type of customer account
account_type	String	Account type
product_code	String	The code of the product offered by the bank that was used to initiate the transfer
sender_product_type	String	Type of product held

Please refer to the AFIS Extensions reference for more details.

On this page

HSBC-4029

Add Reference Data fields to the Outbound workflow schemasâ

Feature FTMT-34648 HSBC-4029

We've added the following fields to Transfers and Digital Activity to be enriched with Account Reference Data information:

New Transfers and Digital Activity Fields

Field Name	Datatype	Entity
sender_business_start_date	Long	Accounts
sender_customer_occupation_status	String	Accounts
sender_customer_reported_income_dbl	Double	Accounts
sender_line_of_business	String	Accounts
sender_product_type	String	Accounts
receiver_business_start_date	Long	Accounts
receiver_customer_occupation_status	String	Accounts
receiver_customer_reported_income_dbl	Double	Accounts
receiver_line_of_business	String	Accounts
receiver_product_type	String	Accounts

We've added the following fields to the Account Reference Data API:

New Fields

Field Name	Datatype	Entity
business_start_date	Long	Accounts
customer_occupation_status	String	Accounts
customer_reported_income_dbl	Double	Accounts
line_of_business	String	Accounts
product_type	String	Accounts

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HSBC-4030

Add new fields to Outbound Payments and Digital Activityâ

Feature FTMT-32720 FTMT-34654 HSBC-4030

We have added the below elements to the `bioc` tuple in Outbound Payments and Digital Activity API schemas:

Field Name	Datatype
trust_score	Integer
mule_account_score	Integer
max_30_day_score	Integer
chall_risk_score	Double
chall_voice_scam_score	Double

The `Payments` and `Payments + Outcomes` schemas were updated with all the above fields.

On this page

HSBC-4034

Add new enrichers for Cards and Non-Mon eventsâ

Feature FTMT-28255 HSBC-4034

We've added the following enrichers to be used in Cards and Non-Mon events:

Field Name	Enricher Name
sic	sicEnrich
trans_reversal_ind	transReversalIndEnrich
digital_wallet_token_type	digitalWalletTokenTypeEnrich
card_customer_present_details	cardCustomerPresentDetailsEnrich
pos_cardholder_ver_method	posCardholderVerMethodEnrich

For more details on the available values and configuration please refer to the Case Manager Enrichers page.

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HSBC-4045

Add new fields to Cards and Non-Monâ

Feature FTMT-34892 HSBC-4045

We've added the following fields to the Cards and Non-Mon endpoints and workflow schemas:

Field Name	Datatype
customer_id_number	String
customer_id_global	String
dw_auth_mthd	String

On this page

HSBC-4133

Change fields required by the Cards Info endpointâ

Feature FTMT-34893 HSBC-4133

We've changed the mandatory fields of the Cards Info endpoint. When calling this endpoint, only the following fields are now required:

- `card_pan`
- `lifecycle_id`
- `event_occurred_at`
- `customer_id`

For more information, please refer to the API reference.

On this page

HSBC-4145

Add Breach Status option to Cards and Non-Mon SLAsâ

Feature FTMT-34982 HSBC-4145

We've enabled the `Set Breach Status` and `Set Initial Breach Status` options in Breach Actions for SLAs in the Cards and Non-Mon channels.

On this page

HSBC-4146

Add previous transactions to Cards `Send 2-WAY SMS` extensionâ

Feature FTMT-34477 HSBC-4146

We've added the `previous_transactions` field to the Cards `Send 2-WAY SMS` extension. This field will be populated by listing the last transaction that fall under the following criteria:

- Same `card_pan` value
- Same `customer_id` value
- Happened in the last 24 hours (calculated from the time the extension was triggered)
- Is not the event that triggered the extension

For more details please refer to AFIS Extensions.

On this page

HSBC-4152

Add Session ID column to Transaction History and Search Pagesâ

Feature FTMT-34971 HSBC-4152

We've added the `Session ID` column (mapped to `session_id) in the following pages:

- Event Search Page: Transfers tab, between `Beneficiary Account` and `Assignee`
- Alerts Page: Transfers tab, between `Beneficiary Account` and `Assignee`
- Event Details Transaction History: Transfers tab, between `Device ID` and `Assignee`
- Customer Details Transaction History: Transfers tab, between `Beneficiary Account` and `Assignee`

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HSBC-4154

Add support for RabbitMQ 3.13.7â

Feature FTMT-27426 HSBC-4154

Feedzai components now support and have been tested against RabbitMQ `3.13.7`. The RabbitMQ `3.13` branch is still under extended commercial support through December 2027.

On this page

HSBC-4175

Add new fields to Cards and Non-Monâ

Feature FTMT-35019 HSBC-4175

We've added the following fields to the Cards and Non-Mon endpoints and workflow schemas:

Field Name	Datatype
digital_wallet_tar_ref_num	String
digital_wallet_tcn_ref_num	String
account_other_balances_2	Object
account_other_balances_2.amount	Number
account_other_balances_2.type	String
card_ond_vec_value_new	String

On this page

HSBC-4177

Fix `Transaction Sub-Type` field inconsistent mappings in Case Managerâ

Bug Fix FTMT-32085 HSBC-4177

We've fixed inconsistent mappings in Case Manager regarding the `Transaction Sub-Type` field. Previously, the column in the transaction history widget was getting its value from `sender_transaction_code`. We have changed it to get its value from `cmd_payment_sub_type` to keep it consistent with the Search and Alerts pages.

On this page

HSBC-4178

Add fdz_channel to the Payment Fulfillment extensionâ

Feature FTMT-33848 HSBC-4178

The following field was added to Case Manager Payment Fulfillment extension:

Field Name	Datatype	Channels
fdz_channel	String	Transfers, Inbound Payments

Please refer to the AFIS Extensions for more details.

On this page

HSBC-4201

Update Case Manager to version 13.30.0â

Feature FTMT-35919 HSBC-4201

We've upgraded Case Manager to version 13.30.0.

The Case Manager's SLA scheduler now performs a graceful shutdown, which ensures its resources are safely and consistently released when the system stops. This fixes an issue where SLAs would not trigger due to a database lock.

On this page

HSBC-4205

Add `customer_country_code` to Cards, Token and Account Reference Data entitiesâ

Feature FTMT-34393 FTMT-34969 HSBC-4205

We've added the `customer_country_code` field to the Cards, Token and Account Reference Data entities.

This changes ensures that all entities have a common field where the tenant can be specified and allow the enrichment of events in multi-tenanted environments.

The default value of the variable `reference_data_tenant_api_payload_field` has been changed to `customer_country_code`.

For more information, refer to the Pulse Configuration Recommendations.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-3.x/release-3.0.0/individual-release-notes/HSBC-4206

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HSBC-4206

Add `alert_number` to Cards and Non-Monâ

Feature FTMT-35849 HSBC-4206

We've added a new `alert_number` field to Cards and Non-Mon, containing the Alert ID for these channels. The Alert ID generation follows the same approach used for the other channels: it's generated by hashing the full lifecycle ID using SHA-256, encoding the hash with Base64, removing any non-alphanumeric characters, and then taking the last 8 characters of the result.

We've also extended the Alert ID search box in Case Manager to search for `alert_id` and `alert_number` and added both fields to the Cards/Non-Mon Event Details page baseline.

On this page

HSBC-4223

Fix Digital Activity status and reasons bulk actionsâ

Bug Fix FTMT-34940 HSBC-4223

We've fixed an issue with the bulk actions for Digital Activity events in the Transaction History widget, which did not update the correct state machine when labelling alerts as Fraud or Not Fraud.

Labelling Digital Activity events through bulk actions now correctly triggers Automated Rules based on the Digital Activity State Machine and its Reasons can be updated.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-3.x/release-3.0.0/individual-release-notes/HSBC-4239

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HSBC-4239

Add Transaction Block Code to Cardsâ

Feature FTMT-30641 FTMT-34449 HSBC-3667
Feature FTMT-34504 FTMT-34504 HSBC-3984 HSBC-4044
Feature FTMT-32330 HSBC-4031
Feature FTMT-29506 FTMT-34942 HSBC-4120
Feature FTMT-32330 HSBC-4143
Feature FTMT-27481 HSBC-4212

We've added a set of changes for both Pulse and Case Manager to introduce the Transaction Block Code feature, for Cards event.

Given this work has been done in multiple tasks, we've outlined the major changes in [Transaction Block Page](#).

On this page

HSBC-4244

Upgrade Reference Data Service to version 3.7.12â

Feature HSBC-4244
Also In: 2.11.2

We've upgraded the Reference Data Service to version 3.7.12, which addresses the `pac4j` CVE-2026-29000 vulnerability.

On this page

HSBC-4259

Upgrade Pulse to version 25.1.9â

We've upgraded Pulse to version 25.1.9.

This upgrade include the following changes:

HSBC-3866
Bug Fix HSBC-3866
Also In: 2.11.2

Fixes an issue where SMJs did not retrieve values from the secondary channel when different event storages were configured.

HSBC-3998
Bug Fix HSBC-3998
Also In: 2.11.2

Fixes an issue where SMJs did not retrieve values from the secondary channel when the workflow partition keys are incompatible or the second channel has a different number of partitions.

This upgrade also adds new monitoring metrics. For more information, please consult Workflow Validation metrics, App Lifecycle Operations metrics, and Pulse Publish metrics.

HSBC-4156
Feature HSBC-4156
Also In: 2.11.2

Includes additional logging to help identify the root cause of an error that occurs when deleting datasources in the Data Science environment.

For these logs to be available, the following classes must be set to the `DEBUG` level:

- com.feedzai.pulse.service.apps.manager.datascience.AppDataScienceDataSourceManagerImpl
- com.feedzai.pulse.service.apps.manager.AbstractModelManager
- com.feedzai.pulse.service.apps.manager.AppSchemaManagerImpl

This can be done during deployment by using the `pulse_loggers` Ansible variable.

warning

After replicating the issue with the above log levels enabled, the loggers should be disabled to avoid excessive logging.

FTMT-26214 / HSBC-4218
Feature FTMT-26214 HSBC-4218
Also In: 2.11.2

Includes extended support for real-time cross-channel metrics. Cross-channel metrics now have a messaging-based implementation which adds support for partitioned PKernel and workflow partitioning. The new implementation can be enabled by setting the following Ansible variable to `false`:

- `cross_channel_metrics_rmi_based_lookup` (default: `true`)

Please check Pulse Configuration Reference for more information.

HSBC-4259

Feature HSBC-4259
Also In: 2.11.2

Upgrades pac4j from version `4.5.7` to `4.5.9` to address critical CVE-2026-29000.

FTMT-27426 / HSBC-4126
Feature FTMT-27426 HSBC-4126

Pulse now supports RabbitMQ `4.2`. As part of this upgrade, it is now possible to configure Pulse to use RabbitMQ `QUORUM` queues.

This is configurable using the `rabbitmq_queue_type` Ansible variable. To ensure backwards compatibility, the default value of the variable is `CLASSIC`.

For more information please refer to the Pulse Configuration Page.

note

Although classic queues are still supported in RabbitMQ 4.2 please note they aren't mirrored which removes high availability. The default value is only meant as an upgrade point from 3.X versions.

On this page

HSBC-4280

Upgrade Case Manager to version 13.31.0â

Feature FTMT-27426 HSBC-4280

We've upgraded Case Manager version to 13.31.0.

Case Manager now supports RabbitMQ 4.2. As part of this upgrade, it is now possible to configure Case Manage to use RabbitMQ `QUORUM` queues.

This is configurable using the `rabbitmq_casemanager_use_quorum_queues` Ansible variable. To ensure backwards compatibility, the default value of the variable is `False`.

For more information please refer to the Case Manager Configuration Page.

note

Although classic queues are still supported in RabbitMQ 4.2 please note they aren't mirrored which removes high availability. The default value is only meant as an upgrade point from 3.X versions.

Upgrade Data Extraction Pipelines to version 2.2.0â

Feature FTMT-27426 HSBC-4280

We've upgraded the Data Extraction Pipelines to version 2.2.0. As part of this upgrade, it is now possible to configure the Data Extraction Pipelines to use RabbitMQ `QUORUM` queues.

This is configurable using the following Ansible variables:

- `rabbitmq_casemanager_queue_type_green`
- `rabbitmq_casemanager_queue_type_blue`

To ensure backwards compatibility, the default value of both variables is `CLASSIC`.

For more information please refer to the Data Extraction Pipelines Configuration Page.

Upgrade ETL Tool to version 18.3.0â

Feature FTMT-27426 HSBC-4280

We've upgraded the ETL extraction tool to version `18.3.0`. As part of this upgrade, it is now possible to configure the ETL tool to use RabbitMQ `QUORUM` queues. Please note, this does not have an impact on the regular ETL extractions used to extract events to GCS buckets for analytics. This will only be applicable to the recovery of events from Pulse to Case Manager.

This is configurable using the `etl_cm_rabbitmq_queue_type` Ansible variable.

To ensure backwards compatibility, the default value of both variables is `CLASSIC`.

For more information please refer to the ETL Configuration Guide.

On this page

Non-Mon Channel

This page highlights the main changes introduced in version 3.0.0 to support the new Non-Monetary channel. These include both Pulse and Case Manager and their corresponding FTMT ticket ids.

Pulseâ

In this section you can find the changes introduced in Pulse to support the new Non-Mon Channel.

Setup custom Alert Storage Service to segregate Non-Mon events from Cards paymentsâ

Feature FTMT-30076 HSBC-3689

We've setup the custom Alert Storage Service to separate Non-Mon events from Cards payments. This service will now set the channel of the event based on the API that received it (similar to what happens with Digital-Activity and Outbound Payments). Cards events will have `fdz_channel` set to `CARDS` while Non-Mon events the `NON-MON` value. This field can then be used to determine to what channel each event belongs when querying the Event Storage tables.

This change also includes a new RabbitMQ queue (`fdz-sq-EVENT_QUEUE_NON-MON`) to publish Non-Mon events to be consumed by Case Manager.

Add Non-Mon permissionsâ

Feature FTMT-27453 FTMT-29259 FTMT-30076 HSBC-3690

We've created the below new permissions for the Non-Mon channel.

- `cm:channel:non-mon:read-only`
- `cm:channel:non-mon:read-write`
- `cm:state-machine:event-status:write:status_non_mon`
- `cm:state-machine:event:read:status_non_mon`
- `cm:state-machine:event-status:write:status_non_mon`
- `cm:state-machine:status-reason:delete:status_non_mon`
- `cm:state-machine:status-reason:read:status_non_mon`
- `cm:state-machine:status-reason:write:status_non_mon`

These can be used to control the user access to Non-Mon in Case Manager.

For more details please refer to the Permissions Page.

Enable Feedback and Info Endpoints for Non-Mon eventsâ

Feature FTMT-28259 HSBC-4046 HSBC-4151

We've added two new API endpoints for Non-Monetary events.

Endpoint URLs:

- `/api/nonmon/{lifecycle_id}/feedback/fraud`
- `/api/nonmon/{lifecycle_id}/info`

This work extends the Non-Mon endpoint delivered in Add Non-Mon Endpoint.

For more information please refer to the API documentation on the Feedback Non-Mon and Info Non-Mon endpoints.

Enable GET Endpoint for Non-Mon eventsâ

Feature FTMT-27479 HSBC-3803

We've added a new API endpoint that allow the retrieval of Non-Monetary events previously submitted. By using the `GET` method with a specific and valid `lifecycle_id`, the full event details can now be accessed.

Endpoint URL: `/api/nonmon/{lifecycle_id}`.

This work extends the Non-Mon endpoint delivered in Add Non-Mon Endpoint.

For more information please refer to the API documentation.

Case Managerâ

In this section you can find the changes introduced in Case Manager to support the new Non-Mon Channel.

Add Non-Mon database supportâ

Feature FTMT-27453 FTMT-30076 HSBC-3798

We've added Non-Mon database support to Case Manager. The changes include:

- New database patches to create all the required resources in the Case Manager Non-Mon database
- A new set of Ansible variables to configure Case Manager's Non-Mon database connection. These follow the same structure as for all other channels by setting the channel suffix to the variable (e.g. `database_casemanager_engine_non_mon`)
- Updated retention scripts with the new table which can be found in Retention Scripts

Please refer to Case Manager Configuration Reference for more information.

Add Non-Mon channelâ

Feature FTMT-27453 FTMT-29259 FTMT-30076 HSBC-3690

We've introduced the Non-Mon channel in Case Manager. Please find below information about the new channel:

- The channel is enabled by default but users require additional permissions to see it, similar to all other channels (see Add Non-Mon permissions)
- Non-Mon event pages are based on Cards, meaning that by default they will be alike. Please note that translations are shared and therefore a change to a Cards translation might affect Non-Mon pages
- Advanced search uses the same fields as Cards
- Logstash notifications will show the same payload as Cards

Add Non-Mon to the reasons filterâ

Feature FTMT-28765 HSBC-3805

We've extended the existing tactical solution where Case Manager hides reasons for channels other than the one of the current event. The solution requires all Non-Mon reasons to have `Non-Mon` as their prefix in order for them to be hidden.

As an example, when marking an Outbound Payments event as Fraud/Not-Fraud in Case Manager, all the reasons that have the prefixes `Cards`, `Inbound` and `Non-Mon` will be hidden. When marking a Non-Mon event as Fraud or Not Fraud, only reasons with the `Non-Mon` prefix will be available.

On this page

HSBC-4353

Add network configuration for DTPâ

Feature HSBC-4353

We've added support to configure the network and subnetwork in which DTP Dataflow worker nodes are created. This can be configured during deployment using the following Ansible variables:

- `dtp_dataflow_network` : sets the Dataflow job network
- `dtp_dataflow_subnetwork` : sets the Dataflow job subnetwork

warning

Note that the two parameters are not required to be set at the same time. For example, when setting `subnetwork` the `network` can be omitted and GCP will determine the network automatically.

For more details on Dataflow networking refer to Specify a network and subnetwork.

We've also added a new property to configure if worker pool machines should be created with public IP addresses. This is available via the `dtp_dataflow_use_public_ips` variable and defaults to `false` .

Please refer to DTP Network Configuration and to the DTP Configuration Reference for more information.

On this page

HSBC-4354

Fix `Move to Unresolved` option visibility for Cards channelâ

Bug Fix HSBC-4354

The `Move to Unresolved` action in the Cards channel's dropdown menu was previously restricted by the `cm:resource:alert:status-bulk:*` permission. We have removed this permission requirement, allowing users to see the "Move to Unresolved" option regardless of their permissions.

On this page

HSBC-4358

Fix database error preventing SLAs from breachingâ

Bug Fix FTMT-36820 HSBC-4358
Also In: 2.11.3

We've fixed an inconsistency between database column sizes that prevented SLAs from breaching when Case Manager was deployed on machines with very long hostnames (70 or more characters).

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-3.x/release-3.0.1/individual-release-notes/HSBC-4366

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HSBC-4366

Fix Reasons dropdown options becoming unavailableâ

Bug Fix FTMT-36949 HSBC-4366

We've fixed an issue which, under certain conditions, caused all options to disappear from the Reasons dropdown when trying to set an event's Block Code or Operational Status.

Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/release-information/release-3.x/release-3.0.1/individual-release-notes/HSBC-4382

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HSBC-4382

Revert `bioc` nested fields orderâ

Bug Fix HSBC-4382
Also In: 2.11.3

We've fixed an issue where the `bioc` nested fields order introduced in version 2.11 caused replica ingestion failures during cluster upgrades.

The field order has been reverted to its pre-2.11 state to prevent similar issues in future upgrades.

info

This issue only affected Outbound, as the field order did not change for other channels.

On this page

HSBC-4407

Fix block code expiration value not computing with offsetâ

Bug Fix FTMT-37370 HSBC-4407

We've fixed an issue where the fields `cm_transaction_block_date` and `cm_transaction_block_expiration` in Case Manager extension payloads were not computed using the configured offset days (default `5`).

On this page

HSBC-4425

Add columns to Transfers Transaction Historyâ

Feature FTMT-36746 HSBC-4425

We've added the following HSBCNet columns to the Transaction History widget in the Event Details page, for the Transfers channel:

Column Name	Field Name
User ID	user_id
Customer Country Code	customer_country_code
Sender Bank Country Code	sender_bank_country_code
Receiver Bank Name	receiver_bank_name
Receiver Bank Country Code	receiver_bank_country_code
Receiver Account Bank ID	receiver_account_bank_id
Payment Held Indicator	sender_transaction_held
Transaction Sign Status	sender_transaction_sign_status
Payment Message Source	payment_message_source
Transaction Notes	text_for_receiver
Customer Business Type	customer_business_type

On this page

HSBC-4426

Add columns to Digital Activity Transaction Historyâ

Feature FTMT-36748 HSBC-4426

We've added the following columns to the Transaction History widget in the Event Details page, for the Digital Activity channel:

Column Name	Field Name
IP Country	device_ip_country_code
IP/Device ISP	device_geo_ip_isp

Additionally, the following columns specific to HSBCNet were added:

Column Name	Field Name
IP Address v6	device_ip_address_v6
Device User-Agent	device_user_agent
Malware Score	pinpoint_risk_score

On this page

HSBC-4427

Add support for adding IPv6 addresses to `PosNeg Listed IP Addresses` from Case Managerâ

Feature FTMT-36830 HSBC-4427

We've enabled the ability to add IPv6 addresses (`device_ip_address_v6`) entries to the `PosNeg Listed IP Addresses` list directly from the event details page in Case Manager. This is the same list already used by IPv4 addresses (`device_ip_address`).